PASSBOOK · CA INTERMEDIATE · GROUP II

Paper 6A

Financial Management

Full Lessons

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32 lessons in 9 chapters · A4 edition · made 2026-10-20 from the PassBook lessons

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CHAPTER 1

Scope and Objectives of Financial Management

1.1 Objectives: profit maximisation, wealth maximisation and value creation

Why profit maximisation is at best a limited objective, how wealth (value) maximisation measures success through cash flows, timing and risk, and where the two goals part company.

In plain words

The record quarter that the market did not like

Tapri Tea Co. Ltd has just announced its best quarter ever. Kesar is ordering sweets for the whole office.

> **Kesar:** Record profit, Dev! I postponed the cold-chain truck overhaul and froze all hiring. Sab adjust ho jayega.
>
> **Dev:** And our share price fell 6% the day we announced it.
>
> **Kesar:** Who sells a share after a record profit?
>
> **Dev:** People who are pricing next year's broken trucks. The market discounts too, Kesar. Even faster than I do.

Kesar was chasing **profit maximisation**: taking each decision so that this period's profit is as large as possible. The shareholders were judging something else, **wealth maximisation**: what their shares are worth today, which depends on all the cash the business will produce in future, how soon it comes and how risky it is. Postponing the overhaul raised this quarter's profit and lowered the value of every future quarter.

Profit is easy to count, but the word hides choices: profit of which period, before or after tax, in total or per share? It ignores risk (a wild gamble and a safe bet with the same profit look equal) and timing (₹1 crore next year and ₹1 crore in ten years look equal). Wealth, measured by the market price of the shares, takes all three into account. That is why the Study Material treats wealth maximisation as the better objective for a world that is uncertain and stretches over many periods, and why it calls the same idea **value creation**.

Think of it like

The mango orchard

Profit maximisation counts this season's mangoes. You can pick every raw mango today and chop the branches for firewood: a record season and a dead tree. Wealth maximisation asks what a buyer would pay for the orchard today. That price counts every future season, how soon each harvest comes, and how likely the storms are.

Key idea

Profit maximisation asks "how much profit?"; wealth maximisation asks "how much are the owners' shares worth?", and so counts cash flows, their timing and their risk. In an uncertain, multi-period world, wealth (value) maximisation is the better objective of financial management.

How it works

Why the finance manager needs a goal at all

Efficient financial management requires the existence of some objectives or goals, because the judgment as to whether or not a financial decision is efficient must be made in the light of some objective. Various objectives are possible; the Study Material takes up two for discussion: **profit maximisation** and **wealth / value maximisation** (SM 1.12).

1. Profit maximisation, and why it is at best a limited objective

It has traditionally been argued that the primary objective of a company is to earn profit, so the objective of financial management is also profit maximisation: the finance manager makes decisions so that the profits of the concern are maximised, and each alternative is judged by whether it gives maximum profit (SM 1.13).

But profit maximisation cannot be the sole objective of a company; it is **at best a limited objective**. If profit is given undue importance, a number of problems can arise; the Study Material discusses four, in this order (SM 1.13):

1. **The term profit is vague.** It does not clarify what exactly it means and conveys a different meaning to different people: profit may be short term or long term; it may be total profit or rate of profit.
2. **It ignores risk.** There is a direct relationship between risk and profit: many risky propositions yield high profit, and the higher the risk, the higher the possibility of profits. If profit maximisation is the only goal, the risk factor is altogether ignored, and the finance manager will accept highly risky proposals too if they give high profits. In practice risk is a very important consideration and has to be balanced with the profit objective.
3. **It ignores the time pattern of returns.** Proposal A may give more profit than proposal B, yet if A's returns begin to flow, say, 10 years later, B may be preferred: its overall profit is lower, but its returns flow earlier and quicker.
4. **It is too narrow.** It fails to take into account social considerations and the obligations to the various interests of workers, consumers and society, and ethical trade practices. If these are ignored, a company cannot survive for long: profit maximisation at the cost of social and moral obligations is a **short-sighted policy**.

2. Wealth maximisation, also called value creation

Shareholders' wealth is the result of **cost-benefit analysis adjusted for timing and risk**, that is, for the time value of money (SM 1.14):

$$\text{Wealth} = \text{PV of benefits} - \text{PV of costs}$$

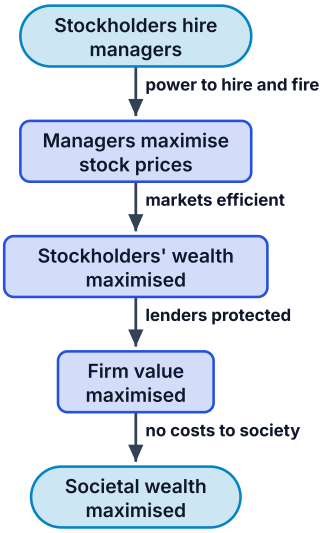
The benefits are measured in **cash flows**, not accounting profit. The **shareholder value maximisation model** holds that the primary goal of the firm is to maximise its **market value**, so business decisions should seek to increase the net present value of the economic profits of the firm. To measure and maximise shareholders' wealth, the finance manager follows three things (SM 1.14):

1. **Cash flow approach, not accounting profit**: count the rupees that actually come in and go out.
2. **Cost-benefit analysis**: set every benefit against every cost of the decision.
3. **Application of the time value of money**: bring each cash flow to its present value, at a rate that reflects its risk. Wealth is the present value of the benefits less the present value of the costs.

How is the wealth of a firm measured? By the market. In Van Horne's words, the value of a firm is represented by the **market price of the company's common stock**: the focal judgment of all market participants, which takes into account present and prospective future earnings per share, the timing and risk of these earnings, the dividend policy of the firm and many other factors. "The market price serves as a performance index or report card of the firm's progress", showing how well management is doing on behalf of the shareholders (SM 1.14).

The Study Material sets out the chain of reasoning behind the goal, with the condition each link depends on (SM 1.15):

Why maximising the share price should maximise everyone's wealth



Each step follows only because of the condition on its arrow: stockholders have absolute power to hire and fire managers (so managers set aside their own interest); markets are efficient; lenders are fully protected from shareholders' actions; no costs are created for society.

Read the arrows as warnings as well as reasons: when managers chase their own goals, the first link breaks (the **agency problem**, in agency problem, environment and distress).

Value creation. The Study Material treats wealth maximisation and value creation under one heading (SM 1.14). ICAI's Suggested Answer to a value creation case puts it this way: value creation refers to making strategic decisions that generate sustainable returns exceeding the cost of capital (MTP Series II for September 2025, Q4(a)). A project creates value only if it earns more than the cost of the funds it uses (the cost of capital); the rupee measure of that is its NPV (net present value).

3. Other goals, and why wealth still comes first

A business may follow other goals too (SM 1.15):

- Achieving a higher growth rate
- Attaining a larger market share
- Gaining leadership in the market in terms of products and technology
- Promoting employee welfare
- Increasing customer satisfaction
- Improving community life, supporting education and research, solving societal problems, etc.

These are important, but the primary goal remains wealth maximisation, as it is critical for the very existence of the business. If it is not met, the public and institutions lose confidence in the enterprise and will not invest further in its growth; and if growth is restricted, the other goals, such as community welfare, will not be fulfilled either (SM 1.16).

4. Profit versus value: where the two goals part company

In any company the management is the decision-taking authority, and as a normal tendency it may pursue its own goals (**profit maximisation**). But where there is significant outside participation (shareholding, lenders and so on), management cannot pursue only its own goals, because of the **constant supervision of the various stakeholders**: employees, creditors, customers, government. Each evaluates management by the fulfilment of its own objective, and the survival of management is threatened if the objective of any of them remains unfulfilled (SM 1.16).

The wealth maximisation objective is generally in accord with the interests of the various groups (owners, employees, creditors and society), so it may be consistent with the management objective of **survival**. Owing to the limitations of profit maximisation (timing, social consideration and so on), in today's real world, which is **uncertain and multi-period**, wealth maximisation is a better objective. Where the time period is short and the degree of uncertainty is not great, the two amount to essentially the same (SM 1.16).

Two ways profit can be bought at the cost of value (SM 1.17): a costly investment may make losses in the short term but substantial profits in the long term, so cutting it raises today's profit; and a firm that wants to show a short-term profit may postpone major repairs or replacement, which is likely to hurt its long-term profitability (Kesar's trucks).

Risk, in ICAI's own illustration. Two products earn ₹10,000 and ₹11,000 a year for five years, ₹50,000 against ₹55,000. Profit maximisation favours the second. But if it is riskier, the choice is not as straightforward as the figures seem: a trade-off exists between risk and return, and shareholders would demand a sufficiently large return to compensate for the greater risk (SM 1.18, Illustration 1).

Formats and formulas

Formula

Shareholders' wealth

$$\text{Wealth} = \text{PV of benefits} - \text{PV of costs}$$

PV = present value. Benefits and costs are cash flows, discounted for timing and risk (SM 1.14).

Formula

Value of a firm

$$V = N \times MP \qquad \text{or} \qquad V = V_e + V_d$$

V = value of the firm; N = number of shares; MP = market price of a share; Ve = value of equity; Vd = value of debt (SM 1.15).

The comparison ICAI prints

Reproduce this table from memory: ICAI's suggested answers copy it row for row (SM 1.17).

Point	Profit maximisation	Wealth maximisation
Objective	Large amount of profits	Highest market value of shares
Advantages	(i) Easy to calculate profits; (ii) easy to determine the link between financial decisions and profits	(i) Emphasises the long-term gains; (ii) recognises risk or uncertainty; (iii) recognises the timing of returns; (iv) considers shareholders' return
Disadvantages	(i) Emphasises the short-term gains; (ii) ignores risk or uncertainty; (iii) ignores the timing of returns; (iv) requires immediate resources	(i) Offers no clear relationship between financial decisions and share price; (ii) can lead to management anxiety and frustration

Definition

Shareholder value maximisation model

The model which holds that the primary goal of the firm is to maximise its market value, and implies that business decisions should seek to increase the net present value of the economic profits of the firm. (SM 1.14)

Worked examples

Worked example **The kiosk and the tea estate**

Tapri Tea Co. Ltd has ₹1,000 lakh to invest today and two five-year proposals:

- **Station kiosks**: 60 more Tapri Express kiosks at railway stations, bringing in ₹320 lakh at the end of every year for five years.
- **Tea estate**: a Darjeeling estate whose new bushes take three years to mature, bringing in nothing in years 1 to 3, then ₹900 lakh in year 4 and ₹950 lakh in year 5.

Treat each year's inflow as its profit before charging the ₹1,000 lakh cost. Tapri's shareholders expect 12% on a proposal of normal risk. Present value factors:

Year	PV factor at 12%	PV factor at 15%
1	0.893	0.870
2	0.797	0.756
3	0.712	0.658
4	0.636	0.572
5	0.567	0.497

Required:

(i) Rank the proposals under profit maximisation, using total profit over the five years.

(ii) Calculate the wealth each creates for shareholders at 12%, and rank them under wealth maximisation.

(iii) The estate's crop depends on the weather, and the shareholders want 15% from it. Recalculate its wealth, and state which limitations of profit maximisation the case shows.

THE WORKING

(i) Profit maximisation

Particulars	Kiosks (₹ lakh)	Tea estate (₹ lakh)
Total inflows, years 1–5	1,600	1,850
Less: Cost	1,000	1,000
Total profit	**600**	**850**

Profit maximisation ranks the **tea estate first**: ₹850 lakh against ₹600 lakh.

(ii) Wealth maximisation at 12%

Wealth = PV of benefits − PV of costs.

- Kiosks: an equal annual flow, so PV = ₹320 lakh × (0.893 + 0.797 + 0.712 + 0.636 + 0.567) = 320 × 3.605 = **₹1,153.60 lakh**.
- Tea estate: PV = 900 × 0.636 + 950 × 0.567 = 572.40 + 538.65 = **₹1,111.05 lakh**.

Particulars	Kiosks (₹ lakh)	Tea estate (₹ lakh)
PV of benefits at 12%	1,153.60	1,111.05
Less: PV of costs	1,000.00	1,000.00
Wealth created	**153.60**	**111.05**

Wealth maximisation ranks the **kiosks first**: the estate's larger profit arrives so late that it is worth less today.



Amber bars: total profit, which ranks the tea estate first. Blue bars: wealth created (PV of benefits less PV of costs), which ranks the kiosks first.

(iii) The estate at 15%

PV = 900 × 0.572 + 950 × 0.497 = 514.80 + 472.15 = ₹986.95 lakh. Wealth created = 986.95 − 1,000 = **−₹13.05 lakh**: at a rate that reflects its risk, the estate destroys value although it shows the largest profit.

Limitations shown. Profit maximisation (a) ignores the time pattern of returns: ₹850 lakh, mostly in years 4 and 5, was ranked above ₹600 lakh spread evenly from year 1; (b) ignores risk: the weather-dependent estate was judged at the same yardstick as the kiosks; and (c) is vague: "profit" here meant total profit over five years, and a rate of profit or a one-year profit would give yet other answers. Wealth maximisation uses cash flows, cost-benefit analysis and the time value of money, so Tapri should choose the **station kiosks**.

Worked example

The kulhad switch

Since 2025, Tapri's cafés in 12 cities have served chai only in kulhads (clay cups) and compostable paper cups, after three municipal corporations banned plastic-lined cups. Packaging cost has risen ₹14 crore a year. At the annual general meeting a shareholder says: "The company exists to maximise profit. Every rupee spent beyond what the law demands in the other cities reduces profit and is against the objective of financial management."

Examine the shareholder's view in the light of the objectives of financial management. (4 marks)

MODEL ANSWER

Relevant concept. Profit maximisation cannot be the sole objective of a company; it is at best a limited objective. It is too narrow: it fails to take into account social considerations and the obligations to workers, consumers and society and ethical trade practices; if these are ignored a company cannot survive for long, and profit maximisation at the cost of social and moral obligations is a short-sighted policy. The objective of financial management is the maximisation of shareholders' wealth, measured by the market value of the shares, which takes into account the timing and risk of all future cash flows.

Facts of the case. The switch costs ₹14 crore a year more than plastic-lined cups. It is required by law in three cities and adopted voluntarily in the rest.

Analysis. The extra cost lowers the current year's accounting profit, which is all the shareholder is looking at. Wealth maximisation looks at the present value of all future benefits less all costs. The voluntary switch may protect future cash flows: it reduces the risk of fines and disruption if more cities ban plastic-lined cups, and supports customer satisfaction and the brand. Goals such as increasing customer satisfaction and improving community life are important, but the primary goal remains wealth maximisation, as it is critical for the very existence of the business enterprise; and the wealth maximisation objective is generally in accord with the interests of owners, employees, creditors and society.

Conclusion. The shareholder's view applies the narrow profit maximisation objective. Judged by wealth maximisation, the switch is consistent with the objective of financial management if the present value of the future benefits it protects exceeds the present value of its extra cost, and the board should evaluate it on that basis, not by its effect on one year's profit.

Worked example **Four claims from the board pack**

State, with brief reasons, whether each statement is true or false. (4 marks)

(i) Where the time period is short and the degree of uncertainty is not great, profit maximisation and wealth maximisation amount to essentially the same.

(ii) Wealth maximisation ignores dividends, since it looks only at the market price of the shares.

(iii) Under wealth maximisation the finance manager measures the benefits of a decision in terms of accounting profit.

(iv) Tapri has 4 crore equity shares. If its share price rises from ₹330 to ₹345 because of a decision, the decision has added ₹60 crore to shareholders' wealth.

MODEL ANSWER

Statement	True or false	Reason
(i)	True	With a short horizon and little uncertainty, timing and risk hardly differ between alternatives, so the two objectives amount to essentially the same (SM 1.16).
(ii)	False	The market price takes into account the timing and risk of earnings and the dividend policy of the firm (Van Horne, SM 1.14).
(iii)	False	Benefits are measured in terms of cash flow; the finance manager follows the cash flow approach, not accounting profit (SM 1.14).
(iv)	True	Value of equity = number of shares × market price: 4 crore × (₹345 – ₹330) = ₹60 crore.

Exam angle

In the exam

Where this topic appeared

In the ICAI documents read for this lesson, the objectives of financial management were asked 22 times, always as theory: in past papers and MTPs in the theory question (Question 4 in the new scheme, Question 6 in the old), for 2 or 4 marks; in RTPs among the miscellaneous questions:

- **Limitations or disadvantages of profit maximisation**: MTP Series I for May 2024 Q4(a) (4 marks); September 2025 Q4(c) (any two, 2 marks, an OR choice); RTP January 2026 Q11(d); old scheme November 2023 Q6(c) (2 marks), RTP May 2022 Q10(a) and RTP July 2021 Q10(i) ("cannot be the sole objective").
- **"Not an operationally feasible criterion"**: MTP Series I for May 2026 Q4(a) (4 marks), MTP Series I for September 2026 Q4(c) (2 marks); a case form, a company raising prices sharply for short-term profit, in MTP Series I for January 2025 Q4(c) (2 marks); old scheme RTP May 2020 Q10(i).
- **Wealth maximisation**: objectives and advantages, January 2025 Q4(b) (4 marks, answered with "highest market value of shares" and the four advantages); old scheme May 2022 Q6(c) (2 marks); superiority over profit maximisation, RTP May 2024 Q8(i), RTP May 2025 Q10(a).
- **Comparison**: advantages and disadvantages of both, RTP September 2024 Q12(a); objective, risk, time horizon and sustainability, May 2026 Q4(c) (2 marks); conflicts in profit versus wealth maximisation, MTP Series II for May 2025 Q4(c) (2 marks); old scheme: distinguish the two, MTP Series II for May 2023 Q6(i); advantages and disadvantages of wealth maximisation, MTP Series I for July 2021 Q6(a); "write two main objectives", MTP Series I for December 2021 Q6(c) (2 marks).
- **Value creation case**: a project with high short-term accounting profit against one with long-term cash flows, MTP Series II for September 2025 Q4(a) (4 marks).
- **True or false**: "maximising market price per share maximises the wealth of shareholders" (true) and "lower risk and higher return is the risk-return trade-off" (false), May 2024 Q4(a)(i)–(ii).

In the exam

How the marks are earned

The Suggested Answers reproduce the Study Material's lists in its order: the four limitations (vague; risk; timing; too narrow), and the comparison table's rows (objective, advantages, disadvantages). A 2-mark answer is the list with one line each; a 4-mark answer adds a sentence of explanation per point and, for "comment" questions, a verdict ("This statement is true because..."). In the May 2026 answer, sustainability was answered as "not sustainable in long term; sustainability is low" for profit maximisation and "critical for the very existence of the business enterprise; sustainability is high" for wealth maximisation.

Trick

V-R-T-N: the four limitations of profit maximisation

Vague, **R**isk ignored, **T**iming ignored, **N**arrow. Write them in this order: it is ICAI's (SM 1.13). Condition: these are the limitations of profit maximisation; the disadvantages in the comparison table are a different list (short-term gains, risk, timing, immediate resources).

Answer frame

"Explain the limitations of profit maximisation" (2 or 4 marks)

1. One line of context: profit maximisation cannot be the sole objective; it is at best a limited objective.
2. The four limitations in ICAI's order, each with its one-line reason: vague (short or long term, total or rate); risk ignored (higher risk, higher possible profit, risky proposals accepted); timing ignored (proposal A's returns 10 years later); too narrow (social considerations, short-sighted policy).

"Distinguish between profit maximisation and wealth maximisation"

Point	Profit maximisation	Wealth maximisation
Objective	Large amount of profits	Highest market value of shares
Measure	Accounting profit	Cash flows, discounted (PV of benefits – PV of costs)
Risk	Ignores risk or uncertainty	Recognises risk or uncertainty
Timing	Ignores the timing of returns	Recognises the timing of returns
Time horizon	Emphasises short-term gains	Emphasises long-term gains
Sustainability	Low: not sustainable in the long term	High: critical for the very existence of the business

"Comment" on a statement or a case

Verdict first ("The statement is true / not correct because..."), then the relevant limitations or advantages from the Study Material, then the facts of the case applied to them, then the conclusion: which objective the decision serves, and which ICAI prefers (wealth maximisation, in an uncertain multi-period world).

Common mistakes

⚠ **Common mistake**

Hall of Shame: "wealth maximisation means making the owners rich at everyone else's cost"

Fix: the Study Material says the opposite: the wealth maximisation objective is generally in accord with the interests of owners, employees, creditors and society, and so may be consistent with management's objective of survival (SM 1.16). Its price is a report card that counts the long term.

⚠ **Common mistake**

Writing the comparison table's disadvantages as the four limitations

"Limitations of profit maximisation" asks for SM 1.13: vague, risk, timing, too narrow. The table's disadvantages (short-term gains, risk, timing, requires immediate resources) are a separate list. **Fix:** answer the list the question names; when the question says "disadvantages", use the table.

⚠ **Common mistake**

Calling the larger total profit the larger wealth

The tea estate made ₹850 lakh of profit and created ₹111.05 lakh of wealth (−₹13.05 lakh at its risk-adjusted 15%). **Fix:** wealth is present value of benefits less present value of costs; a larger profit that comes later or with more risk can be worth less.

⚠ **Common mistake**

Saying wealth maximisation "always" differs from profit maximisation

Fix: where the time period is short and the degree of uncertainty is not great, the two amount to essentially the same (SM 1.16). They part company when time and risk matter.

Practice

Check yourself

1 Case scenario. Kulhad Crafts Ltd, a pottery supplier to Tapri, can either (P) earn ₹40 lakh profit this year by skipping the relining of its kilns, after which output will fall for the next three years, or (Q) earn ₹25 lakh this year after relining and keep output steady. The finance manager picks P because "our objective is maximum profit". Which statement about the finance manager's choice is correct?

- A It is consistent with wealth maximisation, because P gives the higher profit this year
- B It is inconsistent with profit maximisation, because Q gives the higher profit this year
- C It reflects profit maximisation's emphasis on short-term gains; wealth maximisation would compare the present value of each option's cash flows over all four years
- D It is correct under both objectives, because the time period involved is short

2 Asked for "the limitations of profit maximisation", Bunty writes: emphasises short-term gains; ignores risk; ignores the timing of returns; requires immediate resources. What has he done?

- **A** Written the four limitations in the Study Material's order
- **B** Written the "disadvantages" row of ICAI's comparison table, leaving out "profit is vague" and "too narrow"
- **C** Written the disadvantages of wealth maximisation
- **D** Written the advantages of wealth maximisation, turned round

3 A proposal costs ₹500 lakh today and brings ₹300 lakh at the end of year 1 and ₹300 lakh at the end of year 2. The shareholders' required return is 10% (PV factors 0.909 and 0.826). What wealth, in ₹ lakh, does it create? Give two decimals.

Answer: _____

4 Dev's revision slide sorts the rows of ICAI's comparison table. Which of its statements are correct? Select all that apply.

- **A** "Ignores the timing of returns" is a disadvantage of profit maximisation
- **B** "Requires immediate resources" is a disadvantage of wealth maximisation
- **C** "Can lead to management anxiety and frustration" is a disadvantage of wealth maximisation
- **D** "Easy to determine the link between financial decisions and profits" is an advantage of wealth maximisation
- **E** "Considers shareholders' return" is an advantage of wealth maximisation

More than one answer may be right.

5 Tapri's new drive-through format earns a steady 13% on the capital it uses, and Tapri's cost of capital is 14%. The format reports an accounting profit of ₹8 crore a year. In the sense of ICAI's value creation answer, the format:

- **A** Destroys value, because its return is below the cost of capital
- **B** Creates value, because it reports a profit every year
- **C** Creates value, because its profit is sustainable
- **D** Neither creates nor destroys value, because profit and wealth maximisation agree whenever profit is positive

Answers: the answer key at the end of this lesson.

Recap

- Profit maximisation is at best a limited objective: vague, ignores risk, ignores the time pattern of returns, too narrow (in that order).
- Wealth maximisation measures success by the market value of the shares; wealth = PV of benefits − PV of costs.

- To measure wealth: cash flow approach (not accounting profit), cost-benefit analysis, time value of money.
- V = N × MP, or V = Ve + Vd. The market price is the "report card" of the firm (Van Horne).
- Value creation: decisions that earn sustainable returns above the cost of capital (ICAI's MTP answer); the SM treats it with wealth maximisation.
- Wealth maximisation stays the primary goal among growth, market share, leadership, employee welfare, customer satisfaction and community goals.
- In an uncertain, multi-period world wealth maximisation is better; with a short period and little uncertainty the two are essentially the same.
- Comparison table: profit (easy to calculate; short-term, ignores risk and timing, requires immediate resources) against wealth (long-term, risk, timing, shareholders' return; no clear link to share price, management anxiety). Kesar's record quarter cost him 6%.

Answer key · Check yourself

1 **C It reflects profit maximisation's emphasis on short-term gains; wealth maximisation would compare the present value of each option's cash flows over all four years**

Postponing repairs to show a short-term profit is the Study Material's own example of profit bought at the cost of long-term profitability (SM 1.17). The period here runs four years, with output changes after year 1, so timing matters and the two objectives part company.

2 **B Written the "disadvantages" row of ICAI's comparison table, leaving out "profit is vague" and "too narrow"**

The limitations in SM 1.13 are: the term profit is vague; it ignores risk; it ignores the time pattern of returns; it is too narrow. The list Bunty wrote is the comparison table's disadvantages (SM 1.17), a different list that ICAI uses when the question says "disadvantages".

3 **20.5**

PV of benefits = 300 × (0.909 + 0.826) = 300 × 1.735 = 520.50; less the cost of 500.00: wealth created = ₹20.50 lakh. The total profit of ₹100 lakh overstates it because it ignores timing.

4 **A "Ignores the timing of returns" is a disadvantage of profit maximisation; C "Can lead to management anxiety and frustration" is a disadvantage of wealth maximisation; E "Considers shareholders' return" is an advantage of wealth maximisation**

"Requires immediate resources" is the fourth disadvantage of profit maximisation, and the easy link between decisions and profits is profit maximisation's second advantage. Wealth maximisation has the opposite problem: it offers no clear relationship between financial decisions and share price (SM 1.17).

5 **A Destroys value, because its return is below the cost of capital**

Value creation means sustainable returns exceeding the cost of capital (MTP Series II for September 2025, Q4(a)). Earning 13% on funds that cost 14% lowers the present value of the owners' wealth, whatever the accounting profit.

1.2 The finance function, the financial manager and the controller

What financial management is, its two aspects (procuring and using funds), the three long-term decisions and working capital, its scope and evolution, and how the role of the finance executive and CFO has grown.

In plain words

One tapri, three questions

Twenty years ago Tapri Tea Co. was one roadside stall in Indore. Before the first cup was sold, Kesar had to answer four questions: which things to buy (a stove, a cart, kettles), how much they would cost in all, how much cash he would need every day for milk, tea leaves and his helper's wages, and where all that money would come from (his savings, and a loan from his uncle at 2% a month).

> **Kesar:** Back then I did all of it myself, on the back of a matchbox.
>
> **Dev:** And today the same questions need a CFO, a treasurer, a controller and ten managers.
>
> **Kesar:** Same questions? Then why are you so expensive?
>
> **Dev:** Because the matchbox had one stall. We have 40 cities.

Those questions, and one more that arrived with the first profit (how much of it to take home), are the whole of **financial management**: getting money (the **financing decision**), putting it to work in assets (the **investment decision**), and deciding how much of the profit to hand back to the owners (the **dividend decision**). Running the daily cash, stock and credit is **working capital management**. Together these are the **finance functions**, and the value of the firm depends on how well all of them are taken.

Money comes from different sources, each with its own **risk, cost and control**: owners' equity is safe for the firm but expensive; debentures are cheaper but must be repaid with interest come what may. And money raised is only useful if it earns more than it costs, so the finance manager also watches how it is used: no idle cash, no excess stock, no machine that cannot pay its way.

The person in charge has grown too. Once the finance department was called in only for big events (a merger, an expansion). Today the **chief financial officer (CFO)** is a strategic partner of the chief executive, and under the CFO sit a **treasurer** (who raises, invests and guards the money) and a **controller** (who accounts for it, costs it, taxes it and audits it).

Think of it like

The town's water engineer

A town's water engineer brings water in from the river, the borewell and the tanker, each with its own price and reliability (financing); pipes it to the fields and factories that will make the best use of it (investment); and decides how much to release to households now and how much to keep in the dam for next summer (dividend versus retention). A leaking pipe is idle money: water paid for that waters nothing.

Key idea

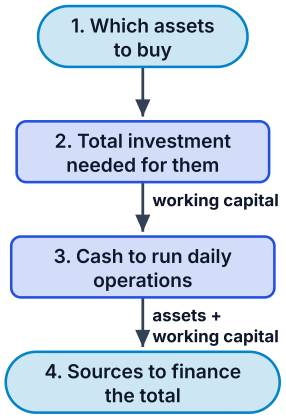
Financial management is the efficient acquisition (financing) and allocation (investment) of funds, with the dividend decision on top: three inter-related long-term decisions plus working capital management. The finance executive who takes them has grown from a fund-raiser for big events into the CEO's strategic partner.

How it works

1. Four stages before the first sale

The Study Material opens with the four stages of decision making every entrepreneur goes through (SM 1.2):

An entrepreneur's four decisions



Stage 1: premises, machinery, equipment. Stage 3 is the working capital requirement (raw material, salaries, wages). Stage 4: share capital, including the entrepreneur's own funds, borrowing from banks, or investment from financial institutions.

While deciding how much to take from each source, the entrepreneur keeps in mind the **cost of capital** of each source (interest, dividend) and tries to keep it low. So financial management is concerned with efficient acquisition (financing) and allocation (investment in assets, working capital etc.) of funds with an objective to make profit (dividend) for the owners. Any business requiring money asks three key questions (SM 1.3):

1. Where to get the money from? (**Financing decision**)
2. Where to invest the money? (**Investment decision**)
3. How much to distribute amongst shareholders to keep them satisfied? (**Dividend decision**)

2. What financial management means

Definition

Financial management

That managerial activity which is concerned with planning and controlling of the firm's financial resources; in other words, acquiring, financing and managing assets to accomplish the overall goal of a business enterprise (mainly to maximise the shareholder's wealth). (SM 1.3)

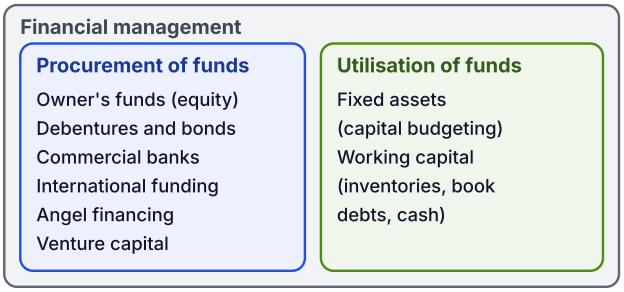
Where positive cash flow matters more than book profit, it can also be defined as planning for the future of a business enterprise to ensure a positive cash flow; some experts call it the science of money management. Two formal definitions (SM 1.3–1.4):

- "Financial Management comprises of forecasting, planning, organizing, directing, co-ordinating and controlling of all activities relating to acquisition and application of the financial resources of an undertaking in keeping with its financial objective."
- **Phillippatus**: "Financial Management is concerned with the managerial decisions that result in the acquisition and financing of short term and long term credits for the firm." It deals with the selection of specific assets (or combinations of assets) and the size and growth of an enterprise, and the analysis is based on the expected inflows and outflows of funds and their effect on managerial objectives.

3. The two aspects: getting funds and using them

There are two basic aspects of financial management: **procurement of funds** and the **effective utilisation** of these funds to achieve business objectives (SM 1.4).

Two aspects of financial management



Procurement. Because funds can be obtained from many sources, procurement is always a complex problem (SM 1.4). New and creative sources keep appearing (trading in **carbon credits** is one, SM 1.5). Funds from different sources have different characteristics in terms of **risk, cost and control**: the cost of funds should be at the minimum, and for that a proper balancing of risk and control must be carried out; the finance manager must also strike a balance between equity and debt so that the funding structure suits the business (SM 1.5).

Source	From the firm's point of view (SM 1.5–1.6)
Equity	Best from the risk point of view: no repayment except in liquidation. Usually the most expensive: shareholders expect more than the prevalent interest rate, and dividends are an appropriation of profit, not an expense under the Income Tax Act. A public issue may dilute existing control.
Debentures	Comparatively cheaper than shares because of their tax advantage: interest is allowed for tax, a dividend comes out of taxed profit. But interest must be paid even in hard times, whether or not there are profits, and the debentures must be repaid as agreed: a high degree of risk.
Banks	Meet long-term and short-term needs. Fund-based: cash credit, overdraft, term loans, working capital term loan, bill purchase/discounting. Non-fund-based: guarantees, letters of credit.
International funding	FDI and FII are the two major routes, besides ADRs and GDRs; procurement must suit the requirements of foreign investors.
Angel financing	Equity from a wealthy individual for a start-up or expansion; sometimes the last option for a start-up that does not qualify for bank funding and is too small for venture capital.

The sources are taught in long-term sources, short-term sources and contemporary and international sources.

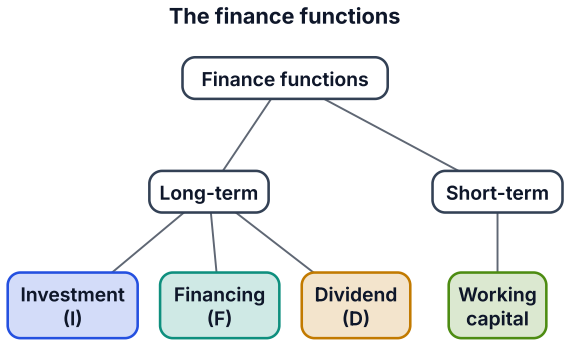
Utilisation. The finance manager must point out where funds are idle or not properly used. All funds are procured at a cost and with some risk; if they do not generate an income higher than the cost of procuring them, there is no point in running the business (SM 1.7). Funds go to:

- **Fixed assets**: invested so that the company can produce at its optimum level without endangering its financial solvency. This needs sound knowledge of **capital budgeting** (investment appraisal), the planning process used to decide whether long-term investments such as new or replacement machinery, new plants, new products and research projects would provide the desired return (SM 1.7; capital budgeting).
- **Working capital**: enough of it, but not too much blocked in inventories, book debts and cash (SM 1.7; working capital).

4. The finance functions: V = f(I, F, D)

The value of a firm depends on its finance functions or decisions (SM 1.8):

$$V = f(I, F, D)$$



The three long-term decisions decide V = f(I, F, D); short-term decisions reduce to working capital management (current assets and current liabilities).

(a) Investment decisions (I) relate to the **selection of assets** in which funds will be invested. Long-term funds go into fixed assets and also current assets, after careful assessment of the projects through capital budgeting; a part of long-term funds is kept for working capital. Asset management policies are laid down for current assets: the **inventory policy** is decided by the production manager and the finance manager in view of production needs, future raw material prices and the availability of funds (SM 1.8).

(b) Financing decisions (F) relate to **acquiring the optimum finance** to meet financial objectives and seeing that fixed and working capital are effectively managed. The finance manager needs (SM 1.8–1.9):

- a good knowledge of the sources of funds and their costs, and a **sound capital structure**: a proper balance between equity capital and debt;
- a clear understanding of the difference between **profit and cash flow**: profit is of little avail unless the organisation has cash to pay for assets and sustain the working capital cycle;
- the evaluation of **risk**: excessive debt carries high risk for equity because of the priority rights of lenders;
- in overseas trading, protection against currency fluctuations by **hedging**, a strategy designed to minimise, reduce or cancel out the risk in another investment (as a shopkeeper hedges the risk of fire with a fire insurance contract).

(c) Dividend decisions (D) relate to **how much and how frequently cash can be paid out** of the profits as income for the owners. An owner is rewarded in two ways, the growth of the capital invested and the cash paid out (drawings for a sole trader, dividends for a company). The dividend decision has two elements: the amount paid out and the amount retained to support growth, the latter being also a financing decision. The level and regular growth of dividends is a significant factor in a company's market value (SM 1.9; dividend policy).

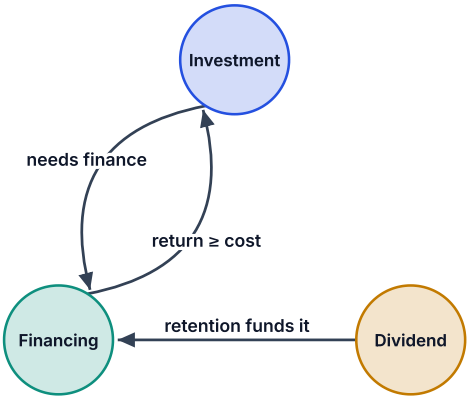
All three are inter-related. The first two arise in any organisation; the dividend decision only in profit-making ones (SM 1.9).

Short-term finance decisions are generally reduced to the management of current assets and current liabilities: **working capital management (WCM)** (SM 1.10).

5. Why the three decisions must be taken together

ICAI's Suggested Answers explain the inter-relationship this way (MTP Series II for May 2024 and for September 2026, Q4(a)): the underlying objective of all three is the same, the **maximisation of shareholders' wealth**, so their joint impact on the market price of the share must be considered and they should be solved jointly.

How the three decisions pull on one another



A new project needs finance; it is accepted only if its return at least covers the cost of that finance; and retained earnings used for internal financing deprive shareholders of dividends.

- **Investment**: long-term funds are invested after careful assessment through capital budgeting and uncertainty analysis, and only a proposal expected to yield at least enough to meet **its cost of financing** is accepted.
- **Financing**: a proper balance between long-term and short-term funds, and within long-term funds a proper mix of loan funds and owners' funds; the optimum mix increases the return to equity shareholders.
- **Dividend**: what part of profit to pay out and what to retain; an optimal dividend pay-out ratio maximises shareholders' wealth.

6. Importance and scope

The importance of financial management cannot be over-emphasised: it is the key to successful business operations, and without proper administration of finance no business enterprise can reach its full potential for growth and success. "Money is to an enterprise, what oil is to an engine." Financial management is all about planning investment, funding the investment, monitoring expenses against budget and managing gains from the investments; it means the management of all matters related to an organisation's finances (SM 1.10). The best way to show its importance is to list the tasks good financial management involves (SM 1.10):

- Taking care not to over-invest in fixed assets
- Balancing cash outflows with cash inflows
- Ensuring a sufficient level of short-term working capital
- Setting sales revenue targets that will deliver growth
- Increasing gross profit by setting the correct pricing for products or services

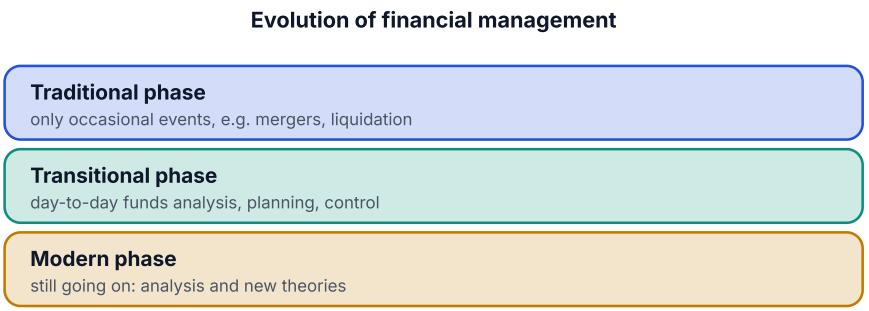
- Controlling general and administrative expenses by finding more cost-efficient ways of running day-to-day operations
- Tax planning that will minimise the taxes the business has to pay

Scope. Based on **Ezra Solomon's** concept, financial management studies (SM 1.11):

1. Determination of the size of the enterprise and its rate of growth.
2. Determining the composition of its assets.
3. Determining the mix of its financing, i.e. the level of debt to equity.
4. Analysis, planning and control of its financial affairs.

7. Three phases of evolution

Financial management evolved as a separate field of study at the beginning of the century and gradually over the past 50 years, in three phases (SM 1.7–1.8):

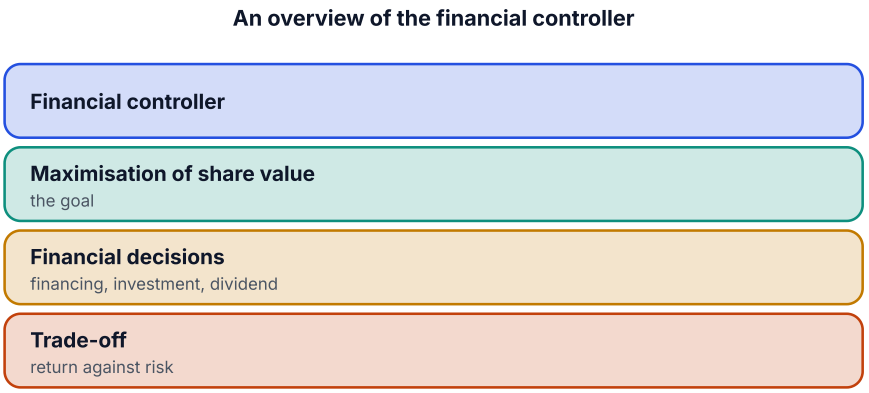


Top to bottom: the three phases in order (SM 1.7–1.8).

- **Traditional phase**: financial management was considered necessary only during occasional events such as takeovers, mergers, expansion and liquidation, and financial decisions kept in mind the needs of outsiders (investment bankers, lenders and the like).
- **Transitional phase**: the day-to-day problems financial managers faced were given importance: funds analysis, planning and control.
- **Modern phase**: still going on. The scope has greatly increased; financial analysis helps decision making, and many theories have been developed on efficient markets, capital budgeting, option pricing, valuation models and other fields.

8. The role of the financial controller

The role of the **financial controller** has changed over the years. Until the middle of the century (in the Study Material's words), its scope was limited to **procurement of funds** under major events in the life of the enterprise such as promotion, expansion or merger. In modern times it includes, besides procurement of funds, the three decisions: investment, financing and dividend (SM 1.11).



The Study Material's figure links the market value, the three financial decisions and the risk-return trade-off (SM 1.12).

In a bid to maximise shareholders' wealth, the financial controller should strive to maximise returns in relation to the given risk, and seek courses of action that avoid unnecessary risks. To ensure maximum return, funds flowing in and out of the firm should be constantly monitored to ensure that they are safeguarded and properly utilised (SM 1.11).

9. The finance executive and the CFO

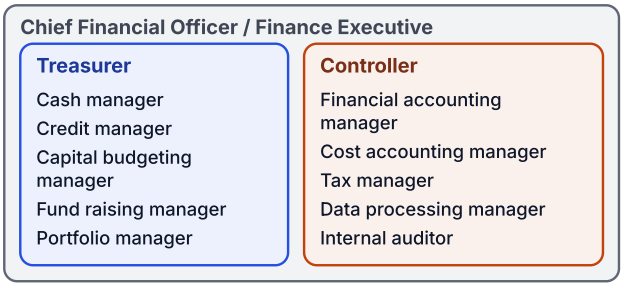
With liberalisation, deregulation and globalisation, the opportunities before finance managers have virtually no limits; the finance executive goes by different names in different organisations, but the role and functions are similar (SM 1.18). ICAI quotes Jeff Thomson, IMA President and CEO: today's CFO team is expected to add value well beyond the traditional roles of cost management, controls and acting as the conscience of the organisation, by serving as the **integration hub** for key business processes, a **catalyst for change** including business transformation, and a **consultant or trusted business advisor** in creating sustainable growth (SM 1.19).

The finance executive's responsibilities include (SM 1.19):

1. **Financial analysis and planning**: determining the proper amount of funds to employ in the firm, i.e. designating the size of the firm and its rate of growth.
2. **Investment decisions**: the efficient allocation of funds to specific assets.
3. **Financing and capital structure decisions**: raising funds on terms as favourable as possible, i.e. determining the composition of liabilities.
4. **Management of financial resources** (such as working capital).
5. **Risk management**: protecting assets.

In a large organisation the finance function may be organised under the CFO in two arms (SM 1.19):

Organisation of the finance function



Blue: the treasurer's side, which raises, invests and safeguards the money. Orange: the controller's side, which records, costs, taxes and checks it.

Today's CFO. The role is no longer confined to accounting, financial reporting and risk management: it is about being a **strategic business partner of the CEO** (SM 1.20).

What a CFO used to do	What a CFO now does
Budgeting	Budgeting
Forecasting	Forecasting
Accounting	Managing M&As
Treasury (cash management)	Profitability analysis (by customer or product)
Preparing internal financial reports for management	Pricing analysis
Preparing quarterly and annual filings for investors	Decisions about outsourcing
Tax filing	Overseeing the IT function
Tracking accounts payable and accounts receivable	Overseeing the HR function
Travel and entertainment expense management	Strategic planning (sometimes overseeing it)
	Regulatory compliance
	Risk management

Worked examples

Worked example **One export plant, three decisions**

Tapri's board is considering a ₹75 crore plant at Indore to make frozen samosas for export. Dev expects it to earn 16% a year; Tapri's cost of capital is 12%. He proposes to fund it with ₹25 crore of this year's profit, cutting this year's dividend from ₹32 crore to ₹7 crore, and ₹50 crore by either 9% non-convertible debentures or a rights issue of equity shares. A director asks why the board cannot simply approve the plant today and "sort out the money and the dividend later".

Explain to the director, with reference to this proposal, how the investment, financing and dividend decisions are inter-related. (4 marks)

MODEL ANSWER

Inter-relationship. The finance functions are divided into three major decisions: investment, financing and dividend. They are inter-related because the underlying objective of all three is the same, the maximisation of shareholders' wealth. Since they are inter-related, their joint impact on the market price of the company's shares must be considered, and they should be solved jointly.

Investment decision. The plant is a long-term investment to be assessed through capital budgeting. It should be accepted only if it is expected to yield at least a return adequate to meet its cost of financing. The expected 16% exceeds the 12% cost of capital, but the cost depends on how the ₹75 crore is raised, so the investment cannot be approved without the financing decision.

Financing decision. The finance manager has to ensure a proper mix of loan funds and owners' funds. Raising ₹50 crore through 9% debentures is cheaper because interest is tax-deductible, but it adds fixed interest and repayment risk; a rights issue avoids that risk but is costlier. The optimum mix increases the return to equity shareholders and maximises their wealth.

Dividend decision. Retaining ₹25 crore of profit is internal financing, so it is also a financing decision: it deprives shareholders of ₹25 crore of dividend now in exchange for growth. The level and regular growth of dividends affect the market price of the share, so the pay-out must be set together with the plant and its financing.

Conclusion. The three decisions are inter-related and must be taken jointly, judging each by its effect on shareholders' wealth. The board should approve the plant, its financing mix and the dividend together.

Worked example **Which decision is it?**

Identify the finance decision (investment, financing, dividend or working capital management) involved in each of these Tapri proposals, giving reasons:

(a) Choosing between one ₹12 crore fully automatic kulhad kiln and two semi-automatic kilns.

(b) Buying US dollars forward to pay, in six months, for espresso machines bought from Italy.

(c) Paying only 30% of this year's profit as dividend so that the rest can fund the delivery app.

(d) Cutting the credit period for corporate catering clients from 60 days to 30 days.

(e) Deciding how many months of Assam tea leaves to stock ahead of an expected price rise, together with the production head.

(f) Choosing between ₹100 crore of debentures and ₹50 crore each of debentures and equity for a new plant.

MODEL ANSWER

Proposal	Decision	Reason
(a)	Investment	Selection of the asset in which funds will be invested, appraised by capital budgeting.
(b)	Financing	Overseas trading exposes the firm to currency fluctuations; hedging the risk is part of the financing decision (SM 1.9).
(c)	Dividend, and also financing	How much to pay out and how much to retain; the retained part finances growth, so it is also a financing decision (SM 1.9).
(d)	Working capital management	A short-term decision on a current asset (receivables).
(e)	Investment	An asset management policy for a current asset: the inventory policy, set by the production manager and the finance manager in view of production needs, future raw material prices and availability of funds (SM 1.8).
(f)	Financing	The mix of debt and equity: a sound capital structure.

On (e): the Study Material places the inventory policy under the investment decision (asset management policies for current assets, SM 1.8), so write "investment"; a line adding that managing the stock day to day is part of working capital management shows you know where both sit.

Worked example **Treasurer or controller?**

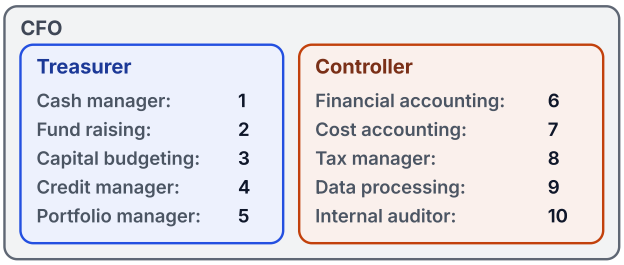
Tapri's new CFO is organising the finance function as the Study Material suggests. Ten jobs are on her list:

1. Tracking the cash position of 40 cities every morning
2. Negotiating a ₹200 crore term loan
3. Appraising the export plant's NPV
4. Checking a new corporate client's creditworthiness
5. Investing surplus cash in liquid funds and bonds
6. Preparing the quarterly financial statements
7. Working out the cost of one kulhad of chai
8. Filing GST and income-tax returns
9. Running the system that records every café's sales
10. Testing whether café cash controls work

Show how the finance function may be organised, and place each job in it.

MODEL ANSWER

Tapri's finance function, with the ten jobs placed



Blue: the treasurer's managers; orange: the controller's. The number beside each manager is the job placed with them.

Under the CFO (finance executive), the **treasurer** manages the money itself: the cash manager (job 1), the fund raising manager (job 2), the capital budgeting manager (job 3), the credit manager (job 4) and the portfolio manager (job 5). The **controller** manages its recording and checking: the financial accounting manager (job 6), the cost accounting manager (job 7), the tax manager (job 8), the data processing manager (job 9) and the internal auditor (job 10).

Exam angle

In the exam

Where this topic appeared

In the ICAI documents read for this lesson, these topics were asked 17 times, always as theory (2 to 4 marks where marks are printed):

- **Inter-relationship of investment, financing and dividend decisions**, as a short note: MTP Series II for May 2024 Q4(a) and MTP Series II for September 2026 Q4(a) (4 marks each, with identical Suggested Answers); old scheme: MTP Series I for November 2019 Q6(a), RTP November 2019 Q10(c), RTP November 2023 Q10(b).
- **The finance functions**: two basic functions, September 2025 Q4(c) (2 marks; the answer gave the definitions of I, F, D and WCM); old scheme: the three finance function decisions, November 2019 Q6(a) (3 marks) and MTP Series I for December 2021 Q6(a) (4 marks).
- **The finance executive and the CFO**: "discuss the role of a chief financial officer", MTP Series I for September 2024 Q4(c) (2 marks, an OR choice, answered with the five responsibilities); key responsibilities of a finance executive in the modern business world, September 2026 (2 marks, Suggested Answers not yet published); emerging issues for the CFO, RTP January 2025 Q11(a); old scheme: the CFO in today's world, November 2020 Q6(a) (4 marks, answered with the "now does" column), and the finance executive's responsibilities, MTP Series I for November 2022 Q6(a) (4 marks).
- **The financial controller**: short note, old scheme MTP Series II for November 2023 Q6(c).
- **Importance of financial management**, as a case of a firm short of cash after expanding: RTP September 2026 Q10(a) (answered with "money is to an enterprise what oil is to an engine" and the seven tasks).
- **Evolution**: the phases, old scheme December 2021 Q6(c) (2 marks).
- **Procurement**: features of international funding and angel financing, RTP May 2026 Q10(a) (taught in contemporary and international sources).

In the exam

How the marks are earned

The Suggested Answers copy the Study Material's lists in its order: the five responsibilities of the finance executive; the CFO's "now does" list; the seven tasks; the definitions of I, F and D. The inter-relationship note has a fixed shape: same objective (shareholders' wealth), so solve jointly; then one paragraph each on investment, financing and dividend; then a one-line conclusion. The case version (RTP September 2026, a firm short of cash after expanding) was answered with the importance paragraph and the seven tasks, word for word, without applying them to the facts; a line tying two or three tasks to the case's cash shortage is a safe addition.

Trick

Treasurer holds the money, controller holds the records

Treasurer: three Cs and two money jobs: **C**ash, **C**redit, **C**apital budgeting; **F**und raising, **P**ortfolio. **Controller**: everything that is recorded or checked: financial accounting, cost accounting, tax, data processing, internal audit. Condition: this is ICAI's chart of how the finance function *may* be organised in a large organisation (SM 1.19); a real company may organise it differently.

Answer frame

"Discuss the role (responsibilities) of the finance executive / CFO"

Open with one sentence: the finance executive plays an important role in the company's goals, policies and financial success. Then the five responsibilities, each with its ICAI gloss: financial analysis and planning (size and rate of growth); investment decisions (efficient allocation of funds to specific assets); financing and capital structure decisions (composition of liabilities); management of financial resources (working capital); risk management (protecting assets). If the question says "in today's world", add the line on the strategic business partner of the CEO and the "now does" list.

"Explain the inter-relationship between investment, financing and dividend decisions"

1. The three decisions; inter-related because their objective is the same, maximisation of shareholders' wealth; consider their joint impact on the share price and solve them jointly.
2. Investment: capital budgeting; accept only if the return at least meets the cost of financing.
3. Financing: balance of long-term and short-term funds; mix of loan and owners' funds; the optimum mix maximises return to equity.
4. Dividend: payout against retention; retained earnings used for internal financing deprive shareholders of dividends; optimal payout maximises wealth.
5. Conclusion: take them jointly, keeping in view their joint effect on shareholders' wealth.

"Distinguish the treasurer's and the controller's functions"

Treasurer	Controller
Cash manager	Financial accounting manager
Credit manager	Cost accounting manager
Capital budgeting manager	Tax manager
Fund raising manager	Data processing manager
Portfolio manager	Internal auditor

Common mistakes

⚠ **Common mistake**

Hall of Shame: the retained profit that was "only a dividend decision"

Bunty classifies "retain 70% of profit to fund new cafés" as a dividend decision and stops. **Fix:** the amount retained to support growth is also a financing decision (SM 1.9). Say both.

⚠ **Common mistake**

Answering "role of the financial controller" with the organisation chart

Fix: SM 1.11's financial controller is the head of finance: once limited to procuring funds for major events, now taking the investment, financing and dividend decisions, maximising returns in relation to the given risk and monitoring the flow of funds. The chart's "Controller" is a different post.

⚠ **Common mistake**

Calling equity "the cheapest source because there is no interest"

Fix: equity is the best source from the **risk** point of view (no repayment except in liquidation) but usually the most **expensive**: shareholders expect more than the interest rate, and dividends are not tax-deductible (SM 1.5). Debentures are cheaper but riskier.

⚠ **Common mistake**

Listing the CFO's old jobs as the "role today"

Accounting, tax filing and tracking payables and receivables are in the "used to do" column. **Fix:** the "now does" list adds managing M&As, profitability and pricing analysis, outsourcing decisions, overseeing IT and HR, strategic planning, regulatory compliance and risk management, on top of budgeting and forecasting.

Practice

Check yourself

1 Case scenario. Mitti Pottery Works Ltd has ₹8 crore of profit after tax. Its board decides to pay ₹2 crore as dividend and use ₹6 crore, with a ₹4 crore bank term loan, to buy a new gas kiln. Which statement is correct?

- **A** Only an investment decision has been taken; the dividend and the loan follow from it automatically
- **B** The ₹6 crore retention is a dividend decision only, and the term loan is a working capital decision
- **C** The three decisions are independent, so each can be judged on its own
- **D** The board has taken an investment decision (the kiln), a financing decision (the loan and the retention) and a dividend decision (the ₹2 crore payout), which are inter-related

2 Which of the following is NOT one of the four aspects of financial management in Ezra Solomon's concept (SM 1.11)?

- **A** Determining the size of the enterprise and its rate of growth
- **B** Preparing the financial statements in accordance with accounting standards
- **C** Determining the mix of financing, i.e. the level of debt to equity
- **D** Determining the composition of the assets of the enterprise

3 In which phase of its evolution was financial management considered necessary only during occasional events such as takeovers, mergers, expansion and liquidation?

- **A** The traditional phase
- **B** The transitional phase
- **C** The modern phase
- **D** The strategic phase

4 Which of these appear in the Study Material's list of what a CFO now does but NOT in what a CFO used to do? Select all that apply.

- **A** Managing M&As
- **B** Tax filing
- **C** Pricing analysis
- **D** Overseeing the IT function
- **E** Budgeting

More than one answer may be right.

5 According to the Study Material, which source of funds is the best from the firm's risk point of view, and why?

A Debentures, because their interest is tax-deductible

B Bank term loans, because banks also give non-fund-based services

C Equity shares, because there is no question of repaying the capital except in liquidation

D Angel financing, because angel investors look for a higher rate of return than traditional investments

Answers: the answer key at the end of this lesson.

Recap

- Financial management is the efficient acquisition (financing) and allocation (investment) of funds, with the objective of making profit (dividend) for owners; mainly to maximise shareholders' wealth.
- Two aspects: procurement of funds (each source differs in risk, cost and control) and effective utilisation (fixed assets via capital budgeting, and working capital without too much blocked in it).
- V = f(I, F, D): investment (selection of assets), financing (optimum finance, sound capital structure, profit versus cash, risk, hedging) and dividend (payout and retention, the retention being also a financing decision); short-term: working capital management.
- The three decisions are inter-related through one objective, shareholders' wealth, and must be taken jointly.
- Scope (Ezra Solomon): size and growth; composition of assets; mix of financing; analysis, planning and control.
- Evolution: traditional (occasional events), transitional (day-to-day problems), modern (analysis and theories).
- Financial controller: once procurement for major events, now the three decisions; maximise returns for the given risk. Finance executive: planning, investment, financing and capital structure, management of financial resources, risk management.
- Under the CFO: treasurer (cash, credit, capital budgeting, fund raising, portfolio) and controller (financial accounting, cost accounting, tax, data processing, internal audit); today's CFO is the CEO's strategic partner. The matchbox has been retired.

Answer key · Check yourself

1 **D The board has taken an investment decision (the kiln), a financing decision (the loan and the retention) and a dividend decision (the ₹2 crore payout), which are inter-related**

Retention is both a dividend and a financing decision (SM 1.9). A term loan for a kiln is long-term financing, not working capital management. The decisions share one objective, shareholders' wealth, and must be taken jointly.

2 **B** **Preparing the financial statements in accordance with accounting standards**

The fourth aspect is the analysis, planning and control of financial affairs. Preparing financial statements is accounting, an input to financial management.

3 **A** **The traditional phase**

SM 1.7: the traditional phase; outsiders' needs (investment bankers, lenders) were kept in mind. The transitional phase gave attention to day-to-day problems; the modern phase, still going on, developed theories. There is no "strategic phase" in the Study Material.

4 **A** **Managing M&As**; **C** **Pricing analysis**; **D** **Overseeing the IT function**

Tax filing is only in the old list; budgeting and forecasting are in both. Managing M&As, pricing analysis and overseeing IT are new (SM 1.20).

5 **C** **Equity shares, because there is no question of repaying the capital except in liquidation**

SM 1.5: equity is best from the risk point of view but usually the most expensive; debentures are cheaper because of the tax advantage but carry a high degree of risk. The other two options state true facts (a bank's non-fund-based services, an angel's return expectation) that say nothing about the firm's risk.

1.3 Agency problem, financial environment, distress and insolvency

Why hired managers may chase their own goals and what that costs shareholders, how lenders and owners respond, where financial management sits among related disciplines, and how distress becomes insolvency.

In plain words

The CEO who loved coupons

When Tapri was one stall, Kesar owned it and ran it: nobody could put Kesar's interests ahead of Kesar's. Since the 2024 listing, Tapri has thousands of shareholders, and none of them pours chai. Its delivery app, TapriNow, is run by a hired chief executive, Arjun, whose bonus is 1% of the app's revenue.

> **Dev:** Arjun's revenue is up 60%. So is his bonus.
>
> **Kesar:** Wonderful! So the app is making money?
>
> **Dev:** It is giving away ₹20 crore of free-chai coupons to buy that revenue. And he has asked for a ₹6 crore office with a rooftop café.
>
> **Kesar:** Whose café is it, his or ours?

That is the **agency problem**. The shareholders (the **principals**) hire managers (their **agents**) to run the firm for them. In theory the managers act in the shareholders' best interest; in reality they may chase their own goals: salary, perks, a bigger bonus. Watching and controlling them costs the shareholders money, the **agency cost**: board reviews, audits of the coupon spend, a new bonus formula. Lenders face the same problem, and protect themselves with **negative covenants**: promises written into the loan, such as "no more borrowing beyond a point".

Two more ideas complete the chapter. The finance manager works inside an environment: accounting feeds the decisions, marketing, production and quantitative methods support them, and stakeholders watch the results. And when cash coming in cannot meet the bills falling due, the firm is in **financial distress**; if that continues until the firm cannot repay its debts, it is **insolvent**.

Think of it like

The driver and the fuel card

You hire a driver and give him a fuel card. He may take the long route to clock overtime: that gap between his interest and yours is the agency problem. A GPS tracker you pay for, a deposit he offers, the trips you cannot take because every detour needs your approval, and the time you spend designing a fair bonus are all costs you would not bear if you drove yourself. Together they are the agency cost.

Key idea

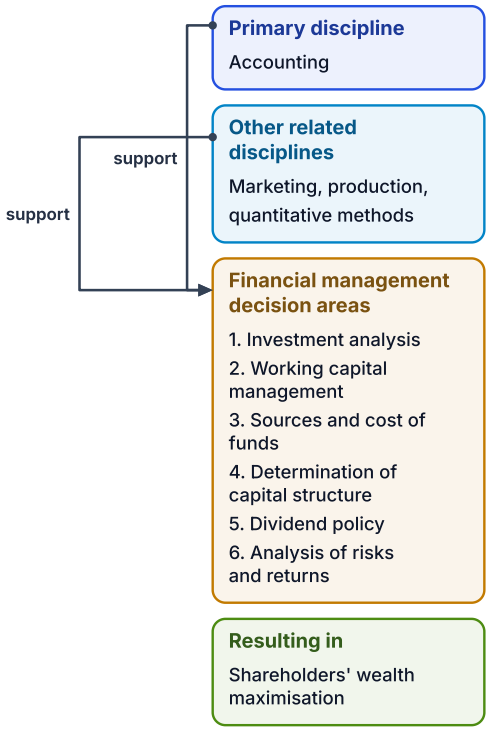
The agency problem is the chance that managers place personal goals ahead of the owners' goals; the extra cost shareholders bear to monitor and control them is the agency cost. Financial distress is cash inflows too small for current obligations; insolvency, the inability to repay debts, is the result of continuous distress.

How it works

1. The environment around the finance manager

Financial management is an integral part of overall management, not a totally independent area: it draws heavily on **economics, accounting, production, marketing and quantitative methods** (SM 1.21). The Study Material draws the relationship like this (SM 1.23):

Impact of other disciplines on financial management



Accounting is the primary discipline the finance manager draws on; marketing, production and quantitative methods support day-to-day decisions only indirectly.

- **Accounting** is a necessary input into the financial management function: financial statements help the finance manager gauge past performance and future directions (SM 1.21–1.22).
- **Marketing and production**: new products, promotion plans and changes in the production process need capital outlays and change the projected cash flows, which the finance manager must evaluate and finance (SM 1.23).
- **Quantitative methods** supply the tools for analysing complex financial problems (SM 1.23).
- **Economics** helps the finance manager know what goes on in the world outside the business (SM 1.23).

Financial management and accounting are closely related but differ in two ways (SM 1.22–1.23):

Point	Accounting	Financial management
Treatment of funds	Accrual principle: revenue when sold, expenses when incurred	Cash flows: revenue when cash is received, expenses when paid
Why it matters	A firm profitable in the accounting sense may still be unable to meet its current obligations (say, receivables not collected)	Cash flow based returns help the finance manager maintain solvency and avoid insolvency
Decision-making	Chief focus: to collect data and present it	Chief focus: financial planning, controlling and decision-making
Sequence	Supplies the data	Financial management begins where accounting ends

The people around it. Where outsiders hold shares or lend money, management is under the constant supervision of stakeholders (employees, creditors, customers, government), each judging it by the fulfilment of its own objective (objectives of financial management, SM 1.16). And the opening of the economy, with liberalisation, deregulation and globalisation, has given finance managers access to global resources and widened their role (the finance function, SM 1.18).

2. The agency problem

In a sole proprietorship or a partnership the owners participate in management. In a company the owners are not active in management, so there is a **separation between owners (shareholders) and managers**. In theory managers should act in the best interest of shareholders; in reality they may try to maximise their individual goals like salary and perks. So there is a **principal–agent relationship** between owners and managers, known as the agency problem (SM 1.24).

Definition

Agency problem

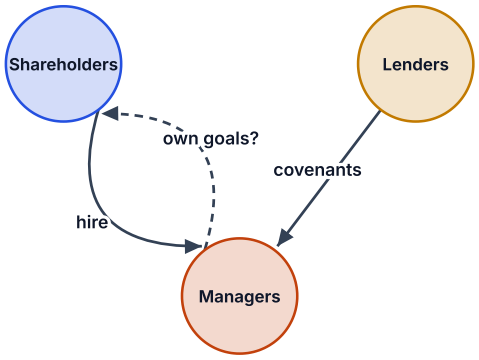
The chances that managers may place personal goals ahead of the goal of owners. (SM 1.24)

Definition

Agency cost

The additional cost borne by the shareholders to monitor the manager and control their behaviour so as to maximise shareholders' wealth. Agency costs are generally of four types: (i) monitoring, (ii) bonding, (iii) opportunity, (iv) structuring. (SM 1.24)

Principals, agent and lender



Shareholders and lenders provide the money; managers run the firm. The dashed edge is the agency problem: managers may pursue salary and perks ahead of the owners' goal. Lenders bind managers with negative covenants.

The four types of agency cost



ICAI's list, in ICAI's order (SM 1.24). The Study Material names the four types without describing them; Under the hood gives their usual meaning.

3. Addressing the agency problem

The agency problem arises when managers' interests are not aligned to the interests of the **debt lenders** and the **equity investors** (SM 1.24).

With lenders. The agency problem of debt lenders is addressed by imposing **negative covenants**, i.e. the managers cannot borrow beyond a point. The Study Material calls this one of the most important concepts of modern finance, applied in the credit risk management of banks, fund raising and valuing distressed companies (SM 1.24).

With shareholders. It can be addressed if the managers' interests are aligned to the shareholders' interests, which is "easier said than done". The efforts made so far (SM 1.25):

1. **Managerial compensation is linked to the profit** of the company to some extent, and also to the **long-term objectives** of the company.
2. **"Employee is also designed** to address the issue with the underlying assumption that maximisation of the stock price is the objective of the investors."
3. **Effective monitoring** can be done.

Note

How the second point is printed

The Study Material (SM 1.25) and every ICAI Suggested Answer that reproduces it print the second point exactly as quoted: the word naming the scheme is missing. Read in context, it describes rewards for employees that are tied to the share price, an employee stock option scheme being the usual example. In an answer, write the point the way ICAI prints it, or as "employee schemes linked to the share price".

4. Financial distress and insolvency

A firm must manage, continuously, the price of its product or service, demand, and the prices of inputs such as raw material and labour. It must also manage the proportion of **debt** very delicately: higher debt means higher interest, and if cash inflows are not sufficient, both short-term and long-term creditors put pressure on the firm. If these factors are not well managed, the firm reaches **distress** (SM 1.21).

Definition

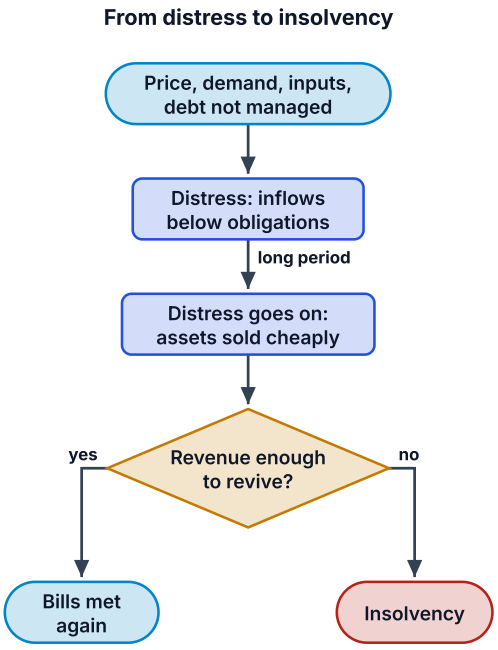
Financial distress

A position where cash inflows of a firm are inadequate to meet all its current obligations. (SM 1.21)

Definition

Insolvency

The inability of a firm to repay various debts; it is a result of continuous financial distress. (SM 1.21)



Distress is a cash position; insolvency is where continuous distress ends when revenue cannot revive the situation (SM 1.21). Rose: insolvency.

If distress continues for a long period, the firm may have to sell its assets, many times at a lower price. When revenue is inadequate to revive the situation, the firm will not be able to meet its obligations and becomes insolvent (SM 1.21). Debt needs the extra care the Study Material asks for: financial leverage helps equity while the return on investment exceeds the cost of debt, and when the cost of debt is more than the return, the firm can be under financial distress, which is why it is called a "double-edged sword" (leverage, FM Study Material 6.18).

Worked examples

Worked example **TapriNow's coupon bonus**

TapriNow Pvt Ltd, Tapri's delivery-app subsidiary, is run by a hired CEO, Arjun Sethi, whose annual bonus is 1% of the app's gross revenue. This year he spent ₹20 crore on free-chai coupons, which raised revenue by 60% and the app's loss by ₹14 crore, and he has proposed a ₹6 crore office fit-out. Tapri's board has appointed an outside firm to review all marketing spend, now requires board approval for any spend above ₹1 crore, and is redesigning Arjun's pay. TapriNow's bank loan agreement prohibits further borrowing while the debt-equity ratio exceeds 1:1.

(i) Explain the agency problem in this case and the agency cost it creates.

(ii) Advise how the agency problem between Tapri and Arjun, and with the bank, can be addressed.

(2 + 2 = 4 marks)

MODEL ANSWER

(i) Agency problem and agency cost. In a company, owners are not active in management, so there is a separation between owners and managers. In theory managers should act in the best interest of shareholders; in reality they may try to maximise their individual goals like salary and perks, so there is a principal–agent relationship, known as the agency problem: the chances that managers place personal goals ahead of the goal of owners.

In the given case, Arjun's bonus depends on gross revenue, not on shareholders' wealth. The coupons raised revenue, and his bonus, while increasing the app's loss by ₹14 crore; the office fit-out serves his perks. His goals and the shareholders' goal have diverged.

The agency problem leads to agency cost: the additional cost borne by the shareholders to monitor the manager and control his behaviour so as to maximise shareholders' wealth. Here it includes the outside review of marketing spend, the delay and board time that the ₹1 crore approval limit imposes, and the cost of redesigning Arjun's pay. ICAI classifies agency costs into four types: monitoring, bonding, opportunity and structuring.

(ii) Addressing the problem. The agency problem between managers and shareholders can be addressed if the managers' interests are aligned to the shareholders' interests:

1. **Compensation**: link Arjun's pay to the profit of the company to some extent and to its long-term objectives, not to gross revenue.
2. **Employee schemes linked to the share price**: designed on the assumption that maximising the stock price is the investors' objective. TapriNow is unlisted, so such a scheme for Arjun would use the shares of its listed parent, Tapri.
3. **Effective monitoring**: the outside review and the approval limit already in place.

The bank's agency problem is addressed by **negative covenants**: managers cannot borrow beyond a point. Its condition that no further borrowing is allowed while the debt-equity ratio exceeds 1:1 is such a covenant, and protects the bank against TapriNow's managers adding risk at the lender's expense.

Worked example **Gokul Doodh: five quarters**

Gokul Doodh Pvt Ltd is a dairy owned by Kesar's cousin, financed largely by a bank term loan and an overdraft. Its cash position for five quarters (₹ crore):

Quarter	Cash inflows	Current obligations falling due
Q1	4.20	3.70
Q2	3.50	3.80
Q3	3.10	3.90
Q4	2.80	4.00
Q5	2.40	4.10

From Q2 the procurement price of milk rose and a hotel chain began paying its bills late. In Q3 the dairy sold two milk tankers, carried at ₹1.10 crore, for ₹0.70 crore to pay suppliers. In Q4 the bank refused to renew the

overdraft. In Q5 the dairy missed two loan instalments, and with demand still falling it sees no way to raise its revenue. Its accounts showed a profit in Q1, Q2 and Q3.

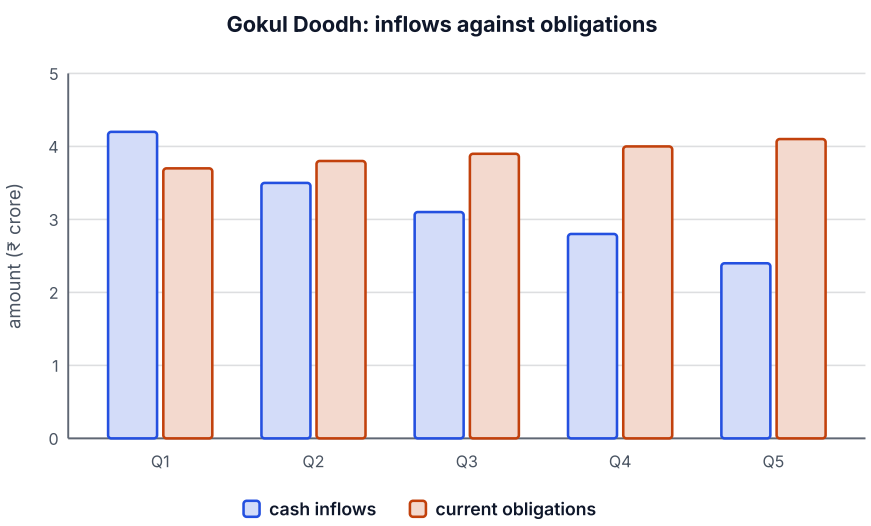
Required:

(i) Identify the quarters in which Gokul Doodh was in financial distress, and the total shortfall.

(ii) State, with reasons, when it became insolvent.

(iii) Explain the consequences of prolonged financial distress, with reference to the case.

THE WORKING



Blue: cash inflows; orange: current obligations. From Q2 the orange bar is the taller one every quarter, and the gap widens.

(i) Distress. Financial distress is a position where cash inflows are inadequate to meet all current obligations.

Quarter	Inflows − obligations (₹ crore)	Position
Q1	4.20 − 3.70 = 0.50	Not in distress
Q2	3.50 − 3.80 = (0.30)	Distress
Q3	3.10 − 3.90 = (0.80)	Distress
Q4	2.80 − 4.00 = (1.20)	Distress
Q5	2.40 − 4.10 = (1.70)	Distress
Total shortfall, Q2–Q5	**(4.00)**	

Gokul Doodh was in financial distress from **Q2 to Q5**, with a total shortfall of **₹4.00 crore**. The accounting profits of Q2 and Q3 do not change this: profit is measured on the accrual basis (the hotel chain's unpaid bills count as sales), while distress is a matter of cash.

(ii) Insolvency. Insolvency is the inability of a firm to repay its various debts, and is a result of continuous financial distress. By Q5 the distress has continued for four quarters, the dairy has missed two loan instalments, and its revenue is inadequate to revive the situation. It became **insolvent in Q5**.

(iii) Consequences of prolonged distress. If distress continues for a long period, the firm may have to sell its assets, many times at a lower price: Gokul Doodh sold tankers carried at ₹1.10 crore for ₹0.70 crore, losing ₹0.40 crore, and lost the capacity they gave. Creditors put pressure on the firm: the bank refused to renew the overdraft. When revenue is inadequate to revive the situation, the firm cannot meet its obligations and becomes insolvent, as Gokul Doodh did in Q5.

Worked example **Profitable and still short of cash**

Tapri's institutional catering division began 2025-26 with no cash of its own. For the year it reports sales of ₹48 crore and expenses of ₹43 crore: a profit of ₹5 crore. All the expenses have been paid in cash, any gap being met by advances from head office. ₹9 crore of the sales were made to a railway caterer that has not paid and is disputing the bills; the rest has been collected. In the first week of April, ₹3 crore falls due to the division's tea suppliers. The division head says: "We made ₹5 crore, so we can pay the ₹3 crore without help from head office."

Comment on the division head's statement, distinguishing between the accounting and the financial management view of funds. (4 marks)

MODEL ANSWER

Relevant concept. Financial management and accounting differ in the treatment of funds. In accounting, funds are measured on the accrual principle: revenue is recognised at the point of sale and not when collected, and expenses when incurred rather than when paid. Accrual-based data do not fully reflect the financial condition of an organisation: one that is profitable in the accounting sense may be unable to meet its current obligations because of a shortage of liquidity, for example from uncollectible receivables. In financial management the treatment of funds is based on cash flows: revenue is recognised only when cash is received and expenses when paid, because the finance manager is concerned with maintaining solvency.

Facts of the case. Accounting profit = ₹48 crore − ₹43 crore = ₹5 crore. Cash received = ₹48 crore − ₹9 crore uncollected = ₹39 crore; cash paid = ₹43 crore.

Analysis. On a cash basis the division has a shortfall of ₹39 crore − ₹43 crore = **₹4 crore**, already met by head office's advances, so it has no cash of its own to pay the ₹3 crore due to suppliers. The ₹5 crore profit includes ₹9 crore of receivables that may never be collected.

Conclusion. The statement is not correct. It reads the accounting profit as if it were cash. The financial manager, whose focus is planning, controlling and decision-making on the basis of cash flows, would arrange funds for the ₹3 crore payment and pursue the ₹9 crore receivable; financial management begins where accounting ends.

Exam angle

In the exam

Where this topic appeared

In the ICAI documents read for this lesson, these topics were asked 15 times, always as theory:

- **Agency problem and agency cost**: explain the agency problem, its consequences and how to overcome it, MTP Series II for September 2024 Q4(a) (4 marks); explain the agency problem and how it can be addressed, MTP Series I for May 2025 Q4(a) (4 marks); what agency cost is, its types, and how to minimise it and align interests, May 2025 Q4(a) (4 marks); old scheme: RTP November 2020 Q10(a), MTP Series I for November 2020 Q6(a), MTP Series II for July 2021 Q6(b), RTP May 2022 Q10(b), MTP Series I for May 2023 Q6(b), MTP Series II for May 2023 Q6(ii), RTP November 2023 Q10(a).
- **Financial distress and insolvency**: short note, MTP Series I for September 2025 Q4(b) (4 marks); define financial distress and the consequences of prolonged distress, January 2026 Q4(c) (2 marks); true or false on its definition, May 2024 Q4(a)(iii); old scheme MTP Series I for May 2022 Q6(a) (4 marks).
- **Financial management and financial accounting**: differentiate, MTP Series I for January 2026 Q4(a) (4 marks); also the chapter's theory question 1.

In the exam

How the marks are earned

Every Suggested Answer on the agency problem is the Study Material's paragraph almost word for word: the separation of owners and managers; the principal–agent relationship; "in a nutshell" definition; agency cost and its four types; then the lender (negative covenants) and the three efforts for shareholders. The September 2024 answer to "consequences" gave agency cost as the consequence. Distress answers give the definition, then the sequence: assets sold at a lower price, revenue inadequate to revive, insolvency. The January 2026 answer for 2 marks was exactly those three sentences.

Trick

"MBOS" for the four agency costs

Monitoring, **B**onding, **O**pportunity, **S**tructuring: ICAI's order (SM 1.24). Condition: the list is of agency costs; the three efforts that address the problem (compensation, employee schemes, monitoring) are a separate list.

Answer frame

"Explain the agency problem and agency cost; how can it be addressed?" (4 marks)

1. **Separation**: owners participate in management in a proprietorship or partnership; in a company they do not, so owners and managers are separate.
2. **Principal–agent relationship**: managers may maximise salary and perks; agency problem = the chances that managers place personal goals ahead of the owners' goal.
3. **Agency cost**: the additional cost borne by shareholders to monitor and control managers; four types: monitoring, bonding, opportunity, structuring.
4. **Addressing it**: debt lenders, negative covenants (no borrowing beyond a point); shareholders, align interests: compensation linked to profit and long-term objectives; employee schemes linked to the share price; effective monitoring.

"Write a short note on financial distress and insolvency"

Factors to manage (price, demand, input prices, the proportion of debt; creditors' pressure) → definition of distress (cash inflows inadequate to meet all current obligations) → if continued, assets sold, often at a lower price → revenue inadequate to revive → definition of insolvency (inability to repay various debts, a result of continuous distress).

"Differentiate financial management and financial accounting"

Two heads, as in the comparison above: **treatment of funds** (accrual against cash flow, with the uncollectible-receivables example and the solvency reason) and **decision-making** (collecting and presenting data against planning, controlling and decision-making), ending with "financial management begins where accounting ends".

Common mistakes

⚠ **Common mistake**

Hall of Shame: "profitable, so not in distress"

Gokul Doodh showed profits in Q2 and Q3 and was in distress in both. **Fix:** distress is about cash inflows against current obligations; accounting profit is on the accrual basis and can coexist with a cash shortfall.

⚠ **Common mistake**

Using "distress" and "insolvency" as synonyms

Fix: distress is a cash shortfall against current obligations; insolvency is the inability to repay debts, the result of distress that continues and that revenue cannot reverse. A firm can come out of distress; insolvency is where distress ends if it does not.

⚠ **Common mistake**

Saying the agency cost is borne by the managers

Fix: ICAI's definition: the additional cost borne by the **shareholders** to monitor and control the managers (SM 1.24).

⚠ **Common mistake**

Forgetting the lender

Answers that discuss only shareholders and managers lose the point on debt. **Fix:** the agency problem arises when managers' interests are not aligned to those of the debt lenders as well as the equity investors; lenders use negative covenants.

Practice

Check yourself

1 Case scenario. Sahyadri Snacks Ltd's lenders insert a clause in its loan agreement that the company will not raise any further borrowing until its interest coverage ratio exceeds 3 times. In ICAI's terms, this clause is:

- **A** A bonding cost borne by the lenders
- **B** A form of managerial compensation linked to profit
- **C** A negative covenant addressing the agency problem of debt lenders
- **D** Evidence that the company is insolvent

2 Which of the following is true according to the Study Material?

- **A** Financial distress is a position where accounting profits are inadequate to meet long-term obligations
- **B** Insolvency is the inability of a firm to repay various debts, and is a result of continuous financial distress
- **C** A firm in financial distress is insolvent from the first quarter of the shortfall
- **D** A firm that shows an accounting profit cannot be in financial distress

3 A firm's cash inflows for a quarter are ₹2.60 crore and its current obligations falling due are ₹3.15 crore. Its accounts show a profit of ₹0.40 crore for the quarter. By how much, in ₹ crore, does it fall short of meeting its current obligations? Give two decimals.

Answer: _____

4 Which of these are among the efforts the Study Material lists for addressing the agency problem between managers and shareholders? Select all that apply.

- **A** Managerial compensation linked to the profit and long-term objectives of the company
- **B** Effective monitoring
- **C** Imposing negative covenants on managers' borrowing
- **D** Employee schemes resting on the assumption that maximising the stock price is the investors' objective

More than one answer may be right.

5 In the Study Material's figure, which discipline is the primary one on which the financial manager draws?

- **A** Economics
- **B** Quantitative methods
- **C** Accounting
- **D** Marketing

Answers: the answer key at the end of this lesson.

Recap

- Agency problem: the chances that managers place personal goals ahead of the owners' goal; it arises from the separation of ownership and management in companies.
- Agency cost: the additional cost borne by shareholders to monitor and control managers; four types: monitoring, bonding, opportunity, structuring.
- Lenders' agency problem: negative covenants (managers cannot borrow beyond a point).
- Shareholders: link pay to profit and long-term objectives; employee schemes tied to the share price; effective monitoring.
- Financial distress: cash inflows inadequate to meet all current obligations; prolonged, it forces asset sales, often at lower prices.
- Insolvency: inability to repay various debts, the result of continuous distress when revenue cannot revive the situation.
- Accounting (accrual) is the primary input; financial management (cash flows; planning, control and decisions) begins where accounting ends; marketing, production and quantitative methods support, economics shows the world outside.
- The driver with the fuel card is cheaper to watch than to replace, but never free.

Answer key · Check yourself

1 **C A negative covenant addressing the agency problem of debt lenders**

SM 1.24: the agency problem of debt lenders is addressed by imposing negative covenants, i.e. managers cannot borrow beyond a point. A covenant does not show insolvency; it protects the lender before trouble starts.

2 **B Insolvency is the inability of a firm to repay various debts, and is a result of continuous financial distress**

Distress is about cash inflows and current obligations; insolvency follows only when distress continues and revenue cannot revive the situation.

3 **0.55**

Shortfall = 3.15 − 2.60 = ₹0.55 crore: the firm is in financial distress whatever its accounting profit. The ₹0.40 crore profit is a distractor, measured on the accrual basis.

4 **A Managerial compensation linked to the profit and long-term objectives of the company; B Effective monitoring; D Employee schemes resting on the assumption that maximising the stock price is the investors' objective**

Negative covenants address the debt lenders' agency problem, not the shareholders'.

5 **C Accounting**

SM 1.23: accounting is the primary discipline; marketing, production and quantitative methods are supportive; economics helps the manager understand the world outside the business.

CHAPTER 2

Financial Analysis and Planning: Ratio Analysis

2.1 Liquidity and capital structure ratios

Can the business pay what falls due this year, who funded it, and how many times do its earnings cover interest and instalments: the liquidity, capital structure and coverage ratios.

In plain words

Three questions from the bank

Tapri Tea Co. Ltd wants a ₹50 crore term loan from Narmada Bank for a new cold-chain fleet. The bank's credit officer has sent back one page: "Please give us your ratios."

> **Kesar:** Our ratio is simple: more chai, more money. Write that.
>
> **Dev:** They want three answers, Kesar-ji. Can we pay what falls due this year? Who paid for the business, we or the lenders? And how many times do our profits cover the interest and the instalments?
>
> **Kesar:** Tell them sab adjust ho jayega.
>
> **Dev:** The bank discounts that sentence to zero.

Dev opens Tapri's balance sheet at 31 March 2026. Within the next year Tapri must pay ₹70 crore (suppliers, unpaid expenses, tax); those are its **current liabilities**. Within the same year it expects ₹150 crore to turn into cash (stock, money owed by customers, cash itself); those are its **current assets**. ₹150 crore against ₹70 crore is ₹2.14 of current assets for every rupee due: that is the **current ratio**, the first test of **liquidity**, the ability to pay short-term liabilities.

Some of that ₹150 crore is stock that must first be sold. Take out stock (and expenses paid in advance, which no supplier will accept as payment) and ₹85 crore is left: ₹1.21 for every rupee due. That stricter test is the **quick ratio**.

The second question is about the long run. The equity shareholders have put in ₹300 crore (equity share capital plus retained profits); lenders ₹110 crore (debentures and a term loan). The **debt-equity ratio** of 0.37 says lenders have ₹37 in the business for every ₹100 of owners' money. Ratios like it, which show the mix of owners' and lenders' funds, are **capital structure ratios**.

The third question is about paying lenders. Tapri's profit before interest and tax is ₹132 crore and its interest ₹12 crore: profits cover interest 11 times. Ratios like this, which compare earnings with fixed charges, are **coverage ratios**.

Think of it like

The household loan file

A bank deciding on your home loan asks the same three things. Can you pay this month's bills from your salary account and what friends owe you (the current ratio), without selling the family gold (the quick ratio)? How much of the house is yours and how much is the bank's (the debt-equity ratio)? How many times does your income cover the interest (interest coverage), and the whole EMI with the principal in it (debt service coverage)?

Key idea

A single figure says little; a figure compared with another says a lot. Liquidity ratios test whether short-term liabilities can be paid from assets that turn into cash within the year; capital structure ratios show how much of the business owners fund and how much lenders; coverage ratios show how many times earnings cover interest, instalments and preference dividend.

How it works

What a ratio is, and where its numbers come from

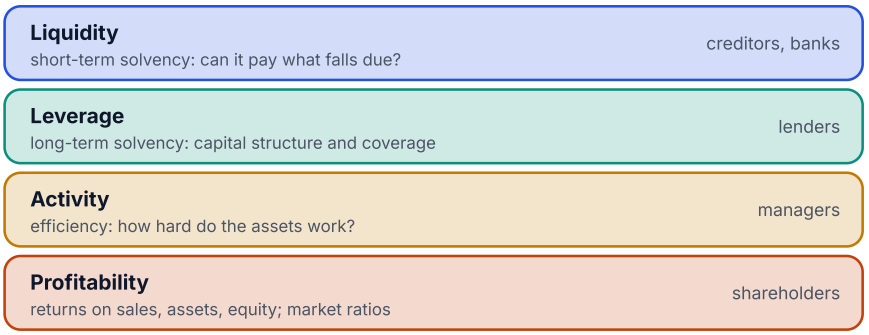
ICAI defines a ratio as "the indicated quotient of two mathematical expressions and as the relationship between two or more things"; a financial (accounting) ratio is a mathematical expression of the relationship between two accounting figures (SM 3.3). **Ratio analysis** rests on the fact that a single accounting figure by itself may not communicate any meaningful information, but expressed relative to some other figure it may. Its calculation basis and objective (SM 3.3):

- **Calculation basis:** a relationship expressed in mathematical terms, between two individual figures or group of figures, connected with each other in some logical manner, selected from the financial statements of the concern.
- **Objective:** all stakeholders (owners, investors, lenders, employees etc.) can draw conclusions about the performance (past, present and future) and the strengths and weaknesses of a firm, and can take decisions in relation to the firm.

A ratio means most when it is compared: with the firm's own previous years (**intra-firm comparison**), with other companies (**inter-firm comparison**), with the industry, or even with the economy in general (SM 3.3). The sources of information for financial statement analysis are (SM 3.4): (i) annual reports, (ii) interim financial statements, (iii) notes to accounts, (iv) statement of cash flows, (v) business periodicals, (vi) credit and investment advisory services.

Four families of ratios

Four families of ratios, and who asks first



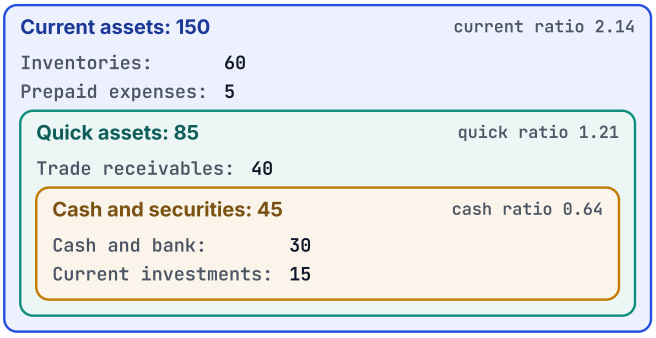
ICAI's classification (SM 3.4). Leverage ratios split into capital structure ratios and coverage ratios; profitability ratios into four groups. The right column is the user who leans on that family most (SM 3.30–3.31).

This lesson covers the first two families. Activity ratios are in activity (turnover) ratios and profitability and market ratios in profitability ratios and the DuPont analysis. ICAI adds a warning to its chart: liquidity ratios should be examined taking the relevant turnover ratios into consideration (SM 3.4), because a current ratio of 2 built on stock that never sells is no comfort.

Liquidity: three tests, each stricter than the last

The terms "liquidity" and "short-term solvency" are used synonymously. Inability to pay short-term liabilities affects a business's credibility and credit rating, and continuous default leads to commercial bankruptcy. Both lack of sufficient liquidity and excess liquidity is bad for the organization (SM 3.5). Tapri's current assets, and the three tests drawn from them:

Tapri, 31 March 2026: three liquidity tests (₹ crore)



Each box drops the least liquid items of the box around it; every ratio divides its box by the same current liabilities of ₹70 crore.

1 **Current ratio** = current assets ÷ current liabilities = 150 ÷ 70 = **2.14**. It asks: "Does your business have enough current assets to meet the payment schedule of its current debts with a margin of safety for possible losses in current assets?" A generally acceptable current ratio is 2:1, but whether a ratio is satisfactory depends on the nature of the business and the characteristics of its current assets and liabilities (SM 3.5–3.6).

2 **Quick ratio** (acid-test ratio) = quick assets ÷ current liabilities, where quick assets = current assets − inventories − prepaid expenses = 150 − 60 − 5 = 85; 85 ÷ 70 = **1.21**. Inventories are left out because they are not near-cash and, in financial difficulty, may sell only at liquidation value (though in a seller's market they are near-cash too). An acid-test of 1:1 is considered satisfactory unless most quick assets are receivables whose collection lags behind the schedule for paying current liabilities (SM 3.6).

3 **Cash ratio** (absolute liquidity ratio) = (cash and bank balances + marketable securities or current investments) ÷ current liabilities = (30 + 15) ÷ 70 = **0.64**. It tests liquidity in terms of cash and marketable securities only (SM 3.7).

4 **Basic defense interval** = quick assets ÷ daily operating expenses. Daily operating expenses = (cost of goods sold + selling, administration and other general expenses − depreciation and other non-cash expenditure) ÷ days in the year. Tapri: (₹540 + ₹228 − ₹36) crore ÷ 360 = ₹2.03 crore a day, so 85 ÷ 2.03 = **41.8 days**: if all revenue stopped tomorrow, Tapri could pay its cash expenses for about 42 days without new financing (SM 3.7).

5 **Net working capital** = current assets − current liabilities = 150 − 70 = **₹80 crore**. It is more a measure of cash flow than a ratio; it should be positive, though in certain business models it may be negative. Bankers watch it over time to judge the ability to weather financial crises, and loans are often tied to a minimum working capital (SM 3.8).

Interactive version: PassBook app, lesson Liquidity and capital structure ratios

Long-term solvency: who funded the business?

Leverage ratios measure the long-term stability and capital structure of the firm. They show the mix of funds provided by owners and lenders and assure long-term lenders about (i) periodic payment of interest during the loan and (ii) repayment of principal on maturity (SM 3.8). They are of two types: capital structure ratios and coverage ratios.

Capital structure ratios show the relative weight of the different sources of funds, which the balance sheet shows only in absolute amounts (SM 3.9). Tapri's long-term side at 31 March 2026 (₹ crore):

Equity and liabilities	₹ crore	Assets	₹ crore
Equity share capital (₹10 shares)	40	Fixed assets (net)	355
Reserves and surplus	260	Current assets	150
8% Preference share capital	25		
10% Debentures	60		
12% Term loan	50		
Current liabilities	70		
Total	**505**	**Total**	**505**

From it: shareholders' equity (equity capital + reserves) = ₹300 crore; total debt (debentures + term loan) = ₹110 crore; total outside liabilities (debt + current liabilities) = ₹180 crore; net assets (fixed assets + net current assets) = 355 + 80 = ₹435 crore, which is also equity 300 + preference 25 + debt 110. ICAI also calls net assets **capital employed** (SM 3.9).

Ratio	Tapri's working	Result
Equity ratio	300 ÷ 435	0.69
Debt ratio	110 ÷ 435	0.25
Debt to equity (total outside liabilities)	180 ÷ 300	0.60
Debt to equity (long-term debt)	110 ÷ 300	0.37
Debt to total assets	180 ÷ 505	0.36
Capital gearing	(25 + 60 + 50) ÷ 300	0.45
Proprietary ratio	(40 + 25 + 260) ÷ 505	0.64

Read them as: owners fund 69% of net assets and debt 25%; ₹60 is owed (₹37 of it borrowed long-term) for every ₹100 the shareholders own; outsiders finance 36% of total assets; fixed-charge capital is ₹0.45 for every rupee of equity; shareholders finance 64% of total assets.

Higher equity and proprietary ratios mean a lower risk for lenders; higher debt, debt-equity, debt to total assets and gearing ratios mean more financial leverage and more risk (SM 3.9–3.11). **Debt equity ratio is the indicator of firm's financial leverage**, and it is very often used for capital structure decisions such as issuing shares or debentures (SM 3.10).

Coverage: how many times do earnings pay the fixed claims?

Coverage ratios measure the firm's ability to service its fixed liabilities: (i) interest on loans, (ii) preference dividend, (iii) amortisation of principal, repayment of loan instalments or redemption of preference capital (SM 3.11). Tapri's year to 31 March 2026 (₹ crore): sales 900, cost of goods sold 540, operating expenses 228 (depreciation 36 included), so EBIT 132; interest 12 (10% on ₹60 crore debentures and 12% on the ₹50 crore term loan); tax at 25%, so profit after tax (EAT) 90; preference dividend 2; equity dividend ₹5.50 a share on 4 crore shares, ₹22; term loan instalment due ₹10.

ICAI lists four, in this order (SM 3.9, 3.11–3.13):

1. **Debt service coverage ratio (DSCR)** = earnings available for debt service ÷ (interest + instalments), where earnings available = EAT + depreciation and other non-cash operating expenses + interest + other adjustments such as loss on sale of a fixed asset: (90 + 36 + 12) ÷ (12 + 10) = 138 ÷ 22 = **6.27 times**. Lenders use it to judge the firm's ability to pay current interest and instalments; normally a DSCR of 1.5 to 2 is satisfactory (SM 3.11–3.12).

2 **Interest coverage ratio** ("times interest earned") = EBIT ÷ interest = 132 ÷ 12 = **11 times**. EBIT is used because interest is a deductible expense, so the tax burden does not affect the ability to pay it. A high ratio means interest can be met even if EBIT falls considerably; a low one indicates excessive use of debt or inefficient operations (SM 3.12).

3 **Preference dividend coverage ratio** = EAT ÷ preference dividend = 90 ÷ 2 = **45 times**: the margin of safety for preference shareholders (SM 3.12–3.13). The **equity dividend coverage ratio** = (EAT − preference dividend) ÷ equity dividend = 88 ÷ 22 = **4 times**.

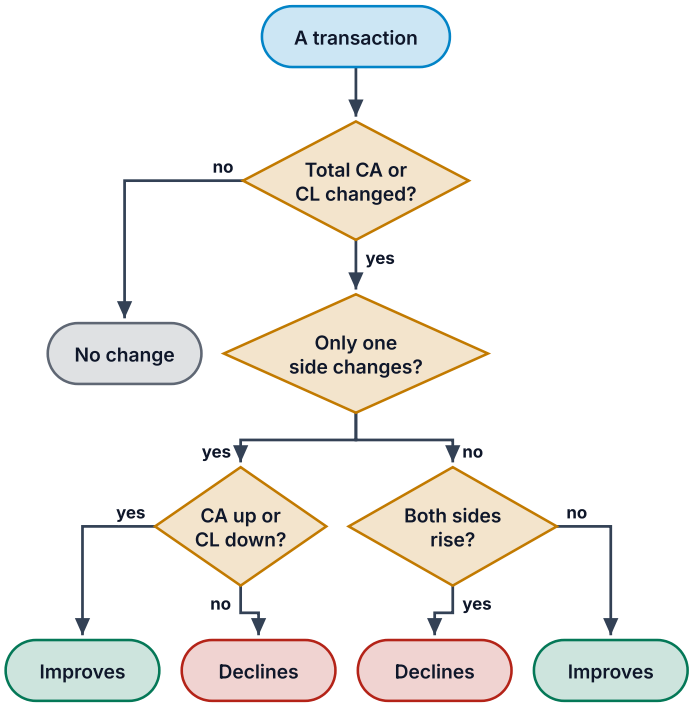
4 **Fixed charges coverage ratio** = (EBIT + depreciation) ÷ (interest + repayment of loan) = (132 + 36) ÷ 22 = **7.64 times**: how many times the cash flow before interest and taxes covers all fixed financing charges; more than 1 is considered safe (SM 3.13).

Interactive version: PassBook app, lesson Liquidity and capital structure ratios

What moves the current ratio

ICAI asks this as a two-mark "improve, decline or remain unchanged" question. Work out what the transaction does to total current assets and total current liabilities, then compare:

Effect of a transaction on the current ratio



CA: current assets; CL: current liabilities. Green: improves; rose: declines. The right-hand branch is for a transaction that moves both sides by the same amount (both rise, or both fall); it assumes the ratio is above 1, as it almost always is in exam questions; below 1 the two answers swap, and at exactly 1 nothing changes. A transaction that moves the two sides by different amounts: recompute the ratio.

The rule behind the right-hand branch: with a current ratio above 1, adding the same amount to both sides pulls the ratio towards 1 (it falls), and removing the same amount from both pushes it away

from 1 (it rises). Repaying a ₹20 crore short-term bank loan from the bank balance takes Tapri from 150 ÷ 70 = 2.14 to 130 ÷ 50 = 2.60. Collecting ₹8 crore from debtors only swaps one current asset for another: 2.14 stays 2.14.

Formats and formulas

Reproduce the two lists and every formula from memory; ICAI's lists are what its answers count as current.

Current assets (SM 3.5)	Current liabilities (SM 3.6)
Inventories	Creditors for goods and services
Sundry debtors	Short-term loans
Cash and bank balances	Bank overdraft
Receivables / accruals	Cash credit
Loans and advances	Outstanding expenses
Disposable investments	Provision for taxation
Any other current assets	Proposed dividend
	Unclaimed dividend
	Any other current liabilities

Definition

Ratio (SM 3.3)

"The indicated quotient of two mathematical expressions and as the relationship between two or more things." A financial or accounting ratio is a mathematical expression of the relationship between two accounting figures.

Formula

Current ratio and quick (acid-test) ratio

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}} \qquad \text{Quick ratio} = \frac{\text{Quick assets}}{\text{Current liabilities}}$$

Quick assets = current assets − inventories − prepaid expenses. ICAI's norms: 2:1 and 1:1 (SM 3.38).

Formula

Cash ratio (absolute liquidity ratio)

$$\text{Cash ratio} = \frac{\text{Cash} + \text{Marketable securities}}{\text{Current liabilities}}$$

Cash = cash and bank balances; current investments may replace marketable securities (SM 3.7).

Σ **Formula**

Basic defense interval (interval measure)

$$\text{BDI} = \frac{C + R + M}{\text{Daily operating expenses}} \qquad = \frac{\text{CA} - \text{Prepaid} - \text{Inventories}}{\text{Daily operating expenses}}$$

C = cash and bank balances, R = net receivables, M = marketable securities; CA = current assets; prepaid = prepaid expenses (SM 3.7).

$$\text{Daily operating expenses} \qquad = \frac{\text{COGS} + S - N}{\text{No. of days in a year}}$$

S = selling, administration and other general expenses; N = depreciation and other non-cash expenditure. The answer is in days; take 360 days unless the question says otherwise.

Σ **Formula**

Net working capital

$$\text{NWC} = \text{CA} - \text{CL}$$

NWC = net working capital; CA = current assets; CL = current liabilities. The Study Material's definition adds " (excluding short-term bank borrowing)" to current liabilities (SM 3.8); see Under the hood.

Σ **Formula**

Equity ratio and debt ratio

$$\text{Equity ratio} = \frac{\text{Shareholders' equity}}{\text{Net assets}} \qquad \text{Debt ratio} = \frac{\text{Total debt}}{\text{Net assets}}$$

Shareholders' equity = equity share capital + reserves and surplus (excluding fictitious assets). Net assets (capital employed) = net fixed assets + net current assets. Total debt = short and long-term borrowings from financial institutions, debentures and bonds, deferred payment arrangements for capital equipment, bank borrowings, public deposits and any other interest-bearing loan (SM 3.9–3.10).

Σ **Formula**

Debt to equity ratio

$$\text{D/E} = \frac{\text{Total outside liabilities}}{\text{Shareholders' equity}} \qquad \text{or} \qquad \frac{\text{Long-term debt}}{\text{Shareholders' equity}}$$

The first counts both current and non-current liabilities; sometimes only interest-bearing long-term debt is used (SM 3.10).

Σ Formula

Debt to total assets, capital gearing and proprietary ratio

$$\text{Debt to total assets} = \frac{\text{Total outside liabilities}}{\text{Total assets}}$$

$$\text{Capital gearing} = \frac{P + D + B}{E + R - L}$$

$$\text{Proprietary ratio} = \frac{\text{Proprietary fund}}{\text{Total assets}}$$

Debt to total assets may use total debt in the numerator. For gearing, P = preference share capital, D = debentures, B = other borrowed funds, E = equity share capital, R = reserves and surplus, L = losses. Proprietary fund = equity share capital + preference share capital + reserves and surplus; total assets exclude fictitious assets and losses (SM 3.10–3.11).

Σ Formula

Coverage ratios

$$\text{DSCR} = \frac{\text{Earnings for debt service}}{\text{Interest} + \text{Instalments}} \qquad \text{Interest coverage} = \frac{\text{EBIT}}{\text{Interest}}$$

Earnings (available) for debt service = EAT + depreciation and other non-cash operating expenses + interest + other adjustments such as loss on sale of fixed assets (SM 3.11–3.12).

$$\text{Preference dividend coverage} = \frac{\text{EAT}}{\text{Preference dividend}}$$

$$\text{Equity dividend coverage} = \frac{\text{EAT} - \text{Preference dividend}}{\text{Equity dividend}}$$

$$\text{Fixed charges coverage} = \frac{\text{EBIT} + \text{Depreciation}}{\text{Interest} + \text{Repayment of loan}}$$

ICAI's norms (SM 3.39): DSCR 2; interest, preference dividend and fixed charges coverage above 1.

ⓘ Note

ICAI's notes for calculating ratios (SM 3.13)

1. EBIT = PBIT (profit before interest and taxes); EAT = PAT (profit after taxes); EBT = PBT (profit before taxes).
2. Ratios shall be calculated based on requirement and availability of information and may deviate from the original formulae. If required, assumptions should be given.
3. The numerator should be taken in correspondence with the denominator, and vice versa.

Worked examples

Worked example **The kulhad maker's loan file**

Malwa Kulhad Udyog Pvt Ltd makes clay cups for Tapri. It has applied to Narmada Bank for a ₹25 lakh loan for a new kiln, and the bank asks for a financial soundness table: each ratio, its norm, and whether it is favourable or adverse. Its statements for 2025-26 (₹ lakh):

Equity and liabilities	₹ lakh	Assets	₹ lakh
Equity share capital (₹10 each)	50	Fixed assets (net)	92
Reserves and surplus	30	Inventories	38
9% Preference share capital	10	Trade receivables	30
10% Term loan	40	Prepaid insurance	2
Trade payables	28	Cash and bank	8
Outstanding wages	6		
Provision for taxation	6		
Total	**170**	**Total**	**170**

Statement of profit (2025-26)	₹ lakh
Sales	240
Less: Cost of goods sold (including depreciation ₹9 lakh)	168
Less: Administrative and selling expenses	36
EBIT	**36**
Less: Interest on term loan	4
Less: Tax at 25%	8
Profit after tax	**24**

The term loan is repayable in instalments of ₹8 lakh a year.

Required:

(i) Calculate the current ratio, quick ratio, cash ratio and basic defense interval (360 days).

(ii) Calculate the debt-equity ratio (long-term debt), capital gearing ratio and proprietary ratio.

(iii) Calculate the interest coverage ratio, DSCR, preference dividend coverage ratio and fixed charges coverage ratio.

(iv) Present the bank's table with ICAI's norms, and comment.

THE WORKING

Working notes

1. Current assets = 38 + 30 + 2 + 8 = **₹78 lakh**; current liabilities = 28 + 6 + 6 = **₹40 lakh**.
2. Quick assets = 78 − 38 − 2 = **₹38 lakh**.
3. Daily operating expenses = (168 + 36 − 9) ÷ 360 = 195 ÷ 360 = **₹0.5417 lakh**.
4. Shareholders' equity = 50 + 30 = **₹80 lakh**; proprietary fund = 50 + 10 + 30 = **₹90 lakh**.
5. Preference dividend = 9% × 10 = **₹0.90 lakh**.

(i) Liquidity

Ratio	Working	Result
Current ratio	78 ÷ 40	1.95
Quick ratio	38 ÷ 40	0.95
Cash ratio	8 ÷ 40	0.20
Basic defense interval	38 ÷ 0.5417	70.15 days

(ii) Capital structure

Ratio	Working	Result
Debt-equity (long-term debt)	40 ÷ 80	0.50
Capital gearing	(10 + 40) ÷ 80	0.625
Proprietary ratio	90 ÷ 170	0.53

(iii) Coverage

Ratio	Working	Result
Interest coverage	36 ÷ 4	9.00 times
DSCR	(24 + 9 + 4) ÷ (4 + 8)	3.08 times
Preference dividend coverage	24 ÷ 0.90	26.67 times
Fixed charges coverage	(36 + 9) ÷ (4 + 8)	3.75 times

(iv) The bank's table

Malwa Kulhad Udyog: financial soundness table

Ratio	Malwa	Norm	Verdict
Current ratio	1.95	2	**Adverse**
Quick ratio	0.95	1	**Adverse**
DSCR	3.08	2	**Favourable**
Interest cover	9.00	> 1	**Favourable**
Pref. dividend cover	26.67	> 1	**Favourable**
Fixed charges cover	3.75	> 1	**Favourable**

Norms from ICAI's summary of ratios (SM 3.38–3.39). Rose: adverse; green: favourable.

Comment. Long-term solvency is comfortable: earnings cover interest nine times and interest plus instalments about three times, against a norm of 2, and lenders fund only ₹50 for every ₹100 of equity. Short-term liquidity is slightly below the norms: the current ratio is 1.95 against 2, and the quick ratio 0.95 against 1, because inventories are ₹38 lakh of the ₹78 lakh of current assets. The firm should be asked about the age and saleability of its stock (the turnover ratios), since it is the weak point of its liquidity.

Worked example **Mehra Masala Traders: what the two ratios hide**

Kesar's cousin runs Mehra Masala Traders, which supplies spice mixes to Tapri. At 31 March 2026 its inventory is ₹12 lakh and its current assets also include prepaid rent of ₹1.20 lakh. Its current ratio is 2.5 and its quick ratio 1.4. Cash operating expenses are ₹1.44 crore a year.

Required:

(i) Calculate current liabilities, current assets, quick assets and working capital.

(ii) Calculate the basic defense interval (360 days).

(iii) The cousin wants to "bring the current ratio down to a sensible 2" by buying stock on credit. Calculate the stock to be bought, and the quick ratio afterwards.

THE WORKING

(i) Current liabilities and current assets

Current ratio: CA = 2.5 CL. Quick ratio: quick assets = CA − inventory − prepaid expenses = 1.4 CL. Subtracting the second from the first:

$$2.5\,\text{CL} - 1.4\,\text{CL} = 12 + 1.20 \qquad 1.1\,\text{CL} = 13.20 \qquad \text{CL} = ₹12\ \text{lakh}$$

CA = 2.5 × 12 = **₹30 lakh**; quick assets = 1.4 × 12 = **₹16.80 lakh** (check: 30 − 12 − 1.20); working capital = 30 − 12 = **₹18 lakh**. Leaving the prepaid rent out of the equation gives CL = 12 ÷ 1.1 = ₹10.91 lakh, and every later figure goes wrong with it.

(ii) Basic defense interval

Daily cash operating expenses = ₹144 lakh ÷ 360 = ₹0.40 lakh. BDI = 16.80 ÷ 0.40 = **42 days**.

(iii) Stock bought on credit

Buying stock of ₹x on credit adds x to both sides:

$$\frac{30 + x}{12 + x} = 2 \qquad 30 + x = 24 + 2x \qquad x = ₹6\ \text{lakh}$$

After the purchase, quick assets are unchanged at ₹16.80 lakh and current liabilities are ₹18 lakh: quick ratio = 16.80 ÷ 18 = **0.93**. The current ratio did come down to 2, but the firm is less liquid, not more sensible: the new stock is not a quick asset and the new creditors must be paid in cash.

Worked example **Seven entries and one current ratio**

Tapri's current assets are ₹150 crore, including inventories of ₹60 crore and prepaid expenses of ₹5 crore, and its current liabilities are ₹70 crore (current ratio 2.14). State, with reasons, whether each transaction below improves, reduces or leaves unchanged the current ratio. Take each transaction separately.

(i) Paid trade payables of ₹10 crore by cheque.

(ii) Bought premix stock of ₹5 crore on credit.

(iii) Accepted a 90-day bill of exchange for ₹6 crore drawn by a supplier, in settlement of the amount owed to him.

(iv) Sold an old delivery van, with a book value of ₹2 crore, for ₹3 crore in cash.

(v) Paid next year's fleet insurance premium of ₹1 crore in advance by cheque. Also state its effect on the quick ratio.

(vi) An instalment of ₹6 crore of the term loan falls due within the next twelve months and is now shown under current liabilities.

(vii) Bought a new delivery van for ₹4 crore in cash.

MODEL ANSWER

Transaction	Effect on CA and CL	Current ratio
(i) Creditors paid ₹10 crore	CA and CL both fall by ₹10 crore; the ratio is above 1	Improves (140 ÷ 60 = 2.33)
(ii) Stock bought on credit ₹5 crore	CA and CL both rise by ₹5 crore; the ratio is above 1	Reduces (155 ÷ 75 = 2.07)
(iii) Bill payable ₹6 crore accepted for a creditor	One current liability (creditors) is replaced by another (bills payable): total CL unchanged	Unchanged (2.14)
(iv) Old van sold for ₹3 crore cash	Cash rises ₹3 crore; the van was a fixed asset	Improves (153 ÷ 70 = 2.19)
(v) Insurance ₹1 crore prepaid	Cash falls, prepaid expenses rise: total CA unchanged	Unchanged (2.14); quick ratio falls from 85 ÷ 70 = 1.21 to 84 ÷ 70 = 1.20, because prepaid expenses are not quick assets
(vi) ₹6 crore of term loan now current	CL rises; CA unchanged	Reduces (150 ÷ 76 = 1.97)
(vii) New van bought for ₹4 crore cash	Cash (current) falls; the van is a fixed asset	Reduces (146 ÷ 70 = 2.09)

Exam angle

In the exam

Where these ratios appeared, May 2024 to September 2026

In the ICAI documents read for this lesson:

- **Descriptive, 5 marks.** January 2026 Q1(a): closing inventory from an inventory holding period, then working capital from a current ratio of 1.20 and a quick ratio of 0.80 (two equations, two unknowns), then the basic defense interval on cash operating expenses (2 + 1 + 2 marks). May 2024 Q1(a): interest coverage, gross profit ratio and current assets from a debt-equity ratio, a return on shareholders' fund and a current ratio.
- **Two-mark theory.** January 2026 Q4(c) (an OR option): whether four transactions improve, reduce or leave unchanged a current ratio of 2:1; the suggested answer is a table of transaction and impact.
- **MCQs.** January 2025 MCQs 6–7 (inventory, receivables, current assets and liabilities from a current ratio of 1.5 and a quick ratio of 1); September 2025 MCQ 8 (stock from current liabilities, current and quick ratios, 1 mark); September 2026 MCQ 6 (percentage change in current liabilities and working capital when the current ratio is moved from 2 to 2.5, 2 marks).
- **RTPs and MTPs.** RTP January 2026 Q5: a bank's "financial soundness report" in the format issue, ratio, numerator, denominator, ideal ratio, company's ratio, favourable or adverse. RTP January 2025 Q4: DSCR with future instalments. RTP January 2026 MCQ 4 and RTP September 2025 MCQ 2: interest coverage from the degree of financial leverage. MTP Series I for September 2026 Q3(b) (3 marks): current liabilities and inventories from net working capital and the current and quick ratios, inside a summarised balance sheet. MTP Series II for September 2026: a five-MCQ case for a loan appraisal (current and quick ratios, receivable and inventory turnovers, long-term debt to total debt and debt-equity, net profit ratio and return on total assets, then what a falling interest coverage with a rising debt-equity ratio means for the lender), all read against industry averages.

In the exam

The traps ICAI set

- Two equations, two unknowns: a current ratio and a quick ratio together give current liabilities only after inventory is known (January 2026 Q1(a)); a current ratio with working capital gives CL = WC ÷ (CR − 1).
- The basic defense interval uses **cash** operating expenses: depreciation and other non-cash items are deducted (January 2026 Q1(a)).
- "Future instalments" are the instalments in the DSCR denominator, not the whole loan (RTP January 2025 Q4).
- The paper's own instruction: "Working notes should form part of the answer."

Trick

Current liabilities in one line

From working capital: CL = working capital ÷ (current ratio − 1), and CA = CL × current ratio; then inventory = CA − (quick ratio × CL) − prepaid expenses. From inventory: CL = (inventory + prepaid expenses) ÷ (current ratio − quick ratio). Conditions: the working capital is CA − all CL, and quick assets exclude exactly inventory and prepaid expenses.

Answer frame

"Discuss liquidity ratios" and "Discuss gearing ratio"

The chapter's own theory questions include "Discuss Stock Turnover ratio and Gearing ratio" and "Discuss the financial ratios for evaluating company performance on ... liquidity position" (SM 3.65). For each ratio write, in this order:

1. What it measures, in ICAI's words (liquidity: ability to pay short-term liabilities; gearing: proportion of fixed interest or dividend bearing capital to equity shareholders' funds).
2. The formula, with the components defined (what is a quick asset; what is in the denominator of gearing).
3. The interpretation and the norm where ICAI gives one (2:1, 1:1; higher gearing may indicate more risk and signifies the advantage of financial leverage to equity shareholders).
4. Who uses it (short-term creditors and banks for liquidity; long-term lenders for leverage).
5. A one-line caution: a ratio needs comparison (past years, competitors, industry), and liquidity ratios should be read with the turnover ratios.

Three answers to "calculate the DSCR"

Loses marks

"DSCR = 3.08."

A bare figure: no formula, no basis, nothing to give method marks for.

Passes

"DSCR = 37 ÷ 12 = 3.08 times."

Right, but the examiner cannot see that 37 is profit after tax + depreciation + interest.

Full marks

Working note for earnings available for debt service (24 + 9 + 4 = 37) and for debt service (4 + 8 = 12); formula; result in "times"; one line against the norm of 2 ("favourable").

Common mistakes

⚠ **Common mistake**

Hall of Shame: the prepaid insurance that paid a supplier

Quick assets taken as current assets − inventory only. Malwa's quick ratio becomes 1.00 instead of 0.95, and "adverse" turns into "satisfactory". **Fix:** deduct inventories **and** prepaid expenses (SM 3.6).

⚠ **Common mistake**

Interest coverage on profit after tax

Dividing PAT (₹24 lakh) by interest gives 6 times instead of 9. **Fix:** the numerator is EBIT, because interest is paid before tax and is tax-deductible.

⚠ **Common mistake**

The DSCR that forgot the principal

Dividing earnings by interest alone (37 ÷ 4 = 9.25) turns the DSCR into an interest coverage ratio. **Fix:** the denominator is interest + instalments due; "debt service" means both.

⚠ **Common mistake**

The preference shares that became equity

Capital gearing taken as (debentures + loans) ÷ (equity + preference + reserves). **Fix:** preference capital carries a fixed dividend, so it goes in the **numerator** with the debt; the denominator is equity share capital + reserves − losses.

⚠ **Common mistake**

"Higher current ratio, better company"

A current ratio of 4 may mean idle cash, slow debtors or unsold stock. **Fix:** say that both too little and too much liquidity are bad (SM 3.5), and check the turnover ratios before praising it.

Practice

Check yourself

1 A wholesaler has cash and bank balances of ₹6 lakh, trade receivables of ₹14 lakh, marketable securities of ₹4 lakh, inventory of ₹20 lakh and prepaid expenses of ₹1 lakh. Its cost of goods sold for the year is ₹1.50 crore and its selling, administration and general expenses ₹30 lakh; depreciation of ₹12 lakh is included in these costs. What is its basic defense interval, in days (360-day year)? Round to one decimal.

Answer: _____

2 Case scenario. Narmada Bank is appraising Tapri's cold-chain subsidiary for a term loan. Its last year: profit after tax ₹18 lakh, depreciation ₹6 lakh, interest ₹4 lakh, and loan instalments of ₹10 lakh falling due. The credit officer must enter the debt service coverage ratio in the appraisal note. Which figure is correct?

A 1.71 times

B 2.00 times

C 7.00 times

D 2.40 times

3 A company has equity share capital ₹30 lakh, reserves ₹20 lakh, 10% preference share capital ₹10 lakh and 12% debentures ₹25 lakh. What is its capital gearing ratio?

- A 0.42
- B 0.50
- C 0.70
- D 0.41

4 A company's current ratio is 2.5. It buys goods of ₹2 lakh on credit. What happens to the current ratio?

- A It rises
- B It falls
- C It does not change
- D It cannot be said without the amounts

5 Which of these does ICAI list as current liabilities? Select all that apply.

- A Bank overdraft
- B Provision for taxation
- C Unclaimed dividend
- D Debentures redeemable after five years
- E Cash credit

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Current ratio = CA ÷ CL (norm 2:1); quick ratio = (CA − inventories − prepaid expenses) ÷ CL (norm 1:1); cash ratio = (cash and bank + marketable securities) ÷ CL.
- Basic defense interval = quick assets ÷ daily cash operating expenses (COGS + selling, admin and general − depreciation, ÷ days): days of survival without revenue.
- Net working capital = CA − CL. Too little and too much liquidity are both bad.
- Debt-equity = total outside liabilities (or long-term debt) ÷ shareholders' equity: the indicator of financial leverage. Say which debt you use.
- Capital gearing = (preference + debentures + other borrowings) ÷ (equity capital + reserves − losses); proprietary ratio = (equity + preference + reserves) ÷ total assets.
- Interest coverage = EBIT ÷ interest; DSCR = (EAT + non-cash expenses + interest) ÷ (interest + instalments), norm 2; fixed charges coverage = (EBIT + depreciation) ÷ (interest + repayment).
- With a current ratio above 1, equal additions to CA and CL lower it; equal payments raise it.
- CL = working capital ÷ (current ratio − 1). (The bank discounts "sab adjust ho jayega" to zero.)

Answer key · Check yourself

1 **51.4 to 51.5**

Quick assets = 6 + 14 + 4 = ₹24 lakh (inventory and prepaid expenses out). Daily operating expenses = (150 + 30 − 12) ÷ 360 = ₹0.4667 lakh. BDI = 24 ÷ 0.4667 = 51.4 days. Keeping inventory and prepaid expenses in gives 96.4 days; forgetting to deduct depreciation gives 48 days.

2 **B** **2.00 times**

DSCR = (EAT + depreciation + interest) ÷ (interest + instalments) = (18 + 6 + 4) ÷ (4 + 10) = 28 ÷ 14 = 2.00. 1.71 leaves interest out of the numerator; 7.00 divides by interest alone; 2.40 compares 24 with the instalments only.

3 **C** **0.70**

Gearing = (preference + debentures) ÷ (equity + reserves) = 35 ÷ 50 = 0.70. 0.42 puts preference capital in the denominator (25 ÷ 60); 0.50 ignores it; 0.41 divides by everything (35 ÷ 85).

4 **B** **It falls**

With the ratio above 1, an equal increase in current assets and current liabilities pulls it towards 1. No amounts are needed for the direction.

5 **A** **Bank overdraft**; **B** **Provision for taxation**; **C** **Unclaimed dividend**; **E** **Cash credit**

ICAI's list (SM 3.6): creditors, short-term loans, bank overdraft, cash credit, outstanding expenses, provision for taxation, proposed dividend, unclaimed dividend and other current liabilities. Debentures due after five years are long-term debt.

2.2 Activity (turnover) ratios

How hard the assets work: sales or cost of goods sold divided by total, fixed, net or current assets and working capital, and the inventory, debtors and creditors turnovers and their days.

In plain words

The godown that was full to the ceiling

It is October, and Tapri's premix godown in Indore is stacked to the ceiling for Diwali. Kesar walks Dev through it like a museum.

> **Kesar:** Look at this, Dev. Sixty crore of premix. Nobody in India has more chai sitting ready.
>
> **Dev:** That is the problem, Kesar-ji. Sitting. How many times did this godown empty and refill last year?
>
> **Kesar:** Ten times? We sold ₹540 crore at cost, and on average ₹54 crore was lying here.
>
> **Dev:** Then each packet waits 36 days for a customer. Stock that sits is cash that sleeps.

Kesar has just computed an **activity ratio**: what the business did in a year (here the **cost of goods sold**, ₹540 crore) divided by what it held to do it (the **average inventory**, ₹54 crore). The answer, 10, is the **inventory turnover**: the stock "turned over" ten times. Divide the 360 days of a year by it and you get the days a rupee of stock waits to be sold: 36.

The same idea works for every asset. Tapri's ₹900 crore of sales came from ₹505 crore of total assets, so each rupee of assets "produced" ₹1.78 of sales. Credit customers took ₹720 crore of goods while owing Tapri ₹36 crore on average: they turned over 20 times and paid in about 18 days. Suppliers sold Tapri ₹432 crore on credit while being owed ₹48 crore on average: Tapri paid them 9 times a year, every 40 days.

Think of it like

The kettle at the tapri

A roadside tapri with one kettle that is refilled and emptied forty times a day earns more than a five-star coffee shop whose ten kettles are refilled four times a day. The kettle is the asset; the refills are its turnover. A big godown, a big factory or a big pile of debtors is only impressive if it keeps emptying and refilling.

Key idea

An activity ratio divides a year's sales (or cost of goods sold) by an asset. The higher it is, the more efficiently the asset is used; and 360 divided by a turnover gives the number of days the asset holds each rupee: days in stock, days to collect from debtors, days taken to pay creditors.

How it works

Asset management ratios

Activity ratios (also called efficiency, performance or turnover ratios) are employed to evaluate the efficiency with which the firm manages and utilises its assets; for this reason they are often called

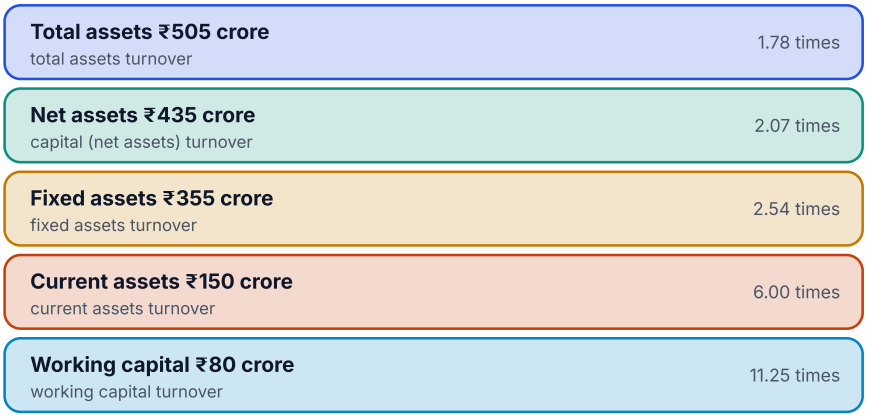
asset management ratios. They usually indicate the frequency of sales with respect to an asset, are calculated with reference to sales or cost of goods sold, and are expressed as a rate or "times" (SM 3.13–3.14). ICAI lists them in this order (SM 3.14):

- (a) Total assets turnover ratio
- (b) Fixed assets turnover ratio
- (c) Capital turnover ratio / net assets turnover ratio
- (d) Current assets turnover ratio
- (e) Working capital turnover ratio, which is further segregated into (i) inventory / stock turnover, (ii) receivables (debtors) turnover and (iii) payables (creditors) turnover.

Same sales, five bases

Tapri's year to 31 March 2026 (the balance sheet from the liquidity lesson): sales ₹900 crore, total assets ₹505 crore, net assets (capital employed) ₹435 crore, fixed assets ₹355 crore, current assets ₹150 crore, working capital ₹80 crore. The same ₹900 crore divided by smaller and smaller bases:

Tapri: one year's sales over five bases



Each ratio is sales of ₹900 crore divided by the base on its row: the narrower the base, the larger the figure, so a turnover means something only next to the same ratio for another year or another firm.

1. **Total assets turnover** = sales (or COGS) ÷ total assets = 900 ÷ 505 = **1.78 times**: the efficiency with which the firm uses its total assets; the higher the ratio, the better (SM 3.14).

2. **Fixed assets turnover** = 900 ÷ 355 = **2.54 times**: the efficiency in using fixed assets. A firm whose plant and machinery are old may show a higher ratio than one that bought them recently, because the old plant stands in the books at a written-down value (SM 3.15).

3. **Capital turnover** (net assets turnover) = 900 ÷ 435 = **2.07 times**: since net assets equal capital employed, it shows the sales generated per rupee of long-term investment; the higher, the more efficient the use of owners' and long-term creditors' funds (SM 3.15).

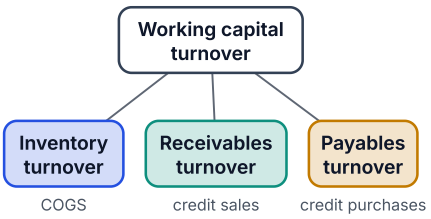
4. **Current assets turnover** = 900 ÷ 150 = **6.00 times** (SM 3.15).

5 **Working capital turnover** = 900 ÷ 80 = **11.25 times**: sales generated for every rupee of working capital. Higher is more efficient, but a very high working capital turnover ratio indicates that the company needs to raise additional working capital for future needs (SM 3.15).

ICAI allows the average of total, fixed, current or net assets or working capital in the denominator, and when averages of total assets, net assets, capital employed or shareholders' funds are available, they may be preferred (SM 3.16, 3.18). Tapri's figures above use closing balances because only one balance sheet is given.

Working capital turnover, split three ways

The three turnovers inside working capital



Under each turnover, its numerator; the denominator is the average balance (stock, debtors, creditors). Each pairs a balance with the flow that passes through it: stock is carried at cost, so it pairs with cost of goods sold; debtors arise from credit sales; creditors from credit purchases.

Tapri's balances (₹ crore): inventory 48 at the start of the year and 60 at the end; trade receivables 32 and 40; trade payables 44 and 52. Credit sales were ₹720 crore (cafés sell for cash), cost of goods sold ₹540 crore, and credit purchases ₹432 crore (the rest of the year's purchases were paid for in cash).

1 **Inventory turnover** = COGS ÷ average inventory, where average inventory = (opening + closing) ÷ 2 = (48 + 60) ÷ 2 = 54. So 540 ÷ 54 = **10 times**, and 360 ÷ 10 = **36 days** in stock. It shows how fast inventory is used or sold: a high ratio is good from the point of view of liquidity; a low ratio means stock is not used, not sold or lost, and stays on a shelf or in the warehouse for a long time (SM 3.16).

2 **Receivables (debtors) turnover** = credit sales ÷ average accounts receivable = 720 ÷ 36 = **20 times**. A low ratio reflects liberal credit terms; a high ratio, rapid collection (SM 3.17).

3 **Average collection period** (receivables velocity) = average receivables ÷ average daily credit sales = 36 ÷ (720 ÷ 360) = 36 ÷ 2 = **18 days**, which is also 360 ÷ 20. It is the average number of days taken to collect a receivable, also called the number of days of receivable or the days' sales in receivables, and it gives unique guidance in setting the credit policy (SM 3.17).

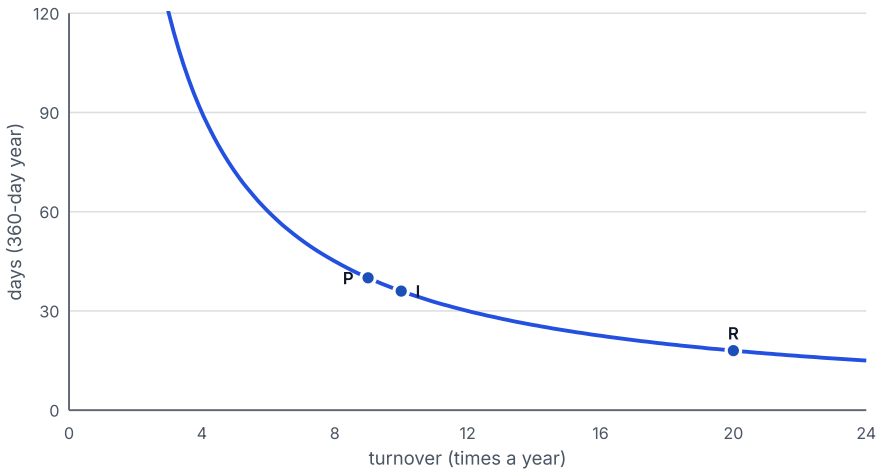
4 **Payables (creditors) turnover** = annual net credit purchases ÷ average accounts payable = 432 ÷ 48 = **9 times**: how fast the company pays its creditors. A low ratio reflects liberal credit terms granted by suppliers; a high ratio, prompt settlement (SM 3.17).

5 **Average payment period** (payables velocity) = average payables ÷ average daily credit purchases = 48 ÷ (432 ÷ 360) = **40 days**, or 360 ÷ 9. The firm compares the credit it receives from suppliers with the credit it gives customers, and with the credit customary in its industry (SM 3.17–3.18).

Tapri takes 40 days to pay its suppliers and gives its customers only 18: its average creditors (₹48 crore) more than cover its average debtors (₹36 crore). ICAI adds that these three ratios are also relevant to examine liquidity (SM 3.18), which is why the liquidity ratios should be read with them.

From times to days

Turnover and days are reciprocals



Blue curve: days = 360 ÷ turnover. Tapri: P, payables (9 times, 40 days); I, inventory (10 times, 36 days); R, receivables (20 times, 18 days). Doubling a turnover halves the days, so a firm that raises its receivables turnover from 10 to 20 cuts its collection period from 36 days to 18.

With months, use 12 ÷ turnover; with weeks, 52 ÷ turnover (SM 3.17). Take the year the question gives: ICAI's own illustrations use both 360 and 365 days.

Interactive version: PassBook app, lesson Activity (turnover) ratios

Formats and formulas

Reproduce these from memory, with the numerator that belongs to each denominator.

Ratio	Numerator	Denominator
Total assets turnover	Sales or COGS	Total assets
Fixed assets turnover	Sales or COGS	Fixed assets
Capital (net assets) turnover	Sales or COGS	Net assets
Current assets turnover	Sales or COGS	Current assets
Working capital turnover	Sales or COGS	Working capital
Inventory turnover	COGS or sales	Average inventory
Raw material inventory turnover	Raw material consumed	Average raw material stock
Receivables turnover	Credit sales	Average receivables
Payables turnover	Net credit purchases	Average payables

∑ **Formula**

Asset-based turnovers

$$\text{Turnover} = \frac{\text{Sales or COGS}}{\text{Asset base}}$$

Asset base = total assets, fixed assets, net assets (capital employed), current assets or working capital; their averages may be used (SM 3.14–3.16).

∑ **Formula**

Inventory turnover

$$\text{Inventory turnover} = \frac{\text{COGS}}{\text{Average inventory}}$$

$$\text{Average inventory} = \frac{\text{Opening stock} + \text{Closing stock}}{2}$$

$$\text{Raw material inventory turnover} = \frac{\text{Raw material consumed}}{\text{Average raw material stock}}$$

ICAI prints the numerator as "Cost of Goods Sold / Sales" (SM 3.16): use COGS unless the question says "based on sales".

∑ Formula

Receivables turnover and average collection period

$$\text{Receivables turnover} = \frac{\text{Credit sales}}{\text{Average receivables}}$$

$$\text{Average collection period} = \frac{\text{Average receivables}}{\text{Average daily credit sales}} = \frac{360 \text{ days}}{\text{Receivables turnover}}$$

Average daily credit sales = credit sales ÷ number of days in the year (say 360); with months or weeks, use 12 or 52 in place of 360 (SM 3.17).

∑ Formula

Payables turnover and average payment period

$$\text{Payables turnover} = \frac{\text{Net credit purchases}}{\text{Average payables}}$$

$$\text{Average payment period} = \frac{\text{Average payables}}{\text{Average daily credit purchases}} = \frac{360 \text{ days}}{\text{Payables turnover}}$$

Net credit purchases are the year's (annual) credit purchases; 12 months or 52 weeks may replace 360 days (SM 3.17).

∑ Formula

Purchases from the trading account

$$\text{Purchases} = \text{COGS} + \text{Closing} - \text{Opening stock}$$

Needed when a question gives sales, gross profit and stocks but not purchases.

ⓘ Note

ICAI's notes for turnover ratios (SM 3.18)

1. Only selling and distribution expenses differentiate cost of goods sold (COGS) and cost of sales (COS); in their absence, COGS = cost of sales.
2. Cost of goods sold or cost of sales can be used to calculate turnover ratios, eliminating the profit part.
3. Averages of total, fixed, current or net assets or working capital can also be taken in the denominator; when average figures are available they may be preferred.
4. Ratios shall be calculated based on requirement and availability of information and may deviate from the original formulae; if required, assumptions should be given.

Worked examples

Worked example **Indore Chai Distributors: two years, one squeeze**

Indore Chai Distributors Pvt Ltd buys premix from Tapri and sells it to grocers. Tapri's sales team suspects the distributor is "building stock and giving grocers too much credit". Its figures (₹ lakh; all purchases on credit; 360-day year):

Particulars	2024-25	2025-26
Sales	1,200	1,500
Of which credit sales	1,080	1,380
Cost of goods sold	1,020	1,290
Opening stock	84	96
Closing stock	96	150
Opening debtors	130	140
Closing debtors	140	250
Opening creditors	80	92
Closing creditors	92	100
Fixed assets	150	160
Total assets	400	570

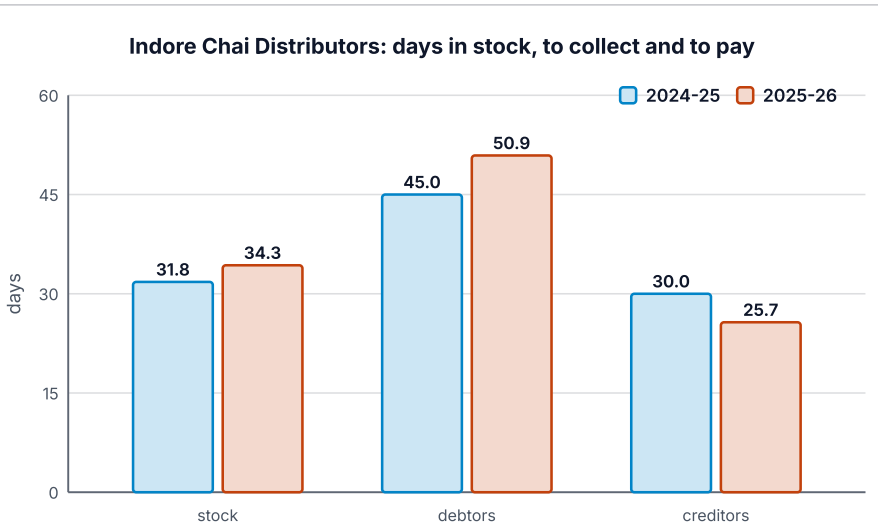
Required:

(i) Calculate for both years the inventory turnover and days in stock, the debtors turnover and average collection period, the creditors turnover and average payment period.

(ii) Calculate the fixed assets and total assets turnover ratios.

(iii) Comment on the changes.

THE WORKING



Sky blue: 2024-25; orange: 2025-26. Stock is held and debtors are carried longer, while creditors are paid sooner: all three move against the distributor's cash.

Working notes

1. Purchases = COGS + closing stock − opening stock: 2024-25: 1,020 + 96 − 84 = **₹1,032 lakh**; 2025-26: 1,290 + 150 − 96 = **₹1,344 lakh**.
2. Averages, (opening + closing) ÷ 2:

Average (₹ lakh)	2024-25	2025-26
Stock	90	123
Debtors	135	195
Creditors	86	96

(i) and (ii) Ratios

Inventory turnover = COGS ÷ average stock; debtors turnover = credit sales ÷ average debtors; creditors turnover = purchases ÷ average creditors; days = 360 ÷ turnover; fixed and total assets turnover = sales ÷ fixed or total assets.

Ratio	2024-25	2025-26
Inventory turnover	1,020 ÷ 90 = 11.33	1,290 ÷ 123 = 10.49
Days in stock	31.8	34.3
Debtors turnover	1,080 ÷ 135 = 8.00	1,380 ÷ 195 = 7.08
Collection period (days)	45.0	50.9
Creditors turnover	1,032 ÷ 86 = 12.00	1,344 ÷ 96 = 14.00
Payment period (days)	30.0	25.7
Fixed assets turnover	1,200 ÷ 150 = 8.00	1,500 ÷ 160 = 9.38
Total assets turnover	1,200 ÷ 400 = 3.00	1,500 ÷ 570 = 2.63

(iii) Comment

1. **Stock**: inventory turnover fell from 11.33 to 10.49 times; stock is held about 2.5 days longer. Closing stock rose from ₹96 lakh to ₹150 lakh (56%) while cost of goods sold rose 26%: the distributor has been building stock, as Tapri suspected.
2. **Debtors**: the collection period lengthened from 45 to about 51 days, so credit to grocers has become more liberal. The average hides the year-end position: closing debtors of ₹250 lakh are 65 days of credit sales.
3. **Creditors**: the payment period shortened from 30 to about 26 days: suppliers (Tapri) are being paid faster, so they finance less of the stock and debtors.
4. **Assets**: fixed assets are used better (8.00 to 9.38 times, sales grew faster than the warehouse and vans), but total assets turnover fell from 3.00 to 2.63 because the growth went into stock and debtors.

Conclusion: sales grew 25%, but more cash is now tied up for longer in stock and debtors while less credit is taken from suppliers. The distributor's liquidity will be under strain unless collections are tightened.

Worked example **Rasoi Packaging: the bank that found the turnover too good**

Rasoi Packaging Pvt Ltd prints Tapri's premix pouches and has applied to its bank to renew a cash credit limit. Its sales for 2025-26 were ₹7.20 crore, 80% on credit, and its cost of goods sold ₹5.40 crore. All purchases are on credit, and the stock at the start of the year was ₹40 lakh. All ratios are on closing balances unless stated; assume a 360-day year.

Ratio	Figure
Total assets turnover (on sales)	2.4 times
Fixed assets turnover (on sales)	4 times
Working capital turnover (on sales)	12 times
Inventory turnover (on COGS and average inventory)	10.8 times
Average collection period (on credit sales)	30 days
Payables turnover (on credit purchases)	20 times

Current liabilities are trade creditors and the bank's cash credit.

Required:

(i) Calculate total assets, fixed assets, current assets, working capital and current liabilities, and the capital (net assets) turnover ratio.

(ii) Calculate the average and closing stock.

(iii) Calculate the debtors and the cash balance.

(iv) Calculate the trade creditors and the cash credit.

(v) The bank's officer says a working capital turnover of 12 is "too good to be comfortable". Comment, and calculate the cash released if the collection period were brought down to 20 days with credit sales unchanged.

THE WORKING

Working notes (₹ lakh)

1. Credit sales = 80% × 720 = **576**.
2. Total assets = sales ÷ total assets turnover = 720 ÷ 2.4 = **300**; fixed assets = 720 ÷ 4 = **180**; current assets = 300 − 180 = **120**.
3. Working capital = 720 ÷ 12 = **60**; current liabilities = 120 − 60 = **60** (current ratio 2).
4. Net assets (capital employed) = total assets − current liabilities = 300 − 60 = 240, so capital turnover = 720 ÷ 240 = **3 times**.

(ii) Stock

Average stock = COGS ÷ inventory turnover = 540 ÷ 10.8 = **₹50 lakh** (360 ÷ 10.8 = 33.3 days). Closing stock = 2 × average − opening = 2 × 50 − 40 = **₹60 lakh**.

(iii) Debtors and cash

Debtors = credit sales × collection period ÷ 360 = 576 × 30 ÷ 360 = **₹48 lakh**. Cash = current assets − stock − debtors = 120 − 60 − 48 = **₹12 lakh** (balancing figure).

(iv) Creditors and cash credit

Credit purchases = COGS + closing stock − opening stock = 540 + 60 − 40 = ₹560 lakh. Trade creditors = 560 ÷ 20 = **₹28 lakh** (about 18 days of purchases). Cash credit = current liabilities − creditors = 60 − 28 = **₹32 lakh** (balancing figure).

(v) Comment

A higher working capital turnover means more sales for every rupee of working capital, but ICAI's interpretation is that a very high working capital turnover ratio indicates that the company needs to raise additional working capital for future needs (SM 3.15). Rasoi runs ₹720 lakh of sales on ₹60 lakh of working capital: it pays its suppliers in about 18 days while it waits 30 days for its customers and holds about 33 days of stock, and it already borrows ₹32 lakh on cash credit. Any growth in sales will need more working capital, so the bank's caution is justified.

At a collection period of 20 days, debtors = 576 × 20 ÷ 360 = ₹32 lakh, so **₹16 lakh** of cash is released (48 − 32), half of the cash credit.

Exam angle

In the exam

Where turnover ratios appeared, May 2024 to September 2026

In the ICAI documents read for this lesson:

- **Descriptive, 5 marks, mostly "find the balance".** September 2026 Q1(b): credit sales, COGS, average inventory, average debtors and proprietary fund from an inventory turnover of 8, a debtors velocity of 36.5 days (365-day year) and the gap between average debtors and average inventory. May 2025 Q1(a): a balance sheet from a raw material inventory turnover, a finished goods holding period, a debtors collection period and a fixed assets turnover. MTP Series I for January 2026: assets and liabilities from fixed assets and capital turnover "based on cost of sales" and inventory, receivable and payable turnovers.
- **Case scenarios and MCQs.** January 2025 MCQ 6 (inventory and receivables from an inventory turnover of 6 and a collection period of 2 months); RTP September 2024 MCQ 2 and RTP September 2026 MCQ 3 (creditors turnover, where purchases must first be derived from sales, gross profit and stocks); RTP January 2026 Q1(v) (working capital turnover on cash cost of sales); MTP Series I for May 2025 MCQs 3–4 (debtors and creditors turnover and working capital turnover); MTP Series II for September 2026 MCQ 2 (turnovers "computed on average balances").
- **Analysis.** The Study Material's case scenario on RNOC Ltd (SM 3.71–3.75: inventory and receivables turnover in days against industry standards of 100 and 90 days, then projected balances). Its figures were set again as EOC Ltd in MTP Series II for May 2024 Q3(a) (8 marks): inventory and receivables turnover in days, then the projected receivables, inventory, payables and long-term debt.

In the exam

The traps ICAI set

- Purchases are not given: purchases = COGS + closing stock − opening stock (RTP September 2024 MCQ 2, RTP September 2026 MCQ 3).
- "Velocity" or "collection period" in months or days must be turned into a turnover (12 ÷ months, 360 ÷ days) before it is used, or used directly as a fraction of the year.
- The year length: 360 in most questions, 365 in September 2026 Q1(b), MTP Series II for May 2026 Q1(c) and the Study Material's Illustrations 1 and 5. Use what is given.
- "Based on sales" or "based on cost of goods sold" decides the numerator; read the brackets.

Trick

Balance from days in one step

Balance = annual flow × days ÷ 360 (or × months ÷ 12). Average debtors = credit sales × collection period ÷ 360; average stock = COGS × days in stock ÷ 360; average creditors = credit purchases × payment period ÷ 360. Conditions: the period is based on the same flow and on average balances, and the year length matches the question's.

Answer frame

"Discuss stock turnover ratio" and "ratios for operating efficiency"

The chapter's theory questions include "Discuss Stock Turnover ratio and Gearing ratio" and "Discuss the financial ratios for evaluating company performance on operating efficiency and liquidity

position aspects" (SM 3.65). For a turnover ratio, write:

1. What it measures: the efficiency with which the firm manages and utilises the asset (activity ratios are asset management ratios).
2. The formula, with the numerator that matches the denominator (COGS for stock, credit sales for debtors, credit purchases for creditors) and the average balance.
3. The conversion to days or months.
4. The interpretation in ICAI's words (a high stock turnover is good for liquidity; a low one means stock stays on the shelf; a low debtors' turnover reflects liberal credit terms).
5. The comparison that gives it meaning: previous years, competitors, the industry.

For **operating efficiency** (SM 3.34): ratio analysis throws light on the degree of efficiency in the management and utilisation of assets; the activity ratios measure this operational efficiency; and the solvency of a firm is, in the ultimate analysis, dependent upon the sales revenues generated by the use of its assets, total as well as its components.

Three answers to "calculate the average collection period"

Loses marks

"ACP = 250 ÷ 1,380 × 360 = 65 days."

Closing debtors used although the opening balance is given; no average, no working.

Passes

"ACP = 195 ÷ 1,380 × 360 = 50.9 days."

Right, but 195 appears from nowhere.

Full marks

Working note: average debtors = (140 + 250) ÷ 2 = 195; average daily credit sales = 1,380 ÷ 360 = 3.83; ACP = 195 ÷ 3.83 = 50.9 days; one line of comment against the previous year or the credit terms.

Common mistakes

⚠ **Common mistake**

Hall of Shame: stock at cost, sales at selling price

Dividing sales by average stock mixes a selling-price numerator with a cost denominator and overstates the turnover (Tapri: 900 ÷ 54 = 16.7 instead of 10). **Fix:** use cost of goods sold for inventory unless the question says "based on sales".

⚠ **Common mistake**

Total sales in the debtors turnover

Using ₹900 crore of sales when only ₹720 crore was on credit makes Tapri's collection period 14.4 days instead of 18: cash sales never create a debtor. **Fix:** credit sales ÷ average receivables; take total sales only when credit sales are not given, and say so.

⚠ **Common mistake**

Cost of goods sold in the creditors turnover

Creditors arise from purchases, not from goods sold. **Fix:** derive purchases (COGS + closing stock − opening stock) and use the credit part of them.

⚠ **Common mistake**

The closing balance that pretended to be an average

When the opening balance is given, the average is (opening + closing) ÷ 2. Using the closing balance alone mis-states the turnover whenever the balance moved during the year. **Fix:** average whenever you can; use the closing balance only when nothing else is given, and state it.

Practice

Check yourself

1 A spice wholesaler's cost of goods sold for the year is ₹45 lakh. Its stock fell from ₹8 lakh at the start of the year to ₹5 lakh at the end. Three-quarters of its purchases are on credit, and its trade creditors were ₹4.50 lakh at the start of the year and ₹6 lakh at the end. What is its average payment period, in days (360-day year)?

Answer: _____

2 Case scenario. Tapri's head of modern trade reports that credit sales to supermarket chains were ₹48 crore, debtors were ₹5 crore at the start of the year and ₹11 crore at the end, and that "supermarkets pay us in about 82 days". Dev, who uses a 360-day year, says the figure does not follow ICAI's method. What is the average collection period on ICAI's formula?

A 82.5 days

B 60 days

C 37.5 days

D 6 days

3 According to ICAI, a very high working capital turnover ratio indicates that the company:

A Holds idle working capital

B Needs to raise additional working capital for future needs

C Has a very high current ratio

D Is selling too little for its assets

4 A company's sales are ₹80 lakh, cost of goods sold ₹60 lakh, opening stock ₹8 lakh and closing stock ₹12 lakh. Nothing is said about the basis. What is its inventory turnover?

- A 6 times
- B 8 times
- C 7.5 times
- D 5 times

5 Which of these statements are correct according to ICAI? Select all that apply.

- A A low debtors' turnover ratio reflects liberal credit terms granted to customers
- B A low creditors' turnover ratio reflects liberal credit terms granted by suppliers
- C A firm with old plant may show a higher fixed assets turnover than one that bought its plant recently
- D A high inventory turnover is bad from the point of view of liquidity

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Activity ratios (efficiency, performance, turnover; "asset management ratios") = sales or COGS ÷ an asset, in "times".
- Total assets, fixed assets, capital (net assets), current assets and working capital turnover: sales or COGS ÷ that base; averages may be used.
- A very high working capital turnover means more working capital will be needed.
- Inventory turnover = COGS ÷ average inventory; raw material turnover = materials consumed ÷ average raw material stock.
- Debtors turnover = credit sales ÷ average receivables; collection period = 360 ÷ turnover (or average receivables ÷ daily credit sales).
- Creditors turnover = credit purchases ÷ average payables; payment period = 360 ÷ turnover. Purchases = COGS + closing − opening stock.
- Balance = annual flow × days ÷ 360. (A full godown is only impressive if it keeps emptying.)

Answer key · Check yourself

1 **60**

Purchases = COGS + closing stock − opening stock = 45 + 5 − 8 = ₹42 lakh (a fall in stock means less was bought than was sold); credit purchases = 75% × 42 = ₹31.50 lakh; average creditors = (4.50 + 6) ÷ 2 = ₹5.25 lakh; payment period = 5.25 ÷ 31.50 × 360 = 60 days (payables turnover 6 times). Using COGS instead of purchases gives 56 days; adding the fall in stock gives 52.5 days; using all purchases gives 45 days.

2 **B** **60 days**

Average debtors = (5 + 11) ÷ 2 = ₹8 crore; turnover = 48 ÷ 8 = 6 times; collection period = 360 ÷ 6 = 60 days. 82.5 days uses closing debtors; 37.5 days uses opening debtors; 6 is the turnover in times, not days.

3 **B** **Needs to raise additional working capital for future needs**

Higher working capital turnover means efficient use of working capital, but a very high ratio means sales are being run on too thin a working capital, so more will be needed (SM 3.15).

4 **A** **6 times**

COGS ÷ average stock = 60 ÷ 10 = 6 times. 8 uses sales; 7.5 uses opening stock alone; 5 uses closing stock alone.

5 **A** **A low debtors' turnover ratio reflects liberal credit terms granted to customers**; **B** **A low creditors' turnover ratio reflects liberal credit terms granted by suppliers**; **C** **A firm with old plant may show a higher fixed assets turnover than one that bought its plant recently**

The first three are ICAI's interpretations (SM 3.15, 3.17). A high inventory turnover is good for liquidity: stock turns into sales quickly (SM 3.16).

2.3 Profitability and market ratios, and the DuPont analysis

Profit as a share of sales, as a return on assets, capital employed and equity, per share and against the share price; and the DuPont split of ROE into margin, turnover and leverage.

In plain words

Same twelve and a half per cent, different sums

Kesar has just read BrewBucks India's annual report.

> **Kesar:** BrewBucks India earns 12.5% on its shareholders' money. Our café company earns 12.5% too. Dev, we are as good as the Americans. Well done!
>
> **Dev:** Present value of that compliment: lower than you think, Kesar-ji. Same answer, not necessarily the same sums.
>
> **Kesar:** Twelve and a half is twelve and a half.
>
> **Dev:** Not when one of the three engines behind it runs on borrowed money, and borrowed money sends a bill every month.

Profit by itself says little: ₹90 crore is a fortune for a tea stall and a disaster for a bank. **Profitability ratios** compare profit with something, and ICAI groups them by that something:

- with **sales**: how many rupees of each ₹100 of sales remain as gross profit, operating profit or net profit (the **margins**);
- with the **investment**: the return on assets, on capital employed and on the shareholders' equity;
- per **share**: earnings and dividend per share, the owner's view;
- with the **market price** of the share: the price-earnings ratio and the yields, the investor's view.

Tapri's year to 31 March 2026: sales ₹900 crore, profit after tax ₹90 crore, a **net profit ratio** of 10%. The equity shareholders' funds are ₹300 crore and they earned ₹88 crore after the preference dividend, a **return on equity** of 29.3%. With 4 crore shares, that is ₹22 a share (**EPS**), and at a market price of ₹330 the share sells at 15 times its earnings (the **P/E ratio**).

Dev's three engines are the **DuPont analysis**: return on equity is the product of three ratios, the margin on sales, the sales produced per rupee of assets, and the assets carried per rupee of equity. The last one grows with debt, so it shows how much of the return is borrowed.

Think of it like

Three ways to fill the glass

A chaiwala can earn more on his own money in three ways: keep more per cup (margin), sell more cups from the same stall (turnover), or set up a bigger stall with a moneylender's cash (the equity multiplier). The first two are skill. The third works only while each cup earns more than the moneylender charges, and the moneylender comes every month either way.

Key idea

Profitability ratios relate profit to sales (margins), to the money invested (ROA, ROCE, ROE), to each share (EPS, DPS) and to the share's price (P/E, yields, market-to-book). Return on investment = margin × turnover, and return on equity = net profit margin × asset turnover × equity multiplier: two companies with the same ROE can be very different businesses.

How it works

Four groups

The profitability ratios measure the profitability or the operational efficiency of the firm. They reflect the final results of business operations; they are some of the most closely watched and widely quoted ratios, and management attempts to maximise them to maximise the firm's value (SM 3.18). ICAI's four groups (SM 3.19):

Group	Ratios
(i) Based on sales	Gross profit ratio; net profit ratio; operating profit ratio; expenses ratios
(ii) Overall return on assets / investments	Return on investment (ROI): return on assets (ROA), return on capital employed (ROCE), return on equity (ROE)
(iii) From the owner's point of view	Earnings per share (EPS); dividend per share (DPS); dividend pay-out ratio
(iv) Market / valuation / investors	Price-earnings ratio; dividend and earnings yield; market value to book value per share; Q ratio

Margins: every line of the profit statement over sales

Tapri's statement of profit for 2025-26, each line as a percentage of sales:

Particulars	₹ crore	% of sales	Ratio
Sales	900	100.00	
Less: Cost of goods sold	540	60.00	COGS ratio
Gross profit	**360**	**40.00**	Gross profit ratio
Less: Administrative expenses	100	11.11	
Less: Selling and distribution expenses	128	14.22	
Operating profit (EBIT)	**132**	**14.67**	Operating profit ratio
Less: Interest	12	1.33	Financial expenses ratio
Profit before tax	**120**	**13.33**	Pre-tax profit ratio
Less: Tax at 25%	30	3.33	
Profit after tax	**90**	**10.00**	Net profit ratio

1. **Gross profit ratio** = gross profit ÷ sales × 100 = **40%**: the percentage of each sale in rupees remaining after payment for the goods sold. It depends on the relationship between price, volume and costs; a high gross profit margin is a favourable sign of good management (SM 3.20).

2. **Operating profit ratio** = operating profit ÷ sales × 100 = **14.67%**, where operating profit = sales − COGS − operating expenses, often referred to as EBIT. It is the percentage of each sale that remains after all costs and expenses except interest and taxes, followed closely by analysts because it focuses on operating results (SM 3.20–3.21).

3. **Expenses ratios**: COGS ratio 60%; **operating expenses ratio** = (administrative + selling and distribution expenses) ÷ sales = 228 ÷ 900 = 25.33%; **operating ratio** = (COGS + operating expenses) ÷ sales = 768 ÷ 900 = **85.33%**, the cost of earning each rupee of sales; **financial expenses ratio** = 12 ÷ 900 = 1.33% (financial expenses exclude taxes, loss due to theft, goods destroyed by fire and the like). Administration and selling and distribution expense ratios are calculated the same way (SM 3.21).

4. **Net profit ratio** = net profit (EAT) ÷ sales × 100 = **10%**; the **pre-tax profit ratio** = EBT ÷ sales × 100 = 13.33%. It finds the proportion of revenue that finds its way into profits after meeting all expenses (SM 3.20).

Since operating profit is what sales leave after COGS and operating expenses, **operating ratio + operating profit ratio = 100%**: 85.33% + 14.67%. ICAI used exactly this in May 2024 Q1(a), where an operating ratio of 85% gave an operating profit ratio of 15%.

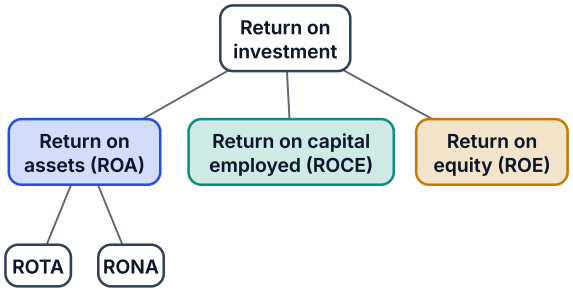
Returns: profit over the money invested

ROI is "the most important ratio of all": the percentage return on funds invested in the business, which tells the owner whether all the effort put into the business has been worthwhile. It splits into two ratios you already know (SM 3.22):

$$\text{ROI} = \frac{\text{Return}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Investment}} \qquad = \text{Profitability ratio} \times \text{Investment turnover}$$

So ROI can be improved by improving the margin, the turnover, or both. "Investment" may be total assets or net assets; funds employed in net assets are called **capital employed**, which is net worth plus debt; and the return may be measured in several ways according to the requirement and the information available (SM 3.23). Tapri (₹ crore): total assets 505; capital employed 435 (equity 300 + preference 25 + debt 110); EBIT 132; tax rate 25%.

The return on investment family



ICAI's chart (SM 3.22): three broad categories of ROI, by the investment in the denominator; ROTA (on total assets) and RONA (on net assets) are the ROA measured on EBIT(1 – t).

Ratio	Return ÷ investment	Tapri
ROA (on profit after tax)	90 ÷ 505	17.82%
ROA (with interest added back)	(90 + 12) ÷ 505	20.20%
Return on total assets, ROTA	132 × (1 − 0.25) ÷ 505	19.60%
Return on net assets, RONA	132 × (1 − 0.25) ÷ 435	22.76%
ROCE, pre-tax	132 ÷ 435	30.34%
ROCE, post-tax	99 ÷ 435	22.76%
ROE	(90 − 2) ÷ 300	29.33%

Check the split for ROCE: operating profit ratio 14.67% × capital turnover 2.069 (900 ÷ 435) = 30.34%. Tapri's ROCE comfortably exceeds the 10–12% it pays on debt, and ICAI's rule is that ROCE should always be higher than the rate at which the company borrows (SM 3.24).

Return on equity = (profit after tax − preference dividend) ÷ net worth (equity shareholders' funds) × 100. It measures how profitably the owners' funds have been utilised, and it is one of the most important indicators of a firm's profitability and potential growth. But "two companies can have the same return on equity, yet one can be a much better business" (SM 3.24–3.25), which is where DuPont comes in.

Interactive version: PassBook app, lesson Profitability and market ratios, and the DuPont analysis

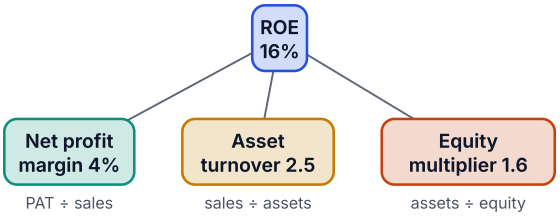
DuPont: one return, three engines

A finance executive at E.I. Du Pont de Nemours and Co. created the DuPont system of financial analysis in 1919; it is still used around the world as the basis of the components that make up return on equity (SM 3.25):

$$\text{ROE} = \frac{\text{Net income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Shareholders' equity}}$$

Tapri Premix Ltd (Tapri's packaged-chai subsidiary, with no preference shares): sales ₹200 crore, profit after tax ₹8 crore, total assets ₹80 crore, financed by ₹50 crore of equity and ₹30 crore of debt and current liabilities.

Tapri Premix: ROE in three parts



4% × 2.5 × 1.6 = 16%. Each child is one engine of the return: margin, turnover and leverage.

1. **Net profit margin** = 8 ÷ 200 = 4%: the after-tax profit generated for each rupee of revenue. It varies across industries, so compare with competitors; it is a safety cushion, and the lower it is, the less room for error: a business with a 1% margin has no room for flawed execution (SM 3.25–3.26).

2. **Asset turnover** (investment or capital turnover) = 200 ÷ 80 = 2.5: how effectively the company converts its assets into sales. It tends to be inversely related to the net profit margin, which lets an investor compare a low-profit, high-volume model with a high-profit, low-volume one (SM 3.26).

3. **Equity multiplier** = 80 ÷ 50 = 1.6: a measure of financial leverage. A company with terrible sales and margins can take on excessive debt and artificially increase its ROE; the multiplier shows what portion of ROE is the result of debt (SM 3.26).

Leave out the multiplier to see what the company would earn if it were completely debt-free, as the Study Material does in its Example 1 (SM 3.27): 4% × 2.5 = 10%. So 10 points of Tapri Premix's 16% come from margins and sales, and the other 6 points from the debt at work in the business. The Study Material's conclusion: of two companies with the same ROE at a comparable valuation, the one whose return arises more from internally generated sales is the more attractive.



Every point on the blue curve earns the same 10% before leverage. Tapri Premix (4% margin, turnover 2.5: low profit, high volume) and Tapri's Reserve tea boutiques (10% margin, turnover 1.0: high profit, low volume) reach it by opposite models; a point below the curve earns less, a point above it more.

Interactive version: PassBook app, lesson Profitability and market ratios, and the DuPont analysis

Per share, and against the share price

Tapri has 4 crore equity shares of ₹10 and paid a dividend of ₹5.50 a share; the market price is ₹330; net worth is ₹300 crore.

1. **Earnings per share** = net profit available to equity shareholders ÷ number of equity shares outstanding = (90 − 2) ÷ 4 = **₹22**. It is the profitability of the firm from the ordinary shareholder's point of view (SM 3.27).

2. **Dividend per share** = total dividend paid to equity shareholders ÷ number of equity shares = **₹5.50**: how much of the profit is distributed per share (SM 3.27). **Dividend pay-out ratio** = DPS ÷ EPS = 5.50 ÷ 22 = **25%**: the extent to which earnings are paid out and, by difference, retained (SM 3.28).

3. **Price-earnings ratio** = market price per share ÷ EPS = 330 ÷ 22 = **15 times**. It indicates the expectation of equity investors about the firm's earnings, and is taken as a summary measure of growth potential, risk characteristics, shareholders' orientation, corporate image and degree of liquidity. It indicates the payback period to investors; a higher P/E could mean the share is over-valued or that investors expect high growth (SM 3.28).

4. **Dividend yield** = DPS ÷ MPS × 100 = 5.50 ÷ 330 = **1.67%**; **earnings yield** (earnings-price ratio) = EPS ÷ MPS × 100 = **6.67%**, the reciprocal of the P/E. Dividend (%) shows the return on the paid-up value of shares; yield (%) is the indicator of true return, with share capital taken at its market value (SM 3.28–3.29).

5. **Market value to book value per share** = 330 ÷ (300 ÷ 4) = 330 ÷ 75 = **4.4**: the market response to the shareholders' investment; the higher, the better the shareholders' position in terms of return and capital gains (SM 3.29).

6 **Q ratio** (proposed by James Tobin) = market value of equity and liabilities ÷ estimated replacement cost of assets. It relates market valuation to intrinsic value: equilibrium is 1; below 1 the stock could be undervalued, above 1 overvalued (SM 3.29).

Interactive version: PassBook app, lesson Profitability and market ratios, and the DuPont analysis

Formats and formulas

Reproduce these from memory. Every "× 100" ratio is a percentage; the others are in times or rupees.

∑ **Formula**

Margins on sales

$$\text{Gross profit ratio} = \frac{\text{Gross profit}}{\text{Sales}} \times 100 \qquad \text{Net profit ratio} = \frac{\text{EAT}}{\text{Sales}} \times 100$$

$$\text{Pre-tax profit ratio} = \frac{\text{EBT}}{\text{Sales}} \times 100 \qquad \text{Operating profit ratio} = \frac{\text{EBIT}}{\text{Sales}} \times 100$$

Operating profit = sales − COGS − operating expenses (SM 3.20–3.21).

∑ **Formula**

Expenses ratios

$$\text{COGS ratio} = \frac{\text{COGS}}{\text{Sales}} \times 100 \qquad \text{Operating expenses ratio} = \frac{A + S}{\text{Sales}} \times 100$$

$$\text{Operating ratio} = \frac{\text{COGS} + \text{Operating expenses}}{\text{Sales}} \times 100$$

$$\text{Financial expenses ratio} = \frac{\text{Financial expenses}}{\text{Sales}} \times 100$$

A = administrative expenses; S = selling and distribution overheads. Financial expenses exclude taxes, loss due to theft, goods destroyed by fire etc. (SM 3.21).

∑ **Formula**

Return on investment

$$\text{ROI} = \frac{\text{Return}}{\text{Investment}} \times 100 = \frac{\text{Return}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Investment}}$$

(SM 3.22)

∑ Formula

Return on assets

$$\text{ROA} = \frac{\text{PAT}}{\text{Average total assets}} \quad \text{or} \quad \frac{\text{PAT} + \text{Interest}}{\text{Average total assets}}$$

$$\text{ROTA} = \frac{\text{EBIT}(1-t)}{\text{Average total assets}} \qquad \text{RONA} = \frac{\text{EBIT}(1-t)}{\text{Average net assets}}$$

PAT = net profit after taxes; t = tax rate. Average tangible assets or average fixed assets may replace total assets in the first two; total assets are sometimes used instead of average assets (SM 3.23).

∑ Formula

Return on capital employed

$$\text{ROCE (pre-tax)} = \frac{\text{EBIT}}{\text{Capital employed}} \times 100$$

$$\text{ROCE (post-tax)} = \frac{\text{EBIT}(1-t)}{\text{Capital employed}} \times 100$$

Sometimes (PAT + interest) ÷ capital employed. Capital employed = total assets − current liabilities = fixed assets + working capital = equity + long-term debt; include intangibles (goodwill, patents, trade marks), exclude fictitious assets (deferred expenses); use average capital employed if available (SM 3.24, 3.30).

∑ Formula

Return on equity and the DuPont model

$$\text{ROE} = \frac{\text{PAT} - \text{Preference dividend}}{\text{Equity shareholders funds}} \times 100$$

$$\text{ROE} = \text{Net profit margin} \times \text{Asset turnover} \times \text{Equity multiplier}$$

Net profit margin = net income ÷ sales; asset (investment, capital) turnover = sales ÷ investment, assets or capital; equity multiplier = investment, assets or capital ÷ shareholders' equity (SM 3.25–3.26).

∑ Formula

Owner's ratios

$$\text{EPS} = \frac{\text{Equity earnings}}{\text{No. of equity shares}} \qquad \text{DPS} = \frac{\text{Equity dividend}}{\text{No. of equity shares}}$$

$$\text{Dividend pay-out ratio} = \frac{\text{DPS}}{\text{EPS}}$$

Equity earnings = net profit available to equity shareholders; equity dividend = total dividend paid to equity shareholders; both are divided by the number of equity shares outstanding. In the absence of preference dividend, PAT is the earnings available to equity shareholders (SM 3.27–3.28, 3.30).

∑ Formula

Market ratios

$$\text{P/E ratio} = \frac{\text{MPS}}{\text{EPS}} \qquad \text{Earnings yield} = \frac{\text{EPS}}{\text{MPS}} \times 100$$

$$\text{Dividend yield} = \frac{\text{DPS}}{\text{MPS}} \times 100 \qquad \text{or} \qquad \frac{\text{Dividend} \pm \text{Price change}}{\text{Initial share price}} \times 100$$

$$\text{MV/BV} = \frac{\text{Share price}}{\text{Book value per share}} \qquad \text{Q ratio} \quad = \frac{\text{Market value of a company}}{\text{Replacement cost of assets}}$$

Share price = the average or the closing share price; book value per share = net worth ÷ number of equity shares; price change = the change in share price over the period. The Q ratio is also written as market value of equity and liabilities ÷ estimated replacement cost of assets (SM 3.28–3.29).

Worked examples

Worked example **Tapri Cold Chain: sales up 50%, return down**

Tapri Cold Chain Pvt Ltd, the refrigerated-transport subsidiary, expanded its fleet in 2025-26 with ₹15 crore of new 10% debentures issued on 1 April 2025. Kesar calls it "our best year yet: sales are up 50%". Its summarised figures (₹ crore; tax rate 25%; there are no preference shares):

Particulars	2024-25	2025-26
Sales	40.00	60.00
Direct costs	28.00	45.00
Operating expenses	6.00	7.50

Balance sheet at 31 March	2025	2026
Equity share capital (₹10 shares)	10.00	10.00
Reserves and surplus	15.00	17.50
10% Debentures	10.00	25.00
Current liabilities	5.00	7.50
Total equity and liabilities	**40.00**	**60.00**
Fixed assets (net)	30.00	46.00
Current assets	10.00	14.00
Total assets	**40.00**	**60.00**

Required:

(i) Prepare the statement of profit for both years, and calculate the gross profit ratio, operating ratio, operating profit ratio and net profit ratio.

(ii) Calculate the pre-tax return on capital employed and the return on equity for both years, using closing balances.

(iii) Split each year's ROE with the DuPont model, and explain what changed.

(iv) Calculate the 2025-26 ROE had the extra ₹15 crore been raised by issuing equity shares instead of debentures, and comment on the use of debt.

THE WORKING

(i) Statement of profit and ratios on sales

Particulars (₹ crore)	2024-25	2025-26
Sales	40.00	60.00
Less: Direct costs	28.00	45.00
Gross profit	**12.00**	**15.00**
Less: Operating expenses	6.00	7.50
EBIT	**6.00**	**7.50**
Less: Interest at 10% on debentures	1.00	2.50
Profit before tax	**5.00**	**5.00**
Less: Tax at 25%	1.25	1.25
Profit after tax	**3.75**	**3.75**

Ratio	2024-25	2025-26
Gross profit ratio	30.00%	25.00%
Operating ratio	85.00%	87.50%
Operating profit ratio	15.00%	12.50%
Net profit ratio	9.38%	6.25%

Gross profit ratio = gross profit ÷ sales; operating ratio = (direct costs + operating expenses) ÷ sales; operating profit ratio = EBIT ÷ sales; net profit ratio = PAT ÷ sales; all × 100. Check: 85% + 15% = 100% and 87.5% + 12.5% = 100%.

(ii) Returns

Capital employed = total assets − current liabilities = 40 − 5 = ₹35 crore (2025) and 60 − 7.50 = ₹52.50 crore (2026). Equity shareholders' funds = share capital + reserves = ₹25 crore and ₹27.50 crore.

Ratio	2024-25	2025-26
ROCE, pre-tax: EBIT ÷ capital employed	6 ÷ 35 = 17.14%	7.50 ÷ 52.50 = 14.29%
ROE: PAT ÷ equity funds	3.75 ÷ 25 = 15.00%	3.75 ÷ 27.50 = 13.64%

(iii) DuPont

Component	2024-25	2025-26
Net profit margin (PAT ÷ sales)	9.375%	6.25%
Asset turnover (sales ÷ total assets)	1.00	1.00
Equity multiplier (total assets ÷ equity)	40 ÷ 25 = 1.60	60 ÷ 27.50 = 2.18
ROE	**15.00%**	**13.64%**

ROE fell from 15% to 13.64% although sales rose 50%. Asset turnover did not change: the new fleet produced sales in proportion. The net profit margin fell from 9.375% to 6.25%, for two reasons: direct costs rose from 70% to 75% of sales, so the operating profit ratio fell from 15% to 12.5%; and interest rose from 2.5% to 4.17% of sales on the new debentures. The higher equity multiplier (1.60 to 2.18) made up for only part of the fall in margin.

(iv) Had the ₹15 crore been raised as equity

Interest stays at ₹1 crore (10% of ₹10 crore); PBT = 7.50 − 1.00 = ₹6.50 crore; PAT = ₹4.875 crore; equity = 27.50 + 15 = ₹42.50 crore; ROE = 4.875 ÷ 42.50 = **11.47%**.

Comment. Debt financing gave the equity shareholders 13.64% instead of 11.47%, because the capital employed still earns a pre-tax 14.29%, more than the 10% paid on the debentures. ROE fell because operations became less profitable, not because of the debt. The cushion is thinner, though: ROCE has come down from 17.14% to 14.29%, and if it falls below the 10% rate of borrowing, the debentures will start to reduce ROE (see trading on equity). Kesar's "best year" is a year of growth with a lower return on equity.

Worked example **Cafés against BrewBucks: whose 12.5% is better?**

Kesar wants to tell the board that Tapri Kulhad Cafés Ltd "earns as well as BrewBucks India". Dev extracts (₹ crore):

Particulars	Tapri Kulhad Cafés	BrewBucks India
Sales	50.00	80.00
Profit after tax	1.875	8.00
Total assets	40.00	80.00
Shareholders' equity	15.00	64.00

Neither company has preference shares.

Required:

(i) Calculate each company's ROE using the DuPont model.

(ii) Calculate the ROE each would earn without the equity multiplier, and the part of ROE that is due to leverage.

(iii) Advise Kesar on his claim.

THE WORKING

(i) DuPont

Component	Tapri Kulhad Cafés	BrewBucks India
Net profit margin (PAT ÷ sales)	1.875 ÷ 50 = 3.75%	8 ÷ 80 = 10.00%
Asset turnover (sales ÷ assets)	50 ÷ 40 = 1.25	80 ÷ 80 = 1.00
Equity multiplier (assets ÷ equity)	40 ÷ 15 = 2.667	80 ÷ 64 = 1.25
ROE	**12.50%**	**12.50%**

(ii) Without the multiplier

Particulars	Tapri Kulhad Cafés	BrewBucks India
Margin × turnover	3.75% × 1.25 = 4.69%	10% × 1.00 = 10.00%
Part of ROE due to leverage	12.50 − 4.69 = 7.81 points	12.50 − 10.00 = 2.50 points

(iii) Advice

The two returns on equity are equal, but they are earned differently. BrewBucks India earns 10% on its sales and assets before leverage, and only 2.5 points of its ROE come from debt. The cafés earn 4.69% from operations and 7.81 points, more than half their ROE, from debt: their margin is thin (3.75%), so there is little room for error, and each ₹1 of equity carries ₹2.67 of assets. The claim is not correct: on equal ROE, the business whose return arises from its own margins and sales is the more attractive. The cafés should work on their margin, or on the sales produced by each rupee of assets, rather than on further borrowing.

Worked example **Chambal Snacks Ltd: the share price talks back**

Chambal Snacks Ltd, a listed snack maker that competes with Tapri's samosas, has 50 lakh equity shares of ₹10 and 9% preference share capital of ₹10 crore. Its share trades at ₹240, at a P/E of 12 times and 1.6 times its book value per share. It pays out 40% of its earnings to equity shareholders. A year ago the share was ₹200.

Required:

(i) Calculate the EPS, DPS, dividend yield and earnings yield.

(ii) Calculate the book value per share, net worth and return on equity.

(iii) Calculate the profit after tax, the equity dividend coverage and the total dividend coverage (profit after tax ÷ all dividends).

(iv) Calculate the yield including the change in share price over the year.

THE WORKING

(i) Per share and yields

- EPS = MPS ÷ P/E = 240 ÷ 12 = **₹20**
- DPS = 40% × ₹20 = **₹8**
- Dividend yield = 8 ÷ 240 × 100 = **3.33%**
- Earnings yield = 20 ÷ 240 × 100 = **8.33%** (= 1 ÷ 12)

(ii) Book value and ROE

- Book value per share = MPS ÷ (MV/BV) = 240 ÷ 1.6 = **₹150**
- Net worth (equity shareholders' funds) = ₹150 × 50 lakh shares = **₹75 crore**
- Earnings for equity = ₹20 × 50 lakh = ₹10 crore; ROE = 10 ÷ 75 = **13.33%** (also EPS ÷ book value per share)

(iii) Profit and dividend cover

- Preference dividend = 9% × ₹10 crore = ₹0.90 crore; PAT = 10 + 0.90 = **₹10.90 crore**
- Equity dividend = ₹8 × 50 lakh = ₹4 crore; equity dividend coverage = (10.90 − 0.90) ÷ 4 = **2.5 times**
- Total dividend coverage = 10.90 ÷ (4 + 0.90) = **2.22 times**

(iv) Yield with the price change

$$\text{Yield} = \frac{\text{Dividend} + \text{Change in price}}{\text{Initial price}} \times 100 \qquad = \frac{8 + (240 - 200)}{200} \times 100 \qquad = 24\%$$

Exam angle

In the exam

Where these ratios appeared, May 2024 to September 2026

In the ICAI documents read for this lesson:

- **Descriptive, 5 marks.** September 2025 Q1(b): PAT, net profit margin, return on assets (after tax), asset turnover and ROE from total assets, a debt share, direct costs and operating expenses (1 mark each). May 2026 Q1(c): total equity dividend, EPS, rate of preference dividend and PAT from a total dividend coverage ratio, a dividend yield, the market price and a P/E. May 2024 Q1(a): gross profit ratio from an operating ratio and a ratio of operating expenses to COGS, with interest coverage from a return on shareholders' fund.
- **MCQs.** September 2024 MCQ 8 (ROE by DuPont from total assets, equity, net profit margin and asset turnover); September 2026 MCQ 8 (total asset turnover from a net profit margin and a return on assets); January 2025 MCQ 2 (market price from EPS and P/E, in an EBIT-EPS case); RTP January 2026 MCQ 3 (the third term of the extended DuPont model).
- **RTPs and MTPs.** RTP January 2025 Q4: EPS and P/E inside a larger ratio question; RTP January 2026 Q5: EPS, market value to book value and P/E, each named by what it measures ("indicates market response of the equity shareholders' investment"). MTP Series II for May 2026 Q1(c): inventory turnover, financial leverage, ROCE, ROE and collection period from published statements. MTP Series II for September 2026: net profit ratio and return on total assets against industry averages.

In the exam

The traps ICAI set

- EPS and ROE use profit **after** preference dividend; P/E then follows from EPS.
- Operating ratio and operating profit ratio add up to 100% (May 2024 Q1(a)).
- "Return on assets (after tax)" in September 2025 Q1(b) meant EBIT(1 – t) ÷ assets; name your basis.
- In DuPont MCQs a distractor may invert the multiplier (equity ÷ assets): 12.5% × 1.5 × 0.625 = 11.72% is one of the wrong options in September 2024 MCQ 8, whose answer is 12.5% × 1.5 × 1.6 = 30%.

Trick

Three shortcuts from ICAI's identities

ROE = ROA (on PAT) × equity multiplier; EPS ÷ book value per share = ROE; earnings yield = 1 ÷ P/E. Conditions: no preference capital or dividend (otherwise use profit after preference dividend and equity shareholders' funds throughout), and the same profit figure and the same balance sheet date on both sides.

Answer frame

"Discuss the DuPont model" (SM 3.65, theory questions 4 and 6)

1. Origin: created in 1919 by a finance executive at E.I. Du Pont de Nemours and Co.; used worldwide as the basis of the components of ROE.
2. The model: ROE = net profit margin × asset (investment) turnover × equity multiplier.
3. Each component, its formula and meaning: net profit margin (profit for each rupee of revenue; a safety cushion); asset turnover (how effectively assets are converted into sales; inversely related to

margin); equity multiplier (financial leverage; shows what portion of ROE results from debt).

4. Use: by examining each input, the sources of a company's ROE can be discovered and compared with competitors; a high ROE from leverage is less attractive than one from margins and turnover.
5. A two-line numerical illustration.

"Discuss any three ratios computed for investment analysis" (SM 3.65, theory question 1)

The Study Material's answer key points to paragraph 3.4.2, the return on investment ratios. For each of ROA, ROCE and ROE: definition, formula with the components named, interpretation (ROCE above the borrowing rate; ROE as an indicator of profitability and potential growth), and who uses it.

Three answers to "calculate the return on equity"

Loses marks

"ROE = 3.75 ÷ 60 = 6.25%."

Total assets in the denominator: that is a return on assets.

Passes

"ROE = 3.75 ÷ 27.50 = 13.64%."

Correct, but the equity figure is unexplained.

Full marks

Working note: equity shareholders' funds = share capital ₹10 crore + reserves ₹17.50 crore = ₹27.50 crore; PAT from the statement of profit; ROE = (PAT − preference dividend) ÷ equity shareholders' funds × 100 = 13.64%; DuPont check if asked.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the preference dividend that went to the equity shareholders

EPS taken as PAT ÷ shares: Chambal's ₹10.90 crore ÷ 50 lakh = ₹21.80 instead of ₹20, and every ratio built on it (P/E, yields, ROE) goes wrong. **Fix:** deduct the preference dividend first; only what is left is available to equity shareholders.

⚠ **Common mistake**

The return that did not match its investment

EBIT over equity, or PAT over total assets labelled "ROCE". **Fix:** returns before interest (EBIT, EBIT(1 − t), PAT + interest) go over assets or capital employed; returns after interest and preference dividend go over equity shareholders' funds.

⚠ **Common mistake**

DuPont with two engines

Multiplying margin and turnover and calling it ROE (the cafés' 4.69%). **Fix:** that product is a return on assets; ROE needs the equity multiplier too.

⚠ **Common mistake**

Fictitious assets in capital employed

Adding deferred expenses or accumulated losses to capital employed lowers ROCE without a single rupee more invested. **Fix:** include intangibles such as goodwill, patents and trade marks; exclude fictitious assets (SM 3.24).

Practice

Check yourself

1 Tapri's kiosk company has sales of ₹75 crore, total assets of ₹50 crore and shareholders' equity of ₹20 crore, and no preference shares. Its ROE is 18%. What is its net profit margin, in per cent?

Answer: _____

2 Case scenario. To lift Tapri Kulhad Cafés' ROE before an investor meeting, Kesar proposes to borrow ₹5 crore and buy back shares of the same amount, with sales, operating profit and the asset base unchanged. Dev warns that this changes only one part of the DuPont equation directly. Which part?

- A Net profit margin, because interest is an operating expense
- B Asset turnover, because borrowing adds to the assets
- C Equity multiplier, because the same assets are financed by less equity
- D None, because a buyback does not affect ROE

3 According to ICAI, a Q ratio of more than 1 could mean that the stock is:

- A Undervalued
- B Overvalued
- C In equilibrium
- D Not quoted

4 A company's profit before tax is ₹80 lakh and the tax rate 25%. It has 10% preference share capital of ₹1 crore and 8 lakh equity shares. What is its EPS?

- A ₹7.50
- B ₹6.56
- C ₹6.25
- D ₹8.75

5 A firm's cost of goods sold is 62% of sales and its operating expenses 23% of sales. Which statements are correct? Select all that apply.

- A Its operating ratio is 85%
- B Its operating profit ratio is 15%
- C Its gross profit ratio is 15%
- D Its gross profit ratio is 38%

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Gross profit, operating profit (EBIT) and net profit ratios = that profit ÷ sales × 100; operating ratio = (COGS + operating expenses) ÷ sales, and operating ratio + operating profit ratio = 100%.
- ROI = margin × investment turnover; ROA = PAT (or PAT + interest, or EBIT(1 − t)) ÷ average assets.
- ROCE = EBIT (pre-tax) or EBIT(1 − t) (post-tax) ÷ capital employed; it should exceed the borrowing rate; intangibles in, fictitious assets out.
- ROE = (PAT − preference dividend) ÷ equity shareholders' funds.
- DuPont: ROE = net profit margin × asset turnover × equity multiplier (assets ÷ equity); the multiplier shows the part of ROE that comes from debt.
- EPS = earnings for equity ÷ equity shares; DPS; pay-out = DPS ÷ EPS.
- P/E = MPS ÷ EPS; dividend yield = DPS ÷ MPS (or dividend ± price change ÷ initial price); earnings yield = EPS ÷ MPS; MV/BV; Q ratio = market value ÷ replacement cost (1 = equilibrium).
- Same ROE, different businesses: ask which engine produced it. (Borrowed money sends a bill every month.)

Answer key · Check yourself

1 **4.79 to 4.81**

Asset turnover = 75 ÷ 50 = 1.5; equity multiplier = 50 ÷ 20 = 2.5; ROE = NPM × 1.5 × 2.5, so NPM = 18% ÷ 3.75 = 4.8%. Check: PAT = 18% × ₹20 crore = ₹3.6 crore, and 3.6 ÷ 75 = 4.8%.

2 **C Equity multiplier, because the same assets are financed by less equity**

Assets are unchanged and equity falls, so assets ÷ equity rises. The margin also falls indirectly (more interest, lower PAT), but the direct lever is the multiplier: ROE rises only while the return on assets exceeds the cost of the new debt.

3 **B Overvalued**

Q = market value of equity and liabilities ÷ replacement cost of assets. Equilibrium is 1; below 1 the stock could be undervalued, above 1 overvalued (SM 3.29).

4 **C** ₹6.25

PAT = 80 × 75% = ₹60 lakh; preference dividend = ₹10 lakh, paid out of profit after tax; earnings for equity = ₹50 lakh; EPS = 50 ÷ 8 = ₹6.25. ₹7.50 forgets the preference dividend; ₹6.56 deducts it before tax, as if it were interest ((80 − 10) × 75% ÷ 8); ₹8.75 uses profit before tax.

5 **A** Its operating ratio is 85%; **B** Its operating profit ratio is 15%; **D** Its gross profit ratio is 38%

Operating ratio = 62% + 23% = 85%; operating profit ratio = 100% − 85% = 15%; gross profit ratio = 100% − 62% = 38%.

2.4 Building financial statements from ratios; limitations of ratios

Rebuilding the trading and profit and loss account and balance sheet from a set of ratios and one rupee figure; who uses ratios and for what; and ICAI's eight limitations of financial ratios.

In plain words

The night of the masala chai

It is 11 pm before Tapri's board meeting. Kesar knocks a full glass of masala chai over the only printout of Kshipra Spice Mills' accounts, the supplier Tapri wants to buy. One page survives: a list of ratios and a single rupee figure.

> **Kesar:** It is gone. The balance sheet, the profit and loss account, everything. Tell the board we will present next month.
>
> **Dev:** We still have the ratios and one rupee figure. Each ratio is an equation. Give me twenty minutes and I will give you both statements back.
>
> **Kesar:** You can rebuild a balance sheet from ratios?
>
> **Dev:** Ratios are just accounts with the rupees taken out. I will put them back in.

A ratio links two figures. Suppose a sheet says "fixed assets are 0.75 of the proprietary fund" and "fixed assets are ₹30 lakh": together they give the proprietary fund, ₹40 lakh. If it also says "the proprietary fund is half of total assets", total assets follow: ₹80 lakh. Each step uses one ratio and one figure already known to find one new figure. When two figures are unknown and two ratios link them (a current ratio and a quick ratio, say), the two equations are solved together. And the last figure, usually cash, is whatever makes the balance sheet balance: the **balancing figure**.

The same evening Ananya, Tapri's head of strategy, warns Kesar about the other half of this lesson: ratios can be computed exactly and still mislead. Kshipra's plant was bought fifteen years ago, its stock of dried chillies peaks after the harvest, and it values that stock on a different method from Tapri's. ICAI lists eight such **limitations of financial ratios**, and ends with a line worth memorising: financial ratios provide clues but not conclusions.

Think of it like

Sudoku with rupees

A ratio question is a Sudoku. The rupee figure is the number printed in the grid; each ratio is a rule that fills one more cell once its neighbour is known. You never start with the cell that has two unknowns: you start where only one is missing, and the last cell is forced by the totals.

Key idea

Rebuilding statements from ratios is solving equations: start from a rupee figure, use each ratio to find one new figure, solve two ratios together when they share two unknowns, call sales "x" when no chain starts from the figure you have, and let total assets = total liabilities fix the balancing figure. Then remember that a ratio, however exact, is a clue for an expert, not a conclusion.

How it works

Seven steps from a ratio sheet to a balance sheet

1. **Sort the data.** Separate the rupee figures (the anchors) from the ratios. Note the basis of each ratio: on sales or on cost of goods sold; closing or average balance; 360 or 365 days; gross profit on sales or on cost.

2. **Write each ratio as an equation** and mark the two figures it links: current ratio = CA ÷ CL links current assets and current liabilities.

3. **Start from an anchor** and use the ratio that links it to exactly one unknown. Keep going: every new figure is an anchor for the next step.

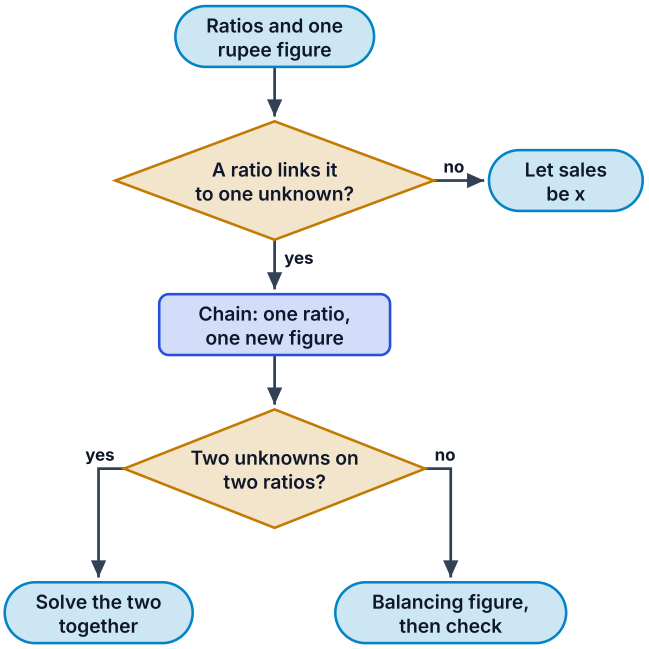
4. **Two unknowns, two equations.** When two ratios share the same two unknowns, solve them together: a current ratio with a quick ratio, a current ratio with working capital, a debt-equity ratio with the total of the balance sheet.

5. **No chain from the anchor?** Let sales be x, write every balance sheet item in terms of x, and solve total assets = total liabilities for x.

6. **Use the identities** that hold in every set of accounts: gross profit = sales − COGS; COGS = opening stock + purchases − closing stock; net worth = share capital + reserves; average = (opening + closing) ÷ 2; and the balance sheet balances.

7. **Balancing figure last, then check.** Let one item (cash, other current assets, purchases, operating expenses) be the balancing figure and label it "(balancing figure)" or "(bal. fig.)". Prepare the statements in the format asked, then recompute two of the given ratios from your statements.

Choosing the route through a ratio question

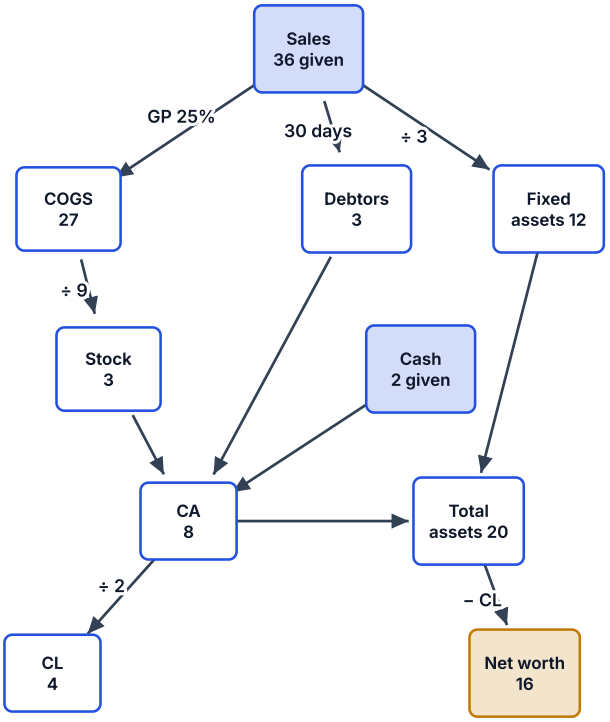


Most questions take the left route from a rupee figure. The right route (let sales be x, write every item in terms of x and solve total assets = total liabilities) is for an anchor such as cash or an investment that enters the balance sheet only through the totals; once x is found, the chain continues as usual. Every route ends with the balancing figure and a check.

A small rebuild, step by step

Tapri's kiosk division has lost its balance sheet. What survives (₹ lakh): sales 36; gross profit 25% of sales; stock turnover 9 (on COGS, closing stock); debtors collection period 30 days (360-day year, all sales on credit); fixed assets turnover 3 (on sales); current ratio 2; cash 2; no long-term debt.

Kiosk division: which ratio gives which figure (₹ lakh)



Blue boxes: the two rupee figures given (sales and cash). Each arrow is one ratio or one addition; amber: net worth, the balancing figure (total assets 20 − current liabilities 4).

Sales ₹36 lakh give COGS of 75% = ₹27 lakh, then closing stock 27 ÷ 9 = ₹3 lakh; debtors 36 × 30 ÷ 360 = ₹3 lakh; fixed assets 36 ÷ 3 = ₹12 lakh. Current assets = 3 + 3 + 2 = ₹8 lakh, so current liabilities = 8 ÷ 2 = ₹4 lakh. Total assets = 12 + 8 = ₹20 lakh, and with no long-term debt, net worth = 20 − 4 = ₹16 lakh. Check: current ratio 8 ÷ 4 = 2.

Two equations, two unknowns

Given	Write	Solve
Current ratio r and working capital W	CA = r × CL; CA − CL = W	CL = W ÷ (r − 1)
Current ratio r, quick ratio q, inventory I (no prepaid)	CA = r × CL; CA − I = q × CL	CL = I ÷ (r − q)
Long-term debt to equity d, balance sheet total T, CL (no preference capital)	Debt = d × Equity	Equity = (T − CL) ÷ (1 + d)
Opening stock O, average stock A	(O + closing) ÷ 2 = A	Closing = 2A − O
Gross profit g% on cost	GP = g% × COGS	GP ÷ sales = g ÷ (100 + g)

Who reads ratios, and for what

Ratio analysis presents facts on a comparative basis and enables inferences about the performance of a firm. ICAI lists the aspects it is relevant to, in this order (SM 3.34–3.36):

- (a) **Liquidity position**: whether the firm can meet its obligations when they become due; particularly useful in credit analysis by banks and other suppliers of short-term loans.
- (b) **Long-term solvency**: the long-term financial viability of a firm, of concern to long-term creditors, security analysts and present and potential owners; measured by leverage and profitability ratios.
- (c) **Operating efficiency**: the degree of efficiency in the management and utilisation of assets, measured by the activity ratios.
- (d) **Overall profitability**: management's integrated view of all the ratios together.
- (e) **Inter-firm comparison**: comparison with competitors and industry averages, a stepping stone to remedial measures. Ratios not only perform a post mortem of operations but also serve as a barometer for the future.
- (f) **Financial ratios for budgeting**: estimating budgeted figures and measuring actual performance against them.

Financial analysis may be horizontal or vertical (SM 3.37). **Horizontal analysis** compares the financial statements of one year with another year or years. **Vertical analysis** analyses the statements of a single year: every item of the profit and loss account is expressed as a percentage of gross sales and every item of the balance sheet as a percentage of total assets; it is useful in inter-firm comparison. Tapri's margins table in the profitability lesson is a vertical analysis.

Limitations of financial ratios

ICAI's list, in ICAI's order (SM 3.36–3.37), each with the Tapri case it fits:

1. **Diversified product lines.** Many businesses operate divisions in quite different industries; ratios on aggregate data cannot be used for inter-firm comparison. Tapri's group ratios mix cafés, premix, a samosa factory and a cold-chain fleet: no single competitor has that mix.
2. **Financial data are badly distorted by inflation.** Historical cost values may be substantially different from true values, and the distortion is carried into the ratios. A café bought in 2015 sits at its 2015 cost.
3. **Seasonal factors** may influence financial data. ICAI's example: a cotton garment company keeps a high inventory from October to January and about a quarter of it the rest of the year, so year-end liquidity and inventory ratios give a biased picture; monthly averages are suggested, but outsiders rarely get them. Tapri's Diwali gift-box stock is the same story.
4. **To give a good shape to the popularly used financial ratios** (window dressing). To flatter ratios such as the current ratio and the debt-equity ratio, a business may make year-end adjustments; such window dressing changes the character of the ratios from what they would have been without it. Repaying ₹10 crore of creditors from the bank on 30 March lifts Tapri's current ratio from 2.14 to 2.33 for one night.

5. **Differences in accounting policies and accounting period** make the data, and the ratios, of two firms non-comparable. BrewBucks India closes its books on 31 December and depreciates on a different method.
6. **No standard set of ratios against which a firm's ratios can be compared.** The industry average is a low standard for a firm that wants to be above average, and too high a standard for a below-average firm.
7. **Difficulty to generalise whether a particular ratio is good or bad.** A low current ratio may be "bad" for liquidity, but a high current ratio may not be "good": it may result from inefficient working capital management.
8. **Financial ratios are inter-related, not independent.** One ratio viewed in isolation may highlight efficiency, while the set may speak differently; such interdependence can be handled through multivariate analysis (analysing the relationship between several variables simultaneously).

Financial ratios provide clues but not conclusions. They are tools only in the hands of experts, because there is no standard ready-made interpretation of financial ratios (SM 3.37).

Formats and formulas

Reproduce the two formats from memory; ICAI's suggested answers present the rebuilt statements in these layouts, with each balancing figure labelled (bal. fig. = balancing figure).

Trading and Profit and Loss Account for the year ended ...

Dr. Particulars	₹
To Opening stock	xxx
To Purchases (balancing figure)	xxx
To Gross profit c/d	xxx
	xxx
To Operating expenses (balancing figure)	xxx
To Interest	xxx
To Provision for tax	xxx
To Net profit	xxx
	xxx

Cr. Particulars	₹
By Sales	xxx
By Closing stock	xxx
	xxx
By Gross profit b/d	xxx
	xxx

Liabilities	₹	Assets	₹
Share capital	xxx	Fixed assets	xxx
Reserves and surplus	xxx	Current assets:	
Long-term debt	xxx	Stock	xxx
Current liabilities	xxx	Debtors	xxx
		Cash (bal. fig.)	xxx
Total	**xxx**	**Total**	**xxx**

∑ **Formula**

The identities behind every rebuild

$$\text{Total assets} = \text{Total liabilities} \qquad \text{GP} = \text{Sales} - \text{COGS}$$

$$\text{COGS} = \text{Opening stock} + \text{Purchases} \qquad - \text{Closing stock}$$

$$\text{Net worth} = \text{Share capital} + \text{Reserves} \qquad \text{Average} = \frac{\text{Opening} + \text{Closing}}{2}$$

Net worth here means equity shareholders' funds; total liabilities include capital unless the question says otherwise.

∑ **Formula**

Balances from ratios

$$\text{Debtors} = \frac{\text{Credit sales} \times \text{Days}}{360} \qquad \text{Stock} = \frac{\text{COGS}}{\text{Stock turnover}}$$

$$\text{CL} = \frac{\text{Working capital}}{\text{Current ratio} - 1} \qquad \text{Fixed assets} = \frac{\text{Sales}}{\text{Fixed assets turnover}}$$

Days = the collection period in days. Use the basis the question gives (sales or COGS; closing or average; 360 or 365 days).

Worked examples

Worked example **Kshipra Spice Mills: rebuild both statements**

From the surviving ratio sheet of Kshipra Spice Mills Pvt Ltd for the year ended 31 March 2026:

Particulars	Figure
Fixed assets (net)	₹48,00,000
Fixed assets to proprietary fund	0.80
Share capital to reserves and surplus	2 : 1
Proprietary ratio (proprietary fund ÷ total assets)	0.60
Current ratio	2.6 : 1
Quick ratio	1.4 : 1
Stock turnover (on COGS and average stock)	6 times
Opening stock	₹20,00,000
Gross profit ratio	20% of sales
Debtors collection period (all sales on credit; closing debtors; 360 days)	36 days
Interest coverage ratio	5 times

The only long-term debt is a 12% term loan. There are no preference shares and no prepaid expenses; current assets are stock, debtors and cash. The tax rate is 25%.

Required:

(i) Prepare the Trading and Profit and Loss Account for the year ended 31 March 2026.

(ii) Prepare the Balance Sheet as at 31 March 2026.

(iii) Verify the current ratio and the interest coverage ratio from your statements.

THE WORKING

Working notes (₹)

1. Proprietary fund = 48,00,000 ÷ 0.80 = **60,00,000**; share capital = 60,00,000 × 2/3 = **40,00,000**; reserves = **20,00,000**.
2. Total assets = 60,00,000 ÷ 0.60 = **1,00,00,000**; current assets = 1,00,00,000 − 48,00,000 = **52,00,000**.
3. Current liabilities = 52,00,000 ÷ 2.6 = **20,00,000**.
4. Term loan = total liabilities − proprietary fund − current liabilities = 1,00,00,000 − 60,00,000 − 20,00,000 = **20,00,000**; interest at 12% = **2,40,000**.
5. Quick assets = 1.4 × 20,00,000 = 28,00,000; closing stock = 52,00,000 − 28,00,000 = **24,00,000**.
6. Average stock = (20,00,000 + 24,00,000) ÷ 2 = 22,00,000; COGS = 6 × 22,00,000 = **1,32,00,000**.
7. Sales = 1,32,00,000 ÷ 80% = **1,65,00,000**; gross profit = **33,00,000**.

8. Debtors = 1,65,00,000 × 36 ÷ 360 = **16,50,000**; cash = 28,00,000 − 16,50,000 = **11,50,000**.
9. EBIT = 5 × 2,40,000 = **12,00,000**; operating expenses = 33,00,000 − 12,00,000 = **21,00,000**.
10. PBT = 12,00,000 − 2,40,000 = 9,60,000; tax at 25% = 2,40,000; net profit = **7,20,000**.
11. Purchases = COGS + closing stock − opening stock = 1,32,00,000 + 24,00,000 − 20,00,000 = **1,36,00,000**.

(i) Trading and Profit and Loss Account of Kshipra Spice Mills Pvt Ltd for the year ended 31 March 2026

Dr. Particulars	₹
To Opening stock	20,00,000
To Purchases (balancing figure)	1,36,00,000
To Gross profit c/d	33,00,000
	1,89,00,000
To Operating expenses (balancing figure)	21,00,000
To Interest on term loan	2,40,000
To Provision for tax	2,40,000
To Net profit	7,20,000
	33,00,000

Cr. Particulars	₹
By Sales	1,65,00,000
By Closing stock	24,00,000
	1,89,00,000
By Gross profit b/d	33,00,000
	33,00,000

(ii) Balance Sheet of Kshipra Spice Mills Pvt Ltd as at 31 March 2026

Liabilities	₹
Share capital	40,00,000
Reserves and surplus	20,00,000
12% Term loan	20,00,000
Current liabilities	20,00,000
Total	**1,00,00,000**

Assets	₹
Fixed assets (net)	48,00,000
Stock	24,00,000
Debtors	16,50,000
Cash (balancing figure)	11,50,000
Total	**1,00,00,000**

(iii) Verification

Current ratio = (24,00,000 + 16,50,000 + 11,50,000) ÷ 20,00,000 = 52,00,000 ÷ 20,00,000 = **2.6**. Interest coverage = 12,00,000 ÷ 2,40,000 = **5 times**. Both agree with the data.

Interactive version: PassBook app, lesson Building financial statements from ratios; limitations of ratios

Worked example **Indore Namkeen: when the only rupee figure is an investment**

Indore Namkeen Co. Ltd sells snacks through Tapri's kiosks. Its accountant has left Dev the following for the year ended 31 March 2026:

Particulars	Figure
Gross profit ratio	20% of sales
Credit sales	75% of sales
Closing stock	1.5 months' cost of goods sold
Closing debtors	2 months' credit sales
Creditors	1 month's cost of goods sold
Fixed assets turnover (on cost of goods sold)	2.4 times
Current ratio	2.5 : 1
Quick ratio	1.5 : 1
Long-term debt to total assets	0.25
Proprietary ratio (proprietary fund ÷ total assets)	0.60
Share capital to reserves	5 : 3
Non-current investments	₹6,00,000

Current assets are stock, debtors and cash; current liabilities are creditors and a bank overdraft. There are no preference shares.

Required:

(i) Calculate the sales for the year.

(ii) Prepare the summarised Balance Sheet as at 31 March 2026, showing working notes.

(iii) Verify the current ratio and the proprietary ratio from your balance sheet.

THE WORKING

No ratio links the investments of ₹6,00,000 to any other single figure: they enter only through total assets. So let sales be **₹x** and write every item in terms of x.

Indore Namkeen: the balance sheet in terms of sales x

Item	In terms of x	From
Fixed assets	x/3	0.8x ÷ 2.4
Stock	0.10x	0.8x × 1.5 ÷ 12
Debtors	0.125x	0.75x × 2 ÷ 12
Current liabilities	0.10x	stock ÷ (2.5 − 1.5)
Current assets	0.25x	2.5 × CL
Cash	0.025x	CA − stock − debtors
Total assets	7x/12 + 6,00,000	FA + investments + CA
Same CL, from totals	0.15 × total assets	1 − 0.60 − 0.25

Working notes (₹)

1. Cost of goods sold = 80% of sales = 0.8x; credit sales = 0.75x.
2. Fixed assets = COGS ÷ 2.4 = 0.8x ÷ 2.4 = x/3. Stock = 0.8x × 1.5 ÷ 12 = 0.10x. Debtors = 0.75x × 2 ÷ 12 = 0.125x.
3. Current ratio: CA = 2.5 CL. Quick ratio: CA − stock = 1.5 CL. So CL = stock ÷ (2.5 − 1.5) = 0.10x, CA = 0.25x and cash = 0.25x − 0.10x − 0.125x = 0.025x.
4. Total assets = x/3 + 6,00,000 + 0.25x = 7x/12 + 6,00,000.
5. Proprietary fund = 0.60 × total assets and long-term debt = 0.25 × total assets, so current liabilities = (1 − 0.60 − 0.25) × total assets = 0.15 × total assets.
6. Equating the two expressions for current liabilities: $0.10x = 0.15\left(\frac{7x}{12} + 6{,}00{,}000\right)$, so $0.0125x = 90{,}000$ and $x = 72{,}00{,}000$.
7. **(i) Sales = ₹72,00,000**; COGS = ₹57,60,000; credit sales = ₹54,00,000.
8. Fixed assets = 24,00,000; stock = 7,20,000; debtors = 9,00,000; cash = 1,80,000; current assets = 18,00,000; total assets = 24,00,000 + 6,00,000 + 18,00,000 = 48,00,000.
9. Proprietary fund = 0.60 × 48,00,000 = 28,80,000: share capital 5/8 = 18,00,000, reserves 3/8 = 10,80,000. Long-term debt = 0.25 × 48,00,000 = 12,00,000.
10. Current liabilities = 0.10 × 72,00,000 = 7,20,000; creditors = 57,60,000 ÷ 12 = 4,80,000; bank overdraft = 7,20,000 − 4,80,000 = 2,40,000 (balancing figure).

(ii) Summarised Balance Sheet of Indore Namkeen Co. Ltd as at 31 March 2026

Liabilities	₹
Share capital	18,00,000
Reserves and surplus	10,80,000
Long-term debt	12,00,000
Creditors	4,80,000
Bank overdraft (balancing figure)	2,40,000
Total	**48,00,000**

Assets	₹
Fixed assets	24,00,000
Non-current investments	6,00,000
Stock	7,20,000
Debtors	9,00,000
Cash	1,80,000
Total	**48,00,000**

(iii) Verification: current ratio = 18,00,000 ÷ 7,20,000 = 2.5; proprietary ratio = 28,80,000 ÷ 48,00,000 = 0.60. Also quick ratio = (18,00,000 − 7,20,000) ÷ 7,20,000 = 1.5, and fixed assets turnover = 57,60,000 ÷ 24,00,000 = 2.4.

Worked example

The board slide that compared too much

Kesar's board slide says: "Tapri beats BrewBucks India on every ratio. Our current ratio is 2.33 against their 1.6, our fixed assets turnover 2.54 against 1.9, and our ROCE is above the industry average, so we are the better company." Ananya points out that:

(a) Tapri's figures are for the whole group: cafés, packaged premix, the samosa factory and the cold-chain fleet; BrewBucks India runs only coffee shops.

(b) Most of Tapri's cafés and the samosa factory were bought between 2012 and 2016 and stand at historical cost; BrewBucks India's outlets are three years old.

(c) Tapri's year ends on 31 March, when its Diwali stock has long been sold; BrewBucks India's ends on 31 December, at the peak of its winter stock. BrewBucks also uses a different depreciation method.

(d) On 30 March Tapri paid ₹10 crore of creditors from the bank, which is why its current ratio is 2.33 and not 2.14.

(e) The "industry average" was taken from a report on café chains, and Tapri's other ratios (inventory turnover, interest coverage) were left off the slide.

Required: Identify the limitations of financial ratios that make Kesar's comparison unreliable, and state what an analyst should do.

MODEL ANSWER

Financial ratios provide clues but not conclusions; they are tools only in the hands of experts, because there is no standard ready-made interpretation of them. Applying the limitations listed in the Study Material to the facts:

1. **Diversified product lines (a).** Ratios calculated on the aggregate data of divisions in different industries cannot be used for inter-firm comparison. Tapri's group ratios should be compared segment by segment: the cafés with BrewBucks India.
2. **Distortion by inflation (b).** Historical cost values may be substantially different from true values. Tapri's old assets at historical cost inflate its fixed assets turnover (2.54) and its ROCE relative to a company with recently acquired outlets.
3. **Seasonal factors (c).** The year-end figures of the two companies fall at different points of the season, so their liquidity and inventory ratios do not represent the average position; average or monthly figures should be used where available.
4. **Giving a good shape to popular ratios: window dressing (d).** The year-end payment to creditors gives a good shape to the current ratio; without it the ratio would be 2.14, not 2.33. The ratio should be read with the reason for the change.
5. **Differences in accounting policies and accounting period (c).** Different year-ends and depreciation methods make the accounting data of the two firms, and their ratios, non-comparable.
6. **No standard set of ratios (e).** An industry average from a different business is not a valid benchmark; even a proper industry average is a low standard for a firm that aims to be above average.
7. **Difficulty to generalise (e).** A high current ratio is not necessarily good; it may reflect inefficient working capital management.
8. **Ratios are inter-related (e).** Selected ratios in isolation may highlight efficiency while the full set speaks differently; all ratios, including inventory turnover and interest coverage, should be analysed together.

Conclusion: the claim that Tapri is "the better company" does not follow from the slide. The analyst should compare like with like (segment figures, the same period and adjusted accounting policies), use average figures to remove seasonal effects, disclose year-end adjustments, choose a relevant benchmark, and read the whole set of ratios together before drawing a conclusion.

Exam angle

In the exam

Where these questions appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, questions that rebuild or complete statements from ratios appeared in four past papers and seven RTPs:

- **Past papers, 5 marks each.** May 2024 Q1(a) (interest coverage, gross profit ratio and current assets, the last from the balance sheet identity); September 2024 Q1(c) (complete a balance sheet from inventory turnover, total assets turnover, long-term debt to shareholders' fund, collection period, gross profit ratio and current ratio); May 2025 Q1(a) (a balance sheet in a given format from raw material and finished goods holding periods, collection period, proprietary ratio and fixed assets turnover); September 2026 Q1(b) (credit sales, COGS, average inventory and debtors and proprietary fund, with a capital gearing ratio of 40%).
- **RTPs.** May 2024 Q1 (balance sheet and income statement from margins, PBIT to PBT and turnovers); September 2024 Q4 (complete a balance sheet from debt to total assets, quick ratio and holding periods); May 2025 Q4 (balance sheet with a capital gearing ratio and depreciation on opening WDV); September 2025 Q4 (operating expenses, then the balance sheet) and May 2026 Q4 (operating expenses, share capital and reserves, closing stock and fixed assets), both starting from a return on net worth; January 2026 Q5 (a balance sheet from capital-structure percentages, then a table of ratios for a bank); September 2026 Q4 (profit and loss statement and balance sheet from percentages of cost of goods sold, the Study Material's Practical Problem 3 set again).
- **MTPs and MCQs.** MTP Series I for January 2025 Q2(a) (trading, profit and loss account and balance sheet); MTP Series I for September 2025 Q1(b) (quick ratio, fixed assets turnover, proprietary ratio and EPS); MTP Series I for September 2026 Q3(b) (a summarised balance sheet from net working capital, bank overdraft, fixed assets to proprietors' fund and the current and quick ratios, 3 marks: the figures of the Study Material's Practical Problem 6); January 2025 MCQ 7 (current assets and liabilities from two ratios).
- **Limitations.** "Explain briefly the limitations of financial ratios" is the chapter's own theory question 5 (SM 3.65); in the old scheme it was asked as MTP Series II for May 2022 Q6(b) for 4 marks, answered with the list of limitations and the closing line on clues and conclusions.

In the exam

The traps ICAI set

- A ratio on cost of goods sold used on sales (or the reverse); gross profit given on cost.
- An opening stock that forces the average-stock route (Kshipra).
- An absolute figure that only enters through the totals, forcing "let sales be x" (the Study Material's Practical Problem 11).
- Items that must be classified before the balance sheet balances: outstanding interest as a current liability, preference capital outside debt-equity, bills receivable with debtors.

Interactive version: PassBook app, lesson Building financial statements from ratios; limitations of ratios

Trick

DISWANDI: the eight limitations in ICAI's order

Diversified product lines, **I**nflation, **S**easonal factors, **W**indow dressing, **A**ccounting policies and period, **N**o standard set of ratios, **D**ifficulty to generalise, **I**nter-related ratios. The magic wand that does not work: ratios give clues, not conclusions.

Answer frame

"Explain briefly the limitations of financial ratios"

1. One line for each of the eight limitations, in ICAI's order, with its explanation (aggregate data of diverse divisions; historical cost under inflation; seasonal stock and year-end figures; year-end window dressing of popular ratios; non-comparable policies and periods; no standard set of ratios; a high or low ratio not good or bad in itself; inter-related ratios and multivariate analysis).
2. The closing line: financial ratios provide clues but not conclusions; they are tools only in the hands of experts, as there is no standard ready-made interpretation.
3. For a case ("Comment", "Advise"), apply each relevant limitation to the facts, then conclude what the analyst should do.

Three answers to a "prepare the balance sheet" question

Loses marks

A balance sheet with the right figures and no working notes.

If one figure is wrong, nothing in the chain earns method marks, and the examiner cannot see which ratio was used.

Passes

Working notes for most figures, but the balancing figure is not labelled and the statement is not in the format asked.

The examiner has to search for how cash was found.

Full marks

Numbered working notes in the order of the chain, each with the ratio, the figures and the result; assumptions stated (closing or average, 360 days); statements in the asked format with " (balancing figure)" labelled; two ratios re-checked.

Common mistakes

Common mistake

Hall of Shame: the balance sheet that balanced by accident

Making cash the balancing figure before checking the other ratios: the sheet balances, but the current ratio no longer agrees with the data. **Fix:** find every figure a ratio gives first; only the one figure no ratio gives may balance the sheet, and recompute two ratios at the end.

Common mistake

Gross profit on cost used as gross profit on sales

With sales of ₹30,00,000 and "gross profit 25% of cost", reading 25% as a share of sales gives COGS of ₹22,50,000 instead of ₹24,00,000, and a stock turnover of 8 then gives stock of ₹2,81,250 instead of ₹3,00,000. **Fix:** convert first: 25 ÷ (100 + 25) = 20% of sales, so COGS = 80% of sales.

⚠ **Common mistake**

Closing stock where the ratio uses the average

Kshipra's stock turnover of 6 applies to the average stock of ₹22,00,000, so COGS = ₹1,32,00,000. Multiplying the closing stock instead (6 × 24,00,000 = ₹1,44,00,000) overstates COGS, sales (₹1,80,00,000) and debtors (₹18,00,000), and understates cash (₹10,00,000). **Fix:** when an opening stock is given, work with the average, (opening + closing) ÷ 2; when the average is what you find, closing = 2 × average − opening.

⚠ **Common mistake**

The limitations answer that listed but did not apply

In a case, a memorised list of eight limitations earns little if the facts are never used. **Fix:** name the limitation, quote the fact from the case that shows it, and say what the analyst should do.

Practice

Check yourself

1 A company's balance sheet total is ₹80 lakh, of which current liabilities are ₹16 lakh. Its long-term debt to equity ratio is 0.6 : 1, its share capital to reserves ratio 3 : 1, and it has no preference shares. What are its reserves, in rupees?

Answer: _____

2 Case scenario. A supplier's ratio sheet says: sales ₹40,00,000; gross profit 25% on cost; stock turnover (on cost of goods sold and average stock) 8 times; opening stock ₹3,00,000. Bunty, the article assistant checking the figures for the audit file, writes closing stock = ₹4,00,000 "because COGS ÷ 8". Which closing stock should go into the balance sheet?

- A ₹4,00,000
- B ₹4,50,000
- C ₹5,00,000
- D ₹7,00,000

3 Which of the following is not one of ICAI's limitations of financial ratios?

- A Seasonal factors may influence financial data
- B Ratios cannot be calculated from interim financial statements
- C Financial data are badly distorted by inflation
- D Financial ratios are inter-related, not independent

4 Case scenario. For the board, Ananya takes the 2025-26 accounts of Tapri and of BrewBucks India and expresses every item of each profit and loss account as a percentage of gross sales, and every balance sheet item as a percentage of total assets. Which statement about her work is correct?

A It is horizontal analysis, useful mainly for studying trends over the years

B It is horizontal analysis, because two companies are compared

C It is vertical analysis, which is useful in inter-firm comparison

D It is multivariate analysis of inter-related ratios

5 Gross profit is 20% of cost. What percentage of sales is it? Give the answer to two decimals.
Answer: _____

Answers: the answer key at the end of this lesson.

Recap

- Method: sort anchors from ratios; one ratio, one new figure; two ratios for two unknowns; "let sales be x" when the anchor enters only through the totals; balancing figure last; verify two ratios.
- CL = working capital ÷ (current ratio − 1); with inventory I, CL = I ÷ (current ratio − quick ratio).
- Closing stock = 2 × average stock − opening stock; GP g% on cost = g ÷ (100 + g) of sales.
- Present the rebuilt statements in the asked format with every balancing figure labelled.
- ICAI's six uses of ratio analysis: liquidity, long-term solvency, operating efficiency, overall profitability, inter-firm comparison, budgeting.
- Horizontal analysis compares years; vertical analysis expresses one year's items as % of gross sales or total assets.
- Eight limitations (DISWANDI): diversified lines, inflation, seasonal factors, window dressing, accounting policies and period, no standard set, difficulty to generalise, inter-related ratios.
- Ratios provide clues, not conclusions. (Even after the masala chai.)

Answer key · Check yourself

1 **1000000**

Equity + long-term debt = 80 − 16 = ₹64 lakh, and debt = 0.6 × equity, so equity = 64 ÷ 1.6 = ₹40 lakh (debt ₹24 lakh). Reserves = 1/4 × 40 = ₹10 lakh. Forgetting to take out the current liabilities gives equity of ₹50 lakh and reserves of ₹12.5 lakh.

2 **C ₹5,00,000**

Gross profit 25% on cost is 20% of sales, so COGS = 80% × 40,00,000 = ₹32,00,000; average stock = 32,00,000 ÷ 8 = ₹4,00,000; closing = 2 × 4,00,000 − 3,00,000 = ₹5,00,000. Bunty's ₹4,00,000 is the average, not the closing stock; ₹4,50,000 reads 25% as a share of sales (COGS ₹30,00,000, average ₹3,75,000); ₹7,00,000 uses sales instead of COGS (average ₹5,00,000).

3 **B** **Ratios cannot be calculated from interim financial statements**

Interim financial statements are one of ICAI's sources of financial data (SM 3.4); the other three are in its list of limitations (SM 3.36–3.37).

4 **C** **It is vertical analysis, which is useful in inter-firm comparison**

Analysing the statements of a single year, with each item as a percentage of gross sales or of total assets, is vertical analysis, and ICAI notes it is useful in inter-firm comparison (SM 3.37). Horizontal analysis compares one year's statements with another year or years.

5 **16.66 to 16.67**

GP ÷ sales = 20 ÷ (100 + 20) = 16.67%.

CHAPTER 3

Types of Financing

3.1 Long-term sources: equity, preference, debt and retained earnings

How ICAI classifies sources of finance, and the features, advantages and disadvantages of equity, preference shares, retained earnings, debentures, bonds, institutional and bank loans and bridge finance.

In plain words

Kesar wants a factory on an overdraft

Tapri Tea Co. Ltd wants a samosa factory at Dewas. It will cost ₹100 crore and run for fifteen years.

> **Kesar:** The bank already gives us an overdraft. Use that. Sab adjust ho jayega.
>
> **Dev:** An overdraft is repayable on demand, Kesar-ji. The factory pays for itself over fifteen years. If the bank calls the money back in year two, we sell the fryers?
>
> **Kesar:** Then where does fifteen-year money come from?
>
> **Dev:** From people who will wait fifteen years. There are only a few kinds of them.

Dev's rule is the first rule of this chapter: money needed for a long time must come from **long-term sources**, money needed for a few months from short-term sources. A business has three kinds of financial need: **long-term** (more than 5–10 years: land, buildings, plant, and the working capital it always needs), **medium-term** (one to five years) and **short-term** (up to one year: stock, debtors, cash).

The long-term money comes from four kinds of people. **Equity shareholders** are the owners: they get whatever is left after everyone else is paid, and they vote. **Preference shareholders** get a fixed dividend before the owners, and usually no vote. **Lenders** (debenture holders, bondholders, financial institutions, banks) get fixed interest on fixed dates, whether the factory makes money or not. And **retained earnings** are the owners' own profits, kept in the business instead of being paid out.

Each source trades the same three things against each other: what it **costs** the company, how much **risk** it puts on the company, and how much **control** it takes from the owners.

Think of it like

The family that buys a delivery van

A family shop needs a van. It can bring in a partner, who shares every profit and wants a say (equity). It can take money from an uncle who wants a fixed 10% before the family takes anything, but stays out of the shop's decisions (preference). It can take a bank loan whose instalment is due on the 5th of every month, good month or bad (debt). Or it can use the savings in its own cash box (retained earnings). The partner is the most expensive in a good year and the safest in a bad one; the bank is the cheapest and the least forgiving.

Key idea

Long-term needs are financed from long-term sources. Equity is the costliest but carries no obligation to pay or repay, and dilutes control; debt is the cheapest because interest is tax-deductible, but its interest and repayment are obligatory and it raises financial risk; preference capital sits in between, a hybrid; retained earnings dilute nothing and carry almost no risk.

How it works

1. Three needs and the matching rule

ICAI groups all the financial needs of a business into three categories (SM 2.2–2.3):

1. **Long-term financial needs**: funds required for a period exceeding 5–10 years. All investments in plant, machinery, land and buildings; funds for permanent or **hard-core working capital** should also be procured from long-term sources.
2. **Medium-term financial needs**: funds required for more than one year but not more than 5 years, for example stores and spares, critical spares, tools, dies and moulds.
3. **Short-term financial needs**: funds for current assets such as stock, debtors and cash (working capital requirements), for a period not exceeding the accounting period, i.e. one year.

Basic principle for funding various needs: short-term needs from short-term sources, medium-term needs from medium-term sources and long-term needs from long-term sources. The method of raising funds is decided with reference to the period for which funds are required (SM 2.3).

The rule can change with the nature of the borrower, its stage of development and the nature of its business. ICAI's general pattern (SM 2.3):

Stage	Nature of business	Sources of fund
Early stage	High uncertainty	Equity; mainly angel fund
Early stage	High to moderate uncertainty	Equity; venture capital; debt
Growth stage	Moderate to low uncertainty	Debt; venture capital; private equity
Stable stage	Low uncertainty	Debt

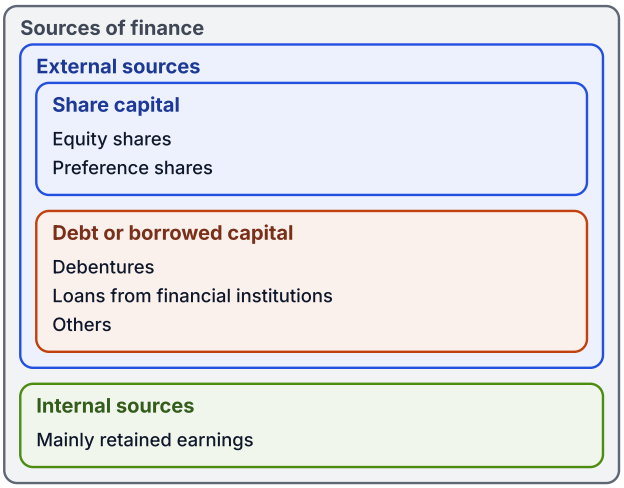
The less certain the business, the more it relies on owners who can wait; the more stable it is, the more it can carry debt.

2. Three ways to classify a source

The Study Material classifies sources of finance in three ways (SM 2.3–2.5).

(i) Based on basic sources: external sources (share capital and debt) and internal sources (mainly retained earnings).

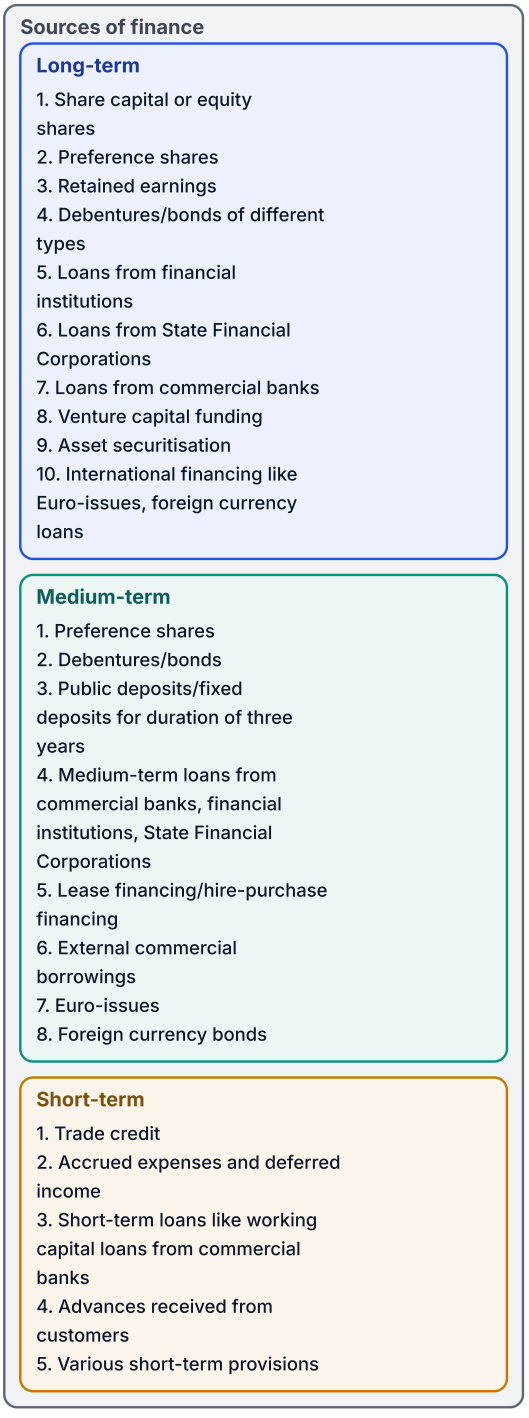
Sources of finance based on basic sources (SM 2.4)



Blue: external sources (share capital in a blue box, debt in an orange one); green: internal sources.

(ii) Based on maturity of payment (SM 2.4). Preference shares and debentures appear twice because their period can be long or medium:

Sources of finance based on maturity of payment (SM 2.4)



Blue box: long-term; teal: medium-term; amber: short-term. Preference shares and debentures appear in both of the first two boxes.

(iii) Based on ownership and control (SM 2.5): **owner's fund** (equity share capital, retained earnings) and **borrowed fund** (debentures, loans from banks, loans from financial institutions, others).

3. Equity share capital: the owners' money

A public limited company may raise funds from promoters or from the investing public by issuing ordinary equity shares. Their characteristics (SM 2.5–2.6):

- A source of **permanent capital**; the holders are called equity shareholders or ordinary shareholders.
- Equity shareholders are practically the **owners** of the company, as they undertake the highest risk.
- They are entitled to dividends **after the income claims of other stakeholders** are satisfied; the dividend is an **appropriation of profits, not a charge** against profits.
- In a winding up, they claim on assets after the claims of the other suppliers of capital have been met.
- The **cost of ordinary shares is usually the highest**, because these shareholders expect a higher return for the highest risk.
- Equity provides a **security to other suppliers of funds**: a lender checks that the debt-equity ratio is comfortable. Types include a new issue, rights issue, bonus shares and sweat equity.

Advantages of raising funds by issue of equity shares (SM 2.6–2.7):

1. It is a **permanent source of finance**. The shares are not redeemable, so the company has no liability for cash outflows on redemption; investors who want to leave sell their shares to others.
2. Equity **increases the company's financial base** and so its borrowing powers.
3. The company is **not legally obliged to pay dividends**: in uncertain times dividends can be reduced or even suspended.
4. The company can further increase its share capital by a **rights issue**.

Disadvantages (SM 2.7):

1. Investors find ordinary shares riskier because of uncertain dividend payments and capital gains.
2. A new issue **reduces the earning per share** of the existing shareholders until profits rise proportionately.
3. A new issue can **reduce the ownership and control** of the existing shareholders.

4. Preference share capital: the hybrid

Preference shares are a special kind of share whose holders enjoy **priority both as regards the payment of a fixed amount of dividend and towards repayment of capital** on winding up (SM 2.7). Their characteristics (SM 2.7–2.8):

- Long-term funds from preference shares can be raised through a public issue.
- They are normally **cumulative**: a dividend unpaid in a year of loss is carried over until there are adequate profits.
- The dividend rate is normally **higher than the rate of interest** on debentures, loans, etc.
- Most carry a stipulated period, at the end of which the funds are repaid.
- Preference share capital is a hybrid form of financing: like equity, its dividend is **not a tax-deductible payment**; like debt, its **rate is fixed**.

- **Cumulative convertible preference shares (CCPs)** carry a cumulative dividend of a specified limit for a period of, say, three years, after which they convert into equity shares: attractive for projects with a long gestation period.
- Preference capital may be redeemed at a pre-decided future date or earlier, out of profits, which lets promoters withdraw capital from a company that has become self-sufficient.

Type of preference share	Salient feature (SM 2.8)
Cumulative	Arrear dividend will accumulate
Non-cumulative	No right to arrear dividend
Redeemable	Redemption should be done
Participating	Can participate in the surplus remaining after payment to equity shareholders
Non-participating	Cannot participate in the surplus after payment of the fixed rate of dividend
Convertible	Option of converting into equity shares

Advantages of raising funds by issue of preference shares (SM 2.8–2.9):

1. **No dilution in EPS** on an enlarged capital base (an equity issue reduces EPS and affects market perception).
2. The **advantage of leverage**, as it bears a fixed charge; non-payment of preference dividend does not force the company into liquidation (the Study Material prints "liquidity"; liquidation is the sense).
3. **No risk of takeover**: preference shareholders have no voting rights except where dividends are in arrears.
4. The dividends are fixed and pre-decided, so preference shareholders cannot share surplus profits as ordinary shareholders can, except participating preference shareholders.
5. Preference capital can be **redeemed** after a specified period.

Disadvantages (SM 2.9):

1. Preference dividend is **not tax-deductible**, so it gives no tax shield: preference shares cost the company more than debt such as debentures.
2. Preference dividends are **cumulative**: unpaid dividends accumulate and must be paid later, and until they are, no dividend can be paid to ordinary shareholders, which could seriously impair the company's reputation.

5. Retained earnings: the money already in the till

Long-term funds can be provided by accumulating the company's profits and **ploughing them back** into the business. These funds belong to the ordinary shareholders and increase the net worth of the company. A public limited company must plough back a reasonable amount of profit every year, keeping in view the legal requirements and its own expansion plans. Such funds **entail almost no risk**, and the control of present owners is **not diluted** (SM 2.10).

Whether to plough back depends on the **rate of return the company generates against the expected cost of equity** (SM 2.10): the subject of the dividend theories.

6. Debentures: the borrowed money with a trust deed

Public limited companies can raise loans from the public by issuing debentures or bonds. Their characteristics (SM 2.10):

- Normally issued in denominations from ₹100 to ₹1,000, at different rates of interest.
- Normally issued on the basis of a **debenture trust deed**, which lists the terms and conditions of the issue.
- Basically instruments for raising **long-term debt capital**.
- Maturity normally varies from **3 to 10 years**, longer for projects with a high gestation period.
- Either **secured or unsecured**; they may or may not be listed on a stock exchange.
- The **cost is quite low**, since interest is charged as an expense before tax.
- For investors, more attractive than preference shares, since **interest is payable whether or not the company makes profits**.

By convertibility (SM 2.11):

1. **Non-convertible debentures**: no conversion feature; repayable on maturity.
2. **Fully convertible debentures**: converted into equity shares on the terms of issue as to price and time; interest is generally lower than on non-convertible debentures because of the attractive conversion feature.
3. **Partly convertible debentures**: carry features of both, in one debenture.

Other type of debenture	Salient feature (SM 2.11)
Bearer	Transferable like negotiable instruments
Registered	Interest payable to the registered person
Mortgage	Secured by a charge on asset(s)
Naked or simple	Unsecured
Redeemable	Repaid after a certain period
Non-redeemable	Not repayable

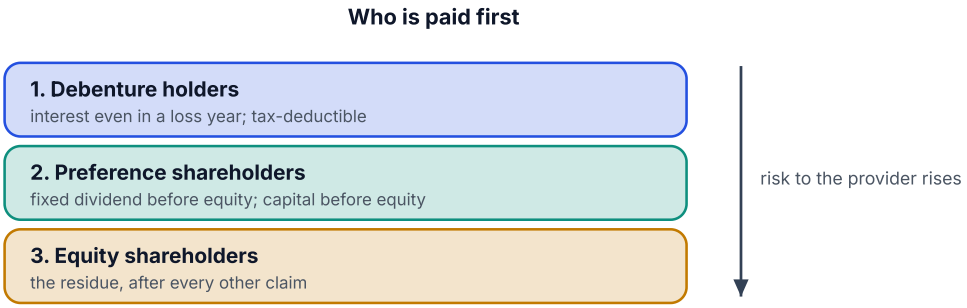
Advantages of raising finance by issue of debentures (SM 2.11):

1. The **cost is much lower** than the cost of preference or equity capital, as the interest is tax-deductible; investors also consider debentures safer and may accept a lower return.
2. Debenture financing **does not result in dilution of control**.
3. In a **period of rising prices**, a debenture issue is advantageous: the fixed monetary outgo decreases in real terms as the price level increases.

Disadvantages of debenture financing (SM 2.12):

1. Debenture interest and the repayment of principal are **obligatory payments**.
2. The **protective covenants** of a debenture issue may be restrictive.
3. Debenture financing **enhances the financial risk** of the firm, because of point 1.
4. Repayment at maturity needs a **large cash outflow** at that time.

A public issue of debentures, and private placement to mutual funds, must be **rated by a credit rating agency** such as CRISIL. The rating weighs the company's track record, profitability, debt servicing capacity, creditworthiness and the perceived risk of lending (SM 2.12).



Interest is payable whether or not there are profits; preference shareholders have priority over equity both in dividend and in repayment of capital on winding up; equity shareholders take dividends after all other income claims and assets after all other suppliers of capital. The later a provider is paid, the more risk it carries: which is why equity costs the most and debt the least (SM 2.6, 2.7, 2.10).

7. Bonds

A **bond** is a fixed income security created to raise funds, through a public issue or a private placement (SM 2.13). Based on call:

- **Callable bonds**: a call option gives the **issuer** the right to redeem the bond before maturity at a predetermined price, the **call price** (generally at a premium).
- **Puttable bonds**: a put option gives the **investor** the right to sell the bond back to the company before maturity.

Indian bonds (SM 2.15–2.16):

Bond	Salient feature
Masala bond	"Masala" means spice: an Indian name for a **rupee-denominated** bond that Indian corporate borrowers sell to investors in overseas markets; issued outside India but denominated in Indian rupees. NTPC raised ₹2,000 crore through masala bonds in 2016
Municipal bonds	Used to finance urban infrastructure. Ahmedabad Municipal Corporation issued a first historical municipal bond in Asia, raising ₹100 crore for part financing a water supply project
Government or treasury bonds	Issued by the Government of India, the Reserve Bank of India, any state government or any other government department

The foreign bonds of SM 2.13–2.15 (FCCB, plain vanilla, drop lock and the rest) are in contemporary and international sources; deep discount, zero coupon, inflation and junk bonds in other sources of finance.

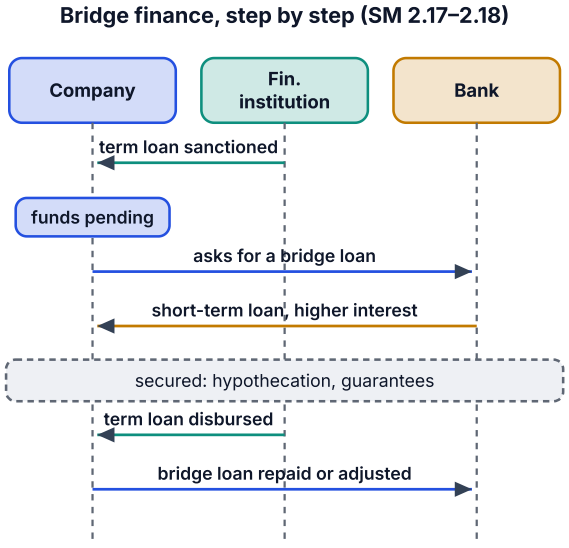
8. Loans from financial institutions and commercial banks

Financial institutions (SM 2.16–2.17). National: IFCI (converted into a public company), the State Financial Corporations, IDBI (converted into a bank), NIDC, ICICI (converted into a bank and privatised), LIC, UTI and IRBI. International: the World Bank/International Bank for Reconstruction and Development (IBRD, 1944), the International Finance Corporation (IFC, 1956) and the Asian Development Bank (ADB, 1966).

Commercial banks primarily cater to the short-term requirements of industry, but have taken an interest in long-term financing in several ways (SM 2.17):

- (a) **Long-term loans** for expansion or setting up new units, repayable over a long period; their liquidity depends on the anticipated income of the borrowers.
- (b) A **working capital term loan (WCTL)**: funding of the portion of working capital that is always required (the minimum level) and is not affected by seasonal requirements.

Bridge finance refers to loans taken by a company, normally from commercial banks, for a short period because of **pending disbursement of loans sanctioned by financial institutions**. Though short-term, it is discussed with long-term sources because it is an important step in the facilitation of long-term loans: once term-lending institutions approve a loan, companies arrange short-term bank loans so as not to lose time in starting their projects (SM 2.17–2.18).



The bridge loan is normally secured by hypothecating movable assets, personal guarantees and demand promissory notes, and is repaid or adjusted out of the term loan as and when the institution disburses it. Its interest is generally higher than that on term loans.

Worked examples

Worked example **The samosa factory: shares, preference or debentures?**

Tapri Tea Co. Ltd needs ₹100 crore for its Dewas samosa factory, which will run for about fifteen years. It has 10 crore equity shares, of which the Mehra family holds 5.20 crore. Its cash flows have been stable for five years, its debt-equity ratio is low, and the tax rate is 25%. Dev has three offers:

(a) a public issue of equity shares at ₹250 each;

(b) 10.5% cumulative redeemable preference shares;

(c) 10% secured non-convertible debentures, redeemable after 7 years, rated by a credit rating agency.

Kesar's only conditions: "The family keeps control, and the taxman helps pay for it." Advise Dev, with reasons, on the most suitable source.

MODEL ANSWER

Relevant points. Equity is a permanent source with no obligation to pay dividends, but it is usually the costliest source and a new issue may reduce the EPS and the ownership and control of existing shareholders. Preference shares cause no dilution in EPS and no risk of takeover, since their holders have no voting rights unless dividends are in arrears, but their dividend is not tax-deductible, so they are costlier than debt. Debentures cost the least because interest is tax-deductible and do not dilute control, but their interest and principal are obligatory, they enhance financial risk, and repayment needs a large cash outflow at maturity.

Facts of the case.

Source	Net cost a year (₹ crore)	Family holding
Equity	Highest	50%
Preference	10.50	52%
Debentures	10.00 − 2.50 = **7.50**	52%

Working: the equity dividend is not obligatory, but equity carries the highest expected return. New shares = ₹100 crore ÷ ₹250 = 40 lakh; family holding = 5.20 crore ÷ 10.40 crore = 50%. Tax saved on interest = ₹10 crore × 25% = ₹2.50 crore.

Analysis.

1. *Control*: the equity issue takes the family from 52% to exactly 50%, no longer a majority. Preference shares and debentures leave it at 52%.
2. *Cost*: the debentures cost 10% × (1 − 0.25) = 7.5% after tax, against 10.5% for preference shares, whose dividend gets no tax shield.
3. *Risk*: stable cash flows and a low debt-equity ratio mean Tapri can carry the obligatory interest and plan for the large cash outflow at redemption in year 7.

Conclusion. Tapri should issue the **10% secured debentures**: they meet both conditions, being the cheapest source after tax without diluting control. If the board wanted to avoid an obligatory payment altogether, the preference shares would be the next choice; the equity issue fails the control condition.

Worked example **Dev's term sheets: name the instrument**

Dev's file holds five term sheets. Identify the instrument or source in each, with a reason.

(a) Tapri may redeem the bond after four years at ₹102 for every ₹100 of face value.

(b) The investor may sell the bond back to Tapri at par at the end of year 3.

(c) A ₹500 crore issue in London and Singapore, denominated in Indian rupees.

(d) A 9% dividend, arrears carried forward, converted into equity shares after three years, for the slow-to-mature tea plantation in Assam.

(e) A ₹30 crore loan from Tapri's bank for four months, because the term loan sanctioned by a financial institution has not yet been disbursed; secured by hypothecation of the delivery vans and Kesar's personal guarantee.

MODEL ANSWER

(a) Callable bond. The call option gives the issuer the right to redeem the bond before maturity at a predetermined call price, generally at a premium (here ₹102).

(b) Puttable bond. The put option gives the investor the right to sell the bond back to the company before maturity.

(c) Masala bond. A rupee-denominated bond issued outside India by an Indian corporate borrower to investors in overseas markets.

(d) Cumulative convertible preference shares (CCPs). They carry a cumulative dividend of a specified limit for a period, say three years, after which they convert into equity shares; they suit projects with a long gestation period.

(e) Bridge finance. A short-period loan, normally from a commercial bank, taken because the disbursement of a loan sanctioned by a financial institution is pending. It is normally secured by hypothecating movable assets, personal guarantees and demand promissory notes, carries a higher interest rate than the term loan, and is repaid or adjusted out of the term loan when disbursed.

Exam angle

In the exam

Where these sources appeared

In the ICAI documents read for this lesson, the long-term sources were examined 20 times in 17 documents, all as descriptive questions (2 to 4 marks where marks are printed).

- **Preference shares.** Any four types with their salient features: MTP May 2026 Series I Q4(b) (4 marks); features of the types: RTP September 2024 Q12(b); old-scheme MTP December 2021 Series II Q6(a). The suggested answers reproduce the six-row table of SM 2.8. A case: an instrument with fixed periodic returns, no voting rights, "characteristics of both equity and debt", ranking above equity and below debenture holders in liquidation: MTP September 2025 Series I Q4(c) (2 marks; answer: preference shares).
- **Debentures and bonds.** Two advantages and two disadvantages of raising finance by debentures: September 2024 Q4(a) (4 marks). Four features of debentures or bonds: MTP May 2025 Series II Q4(b) (4 marks). "Under which circumstances raising funds through the issue of debentures is advantageous": September 2026 Q4(a) (4 marks; suggested answers not yet published). Characteristics of debentures: old-scheme MTP July 2021 Series I Q6(b). "Financing a business through borrowing is cheaper than using equity. Briefly explain": old-scheme MTPs for May 2020 and November 2023, answered with the three advantages of debentures.
- **Callable and puttable bonds.** September 2025 Q4(b), together with masala and drop lock bonds (4 marks); MTP September 2026 Series II Q4(c) (2 marks); old-scheme MTPs December 2021 and May 2022.
- **Bridge finance.** Key features: May 2025 Q4(b) (4 marks), answered with exactly the four points of SM 2.17–2.18; MTP September 2026 Series II Q4(c) (2 marks); old-scheme December 2021 Q6(a), a short note with clean packing credit.
- **A firm that cannot issue shares.** Two long-term sources for a partnership firm: old-scheme MTP May 2022 Series II Q6(a), answered with loans from commercial banks (including the working capital term loan) and lease financing.
- **Matching features to sources.** RTP September 2025 Q10(a): "the most expensive source of funds" (equity), "entails a high degree of risk since they have to be repaid as per the terms of agreement" (debentures).

In the exam

The wording traps

- "Most expensive" is equity, even though it has no obligatory payment; "most risky to the company" is debt.
- "Hybrid" is preference capital, and ICAI gives the reasons: dividend not tax-deductible (like equity), fixed rate (like debt).
- A callable bond's option belongs to the **issuer**, a puttable bond's to the **investor**.
- Bridge finance is short-term but is taught with long-term sources: write why.

Trick

C-R-C for any source

For any "advantages and disadvantages" question, run the source through **Cost** (tax-deductible or not), **Risk** (obligatory payment and repayment, or not) and **Control** (voting dilution, or not), then add its own special points (rights issue for equity; covenants and rising prices for debentures). It recovers ICAI's lists for equity, preference and debentures; it does not replace them.

Answer frame

"Distinguish between preference shares and debentures" (SM 2.12)

Basis	Preference shares	Debentures
Ownership	Preference share capital is a special kind of share	A debenture is a type of loan which can be raised from the public
Payment of dividend / interest	Priority as regards a fixed dividend and repayment of capital on winding up	Carries a fixed percentage of interest
Nature	A hybrid form of financing, with some characteristics of equity and some of debt	An instrument for raising long-term capital with a fixed period of maturity

Add, for full marks, the points the rest of the chapter supports: preference dividend is not tax-deductible while debenture interest is; debenture interest is payable whether or not there are profits.

"Distinguish between equity shares and preference shares" (SM 2.9)

Basis	Equity share	Preference share
Dividend payment	Paid after preference dividend	Preferred over equity dividend
Rate of dividend	Fluctuating	Fixed
Convertibility	Not convertible	Convertible
Voting rights	Full voting rights	Very limited voting rights

"Discuss the advantages of equity shares" and its cousins

Write the heading, then ICAI's points in ICAI's order, one line of explanation each: four for equity (permanent; financial base; no legal obligation to pay dividends; rights issue), five for preference shares, three for debentures. For "disadvantages": three for equity, two for preference shares, four for debentures. A four-mark question wants four points; a two-mark question wants the definition and two points.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the factory on an overdraft

Funding plant, buildings or hard-core working capital from short-term borrowings because they are quick to arrange. **Fix:** the basic principle matches the period: long-term needs from long-term sources. An overdraft is repayable on demand.

⚠ **Common mistake**

"Retained earnings are free money"

They carry almost no risk and dilute no control, but they belong to the equity shareholders, who could have had them as dividends. **Fix:** the decision to plough back depends on the return the company generates against the expected cost of equity (SM 2.10).

⚠ **Common mistake**

Callable and puttable swapped

Writing that a callable bond lets the investor demand early repayment. **Fix:** call = the issuer redeems early (at the call price, generally a premium); put = the investor sells it back.

⚠ **Common mistake**

Preference dividend shown as a tax saving

Treating preference dividend like interest. **Fix:** preference dividend, like equity dividend, is not tax-deductible; only debt interest gives a tax shield.

Practice

Check yourself

1 Case scenario. Tapri Cold Chain Pvt Ltd, Tapri's cold-chain subsidiary, plans ₹40 crore of 9% cumulative redeemable preference shares for its refrigerated trucks. Its finance manager's slide makes four claims. Which one is correct?

- A The preference dividend will cut the tax bill, just as debenture interest would
- B If the dividend is skipped in a loss year, an equity dividend can still be paid next year without clearing the arrears
- C Preference shares can never be redeemed, so the money is permanent
- D The preference shareholders will have no vote, except where their dividend is in arrears

2 Which of the following is NOT a disadvantage of debenture financing listed by ICAI?

- **A** Interest and repayment of principal are obligatory payments
- **B** Protective covenants may be restrictive
- **C** It dilutes the control of existing shareholders
- **D** A large cash outflow is needed at maturity

3 Tapri issues 50,000 debentures of ₹1,000 each at 8% interest. The tax rate is 25%. What is the annual interest cost to Tapri after tax, in rupees?

Answer: _____

4 A bond gives its holder the right to sell it back to the company before maturity. It is a:

- **A** Callable bond
- **B** Puttable bond
- **C** Masala bond
- **D** Municipal bond

5 Which of these are features of bridge finance? Select all that apply.

- **A** It is taken because the disbursement of a loan sanctioned by a financial institution is pending
- **B** It is normally secured by hypothecation of movable assets, personal guarantees and demand promissory notes
- **C** Its rate of interest is generally lower than that on term loans
- **D** It is repaid or adjusted out of the term loan when disbursed

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Fund each need from a source of matching period: long-term (more than 5–10 years), medium-term (1–5 years), short-term (up to one year).
- Three classifications: basic sources (external, internal), maturity of payment, ownership and control (owner's fund, borrowed fund).
- Equity: permanent, owners, residual claim, costliest; advantages: permanent, financial base, no legal obligation to pay dividends, rights issue; disadvantages: risky to investors, EPS dilution, control dilution.
- Preference: priority in fixed dividend and capital; hybrid (dividend not tax-deductible, rate fixed); usually cumulative; six types; no EPS dilution, leverage, no takeover risk.
- Retained earnings: ploughed-back profits of equity shareholders; almost no risk, no dilution; retain if the company's return beats the cost of equity.

- Debentures: trust deed, 3–10 years, secured or unsecured, rated; cheapest (interest tax-deductible), no dilution, good in rising prices; but obligatory, covenants, financial risk, lump-sum repayment.
- Callable: issuer may redeem early at the call price; puttable: investor may sell back. Masala: rupee bond sold abroad.
- Bridge finance: short bank loan pending disbursement of a sanctioned term loan; secured by hypothecation, guarantees, promissory notes; costlier; repaid out of the term loan. (No fryers were sold.)

Answer key · Check yourself

1 **D The preference shareholders will have no vote, except where their dividend is in arrears**

No voting rights except where dividends are in arrears (SM 2.9). The dividend is not tax-deductible, so there is no tax shield; cumulative arrears must be paid before any equity dividend; and these shares are redeemable after a specified period.

2 **C It dilutes the control of existing shareholders**

Debentures do not dilute control: that is one of their advantages. The other three are on ICAI's list, with the enhanced financial risk.

3 **3000000**

Interest = 50,000 × ₹1,000 × 8% = ₹40,00,000. It is tax-deductible, so the cost after tax is ₹40,00,000 × (1 − 0.25) = ₹30,00,000.

4 **B Puttable bond**

The put option (the right to sell) belongs to the investor. A callable bond's option belongs to the issuer.

5 **A It is taken because the disbursement of a loan sanctioned by a financial institution is pending; B It is normally secured by hypothecation of movable assets, personal guarantees and demand promissory notes; D It is repaid or adjusted out of the term loan when disbursed**

Bridge finance generally costs more than the term loan it bridges; the other three are ICAI's features.

3.2 Venture capital, lease financing, sale and lease back and other sources

Venture capital and its four methods, debt securitisation through an SPV, operating and financial leases and the other lease types, and ICAI's thirteen other sources from seed capital to junk bonds.

In plain words

Three problems on Dev's desk

Three people want money from Dev on the same Monday, and none of them fits a share issue or a bank loan.

> **Kesar:** My nephew Rohan's KulhadKart presses clay cups by machine. No sales for two years, but huge potential. Give him a loan.
>
> **Dev:** No bank lends to two years of no sales, Kesar-ji. He needs someone who buys a stake, waits, and sits on his board to teach him.
>
> **Ritu:** Meanwhile the cold-chain trucks are obsolete every three years, the kulhad kiln lasts ten, and you want to sell the Indore head office but keep working in it.
>
> **Dev:** Rent the trucks, finance-lease the kiln, sell the office and lease it back. Sab adjust ho jayega, for once.

Rohan needs **venture capital**: money, usually equity, for a new and highly risky venture whose entrepreneur has the idea but lacks experience and funds. The venture capitalist also brings sales strategy, contacts and management expertise.

Ritu's three problems are all **leasing**: the owner of an asset (the **lessor**) lets a user (the **lessee**) use it for a period in return for rent. A short, cancellable lease where the owner keeps the risks of ownership, like the trucks, is an **operating lease**. A long lease for most of the asset's life, where the user carries the risks and the rent repays the owner's whole cost with a profit, like the kiln, is a **financial lease**: "a loan in disguise". Selling an asset and leasing it straight back, like the head office, is a **sale and lease back**: the seller gets cash and keeps the asset.

Meanwhile the finance company that lent Tapri's franchisees their van loans has run out of money to lend. It sells the loans to a separate entity that turns them into securities for investors: **debt securitisation**.

Think of it like

The car, three ways

Hire a self-drive car for a weekend: the rental company insures and services it, you can return it early, and it is rented to someone else on Monday. That is an operating lease. Take a car on a five-year lease where you pay for insurance and servicing and the payments add up to its price plus interest: a loan in disguise, a financial lease. Sell your own car to a dealer and rent it back to keep driving it: sale and lease back. And if the dealer borrows most of the price from a bank, which you pay directly, the lease is leveraged.

Key idea

Venture capital is long-term, mostly equity finance for new, high-risk, growth-oriented firms, with management support. A financial lease passes the risks and rewards of ownership to the lessee and is non-cancellable; an operating lease leaves them with the lessor and is cancellable. Securitisation pools illiquid loans into marketable securities through a special purpose vehicle.

How it works

1. Venture capital financing

Definition

Venture capital financing

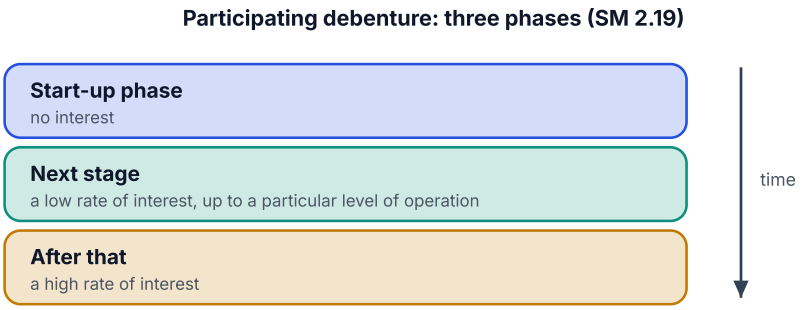
The financing of new high risky venture promoted by qualified entrepreneurs who lack experience and funds to give shape to their ideas. In broad sense, the venture capitalist makes investment to purchase equity or debt securities from inexperienced entrepreneurs who undertake highly risky ventures with potential to succeed in future. (SM 2.18)

Characteristics (SM 2.18):

1. It is basically an **equity finance in new companies**.
2. It is a **long-term investment in growth-oriented small/medium firms**.
3. Apart from funds, the investor provides **support in the form of sales strategy, business networking and management expertise**, enabling the growth of the entrepreneur.

Methods of venture capital financing (SM 2.19):

1. **Equity financing.** Venture capital undertakings need funds for a long period but may not give returns in the initial stages, so finance is generally provided as equity share capital. The equity contribution of the venture capital firm does not exceed 49% of the total equity capital, so that effective control and ownership remain with the entrepreneur.
2. **Conditional loan.** Repayable in the form of a **royalty** after the venture is able to generate sales; **no interest** is paid. In India royalty ranges between 2 and 15 per cent, depending on the gestation period, cash flow patterns, risk and other factors. Some financiers give the choice of a high rate of interest (which could be well above 20 per cent) instead of royalty, once the venture is commercially sound.
3. **Income note.** A hybrid security combining a conventional loan and a conditional loan: the entrepreneur pays **both interest and royalty on sales**, at substantially low rates. IDBI's VCF provides funding equal to 80–87.50% of the project's cost for commercial application of indigenous technology.
4. **Participating debenture.** Charges in three phases:



The charge grows as the venture grows, so the entrepreneur pays most when the business can afford it.

Where venture capital sits among a firm's stages (early stage: equity, venture capital, debt; growth stage: debt, venture capital, private equity) is in the long-term sources lesson (SM 2.3).

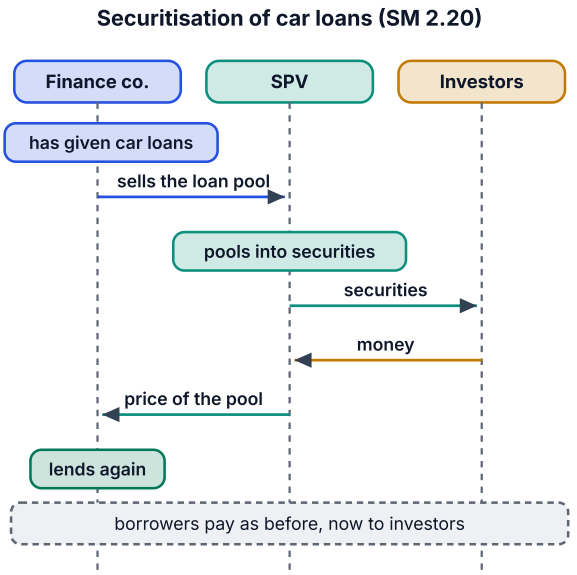
2. Debt securitisation

Definition

Securitisation

A process in which illiquid assets are pooled into marketable securities that can be sold to investors. The process leads to the creation of financial instruments that represent ownership interest in, or are secured by a segregated income producing asset or pool of assets. These assets are generally secured by personal or real property such as automobiles, real estate, or equipment loans but in some cases are unsecured. (SM 2.19–2.20)

ICAI's example (SM 2.20): a finance company has given many car loans and needs money to give more. Selling individual loans is not feasible without a liquid secondary market. Instead it **sells its existing car loans to a Special Purpose Vehicle (SPV)**, which pays the company (which lends again) and **pools the loans into marketable securities** issued to investors. The finance company raises funds and gets the loans **off its balance sheet**; investors get a liquid investment in a diversified pool of car loans. The original borrowers may not even be aware: they go on paying as before, but the payments now go to the new investors.



The car-loan borrowers may not even know: they keep paying their instalments, which now belong to the investors who bought the securities.

The process, as ICAI's suggested answers write it. For "What is debt securitisation? Explain the basics of debt securitisation process" (MTP May 2025 Series I Q4(b); MTP September 2026 Series I Q4(a)), the suggested answers call securitisation a method of **recycling of funds**, especially beneficial to financial intermediaries to support lending volumes (assets generating steady cash flows, such as housing finance, auto loans and credit card receivables, are packaged together and securities issued against the pool), and set out three functions:

1. **The origination function**: a borrower seeks a loan from a finance company or bank; the creditworthiness of the borrower is evaluated and a contract is entered into with a repayment schedule over the life of the loan.
2. **The pooling function**: similar loans or receivables are clubbed together to create an underlying pool of assets, transferred in favour of the **SPV**, which acts as a trustee for investors.
3. **The securitisation function**: the SPV structures and issues securities on the basis of the asset pool, carrying a coupon and an expected maturity; they are generally sold to investors (pension funds, mutual funds, insurance funds) through merchant bankers.

The process is generally **without recourse**: investors bear the credit risk, and the issuer is obliged to pay them only if the cash flows are received from the collateral. The originator's benefit is that assets are shifted off the balance sheet, giving it **off-balance sheet funding**. These three functions are not in the Study Material's text, but they are what ICAI's own answers give marks for.

3. Lease financing

Definition

Leasing

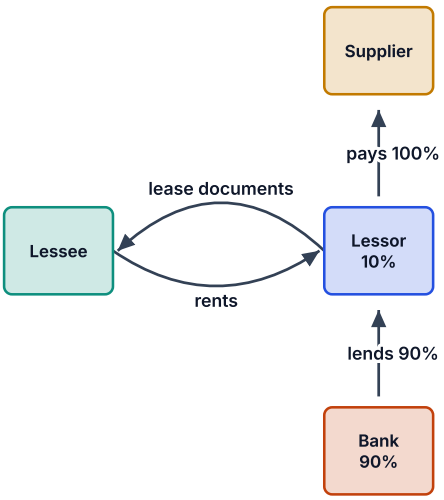
A general contract between the owner and user of the asset over a specified period of time. The asset is purchased initially by the lessor (leasing company) and thereafter leased to the user (lessee company) which pays a specified rent at periodical intervals. Leasing is an alternative to the purchase of an asset out of own or borrowed funds, and lease finance can be arranged much faster than term loans from financial institutions. (SM 2.20)

(a) Operating lease (SM 2.21):

- The lessor gives the lessee the **right to use** the asset; the **ownership right remains with the lessor**. The lessee pays fixed periodic rentals.
- The **lessor bears the insurance, maintenance and repair costs**.
- The lease period is **short**, so the lessor may not recover the cost in the initial lease and tends to lease the asset to **more than one lessee**. These are normally **callable and cancellable with proper notice**; the term is shorter than the asset's economic life.
- It attracts companies that **continually update or replace equipment** and want to return it at lease end and **avoid technological obsolescence**.

The Study Material pictures a lease in which the leasing company invests 10% and a bank funds 90% (SM 2.21):

The leasing arrangement in the Study Material (SM 2.21)



The lessor (the leasing company) invests 10% and pays the equipment supplier 100% of the price; the bank, the 90% funder, lends the rest under a principal and agency agreement; the lessee's rents amortise the bank's debt.

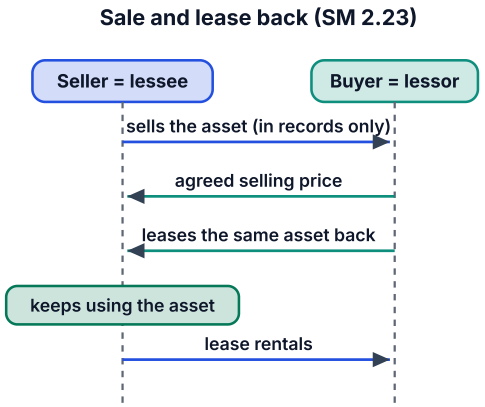
(b) Financial lease (SM 2.22): long-term and **non-cancellable**; the lease period is generally the **full economic life** of the asset, so it finances the use of equipment for the major part of its useful life. The lessee has the right to use the asset while the lessor retains **legal title**; the **lessee bears insurance, maintenance and other related costs**. It is also called a **capital lease**, which is nothing but a loan in

disguise: a contract involving payments over an obligatory period of specified sums sufficient in total to amortise the capital outlay of the lessor and give some profit.

4. Other types of leases

ICAI's "explain four other types of leases" questions want these four, in this order (SM 2.23–2.24):

1. **Sale and lease back.** The owner of an asset sells it to a party (the buyer), who in turn leases the same asset back to the owner for lease rentals. The asset is **not physically exchanged**; it all happens in records only. The main advantage: the lessee can satisfy himself completely regarding the quality of the asset, and after possession convert the sale into a lease agreement. The **seller assumes the role of lessee** and the **buyer the role of lessor**: the seller gets the agreed selling price, the buyer the lease rentals.
2. **Leveraged lease.** A **third party is involved besides the lessor and the lessee**: the lessor borrows part of the purchase cost (say 80%) from a lender, and the asset is held as security for the loan. The lender is paid off from the lease rentals **directly by the lessee**, and the surplus goes to the lessor. The lessor is entitled to claim the depreciation allowance.
3. **Sales-aid lease.** The lessor ties up with a **manufacturer** to market the manufacturer's product through its own leasing operations. In consideration of the aid in sales, the manufacturer may grant the lessor credit or a commission, so the lessor earns from both the lessee and the manufacturer.
4. **Close-ended and open-ended leases.** In a close-ended lease the asset is transferred to the lessor at the end of the lease, and the risk of obsolescence, residual value, etc. remain with the lessor as legal owner. In an open-ended lease the **lessee has the option of purchasing** the asset at the end of the lease period.



The asset never moves: the seller becomes the lessee and gets the price; the buyer becomes the lessor and gets the rentals.

Lease or buy, from the lessee's side (SM 2.24): leasing eliminates the immediate cash outflow, and lease rentals can be deducted in computing total income under the Income-tax Act; buying gives the depreciation allowance (including additional depreciation) and tax-deductible interest on borrowed capital. The two must be evaluated before deciding; practical lease problems are covered at the Final level.

5. Other sources of financing

The Study Material's thirteen "other sources" (SM 2.32–2.34), in its order:

Source	Key features
Seed capital assistance	IDBI scheme for professionally or technically qualified entrepreneurs, or persons with relevant experience, skills and entrepreneurial traits, who lack adequate financial resources (details below)
Internal cash accruals	Profit-making companies investing part of accumulated reserves or cash profits in capital assets for expansion or diversification: the surplus from operations after all contractual, statutory and working requirements
Unsecured loans	From promoters to meet the promoters' contribution norm; subordinate to institutional loans (details below)
Deferred payment guarantee	The machinery supplier allows payment over time, against a bank guarantee by the buyer; the entire cost is financed; no moratorium, so advisable only for an existing profit-making company
Capital incentives	Backward area development incentives: a lump-sum subsidy and exemption from or deferment of sales tax and octroi (details below)
Deep discount bonds	A form of zero-interest bond, sold at a discount, face value paid on maturity; no interest payout during the lock-in period
Secured premium notes	Issued with a detachable warrant; redeemable after a notified period of, say, 4 to 7 years; the warrant is converted into equity within the period notified
Zero interest fully convertible debentures	No interest; compulsorily and automatically converted into equity at a predetermined price after a specified period; if the market price is very high, investors get shares at a lower rate
Zero coupon bonds	No interest; sold at a discount; the difference between the discounted value and the face value is the investor's interest
Option bonds	Cumulative and non-cumulative bonds, interest payable on maturity or periodically; redemption premium offered to attract investors
Inflation bonds	Interest adjusted for inflation: at 11% interest and 5% inflation, the investor earns 16%
Floating rate bonds	Interest floats with market conditions; an ideal instrument for issuers to hedge interest-rate volatility; popular as a money market instrument (IDBI, ICICI)
High yield (junk) bonds	Rated below investment grade; low credit rating and high risk of default, so investors get higher interest: hence "high-yield"

The details ICAI's answers quote (SM 2.32–2.33):

- **Seed capital assistance**: all projects eligible for IDBI assistance, directly or indirectly through refinance, are eligible. It is **interest free** but carries a **service charge of one per cent per annum for the first five years** and at an increasing rate thereafter; IDBI may charge interest if the company's financial position and profitability permit. The repayment schedule depends on the unit's repaying capacity, with an **initial moratorium up to five years**.
- **Unsecured loans**: the interest rate should be less than or equal to that on institutional loans; interest can be paid only after institutional dues; they cannot be repaid without the prior approval of the financial institutions; they are **considered part of the equity** for the debt-equity ratio.

- **Capital incentives**: the quantum depends on the degree of backwardness of the location. The special capital incentive, a lump-sum subsidy sanctioned as a percentage of the fixed capital investment subject to a ceiling, forms part of the long-term means of finance, but the project's viability must not depend on incentives: institutions appraise the project per se. Incentives are released only after compliance with the scheme, in initial effective steps and final effective steps.

Worked examples

Worked example **Three leases at Tapri**

Tapri has three lease offers. Identify the type of lease in each, with reasons, and state who bears the risk of obsolescence.

(a) Billing tablets for 400 kiosks for eighteen months from TabRent Services, cancellable at one month's notice. TabRent insures and repairs them and rents the returned tablets to other retailers; a tablet lasts five years.

(b) A premix blending plant for eleven years of its twelve-year life from Narmada Leasing Ltd, non-cancellable by either party. Tapri insures and maintains it, and the rentals recover Narmada's full cost with a profit.

(c) Espresso machines for the new café chain from Kaveri Leasing, which has a tie-up with the machine maker: the maker pays Kaveri a commission on every machine it places on lease.

MODEL ANSWER

(a) Operating lease. The lease is short (eighteen months of a five-year life) and cancellable with notice; the lessor bears insurance and repairs and leases the same tablets to other users (a non-payout lease). The risk incident to ownership, including **obsolescence, stays with the lessor**, TabRent Services.

(b) Financial lease. The lease covers the major part (eleven of twelve years) of the plant's economic life and is non-cancellable; the lessee bears insurance and maintenance; the rentals amortise the lessor's capital outlay and give a profit (a full payout lease). The lessor is only a financier and the legal owner. The risks and rewards incident to ownership, including **obsolescence, pass to the lessee**, Tapri.

(c) Sales-aid lease. The lessor has a tie-up with the manufacturer to market its machines through its leasing operations, and earns both from the lessee (rentals) and from the manufacturer (commission). The risk of obsolescence depends on whether the lease is operating or financial in its terms, which the facts do not state.

Worked example **The packaging line that changed owners but not floors**

Tapri bought an automated packaging line for its Dewas samosa factory last year for ₹30 crore. Short of funds for its café expansion, it has an offer from Dhanvantari Leasing Ltd: Dhanvantari will buy the line for ₹28 crore and lease it back to Tapri for six years. Dhanvantari will borrow 75% of the price from Shivneri Bank against the line; Tapri will pay its rentals directly to the bank, and the surplus will go to Dhanvantari. At the end of the six years Tapri may buy the line at a price agreed today.

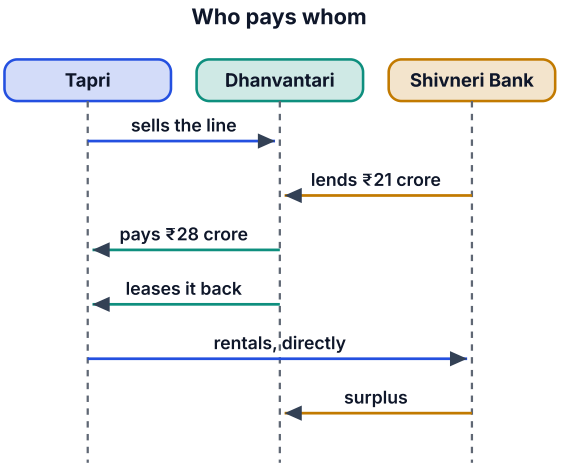
Required:

(i) Explain the arrangement, the role each party assumes and its main advantage as the Study Material states it.

(ii) Identify the second type of lease involved, with the amount Dhanvantari borrows, and state who is entitled to the depreciation allowance.

(iii) State, with reasons, whether the lease is close-ended or open-ended.

MODEL ANSWER



Sale and lease back between Tapri and Dhanvantari; the bank in the middle makes it leveraged.

(i) Sale and lease back. Under this type of lease, the owner of an asset sells the asset to a party (the buyer), who in turn leases back the same asset to the owner in consideration of lease rentals. The asset is not physically exchanged; it all happens in records only, so the line never leaves the Dewas floor. Tapri, the seller, assumes the role of **lessee**; Dhanvantari, the buyer, the role of **lessor**. Tapri gets the agreed selling price of ₹28 crore; Dhanvantari gets the lease rentals. The main advantage stated in the Study Material (SM 2.23): the lessee can satisfy himself completely regarding the quality of the asset and, after possession, convert the sale into a lease agreement. Tapri has run this line for a year.

(ii) Leveraged lease. A third party, Shivneri Bank, is involved besides the lessor and the lessee. The lessor borrows part of the purchase cost (75% × ₹28 crore = **₹21 crore**) from the lender, and the asset is held as security against the loan. The lender is paid off from the lease rentals directly by the lessee, and the surplus after meeting the lender's claims goes to the lessor. **The lessor, Dhanvantari,** is entitled to claim the depreciation allowance.

(iii) Open-ended lease. In an open-ended lease the lessee has the option of purchasing the asset at the end of the lease period, as Tapri has here. In a close-ended lease the asset gets transferred to the lessor at the end of the lease, and the risk of obsolescence, residual value, etc. remain with the lessor as the legal owner.

Worked example **Narmada Leasing prices the vending machines**

Narmada Leasing Ltd buys 50 chai vending machines for ₹15,00,000 to lease to Tapri for five years, the machines' useful life. Narmada expects to sell them for ₹1,00,000 at the end of year 5. It claims depreciation

on the straight line method on cost less this salvage value, so no tax arises on the sale. Lease rentals are taxable in Narmada's hands; the tax rate is 30% and its cost of capital is 11%. Rentals are received at the end of each year. Use PV factors at 11% of 0.901, 0.812, 0.731, 0.659 and 0.593.

Required:

(i) Calculate the present value of the tax savings on depreciation and of the salvage value, and the part of the investment to be recovered from lease rentals.

(ii) Calculate the minimum equal annual lease rental.

(iii) A rival lessor has quoted Tapri ₹4,25,000 a year. Calculate Narmada's net present value at that rental and advise whether it should match the quote.

THE WORKING

Working notes

1. Depreciation = (₹15,00,000 − ₹1,00,000) ÷ 5 = ₹2,80,000 a year; tax saving = ₹2,80,000 × 30% = ₹84,000 a year.
2. PVAF for 5 years at 11% = 0.901 + 0.812 + 0.731 + 0.659 + 0.593 = 3.696.

(i) Investment to be recovered from rentals

Particulars	Amount (₹)
Cost of the machines	15,00,000
Less: PV of tax savings on depreciation (₹84,000 × 3.696)	3,10,464
Less: PV of salvage value (₹1,00,000 × 0.593)	59,300
Investment to be recovered from lease rentals	**11,30,236**

(ii) Equal rentals. Let the annual rental be X. PV of after-tax rentals = X × (1 − 0.30) × 3.696 = 2.5872X.

$$X = \frac{11{,}30{,}236}{2.5872} = ₹4{,}36{,}857$$

Check: ₹4,36,857 × 0.70 = ₹3,05,800 a year after tax, and ₹3,05,800 × 3.696 = ₹11,30,237, the amount to be recovered (the ₹1 is rounding).

(iii) At ₹4,25,000 a year

Particulars	Amount (₹)
PV of after-tax rentals (₹4,25,000 × 0.70 × 3.696)	10,99,560
Less: Investment to be recovered from lease rentals	11,30,236
Net present value	**(30,676)**

> **Advice.** The NPV is negative: at ₹4,25,000 Narmada would not earn its 11% cost of capital. It should not match the quote: ₹4,36,857 a year is its minimum.

Exam angle

In the exam

Where these sources appeared

In the ICAI documents read for this lesson, these topics were examined 29 times in 27 documents, as descriptive questions of 2 to 4 marks where marks are printed, and once as a numerical.

- **Other types of leases** (sale and lease back, leveraged, sales-aid, close-ended and open-ended): four times in two years, 4 marks each: MTP September 2024 Series II Q4(b), MTP May 2025 Series II Q4(a), MTP January 2026 Series II Q4(a) (a case: a supermarket chain's CFO "besides the common finance lease and operating lease"), MTP September 2026 Series I Q4(b). Leveraged lease alone: September 2024 Q4(c) (2 marks). Sale and leaseback benefits and risks for a company with a cash crunch: MTP September 2025 Series II Q4(c) (2 marks).
- **Financial lease versus operating lease**: old-scheme RTPs for November 2020 (Q10(b)) and November 2022 (Q10(a)), answered with the five-point table.
- **Lease rentals as a numerical**: RTP May 2026 Q8 (a lessor's minimum rentals, equal and in a usage ratio, under two depreciation methods).
- **Venture capital**: methods, MTP January 2026 Series I Q4(b) and MTP May 2024 Series I Q4(b) (4 marks each), RTP May 2025 Q10(b), the old-scheme November 2020 paper (4 marks) and RTP December 2021 Q10(b); meaning and characteristics, MTP September 2025 Series I Q2(b) (2 marks) and old-scheme MTP November 2022 Series I Q6(b) (4 marks).
- **Debt securitisation** ("What is debt securitisation? Explain the basics of the process"): MTP May 2025 Series I Q4(b), MTP September 2026 Series I Q4(a) (4 marks each), and old-scheme MTPs for November 2019, July 2021 and May 2023 and RTPs for May 2020 and May 2023. The suggested answers use the origination, pooling and securitisation functions.
- **Other sources**: seed capital assistance, January 2026 Q4(a) (4 marks) and old-scheme MTP November 2022 Series II (2 marks); junk bonds and why they are "high yield", May 2026 Q4(c) (2 marks); secured premium notes, the old-scheme May 2023 paper (2 marks) and old-scheme MTPs for July 2021 and May 2023.

In the exam

The wording traps

- "Other types of leases" means the four of SM 2.23–2.24, not operating and financial leases again.
- A leveraged lease has **three** parties; the lender is paid **directly by the lessee**.
- Venture capital equity "does not exceed 49%": the point is control stays with the entrepreneur.
- A conditional loan pays **royalty, no interest**; an income note pays **both**, at low rates.

Trick

Three lease cousins by who else is in the room

Sale and lease back: the lessee used to own it. Leveraged: a lender is in the room. Sales-aid: a manufacturer is in the room. Close or open-ended: the question is who gets the asset at the end. Condition: this sorts the four other types only; operating versus financial is the five-point test.

Answer frame

"Compare financial lease and operating lease" (SM 2.22–2.23)

Point	Financial lease	Operating lease
Risk and reward	Risk and reward incident to ownership pass to the lessee; the lessor remains only the legal owner	The lessee only gets the use of the asset for a certain time; risk incident to ownership belongs wholly to the lessor
Obsolescence	Lessee bears the risk	Lessor bears the risk
Cancellation	Non-cancellable by either party: the lessor wants his principal back with interest	Kept cancellable by the lessor, who can lease the asset to other willing lessees
Repairs, maintenance	The lessor is only a financier and bears none of these costs	Usually the lessor bears repairs, maintenance or operations
Payout	Usually full payout: one lease repays the cost with interest	Usually non-payout: the lessor leases the same asset again and again

"Explain the methods of venture capital financing"

Four headings in ICAI's order (equity financing, conditional loan, income note, participating debenture), each with its one number: 49% for equity, 2–15% royalty for the conditional loan, 80–87.50% of project cost for IDBI's VCF under the income note, three phases for the participating debenture.

"What is debt securitisation? Explain the process"

Definition (illiquid assets pooled into marketable securities) → the three functions (origination, pooling with transfer to the SPV, securitisation) → without recourse → off-balance sheet funding for the originator. The car-loan example is a good two-line illustration.

"What benefits and risks would you weigh before a sale and leaseback?"

ICAI's suggested answer to MTP September 2025 Series II Q4(c) (2 marks) gives three benefits and two risks, then a conclusion:

- *Benefits*: (i) immediate cash inflow to ease the cash crunch; (ii) continued use of the asset without ownership, avoiding disruption; (iii) off-balance sheet financing improves capital structure ratios.
- *Risks*: (i) loss of ownership may mean higher long-term costs; (ii) lease obligations add to fixed costs and reduce flexibility.
- *Conclusion*: apply it only if the lease terms are favourable and the company lacks better financing options.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the kiln that was "only rented"

Calling a nine-year, non-cancellable kiln lease an operating lease because "Tapri does not own it". **Fix:** ownership stays with the lessor in every lease. Classify by who bears the risks and rewards (here, obsolescence and maintenance), the period against the economic life, cancellability and whether the rentals pay out the full cost.

⚠ **Common mistake**

The venture capitalist who took control

Writing that the venture capital firm takes a majority. **Fix:** its equity contribution does not exceed 49% of the total equity, so that effective control and ownership remain with the entrepreneur.

⚠ **Common mistake**

Securitisation answered with only a definition

Four marks, one definition. **Fix:** the suggested answers give the origination, pooling and securitisation functions, the SPV as trustee, "without recourse" and off-balance sheet funding: write all of them.

⚠ **Common mistake**

Equal rentals divided by the pre-tax annuity

Dividing ₹11,30,236 by 3.696 gives ₹3,05,800, a rental that ignores the tax the lessor pays on it. **Fix:** rentals are taxable, so divide by (1 – tax rate) × PVAF = 0.70 × 3.696 = 2.5872.

Practice

Check yourself

1 Case scenario. Rohan's KulhadKart will have no sales for two years. A venture capital fund offers money with no interest, repayable as a percentage of sales once sales begin, with the option later of a high rate of interest instead once the business is commercially sound. Which method of venture capital financing is this?

- A Equity financing
- B Conditional loan
- C Income note
- D Participating debenture

2 KulhadKart's total equity capital after the investment will be ₹8 crore. Under ICAI's description of equity financing, what is the maximum equity contribution of the venture capital firm, in ₹ crore?

Answer: _____

3 Which of the following is true of a financial lease?

- A The lessor bears the risk of obsolescence
- B It is usually cancellable by the lessee at short notice
- C It is usually a full payout lease, the rentals amortising the lessor's cost with interest
- D The lessor usually bears repairs and maintenance

4 In debt securitisation, which of these is correct?

- A The original borrowers must sign a new loan agreement with the investors
- B The finance company sells the loans to a special purpose vehicle, which issues marketable securities against the pool
- C The finance company keeps the loans on its balance sheet and borrows against them
- D It is generally with recourse to the originator for every default

5 Which of these are features of seed capital assistance? Select all that apply.

- A It is designed by IDBI for qualified entrepreneurs who lack adequate financial resources
- B It is interest free but carries a service charge of one per cent per annum for the first five years
- C It must be repaid within the first five years, with no moratorium
- D The repayment schedule depends on the repaying capacity of the unit

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Venture capital: equity finance in new, high-risk, growth-oriented small and medium firms, with sales, networking and management support.
- Methods: equity (at most 49% of total equity), conditional loan (royalty 2–15%, no interest), income note (interest and royalty, low rates), participating debenture (nil, low, then high interest).
- Securitisation: illiquid loans pooled into marketable securities through an SPV; origination, pooling, securitisation; without recourse; off-balance sheet funding.
- Operating lease: short, cancellable, lessor bears maintenance and obsolescence, non-payout. Financial lease: most of the economic life, non-cancellable, lessee bears the risks, full payout, a loan in disguise.
- Other leases: sale and lease back (seller becomes lessee), leveraged (lender paid directly by lessee), sales-aid (manufacturer pays the lessor), close-ended (asset back to lessor) and open-ended (lessee may buy).
- Seed capital assistance: IDBI, interest free, 1% service charge for five years, moratorium up to five years. Deferred payment guarantee: no moratorium.

- Bonds: deep discount and zero coupon sold at a discount; SPN with detachable warrant (4–7 years); inflation bonds add inflation to interest; junk bonds below investment grade, so high yield.
- Lessor's minimum rental: (cost − PV of depreciation tax shields − PV of salvage) ÷ [(1 − t) × PVAF]. (The kiln stays with Tapri; so does its obsolescence.)

Answer key · Check yourself

1 **B Conditional loan**

A conditional loan is repayable as a royalty after the venture generates sales, with no interest; some financiers offer a high interest rate instead of royalty once the venture is sound. An income note charges both interest and royalty; a participating debenture charges interest in three phases.

2 **3.92**

The venture capital firm's equity contribution does not exceed 49% of the total equity capital: 49% × ₹8 crore = ₹3.92 crore, so that control stays with the entrepreneur.

3 **C It is usually a full payout lease, the rentals amortising the lessor's cost with interest**

In a financial lease the lessee bears obsolescence and maintenance, the lease is non-cancellable, and one lease repays the cost with interest. The other three describe an operating lease.

4 **B The finance company sells the loans to a special purpose vehicle, which issues marketable securities against the pool**

The loans are sold to the SPV, which pools them into securities; the borrowers may not even be aware and go on paying as before. The process is generally without recourse, and the originator gets the assets off its balance sheet.

5 **A It is designed by IDBI for qualified entrepreneurs who lack adequate financial resources**; **B It is interest free but carries a service charge of one per cent per annum for the first five years**; **D The repayment schedule depends on the repaying capacity of the unit**

Seed capital assistance has an initial moratorium of up to five years. No moratorium is the feature of the deferred payment guarantee, which is why that suits only an existing profit-making company.

3.3 Short-term sources of finance

ICAI's ten short-term sources, from trade credit and customer advances to commercial paper, bank advances, packing credit and post-shipment export finance, inter-corporate deposits and public deposits.

In plain words

Diwali in ninety days

Every year Tapri's Diwali season needs four times the usual stock of tea, sugar, kulhads and gift boxes, for about three months.

> **Kesar:** Simple. We issue shares for the Diwali stock.
>
> **Ritu:** Shares are forever, Kesar-ji. The stock is gone by Bhai Dooj.
>
> **Dev:** And half of it is already financed. The tea estate gives us sixty days, the corporate clients pay half when they order their hampers, and the staff are paid on the seventh of next month.
>
> **Kesar:** So the money is free?
>
> **Ritu:** Some of it. Per unit, please, before you celebrate.

A short-term need is one that arises for a period **not exceeding one year**: stock, debtors, cash. By the matching rule of long-term sources, it should be met from **short-term sources**.

Some of them arise on their own as the business runs, with no negotiation: the supplier's credit (**trade credit**), wages and taxes not yet paid (**accrued expenses**), money received for goods not yet delivered (**deferred income**), and **advances from customers**. Others must be arranged: a large, highly rated company can borrow from investors by selling **commercial paper**; banks give **loans, overdrafts and cash credits**; exporters get **packing credit** before they ship and **post-shipment finance** after; a sister company with spare cash can lend through an **inter-corporate deposit**; and the public can lend through **public deposits**.

Think of it like

The sweet shop before Diwali

A sweet shop before Diwali buys khoya and sugar on a month's udhaar from the wholesaler (trade credit), takes advance payment for the big office orders (customer advances), pays its karigars at month end (accrued wages), and draws on a bank limit against the stock in its godown, paying interest only on what it uses (cash credit). By Bhai Dooj the stock is sold, the udhaar paid, the limit back to zero. Nobody sold a share of the shop.

Key idea

Short-term needs are met from short-term sources. Trade credit (without any explicit cost), accrued expenses and deferred income arise spontaneously, and customer advances are cost free; the rest (commercial paper, bank advances, export credit, inter-corporate and public deposits) must be arranged and carry interest.

How it works

1. The spontaneous sources

The Study Material lists ten short-term sources (SM 2.24–2.32). The first three arise in the normal course of business.

(i) Trade credit (SM 2.24) is credit granted by suppliers of goods, etc., as an incident of sale.

- Its usual duration is **15 to 90 days**.
- It **generates automatically** in the course of business and is common to almost all business operations.
- It can be in the form of an **open account** or **bills payable**.
- It is preferred because it is **without any explicit cost**, and as long as the business is a going concern it keeps on rotating.
- It **enhances automatically with the increase in the volume of business**.

(ii) Accrued expenses and deferred (unearned) income (SM 2.24). Accrued expenses are liabilities for services already received, like wages, taxes, interest and dividends; they arise from day-to-day activities and are a **spontaneous source of finance**. Deferred income is money received for goods and services to be provided in the future; it increases liquidity, so it too is a spontaneous source.

(iii) Advances from customers (SM 2.25). Manufacturers and contractors producing costly goods with a long manufacturing or construction time usually demand advance money when they accept the order. It is a **cost free source of finance** and really useful.

> **Note**
>
> **Spontaneous and negotiated sources**
>
> The working capital chapter sorts working capital finance into **spontaneous sources** (which naturally arise in the course of business operations: trade credit, credit from employees, credit from suppliers of services) and **negotiated sources** (which have to be specifically negotiated with lenders: commercial banks, financial institutions, the general public). Among its sources it adds bills payable (spontaneous), and bill discounting and the bill rediscounting scheme (SM 9.102–9.105): see financing of working capital. The cost of availing trade credit, and of not taking it, is in management of payables.

2. The money market instruments

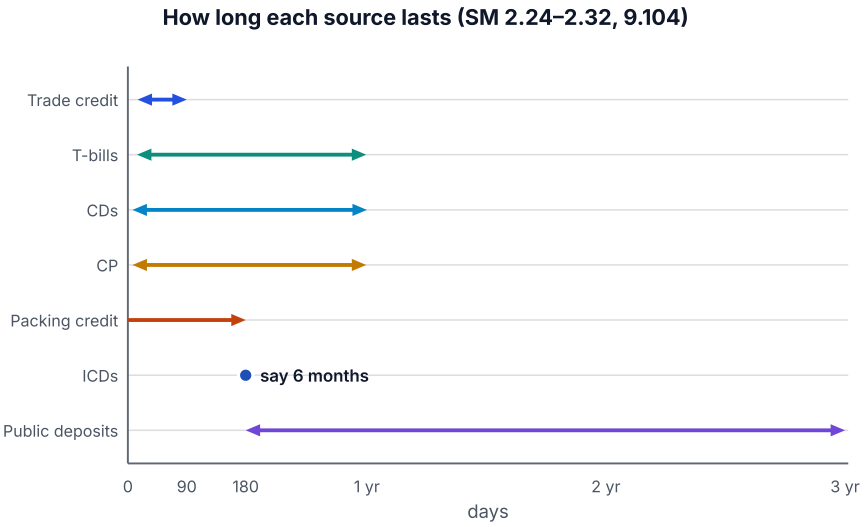
(iv) Commercial paper (CP) (SM 2.25):

- An **unsecured** money market instrument issued in the form of a **promissory note**.
- Introduced by the RBI in **1989**, to enable **highly rated corporate borrowers** to diversify their sources of short-term borrowings and to provide an additional instrument to investors. Besides corporates, **primary dealers and all-India financial institutions** may also issue it.
- Issued in denominations of **₹5 lakh or multiples thereof**; the interest rate is generally linked to the **yield on the one-year government bond**.
- Every eligible issuer must get a **credit rating** from CRISIL, ICRA, CARE, FITCH Ratings India or any other agency specified by the RBI.

The working capital chapter adds that CP matures in a minimum of 7 days to less than one year from the date of issue, and gives its advantages and limitations (SM 9.104–9.105; financing of working capital).

(v) Treasury bills (T-bills) (SM 2.25): a class of **Central Government securities**, issued by the Government of India to meet its short-term borrowing requirements, with maturities ranging **between 14 and 364 days**.

(vi) Certificates of deposit (CD) (SM 2.25): a document of title similar to a time deposit receipt, **issued by a bank**, with a fixed maturity of **not less than 7 days up to a maximum of one year**, except that there is no prescribed interest rate. Its main advantage: the banker is not required to encash the deposit before maturity, and the investor is assured of liquidity because he can **sell the CD in the secondary market**.



Trade credit 15–90 days; T-bills 14–364 days; CDs 7 days to one year; CP 7 days to less than one year (SM 9.104); packing credit liquidated within 180 days of commencement; inter-corporate deposits for a short period, say 6 months; public deposits 6 months to 3 years, which makes them short- and medium-term finance.

3. Bank advances

Banks lend the deposits they receive so that they can be called back when required; lending yields gross revenue out of which interest on deposits and costs are met and a reasonable profit is made. A bank's lending policy is not merely profit motivated but also keeps in mind the **socio-economic development** of the country (SM 2.26). The facilities (SM 2.26–2.28):

1. **Short-term loans.** The entire advance is **disbursed at one time**, in cash or by transfer to the borrower's current account. It is a single advance against securities like shares, government securities, life insurance policies and fixed deposit receipts. Except for interest and charges, no further adjustments are made; repayment is in full or by an agreed schedule.
2. **Overdraft.** The customer may **withdraw in excess of the credit balance in the current account**, up to a fixed limit. Though **repayable on demand**, overdrafts generally continue for long periods through annual renewals of the limits. Interest is charged on **daily balances**; cheque books are provided.

3. **Clean overdrafts.** Granted only to parties which are financially sound and reputed for integrity: the bank relies on the **personal security of the borrower**, with no tangible security. Banks look at the turnover in the account, satisfactory dealings over a considerable period and market reputation, take guarantees from other creditworthy persons, and grant it for a short period only.
4. **Cash credits.** An advance up to a certain limit; the customer **draws only to the extent of his requirements** and deposits surplus funds in the account, and **interest is charged only on the amount actually availed**. Limits are generally sanctioned against tradable goods by **pledge or hypothecation**; though repayable on demand, banks usually do not recall them unless compelled by adverse factors.
5. **Advances against goods.** A reliable source of repayment: goods are in constant demand and turn over quickly, so advances against them are safe and liquid. Goods (agricultural commodities, industrial raw materials, partly finished goods: all forms of movables) are charged to the bank by pledge or hypothecation.
6. **Bills purchased/discounted.** Advances against the security of clean or documentary bills, for approved customers with sanctioned limits, after the banker is satisfied about the creditworthiness of the drawer. Despite the name, the bank holds the bills **only as security**; besides its rights against the parties liable on the bills, it can exercise a pledgee's rights over the goods covered by the documents.

The working capital chapter's forms of bank credit (line of credit, letter of credit, bank guarantees and more) are in financing of working capital.

4. Export finance: before and after the ship sails

Commercial banks provide short-term export finance mainly as **pre-shipment** and **post-shipment** credit, in rupees as well as in foreign currency (SM 2.28). The RBI has eased the flow of export credit through rationalised and liberalised export credit interest rates, flexibility in repayment and prepayment of pre-shipment credit, a special package for large value exporters, export finance for agricultural exports and the Gold Card Scheme, and has allowed banks to source funds from abroad without limit for export credit in foreign currency.

Pre-shipment finance: packing credit (SM 2.28–2.30). Packing credit is an advance by a bank to an exporter for **buying, manufacturing, processing, packing and shipping** goods to overseas buyers. An exporter with a **firm export order** or an **irrevocable letter of credit** in his favour can apply. It must be **liquidated within 180 days from the date of its commencement**, by negotiation of export bills or receipt of export proceeds: it is essentially a short-term advance. Banks normally insist on irrevocable letters of credit, which evidence the arrangement for realising the export proceeds and indicate the finance needed; for customers of long standing, firm contracts may do.

Types of packing credit (SM 2.29–2.30):

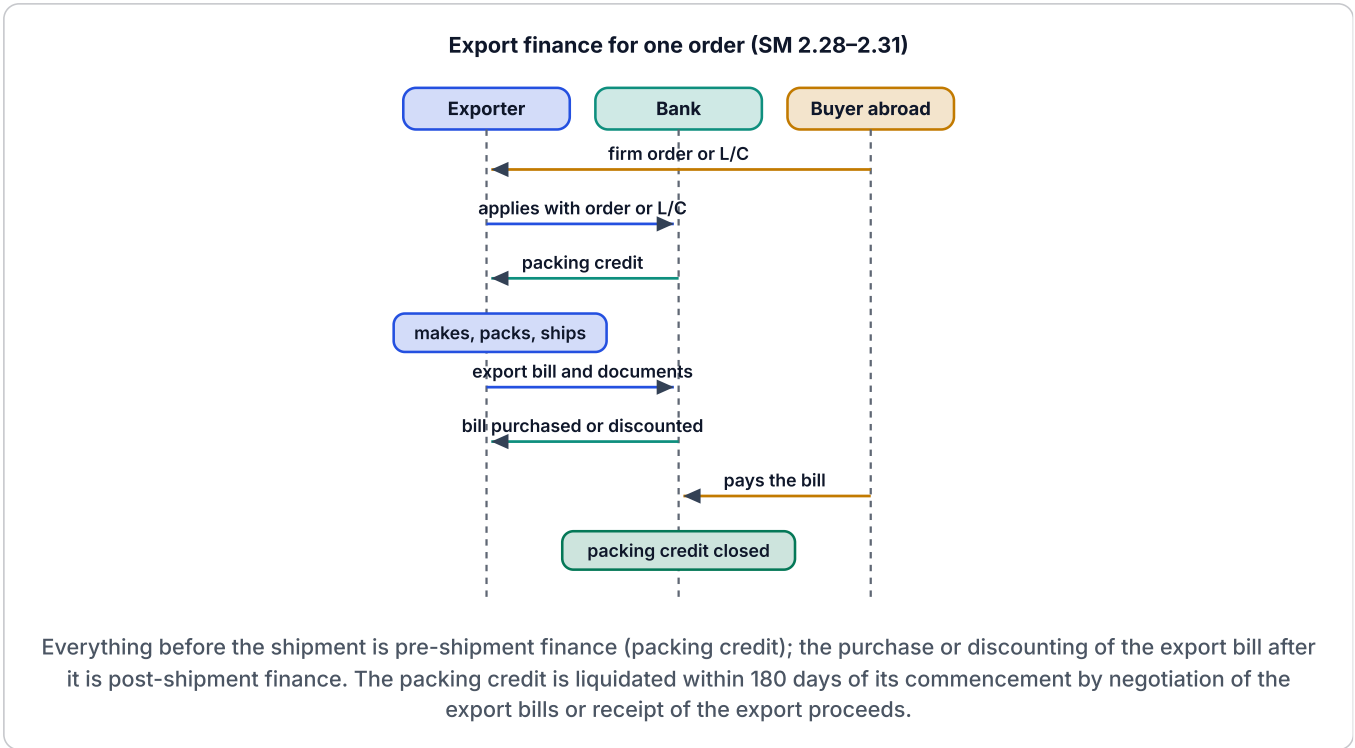
- (a) **Clean packing credit**: made available only on production of a firm export order or a letter of credit, **without any charge or control over raw material or finished goods**. A suitable margin is kept, and the bank obtains **ECGC cover**.
- (b) **Packing credit against hypothecation of goods**: the goods are hypothecated to the bank with a stipulated margin; the exporter submits stock statements with the order or letter of credit, and

then every fortnight and whenever stocks move.

- (c) **Packing credit against pledge of goods**: the exportable finished goods are pledged to the bank with approved clearing agents, who ship them as required; **possession lies with the bank**, under its lock and key.
- (d) **ECGC guarantee**: any loan to an exporter for the manufacture, processing, purchasing or packing of goods for export against a firm order qualifies for the packing credit guarantee of the Export Credit Guarantee Corporation.
- (e) **Forward exchange contract**: if the export bill is to be drawn in a foreign currency, the exporter should enter into a forward exchange contract with the bank, avoiding the risk of a change in the exchange rate.

Post-shipment finance takes these forms (SM 2.30–2.31):

- (a) **Purchase/discounting of documentary export bills**: export bills payable at sight are purchased, and usance export bills discounted, covering confirmed sales and backed by documents including documents of title such as the bill of lading, post parcel receipts or air consignment notes.
- (b) **ECGC guarantee**: post-shipment finance through purchase, negotiation or discount of an export bill qualifies for the post-shipment export credit guarantee, provided the exporter holds ECGC's shipments (comprehensive risks) policy covering both **political and commercial risks**. ECGC fixes a credit limit for each exporter, and its liability is limited to that limit, irrespective of the amount of the policy.
- (c) **Advance against export bills sent for collection**, considering the creditworthiness of the party, the nature of goods, usance and the standing of the drawee.
- (d) **Advance against duty drawbacks, cash subsidy, etc.** receivable against export performance. These are **clean** advances, so precaution is needed; the bank sees that the export bills pass through it, so that it can verify the claims. Such an advance must be liquidated **within 180 days from the date of shipment**.



Everything before the shipment is pre-shipment finance (packing credit); the purchase or discounting of the export bill after it is post-shipment finance. The packing credit is liquidated within 180 days of its commencement by negotiation of the export bills or receipt of the export proceeds.

Banks also (SM 2.31): (i) open letters of credit for approved exporters on their overseas or up-country suppliers; (ii) issue guarantees for waiver of excise duty, due performance of contracts, bonds in lieu of cash security deposits and advance payments; (iii) finance exports on deferred payment terms; (iv) obtain status reports on buyers and trade information through correspondents; and (v) provide economic intelligence on various countries.

5. Inter-corporate deposits and public deposits

(ix) Inter-corporate deposits (ICDs) (SM 2.31): companies can borrow for a short period, **say 6 months**, from other companies which have **surplus liquidity**; the rate varies with the amount and the period. The working capital chapter adds that the rate is higher than the bank rate, depends on the financial soundness of the borrower, and that the source reduces dependence on bank finance (SM 9.104).

(x) Public deposits (SM 2.31–2.32): a very important source of **short-term and medium-term** finance, particularly due to the credit squeeze by the RBI.

- A company can accept public deposits, subject to RBI stipulations from time to time, up to a maximum of 35 per cent of its paid up capital and reserves.
- They may be accepted for a period of **six months to three years**.
- They are **unsecured loans**, and should **not be used for acquiring fixed assets**, since they are repayable within three years; they mainly finance working capital.

⚠ **Careful**

Version-sensitive limits

The 35% limit and the six-month to three-year period are as the Study Material states them (SM 2.31–2.32). Rules on deposits change: verify against the Study Material and Statutory Update for your attempt.

Worked examples

Worked example **Kesar's Diwali shopping list**

For the Diwali season Tapri needs finance for five things. Identify the most suitable short-term source for each, with reasons from its features.

(a) 2 lakh extra kulhads from the potters' co-operative, which will accept Tapri's bills of exchange payable in 60 days.

(b) Twelve custom-built festival kiosks that Tapri's workshop makes for franchisees over ten weeks; the franchisees will pay 40% of the price when they place the order.

(c) ₹25 crore for four months. Tapri is highly rated by a credit rating agency and wants to borrow directly from investors, unsecured, at a rate linked to the one-year government bond yield.

(d) ₹8 crore for six months. Tapri Premix Ltd, Tapri's packaged-chai subsidiary, has surplus cash for exactly that period.

(e) A running limit for buying sugar and packaging as the season goes, secured on that stock, paying interest only on what is used.

MODEL ANSWER

(a) Trade credit, in the form of bills payable. Credit granted by suppliers as an incident of sale, usually for 15 to 90 days, in the form of an open account or bills payable; it arises automatically in the course of business, without any explicit cost, and grows with the volume of business. Sixty-day bills accepted by Tapri are bills payable, inside the usual duration.

(b) Advances from customers. Manufacturers and contractors producing costly goods with a considerable manufacturing or construction time usually demand advance money from customers when they accept the order: a cost free source of finance. The kiosks are costly, custom-built over ten weeks, and the franchisees pay 40% with the order.

(c) Commercial paper. An unsecured money market instrument issued as a promissory note, which enables highly rated corporate borrowers to diversify their short-term borrowings; issued in denominations of ₹5 lakh or multiples, with interest generally linked to the yield on the one-year government bond, after a rating from an agency such as CRISIL, ICRA, CARE or FITCH Ratings India. ₹25 crore is 500 units of ₹5 lakh.

(d) Inter-corporate deposit. Companies can borrow for a short period, say 6 months, from other companies with surplus liquidity, at a rate that varies with the amount and the period.

(e) Cash credit. An advance up to a limit, drawn only to the extent of requirements, with interest charged only on the amount actually availed; generally sanctioned against tradable goods by pledge or hypothecation.

Worked example **The Dubai premix order**

Tapri has a firm order from a Dubai distributor for 40 tonnes of chai premix, payable in US dollars, backed by an irrevocable letter of credit. It needs money now to buy, make and pack the premix, and will ship in ten weeks; the buyer pays 60 days after the bill of lading. It can hypothecate the premix stock to its bank.

Required:

(i) Identify the finance Tapri needs before shipment and after shipment.

(ii) Explain the form of pre-shipment finance that suits Tapri, and the precaution the dollar bill calls for.

(iii) Explain any two forms of post-shipment finance open to Tapri, and the period within which the pre-shipment advance must be repaid.

MODEL ANSWER

(i) Before shipment, **pre-shipment finance**, which generally takes the form of packing credit; after shipment, while the 60-day bill is outstanding, **post-shipment finance**.

(ii) Packing credit against hypothecation of goods. Packing credit is an advance by a bank to an exporter for buying, manufacturing, processing, packing and shipping goods to overseas buyers, available to an exporter with a firm export order or an irrevocable letter of credit, both of which Tapri has. Since Tapri can hypothecate the premix stock, it suits packing credit against hypothecation of goods: the goods are hypothecated to the bank with a stipulated margin, and Tapri submits stock statements with the order or letter of credit, then every fortnight and whenever stocks move. As the export bill will be drawn in US dollars, Tapri should enter into a **forward exchange contract** with the bank to avoid the risk of a change in the exchange rate.

(iii) Post-shipment finance.

1. *Purchase/discounting of documentary export bills*: the bank discounts the usance (60-day) export bill covering the confirmed sale, backed by documents including the bill of lading.
2. *ECGC guarantee*: post-shipment finance through purchase, negotiation or discount of an export bill qualifies for the post-shipment export credit guarantee if Tapri obtains ECGC's shipments (comprehensive risks) policy covering political and commercial risks.

Other forms: advance against export bills sent for collection, and advance against duty drawbacks and cash subsidy. The packing credit must be **liquidated within 180 days from the date of its commencement**, by negotiation of the export bills or receipt of the export proceeds.

Exam angle

In the exam

Where short-term sources appeared

In the ICAI documents read for this lesson, these sources were examined 19 times in 19 documents, mostly as 4-mark descriptive questions.

- **Export finance.** "An Indian exporter has shipped goods ... proceeds yet to be realised. (i) Identify the type of finance (ii) explain any three forms": May 2026 Q4(b) (1 + 3 marks; answer: post-shipment finance, with the forms of SM 2.30–2.31). Forms of post-shipment finance: old-scheme July 2021 paper (4 marks). Types of packing credit: old-scheme MTPs for December 2021 (Series II Q6(b)), May 2022 (Series I Q6(b), 4 marks) and November 2023 (Series II Q6(a)); clean packing credit with bridge finance: old-scheme December 2021 Q6(a).
- **Bank credit.** "Describe any four forms of bank credit": May 2024 Q4(b) (4 marks), answered from this chapter's facilities together with the working capital chapter's forms. Trade credit versus bank overdraft: RTP September 2025 Q10(b).
- **Trade credit and spontaneous sources.** Importance and cost of trade credit and accruals: RTP May 2024 Q8(ii). Which of equity, debt, retained earnings and supplier credit is a short-term source, with its characteristics: MTP September 2025 Series I Q4(c) (2 marks, an OR choice; answer: trade credit). Spontaneous sources: RTP January 2026 Q11(b) and old-scheme November 2023 Q6(b) (4 marks). Costs of availing trade credit: MTP September 2024 Series I Q4(a) (4 marks), answered from the working capital chapter.
- **Commercial paper**, "explain in brief the features": old-scheme MTPs for November 2020, July 2021, November 2022 and November 2023, answered with SM 2.25 word for word. **Inter-corporate deposits and certificates of deposit**: old-scheme MTP November 2022 Series I Q6(c).
- **Public deposits**, as a multiple choice question: "What is the maximum period for which company can accept public deposits?" (MTP May 2024 Series I, Q4 of Part I, 1 mark; answer: 3 years).

In the exam

The wording traps

- "Without any explicit cost" (trade credit) is not "without cost": ICAI's answers point to its implicit cost.
- Packing credit: 180 days **from commencement**; advance against duty drawback: 180 days **from the date of shipment**.
- Clean packing credit has **no charge** over the goods; it still needs a firm order or letter of credit, a margin and ECGC cover.
- T-bills are issued by the **Government of India**, CDs by **banks**, CP by **highly rated corporates**, primary dealers and all-India financial institutions.

Trick

Pre is the factory, post is the ship's paper

Pre-shipment finance pays for buying, making and packing the goods (packing credit, in its five types); post-shipment finance is raised on the export bill and its documents once the goods have sailed (purchase or discount, ECGC guarantee, collection, duty drawback). Condition: it sorts export credit by the moment of shipment; the other facilities (letters of credit, guarantees, status reports) are neither.

Answer frame

"Show the difference between trade credit and bank overdraft" (RTP September 2025 Q10(b))

Point	Trade credit	Bank overdraft
Meaning	Credit extended by suppliers, allowing the firm to delay payment for goods or services	The customer may withdraw in excess of the credit balance in the current account, up to a limit
Period	Usually 15 to 90 days	Repayable on demand, but generally continues for long periods by annual renewal
Cost	A type of interest-free loan, so failing to avail it has an interest cost	Interest charged on daily balances
How it arises	Automatically, in the course of business	Applied for; operated like cash credit and current accounts
Growth	Grows automatically with the volume of business	A fixed limit, within which the borrower draws

"Describe the types of packing credit" and "Explain the forms of post-shipment finance"

The five types in ICAI's order (clean; against hypothecation; against pledge; ECGC guarantee; forward exchange contract), each with its distinguishing line; then the 180-day rule. For post-shipment, the four forms (purchase/discounting of documentary export bills; ECGC guarantee; advance against bills for collection; advance against duty drawbacks and cash subsidy), then the 180-days-from-shipment rule for the last. A 1 + 3 question like May 2026 Q4(b) wants the identification first, in one line.

"Explain in brief the features of commercial paper"

Unsecured promissory note → RBI scheme of 1989 for highly rated corporate borrowers (and primary dealers, all-India financial institutions) → ₹5 lakh or multiples → interest linked to the one-year government bond yield → compulsory credit rating (CRISIL, ICRA, CARE, FITCH Ratings India or another agency specified by the RBI).

Common mistakes

⚠ **Common mistake**

Hall of Shame: shares for the Diwali stock

Financing a three-month seasonal stock with a share issue or a long-term loan. **Fix:** short-term needs from short-term sources; for permanent, hard-core working capital the long-term sources are right, but for the seasonal peak they leave idle, costly money after Bhai Dooj.

⚠ **Common mistake**

Clean packing credit "secured on the goods"

Writing that clean packing credit is secured by hypothecation. **Fix:** clean packing credit has no charge or control over raw material or finished goods; hypothecation and pledge are the second and third types.

⚠ **Common mistake**

The overdraft and the cash credit treated as one

Fix: an overdraft is a limit on a current account with interest on daily balances; a cash credit is a limit sanctioned against tradable goods by pledge or hypothecation, interest on the amount actually availed. Both are repayable on demand.

⚠ **Common mistake**

Trade credit called "free"

Fix: "without any explicit cost" (SM 2.24), but with an implicit cost passed on in the price or the lost cash discount.

Practice

Check yourself

1 Case scenario. Indore Spice Exports has a firm export order from a buyer in Muscat. It needs an advance to buy and pack cumin, but does not want to give the bank any charge over the cumin or the finished packs. Its bank agrees, keeps a margin and obtains ECGC cover. Which facility is this?

- A Clean packing credit
- B Packing credit against hypothecation of goods
- C Packing credit against pledge of goods
- D Purchase of documentary export bills

2 Kulhad Crafts Ltd has paid-up capital of ₹4 crore and reserves of ₹6 crore. Under the limit stated in the Study Material, what is the maximum it can accept as public deposits, in ₹ crore?

Answer: _____

3 Which of these does the Study Material describe as a cost free source of finance?

- A Commercial paper
- B Advances from customers
- C Cash credit
- D Inter-corporate deposits

4 Treasury bills are issued by the Government of India with maturities of:

- A 7 days to one year
- B 14 to 364 days
- C 15 to 90 days
- D Six months to three years

5 Which of these are features of a cash credit? Select all that apply.

- **A** The customer draws only to the extent of his requirements
- **B** Interest is charged on the amount actually availed
- **C** Limits are generally sanctioned against tradable goods by pledge or hypothecation
- **D** The entire advance is disbursed at one time

More than one answer may be right.

6 An advance against duty drawback receivable must be liquidated within:

- **A** 180 days from the commencement of the advance
- **B** 180 days from the date of shipment of the goods
- **C** 90 days from the date of the export order
- **D** One year from the date of shipment

Answers: the answer key at the end of this lesson.

Recap

- Ten short-term sources (SM order): trade credit, accrued expenses and deferred income, advances from customers, commercial paper, treasury bills, certificates of deposit, bank advances, export finance, inter-corporate deposits and public deposits.
- Trade credit: 15–90 days, open account or bills payable, automatic, no explicit cost, grows with business. Accruals and deferred income: spontaneous. Customer advances: cost free.
- CP: unsecured promissory note, RBI 1989, highly rated issuers, ₹5 lakh multiples, linked to the one-year government bond yield, rated. T-bills: Government of India, 14–364 days. CD: bank, 7 days to one year, saleable in the secondary market.
- Bank advances: loan (all at once), overdraft (daily balances), clean overdraft (personal security), cash credit (interest on amount availed, against goods), advances against goods, bills purchased/discounted (bills held as security).
- Packing credit: firm order or irrevocable L/C; liquidated within 180 days of commencement; five types (clean, hypothecation, pledge, ECGC guarantee, forward contract).
- Post-shipment: bills purchased/discounted, ECGC guarantee, advance against bills for collection, advance against duty drawback (within 180 days of shipment).
- ICDs: say 6 months, from companies with surplus liquidity. Public deposits: up to 35% of paid-up capital and reserves, 6 months to 3 years, unsecured, not for fixed assets. (The stock is gone by Bhai Dooj; so is the cash credit.)

Answer key · Check yourself

1 **A Clean packing credit**

Clean packing credit is given only on a firm export order or letter of credit, without any charge or control over raw material or finished goods, with a margin and ECGC cover. Hypothecation and pledge give the bank a charge; purchase of export bills is post-shipment finance.

2 **3.5**

Up to 35 per cent of paid-up capital and reserves: 35% × (₹4 crore + ₹6 crore) = ₹3.5 crore, for six months to three years, and not for fixed assets.

3 **B Advances from customers**

Advances from customers are "a cost free source of finance" (SM 2.25). CP carries interest linked to the one-year government bond yield; cash credit and ICDs carry interest.

4 **B 14 to 364 days**

T-bills: 14 to 364 days. Seven days to one year is a certificate of deposit, 15 to 90 days trade credit, and six months to three years public deposits.

5 **A The customer draws only to the extent of his requirements**; **B Interest is charged on the amount actually availed**; **C Limits are generally sanctioned against tradable goods by pledge or hypothecation**

Disbursing the entire advance at once describes a short-term loan account; the other three are the cash credit.

6 **B 180 days from the date of shipment of the goods**

The duty drawback advance (post-shipment) runs 180 days from the date of shipment; it is the packing credit (pre-shipment) that runs 180 days from its commencement.

3.4 Contemporary and international sources of finance

Raising money abroad (ECBs, Euro and foreign bonds, ESG bonds, ADRs, GDRs and IDRs, the foreign bonds of ICAI's table) and the contemporary sources: crowdfunding, P2P lending and angel finance.

In plain words

Tapri goes abroad

Tapri is opening kulhad cafés in Dubai and London, and Kesar wants every new café to run on solar power.

> **Kesar:** I want Tapri on Nasdaq. Imagine the photo.
>
> **Dev:** I am imagining it, Kesar-ji. The present value of that photo is lower than you think. Also, we can borrow dollars without selling a single share.
>
> **Ananya:** And the solar cafés can pay for themselves: investors queue up for green bonds.
>
> **Kesar:** And Rohan says KulhadKart will raise money from "the crowd". Which crowd?
>
> **Dev:** The internet, Kesar-ji. Thousands of people, a few thousand rupees each.

A company today is not limited to Indian banks and Indian investors. It can **borrow abroad** (a foreign currency loan from a bank, or an **external commercial borrowing** from any non-resident lender), **sell bonds abroad** (a **Euro bond**, in a currency other than that of the country where it is sold, or a **foreign bond**, in that country's own currency), and **sell shares abroad** without listing the shares themselves: a bank abroad issues **depository receipts**, each representing a number of its shares, which trade on a foreign exchange (**ADRs** in the US, **GDRs** mainly in Europe). A foreign company can do the reverse in India through **Indian Depository Receipts**.

At the other end of the scale, a start-up that no bank will lend to can turn to **crowdfunding**: raising money from a large group of people through an online platform, in exchange for shares (**equity crowdfunding**), as loans with interest (**peer-to-peer lending**), or for nothing at all (**donation**).

Think of it like

The cloakroom token

At a wedding hall you hand in your coat and get a token. The coat stays in the cloakroom; the token can be handed to anyone, and whoever holds it can claim the coat. A depository receipt is that token: the company's shares stay with a bank, and the receipts, each standing for a fixed number of shares, are bought and sold on an exchange in another country, in that country's money.

Key idea

International financing reaches foreign capital three ways: loans (foreign currency loans, ECBs), bonds (Euro bonds, foreign bonds, ESG bonds and their variants) and equity (ADRs, GDRs; IDRs the other way round). Contemporary sources such as crowdfunding raise small sums from many people through online platforms, in exchange for equity, for interest, or for nothing.

How it works

1. International funding and angel financing: chapter 1's view

The scope chapter introduces two of these sources among the ways of procuring funds (SM 1.6; the finance function):

- **International funding.** With liberalisation and globalisation, a business enterprise has options to raise capital from international markets also. **Foreign Direct Investment (FDI)** and **Foreign Institutional Investors (FII)** are two major routes for raising funds from foreign sources, besides **ADRs and GDRs**. The mechanism of procuring funds has to be modified in the light of the requirements of foreign investors.
- **Angel financing.** A form of equity financing where an **angel investor**, a wealthy individual, provides capital for start-up or expansion in exchange for an ownership/equity in the company. Angels have idle cash and look for a higher rate of return than traditional investments give, and typically invest around 25 to 60 per cent to help a company get started. This source is sometimes the last option for start-ups which do not qualify for bank funding and are **too small for venture capital** financing.

2. Where external finance comes from abroad

The sources of external financing include (SM 2.35–2.36):

1. **Commercial banks**: like domestic loans, banks all over the world extend **foreign currency (FC) loans** for international operations, and allow overdrawing over and above the loan amount.
2. **Development banks**: long- and medium-term loans, including FC loans. Many national agencies offer concessions to foreign companies to invest in their country and to finance exports from it, e.g. the EXIM Bank of the USA.
3. **Discounting of trade bills**: a short-term financing method, widely used in Europe and Asian countries for domestic and international business.
4. **International agencies**: the International Finance Corporation (IFC), the International Bank for Reconstruction and Development (IBRD), the Asian Development Bank (ADB), the International Monetary Fund (IMF), etc.
5. **International capital markets**, such as London, cater to organisations borrowing in rupees and foreign currency. FC is available under **four main systems**: the Euro-currency market, export credit facilities, bonds issues, and financial institutions.

The **Euro-currency market** began with **dollar-denominated bank deposits and loans in Europe**, particularly in London. Euro-dollar deposits are dollar time deposits at foreign branches of US banks and at some foreign banks; banks based in Europe accept dollar deposits and make dollar loans. Funds are made available as loans through syndicated Euro-credit of instruments such as FRNs and FR certificates of deposit (SM 2.36).

3. External commercial borrowings

External commercial borrowings (ECBs) are commercial loans (bank loans, buyers' credit, suppliers' credit, securitised instruments such as floating rate notes and fixed rate bonds) availed from **non-resident lenders**, with a `minimum average maturity of 3 years` (SM 2.36).

- Raised from internationally recognised sources: (i) international banks, (ii) international capital markets, (iii) multilateral financial institutions such as the IFC and ADB, (iv) export credit agencies, (v) suppliers of equipment, (vi) foreign collaborators and (vii) foreign equity holders.
- Two routes. Under the **automatic route**, no RBI or Government approval is needed; under the **approval route**, it is. Companies registered under the Companies Act and NGOs engaged in micro finance are eligible for the automatic route; financial institutions and banks dealing exclusively in infrastructure or export finance, and those which participated in the textile and steel sector restructuring packages approved by the government, take the approval route.

> ⚠ **Careful**
>
> **ECB rules change**
>
> The maturity and the route conditions are as the Study Material states them (SM 2.36). Verify against the Study Material and Statutory Update for your attempt.

4. The other instruments of the international market

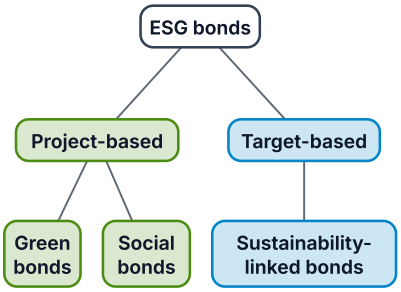
In ICAI's order (SM 2.36–2.38):

Instrument	What it is
Euro bonds	Debt instruments **not denominated in the currency of the country in which they are issued**, e.g. a Yen note floated in Germany; generally in bearer form, without the investor's name or country, so attractive to investors seeking privacy
Foreign bonds	Issued by **foreign corporations or governments**, **denominated in the currency of the country where issued**, e.g. a British firm placing dollar bonds in the USA; exposed to default risk, and to exchange rate risk for investors whose home currency differs
Fully hedged bonds	Foreign bonds whose currency risk is eliminated by selling the entire stream of principal and interest payments in forward markets
Medium term notes (MTN)	A programme under which several lots of bonds, with different coupons, currencies, etc., are issued when market opportunities suit; documentation and regulatory approvals are taken once
Floating rate notes (FRN)	Issued up to seven years' maturity; interest rates adjusted to reflect the prevailing exchange rates; cheaper money than foreign loans
Euro commercial papers (ECP)	Short-term money market instruments, maturity less than one year, usually designated in US dollars
Foreign currency option	The right, not the obligation, to buy or sell foreign currency at a specified price on or before a specified date; a hedge against financial and economic risks
Foreign currency futures	Obligations, not rights, to buy or sell a specified foreign currency in the present for settlement at a future date
Foreign Euro bonds	Local names for such bonds: **Yankee bonds** (US), Swiss Frances (Switzerland), **Samurai bonds** (Tokyo), **Bulldogs** (UK)
Euro convertible bonds	Convertible into a pre-determined number of equity shares, usually at a premium; fixed interest; may include a call option (issuer may redeem early) or a put option (holder may sell back at a set date and price)

Instrument	What it is
Euro convertible zero bonds	No interest; converted on maturity at a pre-determined price; usually five years; treated as a deferred equity issue
Euro bonds with equity warrants	Coupon at market rates; detachable warrants; the pure bonds trade at a discount; suit fixed income funds seeking regular income
ESG-linked bonds	Bonds carrying the issuer's responsibility to prioritise optimal environmental, social and governance factors; investing in them is socially responsible investing

ESG bonds are project-based (green and social bonds) or target-based (sustainability-linked bonds) (SM 2.38–2.39):

ESG bonds (SM 2.39)



Green: project-based bonds, whose proceeds go to specific projects; sky blue: target-based bonds, whose proceeds go to general corporate purposes to achieve key performance indicators.

- **Green bonds**: the most popular ESG bonds, issued by a financial, non-financial or public institution, whose proceeds finance **green projects** with a positive environmental and/or climate impact, including eco-friendly technology. India is the second-largest green bond market; Ghaziabad Municipal Corporation became the first municipal corporation to raise ₹150 crore through a green bond, in 2021.
- **Social bonds**: finance socially impactful projects (human rights, equality, animal welfare, etc.); "vaccine bonds" are an example.
- **Sustainability-linked bonds (SLBs)**: a combination of green and social bonds whose proceeds are **not meant for a specific project** but for general corporate purposes, to achieve **key performance indicators (KPIs)**. UltraTech Cement raised US$400 million through India's first SLB in 2021, aiming to reduce carbon emissions over the bond's 10-year life.

5. The foreign bonds table

The long-term sources section has its own table of foreign bonds (SM 2.13–2.15), which RTP September 2026 Q10(b) asked for in full:

Foreign bond	Salient features
Foreign currency convertible bond (FCCB)	A very low rate of interest; the issuer gets foreign currency at a very low cost; the risk is that if the bond has to be redeemed at maturity, the issuer must pay and may not have the money then
Plain vanilla bond	The issuer pays the principal along with interest; no options; issued as a discounted bond or a coupon-bearing bond
Convertible floating rate note (FRN)	The holder may convert it into a longer-term debt security with a specified coupon; protects the investor against falling interest rates; the long-term security can be sold for a profit; capital gain is not applicable to an FRN
Drop lock bond	An FRN with a normal floating rate, automatically converted into a fixed rate bond if interest rates fall below a predetermined level; the fixed rate stays till maturity. A convertible FRN is a long option structure; a drop lock bond a short option structure
Variable rate demand obligations	A normal FRN with a nominal maturity, which the holder can sell back to the trustee at par plus accrued interest: an exit option, so more liquid than a normal FRN
Yield curve note (YCN)	A structured debt security whose yield rises when prevailing interest rates fall and falls when they rise; used to hedge interest rate risk; works like an inverse floater
Euro bond	Issued or traded in a country using a currency other than the one in which it is denominated, outside the jurisdiction of the central bank issuing that currency; issued by multinationals (a British company issuing a dollar Eurobond in Germany). The term has nothing to do with the euro
Emerging market bond	Issued by developing countries; higher yields than US corporate and Treasury bonds; credit default swaps protect bondholders against default; higher risk than domestic debt

The **masala bond**, a rupee-denominated bond sold to overseas investors, is listed among the Indian bonds: see long-term sources.

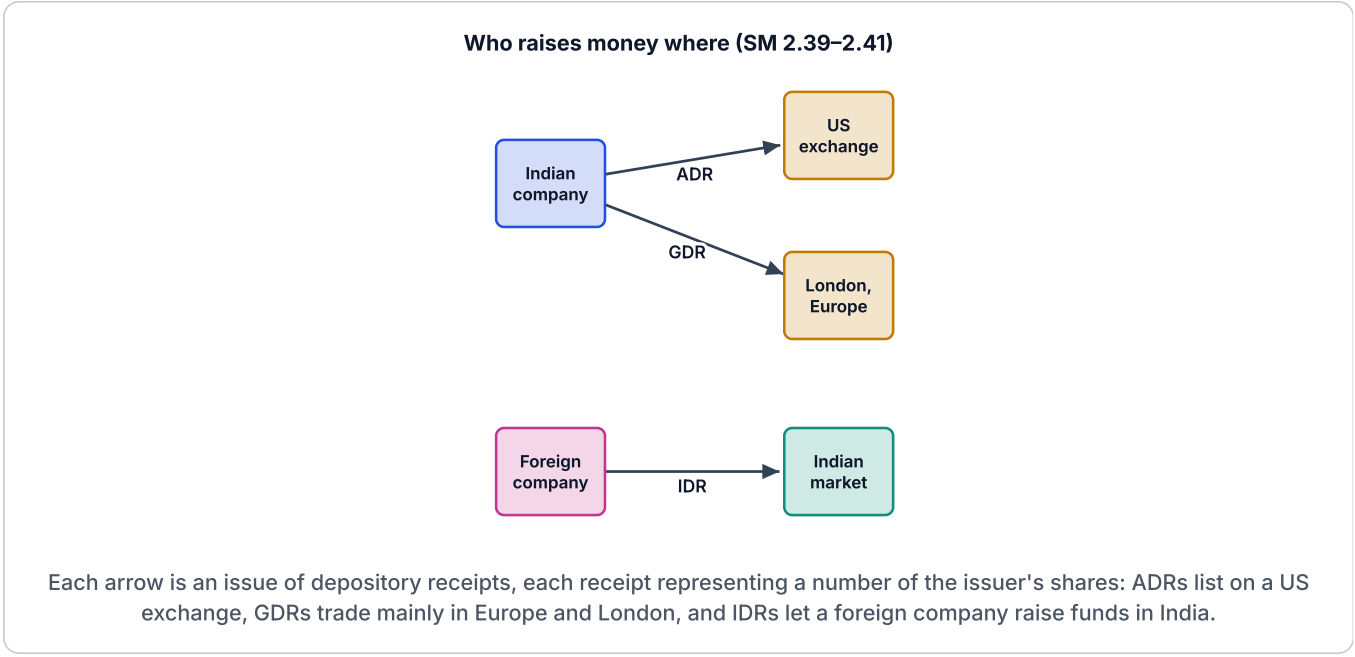
6. Euro issues by Indian companies: ADR, GDR, IDR

Euro issuance refers to the sources of funding that can be accessed to raise money outside the home country in foreign currency. Indian companies may raise foreign currency through ordinary equity shares via **GDRs/ADRs** and/or **FCCBs**, issued to foreign investors (institutional investors or individuals, including NRIs, residing abroad). Such investment is treated as **Foreign Direct Investment**. The government guidelines on these issues are covered under the Foreign Currency Convertible Bonds and Ordinary Shares (through depositary receipt mechanism) Scheme, 1993 and the notifications issued after it (SM 2.39–2.40).

- **American Depository Receipts (ADRs)**: securities offered by non-US companies who want to list on any US exchange. Each ADR represents a certain number of the company's regular shares, and lets US investors buy them without the costs of investing directly in a foreign stock exchange (SM 2.40).
- **Global Depository Receipts (GDRs)**: negotiable certificates held in the bank of one country representing a specific number of shares of a stock traded on the exchange of another country,

used to raise capital in dollars or euros; mainly traded in European countries, particularly London (SM 2.40).

- **Indian Depository Receipts (IDRs)**: the depository receipt mechanism applied in the Indian capital market: **foreign companies** issue IDRs to raise funds from the Indian market on the same lines as an Indian company uses ADRs/GDRs to raise foreign capital. IDRs are listed and traded in India like other Indian securities (SM 2.41).



Each arrow is an issue of depository receipts, each receipt representing a number of the issuer's shares: ADRs list on a US exchange, GDRs trade mainly in Europe and London, and IDRs let a foreign company raise funds in India.

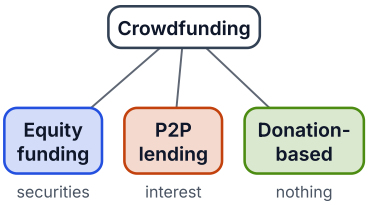
Why Indian companies preferred GDRs to ADRs (SM 2.40): the US market exposes them to a **higher level of responsibility** than a European listing in **disclosure, costs, liabilities and timing**. The regulations are more stringent and onerous, even for companies already listed and held by retail investors at home. The most onerous aspect is providing **full, half-yearly and quarterly accounts in accordance with, or at least reconciled with, US GAAPs**.

ADRs/GDRs and the Indian scenario (SM 2.40): Indian companies are shedding their reluctance to tap the US markets. **Infosys** was the first Indian company listed on Nasdaq, in 1999; the first Indian firm to issue a sponsored GDR or ADR was **Reliance Industries**. Others listed overseas include Wipro, MTNL, State Bank of India, Tata Motors and Dr. Reddy's Lab.

7. Contemporary sources of funding

Crowd funding (SM 2.41): raising money for an individual or organisation from a group of people to fund a project, typically **via the internet** (social media and crowdfunding websites), from family, friends, strangers, corporates and many more, in exchange for **equity** (equity funding), **loans** (P2P lending) or **nothing at all** (donation). It also helps a start-up **substantiate demand for its product before entering production**. **Three parties** are involved: the **fund raiser, the mediator and the fund investor**; the platforms (mediators) may charge processing fees, transaction fees, etc., as a fixed amount, a percentage or both.

Crowdfunding by what the crowd gets (SM 2.41–2.42)



Under each kind, what the investor receives in return. Start-up funding may use equity funding, P2P lending or both.

- **Equity funding** (SM 2.41): investors invest money in an organisation and receive its **securities** in return, each entitled to a stake depending on the investment. Its digital nature targets a **large number of investors with small contributions**; mostly adopted by start-ups. Platforms: StartEngine, EquityNet, SeedInvest, etc.
- **Peer-to-peer (P2P) lending** (SM 2.41–2.42): the category of crowdfunding where **lenders match with borrowers** to provide **unsecured loans** through an online platform, repaid with interest, with a risk of default (just as banks bear in conventional lending). Investors visit the platforms and choose among borrowers considering risk and return. Platforms: i2iFunding, Lendbox, Faircent, RupeeCircle, etc.
- **Start-up funding** (SM 2.42): a newly formed start-up finds bank loans difficult, leaving crowdfunding as one of its sources: equity funding, P2P lending or both.
- **Donation-based crowdfunding** (SM 2.42): a large group of people donate money as charity for a cause, with **no expectation of ownership or debt**. Platforms: GoFundMe (medical needs, education), Ketto (medical needs), FuelADream (charity projects, new ideas).

Worked examples

Worked example **Dev's world tour: name the instrument**

Dev is reviewing six proposals. Identify each instrument, with a reason.

(a) A US$50 million loan from a Singapore bank to Tapri, average maturity five years, raised without RBI or Government approval.

(b) Certificates, each representing four Tapri shares held by a bank, traded on the London market.

(c) A Swiss franc-denominated Tapri bond sold to investors in Singapore, in bearer form.

(d) Dollar bonds placed by Tapri with investors in the USA.

(e) A bond whose proceeds Tapri may use for any corporate purpose, issued on a commitment to meet a stated key performance indicator: cutting its delivery fleet's carbon emissions over the bond's life.

(f) A floating rate note that turns into a fixed rate bond, until maturity, if interest rates fall below 6%.

MODEL ANSWER

(a) External commercial borrowing, automatic route. ECBs are commercial loans availed from non-resident lenders with a minimum average maturity of 3 years; under the automatic route no RBI or Government approval is needed, and companies registered under the Companies Act are eligible for it.

(b) Global Depository Receipts. Negotiable certificates held in the bank of one country representing a specific number of shares of a stock traded on the exchange of another country, mainly traded in European countries and particularly in London.

(c) Euro bond. A debt instrument not denominated in the currency of the country in which it is issued (Swiss francs in Singapore, like the Study Material's yen note floated in Germany), generally in bearer form, attractive to investors seeking privacy. The prefix "euro-" refers to operating outside the jurisdiction of the central bank that issues the currency; it has nothing to do with the euro (SM 2.15).

(d) Foreign bond (in the US, a Yankee bond). A debt instrument issued by a foreign corporation and denominated in the currency of the country where it is issued, like ICAI's example of a British firm placing dollar bonds in the USA.

(e) Sustainability-linked bond (SLB). A target-based ESG bond, a combination of green and social bonds, whose proceeds are not meant for a specific project but for general corporate purposes, to achieve key performance indicators (KPIs). It is not a green bond: a green bond's proceeds are used to finance green projects.

(f) Drop lock bond. A floating rate note with a normal floating rate, automatically converted into a fixed rate bond if interest rates fall below a predetermined level; the fixed rate stays until maturity.

Worked example **Kesar wants Nasdaq; BrewBucks wants India**

(i) Kesar wants Tapri to raise US$200 million by listing on a US exchange "for the photo". Dev prefers a GDR issue in London. Advise, with reasons.

(ii) BrewBucks, the American coffee chain, wants to raise money from Indian investors and have its securities traded in India. Explain the instrument it can use.

MODEL ANSWER

(i) An ADR is a security offered by a non-US company that wants to list on a US exchange; each ADR represents a certain number of the company's regular shares. A GDR is a negotiable certificate held in the bank of one country representing a specific number of shares traded on the exchange of another country, mainly in Europe and particularly London.

Indian companies have preferred GDRs to ADRs because the US market exposes them to a higher level of responsibility than a European listing in the areas of disclosure, costs, liabilities and timing. The regulations are more stringent and onerous, even for companies already listed and held by retail investors in their home country, and the most onerous aspect is providing full, half-yearly and quarterly accounts in accordance with, or at least reconciled with, US GAAPs.

Advice: unless Tapri needs US investors specifically (and Infosys, the first Indian company on Nasdaq, shows it can be done), a **GDR issue in London** raises the same foreign capital with a lighter disclosure and compliance burden. Either way, the investment is treated as Foreign Direct Investment.

(ii) Indian Depository Receipts. The depository receipt mechanism has been applied in the Indian capital market through IDRs: a foreign company issues IDRs to raise funds from the Indian market on the same lines as an Indian company uses ADRs/GDRs to raise foreign capital. The IDRs are listed and traded in India in the same way as other Indian securities.

Worked example **KulhadKart asks the crowd**

Rohan's KulhadKart has three plans: (1) sell a 10% stake to about 2,000 small investors through an online platform; (2) borrow ₹40 lakh, unsecured, from individual lenders matched with it on an online platform, repayable with interest; (3) collect money online for a free chai kitchen for flood-hit families, promising donors nothing in return.

Required: Identify the form of each plan, name the parties to the process, and explain why crowdfunding suits a start-up like KulhadKart, which no bank will lend to, and what else it can do for Rohan before he starts production.

MODEL ANSWER

Forms. (1) is **equity crowdfunding**: investors put money into the organisation and receive its securities, each entitled to a stake depending on the investment, a large number of investors with small contributions. (2) is **peer-to-peer (P2P) lending**: lenders matched with borrowers provide unsecured loans through an online platform, repaid with interest, with a risk of default. (3) is **donation-based crowdfunding**: people donate money as charity for a cause with no expectation of any ownership or debt.

Parties. Three parties are involved: the **fund raiser** (KulhadKart), the **mediator** (the platform, which may charge processing or transaction fees, as a fixed amount, a percentage or both) and the **fund investor**.

Why crowdfunding. A start-up company, being newly formed, needs funds before starting any project, but finds it difficult to manage loans from banks, leaving crowdfunding as one of its sources of finance: it can raise money from a large group of people, in the form of equity funding, P2P lending or both (SM 2.42). Crowdfunding also helps a start-up to **substantiate demand for its product before entering into production** (SM 2.41): if 2,000 people buy a stake in KulhadKart, Rohan knows the machine-pressed kulhad has buyers before he starts production.

Exam angle

In the exam

Where these sources appeared

In the ICAI documents read for this lesson, these sources were examined 21 times in 19 documents, as descriptive questions of 2 to 4 marks and one true-or-false statement.

- **Foreign bonds and international instruments.** "Explain various foreign bonds" for an Indian infrastructure company: RTP September 2026 Q10(b) (answered with the eight-row table of SM 2.13–2.15). Fully hedged bonds, MTNs, FRNs and ECPs: MTP September 2024 Series I Q4(b) (4 marks). Masala and drop lock bonds, with callable and puttable: September 2025 Q4(b) (4 marks). Plain vanilla bond: old-scheme MTP November 2022 Series I Q6(c). Masala bond: old-scheme MTP July 2021 Series I.
- **Samurai and Bulldog bonds**: features of the Bulldog bond, RTP January 2026 Q11(a) and old-scheme November 2023 Q6(a) (4 marks); features of the Samurai bond, old-scheme MTP May 2022 Series I Q6(c) (an OR choice) and RTP November 2022 Q10(b); similarities and differences of the two, old-scheme RTP May 2023 Q10(a).
- **Green bonds**: "Describe any two features of green bond", September 2026 Q4(c) (2 marks; suggested answers not yet published).
- **Depository receipts**: IDRs, MTP May 2024 Series II Q4(c) (2 marks); ADRs versus GDRs, the old-scheme May 2022 paper (2 marks).
- **International funding and angel financing** (from SM 1.6): MTP May 2026 Series II Q4(a) (4 marks) and RTP May 2026 Q10(a); matching "options to raise capital from international markets" and "the last option for startups which doesn't qualify for bank funding": RTP September 2025 Q10(a); "Explain angel financing": September 2024 Q2(b) (2 marks); a true-or-false on angel investors: May 2024 Q4(a)(iv) (true).
- **Crowdfunding**: features, May 2024 Q4(c) (2 marks); crowdfunding versus P2P lending, MTP September 2025 Series II Q4(b) (4 marks), answered that P2P lending is a part of crowdfunding.
- The chapter's own MCQs (SM 2.44–2.45) test masala bonds of the IFC and the two ECB routes.

In the exam

The wording traps

- A Euro bond is defined by **currency against country**, not by Europe; a foreign bond by a **foreign issuer** in the local currency.
- ADRs are for US exchanges, GDRs mainly Europe and London, IDRs for **foreign companies in India**.
- ECB: **non-resident** lenders, **minimum average maturity of 3 years**, automatic **and** approval routes.
- Green and social bonds are **project-based**; SLB proceeds are for **general corporate purposes**, tied to KPIs.

Trick

ADR, GDR, IDR by the first letter

A for America, **G** for global (in practice Europe and London), **I** for India, where the issuer is foreign. Condition: the first two are issued by non-US (for ADRs) or Indian companies raising capital abroad; the IDR runs the other way.

Answer frame

"Distinguish between ADRs and GDRs" (the old-scheme May 2022 suggested answer)

Basis	ADR	GDR
Meaning	A negotiable instrument issued by a US bank, representing a non-US company's stock traded on a US stock exchange	A negotiable instrument issued by an international depository bank, representing a foreign company's stock traded worldwide
Issued in	The US domestic capital market	The European capital market
Listed in	The American stock exchange	Non-US stock exchanges
Relevance	Foreign companies trade in the US stock market	Foreign companies trade in any country's stock market other than the US

"Discuss the features of international funding and angel financing"

Two headings. International funding: liberalisation and globalisation; FDI and FII as the two major routes, besides ADRs and GDRs; procurement adapted to foreign investors' requirements. Angel financing: equity financing by a wealthy individual for start-up or expansion in exchange for equity; seeks a higher return than traditional investments; typically 25 to 60 per cent; sometimes the last option for start-ups that do not qualify for bank funding and are too small for venture capital.

"Explain the features of crowd funding"

Definition (from a group of people, typically via the internet) → three forms (equity, P2P loans, donation) → helps substantiate demand before production → three parties (fund raiser, mediator, fund investor) → platform fees.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the Euro bond issued in euros

Calling any bond in euros a Euro bond, or a dollar bond sold in the US by a British firm a Euro bond. **Fix:** a Euro bond is not denominated in the currency of the country where it is issued (the term has nothing to do with the euro); the British firm's dollar bond in the USA is a foreign bond (a Yankee bond).

⚠ **Common mistake**

IDRs for Indian companies going abroad

Fix: Indian companies go abroad with ADRs and GDRs; IDRs are issued in India by foreign companies.

⚠ **Common mistake**

Crowdfunding and P2P lending as rivals

Fix: P2P lending is a category of crowdfunding (the loan form); crowdfunding also covers equity funding and donations.

⚠ **Common mistake**

The green bond that funds anything

Fix: green and social bonds finance specific projects; it is the sustainability-linked bond whose proceeds go to general corporate purposes, tied to KPIs.

Practice

Check yourself

1 Case scenario. Over the next two years Tapri will need foreign debt several times, in different currencies and at different coupons. Dev wants to issue it lot by lot, choosing the timing of each lot as market opportunities arise, but to complete the documentation and the regulatory approvals only once. Which instrument fits?

- A Fully hedged bonds
- B Euro convertible zero bonds
- C A medium term note (MTN) programme
- D Euro commercial paper

2 As the Study Material describes the ECB routes, which of these borrowers must take the approval route?

- **A** A company registered under the Companies Act
- **B** An NGO engaged in micro finance activities
- **C** Any borrower whose ECB has an average maturity of at least 3 years
- **D** A financial institution dealing exclusively in infrastructure finance

3 A Japanese company sells US dollar bonds to investors in Germany. The bonds are:

- **A** Euro bonds
- **B** Samurai bonds
- **C** Yankee bonds
- **D** Bulldog bonds

4 Which of these statements about ESG bonds are correct? Select all that apply.

- **A** Green bonds and social bonds are project-based
- **B** Proceeds of sustainability-linked bonds are for general corporate purposes, to achieve KPIs
- **C** Social bonds finance eco-friendly technology only
- **D** India is described as the second-largest green bond market

More than one answer may be right.

5 The three parties in a crowdfunding process are:

- **A** The issuer, the depository bank and the stock exchange
- **B** The fund raiser, the mediator and the fund investor
- **C** The lessor, the lessee and the lender
- **D** The originator, the SPV and the investors

Answers: the answer key at the end of this lesson.

Recap

- External finance abroad: commercial banks (FC loans), development banks, discounting of trade bills, international agencies (IFC, IBRD, ADB, IMF), international capital markets (Euro-currency market, export credit, bond issues, financial institutions).
- ECB: commercial loans from non-resident lenders, minimum average maturity 3 years, seven recognised sources, automatic route (no approval) or approval route.
- Euro bond: not in the currency of the country of issue (often bearer), while a foreign bond is a foreign issuer's bond in the local currency (Yankee, Samurai, Bulldog).

- MTN: many lots, one documentation; FRN: up to 7 years; ECP: under one year, usually US dollars; fully hedged bonds: forward sale of all payments.
- ESG bonds: green and social (project-based), sustainability-linked (target-based, KPIs, general purposes).
- Foreign bonds table: FCCB, plain vanilla, convertible FRN, drop lock, variable rate demand obligations, yield curve note, Euro bond, emerging market bond.
- ADR (US exchange), GDR (Europe, London), IDR (foreign company in India); FDI; Indian companies preferred GDRs because US disclosure is more onerous (US GAAP quarterly accounts).
- Crowdfunding: equity, P2P (unsecured loans with interest), donation; three parties; platforms charge fees. Angel: wealthy individual's equity, 25–60%, when banks will not lend and VC will not look. (Kesar got his photo, in London.)

Answer key · Check yourself

1 **C** **A medium term note (MTN) programme**

Under an MTN programme several lots of bonds, with different coupons, currencies, etc., are issued, the timing of each decided by market opportunities, with the documentation and approvals taken at one point of time (SM 2.37). Fully hedged bonds remove currency risk by selling the payments forward; Euro convertible zero bonds convert into equity at maturity; ECPs are money market instruments of less than one year.

2 **D** **A financial institution dealing exclusively in infrastructure finance**

Companies registered under the Companies Act and NGOs engaged in micro finance are eligible for the automatic route (no RBI or Government approval). Financial institutions and banks dealing exclusively in infrastructure or export finance, and those in the approved textile and steel restructuring packages, take the approval route. The 3-year minimum average maturity applies to every ECB; it does not choose the route.

3 **A** **Euro bonds**

The bonds are not denominated in the currency of the country where they are issued (dollars in Germany): Euro bonds. Samurai bonds are yen bonds in Tokyo by non-Japanese issuers; Yankee bonds are dollar bonds in the US by foreign issuers; Bulldogs are sterling bonds in the UK.

4 **A** **Green bonds and social bonds are project-based**; **B** **Proceeds of sustainability-linked bonds are for general corporate purposes, to achieve KPIs**; **D** **India is described as the second-largest green bond market**

SM 2.38–2.39: green and social bonds are project-based, SLBs target-based. Social bonds fund social concerns (human rights, equality, animal welfare); it is green bonds that fund environmental projects.

5 **B** **The fund raiser, the mediator and the fund investor**

SM 2.41. The other sets belong to depository receipts, a leveraged lease and securitisation.

CHAPTER 4

Cost of Capital

4.1 Cost of debt and of preference capital

What a loan, a debenture or a preference share really costs: net proceeds, after-tax interest, the approximation formula, YTM, convertibles and preference capital.

In plain words

Kesar finds the "cheap" money

Tapri Tea Co. Ltd needs ₹60 crore for 40 new kulhad cafés and a cold-chain fleet, and Kesar has two offers on his desk.

> **Kesar:** The bank wants 12% interest. Preference shareholders want only 10% dividend. Preference is cheaper. Done!
>
> **Dev:** Interest is a tax-deductible expense. At our 25% tax rate the taxman pays a quarter of every interest rupee, so the loan really costs 12% × 0.75 = 9%. Preference dividend is paid out of profit after tax: 10% stays 10%.
>
> **Kesar:** So the expensive one is the cheap one?
>
> **Dev:** And our old 9% debentures trade at ₹90, not ₹100. Anyone buying them today expects more than 9%. A source costs what its providers expect, not what is printed on it.

Every rupee a company raises has a price: the **cost of capital**, the return the providers of the money expect as compensation for it (SM 4.2). For a loan or a debenture it is the **cost of debt (Kd)**; for preference shares, the **cost of preference capital (Kp)**.

Three things decide it. First, what the company actually receives: a ₹100 debenture sold at ₹96, with ₹2 of brokerage, brings in only ₹94, its **net proceeds**. Second, what it pays out: interest every year and the redemption amount at the end. Third, tax: interest saves tax, a preference dividend does not. The cost is the rate at which what comes in today equals the present value of what goes out later, after tax.

Think of it like

The "no-cost EMI"

A shop sells a ₹30,000 phone on a "no-cost EMI" of ₹2,500 a month for a year, plus a ₹1,500 processing fee on day one. The sticker says 0%, yet you pay ₹31,500 for ₹30,000 of phone: the real rate is the one that equates what you got with what you pay. A debenture's coupon is the sticker; its cost is the real rate after the discount, the brokerage and any redemption premium. And if the government refunded a quarter of every interest rupee you paid, your real rate would drop by a quarter: that refund is the tax shield.

Key idea

The cost of debt is the rate at which the net proceeds received today equal the present value of the after-tax interest and the redemption value. Interest is tax-deductible, so Kd is an after-tax cost; preference dividend is not, so Kp gets no tax adjustment.

How it works

Cost is what the market expects

ICAI's own example (SM 4.3): a company issues 9% debentures, but investors expect 10%, so they subscribe only at a discount and the cost is not 9% but 10%. Commission and brokerage paid to intermediaries push it above 10%; the tax deductibility of interest pulls it down. To find the cost, identify three kinds of cash flow (SM 4.3–4.4):

1. the inflow of the amount received at the beginning;
2. the outflows of payment of interest, dividend, redemption amount and so on;
3. the inflow of tax benefit on interest, or the outflow of payment of dividend tax.

The rate at which the present value of the outflows equals the inflow is an IRR. In investment decisions the IRR is an income, because an outflow comes first; here an inflow comes first, so the IRR represents cost (SM 4.4). The shortcut formulas are easier, but they give approximate answers.

What a debenture carries

Five features decide its cash flows (SM 4.5–4.6): **face value** (interest is calculated on it), **interest (coupon) rate** (fixed, except for zero coupon and deep discount bonds), **maturity period** (taken as infinite for irredeemable debentures), **redemption value** (fixed by the debt covenants; it may differ from face value) and the **tax shield**: interest is allowed as an expense for corporate tax, so it saves tax. In ICAI's Example 1, ₹40 lakh of interest at 35% tax saves ₹14 lakh, so the effective interest is ₹26 lakh.

A loan from a financial institution is costed like a redeemable debenture (SM 4.5); a term loan taken and repaid at par costs simply I(1 − t), as the Suggested Answers compute it (May 2024: 11% × 0.75 = 8.25%; January 2025: 12% × 0.70 = 8.4%).

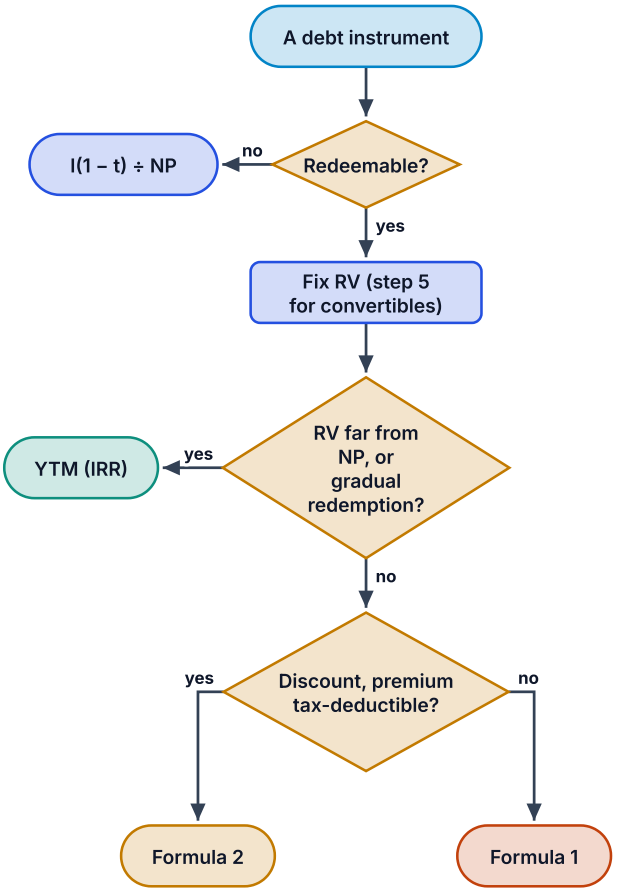
Six steps from a term sheet to a cost

1. **Net proceeds (NP)** = issue price − flotation cost. No issue price given: use the current market price, which is also the figure for existing debt. No issue expenses given: take them as zero (SM 4.7).
2. **After-tax interest** = I(1 − t), with I the coupon on the face value.
3. **Irredeemable debentures**: Kd = I(1 − t) ÷ NP.
4. **Redeemable debentures**: to the after-tax interest add the gap between the redemption value (RV) and NP spread over the remaining life, (RV − NP) ÷ n, and divide by the average of RV and NP (the **approximation method**); or find the **yield to maturity (YTM)**, the IRR of the cash flows.

5 **Convertible debentures**: RV is the higher of the cash value and the value of the shares at maturity, as holders choose the more attractive option (SM 4.15).

6 **Preference shares**: the same formulas with the dividend in place of interest, and no tax adjustment.

Which formula for a debt instrument?

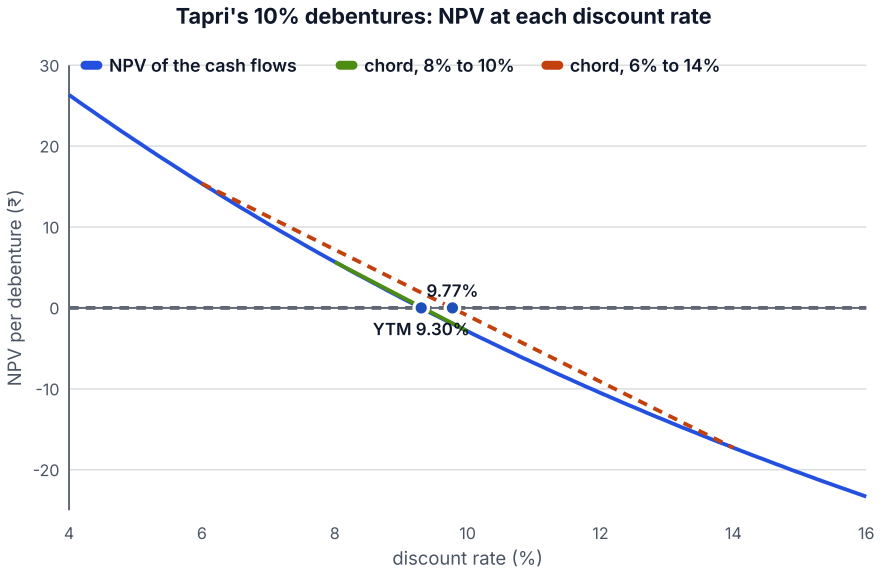


Formula 1 taxes only the interest; Formula 2 also gives tax relief on the discount and premium (both in Formats and formulas). With no information on that, either may be used with a logical assumption (SM 4.9). The approximations lose accuracy as RV and NP move apart and do not suit gradual redemption; the YTM is the superior method (SM 4.12).

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The YTM: the debenture's own IRR

The YTM is the internal rate of return at which the current price of a debt equals the present value of all its cash flows (SM 4.11). Take Tapri's existing 10% debentures: quoted at ₹92, redeemable at par in 6 years, tax 25%. Year 0 is ₹92 received (the price, as for existing debt); years 1 to 6 carry ₹7.50 of after-tax interest; year 6 also carries the ₹100 redemption. Discount these at two rates, one giving a positive and one a negative NPV, and interpolate between them (SM 4.11–4.12).



Blue: the NPV curve; it cuts zero at the true YTM, 9.30%. Interpolation puts a straight chord between two trial rates instead: the green one from 8% to 10% hugs the curve and cuts zero at 9.33%; the dashed orange one from 6% to 14% cuts it at 9.77%. The wider the gap between the two rates, the less accurate the answer (SM 4.12).

Preference capital

Preference shareholders are paid a dividend at a specified rate on face value. Paying it is not mandatory, but it has priority over the equity dividend, and it is an appropriation of after-tax profit, so it does not reduce the tax liability (SM 4.16). An **irredeemable preference share** is a perpetuity: Kp = dividend ÷ net proceeds. A **redeemable preference share** uses the approximation formula with the dividend in place of after-tax interest, or the discount rate that equates the net proceeds with the present value of future dividends and redemption (SM 4.18).

Formats and formulas

Reproduce each formula, its symbols and the YTM layout from memory.

∑ **Formula**

Irredeemable debentures (SM 4.7)

$$K_d = \frac{I\,(1-t)}{NP}$$

Kd = cost of debt after tax; I = annual interest payment; NP = net proceeds of debentures or current market price; t = tax rate.

∑ **Formula**

Redeemable debentures: Formula 1, only interest tax-deductible (SM 4.8)

$$K_d = \frac{I\,(1-t) + \dfrac{RV - NP}{n}}{\dfrac{RV + NP}{2}}$$

RV = redemption value; n = remaining life of the debentures.

∑ Formula

Redeemable debentures: Formula 2, discount and premium also tax-deductible (SM 4.9)

$$K_d = \frac{\left(I + \frac{RV - NP}{n}\right)(1 - t)}{\frac{RV + NP}{2}}$$

∑ Formula

YTM by interpolation (SM 4.12)

$$K_d = L + \frac{NPV_L}{NPV_L - NPV_H} \times (H - L)$$

L and H = the lower and higher trial rates; NPV_L and NPV_H = the NPVs of the cash flows at those rates.

∑ Formula

Zero coupon and deep discount bonds (SM 4.12)

$$RV = NP \times (1 + K_d)^n$$

No interest is paid, so the whole return is the gap between issue price and redemption value.

∑ Formula

Value of an amortised bond (SM 4.13)

$$V_B = \sum_{t=1}^{n} \frac{C_t}{(1 + K_d)^t}$$

C_t = principal repaid in year t + interest on the amount outstanding; the cost of such a bond is the rate that equates these uneven flows with its net proceeds.

∑ Formula

Convertible debentures: redemption value (SM 4.15)

$$RV = \text{higher of} \qquad \left(\text{cash value},\ N \times P_0(1 + g)^n\right)$$

N = shares per debenture; P_0 = current share price; g = its expected growth rate; n = years to maturity. Then apply Formula 1 or the YTM.

Σ Formula

Preference shares (SM 4.17–4.18)

$$K_p = \frac{PD}{P_0} \qquad K_p = \frac{PD + \dfrac{RV - NP}{n}}{\dfrac{RV + NP}{2}}$$

PD = annual preference dividend; P_0 or NP = net proceeds from the issue. The first is for irredeemable shares, the second for redeemable ones.

Definition

Flotation cost

The new issue of a security (debt or equity) involves some expenditure in the form of underwriting or brokerage fees, legal and administrative charges, registration fees, printing expenses etc. The sum of all these costs is known as floatation cost. (SM 4.7)

Definition

Net proceeds

Issue price less issue expenses or floatation cost. If the issue price is not given, it may be assumed equal to the current market price; if issue expenses are not given, they are taken as zero. (SM 4.7)

Definition

Yield to maturity (YTM)

The annual return of an investment from the current date till maturity date: the internal rate of return at which the current price of a debt equals the present value of all cash flows. (SM 4.11)

The YTM working, ICAI's layout

Year	Cash flow	PV at L (₹)	PV at H (₹)
0	NP (new issue) or P0 (existing debt)	(NP) × 1.000	(NP) × 1.000
1 to n	Interest net of tax, I(1 – t)	× PVAF (L, n)	× PVAF (H, n)
n	Redemption value, RV	× PVF (L, n)	× PVF (H, n)
	NPV	**+ (NPV at L)**	**– (NPV at H)**

ICAI prints each discount factor in its own column (six columns in all); write it that way on paper. Then interpolate.

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Worked examples

Worked example **Tapri's new debenture, priced three ways**

Tapri Tea Co. Ltd plans to issue 11% debentures of ₹100 each at a discount of 4%, redeemable at a premium of 5% after 8 years. Flotation cost is 2% of the face value, and the tax rate is 25%. Tapri's existing 9% irredeemable debentures of ₹100 each are quoted at ₹90.

PV factors: PVAF (8%, 8 years) 5.747 and PVF (8%, year 8) 0.540; PVAF (10%, 8 years) 5.335 and PVF (10%, year 8) 0.467.

Required:

(i) Calculate the cost of the new debentures by the approximation method, taking only the interest as tax-deductible.

(ii) Recalculate it if the discount on issue and the premium on redemption are also tax-deductible.

(iii) Calculate the cost of the new debentures by the YTM method.

(iv) Calculate the current cost of the existing irredeemable debentures.

THE WORKING

Working note. NP = ₹100 − ₹4 discount − ₹2 flotation = **₹94**; RV = **₹105**; n = 8; I = ₹11; I(1 − t) = ₹11 × 0.75 = **₹8.25**; (RV − NP) ÷ n = ₹11 ÷ 8 = ₹1.375; (RV + NP) ÷ 2 = ₹99.50.

(i) Approximation method, Formula 1

$$K_d = \frac{8.25 + \dfrac{105 - 94}{8}}{\dfrac{105 + 94}{2}} = \frac{9.625}{99.50} = 0.0967 \text{ or } 9.67\%$$

(ii) Formula 2, discount and premium tax-deductible

$$K_d = \frac{(11 + 1.375)\,(1 - 0.25)}{99.50} = \frac{9.281}{99.50} = 0.0933 \text{ or } 9.33\%$$

(iii) YTM method

Year	Cash flow (₹)	PV @ 8% (₹)	PV @ 10% (₹)
0	(94.00)	(94.00)	(94.00)
1 to 8	8.25	47.41	44.01
8	105.00	56.70	49.04
NPV		**+10.11**	**−0.95**

$$K_d = 8\% + \frac{10.11}{10.11 - (-0.95)} \times (10\% - 8\%) \qquad = 8\% + 1.83\% = 9.83\%$$

(iv) Existing irredeemable debentures, on the current market price:

$$K_d = \frac{9 \times (1 - 0.25)}{90} = \frac{6.75}{90} = 7.50\%$$

Comment. The approximation (9.67%) is 0.16 percentage points below the YTM (9.83%; a financial calculator gives 9.81%); the YTM is the superior method (SM 4.12). The old debentures cost 7.50%, not 6.75%: they trade below par, so today's investors expect more than the coupon. Rounding: factors to three decimals as given (the annuity factors are the table's, rounded once; adding the eight yearly factors gives 5.746 and 5.334 and the same 9.83%), present values to two, rates to two.

Worked example

Kulhad Capital: a convertible, a term loan and two preference shares

Dev is pricing four sources for the café expansion. The tax rate is 25%.

(A) 9% convertible debentures of ₹100 each, to be issued at ₹98 with a flotation cost of ₹1 per debenture, maturing in 5 years. At maturity each holder may take ₹110 in cash or 3 equity shares. Tapri's share is quoted at ₹28 and is expected to grow at 5% a year.

(B) A 10.5% term loan of ₹5 crore from Indore Sahakari Bank, repayable in one sum after 4 years. The bank deducts a processing fee of 2% of the loan when it pays out.

(C) 9% irredeemable preference shares of ₹100 each, issued at par years ago and now quoted at ₹75.

(D) New 10% preference shares of ₹100 each, to be issued at par with a flotation cost of 3%, redeemable at a premium of 5% after 6 years.

Required: Calculate the cost of each source, using the approximation method where a formula is needed.

THE WORKING

(A) Convertible debentures

Value of 3 shares after 5 years = ₹28 × $(1.05)^5$ × 3 = ₹35.736 × 3 = ₹107.21. This is lower than the cash option of ₹110, so holders will take the cash: **RV = ₹110**. NP = ₹98 − ₹1 = ₹97; n = 5; I(1 − t) = ₹9 × 0.75 = ₹6.75.

$$K_d = \frac{6.75 + \dfrac{110 - 97}{5}}{\dfrac{110 + 97}{2}} \qquad = \frac{6.75 + 2.60}{103.50} = 0.0903 \text{ or } 9.03\%$$

(B) Term loan. The fee means Tapri receives only ₹98 for every ₹100 it repays, so the loan is costed like a redeemable debenture (SM 4.5): NP = ₹98, RV = ₹100, n = 4, I(1 − t) = ₹10.50 × 0.75 = ₹7.875.

$$K_d = \frac{7.875 + \dfrac{100 - 98}{4}}{\dfrac{100 + 98}{2}} = \frac{8.375}{99} = 0.0846 \text{ or } 8.46\%$$

Without the fee it would be 10.5% × 0.75 = 7.875%.

(C) Irredeemable preference shares, on the current market price: Kp = ₹9 ÷ ₹75 = **12.00%**.

(D) Redeemable preference shares

NP = ₹100 − 3% = **₹97**; RV = **₹105**; n = 6.

$$K_p = \frac{10 + \dfrac{105 - 97}{6}}{\dfrac{105 + 97}{2}} = \frac{11.333}{101} = 0.1122 \text{ or } 11.22\%$$

Comment. Both preference issues (11.22% and 12.00%) cost more than either debt (9.03% and 8.46%): there is no tax shield on a dividend. The convertible's conversion option is worth less than the cash at maturity, so its cost is set by the ₹110 redemption, exactly as for a redeemable debenture.

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Exam angle

In the exam

Where these costs appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, a cost of debt or of preference capital came up 16 times in 13 documents, usually for 1–3 marks inside a 5–8 mark question that also asks for the cost of equity or the WACC:

- **Convertible debentures**: May 2024 Q2(a) (6 marks), January 2025 Q2(a) (7 marks), January 2026 MCQs 2–3, RTP January 2026 Q6 (May 2024's question again), MTP Series II for May 2025 Q1(b) (5 marks, approximation and YTM, with 75% of holders converting).
- **Theory**: "EXPLAIN any four Features of Debentures or Bonds", MTP Series II for May 2025 Q4(b) (4 marks); the Suggested Answer is the Study Material's five features almost word for word.
- **YTM**: September 2024 Q2(a) (8 marks, debentures); May 2025 Q2(a) (8 marks, preference shares, then the debenture rate solved backwards from a given WACC of 14%).
- **Irredeemable preference and reverse problems**: September 2025 Q1(a) (5 marks) and Q3(b) (3 marks: the issue price from a given Kd of 14.58%).
- **MCQs**: January 2026 MCQ 1 (Kp net of 4% flotation), MTP Series I for May 2025 MCQ 8 (issue at a premium), MTP Series I for September 2024 MCQ 3 (an amortised debenture by IRR), September 2026 MCQ 1 (a tax holiday).
- **RTPs**: January 2025 Q5, May 2026 Q5 (Formula 2, then the tax rate solved backwards).

In the exam

The traps ICAI used

- **Which price is NP for an existing security?** September 2025 Q1(a): the Suggested Answer used the issue price, ₹45 (Kp 17.78%), and the market price, ₹55 (14.55%), as an alternative. January 2025 Q2(a): market value ₹108 (19.52%), alternative ₹100 (20.55%). January 2026 MCQ 3's key (30.74%) uses ₹100 though a market price of ₹150 is given. The Study Material uses the current market price for existing debt (SM 4.10): say which you use.
- **A missing redemption value**: May 2024 Q2(a)'s preference shares, "redeemable after eight years", were solved two ways (7% and 17%); RTP January 2026 Q6 repeats it with "at par".
- **Remaining life**: debentures issued on 1.4.2023 for 6 years and costed on 31.3.2024 have n = 5 (May 2024 Q2(a) and RTP January 2026 Q6).
- **A tax holiday** (September 2026 MCQ 1 asks for the after-tax cost of debt during and after one): with no tax there is no shield, so during the holiday Kd is the pre-tax rate.
- **Annuity factors**: ICAI's answers do it both ways. September 2024 Q2(a) gave yearly factors and the answer added them (3.790 at 10% for 5 years, where the annuity table prints 3.791); May 2025 Q2(a) gave yearly factors but used the table's 2.487 and 2.322 for 3 years (the yearly factors add up to 2.486 and 2.321). Say which you use; the interpolated rate barely moves.

Trick

When NP equals RV

If a debt is issued at par (net of costs) and redeemed at par, (RV − NP) is zero and both the approximation and the YTM give exactly I(1 − t) ÷ NP: the term-loan case. Condition: interest paid annually on the face value.

Answer frame

EXPLAIN the YTM approach of calculating cost of debt (SM 4.11–4.12), in this order:

1. Kd is found by discounting the relevant cash flows using the internal rate of return; YTM is the annual return from the current date till maturity, the IRR at which the current price equals the present value of all cash flows.
2. Cash flows: year 0, net proceeds (new issue) or current market price (existing debt); years 1 to n, interest net of tax, I(1 − t); year n, redemption value.
3. Steps: identify the cash flows; calculate NPVs at two discount rates; calculate the IRR by interpolation.
4. It is superior to the approximation method and preferred in finance; the wider the gap between the two rates, the lower the accuracy.

DISCUSS amortisation of bond (SM 4.13): principal repaid every year rather than at maturity; interest on the outstanding amount, so the cash flows are uneven; value = the sum of each year's interest plus principal discounted at Kd; a two-line example.

EXPLAIN any four features of debentures or bonds (4 marks in MTP Series II for May 2025; SM 4.5–4.6). One heading and two lines each, in ICAI's order:

1. **Face value**: the denominated value of the debenture; interest is calculated on it (9% debentures of ₹100: interest @ 9% on ₹100).
2. **Interest (coupon) rate**: a fixed rate on the face value (except zero coupon and deep discount bonds), paid periodically (annually, semi-annually, etc.).
3. **Maturity period**: a fixed period for redemption; for irredeemable debentures it is not defined and is taken as infinite.
4. **Redemption value**: determined by the debt covenants and paid on the maturity date; it may vary from the face value.
5. **Benefit of tax shield**: interest is allowed as an expense for corporate tax, so it saves tax; the saving is the tax shield.

Write four of the five; ICAI's answer gave all five.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the preference dividend that got a tax refund

Writing Kp = 10% × (1 − 0.25) = 7.5%. **Fix:** preference dividend is an appropriation of after-tax profit and saves no tax; only interest gets the (1 – t).

⚠ **Common mistake**

The face value in the denominator

Tapri's 9% irredeemable debentures quoted at ₹90: 6.75 ÷ 100 = 6.75% is the coupon after tax, not the cost. **Fix:** divide by the net proceeds, or the current market price for existing debt: 6.75 ÷ 90 = 7.50%.

⚠ **Common mistake**

The convertible that forgot to grow

Taking today's share price (4 × ₹24 = ₹96, below the ₹100 cash option) instead of the price at maturity. **Fix:** grow the price to maturity, ₹24 × $(1.06)^6$, multiply by the shares per debenture, then take the higher of that and the cash value.

⚠ **Common mistake**

Interest on the wrong base, life from the wrong date

Interest is the coupon on the **face value**, not on the issue price or NP; n is the **remaining** life on the date of costing, not the original tenure.

Practice

Check yourself

1 Tapri's 12% irredeemable debentures of ₹100 each are quoted at ₹96 and the tax rate is 25%. What is their current cost of debt, in % to two decimals?

Answer: _____

2 Case scenario. Tapri's 12% term loan has an after-tax cost of 9% at today's 25% tax rate. The Finance Bill proposes to cut the corporate tax rate to 22%. Kesar is pleased: "Lower tax, cheaper loan." Which of the following is correct?

- **A** Kesar is right: the after-tax cost falls to 8.64%
- **B** The after-tax cost rises to 9.36%, because each rupee of interest now saves less tax
- **C** The after-tax cost stays at 9%, because the coupon is fixed
- **D** The after-tax cost rises to 12%, because the tax shield is lost

3 Why is the cost of preference shares not multiplied by (1 – t)?

- **A** Because preference dividend is fixed in amount
- **B** Because preference dividend is an appropriation of after-tax profit and does not reduce the tax liability
- **C** Because preference shares are redeemable
- **D** Because preference shareholders pay tax on the dividend themselves

4 A convertible debenture can be redeemed after 5 years for ₹160 in cash or 5 equity shares. The share is quoted at ₹30 and grows at 4% a year; $(1.04)^5 = 1.2167$. What redemption value is used for its cost?

- **A** ₹160.00
- **B** ₹150.00
- **C** ₹182.50
- **D** ₹342.50

5 Which of these lower the net proceeds of a new debenture issue? Select all that apply.

- **A** Underwriting commission
- **B** Issue at a discount
- **C** A premium payable on redemption
- **D** Printing and registration expenses

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Cost of capital: the return expected by the providers of capital; cost is the market's expectation, not the coupon (SM 4.2–4.3).
- NP = issue price − flotation cost; no issue price, use the market price; existing debt, the market price.
- Irredeemable: $Kd = I(1 - t) \div NP$; $Kp = PD \div NP$.
- Redeemable, approximation: $[I(1 - t) + (RV - NP) \div n] \div [(RV + NP) \div 2]$; if the discount and premium are tax-deductible, $[I + (RV - NP) \div n](1 - t) \div$ average.
- YTM: year 0 NP, years 1 to n $I(1 - t)$, year n RV; NPVs at two rates; interpolate. Superior to the approximation.
- Convertible: RV = higher of cash and shares × $Po(1 + g)^n$.
- No tax shield on a preference dividend, or in a tax holiday. (The taxman only chips in on interest.)

Answer key · Check yourself

1 **9.37 to 9.38**

Kd = I(1 − t) ÷ market price = ₹12 × 0.75 ÷ ₹96 = ₹9 ÷ ₹96 = 9.375%. Dividing by the face value gives 9.00%, the coupon after tax, not the cost.

2 **B** **The after-tax cost rises to 9.36%, because each rupee of interest now saves less tax**

Kd = I(1 − t): 12% × (1 − 0.22) = 9.36%. A lower tax rate shrinks the tax shield on interest, so debt becomes dearer after tax; the shield is smaller, not lost. 8.64% is 12% × (1 − 0.28): the three-point change applied the wrong way.

3 **B** **Because preference dividend is an appropriation of after-tax profit and does not reduce the tax liability**

Interest is a tax-deductible expense; a preference dividend comes out of profit after tax (SM 4.16). A fixed amount or redeemability changes nothing about tax.

4 **C** **₹182.50**

Value of the shares = 5 × ₹30 × 1.2167 = 5 × ₹36.50 = ₹182.50, higher than ₹160 in cash, and holders take the more attractive option. ₹150 is the shares at today's price, ignoring growth; ₹342.50 adds the two options.

5 **A** **Underwriting commission**; **B** **Issue at a discount**; **D** **Printing and registration expenses**

NP = issue price − flotation cost; underwriting, printing and registration are flotation costs and a discount lowers the issue price. A redemption premium raises RV, the amount paid back later, not NP.

4.2 Cost of equity and of retained earnings

Why equity and retained earnings are not free: the five ICAI approaches to the cost of equity (dividend price, earnings price, growth, realised yield, CAPM) and the cost of retained earnings.

In plain words

"Shareholders don't send us an interest bill"

Dev is pricing every source of money for Tapri's expansion. Kesar thinks he has spotted a short cut.

> **Kesar:** Debt has interest, preference has a fixed dividend. Equity? We pay what we like, or nothing. And the profits we kept back are our own money. Both are free!
>
> **Dev:** Our shareholders paid ₹240 a share. They are not being charitable: they expect a return on that ₹240, and if we stop delivering it the price falls until the new, lower price gives them the return they want.
>
> **Kesar:** And the profits we kept?
>
> **Dev:** Every rupee we kept is a dividend they did not get. They could have invested it themselves. Free money has the longest invoice.

The **cost of equity (Ke)** is the return equity shareholders expect from the company. Nobody prints it on the share, because the dividend is not fixed, so it is found as a balancing figure: value is performance divided by expectations, so if you know the value (the share price) and the performance (the dividends or earnings), the expectation drops out (SM 4.19). Different assumptions about the performance give different methods; a fifth method, CAPM, builds the expectation from risk instead.

The **cost of retained earnings (Kr)** is the opportunity cost of the dividends shareholders gave up (SM 4.28).

Think of it like

The flat that pays rent

A flat earns ₹3 lakh of rent a year and sells for ₹60 lakh. Buyers paying ₹60 lakh are telling you they expect 5% a year. If the rent is expected to rise 4% every year, they are content with a 5% rent yield because the rising rent adds the other 4%: they expect 9%. That is Gordon's model: Ke = next dividend ÷ price + growth. The share is the flat, the dividend is the rent, and the price says what the market expects.

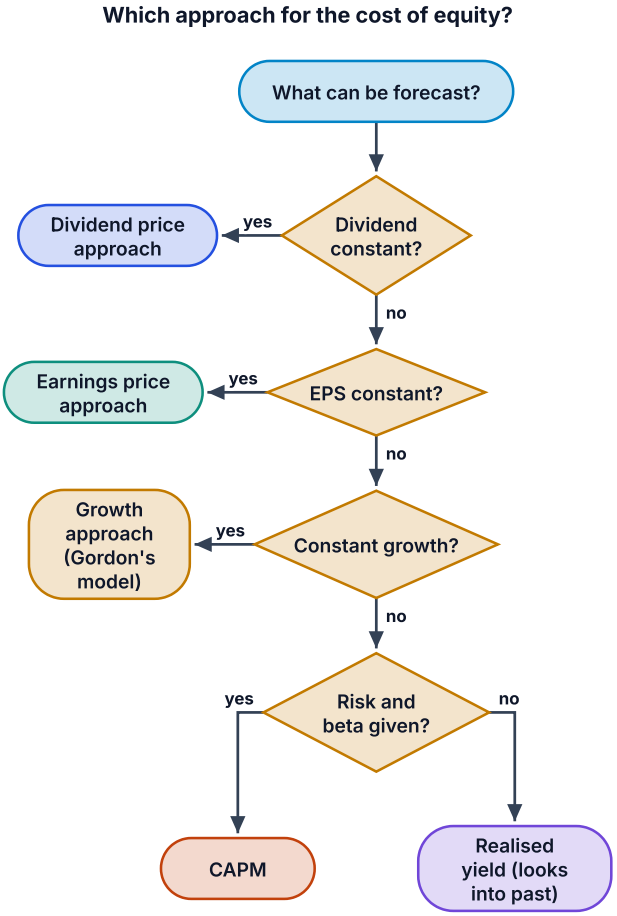
Key idea

Equity has a cost even though no payment is fixed: it is the return shareholders expect, read off the share price (dividend or earnings approaches, Gordon's model, realised yield) or built from risk (CAPM). Retained earnings cost the shareholders' forgone return, normally a little below Ke because there is no flotation cost.

How it works

Five approaches, and when ICAI uses each

The Study Material ties the method to the data (SM 4.19):



ICAI's guide (SM 4.19): constant dividend, dividend price; constant EPS, earnings price; constant growth, Gordon's model; hard to forecast, realised yield; expectation driven by risk, CAPM. The first four find Ke as a balancing figure from the price; CAPM builds it from risk.

The approaches one by one

1 **Dividend price approach** (the dividend valuation model): the dividend per share is expected to stay constant forever, so Ke = expected dividend ÷ market price (ex-dividend) (SM 4.20).

2 **Earnings price approach**: each investor expects a certain earning, distributed or not, so Ke = EPS ÷ market price, assuming EPS stays constant forever. It is similar to the dividend price approach, but it seeks to nullify the effect of changes in the dividend policy (SM 4.20–4.21). ICAI's example: an investor wanting 20% pays ₹150 for a ₹100 share expected to earn 30%.

3 **Growth approach (Gordon's model)**: earnings, dividends and share price all grow at a constant rate g, so Ke = D1 ÷ P0 + g, with D1 = D0(1 + g), the next expected dividend. For a new issue, deduct the flotation cost per share from the price (SM 4.21).

4 **Realised yield approach**: the average return realised in the past few years is taken as the expected return: the IRR that equates the purchase price with the dividends received and the sale price (SM 4.24).

5 **CAPM**: the return investors require for the risk they cannot diversify away: Ke = Rf + β(Rm − Rf) (SM 4.26).

Where g comes from

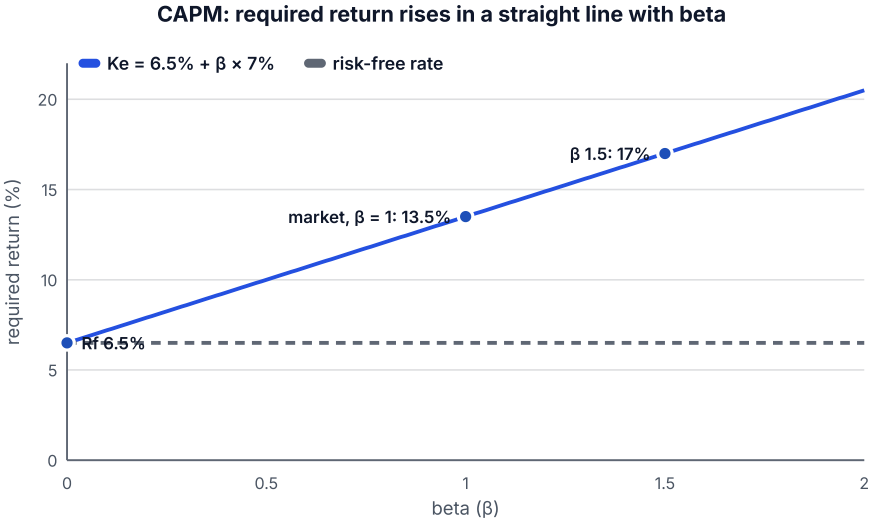
Two methods (SM 4.22–4.23):

- **Average method**: the current dividend $D0 = Dn(1 + g)^n$, where Dn is the dividend n years ago, so $g = \sqrt[n]{(D0 \div Dn)} - 1$. Without a calculator: divide D0 by Dn, find the result in the FVIF table's row for n years, and read the rate at the top of that column. ICAI's Example 4: ₹16.10 ÷ ₹10 = 1.61 in the 5-year row sits under 10%, so g = 10%.
- **Gordon's growth model**: g = b × r, where b is the earnings retention rate (the share of earnings available to equity shareholders not paid as dividend) and r the rate of return on funds invested. Reinvested earnings are what make future dividends grow.

CAPM: paid only for the risk you cannot shake off

A security's risk has two parts (SM 4.26). **Unsystematic risk** is company specific (a strike or lockdown in the company, labour not available); it can be reduced or eliminated by diversification. **Systematic risk** is macro-economic or market specific (inflation, government policy, interest rates) and cannot be diversified away. So under CAPM a business should be concerned solely with the non-diversifiable risk, measured by **beta (β)**, from a regression of the security's returns on the market portfolio's returns.

Investors want two rewards: time value of money (the risk-free rate) and risk (beta times the market risk premium, Rm − Rf) (SM 4.27). Its shortcomings: estimating beta from historical data is unrealistic, and market imperfections may expose investors to unsystematic risk. Despite them, CAPM is useful in calculating the cost of equity even when the entity is suffering losses (SM 4.28).

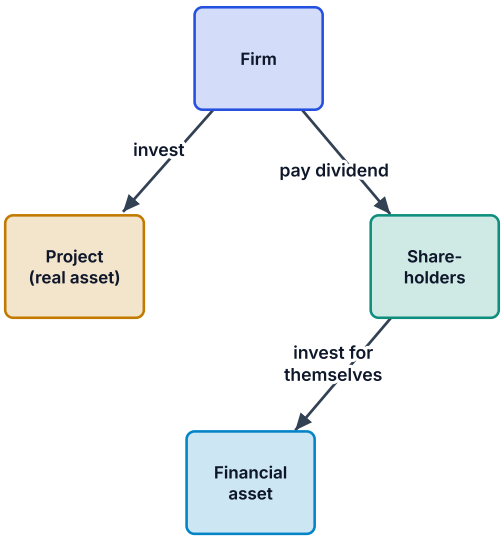


An illustration with Rf = 6.5% and Rm = 13.5%: the market risk premium is 7%. A share with β = 1 moves with the market and needs the market's return; β = 1.5 needs 6.5% + 1.5 × 7% = 17%. The grey dashed line is the time-value part; the gap above it is the risk premium.

Interactive version: PassBook app, lesson Cost of equity and of retained earnings

Retained earnings: the dividend that stayed home

When a company keeps profit instead of paying it out, shareholders lose the chance to invest that cash themselves in financial assets. The return they could have earned there is the cost of the retained earnings (SM 4.28–4.29).



Each box in its own colour. Cash the firm keeps goes into a project (a real asset); paid out as dividend, it lets shareholders invest for themselves in a financial asset. The return they could have earned there is Kr (after the Study Material's figure, SM 4.29).

The same formulas serve for Kr, with one difference: for Ke on a new issue, P is the net proceeds (issue price less flotation cost); for Kr, P is the current market price (sometimes the issue price), and

flotation cost is not used (SM 4.29). Kr is often used interchangeably with Ke, but cost of equity remains higher than the cost of retained earnings normally, because new shares are issued below the market price and carry flotation costs.

Interactive version: PassBook app, lesson Cost of equity and of retained earnings

Formats and formulas

Reproduce each formula and what its symbols mean from memory.

∑ **Formula**

Dividend price approach (SM 4.20)

$$K_e = \frac{D}{P_0}$$

D = expected dividend (also written D1); P_0 = market price of equity (ex-dividend).

∑ **Formula**

Earnings price approach (SM 4.21)

$$K_e = \frac{E}{P}$$

E = current earnings per share; P = market price per share.

∑ **Formula**

Growth approach, Gordon's model (SM 4.21)

$$K_e = \frac{D_1}{P_0} + g \qquad K_e = \frac{D_1}{P_0 - F} + g$$

$D_1 = D_0(1+g)$, the next expected dividend; P_0 = current market price per share; g = constant growth rate of dividend; F = flotation cost per share (second form, for newly issued shares).

∑ **Formula**

Growth rate (SM 4.22–4.23)

$$g = \sqrt[n]{\frac{D_0}{D_n}} - 1 \qquad g = b \times r$$

D_0 = current dividend; D_n = dividend n years ago; b = earnings retention rate; r = rate of return on funds invested.

Σ Formula

Realised yield, year-by-year data (SM 4.25)

$$1 + Y_t = \frac{D_t + P_t}{P_{t-1}} \qquad K_e = \left[(1+Y_1)(1+Y_2)\cdots(1+Y_n)\right]^{1/n} - 1$$

Each year's yield is dividend plus closing price over opening price; Ke is their geometric mean. With a purchase price, a series of dividends and a sale price instead, Ke is the IRR of those flows.

Σ Formula

CAPM (SM 4.26–4.27)

$$K_e = R_f + \beta\,(R_m - R_f)$$

R_f = risk-free rate of return; β = beta coefficient; R_m = rate of return on market portfolio; $(R_m - R_f)$ = market risk premium.

Σ Formula

Cost of retained earnings (SM 4.29)

$$K_r = \frac{D}{P} \qquad K_r = \frac{EPS}{P} \qquad K_r = \frac{D_1}{P_0} + g$$

P = current market price (sometimes the issue price); no flotation cost.

Definition

Systematic and unsystematic risk

Unsystematic risk: company-specific risk related to the company's performance, which can be reduced or eliminated by diversification (diversifiable risk). Systematic risk: the macro-economic or market-specific risk under which a company operates, which cannot be eliminated by diversification (non-diversifiable risk). (SM 4.26)

Definition

Cost of retained earnings

The opportunity cost of dividends foregone by shareholders: the expected rate of return shareholders could have obtained by investing in financial assets. (SM 4.28)

Worked examples

Worked example **Tapri's share, read three ways**

Tapri Tea Co. Ltd's equity shares of ₹10 each are quoted at ₹240 (ex-dividend). The dividend just paid is ₹12 per share; five years ago it was ₹8.17, and the growth is expected to continue. EPS is ₹30. Tapri plans to raise part of its expansion by a new issue at ₹230 per share, with a flotation cost of ₹14 per share, and the rest from retained earnings. FVIF (8%, 5 years) = 1.469.

Required:

(i) Estimate the growth rate of dividends.

(ii) Calculate the cost of the existing equity by Gordon's model.

(iii) Calculate the cost of the new equity shares.

(iv) Calculate the cost of retained earnings.

(v) Calculate the cost of equity by the earnings price approach, and comment on the difference.

THE WORKING

(i) Growth rate (average method)

D0 ÷ D5 = ₹12 ÷ ₹8.17 = 1.469. In the FVIF table's 5-year row, 1.469 sits under 8%, so **g = 8%**. Check: (1.469)^(1/5) − 1 = 8.0%.

(ii) Existing equity

D1 = ₹12 × 1.08 = ₹12.96.

$$K_e = \frac{D_1}{P_0} + g = \frac{12.96}{240} + 0.08 \qquad = 0.054 + 0.08 = 0.134 \text{ or } 13.40\%$$

(iii) New equity shares

$$K_e = \frac{D_1}{P_0 - F} + g = \frac{12.96}{230 - 14} + 0.08 \qquad = \frac{12.96}{216} + 0.08 = 0.06 + 0.08 = 14.00\%$$

(iv) Retained earnings: current market price, no flotation cost:

$$K_r = \frac{12.96}{240} + 0.08 = 13.40\%$$

(v) Earnings price approach

$$K_e = \frac{E}{P} = \frac{30}{240} = 12.50\%$$

Comment. The earnings price approach assumes EPS will stay at ₹30 forever and ignores the 8% growth Tapri's dividends show, so it understates the expectation; with constant growth, Gordon's model is the method the Study Material prescribes (SM 4.19). New equity (14.00%) costs more than retained earnings (13.40%) because it is sold ₹10 below the market price and carries ₹14 of flotation cost.

Worked example **Kulhad Crafts: what the past paid, what risk asks**

Kulhad Crafts Ltd makes Tapri's clay cups. Kesar's neighbour, Mrs. Sunita Joshi, bought its shares at ₹400 each four years ago. She received dividends of ₹24, ₹28, ₹32 and ₹36 per share at the end of years 1 to 4, and has just sold the shares at ₹384.

Dev's brief: the 10-year government bond yields 7%, the market is expected to return 13%, and Kulhad Crafts' beta is 0.9. Kulhad Crafts plans an export division which, its analysts say, will raise its beta to 1.2.

Year	PVF @ 6%	PVF @ 7%
1	0.943	0.935
2	0.890	0.873
3	0.840	0.816
4	0.792	0.763

Required:

(i) Calculate the cost of equity of Kulhad Crafts by the realised yield approach, from Mrs. Joshi's experience.

(ii) Calculate the cost of equity by CAPM, now and after the export division.

(iii) Comment on the difference between (i) and (ii), and state which estimate Dev should use.

THE WORKING

(i) Realised yield: the rate that equates the ₹400 paid with the present value of the dividends and the sale price.

Year	Cash flow (₹)	PV @ 6% (₹)	PV @ 7% (₹)
0	(400.00)	(400.00)	(400.00)
1	24.00	22.63	22.44
2	28.00	24.92	24.44
3	32.00	26.88	26.11
4	36 + 384 = 420.00	332.64	320.46
NPV		**+7.07**	**−6.55**

$$K_e = 6\% + \frac{7.07}{7.07 - (-6.55)} \times (7\% - 6\%) \qquad = 6\% + 0.52\% = 6.52\%$$

(ii) CAPM

Market risk premium = Rm − Rf = 13% − 7% = 6%.

$$K_e = R_f + \beta(R_m - R_f) = 7\% + 0.9 \times 6\% = 12.40\%$$

After the export division: Ke = 7% + 1.2 × 6% = **14.20%**, 1.80 percentage points more, for the extra systematic risk.

(iii) Comment. Mrs. Joshi's four years earned her 6.52%, less than even the 7% risk-free rate, against the 12.40% that Kulhad Crafts' risk requires. The realised yield only records what the past happened to pay: it assumes that the risks faced by the company remain the same, that shareholders continue to expect the same rate of return and that their reinvestment rate is the realised yield, and it is not practical when earnings do not remain stable (SM 4.24); a falling share price and uneven dividends are exactly that case. CAPM prices the systematic risk the shareholders bear today, so Dev should use **12.40%**, and **14.20%** once the export division changes the company's beta.

Exam angle

In the exam

Where the cost of equity appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, Ke or Kr was computed or discussed 18 times in 15 documents:

- **CAPM**, the most frequent: May 2024 Q2(a)(i) (18.32%), January 2025 Q2(a)(i) (18%), May 2025 Q2(a)(i) (20%), January 2026 MCQ 1, May 2026 MCQ 7 (a required return compared with a project's 25%), RTP May 2026 Q5 (Rf and the market risk premium found from two peers, then the company's beta), MTP Series II for September 2026 Q2(a) (4 marks, a company and two competitors).
- **Gordon's model**: September 2025 Q1(a)(i) (2 marks, new issue with ₹4 flotation, 8.75%), May 2025 MCQ 8 (with flotation, 12%), RTP January 2025 Q5 (last year's dividend ₹4, ₹5 flotation, 26.44%), RTP May 2024 Q2 (g of 11% from an EPS trend, Kr 16%), MTP Series II for May 2026 Q3(a) (dividend "@24%" on a ₹100 share), MTP Series I for September 2024 MCQs 1–2 (g from the share price; Gordon and CAPM averaged with weights 0.6 and 0.4, 17.82%).
- **Theory**: "DISCUSS the dividend-price approach" (MTP Series I for May 2024 Q3(b), 2 marks); the dividend price and earnings price approaches (MTP Series II for May 2026 Q4(c), 2 marks); "EXPLAIN any four methods" (RTP January 2025 Q11(b)).

In the exam

The traps ICAI used

- **Cum-dividend price** (September 2024 MCQ 7): a share at ₹75 cum-dividend with ₹6 about to be paid is ₹69 ex-dividend, and g = ₹7.59 ÷ ₹6 − 1 = 26.5%, so Ke = 7.59 ÷ 69 + 26.5% = 37.5%.
- **D0 or D1?** "Paid last year", "just paid" or "this year" is D0: multiply by (1 + g). "Expected", "next year" is D1.
- **Dividend as a percentage** is of the face value: 24% on ₹100 is ₹24 (MTP Series II for May 2026).
- **Personal tax on retained earnings**: May 2024 Q2(a)(iv) gave a personal tax rate of 20% and the Suggested Answer computed Kr = Ke × (1 − 0.20) = 14.66%, adding that Kr = Ke (18.32%) may also be taken. The Study Material gives no such adjustment (SM 4.28–4.29), and when RTP January 2026 Q6 repeated the question without the personal tax rate, its answer took Kr = Ke = 18.32%. Show the adjustment only if the question gives the rate.
- **The price for Ke**: new issue, issue price less flotation; for Kr, the market price.

Trick

Gordon's two pieces

Ke = dividend yield + growth: 12.96 ÷ 240 = 5.4% yield plus 8% growth = 13.4%. If an MCQ's options include the bare yield (5.4%), it is the "forgot g" distractor. Condition: constant growth forever, with g below Ke.

Answer frame

DISCUSS the dividend price approach and the earnings price approach (2 marks; SM 4.20–4.21):

1. Dividend price approach (dividend valuation model): assumes the dividend per share remains constant forever; cost of equity = expected dividend ÷ market price per share (ex-dividend); Ke = D1 ÷ P0, symbols defined.
2. Earnings price approach: co-relates the earnings of the company with the market price of its share; each investor expects a certain earning, distributed or not; Ke = E ÷ P (EPS ÷ market price); assumes EPS constant forever.
3. The difference: the earnings price approach seeks to nullify the effect of changes in the dividend policy.

EXPLAIN any four methods of computing the cost of equity: RTP January 2025's answer opens with one line, "the rate of return which equates the present value of expected dividends with the market share price", then gives a heading, one or two lines and the formula for each method. Choose four of the dividend price, earnings price, growth (Gordon) and realised yield approaches and CAPM, and add when each is used (SM 4.19).

Common mistakes

Common mistake

Hall of Shame: last year's dividend in Gordon's numerator

₹12 ÷ ₹240 + 8% = 13.0%. The ₹12 was just paid; Gordon's model needs the next one. **Fix:** D1 = D0 × (1 + g) = ₹12.96, and Ke = 13.40%.

Common mistake

Flotation cost charged to retained earnings

Writing Kr = 12.96 ÷ (240 − 14) + 8%. **Fix:** retained earnings involve no issue, so no flotation: use the current market price. Flotation belongs only to the new-issue Ke.

Common mistake

The cum-dividend price used as P0

Dividing by a price that still includes a dividend about to be paid overstates P0. **Fix:** P0 is ex-dividend: subtract the dividend payable from the cum-dividend price first.

⚠ **Common mistake**

Beta times the market return

Ke = Rf + β × Rm (7% + 0.9 × 13% = 18.7%). **Fix:** beta multiplies the premium, Rm − Rf: 7% + 0.9 × 6% = 12.40%. Read the question: "market risk premium" or "equity risk premium" is already Rm − Rf; "market return" is Rm.

Practice

Check yourself

1 A share is quoted at ₹106 ex-dividend. The dividend just paid is ₹5 and dividends grow at 6% a year. What is the cost of equity by Gordon's model, in %?

Answer: _____

2 Case scenario. Kulhad Crafts' share is quoted at ₹265 cum-dividend; the final dividend of ₹15 a share will be paid next week. EPS is ₹25, the company keeps the rest of its earnings and earns 20% on the funds it reinvests, and it will keep this policy. Ritu must state the cost of equity in the board note. Which figure is correct?

A 14.11%

B 14.00%

C 14.48%

D 13.66%

3 Why is the cost of retained earnings normally lower than the cost of new equity?

A Retained earnings belong to the company, not the shareholders

B New shares are issued below the market price and carry flotation costs, which retained earnings do not

C Retained earnings are taxed at a lower rate

D Retained earnings earn no dividend

4 A company retains 40% of its earnings and earns 15% on the funds it invests. By Gordon's growth model, what is its growth rate?

A 15%

B 9%

C 6%

D 40%

5 Which of these risks does a share's beta capture under CAPM? Select all that apply.

- A A rise in interest rates
- B A strike at the company's factory
- C Inflation
- D A change in government policy

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Ke = what equity shareholders expect; found as a balancing figure from the price, or from risk (CAPM).
- Dividend price: D ÷ P0; earnings price: EPS ÷ P (nullifies dividend policy); both assume constancy forever.
- Gordon: Ke = D1 ÷ P0 + g; new issue D1 ÷ (P0 − F) + g; D1 = D0(1 + g).
- g = $\sqrt[n]{(D0 \div Dn)} - 1$ (or the FVIF table), or g = b × r.
- Realised yield: the IRR of purchase price, dividends and sale price, or the geometric mean of yearly yields.
- CAPM: Ke = Rf + β(Rm − Rf); beta prices systematic risk only; usable even when the entity is suffering losses.
- Kr: same formulas at the market price, no flotation; normally Kr < Ke. (Free money has the longest invoice.)

Answer key · Check yourself

1 **10.9 to 11.1**

D1 = ₹5 × 1.06 = ₹5.30; Ke = 5.30 ÷ 106 + 6% = 5% + 6% = 11%. Using ₹5 instead of D1 gives 10.72%.

2 **C 14.48%**

Retention b = (25 − 15) ÷ 25 = 40%, so g = b × r = 0.40 × 20% = 8%. The ex-dividend price is ₹265 − ₹15 = ₹250 and D1 = ₹15 × 1.08 = ₹16.20, so Ke = 16.20 ÷ 250 + 8% = 14.48%. 14.11% divides by the cum-dividend price; 14.00% uses D0 instead of D1; 13.66% makes both mistakes.

3 **B New shares are issued below the market price and carry flotation costs, which retained earnings do not**

Both are the shareholders' expected return; new equity is dearer only because of the lower issue price and the flotation cost (SM 4.29). Retained earnings are the shareholders' forgone dividends, so they do have a cost.

4 **C 6%**

g = b × r = 0.40 × 15% = 6%. 9% uses the payout ratio (60%) instead of the retention ratio.

5 **A** A rise in interest rates; **C** Inflation; **D** A change in government policy

Beta measures systematic risk (inflation, government policy, interest rates), which diversification cannot remove. A strike is company-specific, unsystematic risk, diversified away in a portfolio (SM 4.26).

4.3 WACC and the marginal cost of capital

Why a company's hurdle rate is the weighted average of its sources' costs, how book and market value weights change it, and how the marginal cost of capital rises as new funds get dearer.

In plain words

Kesar picks the cheapest dish on the thali

Dev has priced every source of Tapri's money: debentures 7.70% after tax, preference shares 10.53%, retained earnings 12%, new equity 12.38%. Kesar reads only the first line.

> **Kesar:** Debt costs 7.70%. So any new café that earns 8% is profit. Approve them all!
>
> **Dev:** Kesar-ji, no café is funded by debt alone. If we borrowed for every café, the debt would make the shares riskier and the shareholders would want more. Every rupee we invest comes from the same mix: thirty paise of debt, ten of preference, sixty of equity.
>
> **Ritu:** So the price of the rupee is the mix, weighted.
>
> **Dev:** Per unit, as you always say.

A company rarely raises all its money from one source: it mixes sources to balance financial risk, control and cost. So the cost of its total capital is the **weighted average cost of capital (WACC)**: each source's after-tax cost, weighted by its share of total capital (SM 4.31). It is the company's hurdle rate: a project that earns less than the WACC destroys value, however cheap one of its sources looks.

Two questions then follow. **Which weights?** The amounts in the balance sheet (book value) or what the securities are worth today (market value). **And for new money?** The next rupee may cost more than the last: once this year's retained earnings are used up, new shares must be sold at a lower price with flotation costs, and lenders may charge more beyond a limit. The cost of that additional rupee is the **marginal cost of capital (MCC)**.

Think of it like

The chai blend

Tapri's masala chai blends Assam leaf at ₹400 a kg with Darjeeling at ₹1,200, three kilos to one. A kilo of the blend costs (3 × 400 + 1 × 1,200) ÷ 4 = ₹600: a weighted average, not the cheapest leaf and not a simple average of the two prices. Weigh the leaf in the godown at what it cost years ago and you get book value weights; at today's auction prices, market value weights. And once the cheap Assam lot runs out, the next kilo of blend costs more: that is the marginal cost.

Key idea

WACC is the after-tax cost of each source weighted by its proportion of total capital, and it is the minimum return at which the company creates value. Market value weights reflect today's prices; marginal weights are the proportions of new funds, and the marginal cost rises at each break point where a cheaper source runs out.

How it works

Why the cost of capital matters

Cost of capital is the return expected by the providers of capital. It is also called the **cut-off rate**, **hurdle rate** or **minimum rate of return**, and it is the benchmark for framing debt policies and taking capital budgeting decisions (SM 4.2). The correct figure helps in three decisions (SM 4.2–4.3):

1. **Evaluation of investment options**: future cash flows are converted into present values by discounting them at the relevant cost of capital; each option may have its own cost of capital, so use the one relevant to it.
2. **Financing decision**: choosing between two sources, the finance manager compares their costs and picks the lower, considering financial risk and control as well.
3. **Designing of optimum credit policy**: the cost of allowing a credit period is compared with the benefit from it, with the cost of capital used to find the present values of costs and benefits.

What pushes it up or down

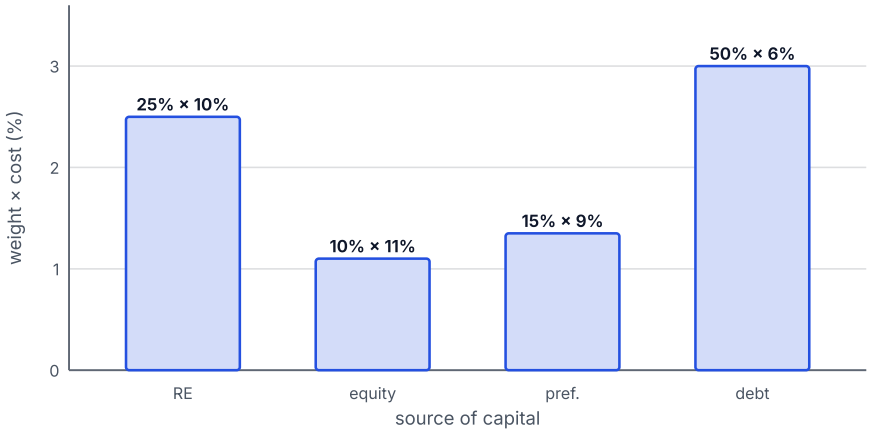
ICAI lists five factors, in this order (SM 4.4–4.5):

1. **Supply and demand**: many exciting business opportunities compete for funding and push the cost of capital up; when opportunities are scarce, investors lend at lower rates.
2. **Investor preferences**: cultural factors and saving habits decide the pool of capital; higher returns generally lead to more savings.
3. **Risk and return**: high-risk ventures must offer higher potential returns, but too high a cost of capital can make a project unprofitable.
4. **Inflation**: investors look for returns that outpace inflation, so that their money keeps its buying power.
5. **Exchange rates**: for multinational companies, their fluctuations add risk and affect the overall cost of capital.

WACC in four steps

1. **Total capital** from all sources: long-term debt + preference share capital + equity share capital + retained earnings.
2. **Proportion of each source**: its amount ÷ total capital.
3. **Multiply** each proportion by that source's after-tax cost (Kd, Kp, Ke, Kr).
4. **Add** the products: the total is the WACC (SM 4.31).

ICAI's Example 5: each source's share of a 7.95% WACC



RE: retained earnings; pref.: preference shares. Each bar is one source's weight times its cost (SM 4.32): 2.50% + 1.10% + 1.35% + 3.00% = 7.95%. Debt has the lowest cost but the tallest bar, because half the capital is debt.

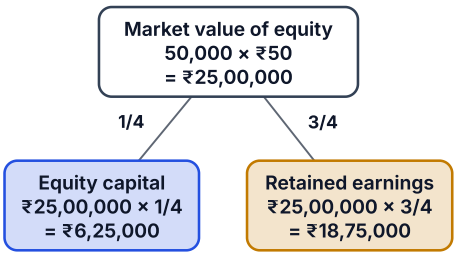
The weighted average is used because the sources are present in different proportions; to be representative, the cost of capital must reflect them (SM 4.32). WACC is the discount rate in discounted cash flow analysis and the hurdle rate against which the return on capital is judged: if a company earns 20% and its WACC is 11%, it creates about 9% of value for its investors; earning less than the WACC, it is shedding value (SM 4.32).

Book value or market value weights

Book value (BV) weights are operationally easy and convenient; reserves such as share premium and retained profits are included in the book value of equity. **Market value (MV)** weights are more correct and represent the firm's capital structure: they represent existing conditions and take into consideration changing market conditions and the current prices of the securities. They are preferable for equity, and better for debt too, though the difference there may not be significant. Under MV, reserves are ignored, because they are in effect incorporated in the market value of equity (SM 4.33).

That creates a puzzle: retained earnings have no market value of their own, yet their cost (Kr) differs from that of equity (Ke). ICAI's answer: apportion the market value of equity shares in the ratio of the book values of paid-up equity capital and retained earnings (SM 4.33).

Illustration 16: splitting the market value of equity



Book values: paid-up equity ₹5,00,000 and retained earnings ₹15,00,000, a ratio of 1 : 3 (SM 4.34). Each part then carries its own cost: 10.41% (Ke) and 10% (Kr), giving a WACC of 10.10%.

The Study Material compares the two on five points, in this order (SM 4.33–4.34):

Point of comparison	Book value (BV)	Market value (MV)
1. Definition	Based on the historical cost of assets and liabilities recorded on the balance sheet	Based on the current market price of equity and debt
2. Reflects current conditions	May not accurately reflect current market conditions or the true economic value	More accurately reflects current market conditions and the value investors place on the company
3. Stability	Generally more stable; does not fluctuate with market conditions	Can be volatile and fluctuate with market conditions
4. Relevance for investors	Less relevant, as it does not consider current market perceptions	More relevant, as it reflects the market's view of the company's value
5. Ease of calculation	Easier, using the financial statements	Requires market data, which can be more complex to gather and calculate

Interactive version: PassBook app, lesson WACC and the marginal cost of capital

The marginal cost of capital

The **marginal cost of capital** is the cost of raising an additional rupee of capital; since capital is raised in substantial amounts in practice, it is the cost of raising new funds over and above the existing. It is the average cost of capital calculated with **marginal weights**: the proportions of funds the firm intends to employ, so the choice between book and market value weights does not arise (SM 4.38).

If a firm raises funds in its usual proportions and the component costs stay unchanged, the marginal and average costs are the same. But component costs may stay constant up to a level of funds and then rise: debt may cost 7% after tax up to ₹10 lakh and 8% between ₹10 lakh and ₹15 lakh; equity becomes dearer once retained earnings run out and new shares with flotation costs are needed. When the component costs start rising, the average cost rises, and the marginal cost rises at a faster rate (SM 4.38).

Each point where a cheaper source runs out is a **break point** (the Study Material does not use the name; its Illustration 18(C) computes the first one):

$$\text{Break point} = \frac{\text{cheaper source available}}{\text{its proportion in new funds}}$$

With retained earnings of ₹11,800 and equity at 80% of new funds, ICAI's Illustration 18 finds that ₹11,800 ÷ 0.80 = ₹14,750 can be spent before new shares must be sold (SM 4.41).

Interactive version: PassBook app, lesson WACC and the marginal cost of capital

Formats and formulas

Reproduce this layout and the formulas from memory; ICAI's Suggested Answers set out every WACC this way.

Source of capital	Amount (₹)	Weight (a)	After-tax cost (b)	(a) × (b)
Equity share capital	xxx	0.xxxx	Ke	xxx
Retained earnings	xxx	0.xxxx	Kr	xxx
Preference share capital	xxx	0.xxxx	Kp	xxx
Debentures / long-term debt	xxx	0.xxxx	Kd	xxx
Total	**xxx**	**1.0000**		**WACC**

Book value weights take the balance sheet amounts. Market value weights take number × market price for each security, with the market value of equity split between share capital and retained earnings in their book value ratio. Marginal weights take the proportions of the new funds.

∑ **Formula**

WACC

$$K_o = W_e K_e + W_r K_r + W_p K_p + W_d K_d$$

K_o = WACC (overall cost of capital); each W = that source's amount ÷ total capital, so the weights add up to 1; each K = that source's after-tax cost.

∑ **Formula**

Market value of equity split (SM 4.33)

$$MV_{EC} = MV_E \times \frac{BV_{EC}}{BV_{EC} + BV_{RE}}$$

MV_E = market value of equity shares (number × price); BV_{EC}, BV_{RE} = book values of paid-up equity capital and retained earnings. The rest of MV_E is the market value of retained earnings.

∑ **Formula**

Marginal cost of capital and break point (SM 4.38; Illustration 18, SM 4.39–4.41)

$$MCC = \sum(\text{proportion of new funds} \times \text{marginal cost of the source})$$

$$\text{Break point} = \frac{\text{cheaper source available}}{\text{its proportion}}$$

Below the first break point, retained earnings and the cheapest debt are used; above it, the dearer component cost replaces the cheaper one.

Definition

Weighted average cost of capital

The weighted average after-tax costs of the individual components of the firm's capital structure: the after-tax cost of each debt and equity is calculated separately and added together to a single overall cost of capital. (SM 4.31)

Definition

Marginal cost of capital

The cost of raising an additional rupee of capital; since capital is raised in substantial amount in practice, it is the cost incurred in raising new funds. It is derived when the average cost of capital is calculated using the marginal weights, which represent the proportion of funds the firm intends to employ. (SM 4.38)

Worked examples

Worked example **Tapri's WACC on two kinds of weights**

The capital structure of Tapri Tea Co. Ltd on 31 March 2026 is:

Particulars	Amount (₹ lakh)
Equity share capital (30 lakh shares of ₹10 each)	300
Retained earnings	300
10% irredeemable preference shares of ₹100 each	100
9% debentures of ₹100 each, redeemable at par after 5 years	300
Total	**1,000**

Market prices: equity share ₹50, preference share ₹95, debenture ₹96. Next year's dividend is expected to be ₹3.00 per share, growing at 6% a year. New equity shares would realise ₹47 per share net of flotation cost. The tax rate is 25%.

Required:

(i) Calculate the cost of each source of finance.

(ii) Calculate the WACC using book value weights.

(iii) Calculate the WACC using market value weights.

THE WORKING

(i) Specific costs

- Equity shares (net proceeds of a new issue): Ke = ₹3.00 ÷ ₹47 + 6% = 6.38% + 6% = **12.38%**.
- Retained earnings (market price, no flotation): Kr = ₹3.00 ÷ ₹50 + 6% = **12.00%**.
- Preference shares: Kp = ₹10 ÷ ₹95 = **10.53%**.
- Debentures (approximation method): Kd = [₹9 × 0.75 + (₹100 − ₹96) ÷ 5] ÷ [(₹100 + ₹96) ÷ 2] = ₹7.55 ÷ ₹98 = **7.70%**.

(ii) WACC, book value weights

Source	Book value (₹ lakh)	Weight	Cost	Weight × cost
Equity shares	300	0.3000	0.1238	0.03714
Retained earnings	300	0.3000	0.1200	0.03600
Preference shares	100	0.1000	0.1053	0.01053
Debentures	300	0.3000	0.0770	0.02310
Total	**1,000**	**1.0000**		**0.10677**

WACC (BV) = 10.68%.

(iii) WACC, market value weights

Working note: market value of equity = 30 lakh × ₹50 = ₹1,500 lakh, split in the book value ratio 300 : 300 = 1 : 1: ₹750 lakh to equity capital and ₹750 lakh to retained earnings. Preference: 1 lakh × ₹95 = ₹95 lakh; debentures: 3 lakh × ₹96 = ₹288 lakh.

Source	Market value (₹ lakh)	Weight	Cost	Weight × cost
Equity shares	750	0.3983	0.1238	0.04931
Retained earnings	750	0.3983	0.1200	0.04780
Preference shares	95	0.0505	0.1053	0.00532
Debentures	288	0.1529	0.0770	0.01177
Total	**1,883**	**1.0000**		**0.11420**

WACC (MV) = 11.42%.

Comment. Equity trades at five times its face value, so market values give equity (the dearest source) almost 80% of the weight against 60% in the books, and the WACC rises from 10.68% to 11.42%. The market value figure is the one that reflects what investors expect today. Convention: equity shares are costed on the net proceeds of a new issue, as in the Study Material's own solutions (SM 4.36, 4.55); if they are costed at the market price instead (Ke = Kr = 12%, as MTP Series II for May 2026 Q3(a) did when no flotation cost was given), the WACC is 10.56% on book and 11.27% on market weights. State the convention you use. Weights and costs to four decimals, products to five (rounding each product to four decimals first would show 10.67%).

Worked example **Thanda Logistics borrows more: the new WACC at market values**

Thanda Logistics Ltd, a listed cold-chain company that carries Tapri's frozen snacks, has 2 lakh equity shares quoted at ₹150 and ₹150 lakh of 10% irredeemable debentures of ₹100 each, quoted at par. Next year's dividend is expected to be ₹12 per share, growing at 4% a year forever. The tax rate is 25%.

Thanda plans to borrow ₹150 lakh more as a 12% term loan at par, secured ahead of the debentures. Its bankers expect that, after the loan, the expected dividend will rise to ₹13.20 with growth unchanged, the

share price will fall to ₹132, and the existing debentures will fall to ₹90 because they now rank behind the loan.

Required:

(i) Calculate the WACC now, using market value weights.

(ii) Calculate the WACC after the loan, using market value weights, and comment.

THE WORKING

Working notes. Now: Ke = ₹12 ÷ ₹150 + 4% = 12.00%; Kd = ₹10 × 0.75 ÷ ₹100 = 7.50%. After the loan: Ke = ₹13.20 ÷ ₹132 + 4% = 14.00%; existing debentures, on their new market price, Kd = ₹7.50 ÷ ₹90 = 8.33%; term loan Kd = 12% × 0.75 = 9.00%.

Market values: equity 2 lakh × ₹150 = ₹300 lakh now and 2 lakh × ₹132 = ₹264 lakh after; debentures 1.5 lakh × ₹100 = ₹150 lakh now and 1.5 lakh × ₹90 = ₹135 lakh after.

(i) WACC now

Source	Market value (₹ lakh)	Weight	Cost (%)	Weighted cost (%)
Equity	300	0.667	12.00	8.00
10% debentures	150	0.333	7.50	2.50
Total	**450**	**1.000**		**10.50**

(ii) WACC after the loan

Source	Market value (₹ lakh)	Weight	Cost (%)	Weighted cost (%)
Equity	264	0.481	14.00	6.73
10% debentures	135	0.246	8.33	2.05
12% term loan	150	0.273	9.00	2.46
Total	**549**	**1.000**		**11.24**

Comment. The WACC rises from 10.50% to 11.24%. The loan itself is cheap (9.00% after tax), but the extra financial risk raises the cost of equity from 12.00% to 14.00% and pushes the old debentures' yield from 7.50% to 8.33%; together they more than offset the cheap tranche. On these figures the loan does not lower Thanda's cost of capital. Each source is costed on its own current market price, and the market values, not the book figures, give the weights.

Worked example **Tapri's ₹2.5 crore expansion: the marginal cost of capital**

Tapri will raise all new funds for its expansion in its target proportions: 40% debt and 60% equity. Next year's EPS is expected to be ₹6.40, of which 40% will be paid as dividend; Tapri earns 15% on the earnings it retains, and the share is quoted at ₹40. Retained earnings of ₹72 lakh are available for the programme; beyond them, new shares can be sold at the market price less a flotation cost of 8%.

Indore Sahakari Bank will lend up to ₹40 lakh at 10% and a further ₹40 lakh at 11%; any more debt must be raised as 12% debentures at par. The tax rate is 25%.

Required:

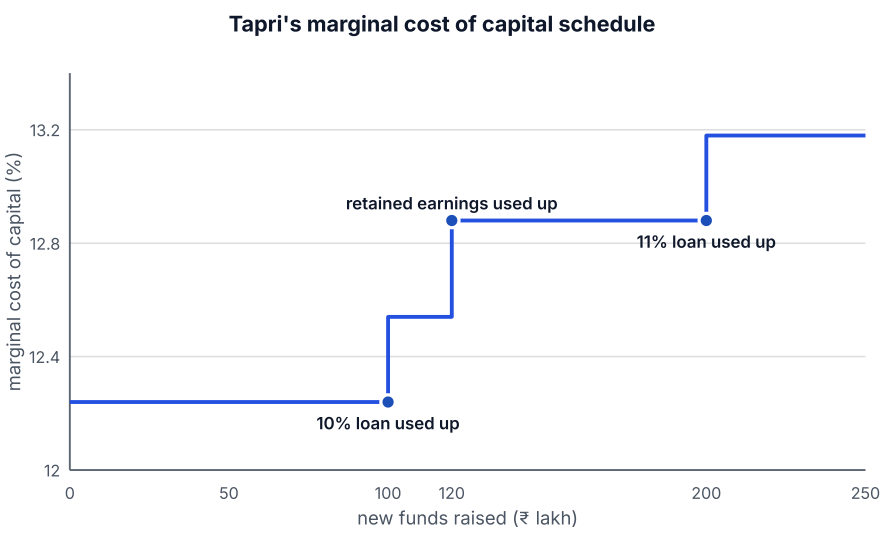
(i) Calculate the after-tax cost of each new source, including the equity from retained earnings and from new shares.

(ii) Determine the break points at which the marginal cost of capital rises.

(iii) Calculate the marginal cost of capital in each range of new funds.

(iv) Calculate the average cost of raising ₹250 lakh. A project needing the last ₹50 lakh of it has an IRR of 12.90%: should it be undertaken?

THE WORKING



The marginal cost steps up at each break point: 12.24% up to ₹100 lakh, 12.54% to ₹120 lakh, 12.88% to ₹200 lakh, 13.18% beyond.

(i) Specific costs

- Growth (Gordon's growth model): g = b × r = 60% × 15% = **9%**. D1 = 40% × ₹6.40 = ₹2.56.
- Retained earnings: Kr = ₹2.56 ÷ ₹40 + 9% = **15.40%**. New shares: net price = ₹40 × 0.92 = ₹36.80; Ke = ₹2.56 ÷ ₹36.80 + 9% = 6.96% + 9% = **15.96%**.
- Debt after tax: first ₹40 lakh 10% × 0.75 = **7.50%**; next ₹40 lakh 11% × 0.75 = **8.25%**; debentures 12% × 0.75 = **9.00%**.

(ii) Break points (cheaper source available ÷ its proportion of new funds)

- 10% loan: ₹40 lakh ÷ 0.40 = **₹100 lakh**.
- Retained earnings: ₹72 lakh ÷ 0.60 = **₹120 lakh**.
- 11% loan (₹80 lakh of debt in all): ₹80 lakh ÷ 0.40 = **₹200 lakh**.

(iii) Marginal cost of capital (weighted cost: debt × 0.40 + equity × 0.60)

Range of new funds (₹ lakh)	Debt (%)	Equity (%)	MCC (%)
0–100	7.50	15.40	**12.24**
100–120	8.25	15.40	**12.54**
120–200	8.25	15.96	**12.88**
Above 200	9.00	15.96	**13.18**

For example, 0–100: 0.40 × 7.50% + 0.60 × 15.40% = 3.00% + 9.24% = 12.24%.

(iv) Average cost of ₹250 lakh

$$\frac{100 \times 12.24\% + 20 \times 12.54\% + 80 \times 12.88\% + 50 \times 13.18\%}{250} = \frac{3{,}164.2}{250} = 12.66\%$$

The project should **not** be undertaken. The last ₹50 lakh costs 13.18%, the marginal cost of that money, and the project earns 12.90%. Judged against the 12.66% average it would look acceptable, which is why the marginal cost, not the average, is the hurdle for the last project.

Exam angle

In the exam

Where WACC and MCC appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, a WACC or a marginal cost of capital came up 18 times in 18 documents:

- **Market value WACC as the last part of a costing question**: September 2024 Q2(a) (8 marks, 13.39%), January 2025 Q2(a)(iv) (3 marks, 17.94%), September 2025 Q1(a)(iii) (2 marks, 11.59%), January 2026 MCQ 4 (17.02%), RTP January 2025 Q5 (22.52%).
- **Book and market value WACC side by side**: MTP Series II for May 2026 Q3(a) (7 marks: 15.50% and 14.88%), MTP Series I for September 2026 Q1(b) (5 marks; the Study Material's Illustration 17, word for word).
- **New WACC after extra debt**: MTP Series I for May 2025 Q2(a) (6 marks: 9.56% to 11.30%), MTP Series I for January 2025 Q1(b) (5 marks), RTP September 2024 Q5, RTP September 2026 Q1 (case scenario MCQs).
- **Marginal cost of capital**: RTP May 2024 Q2 (14.49%, ₹3,77,875 before new shares, then 15.99%), MTP Series II for September 2025 Q1(c) (5 marks: Illustration 18 with the years moved on a year and new shares at ₹25); the pattern-of-finance version in RTP September 2025 Q5; May 2026 MCQ 5 (the post-tax average cost of old and new debt, 7.98%).
- **Working backwards**: May 2025 Q2(a) (8 marks), RTP May 2026 Q5, MTP Series I for May 2024 Q3(a) (8 marks, interest rates found from a 16% WACC).
- **The Study Material again**: two of these MTP questions are Illustrations 17 and 18 with at most a figure changed, so work every illustration in SM 4.31–4.41 once.

In the exam

The traps ICAI used

- **Market values must be computed**: number of securities × market price (5,00,000 shares × ₹70 in September 2024 Q2(a)), not the book figures with a new label.
- **Retained earnings in market value weights**: split the market value of equity in the book value ratio (MTP Series II for May 2026: ₹9 crore split 6 : 4 into ₹5.40 crore and ₹3.60 crore).
- **Two acceptable answers**: September 2025 Q1(a)(iii) gave 11.59% with the preference cost on its issue price and 10.57% on its market price; January 2025 Q2(a)(iv) printed an alternative too. State the basis you use.
- **After additional debt**, each tranche keeps its own cost and the cost of equity is recomputed at the new price (MTP Series I for May 2025).
- **Cheap new debt, dearer WACC**: RTP September 2026 Q1's five MCQs on one case take every cost on book values (Kp = 11.5 ÷ 100; Kd = 10% × 0.65 = 6.5% for the old debentures and 12% × 0.65 = 7.8% for the new) and add the new debentures as a separate line. Ke rises from 10% to 13% when the price falls from ₹300 to ₹250, so the WACC goes from 9.43% to 11.04%: cheap new debt, dearer WACC, the point the Thanda example makes at market values. RTP September 2024 Q5 is the same case with ₹50 lakh of new debentures that must first be priced at a 13% market yield, with 2% flotation cost.

Trick

Sanity checks on any WACC

The weights must add up to 1, and the WACC must lie between the cheapest and the dearest after-tax cost. Market value weights raise the WACC when equity (the dearest source) trades further above its book value than the other sources do. Conditions: all costs after tax and every source included once.

Answer frame

DISCUSS the meaning of weighted average cost of capital. ILLUSTRATE with an example (SM 4.31–4.32):

1. A company mixes sources to balance financial risk, ownership control and cost, so the cost of total capital is the weighted average of the costs of the individual sources.
2. Definition: the weighted average after-tax costs of the individual components of the firm's capital structure; also called the overall cost of capital; it depends on the capital structure.
3. The four steps (total capital; proportions; proportion × cost; add).
4. A four-line example table (ICAI's: retained earnings 25% × 10%, equity 10% × 11%, preference 15% × 9%, debt 50% × 6% = 7.95%).
5. Use: the discount rate for DCF and NPV, the hurdle rate; returns above WACC create value.

What is the DIFFERENCE between book value and market value weights? The five-point table above, in ICAI's order, with one line on retained earnings (included in book value of equity; under market values, apportioned out of the market value of equity).

DISCUSS marginal cost of capital (SM 4.38): definition; marginal weights = proportions of new funds, so no BV/MV choice; equal to the average cost if proportions and component costs are unchanged;

component costs rise beyond certain levels (debt example, external equity with flotation costs); then the average cost rises and the marginal cost rises faster.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the cheapest source as the hurdle rate

Kesar's 7.70% would accept projects that earn less than the 10.68% to 11.42% the mix costs. **Fix:** the hurdle rate is the WACC of the mix that funds the project, not the cost of the source used this time.

⚠ **Common mistake**

Market value weights with retained earnings at book

Adding retained earnings at ₹300 lakh to equity at its market value of ₹1,500 lakh counts the reserves twice: the market value of equity already includes them. **Fix:** split the market value of equity in the book value ratio of equity capital to retained earnings.

⚠ **Common mistake**

Pre-tax interest in the WACC

Using the 9% coupon for the debentures. **Fix:** every cost in a WACC is after tax: Kd = I(1 – t) or its redeemable form; preference and equity costs need no adjustment.

⚠ **Common mistake**

The break point that forgot the proportion

"Retained earnings are ₹72 lakh, so new shares are needed after ₹72 lakh." **Fix:** retained earnings fund only the equity part (60%) of each rupee: the break point is ₹72 lakh ÷ 0.60 = ₹120 lakh of total new funds.

Practice

Check yourself

1 A company has equity of ₹60 lakh costing 15%, preference shares of ₹10 lakh costing 11% and debt of ₹30 lakh at 12% interest before tax. The tax rate is 25%. What is its WACC, in %?

Answer: _____

2 Case scenario. Kulhad Crafts Ltd, the listed company that makes Tapri's clay cups, shows equity share capital of ₹40 lakh (4 lakh shares) and retained earnings of ₹60 lakh in its balance sheet; its shares trade at ₹75. Dev wants market value weights. What market value should he give the retained earnings?

- A ₹60 lakh
- B ₹180 lakh
- C ₹300 lakh
- D ₹120 lakh

3 When will a firm's marginal cost of capital equal its average cost of capital?

- A When it uses market value weights
- B When it raises funds in its existing proportions and the component costs remain unchanged
- C When it has no debt
- D Never, because new funds always cost more

4 A company will raise new funds as 40% debt and 60% equity. It has ₹48 lakh of retained earnings; after that it must sell new shares. Up to what amount of new funds can it invest before selling new shares?

- A ₹48 lakh
- B ₹120 lakh
- C ₹80 lakh
- D ₹28.8 lakh

5 Which of these are among the five factors that the Study Material lists as affecting the cost of capital? Select all that apply.

- A Inflation
- B Exchange rates
- C The company's dividend payout ratio
- D Supply and demand for funds

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Cost of capital: the return expected by providers of capital; also called the cut-off rate, hurdle rate, minimum rate of return; used for investment evaluation, financing decisions and credit policy.
- Five factors: supply and demand, investor preferences, risk and return, inflation, exchange rates.

- WACC = Σ weight × after-tax cost; weights add up to 1; it is the minimum return that creates value.
- BV weights: balance sheet figures, reserves inside equity. MV weights: number × market price; split the market value of equity between share capital and retained earnings in their book value ratio.
- MCC: the cost of an additional rupee, with marginal weights (proportions of new funds); no BV/MV choice.
- Break point = cheaper source available ÷ its proportion; beyond it, the MCC steps up, faster than the average.
- Each tranche keeps its own cost; after new debt, recompute Ke at the new price. (The blend costs more once the Assam runs out.)

Answer key · Check yourself

1 **12.75 to 12.85**

After-tax Kd = 12% × 0.75 = 9%. WACC = 0.60 × 15% + 0.10 × 11% + 0.30 × 9% = 9.0% + 1.1% + 2.7% = 12.8%. Using the pre-tax 12% for debt gives 13.7%.

2 **B ₹180 lakh**

Market value of equity = 4 lakh × ₹75 = ₹300 lakh, split in the book ratio 40 : 60: ₹120 lakh to equity capital and ₹180 lakh to retained earnings. ₹60 lakh is the book value; ₹300 lakh is all of the equity; ₹120 lakh is the equity capital's share.

3 **B When it raises funds in its existing proportions and the component costs remain unchanged**

The Study Material: when a firm raises funds in a proportional manner and the component costs remain unchanged, there is no difference between the average and the marginal cost of capital (SM 4.38).

4 **C ₹80 lakh**

Break point = retained earnings ÷ equity proportion = ₹48 lakh ÷ 0.60 = ₹80 lakh. ₹120 lakh divides by the debt proportion; ₹28.8 lakh multiplies instead of dividing.

5 **A Inflation**; **B Exchange rates**; **D Supply and demand for funds**

ICAI's five: supply and demand, investor preferences, risk and return, inflation and exchange rates (SM 4.4–4.5). The payout ratio is not among them.

CHAPTER 5

Financing Decisions: Capital Structure

5.1 Capital structure theories: NI, NOI, traditional and Modigliani-Miller

Does borrowing change what a company is worth? NI says yes, NOI and MM (no tax) say no, the traditional view says up to a point, and MM with tax says debt adds debt × tax rate.

In plain words

"Debt is cheaper, so borrow it all?"

Tapri Cold Chain Pvt Ltd, the refrigerated-truck subsidiary of Tapri Tea Co. Ltd, earns ₹30 lakh a year before interest and tax and has no debt. Its shareholders expect 15%, so the company is worth ₹30 lakh ÷ 15% = ₹2 crore. A bank offers loans at 10%.

> **Kesar:** Shareholders want 15%, the bank wants 10%. Replace shares with loans, the average cost falls, and the company is worth more. Sab adjust ho jayega.
>
> **Dev:** Or the shareholders, now standing behind a bank, start wanting 20%, and nothing falls at all. Depends whose theory you believe.
>
> **Kesar:** How many theories are there?
>
> **Dev:** For the exam, four, plus two modern ones. Discounted for overlap, still four.

A company's **value** is its operating profit (EBIT) capitalised at its **overall cost of capital (Ko)**, the weighted average of what lenders want (**Kd**) and what shareholders want (**Ke**). **Capital structure** is the mix of debt and equity behind that value. The whole debate is one question: when the mix changes, does Ko change?

- **Net Income (NI) approach**: yes. Kd and Ke stay put, so more cheap debt pulls Ko down and value up. Borrow as much as possible.
- **Net Operating Income (NOI) approach**: no. Shareholders raise Ke exactly enough to cancel the cheap debt; Ko and value never move.
- **Traditional approach**: up to a point. A little debt lowers Ko; too much makes lenders and shareholders nervous and Ko rises. In between lies an **optimum capital structure**.
- **Modigliani-Miller (MM)**: without taxes, no, and they prove it: if two identical firms were valued differently, investors would sell one, buy the other and pocket the difference (**arbitrage**) until the values matched. With corporate tax, yes: interest saves tax, so a levered firm is worth more by the debt × the tax rate.
- **Trade-off** and **pecking order** theories: why real companies stop borrowing somewhere, and which money they use first.

Think of it like

The samosa and the taxman

Cut a samosa into two pieces or eight: you still have one samosa. That is MM without taxes: slicing the same earnings between lenders and shareholders does not make more earnings. Now seat a taxman at the table who bites only the shareholders' pieces, never the lenders'. The more of the samosa you hand to lenders, the less the taxman eats and the more is left for the people who own the plate. That is MM with tax: the extra value of debt is exactly the tax it saves.

Key idea

A firm is worth its operating profit capitalised at its overall cost of capital, so the debt-equity mix matters only if it changes that overall cost. NI: it falls as debt rises. NOI and MM without tax: it stays constant, because the cost of equity rises just enough to cancel the cheap debt. Traditional: it first falls, then rises, so there is an optimum. MM with tax: borrowing adds value equal to the tax the interest saves, the debt times the tax rate.

How it works

The ground rules every theory assumes

The theories explain the relationship between cost of capital, capital structure and value of the firm under these assumptions (SM 5.5):

1. There are only two kinds of funds used by a firm, debt and equity.
2. The total assets of the firm are given; the degree of leverage can be changed by selling debt to purchase shares or selling shares to retire debt.
3. Taxes are not considered.
4. The dividend payout ratio is 100%.
5. The firm's total financing remains constant.
6. Business risk is constant over time.
7. The firm has perpetual life.

With these, the arithmetic is the same in every theory (SM 5.3):

$$\text{Value of the firm} = \frac{\text{EBIT}}{K_o} \qquad K_o = K_d \times \frac{D}{D+S} + K_e \times \frac{S}{D+S}$$

D and S are the market values of debt and equity; Ko is the WACC built in WACC and the marginal cost of capital. What differs is which rate each theory holds constant.

Theory 1: the Net Income approach, where debt is a free lunch

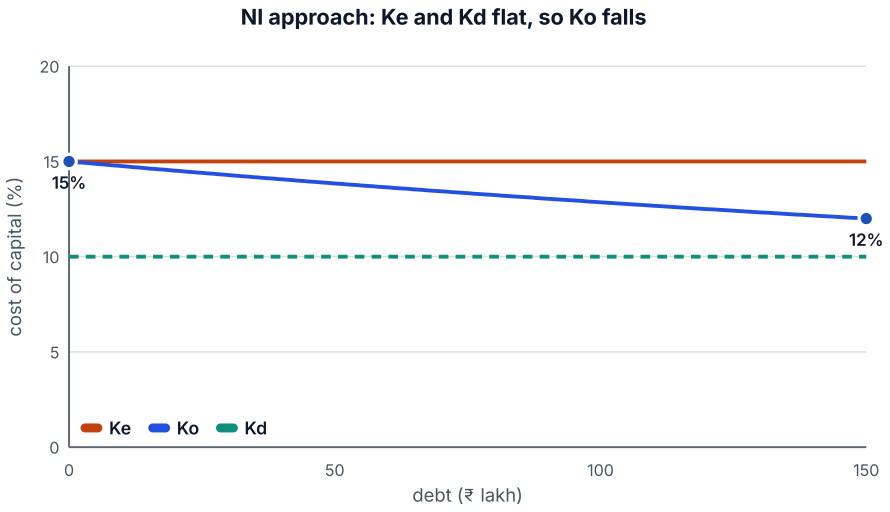
According to the NI approach, capital structure decision is relevant to the value of the firm. An increase in financial leverage will lead to decline in the WACC, while the value of the firm as well as the market price of the ordinary share will increase; a decrease in leverage does the opposite (SM 5.6).

Ke and Kd are assumed not to change with leverage. Equity is valued from the earnings left for shareholders:

$$S = \frac{\text{NI}}{K_e} \qquad V = S + D \qquad K_o = \frac{\text{EBIT}}{V}$$

NI is the earnings available for equity shareholders and Ke the equity capitalisation rate. The theory pleads for the firm to employ as much debt as possible to maximise its value (SM 5.7). Tapri Cold Chain, EBIT ₹30 lakh, Kd 10%, Ke fixed at 15%:

Debt (₹ lakh)	NI = 30 − 10% of D	S = NI ÷ 15%	V = S + D	Ko
0	30	200.00	200.00	15.00%
50	25	166.67	216.67	13.85%
100	20	133.33	233.33	12.86%
150	15	100.00	250.00	12.00%



Tapri Cold Chain under the NI approach. Orange: Ke, fixed at 15%; teal dashed: Kd, fixed at 10%; blue: Ko, sliding from 15% towards Kd as debt replaces equity.

Theory 2: the traditional approach, where there is a sweet spot

As a result of financial leverage up to some point, cost of capital comes down and value of firm increases; beyond that point, reverse trends emerge. The cost of capital depends on the capital structure, and there is an optimal capital structure which minimises it (SM 5.8). Under this approach:

1. The rate of interest on debt remains constant for a certain period and thereafter, with an increase in leverage, it increases.
2. The expected rate of equity shareholders remains constant or increases gradually; after that they start perceiving a financial risk, and from the optimal point the expected rate increases speedily.

3. As a result, the WACC first decreases and then increases. The lowest point on the curve is the optimal capital structure.

Optimum capital structure occurs where value of the firm is highest and the cost of capital is the lowest. The traditional approach strikes a balance between the NI and NOI extremes (SM 5.9). Suppose Tapri Cold Chain's lenders and shareholders react like this as debt rises as a share of total value:

Debt ÷ total value	Kd	Ke	Ko = Kd × w(D) + Ke × w(S)
0%	10.00%	15.00%	15.000%
10%	10.00%	15.00%	14.500%
20%	10.00%	15.50%	14.400%
30%	**10.50%**	**15.75%**	**14.175%**
40%	11.00%	17.50%	14.900%
50%	12.50%	20.00%	16.250%
60%	15.00%	24.00%	18.600%



Orange: Ke, flat, then rising speedily; teal dashed: Kd, flat, then rising; blue: Ko, lowest (14.175%) at 30% debt, the optimal capital structure for this schedule.

Interactive version: PassBook app, lesson Capital structure theories: NI, NOI, traditional and Modigliani-Miller

Theory 3: the Net Operating Income approach, where nothing changes

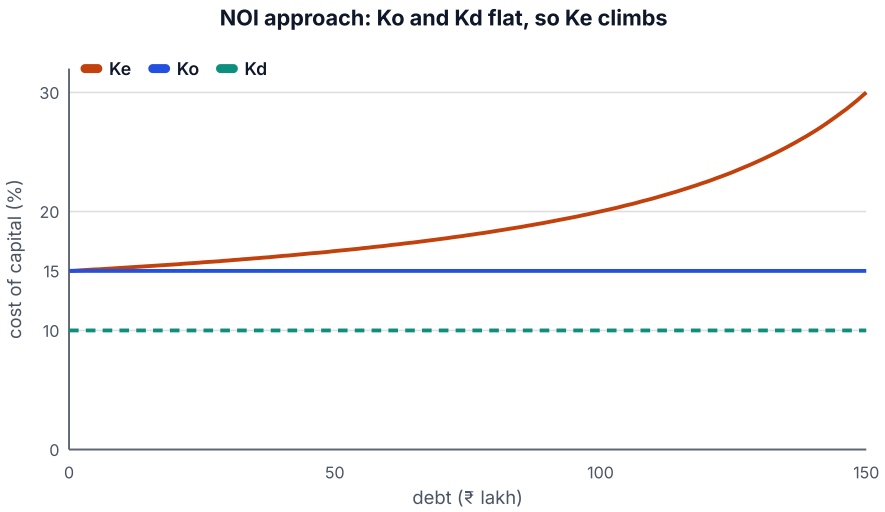
NOI means EBIT. According to this approach, capital structure decisions of the firm are irrelevant: any change in leverage will not lead to any change in the total value of the firm and the market price of shares, as the overall cost of capital is independent of the degree of leverage (SM 5.11). An increase in the use of apparently cheaper debt is offset by an increase in the equity capitalisation rate, because

equity investors seek higher compensation for the greater risk of fixed-return securities in the capital structure. So the firm is valued first, and equity is the balancing figure:

$$V = \frac{\text{EBIT}}{K_o} \qquad S = V - D \qquad K_e = \frac{\text{EBIT} - \text{Interest}}{S}$$

Tapri Cold Chain with Ko fixed at 15%:

Debt (₹ lakh)	V = 30 ÷ 15%	S = V − D	NI	Ke = NI ÷ S
0	200	200	30	15.00%
50	200	150	25	16.67%
100	200	100	20	20.00%
150	200	50	15	30.00%



The same company under the NOI approach. Blue: Ko, fixed at 15%, so value stays ₹2 crore; teal dashed: Kd at 10%; orange: Ke, rising from 15% to 30% to absorb every rupee of cheap debt.

Theory 4: Modigliani-Miller (1958, without tax), with a proof by greed

The NOI approach is definitional or conceptual and lacks behavioural significance; MM provide the behavioural justification for a constant Ko and therefore a constant total value (SM 5.15). In a perfect capital market with no transaction cost and no taxes, the value and cost of capital of a company remain unchanged irrespective of change in the capital structure. Additional assumptions:

- Capital markets are perfect: all information is freely available and there are no transaction costs.
- All investors are rational.
- Firms can be grouped into 'equivalent risk classes' on the basis of their business risk.
- Non-existence of corporate taxes.

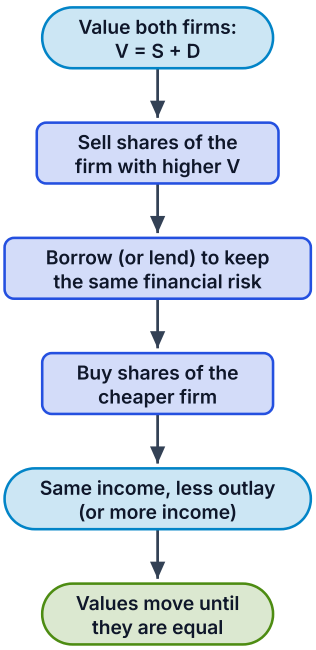
From these, three propositions (SM 5.16):

1. Total market value of a firm is equal to its expected net operating income divided by the discount rate appropriate to its risk class: value of levered firm (Vg) = value of unlevered firm (Vu) = NOI ÷ Ko.
2. A firm having debt in its capital structure has a higher cost of equity than an unlevered firm; Ke includes a risk premium for the financial risk: $Ke = Ko + (Ko - Kd) \times Debt \div Equity$.
3. The structure of the capital (financial leverage) does not affect the overall cost of capital; the cost of capital is only affected by the business risk.

Check it on Tapri Cold Chain at debt ₹100 lakh: 15% + (15% − 10%) × 100 ÷ 100 = 20%, exactly the NOI table's figure.

The arbitrage process. MM's operational justification is arbitrage and the substitution of corporate leverage by personal leverage (SM 5.16–5.17). Arbitrage refers to buying an asset or security at a lower price in one market and selling it at a higher price in another, until equilibrium is attained. If two identical firms differ only in debt and one is valued higher, its shareholders sell, borrow (or lend) on their own account to keep the same risk, buy the cheaper firm, and earn the same return at a lower outlay. Their selling and buying moves both prices until the values are equal.

The arbitrage process, step by step



Borrow on personal account when the levered firm is dearer (SM Illustration 7); lend, buying the levered firm's debt with its equity, when the unlevered firm is dearer (Illustration 9). Green: the equilibrium MM predict.

The value of the levered firm can neither be greater nor lower than that of an unlevered firm: there is a conservation of investment value, since the sum of the parts must equal the whole (SM 5.17). **Shortcoming**: the arbitrage process will fail to work because of imperfections in capital market, existence of transaction cost and presence of corporate income taxes.

Interactive version: PassBook app, lesson Capital structure theories: NI, NOI, traditional and Modigliani-Miller

MM 1963, with tax: the taxman's bite

In 1963 MM incorporated tax: interest is deductible, so the value of the firm will increase, or its cost of capital decrease, where corporate taxes exist; the value of the levered firm exceeds that of the unlevered firm by the amount of debt multiplied by the corporate tax rate (SM 5.17–5.18):

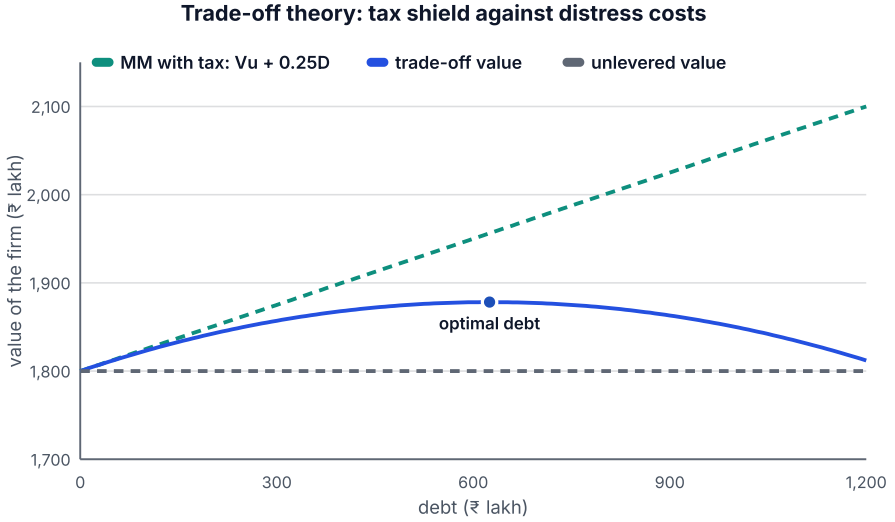
$$V_g = V_u + \text{TB} \qquad K_{eg} = K_{eu} + (K_{eu} - K_d) \times \frac{\text{Debt}\,(1-t)}{\text{Equity}} \qquad K_{og} = K_{eu}\,(1 - tL)$$

TB is the present value of tax shields (debt × t for permanent debt), Keu the cost of equity of the unlevered company, t the tax rate and L = debt ÷ (debt + equity) (SM 5.18, 5.27). Worked Example 3 below applies all three.

Interactive version: PassBook app, lesson Capital structure theories: NI, NOI, traditional and Modigliani-Miller

Two modern theories

Trade-off theory. A company chooses how much debt and equity to use by balancing the costs and benefits: the advantage of debt is its tax benefit; its cost is the cost of financial distress, including bankruptcy costs and non-bankruptcy costs (staff leaving, suppliers demanding disadvantageous payment terms, bondholder/stockholder infighting), plus agency costs from shareholders-managers and shareholders-debt holders conflicts. The marginal benefit of further debt declines as debt increases while the marginal cost increases, which gives an optimal capital structure (SM 5.27–5.28). It explains why corporations are usually financed partly with debt and partly with equity.



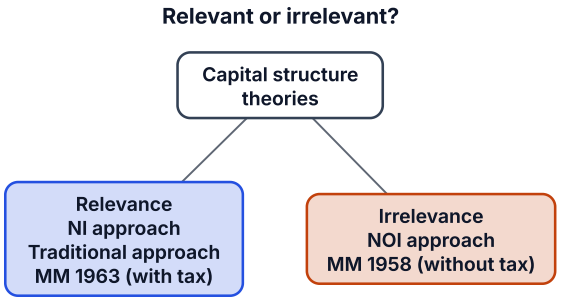
Illustrative: an unlevered firm worth ₹1,800 lakh, tax 25%, and distress costs assumed to grow as 0.0002 × D². Teal dashed: the tax shield alone; blue: value after distress costs, highest at debt of ₹625 lakh.

Pecking order theory. Based on **asymmetric information**: managers know more than investors. Firms prefer to issue debt when they are positive about future earnings; equity is issued when they are doubtful and internal finance is insufficient. The capital structure decision gives higher priority

to sources that reveal the least amount of information; there is no well-defined debt-equity target (SM 5.29). The order:

1. Managers' first choice is to use internal finance.
2. In absence of internal finance, they use secured debt, unsecured debt, hybrid debt etc.
3. Managers issue new equity shares as a last option.

All six on one page



ICAI's own grouping (SM 5.5). Trade-off and pecking order theories explain the mix firms actually choose.

Theory	As debt rises	Message
NI	Kd, Ke constant; Ko falls	use maximum debt
Traditional	Kd, Ke flat, then rising; Ko falls, then rises	an optimum exists
NOI	Kd, Ko constant; Ke rises	mix irrelevant
MM 1958	Ko constant; Ke = Ko + (Ko − Kd) D/E	irrelevant, proved by arbitrage
MM 1963	Ke rises less steeply; Ko falls	value up by D × t
Trade-off	tax benefit against distress and agency costs	an optimum exists
Pecking order	no target ratio	internal funds, then debt, equity last

How a finance manager actually weighs the mix (control, risk, cost, flexibility) is in capital structure planning, and how each mix changes EPS at a given EBIT is in EBIT-EPS analysis.

Formats and formulas

Reproduce these two statements from memory: almost every numerical on this topic is one of them.

Value under the NI approach (Ke given)	₹
EBIT	xxx
Less: Interest on debt	(xxx)
Earnings available for equity holders (NI)	xxx
Equity capitalisation rate (Ke)	x%
Market value of equity (S) = NI ÷ Ke	xxx
Market value of debt (D)	xxx
Total value of firm V = S + D	**xxx**
Overall cost of capital Ko = EBIT ÷ V	x%

Value under the NOI approach or MM without tax (Ko given)	₹
EBIT (NOI)	xxx
Overall cost of capital (Ko)	x%
Value of the firm V = EBIT ÷ Ko	**xxx**
Less: Market value of debt (D)	(xxx)
Market value of equity (S)	xxx
Earnings available for equity holders (EBIT – interest)	xxx
Cost of equity Ke = NI ÷ S	x%

∑ **Formula**

Overall cost of capital

$$K_o = K_d \times \frac{D}{D+S} + K_e \times \frac{S}{D+S} \qquad = \frac{\text{EBIT}}{V}$$

Ko is the WACC; D and S are market values of debt and equity (SM 5.3). With tax, use the post-tax Kd in the weighted average.

∑ **Formula**

MM without tax: cost of equity of a levered firm

$$K_e = K_o + (K_o - K_d) \times \frac{\text{Debt}}{\text{Equity}}$$

Ko equals the Ke of the identical unlevered firm (SM 5.16).

Σ Formula

MM with tax: value of a levered firm

$$V_g = V_u + \text{TB} \qquad V_u = \frac{\text{EBIT}\,(1-t)}{K_{eu}}$$

TB = present value of tax shields; for permanent debt, debt × t (SM 5.18, 5.69).

Σ Formula

MM with tax: cost of equity and WACC of a levered firm

$$K_{eg} = K_{eu} + (K_{eu} - K_d) \times \frac{\text{Debt}\,(1-t)}{\text{Equity}} \qquad K_{og} = K_{eu}\,(1 - tL)$$

Kd here is the pre-tax rate of interest; L = debt ÷ (debt + equity) at market value (SM 5.26–5.27).

Arbitrage in three steps (the layout of ICAI's Suggested Answers): 1. Valuation of firms (EBIT, interest, earnings for equity, Ke, S, D, V); 2. Investment and borrowings (sell, borrow or lend, buy); 3. Change in return (new income, less interest, less old income = incremental income due to arbitrage).

Definition

Arbitrage

Arbitrage refers to buying asset or security at lower price in one market and selling it at a higher price in another market. As a result, equilibrium is attained in different markets. (SM 5.16–5.17)

Worked examples

Worked example **Samosa Junction: one company, two theories**

Samosa Junction Ltd, Tapri's frozen-snacks subsidiary, earns EBIT of ₹18,00,000 a year and has 12% debentures of ₹60,00,000. Ignore taxes.

Required:

(i) Assuming the NI approach with an equity capitalisation rate of 16%, calculate the market value of equity, the value of the firm and the overall cost of capital.

(ii) Assuming the NOI approach with an overall cost of capital of 15%, calculate the value of the firm, the market value of equity and the cost of equity. Verify the cost of equity with MM's formula.

(iii) Recalculate (i) and (ii) if the company issues further debentures to raise its debt to ₹90,00,000 and uses the money to buy back shares. Comment.

THE WORKING

(i) NI approach: statement of value

Particulars	₹
EBIT	18,00,000
Less: Interest (12% × ₹60,00,000)	(7,20,000)
Earnings available for equity holders (NI)	10,80,000
Ke	16%
Market value of equity S = 10,80,000 ÷ 0.16	67,50,000
Market value of debt D	60,00,000
Value of firm V = S + D	**1,27,50,000**

Ko = EBIT ÷ V = ₹18,00,000 ÷ ₹1,27,50,000 = **14.12%**.

(ii) NOI approach: statement of value

Particulars	₹
EBIT	18,00,000
Ko	15%
Value of firm V = 18,00,000 ÷ 0.15	**1,20,00,000**
Less: Market value of debt	(60,00,000)
Market value of equity S	60,00,000
Earnings for equity holders (18,00,000 − 7,20,000)	10,80,000

Ke = ₹10,80,000 ÷ ₹60,00,000 = **18%**. Verification: Ke = Ko + (Ko − Kd) D/E = 15% + (15% − 12%) × 60 ÷ 60 = **18%**.

(iii) Debt raised to ₹90,00,000 (interest ₹10,80,000; NI ₹7,20,000)

Particulars	NI approach	NOI approach
Market value of equity (₹)	45,00,000 (7,20,000 ÷ 16%)	30,00,000 (1,20,00,000 − 90,00,000)
Value of firm (₹)	1,35,00,000	1,20,00,000
Overall cost of capital	13.33%	15.00%
Cost of equity	16.00%	24.00%

Check (NOI): 15% + 3% × 90 ÷ 30 = 24%.

Comment. Under the NI approach the extra debt raises value by ₹7,50,000 and lowers Ko from 14.12% to 13.33%, because Ke is assumed fixed. Under the NOI approach value stays ₹1,20,00,000 and Ko 15%: Ke rises from 18% to 24% and exactly offsets the cheaper debt.

Worked example **Cold-chain twins: the arbitrage that costs interest**

PolarKart Ltd (levered) and FrostLine Logistics Ltd (unlevered), two cold-chain firms that carry Tapri's frozen snacks, are identical in every respect except capital structure. Each earns EBIT of ₹3,00,000 a year. Ignore taxes.

Particulars	PolarKart (₹)	FrostLine (₹)
10% debentures (market value)	10,00,000	Nil
Market value of equity	18,00,000	25,00,000

Bunty's uncle holds 5% of the shares of PolarKart. A bank will lend to him personally at 12%, not at the 10% PolarKart pays.

Required:

(i) Show which company the market values higher.

(ii) Calculate the increase in his annual income if he switches his holding to FrostLine, keeping his risk unchanged.

(iii) Calculate the highest personal borrowing rate at which the switch still pays, and comment.

THE WORKING

(i) Valuation of firms

Particulars	PolarKart (₹)	FrostLine (₹)
EBIT	3,00,000	3,00,000
Less: Interest (10% × ₹10,00,000)	1,00,000	Nil
Earnings available to equity shareholders	2,00,000	3,00,000
Value of equity S	18,00,000	25,00,000
Debt D	10,00,000	Nil
Value of firm V = S + D	**28,00,000**	**25,00,000**

The levered company is valued higher, so he sells its shares and buys shares of the unlevered company. To maintain the level of risk he borrows his proportionate share of the levered company's debt and invests that amount also in the unlevered company.

(ii) Investment and borrowings

Particulars	₹
Sell shares in PolarKart (₹18,00,000 × 5%)	90,000
Borrow money (₹10,00,000 × 5%)	50,000
Buy shares in FrostLine	**1,40,000**

₹1,40,000 buys 1,40,000 ÷ 25,00,000 = 5.6% of FrostLine.

Change in return

Particulars	₹
Income from FrostLine (5.6% × ₹3,00,000)	16,800
Less: Interest on personal loan (₹50,000 × 12%)	6,000
Net income after the switch	10,800
Less: Income from PolarKart (5% × ₹2,00,000)	10,000
Incremental income due to arbitrage	**800**

(iii) Highest personal borrowing rate. The switch pays while ₹16,800 − ₹50,000 × r exceeds ₹10,000, that is while r is below 6,800 ÷ 50,000 = **13.6%**. At PolarKart's own 10% the gain would be ₹1,800.

Comment. Arbitrage works through the substitution of corporate leverage by personal leverage (SM 5.16–5.17), so it assumes investors can borrow on terms close to the company's, as in MM's perfect capital market. The dearer the personal loan, the smaller the gain, and above 13.6% this mispricing would survive.

Worked example

Kulhad Crafts and its ₹80 lakh target

Kulhad Crafts Ltd, Tapri's pottery supplier, is an all-equity company with a market value of ₹1,800 lakh and a cost of capital of 20%. It distributes its entire earnings as dividend and its EBIT is expected to stay constant. The board wants the company's value to rise to ₹1,880 lakh by buying back equity shares and replacing them with 11% debentures of the same amount. The corporate tax rate is 25%.

Required: As per the MM approach,

(i) calculate the amount of debentures needed;

(ii) calculate the overall cost of capital and the cost of equity after the change;

(iii) Dev asks why the company should not replace all its equity with debt. Comment.

THE WORKING

Working note: EBIT. Earnings for equity holders = ₹1,800 lakh × 20% = ₹360 lakh = EAT, so EBIT = ₹360 lakh ÷ (1 − 0.25) = **₹480 lakh**.

(i) Debentures needed. Value of levered firm = value of unlevered firm + tax advantage (D × t). ₹1,880 lakh = ₹1,800 lakh + D × 0.25, so D = ₹80 lakh ÷ 0.25 = **₹320 lakh**.

(ii) Income statement (₹ lakh)

Particulars	All equity	Equity and debt
EBIT	480.00	480.00
Interest on ₹320 lakh @ 11%	—	35.20
EBT	480.00	444.80
Tax @ 25%	120.00	111.20
Income available to equity holders	360.00	333.60

Market value of equity = ₹1,880 lakh − ₹320 lakh = ₹1,560 lakh. Ke = ₹333.60 ÷ ₹1,560 = **21.38%**. Kd = 11% × (1 − 0.25) = 8.25%.

Component	Amount (₹ lakh)	Cost	Weight	WACC
Equity	1,560	21.38%	0.83	17.75%
Debt	320	8.25%	0.17	1.40%
Total	**1,880**		**1.00**	**19.15%**

Weights are 1,560 ÷ 1,880 and 320 ÷ 1,880, rounded to two decimals as ICAI's answers do (unrounded weights give 19.149%, the same 19.15%). Ko has fallen by 0.85% (20% − 19.15%) due to the tax relief on debt interest; Ke has risen by 1.38% due to the financial risk of the debt.

Check with MM's formulas: Ko = 20% × [1 − (0.25 × 320 ÷ 1,880)] = **19.15%**; Ke = 20% + (20% − 11%) × 320 (1 − 0.25) ÷ 1,560 = 20% + 1.38% = **21.38%**.

(iii) Comment. On MM's 1963 model alone every rupee of debt adds its tax saving, so value would be highest with (almost) all debt. That conclusion ignores the costs of financial distress, which rise with debt; the trade-off theory sets them against the tax shield, so value is highest at a moderate level of debt (the trade-off figure above, drawn for a firm of the same ₹1,800 lakh, peaks at debt of ₹625 lakh).

Exam angle

What ICAI asks

In the exam

In the ICAI documents read for this lesson (new scheme, May 2024 to September 2026)

Capital structure theories appeared in 17 of the 32 question sets searched (8 past papers, 8 RTPs, 16 MTPs):

- **Arbitrage, levered and unlevered firms, 4–5 marks**: May 2024 Q2(b) (4 marks, investor holds 12%); MTP Series I for September 2026 Q1(a) (5 marks); RTP May 2024 Q3. All three switch from levered to unlevered and lay out the three-step answer of Example 2.
- **MM with tax, 2–6 marks**: May 2025 Q2(b) (4 marks, all-equity company buys back ₹300 lakh with 12% debentures); September 2025 MCQ 6 (2 marks, market value of equity after replacing ₹325 lakh of equity with 16% debentures: ₹2,100 + ₹325 × 35% − ₹325 = ₹1,888.75 lakh); MTP Series I for January 2026 Q3(a) (6 marks, converting debt to equity, so value falls); RTP September 2024 Q6.
- **MM without and with tax, two companies, 5 marks**: RTP September 2024 Q7, RTP May 2026 Q6, MTP Series I for September 2025: the Study Material's Practical Problem 8 pattern.
- **NI approach, value and Ko**: MTP Series II for January 2025 Q1(b) (5 marks, the statement of Example 1(i)).
- **NOI / MM cost of equity**: MTP Series I for May 2024 Q1(b) (5 marks); MTP Series I for May 2025 MCQ 6 (2 marks).
- **Traditional approach, optimum debt**: MTP Series II for May 2026 Q1(a) and Series II for January 2026 Q1(b) (5 marks each, the same figures); MTP Series II for May 2025 Q2(b) (6 marks, traditional optimum then Ke under MM).
- **Theory, 1–2 marks**: May 2025 Q4(c) and MTP Series I for May 2026 Q4(c) (OR), the order of financing under pecking order theory (May 2025 also asks why it exists); MCQs on whether Ke rises with the debt ratio (MTP Series II for September 2024, MTP Series I for September 2026).

In the exam

The traps in those answers

- MM with tax: equity after the change = Vg − debt. The Suggested Answers subtract the debt that replaced the equity.
- An all-equity value and Ke given, with tax: back out EBIT = (value × Ke) ÷ (1 − t) first, as in May 2025 Q2(b).
- Arbitrage: borrow (or lend) your percentage of the levered firm's debt at the same rate; the answers say "to maintain the level of risk".
- The Suggested Answers round weights to two decimals, so the WACC can differ in the second decimal from the formula Keu (1 − tL): SM Illustration 10 shows 19.11% and 19.10%, and the May 2025 answer prints 16.93% with a note that unrounded weights give 16.91%.

Trick

Which way does the arbitrage run?

Always sell the firm with the **higher** total value V and buy the lower. If the levered firm is dearer, borrow on personal account; if the unlevered firm is dearer, lend (buy the levered firm's debt) in the levered firm's debt-equity proportion. Valid in MM's no-tax world, where the two firms differ only in leverage.

Answer frame

Theory questions and the points that earn marks

"Explain the assumptions of the Modigliani-Miller theory" (SM 5.15): perfect capital markets (information free, no transaction costs); rational investors; equivalent risk classes based on business risk; no corporate taxes; plus the general assumptions (two sources, constant total financing, 100% payout, constant business risk, perpetual life).

"Describe the NOI approach and its assumptions" (SM 5.5, 5.11): NOI = EBIT; capital structure irrelevant; Ko independent of leverage; cheaper debt offset by a rising Ke because equity investors seek compensation for greater risk; Ko and Kd constant, Ke rises; value = EBIT ÷ Ko; the seven general assumptions.

"What is the hierarchy of financing under the pecking order theory, and why does it exist?" (SM 5.29; May 2025): internal finance, then debt (secured, unsecured, hybrid), then new equity last; it exists because of asymmetric information, as managers prefer the source that reveals the least information. ICAI's 2025 answer adds: to minimise the cost of capital, keep financial control and avoid negative market signals.

"Explain the trade-off theory" (SM 5.27–5.28): balance benefit of debt (tax) against costs (financial distress: bankruptcy and non-bankruptcy costs; agency costs); marginal benefit falls and marginal cost rises with debt; optimum capital structure; explains mixed financing.

Three answers to "Calculate the increase in earnings from arbitrage"

Loses marks

"Value of levered firm is higher, so switch. Gain ₹800."

No valuation table, no borrowing: the method marks are gone.

Passes

Valuation table, then sale of ₹90,000 invested in FrostLine.

Forgot the personal loan, so the risk is not the same and the gain comes out wrong.

Full marks

Three headings: valuation of firms; investment and borrowings (borrow 5% of the levered debt "to maintain the level of risk"); change in return. Final line: incremental income due to arbitrage.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the arbitrageur who forgot to borrow

Selling ₹90,000 of levered shares and buying ₹90,000 of unlevered shares compares a risky holding with a safe one. **Fix:** borrow your share of the levered firm's debt (here ₹50,000) and invest it too; the interest on it comes off the new income.

⚠ **Common mistake**

The value of the firm passed off as the value of equity

In MM with tax, ₹1,880 lakh is the value of the whole company. Using it as equity gives Ke = 333.60 ÷ 1,880 = 17.74%, below the unlevered 20%, which is nonsense. **Fix:** equity = Vg − debt = ₹1,560 lakh, and Ke = 21.38%.

⚠ **Common mistake**

NI and NOI swapped

Fixing Ke when the question gives Ko (or the reverse) gives the other theory's answer. **Fix:** read which rate is given: Ke given, start from earnings for equity (NI); Ko given, start from EBIT ÷ Ko (NOI, MM).

⚠ **Common mistake**

Post-tax Kd in the wrong formula

The levered Ke formula with tax uses the pre-tax interest rate (12%) and puts (1 − t) on the debt; the WACC uses the post-tax Kd (9%). Mixing them gives 20% + 11% × 0.2 = 22.2%. **Fix:** pre-tax Kd inside (Keu − Kd); post-tax Kd in the weighted average.

Practice

Check yourself

1 Under the NOI approach, a company has EBIT of ₹12,00,000, an overall cost of capital of 12% and debt of ₹40,00,000 at 9%. Ignoring taxes, what is its cost of equity, in per cent?

Answer: _____

2 Case scenario. Dev models Tapri's planned airport kiosks business as an all-equity company worth ₹500 crore, with a cost of capital of 15% and a constant EBIT. He then tests a version that borrows ₹100 crore of permanent debt and uses it to buy back shares. The corporate tax rate is 25%. What is the overall cost of capital of the levered version under MM with tax?

A 15.00%

B 14.25%

C 14.29%

D 11.25%

3 Which approach holds that cheaper debt is exactly offset by a higher equity capitalisation rate, so the overall cost of capital stays constant?

A Net Income approach

B Traditional approach

C Net Operating Income approach

D Pecking order theory

4 An investor holds 5% of the equity of a levered firm whose debt is ₹20,00,000, and the levered firm is valued higher than its unlevered twin. To switch without changing his risk, how much must he borrow?

- A Nothing: he only sells and buys shares
- B 5% of the levered firm's total value
- C ₹1,00,000, 5% of the levered firm's debt
- D ₹20,00,000, the whole debt

5 Which of these are assumptions of the MM approach (1958, without tax)? Select all that apply.

- A Capital markets are perfect and there are no transaction costs
- B Firms can be grouped into equivalent risk classes
- C The cost of equity stays constant whatever the leverage
- D All investors are rational

More than one answer may be right.

6 Under the pecking order theory, which source does management use last?

- A Retained earnings
- B Secured debt
- C Hybrid debt
- D A fresh issue of equity shares

Answers: the answer key at the end of this lesson.

Recap

- Value of the firm = EBIT ÷ Ko; Ko = Kd × D/V + Ke × S/V. Capital structure matters only if it moves Ko.
- NI: Ke, Kd constant; S = NI ÷ Ke; V = S + D; Ko falls; use maximum debt.
- NOI: Ko constant; V = EBIT ÷ Ko; S = V − D; Ke = (EBIT − I) ÷ S rises; mix irrelevant.
- Traditional: Kd flat then up, Ke gradual then speedy; Ko falls then rises; optimum at the lowest Ko.
- MM 1958: perfect markets, rational investors, equivalent risk classes, no tax; Ke = Ko + (Ko − Kd) D/E; arbitrage with personal leverage equalises values.
- MM 1963: Vg = Vu + D × t; Ke = Keu + (Keu − Kd) D (1 − t)/E; Ko = Keu (1 − tL).
- Trade-off: tax benefit against distress and agency costs gives an optimum. Pecking order: internal funds, debt, equity last.
- One samosa, however you slice it, unless the taxman only eats from one plate.

Answer key · Check yourself

1 **13.9 to 14.1**

V = ₹12,00,000 ÷ 12% = ₹1,00,00,000; S = ₹60,00,000; earnings for equity = ₹12,00,000 − ₹3,60,000 = ₹8,40,000; Ke = 14%. MM check: 12% + 3% × 40 ÷ 60 = 14%.

2 **C** **14.29%**

Vg = ₹500 crore + ₹100 crore × 25% = ₹525 crore, and Ko = Keu × (1 − tD ÷ V) = 15% × (1 − 25 ÷ 525) = 14.29%; equivalently EBIT (1 − t) ÷ Vg = ₹75 crore ÷ ₹525 crore. 15.00% is MM without tax; 14.25% divides by the unlevered ₹500 crore instead of Vg; 11.25% wrongly applies (1 − t) to the whole cost of capital.

3 **C** **Net Operating Income approach**

NOI: Ko and Kd constant, Ke rises with leverage. NI holds Ke constant; traditional lets Ko fall then rise; pecking order is about the order of sources.

4 **C** **₹1,00,000, 5% of the levered firm's debt**

Personal leverage replaces corporate leverage in the same proportion: 5% of ₹20,00,000. Not borrowing changes the risk; that is the commonest slip.

5 **A** **Capital markets are perfect and there are no transaction costs**; **B** **Firms can be grouped into equivalent risk classes**; **D** **All investors are rational**

SM 5.15 lists perfect markets, rational investors, equivalent risk classes and no corporate taxes. A constant Ke is the NI approach; under MM, Ke rises with leverage.

6 **D** **A fresh issue of equity shares**

Internal finance first, then debt (secured, unsecured, hybrid), and new equity as the last option, because it reveals the most information.

5.2 EBIT-EPS analysis, indifference point and financial break-even

How the same EBIT gives different EPS under equity, debt and preference financing; the financial break-even point where EPS is zero; and the indifference point where two plans tie.

In plain words

Three ways to pay for a samosa plant

Tapri Tea Co. Ltd needs ₹20 crore for an automated samosa plant. It has 1.5 crore equity shares and no debt, pays tax at 25%, and expects EBIT (earnings before interest and tax) of ₹30 crore once the plant runs. Dev puts three plans on the table: (A) 50 lakh new shares at ₹40 each; (B) 10% debentures; (C) 11% preference shares.

> **Kesar:** Shares, obviously. No interest, no tension.
>
> **Dev:** And 50 lakh new partners sharing every rupee of profit. Debentures cost ₹2 crore of interest, but the profit stays with 1.5 crore shares.
>
> **Kesar:** So debt always wins?
>
> **Dev:** Only while EBIT stays above ₹8 crore. Below that, the interest eats more than the new partners would have.

The figure that compares the plans is **earnings per share (EPS)**: what is left for equity shareholders after interest, tax and preference dividend, divided by the number of equity shares. Debt and preference shares carry a **fixed financial charge** (interest, preference dividend) that must be paid whatever the EBIT; in exchange, fewer shares share what is left. When the return on the assets is higher than the cost of that fixed-charge money, EPS rises: **favourable financial leverage**, or **trading on equity**. When it is lower, EPS falls.

Two EBIT levels matter. At the **financial break-even point** EBIT just covers the fixed financial charges and EPS is zero. At the **indifference point** two plans give the same EPS: above it the plan with more fixed-charge money gives higher EPS; below it, the plan with more equity does.

Think of it like

The metro card and the monthly pass

A metro card charges per ride; a monthly pass costs a fixed fee and then each ride is cheaper. Ride rarely and the card wins; ride every day and the pass wins. The number of rides at which they cost the same is the indifference point. The pass's fixed fee is interest: pay it with zero rides and you are out of pocket, which is why EPS under debt goes negative below the financial break-even point.

Key idea

Hold EBIT fixed, deduct each plan's interest, tax and preference dividend, and divide by its number of shares: that is EBIT-EPS analysis. The financial break-even point is the EBIT where EPS is zero; the indifference point is the EBIT where two plans' EPS are equal. Expect EBIT above the indifference point: prefer the plan with more debt. Below it: prefer equity.

How it works

Seven steps from EBIT to a decision

ICAI names three analyses for choosing the optimal capital structure: EBIT-EPS-MPS analysis, indifference point analysis and financial break-even point analysis (SM 5.32–5.33). The steps below run all three; the principles that sit beside them (control, risk, flexibility) are in capital structure planning.

1. **Take the same EBIT for every plan** (SM 5.32): the expected EBIT after the new investment.
2. **Deduct interest** on existing and new debt to get EBT, then **tax** to get PAT.
3. **Deduct preference dividend**, existing and new, to get the earnings available to equity shareholders. It comes after tax because it is not tax-deductible.
4. **Divide by the number of equity shares**: existing shares plus new ones, where new shares = amount raised ÷ issue price (face value + premium).
5. **Convert EPS to market price per share (MPS)**: multiply by the price-earnings (PE) ratio or divide by the cost of equity. If the plans have different PE ratios, the plan with the highest MPS is chosen, not the highest EPS (SM 5.35).
6. **Find each plan's financial break-even point**: the EBIT at which its EPS is zero.
7. **Find the indifference point** between plans: the EBIT at which their EPS are equal. Compare it with the expected EBIT.

Tapri's three plans at EBIT of ₹30 crore

A: equity; B: 10% debentures; C: 11% preference shares. Interest under B = 10% × ₹20 crore; preference dividend under C = 11% × ₹20 crore.

₹ crore	A	B	C
EBIT	30.00	30.00	30.00
Less: Interest	—	2.00	—
EBT	30.00	28.00	30.00
Less: Tax @ 25%	7.50	7.00	7.50
PAT	22.50	21.00	22.50
Less: Pref. dividend	—	—	2.20
Equity earnings	22.50	21.00	20.30
Shares (crore)	2.00	1.50	1.50
EPS (₹)	**11.25**	**14.00**	**13.53**

Debt wins at this EBIT. It would beat preference shares even at the same rate (SM Illustration 11: 8% debt against 8% preference shares), because interest is deducted before tax and preference dividend after it. Theoretically the choice is tilted in favour of debt for two reasons: the explicit cost of debt (interest) is generally lower than the preference dividend rate, and interest is tax-deductible, so its real (after-tax) cost is lower (SM 5.33). Here 10% × (1 − 0.25) = 7.5% after tax, against 11% for the preference shares.

The financial break-even point: where EPS hits zero

Financial break-even point is the minimum level of EBIT needed to satisfy all the fixed financial charges, i.e. interest and preference dividends; it is the level of EBIT at which EPS equals zero (SM 5.36):

$$\text{Financial BEP} = \text{Interest} + \frac{\text{Preference dividend}}{1 - t}$$

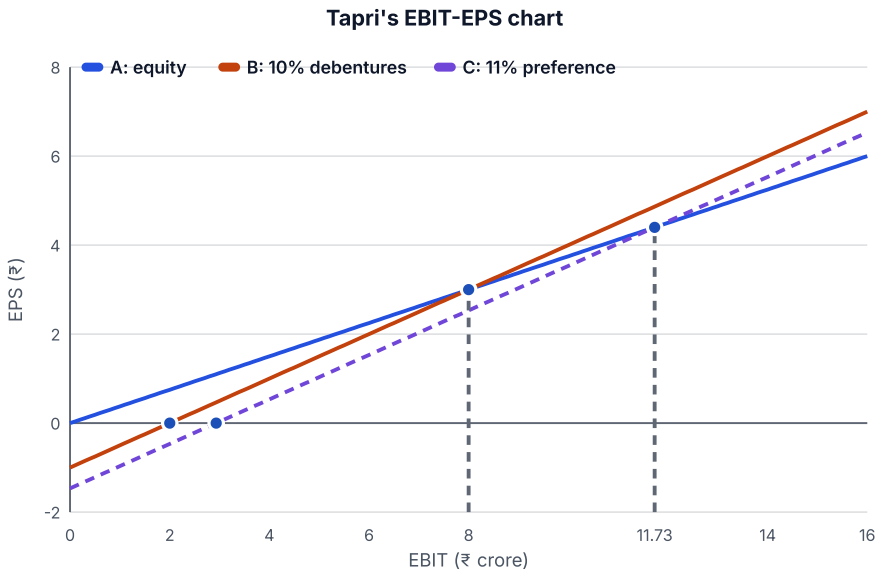
The preference dividend is grossed up because it is paid out of profit after tax. Tapri: plan A, nil; plan B, ₹2 crore; plan C, ₹2.20 crore ÷ 0.75 = ₹2.93 crore. If EBIT is below the financial BEP, EPS is negative; if the expected EBIT is comfortably above it, more fixed-cost financing can be taken; otherwise equity would be preferred.

The indifference point: where two plans tie

Plot each plan's EPS against EBIT: each gives a straight line, starting at its financial BEP. Where two lines intersect is the **EPS equivalency point** or **indifference point**: between those two plans, EPS is the same at that EBIT (SM 5.36–5.37). Algebraically:

$$\frac{(\text{EBIT} - I_1)(1 - t)}{E_1} = \frac{(\text{EBIT} - I_2)(1 - t)}{E_2}$$

Tapri, A against B: EBIT × 0.75 ÷ 2 = (EBIT − 2) × 0.75 ÷ 1.5, so 1.5 EBIT = 2 EBIT − 4, and **EBIT = ₹8 crore** (EPS ₹3.00 under both). A against C, with the preference dividend deducted after tax: 0.75 EBIT ÷ 2 = (0.75 EBIT − 2.2) ÷ 1.5, so **EBIT = ₹11.73 crore** (EPS ₹4.40).



Blue: plan A; orange: plan B; violet dashed: plan C. Dots on the x-axis: the financial BEPs of B (₹2 crore) and C (₹2.93 crore). Grey dashed: the indifference points, A = B at ₹8 crore and A = C at ₹11.73 crore. B and C are parallel, so they never meet. Expected EBIT, ₹30 crore, lies far to the right.

Same number of shares, no single indifference point. If the amount of equity share capital is the same under two plans, one of two situations arises (SM 5.37–5.38):

1. **No indifference point**: if the after-tax cost of the other source is not the same under both plans, one plan is better than the other at all levels of EBIT. Plans B and C both have 1.5 crore shares, and B's after-tax cost (7.5%) is lower than C's (11%): B is better at every EBIT, so B dominates C.
2. **Many indifference points**: if the after-tax cost of the other source is the same under both plans, every EBIT is an indifference point.

Interactive version: PassBook app, lesson EBIT-EPS analysis, indifference point and financial break-even

When leverage turns against you

At an EBIT of ₹6 crore, below both indifference points, the order reverses: EPS is ₹2.25 under A, ₹2.00 under B and ₹1.53 under C. When the rate of return on the total assets is less than the cost of debt, EPS falls with the degree of leverage (SM 5.35–5.36). That is why the expected EBIT must be compared with the indifference point, not just with today's figures: the further the expected EBIT sits above it, and the lower the chance of falling below it, the stronger the case for debt (SM 5.65).

Formats and formulas

Reproduce the EPS statement from memory; every EBIT-EPS answer starts with it.

Statement of EPS under alternative financing plans	Plan I (₹)	Plan II (₹)
EBIT	xxx	xxx
Less: Interest (existing + new)	(xxx)	(xxx)
EBT (PBT)	xxx	xxx
Less: Tax	(xxx)	(xxx)
PAT	xxx	xxx
Less: Preference dividend	(xxx)	(xxx)
Earnings available to equity shareholders	xxx	xxx
Number of equity shares (existing + new)	xxx	xxx
EPS	**xxx**	**xxx**
PE ratio	x	x
MPS = EPS × PE ratio	**xxx**	**xxx**

∑ **Formula**

Earnings per share

$$\text{EPS} = \frac{(\text{EBIT} - I)(1 - t) - \text{PD}}{N}$$

I = interest, t = tax rate, PD = preference dividend, N = number of equity shares (existing + amount raised ÷ issue price).

∑ **Formula**

Financial break-even point

$$\text{Financial BEP} = I + \frac{\text{PD}}{1 - t}$$

If the company has no preference shares, the financial BEP is simply the interest (SM 6.19).

∑ **Formula**

Indifference point between two plans

$$\frac{(\text{EBIT} - I_1)(1 - t)}{E_1} = \frac{(\text{EBIT} - I_2)(1 - t)}{E_2}$$

E_1, E_2 = number of equity shares and I_1, I_2 = interest under plans 1 and 2 (SM 5.37). With preference shares, deduct PD after tax: $(\text{EBIT} - I_2)(1 - t) - \text{PD}_2$ in the numerator.

Formula

Market price per share

$$\text{MPS} = \text{EPS} \times \text{PE ratio} \quad = \frac{\text{EPS}}{K_e}$$

(SM 5.32)

Definition

Financial break-even point

Financial break-even point is the minimum level of EBIT needed to satisfy all the fixed financial charges i.e. interests and preference dividends. It denotes the level of EBIT for which the company's EPS equals zero. (SM 5.36)

Definition

Indifference point

The EBIT at which the EPS lines of two alternatives of financing intersect: between the two given alternatives (regardless of leverage in the financial plans), EPS would be the same at that level of EBIT. Also called the EPS equivalency point. (SM 5.37)

Trick

The one-line indifference formula

Write F for each plan's financial BEP (interest + PD ÷ (1 − t)) and E for its number of shares. Then EBIT at indifference = $(E_1 \times F_2 - E_2 \times F_1) \div (E_1 - E_2)$. Tapri, A against C: (2 × 2.9333 − 1.5 × 0) ÷ 0.5 = ₹11.73 crore. Conditions: the same tax rate in both plans and $E_1 \neq E_2$ (equal shares means no point or every point).

Worked examples

Worked example **Kulhad Crafts: two mixes, one kiln**

Kulhad Crafts Ltd, Tapri's pottery supplier, has 5,00,000 equity shares of ₹10 each and 10% debentures of ₹30,00,000. It needs ₹1,20,00,000 for a new kiln line and can issue equity shares of ₹10 at a premium of ₹10. Two plans are proposed:

Particulars	Plan I (₹)	Plan II (₹)
Equity shares at ₹20	80,00,000	40,00,000
11% term loan	40,00,000	80,00,000

The tax rate is 25%.

Required:

(i) Calculate the EBIT at the indifference point between the two plans and the EPS at that point.

(ii) Calculate the financial break-even point of each plan.

(iii) Calculate the EPS of each plan if EBIT after expansion is ₹36,00,000, and if it is only ₹18,00,000. State which plan is preferable in each case.

THE WORKING

Working notes

1. Equity shares: Plan I = 5,00,000 + ₹80,00,000 ÷ ₹20 = **9,00,000**; Plan II = 5,00,000 + ₹40,00,000 ÷ ₹20 = **7,00,000**.
2. Interest: Plan I = ₹3,00,000 (existing) + 11% × ₹40,00,000 = **₹7,40,000**; Plan II = ₹3,00,000 + 11% × ₹80,00,000 = **₹11,80,000**.

(i) Indifference point

$$\frac{(\text{EBIT} - 7{,}40{,}000)(1 - 0.25)}{9{,}00{,}000} = \frac{(\text{EBIT} - 11{,}80{,}000)(1 - 0.25)}{7{,}00{,}000}$$

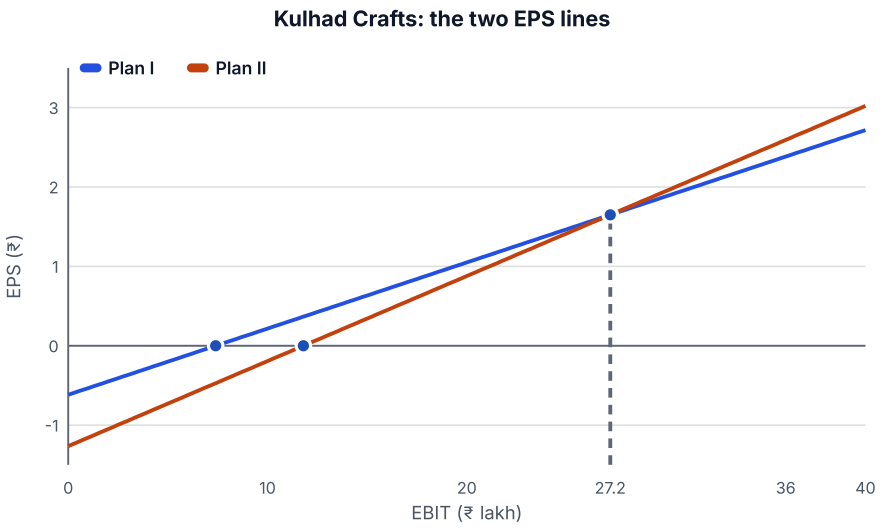
7 EBIT − 51,80,000 = 9 EBIT − 1,06,20,000, so 2 EBIT = 54,40,000 and **EBIT = ₹27,20,000**. EPS = (27,20,000 − 7,40,000) × 0.75 ÷ 9,00,000 = **₹1.65** under both plans.

(ii) Financial break-even point (no preference shares, so = interest): Plan I **₹7,40,000**; Plan II **₹11,80,000**.

(iii) EPS at the two EBIT levels

EBIT of ₹36,00,000	Plan I (₹)	Plan II (₹)
EBIT	36,00,000	36,00,000
Less: Interest	7,40,000	11,80,000
EBT	28,60,000	24,20,000
Less: Tax @ 25%	7,15,000	6,05,000
PAT	21,45,000	18,15,000
Equity shares	9,00,000	7,00,000
EPS	**2.38**	**2.59**

EBIT of ₹18,00,000	Plan I (₹)	Plan II (₹)
EBIT	18,00,000	18,00,000
Less: Interest	7,40,000	11,80,000
EBT	10,60,000	6,20,000
Less: Tax @ 25%	2,65,000	1,55,000
PAT	7,95,000	4,65,000
Equity shares	9,00,000	7,00,000
EPS	**0.88**	**0.66**



Blue: Plan I (more equity); orange: Plan II (more debt). Dots on the x-axis: the financial BEPs, ₹7.4 lakh and ₹11.8 lakh. Grey dashed: the indifference point, ₹27.2 lakh, where both EPS are ₹1.65. Below it Plan I gives the higher EPS; above it, Plan II.

Comment. At EBIT of ₹36,00,000, above the indifference point, Plan II gives the higher EPS: the use of more fixed-cost funds is beneficial and the financial leverage is favourable. At ₹18,00,000, below the indifference point, Plan I is preferable; Plan II's financial BEP of ₹11,80,000 also leaves less room before EPS turns negative.

Worked example **Tapri Kulhad Cafés: EPS against MPS**

Tapri Kulhad Cafés Ltd, Tapri's listed café subsidiary, has a present capital structure of equity share capital of ₹10 crore (1,00,00,000 shares of ₹10 each), reserves of ₹50 crore and 10% debentures of ₹20 crore. Its shares sell at ₹135 at a PE ratio of 15. The tax rate is 25%.

Tapri Kulhad Cafés needs ₹30 crore for 200 new kulhad cafés, which will add ₹7.50 crore a year to EBIT. Alternatives:

(I) Issue equity shares at ₹120 each (premium ₹110) for the whole amount.

(II) Issue equity shares at ₹120 for ₹15 crore and debentures for ₹15 crore.

(III) Issue debentures for the whole ₹30 crore.

New debentures carry 10% interest on the first ₹15 crore and 12% on any amount above it. If total debt exceeds 40% of (debt + net worth) at book value, the PE ratio will fall to 12; otherwise it stays at 15.

Required: Advise which alternative Tapri Kulhad Cafés should choose to maximise the wealth of its shareholders.

THE WORKING

Working note 1: present EBIT

EPS = MPS ÷ PE = ₹135 ÷ 15 = ₹9; PAT = ₹9 × 1 crore shares = ₹9 crore; PBT = ₹9 crore ÷ (1 − 0.25) = ₹12 crore; EBIT = ₹12 crore + interest ₹2 crore = **₹14 crore**. After expansion, EBIT = ₹14 crore + ₹7.50 crore = **₹21.50 crore**.

Working note 2: shares, interest and debt ratio

Particulars	I	II	III
New shares (÷ ₹120)	25,00,000	12,50,000	—
Total shares	1,25,00,000	1,12,50,000	1,00,00,000
New interest (₹ crore)	—	1.50	3.30
Total debt (₹ crore)	20	35	50
Net worth (₹ crore)	90	75	60
Debt ratio	18.18%	31.82%	45.45%
PE ratio	15	15	12

Debt ratio = debt ÷ (debt + net worth). Interest under III = 10% × ₹15 crore + 12% × ₹15 crore = ₹1.50 crore + ₹1.80 crore = ₹3.30 crore; total interest = existing ₹2 crore + new.

Statement of EPS and MPS (₹ crore unless stated)

Particulars	I	II	III
EBIT	21.500	21.500	21.500
Less: Interest (WN 2)	2.000	3.500	5.300
EBT	19.500	18.000	16.200
Less: Tax @ 25%	4.875	4.500	4.050
PAT	14.625	13.500	12.150
Shares (crore)	1.250	1.125	1.000
EPS (₹)	**11.70**	**12.00**	**12.15**
PE ratio	15	15	12
MPS (₹)	**175.50**	**180.00**	**145.80**

> **Advice.** Alternative III gives the highest EPS (₹12.15), but its debt ratio of 45.45% crosses the 40% limit and the PE ratio falls to 12, so its MPS is the lowest. Alternative II gives the highest market price per share (₹180.00) and should be chosen: the objective is to maximise wealth, i.e. MPS, not EPS.

Interactive version: PassBook app, lesson EBIT-EPS analysis, indifference point and financial break-even

Exam angle

What ICAI asks

In the exam

In the ICAI documents read for this lesson (new scheme, May 2024 to September 2026)

EBIT-EPS analysis, the indifference point or the financial BEP appeared in 12 of the 32 question sets searched:

- **Case-scenario MCQs, 2 marks each**: January 2025 Case Scenario I, MCQs 1–5 (EPS, MPS at PE 12 and 15, financial BEP of ₹13,60,000 and ₹15,08,571, indifference point ₹34,40,000, EPS ₹1.30 at that point); MTP Series I for January 2025, MCQs 1–5 (three plans, two indifference points); single MCQs in MTP Series I for May 2024, Series II for January 2026 and Series II for May 2026 (find the preference dividend from a given indifference point).
- **Descriptive, 3–8 marks**: September 2024 Q3(b) (5 marks, EPS and MPS with a lower PE for the levered plan); MTP Series II for September 2024 Q1(a) (5 marks, debt-equity 2:1 insisted on by lenders); MTP Series I for May 2025 Q3(a) (3 marks, indifference with preference shares, plus financial BEP); MTP Series II for September 2026 Q3(a) (8 marks); RTP May 2025 Q6 (advise between two plans on earnings).
- **Theory, 2 marks**: "Discuss financial break-even and EBIT-EPS indifference analysis" (MTP Series I for January 2026); "Explain the relationship between EBIT-EPS-MPS" (MTP Series II for September 2024); "Discuss optimal capital structure and how to analyse it" (MTP Series I for May 2025, Series II for September 2025).

In the exam

The traps in those answers

- New shares are issued at a premium: divide the amount by the issue price, not the face value (January 2025, plan I: ₹30 lakh ÷ ₹25 = 1,20,000 shares).
- Existing interest and existing shares go into every plan (January 2025: the 15% debentures of ₹30 lakh appear in both plans).
- Financial BEP with preference shares grosses the dividend up: ₹11,00,000 + ₹2,86,000 ÷ 0.70 = ₹15,08,571 (January 2025, plan II).
- With different PE ratios, the decision is on MPS (September 2024 Q3(b); SM Illustration 14: "though loan option has higher EPS but equity option has higher MPS").
- "Find the preference dividend from the indifference point" (SM Practical Problem 2; MTP Series II for May 2026, MCQ 7) gives both plans the same number of equity shares. Equate the two EPS at the given EBIT: the answer is always interest × (1 – t) (₹24,000 × 0.70 = ₹16,800 in the MTP), because equal shares and equal after-tax costs make every EBIT an indifference point.
- ICAI's answer to MTP Series II for September 2024 Q1(a) sets the plans out as a statement in terms of E and adds "Presentation of solution may differ": the equation layout of the Study Material is equally acceptable, with working notes either way.

Answer frame

The theory questions

"Discuss financial break-even and EBIT-EPS indifference analysis" (MTP Series I for January 2026; SM 5.36–5.37):

1. Financial BEP: the minimum EBIT needed to satisfy all fixed financial charges (interest and preference dividend); EBIT at which EPS = 0; formula interest + preference dividend ÷ (1 – t).
2. Below it EPS is negative; if expected EBIT is above it, more fixed-cost financing can be taken, otherwise equity is preferred.
3. EBIT-EPS indifference: the EBIT at which EPS is the same under two financing plans; formula $(EBIT - I_1)(1 - t) \div E_1 = (EBIT - I_2)(1 - t) \div E_2$, symbols defined.
4. It is a vital tool for designing the optimal capital structure.

"Explain the relationship between EBIT-EPS-MPS" (MTP Series II for September 2024; SM 5.33): the objective is the highest MPS, which depends on EPS; at a given EBIT, EPS differs with the financing mix because of fixed financial charges; if the return on assets exceeds the cost of fixed-charge funds, EPS rises (favourable leverage, trading on equity), otherwise it falls; debt is preferred to preference shares because of lower cost and tax-deductible interest.

Three answers to "Calculate the indifference point"

Loses marks

"EBIT = ₹27.2 lakh."

No equation, no shares, no interest: nothing to award if a figure slips.

Passes

The equation with figures, but 4,00,000 new shares (₹80 lakh ÷ ₹20) left out the existing 5,00,000.

Method marks only.

Full marks

Working notes for shares (existing + amount ÷ issue price) and interest (existing + new); the equation; the solution; the EPS at the point as a check; a one-line comment on which plan to prefer above and below it.

Common mistakes

⚠ **Common mistake**

Hall of Shame: new shares at face value

₹80 lakh ÷ ₹10 = 8,00,000 new shares when the issue price is ₹20 doubles the dilution and wrecks every EPS. **Fix:** new shares = amount raised ÷ issue price (face value + premium).

⚠ **Common mistake**

Preference dividend treated like interest

Deducting the preference dividend before tax, or adding it to the financial BEP without grossing it up (₹2.20 crore instead of ₹2.93 crore for Tapri's plan C). **Fix:** it comes out of PAT; in the financial BEP, divide it by (1 – t).

⚠ **Common mistake**

The forgotten old loan

Writing only the new interest and the new shares into each plan. **Fix:** every plan carries the existing interest and the existing shares; only the new money differs.

⚠ **Common mistake**

Highest EPS crowned the winner

When the question gives a PE ratio that changes with leverage, the winner is the highest MPS. In Example 2 the all-debt plan has the best EPS and the worst MPS.

Practice

Check yourself

1 Plan X is all equity with 6,00,000 shares. Plan Y has 2,00,000 shares and debt with annual interest of ₹6,00,000. The tax rate is 30%. At what EBIT (₹) are the EPS of the two plans equal?

Answer: _____

2 A company pays interest of ₹4,00,000 and preference dividend of ₹2,10,000. The tax rate is 30%. What is its financial break-even point?

- **A** ₹6,10,000
- **B** ₹7,00,000
- **C** ₹6,73,000
- **D** ₹4,00,000

3 Case scenario. Tapri's indifference point between all-equity plan A and debenture plan B is EBIT of ₹8 crore. After a poor monsoon, Ritu now expects EBIT of only ₹6 crore for the next few years. Which advice is correct?

- **A** Plan B, because debt is always cheaper than equity
- **B** Plan A, because below the indifference point the equity plan gives the higher EPS
- **C** Either plan, because EPS is equal below the indifference point
- **D** Plan B, because its financial break-even point is lower than plan A's

4 Case scenario. Thanda Logistics needs ₹5,00,000 beyond the 50,000 equity shares it will issue under either plan. Plan I raises it as 12% debentures; Plan II as 9% preference shares. The tax rate is 25%. Ritu is asked for the EBIT-EPS indifference point. Which of the following is correct?

- **A** The indifference point is an EBIT of ₹60,000, the interest on the debentures
- **B** Plan I gives the higher EPS at every EBIT, because interest is tax-deductible
- **C** The two plans give the same EPS at every level of EBIT, so every EBIT is an indifference point
- **D** Plan II gives the higher EPS at every EBIT, because preference dividend is not a charge on profit

5 Which statements about EBIT-EPS analysis are correct? Select all that apply.

- **A** At the indifference point the EPS of the two plans are equal
- **B** Two plans with the same number of equity shares but different after-tax costs of the other source have no indifference point
- **C** At the financial break-even point the EPS of the two plans are equal
- **D** Above the indifference point, the plan with more fixed-charge financing gives the higher EPS

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- EPS = [(EBIT − I)(1 − t) − PD] ÷ N; new shares = amount ÷ issue price.
- Same EBIT for every plan; existing interest and shares go into every plan.
- Financial BEP = I + PD ÷ (1 − t): the EBIT at which EPS = 0.

- Indifference point: $(EBIT - I_1)(1 - t) \div E_1 = (EBIT - I_2)(1 - t) \div E_2$; deduct PD after tax where there are preference shares.
- Above the indifference point: more debt gives more EPS. Below: more equity.
- Same number of shares: no indifference point (different after-tax costs) or every EBIT is one (same costs).
- Debt beats preference shares: its rate is usually lower and its interest is tax-deductible, so it wins even at the same rate.
- Different PE ratios: choose the highest MPS, not the highest EPS. (The pass beats the card only if you ride enough.)

Answer key · Check yourself

1 **900000**

EBIT × 0.70 ÷ 6,00,000 = (EBIT − 6,00,000) × 0.70 ÷ 2,00,000, so 2 EBIT = 6 EBIT − 36,00,000 and EBIT = ₹9,00,000.

2 **B ₹7,00,000**

Interest + preference dividend ÷ (1 − t) = ₹4,00,000 + ₹2,10,000 ÷ 0.70 = ₹7,00,000. ₹6,10,000 forgets to gross up; ₹6,73,000 multiplies by 1.3 instead of dividing by 0.7; ₹4,00,000 ignores the preference dividend.

3 **B Plan A, because below the indifference point the equity plan gives the higher EPS**

Below the indifference point the plan with less fixed-charge financing gives the higher EPS (₹2.25 against ₹2.00 at ₹6 crore). Plan B's financial BEP is ₹2 crore, higher than plan A's nil.

4 **C The two plans give the same EPS at every level of EBIT, so every EBIT is an indifference point**

The shares are the same, so only the after-tax cost of the other source matters: debentures ₹60,000 × (1 − 0.25) = ₹45,000; preference dividend 9% of ₹5,00,000 = ₹45,000. Equal after-tax costs mean equal EPS at every EBIT: the Study Material's "many indifference points" case (SM 5.38). Had the after-tax costs differed, the cheaper plan would win at every EBIT and there would be no indifference point. Interest being tax-deductible is already counted in the ₹45,000.

5 **A At the indifference point the EPS of the two plans are equal; B Two plans with the same number of equity shares but different after-tax costs of the other source have no indifference point; D Above the indifference point, the plan with more fixed-charge financing gives the higher EPS**

The financial BEP is where a plan's EPS is zero, not where two plans tie. The other three are the Study Material's rules.

5.3 Capital structure planning, determinants and over- and under-capitalisation

What capital structure is, the control-risk-cost balance and seven principles that shape it, how an optimal structure is found, and the causes, consequences and remedies of over- and under-capitalisation.

In plain words

A wedding hall with no guests, a stall with a queue

Tapri Tea Co. Ltd wants ₹200 crore for 300 new kulhad cafés, and the board meeting has three opinions.

> **Kesar:** Issue shares. No interest, no bank manager, sab adjust ho jayega.
>
> **Dev:** Issue enough shares and the family drops below 51%. Then the board adjusts you.
>
> **Ananya:** And if it is all debt, one bad monsoon and the interest still arrives on the first of the month.

Capital structure is the combination of capitals from different sources of finance: equity shareholders' fund, preference share capital and long-term external debts. Choosing it is a balancing act between **control** (the existing owners keep their majority), **risk** (financial risk stays within tolerable limits) and **cost** (the overall cost of capital stays at a minimum). All three cannot be achieved together, so the finance manager balances them, with one prime objective: to maximise the value of the company.

Two things can go wrong with the amount of capital itself. Kesar's cousin raised ₹60 crore for a sweets factory that runs at about a third of its capacity: its earnings cannot pay a decent dividend and interest, and its share price sinks. That is **over-capitalisation**: more capital than the business needs or can earn on. The original Tapri stall in Indore, on the other hand, ran on ₹50,000 of capital and earned like a factory: a dividend rate no one else matched, and competitors queuing to copy it. That is **under-capitalisation**: less capital in the books than its earning capacity warrants.

Think of it like

The wedding hall and the chai stall

A wedding hall built for a thousand guests that fills two hundred seats a night has too much capital for what it earns: every rupee invested earns very little, and the owner cuts what he pays his backers. That is over-capitalisation. A roadside chai stall with a queue round the block earns a fortune on a few thousand rupees: high returns attract competitors, the staff ask for a raise, and customers feel overcharged. That is under-capitalisation. One is a problem of earnings; the other only of the books catching up.

Key idea

Capital structure is planned to balance control, risk and cost so as to maximise the value of the firm, guided by seven principles (trading on equity, growth and stability of sales, cost, risk, control, flexibility, other considerations). Over-capitalisation (capital more than earnings justify) and under-capitalisation (capital less than earnings warrant) are both bad; over-capitalisation is the more dangerous.

How it works

1. What the capital structure decision decides

Capital structure is the combination of capitals from different sources of finance; the capital of a company consists of equity shareholders' fund, preference share capital and long-term external debts. The source and quantum of capital is decided keeping in mind the following factors (SM 5.2):

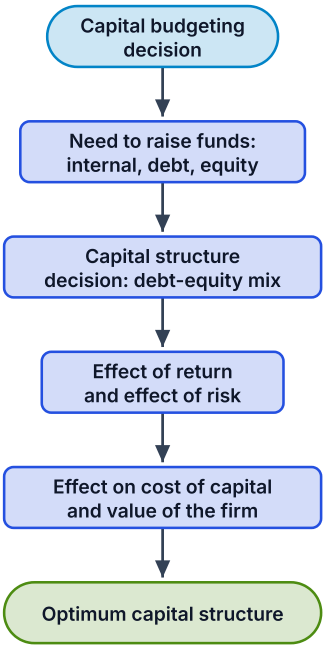
1. **Control**: capital structure should be designed in such a manner that existing shareholders continue to hold majority stake.
2. **Risk**: capital structure should be designed in such a manner that financial risk of a company does not increase beyond tolerable limit.
3. **Cost**: overall cost of capital remains minimum.

The capital structure decision refers to deciding the **forms of financing** (which sources to be tapped), their **actual requirements** (amount to be funded) and their **relative proportions** (mix) in total capitalisation (SM 5.3). Because value of the firm = EBIT ÷ overall cost of capital, the mix decides the weights in the WACC and so the value: capital structure is relevant in maximising value of the firm and minimising overall cost of capital. Whether it really changes the WACC is the question the capital structure theories argue about.

2. The financing decision process

Whenever funds are to be raised to finance investments, a capital structure decision is involved: the demand for funds generates a new capital structure (SM 5.3–5.4).

The financing decision process (SM 5.4)



ICAI's figure, top to bottom: the capital budgeting decision (replacement, modernisation, expansion, diversification) creates the need to raise funds (internal funds, debt, external equity); the capital structure decision takes the existing capital structure and the payout policy into the desired debt-equity mix, whose effect on return and on risk decides the effect on the cost of capital and the value of the firm, and so the optimum capital structure. Green: the goal.

3. The menu of sources

A firm can raise funds from different sources in different proportions (SM 5.30). It can:

- (a) exclusively use debt (in case of existing company), or
- (b) exclusively use equity capital, or
- (c) exclusively use preference share capital (in case of existing company), or
- (d) use a combination of debt and equity in different proportions, or
- (e) use a combination of debt, equity and preference capital in different proportions, or
- (f) use a combination of debt and preference capital in different proportion (in case of existing company).

The choice of the combination of these sources is called the **capital structure mix**.

4. Seven principles for choosing the mix

ICAI's factors affecting capital structure, in ICAI's order (SM 5.30–5.32):

1. **Financial leverage or trading on equity.** The use of long-term fixed interest bearing debt and preference share capital along with equity share capital is called financial leverage or trading on equity. Long-term debt increases EPS if the firm yields a return higher than the cost of debt; preference capital does too, but because interest is deductible for tax, the leverage impact of debt is much more. Leverage can operate adversely if the rate of interest is more than the expected rate of earnings, so it needs caution.

2. **Growth and stability of sales.** Stable sales let a firm raise a higher level of debt, because it will not face difficulty in meeting fixed commitments of interest and repayment. Usually, the greater the rate of growth of sales, the greater the use of debt. If sales are highly fluctuating or declining, debt should be avoided as far as possible.
3. **Cost principle.** The ideal pattern minimises cost of capital and maximises EPS. Debt is cheaper than equity from the point of its cost, and interest is deductible for income tax purpose, whereas dividends are not.
4. **Risk principle.** Reliance is placed more on equity than on excessive use of debt. More debt means a higher commitment of interest, which erodes shareholders' value in unfavourable business situations: as debt increases, financial risk increases.
5. **Control principle.** Existing management control and ownership should remain undisturbed. New equity dilutes the existing control pattern and involves higher cost; more debt causes no dilution in control but a higher degree of financial risk.
6. **Flexibility principle.** Choose a combination that is easier to adjust to future changes in the need for funds. Debt can be interchanged (an 18% loan can be replaced by new debt at 15%); the same option may not be available with equity.
7. **Other considerations.** Nature of industry, timing of issue and competition in the industry. Industries facing severe competition resort to more equity than debt.

A suitable pattern is a satisfactory compromise between these principles, reached by assigning weights to them in terms of the characteristics of the company (SM 5.32).

Principle	Points towards	Because
Trading on equity	debt, if return > cost of debt	EPS rises; interest is tax-deductible
Growth and stability of sales	debt if sales are stable or growing; equity if they fluctuate or decline	fixed interest must be met every year
Cost	debt	cheaper; interest tax-deductible
Risk	equity	more debt, more financial risk
Control	debt	no dilution of control
Flexibility	debt	debt can be replaced by cheaper debt; the same option may not exist for equity
Other considerations	equity in severely competitive industries	nature of industry, timing, competition

5. The optimal capital structure

The objective of financial management is to maximise wealth, so one should choose the capital structure which maximises wealth (SM 5.32); the optimum is where value of the firm is highest and cost of capital is lowest (SM 5.9). Three analyses are used (SM 5.32–5.33):

1. **EBIT-EPS-MPS analysis**: choose the capital structure which maximises market price per share. Start with the same EBIT for all capital structures, calculate EPS, then multiply EPS by the PE ratio or divide it by the cost of equity to arrive at MPS.

2. **Indifference point analysis**: the first analysis considers value at a given EBIT only; to see whether the decision changes when EBIT changes, do indifference point analysis.
3. **Financial break-even point analysis**: financial risk changes with capital structure; though it is already reflected in the PE ratio or cost of equity, it may be considered separately by calculating the financial BEP.

All three are worked in EBIT-EPS analysis.

6. Over-capitalisation: more capital than the business can carry

It is a situation where a firm has more capital than it needs, or in other words assets are worth less than its issued share capital, and earnings are insufficient to pay dividend and interest. It mainly arises when existing capital is not effectively utilised because of a fall in earning capacity, while the company has raised funds more than its requirements. The chief sign of over-capitalisation is the fall in payment of dividend and interest leading to fall in value of the shares (SM 5.45).

Causes (SM 5.45):

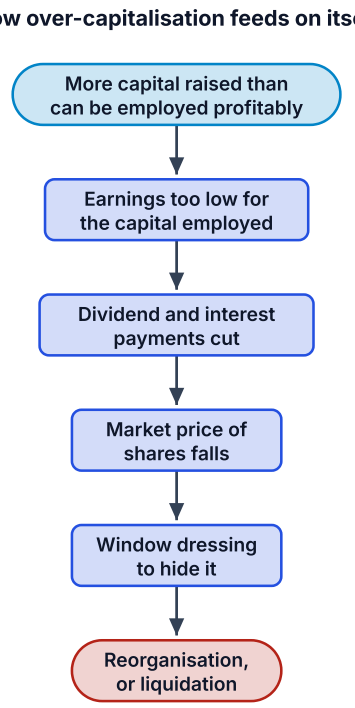
1. Raising more money through issue of shares or debentures than company can employ profitably.
2. Borrowing huge amount at higher rate than rate at which company can earn.
3. Excessive payment for the acquisition of fictitious assets such as goodwill etc.
4. Improper provision for depreciation, replacement of assets and distribution of dividends at a higher rate.
5. Wrong estimation of earnings and capitalisation.

Consequences (SM 5.45):

1. Considerable reduction in the rate of dividend and interest payments.
2. Reduction in the market price of shares.
3. Resorting to "window dressing".
4. Some companies may opt for reorganisation; sometimes the matter gets worse and the company may go into liquidation.

Remedies (SM 5.46):

1. Company should go for thorough reorganisation.
2. Buyback of shares.
3. Reduction in claims of debenture-holders and creditors.
4. Value of shares may also be reduced. This will result in sufficient funds for the company to carry out replacement of assets.



The causes lead to the four consequences in ICAI's order. Rose: the end of the road if nothing is done.

7. Under-capitalisation: less capital than the earnings warrant

It is just reverse of over-capitalisation: a state when actual capitalisation is lower than proper capitalisation as warranted by its earning capacity. It normally happens with companies which have insufficient capital but large secret reserves in the form of considerable appreciation in the values of fixed assets not brought into the books (SM 5.46).

Consequences (SM 5.46):

1. The dividend rate will be higher in comparison to similarly situated companies.
2. Market value of shares will be higher than value of shares of other similar companies, because their earning rate is considerably more than the prevailing rate on such securities.
3. Real value of shares will be higher than their book value.

Effects (SM 5.46–5.47):

1. It encourages acute competition: high profitability encourages new entrepreneurs into the same type of business.
2. High rate of dividend encourages the workers' union to demand high wages.
3. Normally common people (consumers) start feeling that they are being exploited.
4. Management may resort to manipulation of share values.
5. It invites more government control and regulation and higher taxation also.

Remedies (SM 5.47):

1. The shares should be split up. This will reduce dividend per share, though EPS shall remain unchanged.
2. Issue of bonus shares is the most appropriate measure, as this will reduce both dividend per share and the average rate of earning.
3. Revising upward the par value of shares in exchange of the existing shares held by them.

8. Which is worse?

Both are not good. Over-capitalisation is more dangerous to the company, shareholders and the society than under-capitalisation; under-capitalisation can be handled more easily, and it is not an economic problem but a problem of adjusting capital structure. So under-capitalisation is considered less dangerous, but both situations are bad and every company should strive to have a proper capitalisation (SM 5.47).

Point	Over-capitalisation	Under-capitalisation
Meaning	more capital than needed; assets worth less than issued share capital	actual capitalisation lower than proper capitalisation warranted by earning capacity
Typical origin	funds raised beyond requirement; fall in earning capacity	large secret reserves (appreciated fixed assets not in the books)
Dividend rate	falls considerably	higher than similar companies
Market value of shares	falls	higher than similar companies; real value above book value
Remedies	reorganisation; buyback; reduce claims of debenture-holders and creditors; reduce value of shares	split shares; bonus shares (most appropriate); revise par value upward
Danger	more dangerous to the company, shareholders and society	less dangerous: not an economic problem but one of adjusting capital structure

Worked examples

Worked example **Mehra Mithai's empty factory**

Mehra Mithai Ltd, owned by Kesar's cousin, raised ₹40 crore by issuing equity shares and ₹20 crore by 14% debentures to build a sweets factory and buy the brand of a closed sweet-shop chain, paying ₹12 crore for its goodwill. The factory runs at 35% of capacity and the company earns 8% on its capital employed. The dividend has fallen from 18% to 3% in two years and the share trades below its face value. The directors now propose to skip depreciation this year "to show a healthier profit".

Required: Identify the situation the company is in, giving reasons from the facts. State its causes and consequences as seen here, and advise the remedies.

MODEL ANSWER

Meaning. Over-capitalisation is a situation where a firm has more capital than it needs, or in other words assets are worth less than its issued share capital, and earnings are insufficient to pay dividend and interest. Its chief sign is the fall in payment of dividend and interest, leading to a fall in the value of the shares.

Facts and analysis. Mehra Mithai has raised ₹60 crore but employs only 35% of the capacity it built, so the capital is not effectively utilised. It earns 8% on capital employed while paying 14% on debentures. The dividend has fallen from 18% to 3% and the share trades below face value. The company is **over-capitalised**.

Causes visible in the case:

1. Raising more money through issue of shares and debentures than the company can employ profitably (a factory at 35% capacity).
2. Borrowing at a higher rate (14%) than the rate at which the company can earn (8%).
3. Excessive payment for the acquisition of a fictitious asset (₹12 crore of goodwill of a closed chain).
4. Wrong estimation of earnings and capitalisation.
5. The proposal to skip depreciation is improper provision for depreciation, which would worsen the position.

Consequences: considerable reduction in the rate of dividend (18% to 3%); reduction in the market price of shares (below face value); the depreciation proposal is resorting to window dressing; if not corrected, the company may have to opt for reorganisation or may even go into liquidation.

Remedies advised:

1. A thorough reorganisation of the company.
2. Buyback of shares, to bring the capital in line with what the business can employ.
3. Reduction in the claims of debenture-holders and creditors.
4. Reduction in the value of shares, which will leave sufficient funds for replacement of assets.

The directors should not skip depreciation: proper provision is needed, and window dressing does not cure over-capitalisation.

Worked example

Indore Kettle Works and its old land

Indore Kettle Works Ltd, which makes Tapri's brewing equipment, has share capital of ₹1 crore and earns a profit after tax of about ₹90 lakh every year. Its factory land, bought in 1985 for ₹20 lakh, is now worth about ₹40 crore but has never been revalued. It pays a dividend of 60% against 15% by similar companies, and its shares trade at many times their book value. This year two new competitors have set up nearby, the workers' union has demanded a 40% wage rise, and a consumer forum is calling the company's prices exploitative.

Required: Identify the situation, with reasons. Explain the effects the company is already facing, advise the remedies, and comment on whether this situation is more dangerous than over-capitalisation.

MODEL ANSWER

Meaning. Under-capitalisation is a state when the actual capitalisation of a company is lower than its proper capitalisation as warranted by its earning capacity. It normally happens with companies which have insufficient capital but large secret reserves in the form of considerable appreciation in the values of fixed assets not brought into the books.

Facts and analysis. A capital of ₹1 crore earning ₹90 lakh a year, land worth ₹40 crore carried at ₹20 lakh, a dividend rate of 60% against 15% by similar companies, and a market value of shares far above book value: the company is **under-capitalised**.

Effects already visible:

1. Acute competition: high profitability has encouraged two new entrepreneurs into the same business.
2. The high rate of dividend has encouraged the workers' union to demand high wages.
3. Consumers feel they are being exploited.
4. Further effects to guard against: management may be tempted to manipulate share values, and the company may invite more government control and regulation and higher taxation.

Remedies advised:

1. Issue of bonus shares, the most appropriate measure, as it reduces both the dividend per share and the average rate of earning.
2. Splitting up the shares, which reduces dividend per share though EPS remains unchanged.
3. Revising upward the par value of shares in exchange for the existing shares.

Comment. Under-capitalisation is less dangerous than over-capitalisation: it is not an economic problem but a problem of adjusting the capital structure, and it can be handled more easily. Both are bad, and the company should strive for a proper capitalisation.

Worked example **Tapri plans its ₹200 crore**

Tapri Tea Co. Ltd needs ₹200 crore for 300 new cafés. The promoter family holds 52% of the equity. Café sales have grown about 15% a year for five years with little fluctuation. Interest rates are high this year but expected to fall within two years. BrewBucks has just entered Tapri's cities and price competition is severe.

Required: Advise the board on the principles that should guide the choice between debt and equity for this expansion.

MODEL ANSWER

While choosing a suitable financing pattern, the finance manager must reach a satisfactory compromise between the principles of capital structure, assigning weights to them according to the company's characteristics. Applied to Tapri:

1. **Control principle**: the family holds only 52%. A large issue of new equity would dilute the existing control pattern; debt causes no dilution. This favours debt.
2. **Growth and stability of sales**: sales are growing at about 15% and are stable, so Tapri can meet fixed interest commitments; greater growth permits greater use of debt. This favours debt.
3. **Trading on equity and cost principle**: if the cafés earn more than the cost of debt, debt raises EPS, and interest is tax-deductible. This favours debt, provided the expected return exceeds the rate of interest.
4. **Risk principle**: more debt raises financial risk and the fixed interest burden in unfavourable situations. This calls for restraint.
5. **Flexibility principle**: interest rates are expected to fall; debt raised now can later be replaced by cheaper debt, which equity does not allow.
6. **Other considerations**: competition from BrewBucks is severe, and industries facing severe competition resort to more equity than debt.

Advice. A mix: debt for a large part of the ₹200 crore (control, stable growing sales, tax-deductible interest), on terms that allow refinancing when rates fall, with an equity component or internal funds to keep financial risk tolerable given the competition. The final proportion should maximise the value of the firm, tested by EBIT-EPS-MPS, indifference point and financial break-even analysis.

Exam angle

What ICAI asks

In the exam

In the ICAI documents read for this lesson

In the new scheme (May 2024 to September 2026) this topic appeared in 7 of the 32 question sets searched, always as a theory question (2 to 4 marks where marks are shown):

- **Over-capitalisation**: "You are a financial consultant for a company that has a very high capital base but low EPS. Explain over-capitalisation. What are the causes and consequences?" (MTP Series I for January 2025, 4 marks); "Explain in brief the concept of over capitalisation" (MTP Series II for May 2025 Q3(c), OR part, 2 marks); "What are the remedies for over-capitalisation?" (September 2024 Q4, OR part, 2 marks); "What are the causes of over-capitalisation?" (RTP January 2026 Q11(c)).
- **Optimal capital structure**: "What is optimum capital structure?" (MTP Series I for May 2024, 2 marks); "Discuss optimal capital structure and how to analyse it" (MTP Series I for May 2025, 2 marks); "Briefly explain the process for analysing optimal capital structure" (MTP Series II for September 2025, 2 marks).
- **Planning in a case**: a debt-free company raising ₹10 crore: what practical factors would you evaluate before choosing debt, equity or a mix? (MTP Series II for September 2025, 2 marks; ICAI's answer: cost of capital, control, current interest rate and market conditions, cash flow stability).

The old-scheme Paper 8 papers read for practice ask the same things: "The firm has more capital than its requirements. What is this situation called? Give two consequences" or "What are the important factors considered for deciding the source and quantum of capital?" (November 2022, 2 marks); "What are the causes of over-capitalisation?" (November 2023, 2 marks; reused word for word in the RTP for January 2026).

In the exam

What the Suggested Answers give marks for

- ICAI's lists, in ICAI's order and nearly in ICAI's words: five causes, four consequences, four remedies of over-capitalisation; three remedies of under-capitalisation with bonus shares "the most appropriate".
- For a scenario ("more capital than its requirements"), the name of the situation first, then the points.
- For "factors for source and quantum", the three: control, risk, cost.

Trick

Over is cured by shrinking, under by spreading

Over-capitalisation is cured by reducing capital or claims (reorganise, buy back, reduce debenture and creditor claims, reduce share value). Under-capitalisation is cured by spreading the same earnings over more shares or more face value (split, bonus, higher par value). Use it to sort the remedies; the lists themselves must be written in ICAI's words.

Answer frame

"Explain over-capitalisation. What are its causes and consequences?" (4 marks)

Meaning, with the chief sign (falling dividend, interest and share value) → all five causes → all four consequences, each in ICAI's words as listed in section 6 above. In a scenario ("a very high capital

base but low EPS"), name the situation in the first line.

"Write a note on under-capitalisation"

Meaning (with the secret reserves case) → three consequences → five effects → three remedies → one line: less dangerous than over-capitalisation, a problem of adjusting capital structure.

Three answers to "Give two consequences of over-capitalisation" (2 marks)

Loses marks

"The company has too much money and will face losses."

No name for the situation, and "losses" is not one of ICAI's consequences.

Passes

"Over-capitalisation. Dividends fall and so does the share price."

Right ideas, loose words.

Full marks

"The situation is over-capitalisation. Consequences: (i) considerable reduction in the rate of dividend and interest payments; (ii) reduction in the market price of shares."

Common mistakes

⚠ **Common mistake**

Hall of Shame: the bonus issue for an over-capitalised company

Recommending a bonus issue or a share split to a company that cannot pay its dividends. Those spread earnings over more shares: they cure under-capitalisation. **Fix:** over-capitalisation needs reorganisation, buyback, reduction of debenture and creditor claims, or reduction in the value of shares.

⚠ **Common mistake**

The two definitions swapped

Writing "actual capitalisation lower than proper capitalisation warranted by earning capacity" for over-capitalisation: that is under-capitalisation, and the Study Material's own MCQ 11 tests exactly this slip. **Fix:** over = more capital than needed, earnings insufficient; under = capitalisation lower than the earning capacity warrants.

⚠ **Common mistake**

Under-capitalisation read as "short of cash"

Explaining under-capitalisation as a company that cannot pay its bills or lacks working capital. **Fix:** in ICAI's sense it is a company whose earnings are high for its small book capital, often because of secret reserves; its symptoms are a high dividend rate and a high market value.

⚠ **Common mistake**

Only "cost" in a capital structure answer

Arguing that debt is cheaper and stopping there. **Fix:** a full answer balances cost against risk and control (the three factors) and, for "factors affecting", walks through all seven principles, ending in a compromise.

Practice

Check yourself

1 Case scenario. Bunty reviews Chai Kinara Ltd: share capital ₹5 crore, profit after tax ₹4 crore a year, factory land bought decades ago now worth thirty times its book value, a dividend of 50% against an industry average of 12%. What is Chai Kinara's position?

- **A** Over-capitalised, because its assets are worth more than its capital
- **B** Under-capitalised, because its capitalisation is lower than its earning capacity warrants
- **C** Properly capitalised, because it pays a high dividend
- **D** Over-capitalised, because its dividend rate is above the industry average

2 Which measure does the Study Material call the most appropriate remedy for under-capitalisation?

- **A** Buyback of shares
- **B** Reduction in claims of debenture-holders
- **C** Issue of bonus shares
- **D** Thorough reorganisation

3 Which of these are causes of over-capitalisation? Select all that apply.

- **A** Borrowing a huge amount at a rate higher than the company can earn
- **B** Excessive payment for goodwill acquired
- **C** Large secret reserves from appreciated fixed assets
- **D** Wrong estimation of earnings and capitalisation

More than one answer may be right.

4 The promoters of a listed company hold 51% and do not want to lose their majority. Which principle of capital structure points the company towards debt for its next expansion?

- **A** Risk principle
- **B** Control principle
- **C** Flexibility principle
- **D** Other considerations

5 Which statement is correct?

A Under-capitalisation is more dangerous than over-capitalisation because it encourages competition

B Over-capitalisation means the dividend rate is higher than in similar companies

C Over-capitalisation is more dangerous; under-capitalisation is a problem of adjusting the capital structure

D Both are equally dangerous, so neither needs a remedy

Answers: the answer key at the end of this lesson.

Recap

- Capital structure: the combination of equity shareholders' fund, preference share capital and long-term external debts; balance control, risk and cost to maximise value.
- The decision: forms of financing, amounts and proportions in total capitalisation.
- Seven principles in order: trading on equity, growth and stability of sales, cost, risk, control, flexibility, other considerations.
- Optimal structure: highest value, lowest cost of capital; found by EBIT-EPS-MPS, indifference point and financial BEP analysis.
- Over-capitalisation: more capital than needed, earnings insufficient; 5 causes, 4 consequences, 4 remedies; chief sign: falling dividend and interest.
- Under-capitalisation: capitalisation below what earnings warrant (secret reserves); high dividend rate and share value; remedies: split, bonus (most appropriate), higher par value.
- Over-capitalisation is the more dangerous. (The empty wedding hall is the real worry; the busy chai stall only needs bigger books.)

Answer key · Check yourself

1 **B Under-capitalised, because its capitalisation is lower than its earning capacity warrants**

Small book capital, high earnings, secret reserves in appreciated land, a dividend rate far above similar companies: under-capitalisation. Over-capitalisation shows falling dividends and assets worth less than issued capital.

2 **C Issue of bonus shares**

Bonus shares reduce both dividend per share and the average rate of earning (SM 5.47). The other three are remedies for over-capitalisation.

3 **A Borrowing a huge amount at a rate higher than the company can earn; B Excessive payment for goodwill acquired; D Wrong estimation of earnings and capitalisation**

Secret reserves are the typical origin of under-capitalisation. The other three are in ICAI's list of five causes.

4 **B** **Control principle**

New equity dilutes the existing control pattern; debt causes no dilution in control, though it adds financial risk. The risk principle points the other way, towards equity.

5 **C** **Over-capitalisation is more dangerous; under-capitalisation is a problem of adjusting the capital structure**

SM 5.47: over-capitalisation is more dangerous to the company, shareholders and society; under-capitalisation is not an economic problem but one of adjusting the capital structure.

CHAPTER 6

Financing Decisions: Leverages

6.1 Operating, financial and combined leverage

How fixed operating costs magnify a change in sales into a bigger change in EBIT (DOL), fixed financial charges magnify EBIT into EPS (DFL), and both together (DCL = DOL × DFL).

In plain words

Ten per cent more chai, eighty per cent more EPS

Tapri Kulhad Cafés Ltd, Tapri's café subsidiary, sells ₹50 crore a year. Milk, tea and kulhads take 60% of sales; rent, salaries and machine leases are ₹15 crore whatever the sales; interest is ₹2.50 crore; tax is 25%; there are 50 lakh shares. EBIT is ₹5 crore and EPS ₹3.75.

> **Kesar:** Diwali sales are up 10%. So profit is up 10%, right?
>
> **Dev:** EBIT is up 40% and EPS 80%. Fixed costs do not grow with sales, so every extra rupee of contribution falls straight through.
>
> **Kesar:** Then let us open three hundred more cafés!
>
> **Dev:** The lever works both ways. A fall of 12.5% in sales wipes out EPS altogether. Even my optimism is discounted.

Leverage is the influence of one financial variable over another: the percentage change in one divided by the percentage change in the other. Fixed costs are what create it:

- **Operating leverage** links sales and EBIT. Fixed operating costs (rent, salaries) make EBIT change faster than sales. It indicates **business risk**: the uncertainty about future operating income.
- **Financial leverage** links EBIT and EPS. Fixed financial charges (interest, preference dividend) make EPS change faster than EBIT. It indicates **financial risk**: the extra risk shareholders bear because the company uses debt.
- **Combined leverage** links sales and EPS: the two levers multiplied. It indicates total risk.

At Tapri's cafés the operating lever is 4 (EBIT moves four times as fast as sales) and the financial lever 2 (EPS moves twice as fast as EBIT), so the combined lever is 8.

Think of it like

The bicycle gears

Pedal once and the back wheel turns several times: that is a gear ratio. Sales are the pedals, EBIT is the middle sprocket and EPS is the back wheel. Fixed operating costs set the first gear (DOL), fixed interest the second (DFL), and the total ratio is the product of the two (DCL). High gears are glorious going downhill and brutal uphill: when sales fall, the same gears turn a small dip into a crash.

Key idea

Fixed costs are the lever. Fixed operating costs make EBIT move faster than sales (operating leverage, business risk); fixed financial charges make EPS move faster than EBIT (financial leverage, financial risk); the two multiply into combined leverage (total risk). Each degree of leverage is a multiplier on a percentage change, and it magnifies falls exactly as much as rises.

How it works

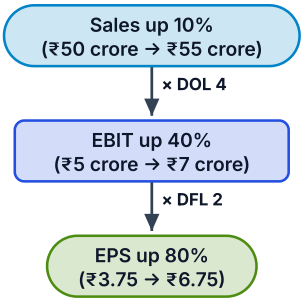
Two risks, three levers

Business risk refers to the risk associated with the firm's operations: the uncertainty about the future operating income (EBIT), i.e. how well can the operating income be predicted. **Financial risk** refers to the additional risk placed on the firm's shareholders because of use of debt, the additional risk a shareholder bears when a company uses debt in addition to equity financing; companies that issue more debt instruments would have higher financial risk (SM 6.2).

$$\text{Leverage of Y with X} = \frac{\Delta Y \div Y}{\Delta X \div X}$$

The three commonly used measures (SM 6.3–6.4): **operating leverage**, the relationship between sales and EBIT, indicating business risk; **financial leverage**, between EBIT and EPS, indicating financial risk; **combined leverage**, between sales and EPS, indicating total risk.

Tapri's cafés: a 10% rise in sales, through both gears



Contribution rises ₹2 crore with no rise in fixed costs, so EBIT rises 40%; interest stays ₹2.50 crore, so EBT rises from ₹2.50 crore to ₹4.50 crore and EPS 80%. DCL = 4 × 2 = 8. A 10% fall gives −40% and −80%.

Operating leverage: the fixed-cost gear

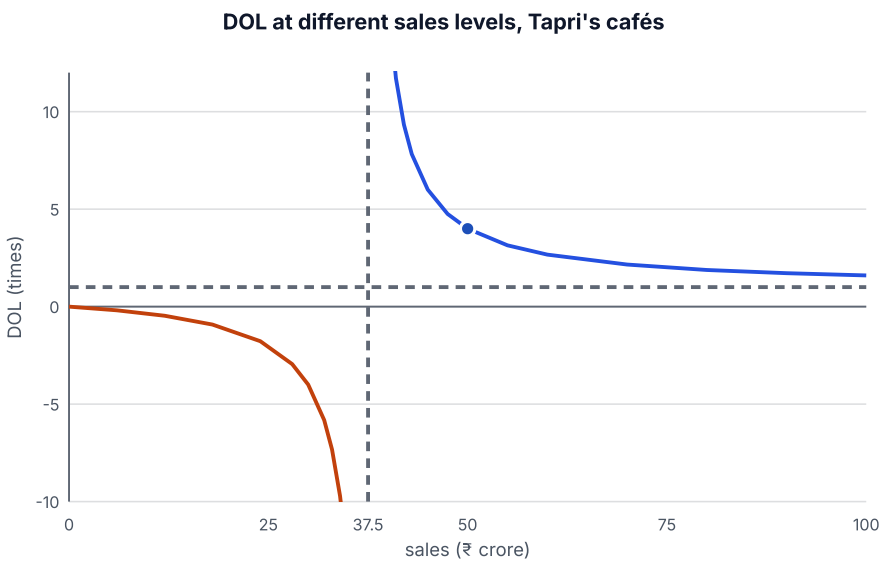
Operating leverage (OL) means the tendency of operating income (EBIT) to change disproportionately with change in sales volume, caused by operating fixed cost, which does not change with sales; it is the use of assets for which a company pays a fixed cost (SM 6.5). It is a function of three factors: (i) amount of fixed cost, (ii) variable contribution margin, and (iii) volume of sales.

The **degree of operating leverage (DOL)** is the percentage change in EBIT with respect to the percentage change in sales quantity. With EBIT = Q (S − V) − F and no change in fixed cost, it simplifies to (SM 6.5–6.6):

$$\text{DOL} = \frac{\%\text{ change in EBIT}}{\%\text{ change in sales}} = \frac{\text{Contribution}}{\text{Contribution} - \text{Fixed cost}} = \frac{\text{Contribution}}{\text{EBIT}}$$

Operating leverage and the break-even point. A firm with high leverage has a higher break-even point; high fixed cost means a high degree of operating leverage (SM 6.7). **Operating leverage and the margin of safety**: multiply and divide the MOS ratio by the P/V ratio and MOS = (contribution − fixed cost) ÷ contribution = EBIT ÷ contribution, so DOL = 1 ÷ margin of safety (SM 6.8). Tapri's cafés: break-even sales = ₹15 crore ÷ 40% = ₹37.50 crore; MOS = (50 − 37.50) ÷ 50 = 25%; DOL = 1 ÷ 0.25 = 4. A higher margin of safety means lower business risk and a lower DOL.

Sign and size of DOL (SM 6.9–6.11): with no fixed cost, DOL = 1 (no operating leverage); above the break-even level it is positive and sales and EBIT move together; at the break-even point, where contribution = fixed cost, it is undefined (infinite); below break-even it is negative and EBIT is negative. DOL can never be between zero and one.



Contribution 40% of sales, fixed cost ₹15 crore; the dot is today, sales ₹50 crore and DOL 4. Grey dashed vertical: the operating BEP, ₹37.5 crore, where DOL is undefined. Blue: above it DOL is positive and falls towards 1 (grey dashed horizontal) as sales grow; orange: below it DOL is negative and rises towards zero at zero sales.

Financial leverage: the fixed-interest gear

Financial leverage (FL) may be defined as "the use of funds with a fixed cost in order to increase earnings per share": funds obtained at a fixed cost in the hope of increasing the return to common stockholders (SM 6.14). The **degree of financial leverage (DFL)** is the ratio of the percentage increase in EPS to the percentage increase in EBIT; FL is also "the ability of a firm to use fixed financial charges

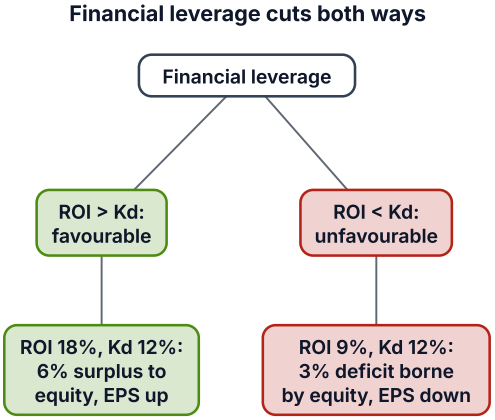
to magnify the effect of changes in EBIT on EPS". Since EPS = [(EBIT − I)(1 − t) − Dp] ÷ number of shares (SM 6.15):

$$\text{DFL} = \frac{\%\text{ change in EPS}}{\%\text{ change in EBIT}} \qquad = \frac{\text{EBIT}}{\text{EBIT} - I - \frac{D_p}{1-t}}$$

Without preference shares, DFL = EBIT ÷ (EBIT − interest) = EBIT ÷ PBT. When DFL is more than one, financial leverage exists. A positive DFL means the firm operates above its **financial break-even point** (the EBIT at which EPS is zero; without preference shares, simply the interest) and EBIT and EPS move in the same direction; at the financial BEP it is undefined; below it, it is negative and EPS is negative (SM 6.15–6.19). Like DOL, DFL can never be between zero and one.

Trading on equity. Financial leverage indicates the use of funds with fixed cost, like long-term debts and preference share capital, along with equity share capital, which is known as trading on equity. A firm has a positive or favourable leverage when its earnings are more than the cost of debt; if earnings are equal to or less than the cost of debt, it is a negative or unfavourable leverage. When the quantity of fixed cost fund is relatively high in comparison to equity capital, the firm is said to be "trading on equity" (SM 6.17).

A double-edged sword. When the cost of the fixed cost fund is less than the return on investment, financial leverage helps to increase return on equity and EPS, and the firm also saves tax on interest. When the cost of debt is more than the return, it affects return on equity and EPS unfavourably and the firm can be under financial distress. Hence financial leverage is called a "double edged sword" (SM 6.18): ROI > interest, favourable, an advantage; ROI < interest, unfavourable, a disadvantage; ROI = interest, neutral.



Green: the favourable edge; rose: the unfavourable edge. The same debt that lifts EPS when the cafés earn 18% drags it down when they earn 9%.

Combined leverage: both gears together

Combined leverage may be defined as the potential use of fixed costs, both operating and financial, which magnifies the effect of sales volume change on the EPS of the firm. The **degree of combined**

leverage (DCL) is the ratio of percentage change in EPS to the percentage change in sales: it indicates the effect the changes in sales will have on EPS (SM 6.19–6.20).

$$\text{DCL} = \text{DOL} \times \text{DFL} \quad = \frac{\text{Contribution}}{\text{EBIT}} \times \frac{\text{EBIT}}{\text{EBT}} \quad = \frac{\text{Contribution}}{\text{EBT}} \quad = \frac{\%\ \text{change in EPS}}{\%\ \text{change in sales}}$$

Combined leverage measures total risk, the combination of operating and financial risk (SM 6.20):

DOL	DFL	Comment
Low	Low	Lower total risk. Cannot take advantage of trading on equity.
High	High	Higher total risk. Very risky combination.
High	Low	Moderate total risk. Not a good combination: lower EBIT due to higher DOL and lower advantage of trading on equity due to low DFL.
Low	High	Moderate total risk. Best combination: higher financial risk is balanced by lower business risk.

Interactive version: PassBook app, lesson Operating, financial and combined leverage

Formats and formulas

Reproduce this statement from memory: ICAI prints it as the "chart showing" the three leverages, with a bracket beside the part of the statement each one covers (SM 6.4). The right-hand column marks where each measure starts and ends.

Profitability statement	₹	Leverage
Sales	xxx	DOL and DCL start
Less: Variable cost	(xxx)	
Contribution	**xxx**	
Less: Fixed cost	(xxx)	
Operating profit / EBIT	**xxx**	DOL ends, DFL starts
Less: Interest	(xxx)	
Earnings before tax (EBT)	**xxx**	
Less: Tax	(xxx)	
Profit after tax (PAT)	xxx	
Less: Preference dividend (if any)	(xxx)	
Net earnings available to equity shareholders	xxx	
Number of equity shares (N)	xxx	
Earnings per share (EPS) = PAT ÷ N	**xxx**	DFL and DCL end

∑ **Formula**

Degree of operating leverage

$$\text{DOL} = \frac{\text{Contribution}}{\text{EBIT}} \quad = \frac{\%\ \Delta\ \text{EBIT}}{\%\ \Delta\ \text{sales}} = \frac{1}{\text{Margin of safety ratio}}$$

Contribution = sales − variable cost; EBIT = contribution − fixed operating cost (SM 6.6, 6.8).

∑ **Formula**

Degree of financial leverage

$$\text{DFL} = \frac{\text{EBIT}}{\text{EBT}} \quad = \frac{\%\ \Delta\ \text{EPS}}{\%\ \Delta\ \text{EBIT}}$$

With preference dividend Dp: DFL = EBIT ÷ [EBIT − interest − Dp ÷ (1 − t)] (SM 6.15).

∑ **Formula**

Degree of combined leverage

$$\text{DCL} = \text{DOL} \times \text{DFL} \quad = \frac{\text{Contribution}}{\text{EBT}} = \frac{\%\ \Delta\ \text{EPS}}{\%\ \Delta\ \text{sales}}$$

(SM 6.19–6.20)

Definition

Business risk and financial risk

Business risk refers to the risk associated with the firm's operations: the uncertainty about the future operating income (EBIT). Financial risk refers to the additional risk placed on the firm's shareholders because of use of debt, i.e. the additional risk a shareholder bears when a company uses debt in addition to equity financing. (SM 6.2)

Definition

Operating leverage

The tendency of operating income (EBIT) to change disproportionately with change in sale volume, caused by operating fixed cost; the employment of an asset with a fixed cost so that enough revenue can be generated to cover all the fixed and variable costs. (SM 6.5)

Definition

Financial leverage

The use of funds with a fixed cost in order to increase earnings per share. (SM 6.14)

DOL, DFL and DCL side by side (SM 6.27)

Point	DOL	DFL	DCL
Shows	business risk	financial risk	total (combined) risk
Depends on	fixed cost	interest and preference dividend	fixed cost, interest and preference dividend
Measures % change in	EBIT from 1% change in sales	EPS from 1% change in EBIT	EPS from 1% change in sales
Example	DOL 3, output +8%: EBIT +24%	DFL 2, EBIT +5%: EPS +10%	DCL 6, sales +8%: EPS +48%
Unique for	each level of output	each level of EBIT	each level of sales
Undefined at	operating BEP	financial BEP	financial BEP

Worked examples

Worked example **Dhaba Dairy: automate the plant?**

Dhaba Dairy Ltd, Tapri's milk supplier, expects sales of ₹40 crore next year. It has 1 crore equity shares of ₹10

and 12% debentures of ₹10 crore, and pays tax at 25%. It is considering automating its processing plant:

Particulars	Present plant	Automated plant
Variable operating cost (% of sales)	65%	50%
Fixed operating cost a year, excluding interest (₹ crore)	6	10
Additional 12% term loan to pay for the automation (₹ crore)	nil	15

Required:

(i) Calculate the EPS and the degrees of operating, financial and combined leverage under each option at expected sales.

(ii) Calculate the percentage change in EPS under each option if sales fall by 10%, and verify it for the automated plant.

(iii) Calculate the sales at which EPS would be zero under each option, and advise.

THE WORKING

Income statements at sales of ₹40 crore (₹ crore)

Particulars	Present plant	Automated plant
Sales	40.00	40.00
Less: Variable operating cost	26.00	20.00
Contribution	14.00	20.00
Less: Fixed operating cost	6.00	10.00
EBIT	8.00	10.00
Less: Interest (12% on ₹10 crore; on ₹25 crore)	1.20	3.00
EBT	6.80	7.00
Less: Tax @ 25%	1.70	1.75
PAT	5.10	5.25
Number of equity shares (crore)	1	1
EPS (₹)	**5.10**	**5.25**

(i) Leverages

Leverage	Present plant	Automated plant
Operating (contribution ÷ EBIT)	14 ÷ 8 = **1.75**	20 ÷ 10 = **2.00**
Financial (EBIT ÷ EBT)	8 ÷ 6.8 = **1.18**	10 ÷ 7 = **1.43**
Combined (contribution ÷ EBT)	14 ÷ 6.8 = **2.06**	20 ÷ 7 = **2.86**

Automation raises EPS a little (₹5.25 against ₹5.10) but raises both leverages: more fixed operating cost and more interest.

(ii) A 10% fall in sales: % change in EPS = DCL × % change in sales.

- Present plant: 2.0588 × (−10%) = **−20.59%**.
- Automated plant: 2.8571 × (−10%) = **−28.57%**.

Verification for the automated plant at sales of ₹36 crore: contribution ₹18 crore; EBIT ₹8 crore; EBT ₹5 crore; PAT ₹3.75 crore; EPS ₹3.75. Fall = (5.25 − 3.75) ÷ 5.25 = **28.57%**.

(iii) Sales at which EPS is zero (no preference shares, so EBT = 0): (fixed operating cost + interest) ÷ P/V ratio.

- Present plant: (₹6 crore + ₹1.2 crore) ÷ 35% = **₹20.57 crore**.
- Automated plant: (₹10 crore + ₹3 crore) ÷ 50% = **₹26 crore**.

Advice. The two options give the same EPS at sales of about ₹38.67 crore (0.35S − 7.2 = 0.50S − 13). Above it automation earns more per share; below it, less, and its EPS falls faster and hits zero ₹5.43 crore of sales sooner. Automation moves Dhaba Dairy towards the high DOL, high DFL combination that the Study Material calls a very risky combination (SM 6.20). With expected sales of ₹40 crore only 3.4% above the point where the options tie, the small gain in EPS does not justify the extra risk unless Dhaba Dairy is confident that sales will stay well above ₹38.67 crore.

Worked example **Tapri Cold Chain: how much debt can the board allow?**

Tapri Cold Chain Pvt Ltd, Tapri's cold-chain subsidiary, has sales of ₹1,60,00,000 at a P/V ratio of 30% and fixed operating costs of ₹24,00,000. It has 4,00,000 equity shares, no preference shares and 10% debt of ₹50,00,000, and pays tax at 25%. The board has set a risk limit: a 15% fall in sales must never cut EPS by more than 45%.

Required:

(i) Calculate the present degrees of operating, financial and combined leverage.

(ii) Calculate the highest financial leverage and the highest interest the limit allows, and the highest total debt if all of it cost 10%.

(iii) The bank will lend any new money at 12%. How much new debt can Tapri Cold Chain raise within the limit? Verify with income statements that a 15% fall in sales then cuts EPS by exactly 45%.

THE WORKING

(i) Present leverages. Contribution = ₹1,60,00,000 × 30% = ₹48,00,000; EBIT = ₹48,00,000 − ₹24,00,000 = ₹24,00,000; interest = 10% × ₹50,00,000 = ₹5,00,000; EBT = ₹19,00,000.

- DOL = ₹48,00,000 ÷ ₹24,00,000 = **2.00**
- DFL = ₹24,00,000 ÷ ₹19,00,000 = **1.26**
- DCL = ₹48,00,000 ÷ ₹19,00,000 = **2.53**

(ii) The limit. DCL = % change in EPS ÷ % change in sales, so the highest DCL allowed = 45% ÷ 15% = **3**. DCL = DOL × DFL, so the highest DFL = 3 ÷ 2 = **1.5**.

DFL = EBIT ÷ (EBIT − interest): 1.5 = ₹24,00,000 ÷ EBT, so EBT must be at least ₹16,00,000 and **interest can be at most ₹8,00,000**. At 10% that is a total debt of **₹80,00,000**, ₹30,00,000 more than now.

(iii) New debt at 12%. Interest room = ₹8,00,000 − ₹5,00,000 = ₹3,00,000, so new debt = ₹3,00,000 ÷ 12% = **₹25,00,000**: the dearer money buys less room.

Particulars	After the new loan (₹)	Sales down 15% (₹)
Sales	1,60,00,000	1,36,00,000
Less: Variable cost (70%)	1,12,00,000	95,20,000
Contribution	48,00,000	40,80,000
Less: Fixed operating cost	24,00,000	24,00,000
EBIT	24,00,000	16,80,000
Less: Interest (5,00,000 + 3,00,000)	8,00,000	8,00,000
EBT	16,00,000	8,80,000
Less: Tax @ 25%	4,00,000	2,20,000
PAT	12,00,000	6,60,000

With the number of shares unchanged, EPS falls in the same proportion as PAT: (12,00,000 − 6,60,000) ÷ 12,00,000 = **45%**, exactly the board's limit. (If the loan is used to buy back shares, EPS changes but the 45% fall does not, because the number of shares is the same in both columns.)

Worked example

Sugar & Spice Foods: leverage with preference shares

Sugar & Spice Foods Ltd has sales of ₹80,00,000, a variable cost ratio of 55% and fixed operating costs of ₹18,00,000. Its capital includes 10% debentures of ₹60,00,000, 9% preference shares of ₹30,00,000 and 3,00,000 equity shares. The tax rate is 25%.

Required:

(i) Calculate EPS and the degrees of operating, financial and combined leverage.

(ii) Calculate the level of sales at which EPS will be zero, and the level at which EBT will be zero.

THE WORKING

Income statement

Particulars	₹
Sales	80,00,000
Less: Variable cost (55%)	44,00,000
Contribution	36,00,000
Less: Fixed cost	18,00,000
EBIT	18,00,000
Less: Interest (10% × ₹60,00,000)	6,00,000
EBT	12,00,000
Less: Tax @ 25%	3,00,000
PAT	9,00,000
Less: Preference dividend (9% × ₹30,00,000)	2,70,000
Earnings for equity shareholders	6,30,000
EPS (÷ 3,00,000 shares)	**₹2.10**

(i) Leverages. Pre-tax preference dividend = ₹2,70,000 ÷ (1 − 0.25) = ₹3,60,000.

- DOL = ₹36,00,000 ÷ ₹18,00,000 = **2.00**.
- DFL = EBIT ÷ [EBT − Dp ÷ (1 − t)] = ₹18,00,000 ÷ (₹12,00,000 − ₹3,60,000) = ₹18,00,000 ÷ ₹8,40,000 = **2.14**.
- DCL = 2.00 × 2.1429 = **4.29** (= ₹36,00,000 ÷ ₹8,40,000).

(ii) Sales levels.

- **EPS zero** means a 100% fall in EPS: sales must fall by 100 ÷ 4.2857 = 23.33%, to ₹80,00,000 × 76.67% = **₹61,33,333**. Check: (fixed cost + interest + pre-tax preference dividend) ÷ P/V ratio = (18,00,000 + 6,00,000 + 3,60,000) ÷ 45% = ₹61,33,333.
- **EBT zero** uses the combined leverage without the preference dividend, contribution ÷ EBT = 36,00,000 ÷ 12,00,000 = 3: sales must fall by 33.33%, to **₹53,33,333**. Check: (18,00,000 + 6,00,000) ÷ 45% = ₹53,33,333.

Note: if preference dividend is ignored in DFL (as some answers do), DFL = 1.50 and DCL = 3.00. State the treatment you use.

Interactive version: PassBook app, lesson Operating, financial and combined leverage

Exam angle

What ICAI asks

In the exam

In the ICAI documents read for this lesson (new scheme, May 2024 to September 2026)

Leverages appeared in all 32 question sets searched: every past paper (8 of 8), every RTP (8 of 8) and every MTP (16 of 16).

- **Past papers, descriptive, usually 5 marks**: May 2024 Q1(b) (EPS, DOL and DFL if sales fall 15% and 20%); September 2024 Q1(a) (contribution, MOS and sales of two companies from CL and FL); January 2025 Q1(b) (EPS, OL, FL, CL from a balance sheet and asset turnover); May 2025 Q1(c) (income statement from P/V ratio, OL 2, FL 1.5 and interest; % change in EPS); September 2025 Q2(a) (7 marks: EBIT, fixed cost, bank loan and its interest from OL 2.4 and CL 3.84; EPS; MPS); January 2026 Q2(a) (with preference shares; sales at which EBT is zero); May 2026 Q1(a) (sales from a 5% rise giving ₹9,000 more EBIT; OL, FL, CL); September 2026 Q1(a) (leverages from year-on-year % changes; debt; FL after expansion).
- **Short questions, 2 marks**: May 2025 MCQ 6 (FL from net worth and debt-equity ratio); May 2025 Q4(c) OR and MTP Series I for May 2026 Q4(c) (comment on DOL-DFL combinations, the table above); January 2026 Q3(b) ("Explain in what ways financial leverage operates as a double edged sword"); MTPs: "Differentiate between business risk and financial risk", "Explain financial leverage as trading on equity".
- **In MTPs**: % change in EBIT and in EBT from a profitability statement (MTP Series II for January 2025 Q1(c), 5 marks); % change in EPS from DOL × DFL × % change in sales; earnings after tax from contribution, OL and CL; DOL from sales and break-even sales; sales from ROCE, DOL and P/V ratio.

In the exam

The traps in those answers

- "Wipe out EPS" means a 100% fall: DCL = 100% ÷ % fall in sales (SM Practical Problem 5).
- Fixed cost in leverage questions is "excluding interest"; when a question says all operating expenses other than depreciation are variable, depreciation is the fixed cost (SM Practical Problem 6).
- With preference shares, ICAI's January 2026 answer used DCL = contribution ÷ (EBT − pre-tax preference dividend) = 3.84 and, as an alternative, contribution ÷ EBT = 3.2, giving different "EBT zero" sales. Name the treatment you use; strictly, the first gives the sales at which EPS is zero.
- Percentage changes are measured from the present level (May 2024 Q1(b)).
- Comments carry marks when asked: MTP Series II for September 2026 says "comment on what each result means", and the Study Material's answer to Practical Problem 7 writes a line on what each leverage indicates. ICAI's January 2025 answer shows leverages to three decimals (1.363, 1.128, 1.538).

Interactive version: PassBook app, lesson Operating, financial and combined leverage

Trick

The leverage ladder

Contribution ÷ EBIT ÷ EBT: each leverage is "the line above ÷ the line below" in the profitability statement, and DCL skips the middle rung (contribution ÷ EBT). With preference shares, the bottom rung is EBT − Dp ÷ (1 − t). Valid at any level of sales away from the break-even points.

Answer frame

"Differentiate between business risk and financial risk"

The points of ICAI's answer to MTP Series I for January 2026 Q2(b), with the Study Material's definitions (SM 6.2):

Basis	Business risk	Financial risk
Meaning	risk associated with the firm's operations; unavoidable, because of the environment in which the firm operates	additional risk placed on the firm's shareholders as a result of debt use in financing
Represented by	variability of EBIT, influenced by revenues and expenses (demand for products, variations in prices, proportion of fixed cost in total cost)	the use of debt: companies issuing more debt instruments have higher financial risk than companies financed mostly by equity
Indicated by	operating leverage (SM 6.3)	financial leverage (SM 6.3); measured by ratios such as the financial leverage multiplier and total debt to assets ratio

"Explain financial leverage as a double-edged sword" (2 marks)

When the cost of fixed cost funds is less than the return on investment, financial leverage increases return on equity and EPS, with a tax saving on interest; when the cost of debt is more than the return, it affects return on equity and EPS unfavourably and the firm can be under financial distress. Hence "double edged sword": ROI > interest favourable, ROI < interest unfavourable, ROI = interest neutral.

Three answers to "Calculate the operating leverage"

Loses marks

"OL = 2.33."

No statement: an error in sales or contribution leaves nothing to mark.

Passes

Income statement, then OL = 10.5 ÷ 4.5 = 2.33.

Correct, but no comment where the question asks to "comment".

Full marks

Working note for sales; income statement to EBIT; formula, figures and result to two decimals; one line on what it indicates (business risk, % change in EBIT for 1% change in sales).

Common mistakes

⚠ **Common mistake**

Hall of Shame: interest in the fixed cost

Adding the ₹1.2 crore of debenture interest to Dhaba Dairy's ₹6 crore fixed operating cost gives "EBIT" of ₹6.8 crore and a DOL of 14 ÷ 6.8 = 2.06: the combined leverage passed off as operating leverage (the true DOL is 1.75). **Fix:** operating leverage uses fixed operating cost only; interest belongs below EBIT, in financial leverage.

⚠ **Common mistake**

DOL multiplied when it should be divided

"A 15% fall in sales may cut EPS by at most 45%, so DCL = 15% × 45%." **Fix:** DCL = % change in EPS ÷ % change in sales = 45 ÷ 15 = 3, and DFL = DCL ÷ DOL. A fall that "wipes out EPS" is a 100% change: a 25% fall that wipes it out means DCL = 100 ÷ 25 = 4.

⚠ **Common mistake**

Preference dividend ignored without saying so

Using DFL = EBIT ÷ EBT when the company has preference shares, with no note. **Fix:** use EBIT ÷ [EBT − Dp ÷ (1 − t)] (SM 6.15), or state that you have ignored it, as ICAI's alternative answers do.

⚠ **Common mistake**

A DOL of 0.6

DOL can never be between zero and one. A figure like 0.6 means sales and EBIT are swapped (EBIT ÷ contribution). **Fix:** DOL = contribution ÷ EBIT, which is at least 1 above the break-even point.

Practice

Check yourself

1 A company has sales of ₹40,00,000, variable cost of 60%, fixed operating cost of ₹10,00,000 and interest of ₹2,00,000. What is its degree of combined leverage? (Answer to two decimals.)

Answer: _____

2 Case scenario. Ritu reports that Tapri's packaged premix division has a margin of safety of 20% and a financial leverage of 1.5. Kesar wants to know how much EPS will rise if premix sales rise by 6%. Which figure should Ritu give?

- A 9%
- B 45%
- C 30%
- D 1.8%

3 Which statement about the degree of operating leverage is correct?

- A It is highest when sales are far above the break-even point
- B It can lie between 0 and 1 for a firm with low fixed costs
- C It is undefined at the break-even point and negative below it
- D It depends on the interest the firm pays

4 A company's DOL is 3 and its DCL is 4.5. Its interest is ₹2,00,000. What is its EBIT, in rupees?

Answer: _____

5 Which combination of operating and financial leverage does the Study Material call the best?

- A High DOL, high DFL
- B High DOL, low DFL
- C Low DOL, high DFL
- D Low DOL, low DFL

Answers: the answer key at the end of this lesson.

Recap

- Business risk: uncertainty about EBIT (operating leverage). Financial risk: extra risk to shareholders from debt (financial leverage).
- DOL = contribution ÷ EBIT = % Δ EBIT ÷ % Δ sales = 1 ÷ MOS; function of fixed cost, contribution margin and sales volume.
- DFL = EBIT ÷ EBT = % Δ EPS ÷ % Δ EBIT; with preference shares, EBIT ÷ [EBT − Dp ÷ (1 − t)].
- DCL = DOL × DFL = contribution ÷ EBT = % Δ EPS ÷ % Δ sales.
- No fixed cost: DOL = 1; at the operating BEP: undefined; below it: negative. Same for DFL around the financial BEP. Never between 0 and 1.
- Financial leverage is a double-edged sword: ROI > Kd favourable, ROI < Kd unfavourable.
- Best combination: low DOL, high DFL. Riskiest: high DOL, high DFL.
- "Wipe out EPS" = a 100% fall: DCL = 100 ÷ % fall in sales. (Gears up, gears down.)

Answer key · Check yourself

1 **3.98 to 4.02**

Contribution = ₹16,00,000; EBIT = ₹6,00,000; EBT = ₹4,00,000. DCL = 16 ÷ 4 = 4.00 (DOL 2.67 × DFL 1.50).

2 **B 45%**

DOL = 1 ÷ MOS = 1 ÷ 0.20 = 5; DCL = 5 × 1.5 = 7.5; % change in EPS = 7.5 × 6% = 45%. 9% uses only DFL; 30% only DOL; 1.8% multiplies MOS by 6% and 1.5.

3 **C It is undefined at the break-even point and negative below it**

At break-even, contribution = fixed cost and EBIT = 0, so DOL is undefined; below it, EBIT is negative and so is DOL. Far above break-even it approaches 1; it is never between 0 and 1; interest affects DFL, not DOL.

4 **600000**

DFL = 4.5 ÷ 3 = 1.5 = EBIT ÷ (EBIT − 2,00,000), so 1.5 EBIT − 3,00,000 = EBIT and EBIT = ₹6,00,000.

5 **C** **Low DOL, high DFL**

Low DOL with high DFL: moderate total risk, the higher financial risk balanced by lower business risk, and the firm still benefits from trading on equity. High-high is very risky; high-low is "not a good combination"; low-low cannot take advantage of trading on equity.

CHAPTER 7

Investment Decisions

7.1 Capital budgeting and estimating relevant cash flows

Why capital budgeting decisions matter, the types of investment decision, and how to build a project's post-tax incremental cash flows: initial outflow, interim flows and the terminal year.

In plain words

Five "costs" Kesar wanted to add

Kesar wants a ₹50 lakh frozen-samosa line at the Indore factory, and Dev must list the cash it will bring in and send out. Kesar has helped.

> **Kesar:** Put everything in, Dev. The consultant's ₹1.5 lakh fee, a share of head-office rent, the loan interest, the depreciation. Sab adjust ho jayega.
>
> **Dev:** The fee is paid whether we build or not. Head office costs the same either way. The interest is already inside the 12% I discount at. And depreciation is not cash.
>
> **Kesar:** Then what do you count?
>
> **Dev:** Only what changes in our bank account because we say yes. After tax.

Choosing long-term investments is **capital budgeting**, and its raw material is a project's **incremental cash flows**: the cash that comes in or goes out only because the project is accepted, after tax.

Some items fool you. The line will take over floor space Tapri now lets to a courier company, so the rent given up is a real cost though no cheque is written: an **opportunity cost**. Depreciation is never paid to anyone, but it lowers the tax bill, and that saving is cash. Money tied up in stock and debtors (**working capital**) goes in at the start and comes back at the end.

Think of it like

Two photographs of the bank account

Photograph Tapri's bank balance every year with the samosa line, and again without it. The relevant cash flows are the differences between the two photos. The consultant's fee, already paid, shows up in both photos and cancels out; so does head-office rent. The courier's rent shows up only in the "without" photo, so losing it counts.

Key idea

Only post-tax incremental cash flows count: what changes in cash because the project is accepted. Depreciation counts only through the tax it saves; sunk costs, allocated overheads and financing costs stay out.

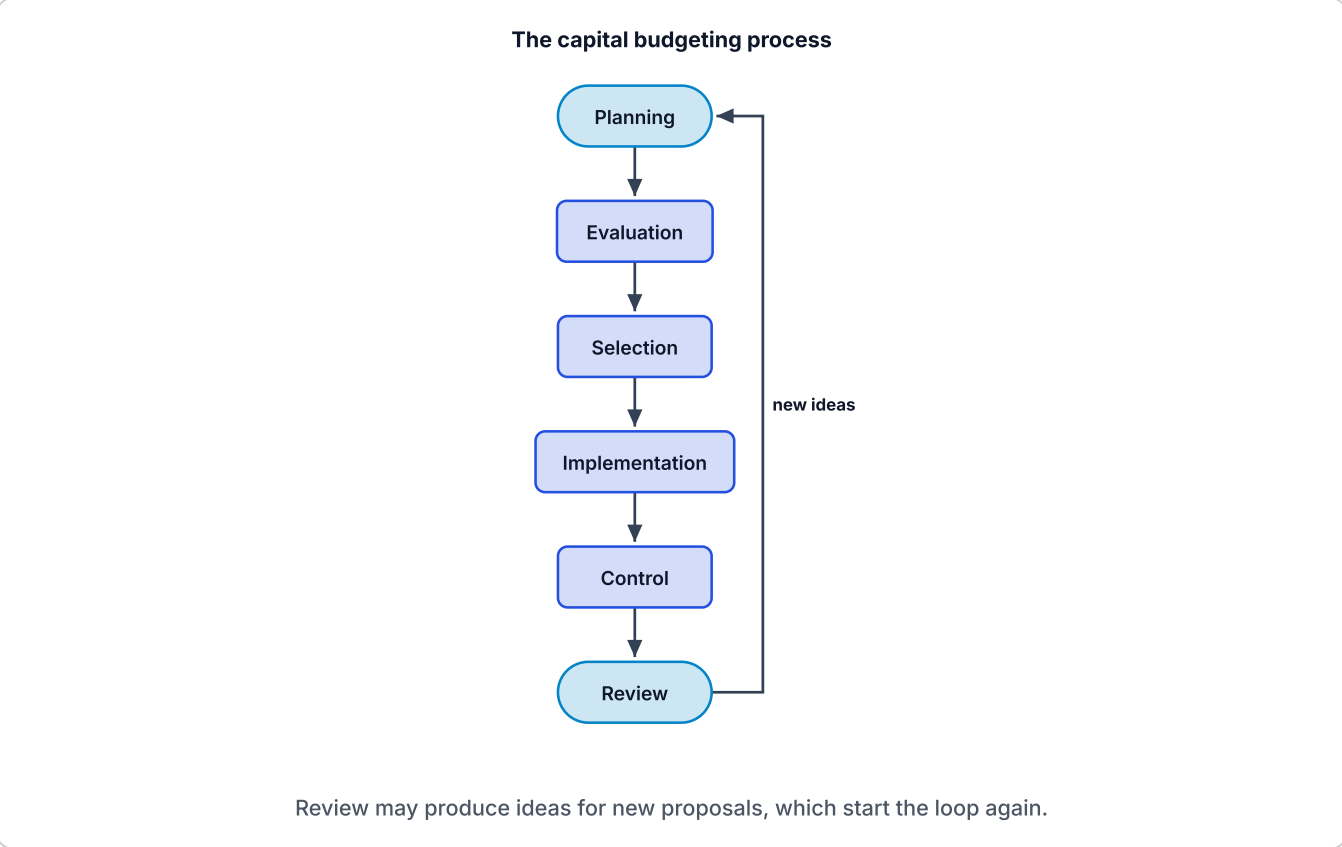
How it works

Why these decisions keep a CFO awake

Capital budgeting identifies projects strategic to the business, estimates and evaluates post-tax incremental cash flows for each, and selects the proposal that maximises the return to investors (SM 7.3). The decisions are critical for four reasons, in ICAI's order (SM 7.3):

1. **Substantial investment**: investment decisions serve long-term objectives and the existence of the organisation, need substantial capital, and the size of capital and timing of cash flows decide the sources of finance; so they need thorough study and planning.
2. **Long time period**: the decision has its effect over a long period; it affects future benefits and costs and influences the rate and direction of the firm's growth.
3. **Irreversibility**: most investment decisions are irreversible; once implemented, reversing them is very difficult and economically not possible (upfront payments, contractual obligations, technological impossibilities).
4. **Complex decisions**: they involve assessing future events that are difficult to predict, and it is difficult to estimate all the benefits and costs in quantitative terms.

Six phases, one loop



Review may produce ideas for new proposals, which start the loop again.

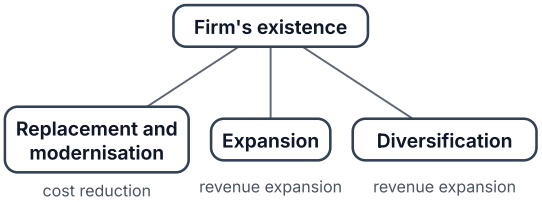
In ICAI's words (SM 7.4–7.5):

1. **Planning**: potential investment opportunities are identified; their potential effect on the firm and management's ability to exploit them are assessed; little-merit ones are rejected and promising ones advanced as proposals.

2. **Evaluation**: the proposal's investment, inflows and outflows are determined and appraised by a technique, from payback and ARR to discounted cash flow, chosen to give the best decision in the circumstances.
3. **Selection**: considering the returns and risks of the projects and the cost of capital, the projects that maximise shareholders' wealth are chosen.
4. **Implementation**: the firm acquires the necessary funds, purchases the assets and begins the project.
5. **Control**: progress is monitored through feedback reports: capital expenditure progress reports, performance reports comparing actual performance against plans, and post-completion audits.
6. **Review**: when the project terminates, or even before, it is reviewed to explain its success or failure; this may change planning and evaluation procedures and produce ideas for new proposals.

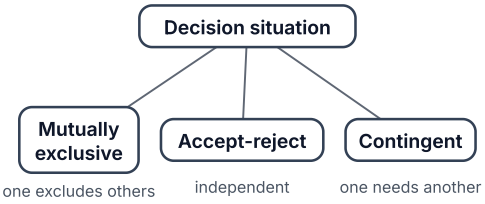
Two ways to sort a decision

On the basis of the firm's existence



Replacement: the economic life is over. Modernisation: the asset is technologically outdated. Both are cost reduction decisions (SM 7.6).

On the basis of the decision situation



Semi-automatic or highly automatic machine: mutually exclusive. A factory in a remote area that also needs roads and staff houses: contingent (SM 7.6–7.7).

Mutually exclusive decisions: acceptance of one proposal will exclude the acceptance of the other alternative proposals. **Accept-reject decisions**: the proposals are independent and do not compete; those giving a higher return than the desired rate are accepted and the rest rejected. **Contingent decisions**: the proposals are dependable, so investment in one requires investment in one or more others (SM 7.6–7.7).

The procedure has three steps (SM 7.7): estimate each project's cash flows over its entire life; evaluate each alternative by the decision criteria; determine the minimum required rate of return, i.e. the WACC, as the discount rate.

What goes into the cash flows

Cash flows, not accounting profit, count: their timing may differ, and depreciation has no outflow (SM 7.7). The project's life is set by technological obsolescence, physical deterioration and decline in demand for the product (SM 7.8). Six items need care, in ICAI's order (SM 7.8–7.10):

1. **Depreciation**: non-cash, but it reduces tax; the **tax shield** (depreciation × tax rate) is a cash inflow. ₹9,00,000 of depreciation at 25% saves ₹2,25,000.

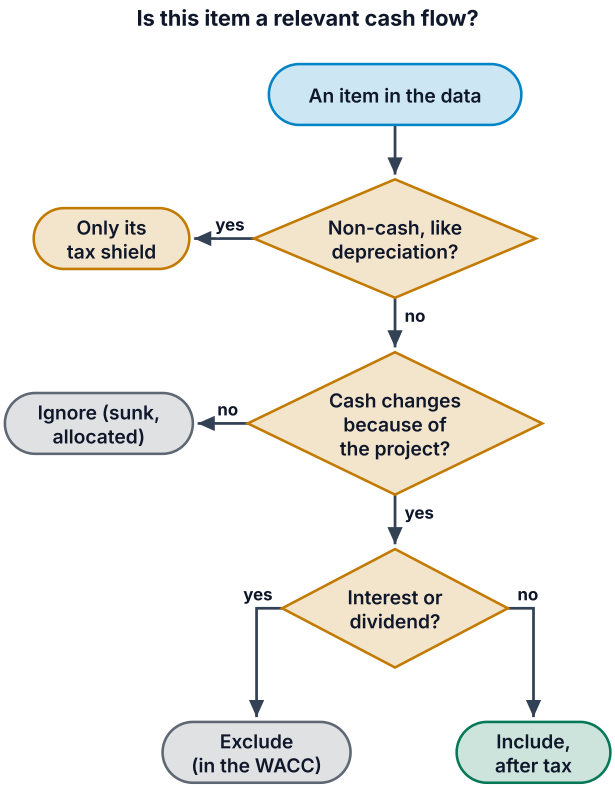
2. **Opportunity cost**: a benefit given up; included in outflows, at the start or during the project. Land that could be sold for ₹10 crore today is a ₹10 crore outlay, whatever it cost.

3. **Sunk cost**: already incurred and irreversible, so excluded, like a project report fee already paid.

4. **Working capital**: the initial requirement is an outflow and its release at the end an inflow; extra working capital later is an outflow, a reduction an inflow. No depreciation on it; if nothing is said, the full amount is released at the end.

5. **Allocated overheads**: existing costs spread over the proposal are not cash flows; only an increase caused by it is.

6. **Additional capital investment** during the project is an outflow in that year.



Amber: depreciation and other non-cash items enter only as the tax they save. Grey ends: left out (sunk costs, allocated overheads, interest and dividends). Green: in the cash flows, net of tax. Opportunity costs and working capital pass every test.

The cash flows fall into three categories (SM 7.10–7.12): the **initial cash outflow** (year 0), the **interim cash flows** of each year of operation, and the **terminal-year net cash flow**, which adds salvage, the tax effect of the sale and the release of working capital. Each has an ICAI format, below.

Three principles

6.1 Block of assets and depreciation. For tax, depreciation is charged on a **block of assets**: a group of assets in one class depreciated at the same rate (SM 7.12–7.13). In the year an asset is sold:

- **Only asset in the block** (the block ceases): no depreciation that year; WDV at the beginning of the year minus sale value is a short-term capital loss (tax benefit), or a gain (tax outflow) if the sale value is higher (SM 7.13–7.14).
- **More than one asset**: depreciation that year on (opening WDV − sale value); none if the sale value exceeds the opening WDV (SM 7.14).

Both cases are worked in the kiln example below.

6.2 Exclusion of financing costs. Interest on long-term debt and dividends are excluded because they are reflected in the WACC; deducting them too means the cost of long-term funds will be counted twice (SM 7.14). If profit was struck after interest, add back interest × (1 − tax rate).

6.3 Post-tax principle. The discounting rate and the cash flows, both must be post-tax only (SM 7.16).

Formats and formulas

Reproduce these formats from memory; ICAI's suggested answers follow them line by line.

Initial cash outflow: a fresh proposal or an expansion (SM 7.10)

Particulars	Amount	Amount
Cost of new asset(s)		xxx
Add: Installation/set-up costs	xxx	
Add: Investment in working capital	xxx	xxx
Initial cash outflow		**xxx**

Initial cash outflow: a replacement decision (SM 7.10–7.11)

Particulars	Amount	Amount
Cost of new asset(s)		xxx
Add: Installation/set-up costs	xxx	
Add/(less): Increase (decrease) in net working capital level	xxx	
Less: Net proceeds from sale of old assets	(xxx)	
Add/(less): Tax expense (saving/loss) due to sale of old asset	xxx	xxx
Initial cash outflow		**xxx**

Interim cash flow: a new project (SM 7.11)

Particulars	Amount	Amount
Profit after tax (PAT)		xxx
Add: Non-cash expenses (e.g. depreciation)	xxx	
Add/(less): Net decrease (increase) in working capital	xxx	xxx
Interim net cash flow for the period		**xxx**

Interim cash flow: a replacement decision (SM 7.11–7.12)

Particulars	Amount
Net increase (decrease) in operating revenue	xxx
Add/(less): Net decrease (increase) in operating expenses	xxx
Net changes in income before taxes	xxx
Add/(less): Net decrease (increase) in taxes	xxx
Net change in income after taxes	xxx
Add/(less): Net decrease (increase) in depreciation charges	xxx
Incremental net cash flow for the period	**xxx**

The format is reproduced as the Study Material prints it. In the arithmetic, the change in depreciation is deducted (as an operating expense) before the change in taxes is worked out; the last adjustment then reverses it, because depreciation is not cash: an increase in depreciation is added back, a decrease deducted. The pouch-packer example below shows the lines.

Terminal-year net cash flow (SM 7.12)

Particulars	Amount	Amount
Final salvage value (disposal costs) of asset		xxx
Add: Interim cash flow	xxx	
Add/(less): Tax savings (tax expenses) due to sale or disposal of asset (including depreciation)	xxx	
Add: Release of net working capital	xxx	xxx
Terminal year net cash flow		**xxx**

Statement showing the calculation of cash inflow after tax (CFAT) (SM 7.16)

Particulars	(₹)	(₹)
Sales value		xxx
Less: Variable cost		xxx
Contribution		**xxx**
Less: Fixed cost: (a) fixed cash cost (excluding interest)	xxx	
(b) Depreciation	xxx	xxx
Earning before tax (EBT)		xxx
Less: Tax		xxx
Earning after tax (EAT)		xxx
Add: Depreciation		xxx
Cash inflow after tax (CFAT)		**xxx**

∑ **Formula**

Tax shield on depreciation

$$\text{Tax shield} = \text{Depreciation} \times \text{Tax rate}$$

∑ **Formula**

Profit after tax excluding financing cost

$$\text{PBIT} \times (1 - t) \quad = \text{PAT} + \text{Interest} \times (1 - t)$$

PBIT = profit before interest and tax, PAT = profit after tax (struck after interest), t = tax rate (SM 7.15).

∑ **Formula**

Terminal year: only asset in the block, or other assets in it

$$\text{STCL (STCG)} = W_0 - S \qquad \text{Depreciation} = (W_0 - S) \times r$$

W_0 = WDV at the beginning of the year of sale, S = sale value, r = depreciation rate. The first when the block ceases; the second when it continues and S is below W_0 (SM 7.13–7.14).

Definition

Capital budgeting

The process of evaluating and selecting long-term investments that are in line with the goal of investor's wealth maximization. (SM 7.69)

Definition

Sunk cost

An outlay of cash that has already been incurred in the past and cannot be reversed in present. (SM 7.9)

Definition

Opportunity cost

Foregoing of a benefit due to choosing an alternative investment option. (SM 7.9)

Worked examples

Worked example **The samosa line that took the courier's floor**

Tapri Tea Co. Ltd is considering a frozen-samosa line at its Indore factory (an expansion decision). Ritu has collected the data:

Particulars	Details
Machine; installation	₹48,00,000; ₹2,00,000
Life; salvage at the end of year 5	5 years; ₹5,00,000
Depreciation	straight-line, for books and tax
Packs sold	3,00,000 in year 1; 4,00,000 a year in years 2–5
Selling price; variable cost	₹45 and ₹30 a pack
Fixed cash costs of the line (excluding depreciation)	₹30,00,000 a year
Insurance taken out for the line	₹1,00,000 a year
Head-office overheads allocated to the line	₹4,00,000 a year
Floor space now let to a courier (lease to be ended)	rent ₹2,00,000 a year
Feasibility report fee, paid last month	₹1,50,000
Working capital	₹6,00,000 at the start; ₹2,00,000 more at the end of year 2; all released at the end of year 5
Interest on the term loan for the machine	₹4,50,000 a year

Tax is 25% and the cost of capital 12%. PV factors at 12% for years 1–5: 0.893, 0.797, 0.712, 0.636, 0.567.

Required:

(i) Identify the items that are not relevant cash flows, with reasons.

(ii) Calculate the initial cash outflow.

(iii) Calculate the cash flows for years 1 to 5, including the terminal year.

(iv) Calculate the NPV and advise.

THE WORKING

(i) Items left out

1. Feasibility report fee ₹1,50,000: a sunk cost, already paid.
2. Head-office overheads ₹4,00,000: allocated, not incremental. (The new insurance of ₹1,00,000 is incremental and stays in.)
3. Interest ₹4,50,000: a financing cost, already in the 12% cost of capital.

The rent given up, ₹2,00,000 a year, is an opportunity cost and is included; the rent would have been taxed, so it is deducted before tax.

(ii) Initial cash outflow

Particulars	Amount (₹)	Amount (₹)
Cost of new asset		48,00,000
Add: Installation	2,00,000	
Add: Investment in working capital	6,00,000	8,00,000
Initial cash outflow		**56,00,000**

(iii) Working note 1. Depreciation = (₹48,00,000 + ₹2,00,000 − ₹5,00,000) ÷ 5 = **₹9,00,000** a year. The book value at the end of year 5 equals the salvage, so the sale has no tax effect.

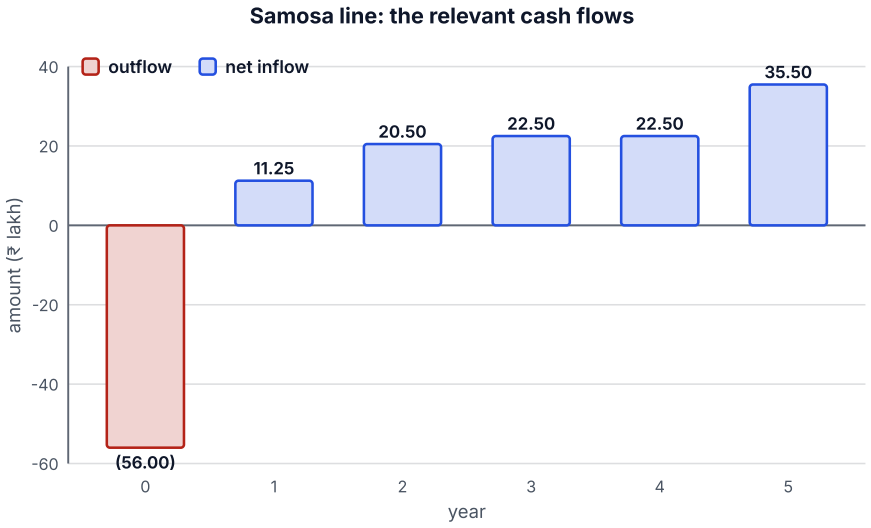
Working note 2. CFAT

Particulars	Year 1 (₹)	Years 2–5 (₹)
Sales (₹45 a pack)	1,35,00,000	1,80,00,000
Less: Variable cost (₹30 a pack)	90,00,000	1,20,00,000
Contribution	**45,00,000**	**60,00,000**
Less: Fixed cash cost (30,00,000 + insurance 1,00,000 + rent foregone 2,00,000)	33,00,000	33,00,000
Less: Depreciation	9,00,000	9,00,000
Earning before tax	3,00,000	18,00,000
Less: Tax @ 25%	75,000	4,50,000
Earning after tax	2,25,000	13,50,000
Add: Depreciation	9,00,000	9,00,000
CFAT	**11,25,000**	**22,50,000**

Terminal year (year 5). Salvage ₹5,00,000 + interim cash flow ₹22,50,000 + tax on sale nil + release of working capital ₹8,00,000 = **₹35,50,000**.

(iv) NPV

Year	Cash flow	Amount (₹)	PV factor @ 12%	PV (₹)
0	Initial cash outflow	(56,00,000)	1.000	(56,00,000)
1	CFAT	11,25,000	0.893	10,04,625
2	CFAT 22,50,000 less additional working capital 2,00,000	20,50,000	0.797	16,33,850
3	CFAT	22,50,000	0.712	16,02,000
4	CFAT	22,50,000	0.636	14,31,000
5	Terminal-year net cash flow	35,50,000	0.567	20,12,850
	Net present value			**20,84,325**



Rose: the initial outflow; blue: net inflows. Year 2 is lower by the extra working capital; year 5 is higher by salvage and the release of working capital.

Advice. The NPV is positive (₹20,84,325), so the samosa line should be accepted. (NPV itself is the subject of the NPV lesson.)

Worked example **The pouch packer that wanted a pension**

Tapri's chai premix plant fills its pouches on a packing machine bought 3 years ago for ₹15,00,000 and depreciated on the straight-line method over 8 years down to a salvage value of ₹3,00,000. It can be sold now for ₹8,50,000. If it is kept, it will run for its remaining 5 years, will need a major overhaul of ₹3,00,000 at the end of year 2 (charged to revenue and allowed for tax), and will fetch its salvage value of ₹3,00,000 at the end of year 5.

A new high-speed packer costs ₹32,00,000 plus ₹1,00,000 for installation, lasts 5 years, has a salvage value of ₹5,00,000 and is depreciated on the straight-line method. It needs no overhaul. Its faster speed lets Tapri sell 2,40,000 more pouches a year at a contribution of ₹2.50 a pouch; film wastage falls by ₹1,90,000 a year; power and the maintenance contract rise by ₹1,32,000 a year. Because it changes over between pack sizes in minutes, Tapri can carry ₹1,50,000 less packing-film stock from the day it is installed; the reduction reverses at the end of year 5.

Tax is 25%; tax depreciation equals book depreciation; a loss on sale of the old packer is allowed for tax at 25% in year 0, as Tapri has other taxable profits. The cost of capital is 12%; PV factors at 12% for years 1–5: 0.893, 0.797, 0.712, 0.636, 0.567.

Required:

(i) Calculate the net initial cash outflow.

(ii) Calculate the incremental cash flow after tax for each year.

(iii) Calculate the terminal-year net cash flow and the NPV of the replacement, and advise.

THE WORKING

Working note 1. Old packer. Depreciation (₹15,00,000 − ₹3,00,000) ÷ 8 = ₹1,50,000 a year; book value = ₹15,00,000 − 3 × ₹1,50,000 = **₹10,50,000**. Loss on sale = ₹10,50,000 − ₹8,50,000 = ₹2,00,000; tax saving @ 25% = **₹50,000**.

Working note 2. Depreciation. New packer (₹32,00,000 + ₹1,00,000 − ₹5,00,000) ÷ 5 = ₹5,60,000; old ₹1,50,000; **incremental depreciation ₹4,10,000**.

Working note 3. Operating changes a year. Additional contribution 2,40,000 × ₹2.50 = ₹6,00,000; film wastage saved ₹1,90,000; extra power and maintenance ₹1,32,000. In year 2 the replacement also avoids the ₹3,00,000 overhaul, a decrease in operating expenses of that year.

(i) Net initial cash outflow

Particulars	Amount (₹)	Amount (₹)
Cost of new asset		32,00,000
Add: Installation	1,00,000	
Less: Decrease in net working capital	(1,50,000)	
Less: Net proceeds from sale of old asset	(8,50,000)	
Less: Tax saving on loss on sale of old asset (WN 1)	(50,000)	(9,50,000)
Net initial cash outflow		**22,50,000**

(ii) Incremental cash flow after tax

Particulars	Years 1, 3, 4, 5 (₹)	Year 2 (₹)
Increase in operating revenue: contribution	6,00,000	6,00,000
Add: Decrease in operating expenses (WN 3)	58,000	3,58,000
Less: Increase in depreciation (WN 2)	(4,10,000)	(4,10,000)
Net change in income before taxes	2,48,000	5,48,000
Less: Increase in taxes @ 25%	(62,000)	(1,37,000)
Net change in income after taxes	1,86,000	4,11,000
Add: Increase in depreciation charges	4,10,000	4,10,000
Incremental net cash flow	**5,96,000**	**8,21,000**

The decrease in operating expenses is wastage ₹1,90,000 less power and maintenance ₹1,32,000 = ₹58,000; year 2 adds the ₹3,00,000 overhaul avoided.

(iii) Terminal year (year 5)

Particulars	Amount (₹)	Amount (₹)
Final salvage value of new packer		5,00,000
Less: Salvage value of old packer forgone		(3,00,000)
Add: Interim cash flow	5,96,000	
Add/(less): Tax on sale (both at book value)	nil	
Less: Working capital rebuilt (the reduction reverses)	(1,50,000)	4,46,000
Terminal year net cash flow		**6,46,000**

Year	Cash flow (₹)	PV factor @ 12%	PV (₹)
0	(22,50,000)	1.000	(22,50,000)
1	5,96,000	0.893	5,32,228
2	8,21,000	0.797	6,54,337
3	5,96,000	0.712	4,24,352
4	5,96,000	0.636	3,79,056
5	6,46,000	0.567	3,66,282
Net present value			**1,06,255**

Advice. The incremental NPV is positive, so the old packer should be replaced.

Two items decide it. Leave out the overhaul avoided and the NPV becomes ₹1,06,255 − ₹3,00,000 × 0.75 × 0.797 = **−₹73,070**: a good replacement rejected. Leave out the old packer's salvage forgone and

it is overstated at ₹2,76,355. Both are incremental cash flows: what happens with the new packer minus what would have happened with the old one.

Assumption stated: the loss on sale is tax-deductible in year 0. Without that saving the outflow is ₹23,00,000 and the NPV ₹56,255; the advice does not change.

Worked example **The kulhad kiln and its block**

Tapri bought a gas kiln for its kulhad unit for ₹8,00,000. For tax it is depreciated at 15% on the written down value (WDV) method. The kiln was sold at the end of year 3, and the tax rate is 25%. The kiln was the only asset in its block, and Ritu's tax working for year 3 shows a tax benefit of ₹69,500 on the sale.

Required:

(i) Calculate the price at which the kiln was sold.

(ii) Had the block held other machines and continued, calculate the tax effect in year 3 at the same sale price, and the part of the kiln's loss that would stay in the block for later years.

(iii) State the sale price at which the year-3 tax effect would be the same in both cases.

THE WORKING

WDV schedule

Year	Opening WDV (₹)	Depreciation @ 15% (₹)	Closing WDV (₹)
1	8,00,000	1,20,000	6,80,000
2	6,80,000	1,02,000	5,78,000

WDV at the beginning of year 3 = **₹5,78,000**.

(i) Only asset in the block: the block ceases, no depreciation in year 3

Particulars	Amount (₹)
Tax benefit on short-term capital loss (given)	69,500
Short-term capital loss (₹69,500 ÷ 25%)	2,78,000
WDV at the beginning of year 3	5,78,000
Less: Short-term capital loss	(2,78,000)
Sale value of the kiln	**3,00,000**

(ii) Other assets in the block: depreciation in year 3

Depreciation = (₹5,78,000 − ₹3,00,000) × 15% = ₹41,700; tax benefit = ₹41,700 × 25% = **₹10,425** in year 3.

The rest of the ₹2,78,000, that is ₹2,78,000 − ₹41,700 = **₹2,36,300**, stays in the block's WDV and is written off through depreciation in later years. The loss is not lost; its tax saving arrives later, which is

worth less in present value terms.

(iii) At a sale price equal to the opening WDV, **₹5,78,000**, there is no capital loss or gain if the block ceases and nothing to depreciate if it continues: the year-3 tax effect is nil in both cases. Below that price the only-asset case gives the larger year-3 benefit; above it the only-asset case gives a tax outflow on the short-term capital gain, while a continuing block charges no depreciation (SM 7.13–7.14).

In every case the cash received on sale is a separate inflow in year 3; only its tax effect differs.

Exam angle

In the exam

Where cash-flow estimation appeared, May 2024 to September 2026

In the documents read for this lesson:

- **Replacement, 5 to 8 marks**: May 2024 Q3(a) (7 marks: installation, testing, a subsidy at the end of year 1); September 2025 Q1(c) (5 marks: tax of ₹52,800 on the profit on sale of the old machine added to the outflow, with an alternative answer without it); MTP Series I for May 2024 Q2(a) (8 marks: WDV tax shields, a capital loss); September 2026 Q3(a) (6 marks: 20% depreciation on old and new machine, extra working capital, benefits that continue after year 5).
- **New projects**: January 2025 Q3(a) (6 marks: working capital as debtors + stock − creditors); MTP Series II for January 2025 Q3 (10 marks: EBT after interest, an allocated common cost, sunk report and R&D fees); RTP January 2025 Q9 (WDV, only asset in the block).
- **MCQs, 2 marks each, often five on one case**: MTP Series I for September 2026 (disposal charges saved are a benefit, research spend ignored); September 2026 Case Scenario I (a tax holiday and a project loan; suggested answers not yet published, but the payback option of 3.76 years and the NPV option of ₹362.768 lakh both work only with cash flows before loan interest, ₹220 lakh and ₹184 lakh); MTP Series II for September 2026 MCQ 6 (cash outlay = cost + shipping and installation).

In the exam

The traps ICAI used

- "Fixed cost (including depreciation)": take depreciation out to get the cash fixed cost (January 2025 Q3(a)).
- "Earnings before tax but after deducting interest": add the interest back before taxing, or restart from PBIT (MTP Series II, January 2025 Q3).
- Costs paid "in advance (at the beginning)" of a year move to the end of the previous year (MTP Series I, September 2024 Q3(a)).
- "Only asset in the block": the Study Material charges no depreciation in the year of sale and uses the opening WDV (SM 7.13). Two ICAI answers (RTP January 2025 Q9; MTP Series I for September 2024 Q3(a)) charged a year's depreciation and used the closing WDV. Follow the Study Material and state it.
- Working capital "in year 1": taken at the start of the year, with the end-of-year alternative noted (January 2025 Q3(a)).

Trick

Salvage at book value means no tax

With straight-line depreciation down to the estimated salvage, selling at that salvage leaves no gain or loss, so no tax on sale. A tax effect arises only when the sale value differs from the tax WDV.

Answer frame

The chapter's own theory questions include "Discuss in detail the capital budgeting process" and "Classify various types of capital investment decisions" (SM 7.74). Write:

- **Purpose**: substantial investment; long time period; irreversibility; complex decisions; a sentence each.
- **Process**: planning, evaluation, selection, implementation, control (the three reports), review; a line each.
- **Types**: by firm's existence (replacement and modernisation, expansion, diversification; cost reduction versus revenue expansion) and by situation (mutually exclusive, accept-reject, contingent), each with an example.
- **Treatment of an item**: the principle (incremental, post-tax), the item's treatment in ICAI's words, a two-line example.

Loses marks	Passes	Full marks
Bare figures: one slip and no method marks survive.	The CFAT and NPV tables, but the sunk fee and the interest are silently dropped.	Items excluded, with reasons; initial outflow, CFAT and terminal year in ICAI's formats; assumptions stated (timing, tax on losses); a one-line advice.

Common mistakes

Common mistake

Hall of Shame: the feasibility fee Kesar paid twice

Adding the ₹1,50,000 report fee to the outflow and the allocated head-office overheads to the yearly costs. **Fix:** the fee is sunk and the overheads only allocated. In the samosa example, including them with the interest turns an NPV of ₹20,84,325 into about −₹3.64 lakh: a good project rejected.

Common mistake

The interest that was charged twice

Deducting loan interest from the cash flows, then discounting at the WACC. **Fix:** the WACC already contains the cost of debt; take cash flows before interest.

⚠ **Common mistake**

The working capital that never came home

Forgetting its release in the terminal year, or depreciating it. **Fix:** working capital comes back in full at the end unless the question says otherwise; no depreciation is provided on it.

⚠ **Common mistake**

Depreciation in the year the block died

Charging a year's depreciation on the only asset in its block in the year it is sold, and also taking a capital loss. **Fix:** no depreciation that year; opening WDV less sale value is the short-term capital loss.

Practice

Check yourself

1 A Tapri kiosk earns ₹8,00,000 a year before depreciation and tax. Depreciation is ₹3,00,000 and tax 25%. What is its annual cash inflow after tax (₹)?

Answer: _____

2 Case scenario. Tapri's Pune café will replace its espresso machine. The new machine costs ₹12,00,000 plus ₹50,000 installation. The old machine, with a tax WDV of ₹1,20,000, sells for ₹2,00,000, and tax of 25% is payable on the gain. Working capital rises by ₹1,00,000. A market survey for the café cost ₹40,000 last month. Which is the initial cash outflow?

A ₹11,70,000

B ₹12,10,000

C ₹11,30,000

D ₹10,70,000

3 A project's earnings before tax are ₹10,00,000 after ₹2,00,000 interest on the loan that finances it. Depreciation is ₹3,00,000 and tax 30%. It is discounted at the WACC. Which annual cash inflow is used?

A ₹10,00,000

B ₹11,40,000

C ₹12,00,000

D ₹15,00,000

4 Both replacement and modernisation decisions are called:

- **A** Cost reduction decisions
- **B** Revenue expansion decisions
- **C** Contingent decisions
- **D** Accept-reject decisions

5 Which of these are relevant cash flows for Tapri's new cold store? Select all that apply.

- **A** Rent Tapri gives up on a godown the cold store will occupy
- **B** Head-office costs allocated to the cold store
- **C** Working capital released at the end of the project
- **D** A consultancy fee paid last year for the site study
- **E** Tax saved because of depreciation on the cold store

More than one answer may be right.

6 A machine, the only asset in its block, has a WDV of ₹4,00,000 at the beginning of its last year and is sold for ₹1,00,000. Tax is 30%. What is the tax benefit in that year (₹)?

Answer: _____

Answers: the answer key at the end of this lesson.

Recap

- Capital budgeting: identify projects, estimate post-tax incremental cash flows, select the proposal that maximises return.
- Critical because of substantial investment, long time period, irreversibility, complex decisions.
- Process: planning, evaluation, selection, implementation, control, review.
- Types: replacement and modernisation (cost reduction), expansion, diversification (revenue expansion); mutually exclusive, accept-reject, contingent.
- In: opportunity cost, working capital (out, then released in full), additional investment, incremental overheads, depreciation's tax shield. Out: sunk cost, allocated overheads, interest and dividends.
- Cash flows: initial outflow, interim CFAT (EAT + depreciation), terminal year (salvage, tax on sale, release of working capital).
- Only asset in the block: no depreciation in the year of sale; STCL or STCG on (opening WDV − sale value). Other assets: depreciation on that difference.
- Cash flows and discount rate both post-tax; financing costs live in the WACC. (Two photographs, one difference.)

Answer key · Check yourself

1 **675000**

EBT ₹5,00,000; tax ₹1,25,000; EAT ₹3,75,000; add back depreciation: ₹6,75,000. Shortcut: ₹8,00,000 × 0.75 + ₹3,00,000 × 0.25.

2 **A ₹11,70,000**

₹12,00,000 + ₹50,000 + ₹1,00,000 − ₹2,00,000 + tax on the ₹80,000 gain (₹20,000) = ₹11,70,000. The others add the sunk survey, deduct the tax, or forget the working capital.

3 **B ₹11,40,000**

PBIT ₹12,00,000 × 0.70 + depreciation ₹3,00,000 = ₹11,40,000. ₹10,00,000 keeps the interest in; ₹12,00,000 adds it back without its tax saving.

4 **A Cost reduction decisions**

ICAI's term (SM 7.6). Expansion and diversification are revenue expansion decisions; contingent and accept-reject classify by situation.

5 **A Rent Tapri gives up on a godown the cold store will occupy**; **C Working capital released at the end of the project**; **E Tax saved because of depreciation on the cold store**

The first, third and fifth change cash because of the project; allocated costs and a sunk fee do not.

6 **90000**

The block ceases: no depreciation; short-term capital loss ₹3,00,000 × 30% = ₹90,000.

7.2 Payback, discounted payback and ARR

How long a project takes to return its cash (payback, and discounted payback in today's rupees), the payback reciprocal, and the accountant's average rate of return, with their decision rules and limits.

In plain words

"When do I get my money back?"

Tapri is choosing a format for its Lucknow cafés, and Kesar has one question.

> **Kesar:** Forget the jargon, Dev. When do I get my ₹24 lakh back?
>
> **Dev:** In four years, if the cash comes in as planned. But rupees in year four are worth less than rupees today.
>
> **Ritu:** And the books will show a profit only after depreciation. Per year, please, not per lifetime.

Kesar's question is the **payback period**: the time a project takes to recover the cash put into it. It is easy to grasp, and a short payback means less waiting and less risk. But it treats a rupee received in year four as equal to a rupee today, and it stops looking the day the money is back: a format that pays back fast and then dies looks better than one that pays back slowly and earns for twenty years.

Dev's fix is the **discounted payback period**: the same count, using the present value of each year's cash. Ritu's view is the **accounting rate of return (ARR)**: the average yearly profit, after depreciation, as a percentage of the money invested.

Think of it like

The cousin's loan

You lend your cousin ₹1 lakh for his shop. Payback asks how soon the ₹1 lakh is back. Discounted payback asks how soon it is back in rupees worth what they were when you lent them. ARR asks what average yearly income the loan earned, by the accountant's book. Only the second cares when the money came; only the third looks past the day you were repaid.

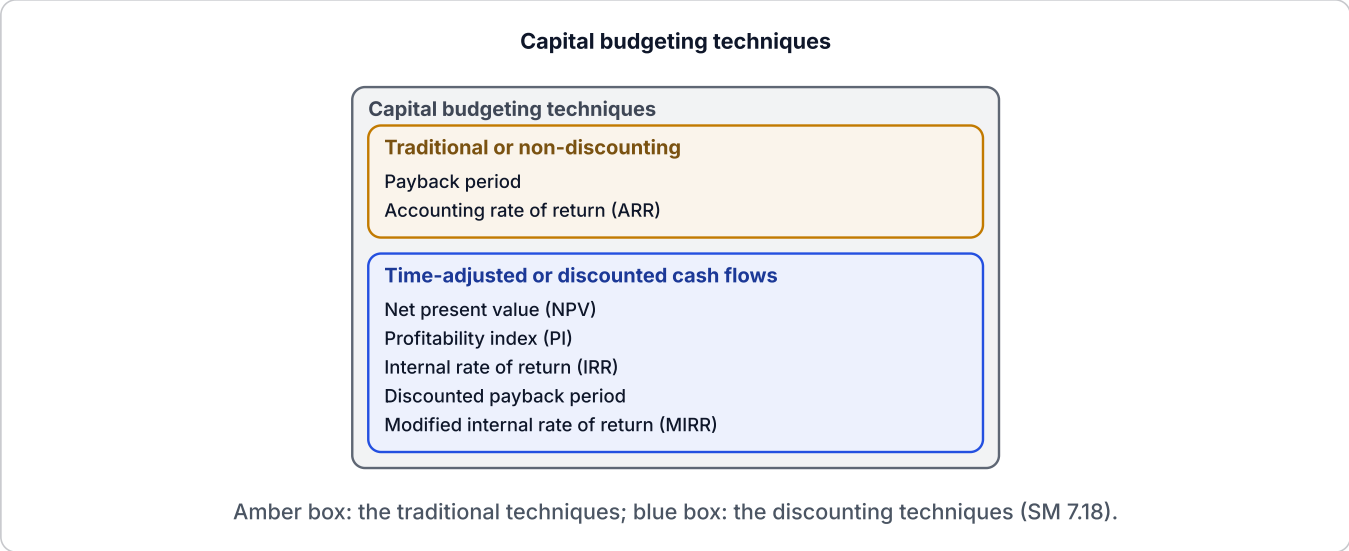
Key idea

Payback measures how fast the initial cash outlay is recovered: a guide to liquidity and risk, blind to the time value of money and to every cash flow after payback. Discounted payback cures the first blindness, not the second. ARR covers the whole life but uses accounting profit and ignores the time value of money.

How it works

Two families of techniques

ICAI divides capital budgeting techniques into two families (SM 7.17–7.18). The first two do not discount future cash flows; this lesson covers them and the discounted payback period.



Amber box: the traditional techniques; blue box: the discounting techniques (SM 7.18).

Payback period

The payback period is the length of time required for the cumulative total net cash flows from the investment to equal the total initial cash outlays (SM 7.18–7.19). The steps: (a) determine the total initial capital investment (cash outflow); (b) estimate the annual expected after-tax cash flows.

1. **Uniform cash flows**: divide the total initial investment by the annual after-tax cash flow. ₹24,00,000 ÷ ₹6,00,000 a year = 4 years.

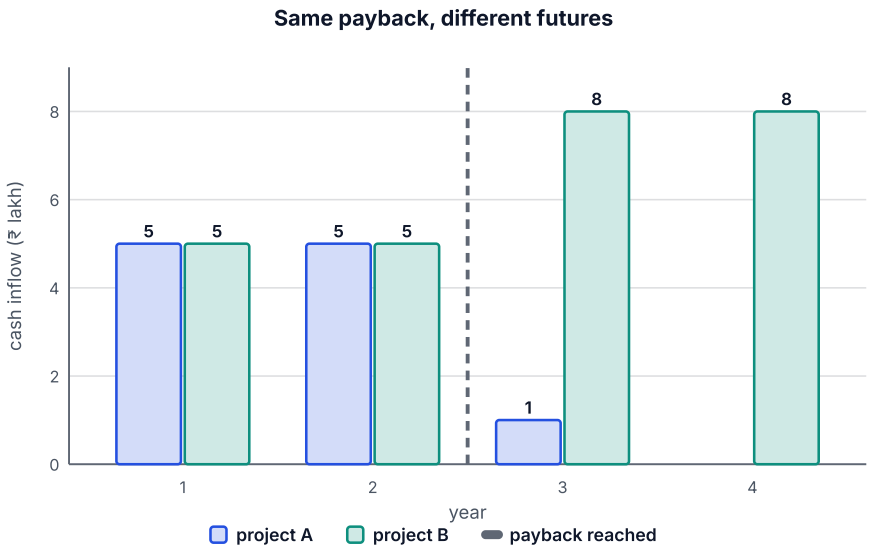
2. **Non-uniform cash flows**: add the cash inflows year by year. Where the cumulative total equals the outlay, that is the payback; otherwise find the year in which it falls and add the fraction of that year: unrecovered amount ÷ that year's cash inflow.

3. Tapri's delivery e-bikes cost ₹6,00,000 and bring ₹2,00,000, ₹2,50,000, ₹2,00,000 and ₹1,00,000. Two years recover ₹4,50,000; the remaining ₹1,50,000 is ¾ of year 3's ₹2,00,000. **Payback = 2.75 years.**

The fraction assumes the year's cash arrives evenly through the year.

Advantages (SM 7.21): it is easy to compute; it is easy to understand, as a quick estimate of the time needed to recoup the cash invested; the length of the payback period can serve as an estimate of a project's risk (the longer, the riskier, since long-term predictions are less reliable), which matters with high obsolescence risk, as in software, or when an organisation is short on cash.

Limitations (SM 7.21): it ignores the time value of money; it fails to consider an investment's total profitability, ignoring cash flows after the payback period; and its emphasis on short payback periods ignores long-term projects.



Blue: project A; teal: project B; both cost ₹10 lakh. Each recovers it in exactly 2 years (grey dashed line), so payback ranks them equal, though B then earns ₹16 lakh more and A only ₹1 lakh.

Payback reciprocal

The payback method gives no cut-off period for decisions. Its reciprocal, average annual cash inflow ÷ initial investment, is argued to be a close approximation of the internal rate of return if the life of the project is at least twice the payback period and the project generates equal annual cash inflows (SM 7.21–7.22). A ₹24 lakh café returning ₹6 lakh a year has a payback reciprocal of 25%.

Accounting (book) rate of return

The ARR measures the average annual net income of the project (incremental income) as a percentage of the investment (SM 7.22). The numerator is the average annual net income over the useful life, after depreciation. The denominator can be either the **initial investment** (including installation cost) or the **average investment**, the average amount of funds blocked over the life: ½ (initial investment − salvage value) + salvage value, plus any additional working capital (SM 7.24). Some organisations prefer the initial investment because it is objective, not influenced by the depreciation method or the salvage estimate; either may be used, but the same method for all investments under consideration (SM 7.24). The rate is compared with a minimum, the **cut-off rate** (SM 7.26).

Advantages (SM 7.24–7.25): it uses readily available data routinely generated for financial reports; it may mirror the method used to evaluate performance, ensuring consistency; it considers all net incomes over the entire life and gives a measure of profitability.

Limitations (SM 7.25): it ignores the time value of money; it uses accounting numbers that depend on accounting procedures such as the depreciation method; it ignores cash flows, though net cash flow is a better measure of performance; and the book value of the asset ignores commitments of working capital and other outlays.

Discounted payback period

The discounted payback period is the payback period computed on cash flows discounted at a predetermined rate (SM 7.42). It is considered superior to simple payback because it takes the time value of money into account. The Study Material's example: a ₹30,000 outlay returning ₹6,000 a year pays back in 5 years, but discounted at 15% it takes about 10 years; as the required rate of return increases, the distortion between simple and discounted payback grows (SM 7.42–7.43). Like simple payback, it ignores the cash flows after the discounted payback period; the shorter the period, the better (ICAI's answer to MTP Series II for May 2024 Q4(c)).

Interactive version: PassBook app, lesson Payback, discounted payback and ARR

Decision rules

Technique	Independent project	Mutually exclusive projects
Payback	Accept if payback ≤ maximum acceptable payback period; reject if longer	Least payback period
ARR	Accept if ARR ≥ minimum acceptable rate of return; reject if lower	Maximum ARR

(SM 7.52)

Formats and formulas

Reproduce these from memory.

∑ **Formula**

Payback period, uniform cash flows

$$\text{Payback period} = \frac{I}{C}$$

I = total initial capital investment; C = annual expected after-tax net cash flow (SM 7.19).

∑ **Formula**

Payback period, non-uniform cash flows

$$\text{Payback} = Y + \frac{U}{C_{Y+1}}$$

Y = full years before the outlay is recovered; U = unrecovered outlay (outlay − cumulative inflows of those Y years); C with subscript Y + 1 = cash inflow of the next year.

∑ **Formula**

Payback reciprocal

$$\text{Payback reciprocal} = \frac{\bar{C}}{I} \times 100$$

C̄ = average annual cash inflow; I = initial investment (SM 7.22).

Σ Formula

Accounting rate of return

$$\text{ARR} = \frac{\text{Average annual net income}}{\text{Investment}}$$

Expressed as a percentage (× 100). Net income = profit after depreciation (and after tax where tax is given). Investment = initial or average investment, as the question says (SM 7.22).

Σ Formula

Average investment

$$\tfrac{1}{2}(I - S) + S + W$$

I = initial investment, S = salvage value, W = additional working capital (SM 7.24). With no working capital this is (I + S) ÷ 2.

The three versions of ARR (SM 7.23–7.24)

Version	Numerator	Denominator
1. Annual basis	Profit after depreciation of each year	Investment at the beginning of that year; then average the yearly rates
2. Total investment basis	Average annual profit	Investment in the beginning
3. Average investment basis	Average annual profit	Average investment

Discounted payback: the working table (SM 7.43)

Year	Cash flow (₹)	PV factor	PV (₹)	Cumulative PV (₹)
1	xxx	x.xxx	xxx	xxx
2	xxx	x.xxx	xxx	xxx
…				

Discounted payback = years before the cumulative PV reaches the outlay + unrecovered amount ÷ that year's PV.

Definition

Payback period

Time required to recover the initial cash-outflow is called pay-back period. (SM 7.18)

Definition

Discounted payback period

The payback period calculated after discounting the cash flows by a predetermined rate. (SM 7.20, 7.42)

Worked examples

Worked example **Kulhad Corner or Express Kiosk?**

Tapri can open its Lucknow outlets in one of two formats. Each costs ₹24,00,000, has no salvage value, and is depreciated on the straight-line method over its life. The cash flows after tax are:

Format	Life	Annual cash flows after tax
Kulhad Corner	8 years	₹6,00,000 every year
Express Kiosk	5 years	₹9,00,000, ₹8,00,000, ₹5,00,000, ₹4,00,000, ₹2,00,000

Required:

(i) Calculate the payback period of each format, and the payback reciprocal of Kulhad Corner.

(ii) Calculate the ARR of each format on initial investment and on average investment.

(iii) Rank the formats by each method and comment.

THE WORKING

(i) Payback

Kulhad Corner: ₹24,00,000 ÷ ₹6,00,000 = **4 years**. Payback reciprocal = ₹6,00,000 ÷ ₹24,00,000 × 100 = **25%**; the life (8 years) is twice the payback, so ICAI's condition for using it as a rough estimate of the IRR holds.

Express Kiosk:

Year	Cash flow (₹)	Cumulative (₹)
1	9,00,000	9,00,000
2	8,00,000	17,00,000
3	5,00,000	22,00,000
4	4,00,000	26,00,000
5	2,00,000	28,00,000

Payback = 3 years + (₹24,00,000 − ₹22,00,000) ÷ ₹4,00,000 = **3.5 years**.

(ii) ARR

Particulars	Kulhad Corner (₹)	Express Kiosk (₹)
Total cash flows	48,00,000	28,00,000
Less: Total depreciation (= cost, no salvage)	24,00,000	24,00,000
Total profit	24,00,000	4,00,000
Average annual profit	3,00,000 (÷ 8)	80,000 (÷ 5)
ARR on initial investment (₹24,00,000)	**12.50%**	**3.33%**
ARR on average investment (₹12,00,000)	**25.00%**	**6.67%**

Average investment = ½ (₹24,00,000 − 0) + 0 = ₹12,00,000.

(iii) Ranking and comment

Method	First	Second
Payback	Express Kiosk (3.5 years)	Kulhad Corner (4 years)
ARR (either base)	Kulhad Corner	Express Kiosk

Payback prefers the kiosk because it recovers the outlay half a year sooner, but it ignores what follows: after payback the kiosk earns only ₹4,00,000 more, Kulhad Corner ₹24,00,000 more. ARR looks at the whole life and reverses the ranking. Neither method considers the time value of money; a discounted technique (NPV) should make the final choice.

Worked example **The refrigerated vans and Kesar's four-year rule**

Tapri's cold-chain division plans to buy refrigerated vans for ₹60,00,000, with a salvage value of ₹6,00,000 at the end of 5 years. They need working capital of ₹4,00,000 at the start, released at the end of year 5. Depreciation is on the straight-line method; tax is 25%; the cost of capital is 12%.

Year	Earnings before depreciation and tax (₹)
1	18,00,000
2	22,00,000
3	26,00,000
4	24,00,000
5	20,00,000

PV factors at 12%: 0.893, 0.797, 0.712, 0.636, 0.567. Tapri's policy: maximum payback 4 years; minimum ARR 20% on average investment.

Required:

(i) Calculate the cash flows after tax.

(ii) Calculate the payback period and the discounted payback period.

(iii) Calculate the ARR on average investment.

(iv) Advise whether the vans meet Tapri's policy.

THE WORKING

Working note. Depreciation = (₹60,00,000 − ₹6,00,000) ÷ 5 = ₹10,80,000 a year. Initial outlay = ₹60,00,000 + ₹4,00,000 = ₹64,00,000.

(i) Cash flows after tax

PBT = EBDT − depreciation ₹10,80,000; CFAT = PAT + depreciation.

Year	PBT (₹)	Tax @ 25% (₹)	PAT (₹)	CFAT (₹)
1	7,20,000	1,80,000	5,40,000	16,20,000
2	11,20,000	2,80,000	8,40,000	19,20,000
3	15,20,000	3,80,000	11,40,000	22,20,000
4	13,20,000	3,30,000	9,90,000	20,70,000
5	9,20,000	2,30,000	6,90,000	17,70,000

Year 5 also brings salvage ₹6,00,000 and working capital ₹4,00,000: a total inflow of ₹27,70,000.

(ii) Payback and discounted payback

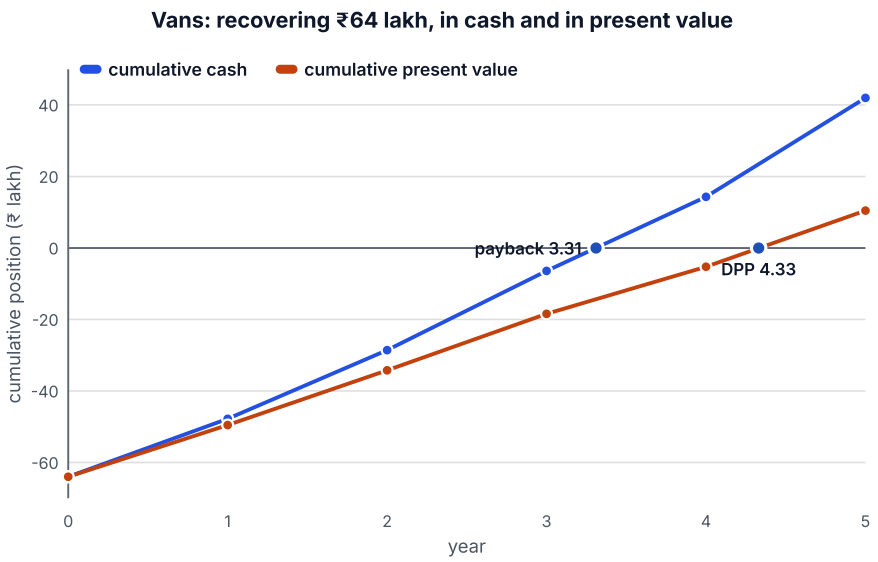
Year	Cash flow (₹)	Cumulative (₹)
1	16,20,000	16,20,000
2	19,20,000	35,40,000
3	22,20,000	57,60,000
4	20,70,000	78,30,000
5	27,70,000*	1,06,00,000

Year	PV factor @ 12%	PV (₹)	Cumulative PV (₹)
1	0.893	14,46,660	14,46,660
2	0.797	15,30,240	29,76,900
3	0.712	15,80,640	45,57,540
4	0.636	13,16,520	58,74,060
5	0.567	15,70,590*	74,44,650

*Includes salvage and the release of working capital, as ICAI's answers do.

$$\text{Payback} = 3 + \frac{64{,}00{,}000 - 57{,}60{,}000}{20{,}70{,}000} = 3.31 \text{ years}$$

$$\text{Discounted payback} = 4 + \frac{64{,}00{,}000 - 58{,}74{,}060}{15{,}70{,}590} = 4.33 \text{ years}$$



Blue: cumulative cash, crossing zero at 3.31 years. Orange: cumulative present value at 12%, crossing at 4.33 years. The orange line ends ₹10.45 lakh above zero: the NPV.

(iii) ARR on average investment

Average annual PAT = (₹5,40,000 + ₹8,40,000 + ₹11,40,000 + ₹9,90,000 + ₹6,90,000) ÷ 5 = ₹42,00,000 ÷ 5 = **₹8,40,000**.

Average investment = ½ (₹60,00,000 − ₹6,00,000) + ₹6,00,000 + working capital ₹4,00,000 = **₹37,00,000**.

The working capital stays blocked for the whole life and comes back in full, so it is added at its full amount, like the salvage. Treating it as part of the outlay gives the same figure: ½ (₹64,00,000 − ₹10,00,000) + ₹10,00,000 = ₹37,00,000. (The Study Material's formula adds "additional working capital", SM 7.24; state the working capital you include.)

ARR = ₹8,40,000 ÷ ₹37,00,000 × 100 = **22.70%**.

(iv) Advice

Test	Policy	Result	Meets policy?
Payback	≤ 4 years	3.31 years	Yes
ARR (average investment)	≥ 20%	22.70%	Yes
Discounted payback	≤ 4 years	4.33 years	No

The vans meet Tapri's stated policy (payback and ARR). Once the time value of money is counted they take 4.33 years to recover their cost in present-value terms, beyond four years; but they do recover it within their life, and the cumulative present value ends ₹10,44,650 above the outlay, a positive NPV. Dev would accept the vans and suggest a discounted payback limit in the policy.

Worked example **Two vending machines, three ARRs**

Tapri must choose one tea-vending machine for its office-tower kiosks. Both cost ₹4,80,000, last 4 years and are depreciated on the straight-line method.

Particulars	Kulhad-Mini	Steel-Pro
Salvage value at the end of year 4 (₹)	nil	1,60,000
Profit after depreciation and tax, year 1 (₹)	30,000	64,000
Year 2 (₹)	50,000	64,000
Year 3 (₹)	70,000	64,000
Year 4 (₹)	90,000	64,000

Required:

(i) Calculate the ARR of each machine by each of ICAI's three versions.

(ii) Rank the machines under each version and comment on which basis Tapri should use.

THE WORKING

Depreciation: Kulhad-Mini ₹4,80,000 ÷ 4 = ₹1,20,000 a year; Steel-Pro (₹4,80,000 – ₹1,60,000) ÷ 4 = ₹80,000 a year.

Version 1: annual basis (profit ÷ investment at the beginning of the year)

Year	Kulhad-Mini: investment (₹)	ARR	Steel-Pro: investment (₹)	ARR
1	4,80,000	6.25%	4,80,000	13.33%
2	3,60,000	13.89%	4,00,000	16.00%
3	2,40,000	29.17%	3,20,000	20.00%
4	1,20,000	75.00%	2,40,000	26.67%
Average ARR		**31.08%**		**19.00%**

Version 2: total investment basis. Average annual profit: Kulhad-Mini ₹2,40,000 ÷ 4 = ₹60,000; Steel-Pro ₹64,000.

- Kulhad-Mini: ₹60,000 ÷ ₹4,80,000 × 100 = **12.50%**
- Steel-Pro: ₹64,000 ÷ ₹4,80,000 × 100 = **13.33%**

Version 3: average investment basis.

- Kulhad-Mini: average investment = (₹4,80,000 + 0) ÷ 2 = ₹2,40,000; ARR = ₹60,000 ÷ ₹2,40,000 × 100 = **25.00%**
- Steel-Pro: average investment = (₹4,80,000 + ₹1,60,000) ÷ 2 = ₹3,20,000; ARR = ₹64,000 ÷ ₹3,20,000 × 100 = **20.00%**

(ii) Ranking

Version	Kulhad-Mini	Steel-Pro	First
1: annual basis	31.08%	19.00%	Kulhad-Mini
2: total investment	12.50%	13.33%	Steel-Pro
3: average investment	25.00%	20.00%	Kulhad-Mini

Comment. The answer depends on the base. Kulhad-Mini wins on versions 1 and 3 only because it has no salvage, so its book value (the denominator) falls faster; its year-4 rate of 75% is earned on a book value of just ₹1,20,000. On the initial investment, which is the same for both, Steel-Pro earns more profit a year and comes first. The Study Material notes that some organisations prefer the initial investment because it is objectively determined and is not influenced by the choice of the depreciation method or the estimation of the salvage value, and that the same method must be used for all investments under consideration (SM 7.24). ARR also ignores timing: Steel-Pro's profits arrive earlier, which no version of ARR rewards. A discounted method settles the choice.

Interactive version: PassBook app, lesson Payback, discounted payback and ARR

Exam angle

In the exam

Where payback and ARR appeared, May 2024 to September 2026

The 12 ICAI documents read for this lesson used them like this, usually as one part of a larger capital budgeting question:

- **Theory, 2 marks**: "Discuss the relevance of payback reciprocal in capital budgeting decisions" (May 2024 Q4(c)); "Explain the concept of discounted payback period" (MTP Series II for May 2024 Q4(c)); "Write a short note on cut-off rate" (MTP Series I for May 2024 Q2(b)).
- **Descriptive parts**: discounted payback in May 2024 Q3(a)(ii) and May 2025 Q3(a)(iv) (1 mark), and for two machines in RTP September 2026 Q7; ARR on average investment with NPV in January 2026 Q2(b) (5 marks).
- **MCQs, 1–2 marks**: payback reciprocal (MTP Series II for May 2024 MCQ 4: the Study Material's own numbers); payback from the IRR (MTP Series I for January 2025 MCQ 7); payback, ARR and PI of two machines (RTP May 2025 Q1; MTP Series II for May 2026 MCQs 1–3); payback with a tax holiday (September 2026 MCQ 4).

In the exam

The traps ICAI used

- ARR "based on average investment" (January 2026 Q2(b)): average investment = (₹24,00,000 + ₹2,40,000) ÷ 2, with ARR on average PAT.
- RTP January 2026 Q9 worked backwards: the initial investment from the MIRR and the terminal value (₹5,00,000), the annual cash flow from the simple payback of 5 years (₹1,00,000), and the WACC from the discounted payback of 7 years (PVAF for 7 years = 5, so 9%); then NPV, PI and ARR (profit after depreciation ÷ initial investment, 7.5%).
- Salvage and working capital released are added to the last year's inflow in the discounted payback table, marked "includes salvage value" (SM 7.89; May 2025 Q3(a)).
- State the denominator of ARR you use; the answers accept initial or average investment when the question is silent (MTP Series I for September 2024 Q3(a)).

Trick

Discounted payback is never shorter

With an outlay at the start, positive inflows and a positive discount rate, every present value is smaller than its cash flow, so the discounted payback is always at least the simple payback. A discounted payback shorter than the payback is an arithmetic slip.

Answer frame

Relevance of payback reciprocal (May 2024 Q4(c), ICAI's points): reciprocal of the payback is a close approximation of the IRR if the life is at least twice the payback period and the cash inflows are equal; a helpful tool for quick estimation of the rate of return; formula: average annual cash flow ÷ initial investment, or 1 ÷ payback period.

Concept of discounted payback period (MTP Series II for May 2024 Q4(c)): payback is the time to recover the investment from project cash flows, focused on liquidity, ignoring time value and profitability; discounted payback uses cash flows discounted at the cost of capital; the period in which discounted cash flows equal the initial outflow; shorter is better; it still ignores cash flows after that period.

Short note on cut-off rate (MTP Series I for May 2024 Q2(b), ICAI's points): the minimum rate of return the management wishes to have from any project; usually based on the cost of capital; the management gains only if a project returns more than the cut-off rate; so it can be used as the discount rate or opportunity cost rate. Add the Study Material's example: a proposal yielding less than the cut-off rate is automatically rejected (SM 7.26).

Advantages and limitations of payback or ARR: ICAI's lists above, in ICAI's order, a line each.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the payback that rounded itself up

Writing "payback: in the 4th year" or "4 years" when three years recover ₹57,60,000 of ₹64,00,000. **Fix:** add the fraction: unrecovered ₹6,40,000 ÷ year 4's ₹20,70,000 = 0.31, so 3.31 years. Divide by that year's cash flow, not by the cumulative total.

⚠ **Common mistake**

ARR on cash instead of profit

Putting the average cash flow after tax in the ARR numerator: ₹19,20,000 ÷ ₹37,00,000 = 51.89% for the vans. **Fix:** ARR uses average annual net income after depreciation (₹8,40,000), giving 22.70%.

⚠ **Common mistake**

The average investment that lost its salvage

Taking half the initial investment (₹30,00,000) as the average investment. **Fix:** ½ (initial − salvage) + salvage + working capital = ₹37,00,000; salvage and working capital stay blocked to the end.

⚠ **Common mistake**

The discount that shortened payback

Discounting the outlay as well, or reading PV factors from the wrong row, and getting a discounted payback below the simple payback. **Fix:** only future inflows are discounted; the year-0 outlay stays at face value, and the discounted payback is never shorter.

Practice

Check yourself

1 A Tapri app feature costs ₹9,00,000 and brings cash flows after tax of ₹3,00,000, ₹4,00,000, ₹3,00,000 and ₹2,00,000 in years 1 to 4. What is its payback period in years (2 decimals)?

Answer: _____

2 Case scenario. Tapri's e-rickshaw fleet costs ₹10,00,000 and brings ₹4,00,000 a year for 4 years. The cost of capital is 10% (PV factors 0.909, 0.826, 0.751, 0.683). Ritu presents the payback and discounted payback to the board. Which of the following is correct?

A Payback 2.50 years; discounted payback 3.02 years

B Payback 2.50 years; discounted payback 2.27 years

C Payback 3.02 years; discounted payback 2.50 years

D Payback 2.50 years; discounted payback 4.00 years

3 Which of the following is NOT a limitation of the payback period method?

- A It ignores the time value of money
- B It ignores cash flows after the payback period
- C Its emphasis on short payback periods ignores long-term projects
- D It depends on accounting procedures such as the depreciation method

4 A machine costs ₹8,00,000, has no salvage value and a 4-year life, and is depreciated on the straight-line method. Its cash flow after tax is ₹2,60,000 a year. What is its ARR on average investment?

- A 65%
- B 32.5%
- C 15%
- D 7.5%

5 A project costs ₹10,00,000, has a salvage value of ₹2,00,000 and needs additional working capital of ₹1,00,000 during its life. What is its average investment (₹)?

Answer: _____

Answers: the answer key at the end of this lesson.

Recap

- Payback = initial investment ÷ annual cash flow after tax (uniform); otherwise cumulate and add the fraction: unrecovered ÷ that year's cash flow.
- Payback: easy, a guide to risk and liquidity; ignores time value, cash flows after payback, long-term projects.
- Payback reciprocal = average annual cash inflow ÷ initial investment; approximates IRR only if life ≥ 2 × payback and inflows are equal.
- ARR = average annual net income (after depreciation) ÷ initial or average investment; the same base for every project.
- Average investment = ½ (initial − salvage) + salvage + additional working capital.
- Discounted payback: payback on discounted cash flows; considers time value; still ignores later cash flows; never shorter than payback.
- Accept: payback ≤ maximum acceptable; ARR ≥ minimum acceptable (cut-off). Mutually exclusive: least payback, highest ARR.
- Last year of a payback or discounted payback table: add salvage and working capital released to that year's inflow. (The cousin's loan: how soon back, how soon back in today's rupees, what it earned by the book.)

Answer key · Check yourself

1 **2.66 to 2.67**

Two years recover ₹7,00,000; the remaining ₹2,00,000 is 2/3 of year 3's ₹3,00,000. Payback = 2.67 years.

2 **A Payback 2.50 years; discounted payback 3.02 years**

Payback = ₹10,00,000 ÷ ₹4,00,000 = 2.5 years. PVs: 3,63,600; 3,30,400; 3,00,400; 2,73,200; three years recover ₹9,94,400, so DPP = 3 + 5,600 ÷ 2,73,200 = 3.02 years. A DPP below the payback is impossible; the third option swaps them.

3 **D It depends on accounting procedures such as the depreciation method**

The last is a limitation of ARR (SM 7.25). Payback works on cash flows; the other three are ICAI's limitations of payback (SM 7.21).

4 **C 15%**

Average profit = ₹2,60,000 − ₹2,00,000 depreciation = ₹60,000; average investment = ₹4,00,000; ARR = 15%. 65% puts cash in the numerator; 7.5% divides profit by the initial investment.

5 **700000**

½ (₹10,00,000 − ₹2,00,000) + ₹2,00,000 + ₹1,00,000 = ₹7,00,000 (SM 7.24).

7.3 NPV, profitability index and equivalent annual cost

Bring every cash flow to today at the cost of capital: NPV and its decision rule, the profitability index for ranking, and equivalent annual figures for projects with unequal lives.

In plain words

Dev's favourite three letters

> **Ritu:** Nice tie, Dev.
>
> **Dev:** Thank you. A compliment received next year would be worth less, of course.
>
> **Kesar:** Focus. The app upgrade costs ₹10 lakh and brings back ₹3, 4, 4 and 3 lakh over four years. ₹14 lakh back, ₹4 lakh profit. Done?
>
> **Dev:** Not yet. Bring every rupee back to today at our 12% cost of capital, then subtract the ₹10 lakh. What is left is the NPV.

A rupee today is worth more than a rupee next year, because today's rupee could be invested and earn the cost of capital meanwhile. So a future cash flow is shrunk to its **present value** before it is added to anything. At 12%, the ₹3 lakh of year 1 is worth ₹2.68 lakh today and the ₹3 lakh of year 4 only ₹1.91 lakh. The four inflows together are worth ₹10.62 lakh today.

The **net present value (NPV)** is that ₹10.62 lakh minus the ₹10 lakh paid: ₹0.62 lakh, not ₹4 lakh. It is positive, so the upgrade earns more than the 12% the money costs, and Tapri's owners are ₹62,300 richer in today's money. The same comparison as a ratio, ₹10.62 lakh ÷ ₹10 lakh = 1.06, is the **profitability index**. And when two machines last different numbers of years, Dev turns each NPV, or each cost, into an equal amount per year before comparing: the **equivalent annual** figure.

Think of it like

Converting currencies

You cannot add dollars, dirhams and rupees until you convert them into one currency. Rupees of 2027, 2028 and 2030 are different currencies too; the discount factor is the exchange rate into today's rupees. NPV is what is left after converting everything and paying the bill.

Key idea

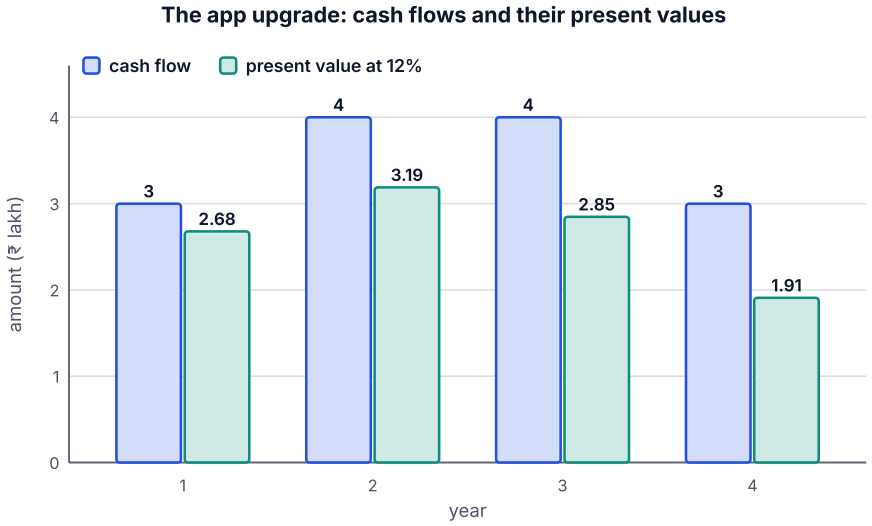
NPV converts every cash flow into today's rupees at the cost of capital and nets them: accept a project when NPV ≥ 0, because a positive NPV is the addition to shareholders' wealth. PI is the same test as a ratio; equivalent annual figures make projects with different lives comparable.

How it works

What a future rupee is worth today

Discounting techniques consider the time value of money and discount cash flows to their present value (SM 7.26). The discount rate is, in theory, the return the firm would earn by investing the same funds in the best available alternative investment of the same risk. That is hard to find in practice, so

a firm may set a minimum rate of return that all projects must meet (based on an industry average or the cost of other investment opportunities), and many firms use the overall cost of capital, the WACC (SM 7.26–7.27). The present value factor for year t at rate k is $1 \div (1 + k)^t$; ICAI's tables give it to three decimals (0.893, 0.797, 0.712, 0.636 at 12%).



Blue: the cash flow; teal: its present value at 12%. The later the rupee, the more it shrinks. The teal bars add up to ₹10.62 lakh against the ₹10 lakh paid today.

Net present value

The net present value of a project is the amount, in current value, the investment earns after paying cost of capital in each period (SM 7.27). When further investment is needed during the life, NPV = present value of cash inflows − present value of cash outflows. ICAI's steps (SM 7.28):

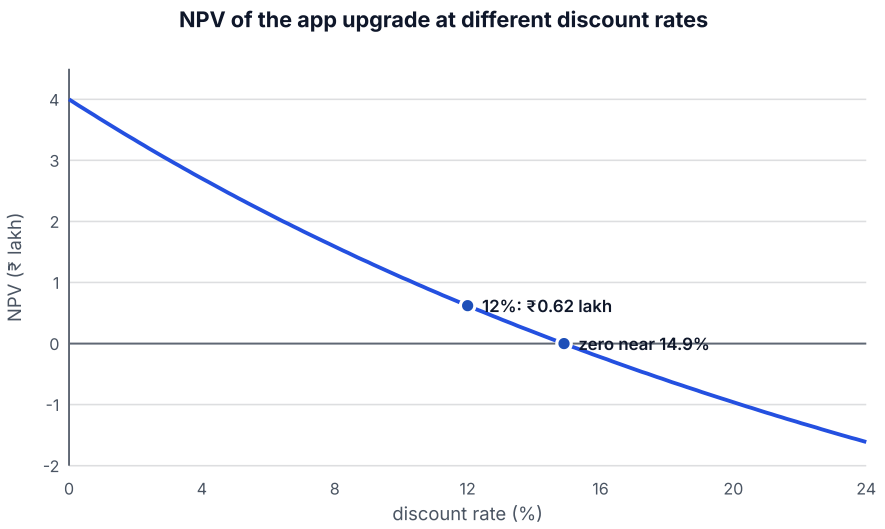
1. Determine the net cash inflow in each year of the investment (relevant cash flows).
2. Select the desired rate of return, discounting rate or weighted average cost of capital.
3. Find the discount factor for each year at that rate.
4. Multiply each year's net cash flow by its discount factor: the present value of the cash flows.
5. Total the present values: the NPV.

Decision rule (SM 7.28): if NPV ≥ 0, accept the proposal; if NPV < 0, reject it. Between mutually exclusive projects, select the one with the higher NPV.

Advantages (SM 7.30): it takes into account the time value of money; the whole stream of cash flows is considered; the NPV can be seen as the addition to the wealth of shareholders, in conformity with basic financial objectives; and since it expresses cash flows in current rupees, NPVs of different projects can be compared, each project evaluated on its own merit.

Limitations (SM 7.30): it involves difficult calculations; it needs forecasts of cash flows and of the discount rate, and is only as accurate as they are; and it is an absolute measure, ignoring differences in initial outflows and size when mutually exclusive projects are compared.

The higher the discount rate, the lower the present values and the NPV:



At 0% the NPV is the undiscounted ₹4 lakh; at Tapri's 12% it is ₹0.62 lakh; near 14.9% it reaches zero. That rate is the project's IRR, the subject of the next lesson.

Profitability index

The **profitability index (PI)**, also called the desirability factor or present value index, compares proposals with different cash flows: sum of discounted cash inflows ÷ initial cash outlay or total discounted cash outflow, as the case may be (SM 7.31). **Decision rule**: if PI ≥ 1, accept; if PI < 1, reject. In ICAI's illustration, a project with the largest inflows (₹1,00,30,000 against an outflow of ₹1,00,20,000) has the lowest desirability factor, 1.001: PI helps rank projects (SM 7.31–7.32).

Since PI is an extension of NPV it has the same advantages and limitations, and some of its own (SM 7.32). **Advantages**: it uses the time value of money; as present value of inflows ÷ present value of outflows, it is a relative measure of profitability. **Limitations**: it fails as a guide in resolving capital rationing where projects are indivisible; once a single large project with high NPV is selected, several small projects with a higher total NPV are excluded; and a project with a lower PI may generate cash that lets another project start a year or two later, for a higher total NPV. So PI cannot be used indiscriminately; all alternatives must be worked out.

NPV ≥ 0 and PI ≥ 1 are the same test, so they always agree on accepting or rejecting a single project. They can **rank** mutually exclusive projects differently: NPV in absolute rupees, PI as a ratio (ICAI's RTP for January 2025, Q11(c)). The Study Material's PI section says the project with the higher PI is selected among mutually exclusive projects (SM 7.31); its summary table narrows this to "when Net Present Value is same", and otherwise picks the project with the highest positive NPV (SM 7.52). Where its illustrations meet a ranking conflict (NPV against IRR) they follow NPV, which "generally gives consistent results in conformity with the wealth maximization principle" (SM 7.59). So when NPV and PI rank mutually exclusive projects differently, decide on NPV and say why. Under capital rationing, ranking by PI comes into its own: see capital rationing.

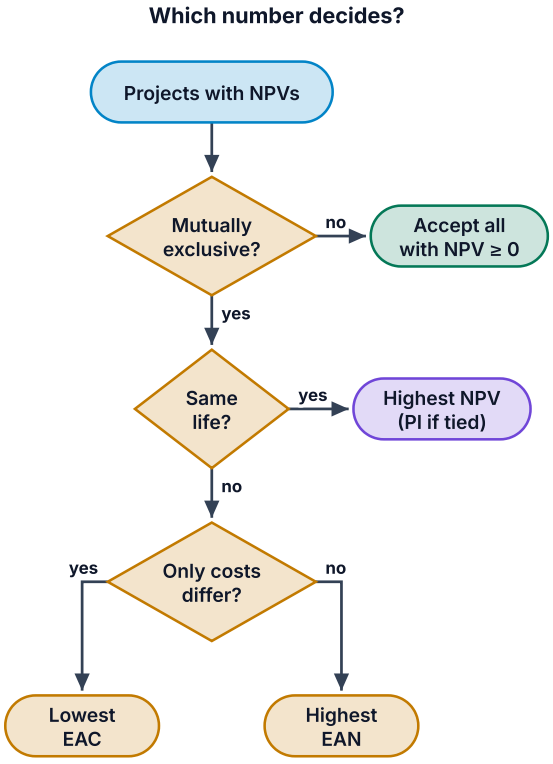
Interactive version: PassBook app, lesson NPV, profitability index and equivalent annual cost

When lives differ

Comparing the NPV of a 3-year machine with that of a 6-year machine is unfair: the 3-year one could be bought again and earn a second NPV. For replacing an old asset, or choosing between mutually exclusive proposals with unequal lives, ICAI gives two methods (SM 7.54–7.56):

- **Replacement chain (common life) method**: repeat the shorter project until the lives match, then compare NPVs over the common life. Its drawback: a 3-year and a 5-year project need a 15-year analysis.
- **Equivalent annualised criterion**: (a) compute each NPV at the WACC or discount rate; (b) compute the present value annuity factor (PVAF) of that rate for each project's life; (c) divide (a) by (b) and compare. The higher equivalent annual NPV wins.

When the alternatives do the same job and only their costs differ, the same idea gives the **equivalent annual cost**: present value of all costs (net of salvage) ÷ PVAF for the life. The lower equivalent annual cost wins (SM 7.93, Test Your Knowledge practical problem 6; SM 7.100, problem 12).



EAC: equivalent annual cost; EAN: equivalent annualised NPV. Green: independent projects; violet: equal lives; amber: unequal lives. Under capital rationing, independent projects are ranked instead (next lessons).

Decision rules (SM 7.52)

Technique	Independent project	Mutually exclusive projects
NPV	Accept if NPV ≥ 0; reject if NPV < 0	Highest positive NPV
PI	Accept if PI ≥ 1; reject if PI < 1	When NPV is the same, highest PI

Formats and formulas

Reproduce these from memory.

∑ **Formula**

Net present value

$$\text{NPV} = \sum_{t=1}^{n} \frac{C_t}{(1+k)^t} - I$$

C_t = cash flow of year t, k = discount rate, n = life of the project, I = investment (SM 7.27–7.28). With outflows later in the life: NPV = PV of cash inflows − PV of cash outflows.

∑ **Formula**

Profitability index

$$\text{PI} = \frac{\text{Sum of discounted cash inflows}}{\text{Discounted cash outflow}}$$

The denominator is the initial cash outlay or the total discounted cash outflow, as the case may be (SM 7.31).

∑ **Formula**

Equivalent annualised NPV and equivalent annual cost

$$\text{EAN} = \frac{\text{NPV}}{\text{PVAF}_{k,n}} \qquad \text{EAC} = \frac{\text{PV of costs}}{\text{PVAF}_{k,n}}$$

PVAF = present value annuity factor at the discount rate k for the project's own life n (SM 7.56). Higher EAN or lower EAC wins.

The NPV statement (SM 7.29)

Year	Net cash flow (₹)	PV factor @ k	Discounted cash flow (₹)
0	(xxx)	1.000	(xxx)
1	xxx	x.xxx	xxx
…			
Net present value			**xxx**

For equal yearly flows, one line: cash flow × PVAF for years 1 to n.

Definition

Net present value

The net present value of a project is the amount, in current value of amount, the investment earns after paying cost of capital in each period. (SM 7.27)

Worked examples

Worked example **Two packaging lines, one floor**

Tapri's premix plant has room for one sachet-packing line. Line P costs ₹80,00,000 and has a salvage value of ₹5,00,000 after 5 years; line Q costs ₹30,00,000 with no salvage after 5 years. Both are depreciated on the straight-line method; tax is 25% and the cost of capital 12% (PV factors 0.893, 0.797, 0.712, 0.636, 0.567). Earnings before depreciation and tax:

Year	Line P (₹)	Line Q (₹)
1	24,00,000	12,00,000
2	29,00,000	14,00,000
3	34,00,000	16,00,000
4	31,00,000	14,00,000
5	26,00,000	12,00,000

Required:

(i) Calculate the NPV and the profitability index of each line.

(ii) Which line should Tapri install? Would the answer change if the two lines were independent and could both be installed?

THE WORKING

Working note. Depreciation: P = (₹80,00,000 − ₹5,00,000) ÷ 5 = ₹15,00,000; Q = ₹30,00,000 ÷ 5 = ₹6,00,000. CFAT = (EBDT − depreciation) × 0.75 + depreciation.

Line P

Year	CFAT (₹)	PV factor @ 12%	PV (₹)
1	21,75,000	0.893	19,42,275
2	25,50,000	0.797	20,32,350
3	29,25,000	0.712	20,82,600
4	27,00,000	0.636	17,17,200
5	23,25,000 + salvage 5,00,000 = 28,25,000	0.567	16,01,775
	PV of inflows		**93,76,200**

NPV = ₹93,76,200 − ₹80,00,000 = **₹13,76,200**; PI = ₹93,76,200 ÷ ₹80,00,000 = **1.172**.

Line Q

Year	CFAT (₹)	PV factor @ 12%	PV (₹)
1	10,50,000	0.893	9,37,650
2	12,00,000	0.797	9,56,400
3	13,50,000	0.712	9,61,200
4	12,00,000	0.636	7,63,200
5	10,50,000	0.567	5,95,350
	PV of inflows		**42,13,800**

NPV = ₹42,13,800 − ₹30,00,000 = **₹12,13,800**; PI = ₹42,13,800 ÷ ₹30,00,000 = **1.405**.

(ii) Decision

	Line P	Line Q
NPV (₹)	13,76,200	12,13,800
PI	1.172	1.405
Rank by NPV	1	2
Rank by PI	2	1

The lines are mutually exclusive and the NPVs differ, so the higher NPV decides: **install line P**, which adds ₹1,62,400 more to shareholders' wealth. PI favours Q because it is a ratio and Q is smaller. If the lines were independent, both would be accepted: each has NPV > 0 and PI > 1.

Interactive version: PassBook app, lesson NPV, profitability index and equivalent annual cost

Worked example **The kulhad machines and the price tag that moves**

Tapri must choose one of two kulhad-moulding machines (mutually exclusive). The cost of capital is 10%.

Particulars	Machine K4	Machine K8
Cost (₹)	12,00,000	22,00,000
Life	4 years	8 years
Cash flow after tax, every year (₹)	4,80,000	4,90,000

PVAF at 10%: 4 years 3.170; 8 years 5.335. PVF at 10%, year 4: 0.683. The supplier has told Ritu that a K4 bought at the end of year 4 will cost ₹15,60,000; its cash flows would be the same as the first one's.

Required:

(i) Calculate the NPV and PI of each machine.

(ii) Recommend a machine using the equivalent annualised criterion.

(iii) Evaluate the two machines over a common life of 8 years by the replacement chain method, using the supplier's price for the second K4, and advise.

THE WORKING

(i) NPV and PI

Particulars	K4 (₹)	K8 (₹)
PV of inflows	15,21,600	26,14,150
Less: Cost	12,00,000	22,00,000
NPV	**3,21,600**	**4,14,150**
PI	1.27	1.19

PV of inflows: K4 ₹4,80,000 × 3.170; K8 ₹4,90,000 × 5.335. On NPV alone K8 looks better, but it runs twice as long. (Annuity factors as given; summing the rounded yearly factors gives 3.169 and 5.334, which lowers the NPVs by ₹480 and ₹490 and changes nothing below.)

(ii) Equivalent annualised criterion

Particulars	K4	K8
NPV (₹)	3,21,600	4,14,150
PVAF at 10% for the machine's life	3.170	5.335
Equivalent annualised NPV (₹)	**1,01,451**	**77,629**

On this criterion K4 earns more per year of service.

(iii) Replacement chain over 8 years

The second K4 is not a repeat of the first: it costs ₹15,60,000, not ₹12,00,000.

Particulars	Amount (₹)
NPV of the first K4 (year 0)	3,21,600
NPV of the second K4 at the end of year 4 (15,21,600 − 15,60,000)	(38,400)
PV of the second K4's NPV (× 0.683)	(26,227)
NPV of the K4 chain over 8 years	**2,95,373**
NPV of K8 over 8 years	4,14,150



Blue: K4 and its dearer replacement; in year 4 the ₹4.8 lakh inflow and the second ₹15.6 lakh purchase net to −₹10.8 lakh. Orange: K8.

Recommendation. Machine K8. Over the same 8 years it adds ₹4,14,150 against ₹2,95,373 for the K4 chain. The equivalent annualised NPV pointed to K4 because it spreads each machine's own NPV over its own life, which compares the two fairly only if K4 can be bought again on the same terms. At the old price of ₹12,00,000 the chain would be worth ₹3,21,600 × 1.683 = ₹5,41,253 and would agree with the equivalent annualised NPV; K4 stays ahead only while its replacement costs less than about ₹13,86,000. When the second purchase is known to differ, lay out the common-life cash flows and decide on their NPV.

Worked example **Zippy or Volt: scooters that only cost**

Tapri's delivery arm will buy electric scooters of one brand. Both do the same job; ignore tax. The cost of capital is 12%.

Particulars	Zippy	Volt
Cost per scooter (₹)	1,20,000	80,000
Life	4 years	3 years
Running cost a year (₹)	30,000	40,000
Salvage at the end of life (₹)	20,000	nil

PVAF at 12%: 4 years 3.037; 3 years 2.402. PVF at 12%, year 4: 0.636.

Required: Kesar compares the present value of each scooter's costs over its life and picks the cheaper. Calculate the equivalent annual cost of each and advise.

THE WORKING

Particulars	Zippy (₹)	Volt (₹)
Cost now	1,20,000	80,000
Add: PV of running costs	30,000 × 3.037 = 91,110	40,000 × 2.402 = 96,080
Less: PV of salvage	20,000 × 0.636 = (12,720)	nil
PV of costs over the life	**1,98,390**	**1,76,080**
PVAF for the life	3.037	2.402
Equivalent annual cost	**65,324**	**73,306**

Kesar's comparison of ₹1,98,390 with ₹1,76,080 sets four years of Zippy against three years of Volt. Per year of service Zippy costs ₹65,324 and Volt ₹73,306: **buy Zippy**, the lower equivalent annual cost, saving about ₹7,982 a scooter every year. (Factors as given; the four rounded yearly factors at 12% sum to 3.038, which puts Zippy at ₹65,313.)

Exam angle

In the exam

Where NPV, PI and equivalent annual figures appeared, May 2024 to September 2026

In the documents read for this lesson, NPV was the end point of most capital budgeting numericals:

- **Descriptive, 6 to 7 marks**: NPV and PI of a replacement (May 2024 Q3(a), 7 marks) and of a new plant (January 2025 Q3(a), 6 marks); NPV, PI, discounted payback and MIRR from an IRR (May 2025 Q3(a), 6 marks); NPV and PI of two machines with lives of 5 and 7 years, then the choice on the **equivalent annualised criterion** (May 2026 Q2(a), 7 marks: machine M won on EAN although N had the higher NPV).
- **MCQs, 2 marks each**: equivalent annual NPV of two proposals with lives of 5 and 3 years (May 2025 MCQ 7); NPV and equivalent annual cost of two warehouses (MTP Series I for September 2025, five MCQs); equivalent annual cost of buying two scanners or renting one (MTP Series II for September 2025, five MCQs); which project NPV and PI recommend (RTP January 2026 MCQ 2); PI of two machines (RTP May 2025 Q1(v)).
- **Theory**: "Do the profitability index and the NPV criterion lead to the same acceptance-rejection and ranking decisions? In what situations will they give conflicting results?" (RTP January 2025 Q11(c)).

In the exam

The traps ICAI used

- A subsidy received at the end of year 1 (May 2024 Q3(a)): shown as a reduction of the outlay, PI = 1.224; shown as a year-1 inflow, PI = 1.195. The NPV is ₹4,71,640 either way: state your treatment, since PI depends on it.
- Working capital "in year 1" (January 2025 Q3(a)): taken at year 0, PI 4.62; at the end of year 1, 4.70.
- Two mutually exclusive machines with lives of 5 and 6 years in RTP September 2026 Q7: ICAI's answer recommended Machine I because it ranked first on two of the three methods asked (discounted payback and IRR). On NPV (₹1,18,074 against ₹86,909), the Study Material's rule for mutually exclusive projects, Machine II comes first, and also on equivalent annualised NPV (₹28,714 against ₹24,108, dividing by 4.112 and 3.605, the sums of the RTP's yearly factors at 12%; these figures are our working, not the RTP's). When lives differ and the question asks which to choose, show the equivalent annual figure and state the rule you follow.

Trick

NPV from PI, PI from NPV

PI = 1 + NPV ÷ outlay, and NPV = (PI − 1) × outlay, whenever the whole outlay is at year 0. RTP January 2026's MCQ needs only this: NPV 15 on 60 is a PI of 1.25.

Answer frame

"Discuss NPV. How is it calculated?" (SM 7.74, theory question 2): definition (the amount, in current value, the investment earns after paying cost of capital in each period); formula; ICAI's five steps; decision rule (accept if NPV ≥ 0; highest NPV among mutually exclusive projects); advantages and limitations, a line each.

"Describe the advantages and disadvantages of profitability index" (SM 7.74, question 5): the formula and its other names; two advantages; three limitations; the conclusion that PI cannot be used indiscriminately.

PI and NPV: same decisions? (RTP January 2025 Q11(c), ICAI's points): in most situations the same accept-reject decision (PI > 1 means NPV > 0); conflict in ranking mutually exclusive projects; a two-project example (larger NPV, smaller PI); reason: NPV is absolute, PI a ratio; PI is the better technique under capital rationing.

Common mistakes

Common mistake

Hall of Shame: the ₹4 lakh profit that was ₹62,300

Adding the app upgrade's ₹14 lakh of inflows and subtracting ₹10 lakh. **Fix:** discount each year's cash flow first. At 12% the inflows are worth ₹10,62,300 and the NPV is ₹62,300.

⚠ **Common mistake**

The eight-year machine that won on paper

Choosing K8 because its NPV (₹4,14,150) beats K4's (₹3,21,600). That sets eight years of one machine against four of the other. **Fix:** with unequal lives, compare equivalent annualised NPVs or NPVs over a common life. Here the two disagree, because K4's replacement costs more, and the common-life chain built on the supplier's price decides: K8, at ₹4,14,150 against ₹2,95,373.

⚠ **Common mistake**

Picking by PI when NPV should decide

Choosing line Q for its PI of 1.405. **Fix:** between mutually exclusive projects, the highest NPV is selected; PI decides only when NPVs are the same, or when capital is rationed.

⚠ **Common mistake**

The cheaper scooter that was dearer

Comparing PVs of cost over different lives (₹1,98,390 for four years against ₹1,76,080 for three). **Fix:** divide each by the PVAF for its own life; the lower equivalent annual cost wins.

Practice

Check yourself

1 A Tapri kiosk costs ₹5,00,000 and brings ₹2,00,000 a year for 3 years. The cost of capital is 10% (PVAF 2.487). What is its NPV (₹)?

Answer: _____

2 Case scenario. Dev must pick one of two café formats for a site (mutually exclusive). Format M: PV of inflows ₹54 lakh, outlay ₹40 lakh now. Format N: PV of inflows ₹31.2 lakh, outlay ₹12 lakh now and ₹12 lakh at the end of year 1 (PVF at 12%: 0.893). Which of the following is correct?

A. N, because its PI is higher than M's

B. M, because both its NPV and its PI are higher than N's

C. M, because its NPV is higher, although N's PI is higher

D. Either, because both PIs exceed 1

3 Machine A costs ₹3,00,000, lasts 3 years and costs ₹50,000 a year to run; machine B costs ₹4,00,000, lasts 5 years and costs ₹40,000 a year. Both do the same job; at 10%, PVAF is 2.487 for 3 years and 3.791 for 5 years; ignore tax and salvage. Which should Tapri buy?

A. A, because the PV of its costs is lower

B. B, because its equivalent annual cost is lower

C. A, because its equivalent annual cost is lower

D. B, because the PV of its costs is lower

4 A project has an NPV of ₹1,20,000 over 4 years at 12% (PVAF 3.037). What is its equivalent annualised NPV (₹, nearest rupee)?

Answer: _____

5 Which statements about the profitability index are correct? Select all that apply.

- A It is also called the desirability factor or present value index
- B For a single project with its whole outlay at the start, PI ≥ 1 exactly when NPV ≥ 0
- C It is an absolute measure, so it favours larger projects
- D It fails as a guide in capital rationing where projects are indivisible

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Discount every cash flow to today at the cost of capital (WACC); PV factor = $1 \div (1 + k)^t$.
- NPV = PV of cash inflows − PV of cash outflows; accept if NPV ≥ 0; among mutually exclusive projects, the highest NPV.
- NPV is the addition to shareholders' wealth; its limitations: hard calculations, forecasts of cash flows and rate, an absolute measure.
- PI (desirability factor, present value index) = discounted inflows ÷ discounted outflow; accept if PI ≥ 1; it ranks by ratio and fails with indivisible projects under rationing.
- NPV and PI agree on accept-reject; they can conflict on ranking; with NPVs different, NPV decides.
- Unequal lives: replacement chain (common life) or equivalent annualised criterion, NPV ÷ PVAF for each life; the higher wins.
- Costs only: equivalent annual cost = PV of costs ÷ PVAF; the lower wins.
- Use the PV factors the question gives and say which: a printed annuity factor (4.111 at 12% for 6 years) and the sum of three-decimal yearly factors (4.112) differ in the last digits. (Convert the currencies first: a 2030 rupee is not a 2026 rupee.)

Answer key · Check yourself

1 **-2600**

PV of inflows = ₹2,00,000 × 2.487 = ₹4,97,400; NPV = ₹4,97,400 − ₹5,00,000 = −₹2,600. ₹6,00,000 comes back in cash, but not in today's rupees: reject.

2 **C M, because its NPV is higher, although N's PI is higher**

N's outflows have a PV of 12 + 12 × 0.893 = ₹22.716 lakh, so its NPV is ₹8.484 lakh and its PI 31.2 ÷ 22.716 = 1.37; M's NPV is ₹14 lakh and its PI 54 ÷ 40 = 1.35. For mutually exclusive projects whose NPVs differ, the highest NPV is selected (SM 7.52). "Both higher" is what you get by leaving N's year-1 outlay undiscounted (PI 31.2 ÷ 24 = 1.30).

3 **B** **B, because its equivalent annual cost is lower**

PV of costs: A ₹4,24,350, B ₹5,51,640, over different lives. EAC: A ₹4,24,350 ÷ 2.487 = ₹1,70,627; B ₹5,51,640 ÷ 3.791 = ₹1,45,513. B is cheaper per year.

4 **39500 to 39520**

₹1,20,000 ÷ 3.037 = ₹39,513.

5 **A** **It is also called the desirability factor or present value index**; **B** **For a single project with its whole outlay at the start, PI ≥ 1 exactly when NPV ≥ 0**; **D** **It fails as a guide in capital rationing where projects are indivisible**

PI is a relative measure (SM 7.32), which is why it can favour smaller projects; the other three are ICAI's statements or follow from PI = 1 + NPV ÷ outlay.

7.4 IRR, MIRR and the NPV-IRR conflict

The discount rate at which NPV becomes zero, found by interpolation; MIRR, which reinvests at the cost of capital; and why NPV and IRR can rank mutually exclusive projects differently.

In plain words

"Just tell me the percentage, Dev"

> **Kesar:** Dev, the board does not want a lecture on present values. The app upgrade: what return does it give? One number, in per cent.
>
> **Dev:** At our 12% cost of capital its NPV is ₹62,300, so it earns more than 12%. To find how much more, I keep raising the discount rate until the NPV falls to zero.
>
> **Ritu:** And where does it fall to zero?
>
> **Dev:** Just under 15%. That is the project's own rate of return.

The app upgrade from the NPV lesson costs ₹10 lakh and brings back ₹3, 4, 4 and 3 lakh over four years. Discounted at 12%, the inflows are worth ₹62,300 more than the outlay. At 14% they are worth only ₹18,300 more; at 15% they fall ₹1,800 short. Somewhere in between, at about 14.9%, the present value of the inflows exactly equals the ₹10 lakh paid. That rate is the **internal rate of return (IRR)**: "internal" because it comes from the project's own cash flows, not from the market. Tapri accepts the upgrade because its IRR is above the 12% the money costs, the **cut-off rate**.

Two things can go wrong with that single percentage. First, IRR quietly assumes that every rupee the project throws off is reinvested at 14.9%, though Tapri can only count on 12%. The **modified internal rate of return (MIRR)** fixes this: it carries each inflow forward to year 4 at the cost of capital (₹16.71 lakh in all) and asks what single rate turns ₹10 lakh into that sum: 13.70%. Second, when two projects compete for the same site, a percentage can mislead: a small project with a high IRR can add fewer rupees to the owners' wealth than a large one with a lower IRR. Then NPV decides.

Think of it like

The project's own FD rate

A bank fixed deposit pays you interest at a stated rate. Imagine a deposit that, for ₹10 lakh placed today, paid back exactly ₹3, 4, 4 and 3 lakh over four years. The interest rate on that deposit, about 14.9%, is the project's IRR. If your money costs you 12%, a 14.9% deposit is worth taking. MIRR is the same deposit with an honest footnote: the interest you receive along the way is redeposited at the 12% you can actually get.

Key idea

IRR is the discount rate at which NPV is zero. Accept a project when its IRR is at least the cost of capital: for a single project this always agrees with NPV. When mutually exclusive projects differ in size, timing or life, IRR and NPV can rank them differently, and the NPV ranking is the one consistent with wealth maximisation.

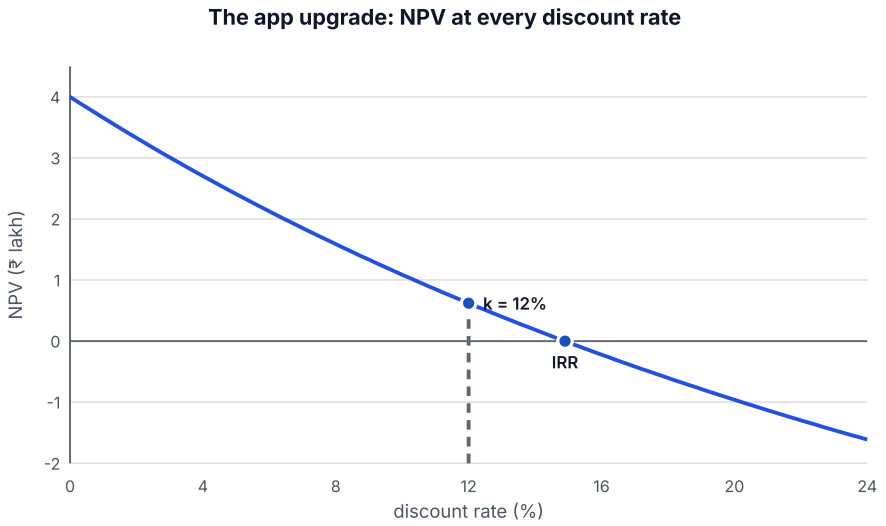
How it works

Definition

Internal rate of return

Internal rate of return for an investment proposal is the discount rate that equates the present value of the expected cash inflows with the initial cash outflow. (SM 7.32)

Like NPV, the method considers the time value of money, the initial investment and all cash flows. Unlike NPV, it does not start from the desired rate of return: it estimates the rate that makes the present value of the inflows equal to the investment, and then compares that IRR with a criterion rate of return, the organisation's desired rate for evaluating capital investments (SM 7.32–7.33).



The NPV profile: NPV falls as the discount rate rises. Where the blue curve crosses zero, at about 14.9%, is the IRR. Any cost of capital k to the left of it (such as 12%, grey dashed, NPV ₹0.62 lakh) gives a positive NPV.

Two ways to find the rate

The procedure depends on the pattern of the cash flows (SM 7.33–7.34).

Scenario 1: uniform cash inflows. Total initial investment = annual cash inflow × annuity discount factor (PVAF) for the life, so:

1. Compute A = initial investment ÷ annual cash inflow. This is the approximate payback period, also called the **fake payback period**.
2. Look for A in the PVAF table in the row for the project's life. If it is there exactly, that rate is the IRR; usually it falls between two rates.
3. Discount the cash flows at those two rates (NPV at each).
4. Interpolate between them with the formula below.

Scenario 2: uneven cash inflows. Trial and error:

1. Discount the cash flows at any random rate, say 10%, 15% or 20%.
2. If the NPV is negative, discount again at a lower rate to make it positive; if positive, at a higher rate to make it negative.
3. Interpolate between the rate with a positive NPV (the lower rate, LR) and the rate with a negative NPV (the higher rate, HR).

$$\text{IRR} = LR + \frac{\text{NPV at } LR}{\text{NPV at } LR - \text{NPV at } HR} \times (HR - LR)$$

Because NPV at HR is negative, the denominator is the **sum** of the two NPVs without their signs. For the app upgrade (three-decimal factors): NPV at 14% = +₹18,300, at 15% = −₹1,800, so IRR = 14% + 18,300 ÷ 20,100 × 1% = **14.91%** (the exact rate is 14.90%).

Interactive version: PassBook app, lesson IRR, MIRR and the NPV-IRR conflict

The acceptance rule

The IRR is compared with the required rate of return, the **cut-off rate** (SM 7.37–7.38):

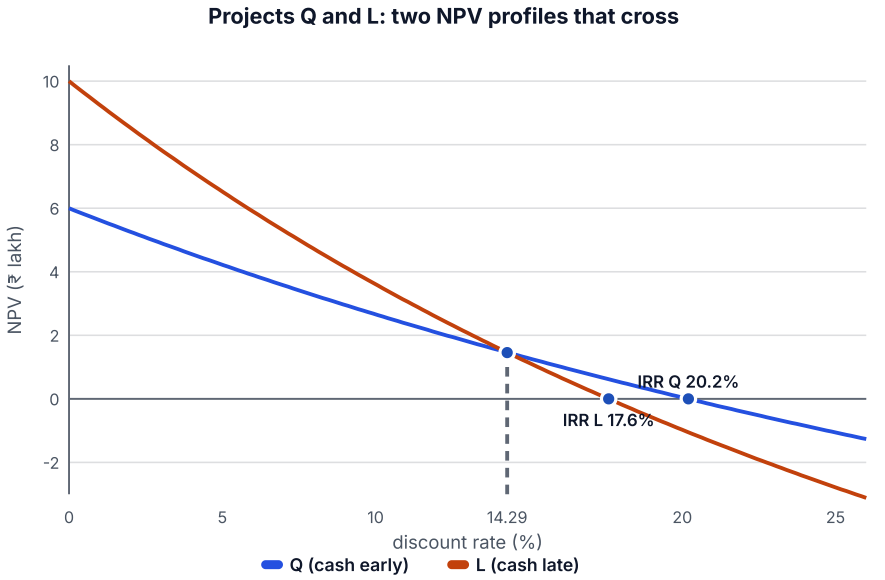
Condition	Decision
IRR ≥ cut-off rate or WACC	Accept the proposal
IRR < cut-off rate or WACC	Reject the proposal

At IRR = cut-off rate the firm is indifferent. For one project, IRR ≥ cost of capital and NPV ≥ 0 are the same test: the NPV profile is above zero exactly to the left of the IRR.

Mutually exclusive projects: where the percentage misleads

Projects are **mutually exclusive** when the selection of one precludes the selection of the others, for example a plot of land that can be used either for project S or for project L (SM 7.38). The trouble is that IRR is expressed in percentage and does not take into account the scale of investment or the quantum of money earned.

Tapri's Nagpur plot can take either a quick-service kiosk project Q or a premix warehouse L, each costing ₹20 lakh. Q returns ₹16, 8 and 2 lakh over three years; L returns ₹2, 10 and 18 lakh.



Blue: Q; orange: L. The dot on the grey dashed line is the crossover rate, 14.29%, where both NPVs are ₹1.46 lakh. Left of it L has the higher NPV; right of it Q does. Q's IRR is higher whatever the cost of capital. At a 10% cost of capital NPV picks L (₹3.61 lakh against ₹2.66 lakh) while IRR picks Q.

As long as the cost of capital is above the **crossover rate**, the rate at which the two NPVs are equal, NPV and IRR both pick the same project. Below it, NPV ranks one project first and IRR the other (SM 7.38). The Study Material's Example 8 shows the size version: project A (₹1,00,000 for ₹1,50,000 a year later) earns 50% and project B (₹5,00,000 for ₹6,25,000) only 25%, yet at 10% B's NPV is ₹68,180 against A's ₹36,360. Choosing A gives up ₹31,820 of shareholders' wealth (SM 7.39). And it is not safe to choose by IRR even when the larger project also has the higher IRR (Example 9, SM 7.39–7.40).

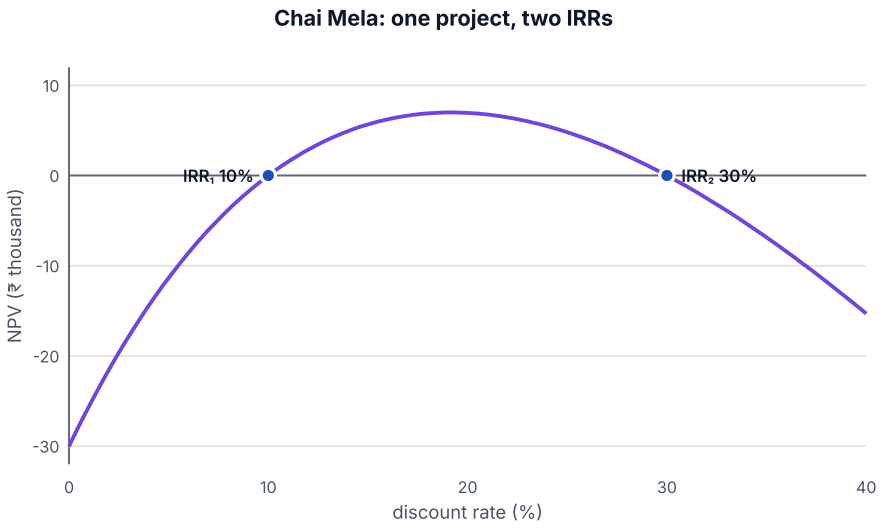
Why they disagree: the reinvestment assumption

The NPV technique assumes that all cash flows can be reinvested at the discount rate used for calculating the NPV, a logical assumption since NPV accepts every project that beats that rate. The IRR technique assumes that all cash flows are reinvested at the project's IRR. So IRR favours projects with heavy cash flows in the early years over projects with larger cash flows in the later years (SM 7.40–7.41): Q over L above.

Interactive version: PassBook app, lesson IRR, MIRR and the NPV-IRR conflict

Multiple IRRs

When project cash flows change signs during the life, for example an initial outflow, then inflows, then a major outflow, there may be more than one IRR. If the cost of capital is less than both IRRs a decision can be made easily; otherwise the IRR rule may mislead, because the project should be invested in only if the cost of capital lies **between IRR_1 and IRR_2** (SM 7.41).



Tapri pays ₹10 lakh for a stadium festival, collects ₹24 lakh from stall-holders in year 1 and pays ₹14.3 lakh to restore the ground in year 2. The signs change twice (−, +, −), so the violet NPV curve crosses zero at 10% and at 30%: the project adds value only for a cost of capital between the two.

What IRR does well, and where it fails

Advantages of IRR (SM 7.41–7.42):

- It makes use of the concept of time value of money.
- All the cash flows in the project are considered.
- It is easier to use: comparing it with the cost of capital gives an instantaneous understanding of desirability.
- It helps in achieving the objective of maximisation of shareholders' wealth.

Limitations of IRR (SM 7.42):

- The calculation is tedious if there is more than one cash outflow interspersed between the inflows; there can be multiple IRRs, whose interpretation is difficult.
- It creates a peculiar situation when two projects with different inflow/outflow patterns are compared.
- It assumes all future cash inflows are reinvested at a rate equal to the IRR, ignoring the firm's ability to reinvest in a portfolio of different rates.
- With mutually exclusive projects of considerably different cash outlays, a project with a larger fund commitment but lower IRR contributes more absolute NPV and increases shareholders' wealth more; decisions based only on IRR may then not be correct.

MIRR: the reinvestment problem, repaired

The **modified internal rate of return (MIRR)**, also called the **terminal value method**, addresses some of these deficiencies: it eliminates multiple IRR rates, addresses the reinvestment rate issue and produces results consistent with the NPV method (SM 7.43).

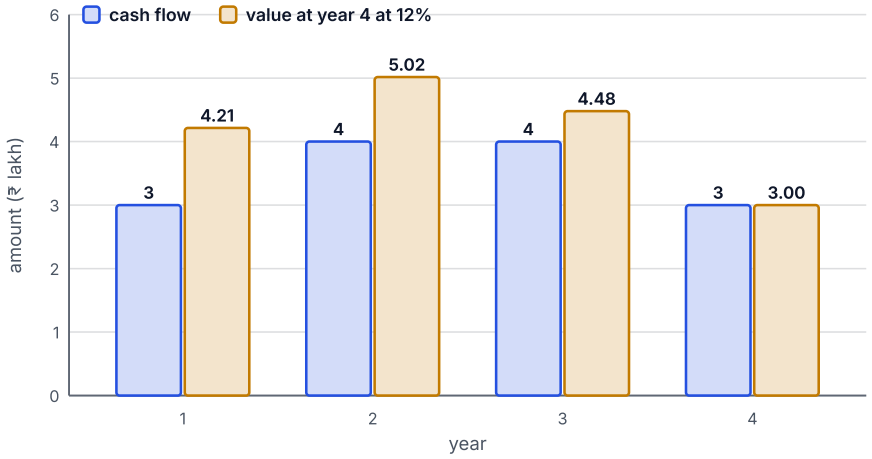
1 Bring every cash flow except the initial investment to the terminal (last) year at an appropriate rate, usually the cost of capital: an inflow of year t is multiplied by $(1 + k)^{n-t}$, so the last year's inflow by 1.

2 Add them: a single cash inflow in the terminal year, the **terminal value**.

3 The MIRR is the discount rate that equates the present value of this terminal inflow with the single outflow of year 0.

4 Decide as with IRR: accept if MIRR is larger than the required rate of return, reject if lower (SM 7.44).

The app upgrade: each inflow carried to year 4 at 12%



Blue: the cash flow; amber: its value at the end of year 4, reinvested at 12%. The earlier the rupee, the longer it grows. The amber bars add up to the terminal value, ₹16.71 lakh, against ₹10 lakh paid today: MIRR is the fourth root of 1.6712, less 1, which is 13.70%.

MIRR (13.70%) sits between the cost of capital (12%) and the IRR (14.90%): reinvesting at 12% instead of 14.9% pulls the return down.

Interactive version: PassBook app, lesson IRR, MIRR and the NPV-IRR conflict

NPV and IRR side by side

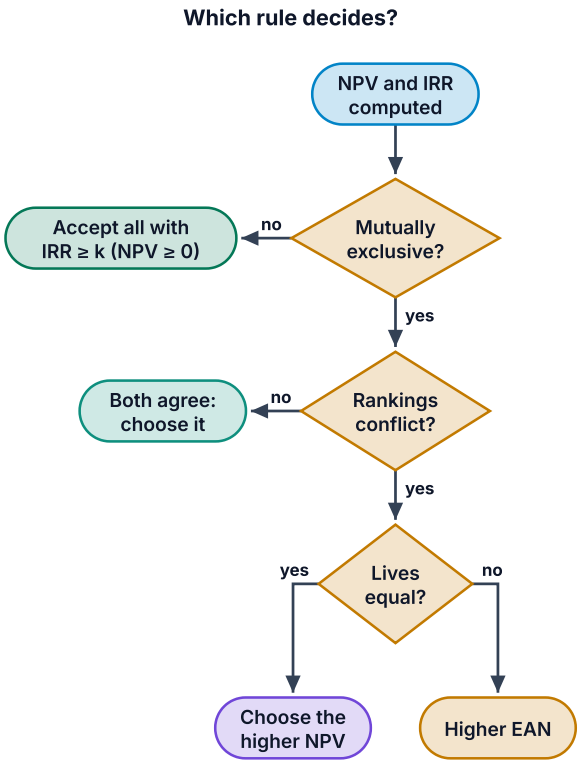
Similarities (SM 7.45): both are discounted cash flow methods that consider the time value of money, and both consider all cash flows over the expected useful life.

Where they reach different conclusions (SM 7.45–7.51), with the Study Material's own illustrations:

Scenario	Why they differ	ICAI's figures
Scale or size disparity	IRR is relative, NPV absolute	Ill. 10: NPV ₹25,050 vs ₹59,300; IRR 24.26% vs 21.48%
Time disparity in cash flows	Early cash flows favoured by IRR	Ill. 11: NPV ₹51,950 vs ₹53,350; IRR 24.87% vs 17.60%
Unequal lives	Short life, high IRR, less NPV	Ill. 12: NPV ₹1,69,750 vs ₹3,36,400; IRR 50% vs 43.07%

Which to follow. Between the two, the NPV rule generally gives consistent results in conformity with the wealth maximization principle, so on a conflict between mutually exclusive projects the Study Material's illustrations select by NPV (SM 7.39, 7.59). For independent projects with no capital rationing either method can be used: the same set of projects is accepted (SM 7.59). When lives differ, compare equivalent annualised NPVs or use a common life (equivalent annualised criterion, SM 7.55–7.56).

ICAI's summary table (SM 7.52) gives the IRR rule as: independent project, accept when IRR ≥ K and reject when IRR < K; mutually exclusive projects, the project with the maximum IRR. That is the rule when IRR is the method asked for. When a question asks you to choose after computing both, and they conflict, say so and follow NPV, giving the reason above.



k: the cost of capital or cut-off rate; EAN: equivalent annualised NPV. Green: independent projects, where IRR and NPV accept the same set; teal: no conflict; violet and amber: a conflict, settled by NPV, annualised when lives differ.

Formats and formulas

Reproduce these from memory.

∑ **Formula**

Internal rate of return

$$\sum_{t=1}^{n} \frac{C_t}{(1+r)^t} - I = 0$$

r is the IRR: the rate at which the present value of the inflows C_t equals the investment I (SM 7.32).

Σ Formula

Fake payback factor (uniform inflows)

$$A = \frac{\text{Total initial investment}}{\text{Annual (equal) cash inflow}}$$

Look up A in the PVAF table in the row for the project's life (SM 7.33).

Σ Formula

Interpolation

$$\text{IRR} = LR + \frac{\text{NPV at } LR}{\text{NPV at } LR - \text{NPV at } HR} \times (HR - LR)$$

or, with present values,

$$\text{IRR} = LR + \frac{\text{PV at } LR - CI}{\text{PV at } LR - \text{PV at } HR} \times (HR - LR)$$

LR = lower rate, HR = higher rate, CI = capital investment (SM 7.33).

Σ Formula

Modified internal rate of return

$$\text{MIRR} = \left(\frac{\text{TV}}{\text{Initial outflow}}\right)^{1/n} - 1$$

TV, the terminal value: each inflow compounded to year n at the cost of capital, $(1 + k)^{n-t}$. Alternatively, find outflow ÷ terminal value in the year-n row of the PV table (SM 7.45).

Definition

Modified internal rate of return

Under this method, all cash flows, apart from the initial investment, are brought to the terminal value using an appropriate discount rate (usually the Cost of Capital). This results in a single stream of cash inflow in the terminal year. The discount rate which equates the present value of the terminal cash inflow to the zero year outflow is called the MIRR. (SM 7.43)

IRR working (one table for each trial rate, as in SM 7.35–7.36)

Year	Cash flow (₹)	PV factor @ r	Present value (₹)
0	(xxx)	1.000	(xxx)
1 … n	xxx	x.xxx	xxx
NPV at r			**xxx**

Then the interpolation line, with LR, HR and both NPVs written out.

MIRR working (SM 7.44)

Year	Cash flow (₹)	Reinvestment factor @ k	Terminal value (₹)
1	xxx	$(1 + k)^{n-1}$	xxx
...			
n	xxx	1.000	xxx
Total terminal value			**xxx**

Decision rules

Technique	Independent project	Mutually exclusive projects
IRR	Accept if IRR ≥ k; reject if IRR < k	Maximum IRR (SM 7.52); on a conflict with NPV, the higher NPV
MIRR	Accept if MIRR > required rate; reject if lower	As IRR

Worked examples

All working uses ICAI's three-decimal PV factors, and rates are rounded to two decimals.

Worked example **The kulhad press and its scrap value**

Tapri plans an automatic kulhad press costing ₹12,60,000, with a life of 5 years and a salvage value of ₹1,00,000 at the end. It is depreciated on the straight-line method and will earn ₹4,40,000 a year before depreciation and tax. The tax rate is 25%, and Tapri's cut-off rate for automation projects is 17%.

Rate	PVAF, 5 years	PVF, year 5
15%	3.352	0.497
16%	3.274	0.476
17%	3.199	0.456
18%	3.127	0.437
19%	3.058	0.419

Required:

(i) Calculate the annual cash inflow after tax.

(ii) Use the fake payback factor to locate a starting rate, and state why it cannot give the IRR directly here.

(iii) Compute the internal rate of return.

(iv) Advise Tapri.

THE WORKING

(i) Cash inflow per annum

Particulars	Amount (₹)
Earnings before depreciation and tax	4,40,000
Less: Depreciation (₹12,60,000 – ₹1,00,000) ÷ 5	(2,32,000)
Earnings before tax	2,08,000
Less: Tax @ 25%	(52,000)
Profit after tax	1,56,000
Add: Depreciation	2,32,000
Cash inflow	**3,88,000**

(ii) Starting rate. Fake payback factor = ₹12,60,000 ÷ ₹3,88,000 = 3.247, which lies between 3.274 (16%) and 3.199 (17%). This shortcut works only when every inflow is equal. Here year 5 also brings the ₹1,00,000 salvage, so the cash flows are not an annuity: the true IRR is higher than the rate read from the table, and the trial must include the salvage.

(iii) IRR. Discount the inflows and the salvage. At 17% the NPV is still positive, so try 18%:

Particulars	17%	18%
PV of annual inflows (₹3,88,000 × PVAF)	12,41,212	12,13,276
PV of salvage (₹1,00,000 × PVF)	45,600	43,700
Less: Initial outlay	(12,60,000)	(12,60,000)
NPV (₹)	**26,812**	**(3,024)**

$$\text{IRR} = 17\% + \frac{26{,}812}{26{,}812 + 3{,}024} \times (18\% - 17\%) \qquad = 17\% + 0.90\% = 17.90\%$$

(iv) Advice. The IRR of 17.90% exceeds the cut-off rate of 17%: **buy the press** (its NPV at 17% is ₹26,812, positive).

Leaving out the salvage gives NPVs of ₹10,312 at 16% and −₹18,788 at 17%, an IRR of 16 + 10,312 ÷ 29,100 = **16.35%**, below the cut-off: a profitable press rejected for the sake of one forgotten year-5 inflow.

Worked example **The dealer's 14% and the buy-back clause**

A dealer offers Tapri a fleet of cold-chain vans for ₹12,93,000 and promises "a 14% return", provided Tapri sells the vans back to him at the end of year 4. Ritu's route plan gives the inflows below; the buy-back price is still to be agreed. Tapri's cost of capital is 10%.

Particulars	Amount
Price of the fleet, paid now	₹12,93,000
Cash inflow, year 1	₹3,00,000
Cash inflow, year 2	₹4,00,000
Cash inflows, years 3 and 4	₹5,00,000 a year
Buy-back at the end of year 4	to be agreed

Year	PVIF @ 14%	PVIF @ 10%	FVIF @ 10%	FVIF @ 14%
1	0.877	0.909	1.100	1.140
2	0.769	0.826	1.210	1.300
3	0.675	0.751	1.331	1.482
4	0.592	0.683	1.464	1.689

Required:

(i) Calculate the buy-back price at which the dealer's 14% is the internal rate of return.

(ii) At that price, calculate the net present value and the profitability index at the cost of capital.

(iii) Calculate the MIRR reinvesting at the cost of capital, and Dev's MIRR reinvesting at 14%. Comment on what the two show.

(iv) The dealer then offers ₹50,000 off the price if Tapri gives up the buy-back and scraps the vans for nothing. Advise.

THE WORKING

(i) Buy-back price. At the IRR the NPV is zero: the price equals the present value at 14% of the inflows and the buy-back.

Year	Cash inflow (₹)	PVF @ 14%	PV (₹)
1	3,00,000	0.877	2,63,100
2	4,00,000	0.769	3,07,600
3	5,00,000	0.675	3,37,500
4	5,00,000	0.592	2,96,000
	PV of route inflows		**12,04,200**

PV the buy-back must supply = ₹12,93,000 − ₹12,04,200 = ₹88,800; buy-back price = ₹88,800 ÷ 0.592 = **₹1,50,000**. Year 4's cash inflow is therefore ₹6,50,000.

(ii) NPV and PI at 10%

Year	Cash inflow (₹)	PVF @ 10%	PV (₹)
1	3,00,000	0.909	2,72,700
2	4,00,000	0.826	3,30,400
3	5,00,000	0.751	3,75,500
4	6,50,000	0.683	4,43,950
	PV of inflows		**14,22,550**

NPV = ₹14,22,550 − ₹12,93,000 = **₹1,29,550**. Profitability index = ₹14,22,550 ÷ ₹12,93,000 = **1.10**.

(iii) MIRR. Each inflow is reinvested to the end of year 4; year 4's inflow takes a factor of 1.

Year	Cash inflow (₹)	Factor @ 10%	Terminal value (₹)
1	3,00,000	1.331	3,99,300
2	4,00,000	1.210	4,84,000
3	5,00,000	1.100	5,50,000
4	6,50,000	1.000	6,50,000
Total terminal value			**20,83,300**

$$\text{MIRR} = \left(\frac{20{,}83{,}300}{12{,}93{,}000}\right)^{1/4} - 1 \qquad = (1.6112)^{1/4} - 1 = 12.66\%$$

Or: ₹12,93,000 ÷ ₹20,83,300 = 0.621, which in the year-4 row of the PV table lies between 12% (0.636) and 13% (0.613): about 12.7%.

Reinvesting at 14%: terminal value = 3,00,000 × 1.482 + 4,00,000 × 1.300 + 5,00,000 × 1.140 + 6,50,000 × 1.000 = ₹21,84,600.

$$\text{MIRR at } 14\% = \left(\frac{21{,}84{,}600}{12{,}93{,}000}\right)^{1/4} - 1 \qquad = 14.01\%$$

Comment. Reinvesting at the IRR gives back the IRR itself (14%; the 0.01% is factor rounding). That is the reinvestment assumption hidden inside IRR, made visible. Tapri can only count on reinvesting at its 10% cost of capital, so the realistic return on the fleet is the MIRR of 12.66%: above 10%, consistent with the positive NPV, but not the 14% the dealer advertises.

(iv) The ₹50,000 discount. Incremental effect at 10%: ₹50,000 saved now, less the buy-back given up, ₹1,50,000 × 0.683 = ₹1,02,450. The offer lowers the NPV by **₹52,450**: decline it and keep the buy-back clause.

Worked example

One plot, two plans: the kiosks or the café

Tapri owns one vacant plot in Bhopal. It can build a cluster of chai kiosks (A) or a kulhad café (B), not both. The cost of capital is 12%.

Year	Project A (₹ lakh)	Project B (₹ lakh)
0	(10)	(30)
1	7	6
2	6	18
3	3	22

PV factor	Year 1	Year 2	Year 3
12%	0.893	0.797	0.712
15%	0.870	0.756	0.658
20%	0.833	0.694	0.579
25%	0.800	0.640	0.512
30%	0.769	0.592	0.455
35%	0.741	0.549	0.406

Required:

(i) Calculate the NPV of each project at 12%.

(ii) Calculate the IRR of each project.

(iii) Which project should Tapri choose? Give reasons.

(iv) Calculate the rate at which both projects have the same NPV, and state over what range of cost of capital NPV and IRR would rank them the same way.

THE WORKING

(i) NPV at 12% (₹ lakh)

Year	PVF @ 12%	PV of A	PV of B
1	0.893	6.251	5.358
2	0.797	4.782	14.346
3	0.712	2.136	15.664
	PV of inflows	**13.169**	**35.368**
	Less: Outlay	10.000	30.000
	NPV	**3.169**	**5.368**

(ii) IRR. A is still positive at 30% (PV 10.300, NPV +0.300) and negative at 35% (PV 9.699, NPV −0.301):

$$\text{IRR}_A = 30\% + \frac{0.300}{0.300 + 0.301} \times 5\% = 32.50\%$$

B is positive at 20% (PV 30.228, NPV +0.228) and negative at 25% (PV 27.584, NPV −2.416):

$$\text{IRR}_B = 20\% + \frac{0.228}{0.228 + 2.416} \times 5\% = 20.43\%$$

(iii) Choice

	Project A	Project B
NPV @ 12% (₹)	3,16,900	5,36,800
IRR	32.50%	20.43%
Rank by NPV	2	1
Rank by IRR	1	2

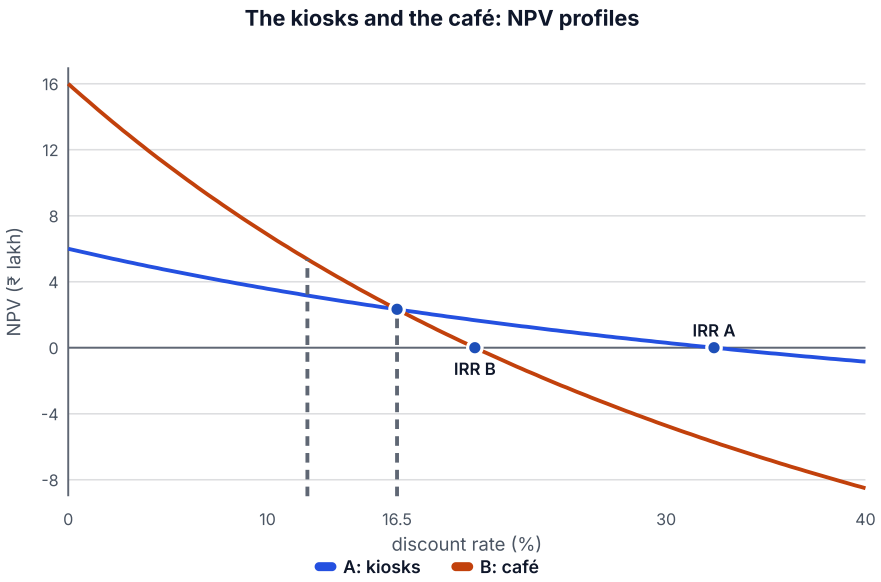
The rankings conflict because the projects differ in size (₹10 lakh against ₹30 lakh) and in timing (A's cash comes early). The projects are mutually exclusive, so the decision follows NPV, which is consistent with wealth maximisation: **choose the café (B)**. It adds ₹2,19,900 more to shareholders' wealth. A earns a higher percentage, but on a third of the money; the extra ₹20 lakh that B needs earns more than the 12% it costs (16.5%, part (iv)).

(iv) Crossover rate. It is the IRR of the difference in cash flows, B − A: (20), (1), 12, 19.

At 15%: −20 − 0.870 + 9.072 + 12.502 = +0.704; at 20%: −20 − 0.833 + 8.328 + 11.001 = −1.504.

$$\text{Crossover rate} = 15\% + \frac{0.704}{0.704 + 1.504} \times 5\% = 16.59\%$$

(The exact crossover rate is 16.50%.)



Blue: A; orange: B. Grey dashed lines: the 12% cost of capital and the 16.5% crossover rate (the dot, where both NPVs are ₹2.33 lakh). Left of 16.5% B has the higher NPV, so NPV and IRR disagree; between 16.5% and B's IRR of 20.4% both prefer A; above 20.4% only A is acceptable.

Below 16.5%, NPV ranks B first while IRR ranks A first. At a cost of capital above 16.5% (up to 20.43%), both rank A first; above 20.43% B is rejected outright. At Tapri's 12% the conflict is real, and NPV settles it.

Exam angle

In the exam

Where IRR and MIRR appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, IRR or MIRR came up in 10 new-scheme documents and 2 old-scheme ones:

- **Descriptive**: initial investment from an IRR of 12%, then NPV, PI, discounted payback and MIRR (May 2025 Q3(a); 2 of the 6 marks for MIRR). NPV, IRR (18.89%, between 16% and 20%) and desirability factor with WDV depreciation and a capital loss (RTP January 2025 Q9). IRR of a replacement decision (RTP May 2024 Q5). DPP, NPV and IRR of two mutually exclusive machines (RTP September 2026 Q7).
- **MCQs**: IRR from NPVs of ₹50.71 lakh at 16% and (₹26.84 lakh) at 20%, 18.62%, then accept at a cost of capital of 18.50% (May 2026 MCQ 8); cost of a project from its IRR, then payback (MTP Series I for January 2025, MCQ 7, 2 marks); IRR in a start-up case with a terminal value (RTP September 2025, case 1(v)).
- **Theory, 2 to 4 marks**: define MIRR (MTP Series I for January 2025 Q4(c), 2 marks); short note on IRR (MTP Series II for May 2025 Q3(b), 2 marks); acceptance rule of IRR (RTP May 2026 Q10(c)); limitations of IRR (old scheme, May 2022 Q6(a), 4 marks); NPV versus IRR (old scheme, MTP Series II for November 2022 Q6(a)).
- **A reverse MIRR**: MIRR 10.9% and a terminal value of ₹11,43,589 given, investment backed out as ₹5,00,000 (RTP January 2026 Q9).

In the exam

The traps ICAI used

- Reverse questions: "IRR 12%" means the investment is the PV of the inflows at 12% (May 2025 Q3(a); MTP Series I for January 2025 MCQ 7). Include salvage in the last year.
- MIRR with a terminal value: the last year's inflow takes factor 1. May 2025 Q3(a) printed FVIFs up to year 4 although only years 1–3 are needed.
- RTP September 2026 Q7 ranked Machine I first on discounted payback and IRR and Machine II first on NPV (₹1,18,074 against ₹86,909), and recommended Machine I on the majority of rankings. The Study Material's rule for mutually exclusive projects is the higher NPV, and with lives of 5 and 6 years the equivalent annualised NPV (SM 7.56): both point to Machine II. If you meet such a question, state the rule you follow.

Trick

The two-line IRR sanity check

NPV positive at a rate means the IRR is above that rate; NPV negative means below. So the IRR must lie between LR and HR, and nearer the rate whose NPV is smaller in size. Condition: conventional cash flows (one sign change). If your answer falls outside the bracket, the signs in the denominator are wrong.

Answer frame

"Write a short note on IRR" (MTP Series II for May 2025, 2 marks; ICAI's points): the rate at which discounted cash inflows equal discounted cash outflows, the rate that discounts the cash flows to zero; it can be stated as discounted inflows ÷ discounted outflows = 1; found by trial and error, since the rate is unknown while the cash flows are known; compared with the cut-off rate: accept if more, otherwise reject.

"Identify the limitations of IRR" (old scheme May 2022, 4 marks): the four points of SM 7.42 in ICAI's order (tedious with interspersed outflows and multiple IRRs; peculiar results with different inflow/outflow patterns; reinvestment at the IRR; mutually exclusive projects with different outlays).

"Describe MIRR" (SM theory question 6; MTP Series I for January 2025, 2 marks): what it addresses (multiple IRRs, the reinvestment rate, consistency with NPV); the terminal value method in ICAI's words; the single outflow and single inflow; the decision rule.

"Distinguish between NPV and IRR"

Point	NPV	IRR
Result	Absolute amount (₹)	Percentage
Discount rate	Cost of capital, given	Found; then compared with the cost of capital
Reinvestment assumed at	The discount rate	The project's IRR
Multiple results	One NPV at a given rate	Several IRRs possible when signs change
Mutually exclusive projects	Highest NPV; consistent with wealth maximisation	May conflict with NPV (size, timing, lives)

Common mistakes

⚠ **Common mistake**

Hall of Shame: the minus sign that sank the IRR

NPV is ₹2,400 at 18% and −₹1,600 at 20%. Bunty's classmate writes 18% + 2,400 ÷ (2,400 − 1,600) × 2% = 24%, outside the bracket. **Fix:** the denominator is NPV at LR − NPV at HR = 2,400 − (−1,600) = 4,000, so IRR = 18% + 1.2% = 19.2%.

⚠ **Common mistake**

Choosing the mutually exclusive project with the higher IRR

The kiosks' 32.50% beats the café's 20.43%, so the hurried answer picks the kiosks and gives up ₹2,19,900 of NPV. **Fix:** with mutually exclusive projects that conflict, choose the higher NPV and give the reason: NPV is consistent with wealth maximisation; IRR ignores scale and assumes reinvestment at the IRR.

⚠ **Common mistake**

Compounding the last year's inflow, or the outlay

In MIRR, the year-n inflow is multiplied by 1, not by (1 + k); the year-0 outlay is not compounded at all, it is what the terminal value is compared with. **Fix:** an inflow of year t takes $(1 + k)^{n-t}$; check that the year-1 inflow has the largest factor.

⚠ **Common mistake**

Reading the fake payback factor as the answer

For uneven cash flows, or equal inflows plus a salvage value, investment ÷ inflow only suggests where to start. In the kulhad press, reading 3.247 off the PVAF row gives 16.35% and a wrong rejection; with the salvage the IRR is 17.90%. **Fix:** discount the actual cash flows at two rates and interpolate.

Practice

Check yourself

1 A project's NPV is ₹45,000 at 14% and −₹15,000 at 16%. What is its IRR by interpolation, in %?

Answer: _____

2 Tapri invests ₹5,00,000 today. Its inflows, reinvested at the 10% cost of capital, are worth ₹8,81,171 at the end of year 5. What is the MIRR, in % (two decimals)?

Answer: _____

3 Case scenario. Tapri's samosa factory has room for one new fryer. Fryer X costs ₹15 lakh, has an NPV of ₹4.2 lakh and an IRR of 26%; fryer Y costs ₹40 lakh, has an NPV of ₹6.8 lakh and an IRR of 19%. Both last five years; the cost of capital is 11%. Kesar asks Dev which fryer to buy. Which of the following is correct?

- **A** Fryer X, because its IRR is higher
- **B** Fryer Y, because between mutually exclusive projects the higher NPV is consistent with wealth maximisation
- **C** Both fryers, because both IRRs exceed 11%
- **D** Neither, until the crossover rate equals the cost of capital

4 Which pairing of a technique with the reinvestment rate it implicitly assumes is correct?

- **A** IRR: the cost of capital
- **B** IRR: the project's own IRR; NPV: the discount rate used
- **C** NPV: the project's IRR; MIRR: the IRR
- **D** MIRR: the project's IRR

5 Which statements about multiple IRRs are correct? Select all that apply.

- **A** They can arise when cash flows change sign more than once, such as an outflow at the end of the life
- **B** With two IRRs, the project adds value only if the cost of capital lies between them
- **C** They arise whenever cash inflows are uneven
- **D** MIRR eliminates the problem of multiple IRR rates

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- IRR: the discount rate that equates the PV of the expected cash inflows with the initial cash outflow (NPV = 0).

- Uniform inflows: fake payback = investment ÷ annual inflow, read in the PVAF row for the life, then interpolate. Uneven inflows: trial and error at two rates, then interpolate.
- IRR = LR + NPV at LR ÷ (NPV at LR − NPV at HR) × (HR − LR); the IRR lies between LR and HR.
- Accept if IRR ≥ cut-off rate (WACC), reject if lower; for one project this always agrees with NPV ≥ 0.
- NPV and IRR can conflict for mutually exclusive projects: size disparity, time disparity, unequal lives. Follow NPV (wealth maximisation); for unequal lives, the equivalent annualised NPV.
- Reinvestment: NPV at the discount rate, IRR at the IRR; sign changes can give multiple IRRs.
- MIRR (terminal value method): compound inflows to year n at the cost of capital; MIRR = $(\text{TV} \div \text{outlay})^{1/n} - 1$; accept if MIRR > required rate.
- State your factors: once-rounded annuity factors and summed yearly factors differ in the last digit. (A project's own FD rate is impressive only if you can redeposit at it.)

Answer key · Check yourself

1 **15.5**

IRR = 14% + 45,000 ÷ (45,000 + 15,000) × 2% = 14% + 1.5% = 15.5%.

2 **11.95 to 12.05**

8,81,171 ÷ 5,00,000 = 1.7623, whose fifth root is 1.1200: MIRR = 12.00%. It exceeds the 10% cost of capital: accept.

3 **B Fryer Y, because between mutually exclusive projects the higher NPV is consistent with wealth maximisation**

The fryers are mutually exclusive and the rankings conflict (scale disparity); NPV decides. Both would be accepted only if they were independent.

4 **B IRR: the project's own IRR; NPV: the discount rate used**

NPV assumes reinvestment at the discount rate, IRR at the IRR (which is why it favours early cash flows), and MIRR at an appropriate rate, usually the cost of capital.

5 **A They can arise when cash flows change sign more than once, such as an outflow at the end of the life; B With two IRRs, the project adds value only if the cost of capital lies between them; D MIRR eliminates the problem of multiple IRR rates**

Sign reversals, not unevenness, cause multiple IRRs (SM 7.41); MIRR produces a single rate (SM 7.43).

7.5 Capital rationing

When every project has a positive NPV but the budget cannot fund them all: rank divisible projects by NPV per rupee, and test combinations of indivisible ones for the highest total NPV.

In plain words

Five good ideas, one cheque book

Tapri's capital expenditure committee meets every March. This year five proposals have a positive NPV at the 12% cost of capital, together needing ₹165 lakh. The board has capped next year's capital spending at ₹100 lakh.

> **Kesar:** All five make money. Why can't we do all five?
>
> **Dev:** Because the board gave us ₹100 lakh, not ₹165 lakh. So the question changes: not "which projects are good?" but "which ₹100 lakh of projects add the most NPV?"
>
> **Ritu:** Then start with the biggest NPV, surely. The Pune café adds ₹17.5 lakh.
>
> **Dev:** It also eats half the budget. Per rupee, it is the second-worst on the list.

When a firm cannot take up every positive-NPV project because funds are limited, it faces **capital rationing**. The aim stays the same, maximise shareholders' wealth, but now within a budget: choose the package of projects with the **highest total NPV** that fits.

How you build that package depends on one question. If a project can be taken up in part (half a fleet of vans, a third of the solar panels), it is **divisible**: rank the projects by NPV per rupee invested and fill the budget from the top, taking a part of the last one. If a project is all or nothing (a café is either built or not), it is **indivisible**: try the combinations that fit and pick the one with the highest total NPV. The two answers can differ: Tapri's best whole-project package leaves out the third-ranked project.

Think of it like

The airline baggage limit

You may check in 20 kg. If you are packing loose kaju katli, pack the sweets with the most value per kilo first and fill the last gap with part of a box. If you are packing whole items, a pressure cooker, a laptop and a carton of books, the item with the best value per kilo can leave an awkward gap; you have to try the sets that fit and take the most valuable set.

Key idea

Under capital rationing, maximise the total NPV the budget can buy. Divisible projects: rank by NPV per rupee (the profitability index) and fill the budget, the last project in part. Indivisible projects: examine the combinations within the budget and select the one with the highest total NPV, even if it skips a higher-ranked project.

How it works

Definition

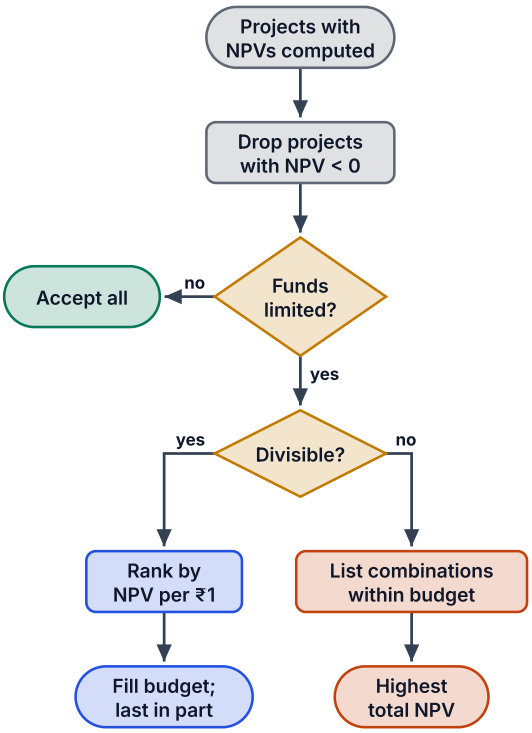
Capital rationing

Capital rationing refers to a situation where funds are restricted and the management has to choose from amongst available alternative investments. (SM 7.73, MCQ 14; answer SM 7.86)

If a project has a positive NPV, it should be accepted with an objective of maximisation of wealth of shareholders. But resource (capital) constraints may force a firm to select some projects among several, all having positive NPVs. Two scenarios decide the method (SM 7.53):

1. **Projects are independent of each other and divisible**: the NPV rule is modified, and projects are ranked on the basis of the **'NPV per rupee of capital'** method.
2. **Projects are not divisible**: projects are ranked on the basis of **absolute NPV** and mixed up to the point the available resources are exhausted; the combinations that fit are compared.

Capital rationing: which method?



Blue: divisible projects, ranked by NPV per rupee; orange: indivisible projects, compared as whole combinations. In both, a project with a negative NPV never enters, even to use up spare funds.

Divisible projects: the most NPV per rupee first

1. Compute each project's NPV at the cost of capital and drop any with a negative NPV.
2. Compute **NPV per ₹1 invested** = NPV ÷ investment, and rank the projects on it.

3 Take projects in that order until the next one no longer fits; take the fraction of it that the remaining funds allow, with the same fraction of its NPV.

Tapri's five proposals are the app upgrade (A), cold-chain vans (V), a kulhad press line (K), the Pune café (C) and a solar roof (S). Budget ₹100 lakh; amounts in ₹ lakh, NPVs at 12%:

Project	Investment	NPV	NPV per ₹1	Rank
A	20	10.0	0.50	1
V	30	13.5	0.45	2
K	40	16.0	0.40	3
C	50	17.5	0.35	4
S	25	7.5	0.30	5

A, V and K use ₹90 lakh and add ₹39.5 lakh of NPV. The last ₹10 lakh buys one-fifth of C, worth 17.5 × 10 ÷ 50 = ₹3.5 lakh. **Total NPV: ₹43.0 lakh**; S is not taken up.

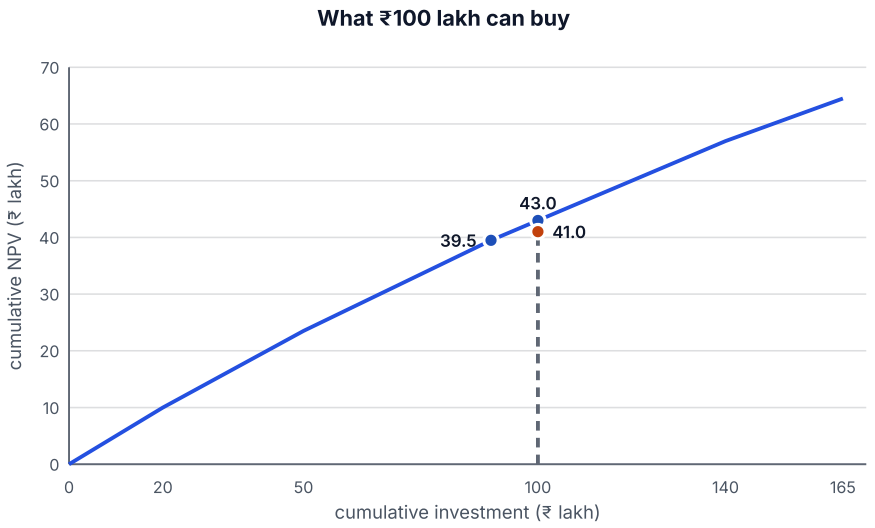
Ranking by NPV per rupee is ranking by the profitability index (PI = 1 + NPV ÷ investment when the whole outlay is made at the start), so both give the same order. That is why ICAI calls PI a better evaluation technique than NPV in a situation of capital rationing (RTP January 2025 Q11(c)).

Indivisible projects: whole combinations

A café cannot be one-fifth built. With whole projects only, the order above suggests A + V + K (₹90 lakh, NPV ₹39.5 lakh), leaving ₹10 lakh idle. Other combinations that fit:

Combination	Investment (₹ lakh)	Total NPV (₹ lakh)
A + V + K	90	39.5
A + V + C	**100**	**41.0**
K + V + S	95	37.0
A + C + S	95	35.0
K + C	90	33.5

A + V + C wins: it drops K, ranked third, for C, ranked fourth, because C uses the whole budget. The Study Material's Illustration 13 makes the same point: E + B + D (NPV ₹49.2 thousand) beats E + B + C (₹48.1 thousand) "despite D ranking lower than C" (SM 7.54).



The blue line adds the projects in order A, V, K, C, S; each segment's slope is that project's NPV per rupee, so the line gets flatter. The grey dashed line is the ₹100 lakh budget. Divisible projects reach ₹43.0 lakh, on the line at the budget; the best whole-project package, A + V + C (orange dot), reaches ₹41.0 lakh, more than the ₹39.5 lakh of A + V + K (the blue dot at ₹90 lakh).

Where the profitability index falls short

PI ranks divisible projects correctly. The Study Material lists its limits (SM 7.32): it **fails as a guide in resolving capital rationing where projects are indivisible**; once a single large project with high NPV is selected, the possibility of accepting several small projects which together may have higher NPV is excluded; and a project with a lower PI may generate cash that lets another project start a year or two later, for a higher total NPV. So the PI approach cannot be used indiscriminately: all other alternatives have to be worked out.

Without rationing, none of this arises: for independent projects either IRR or NPV can be used, since the same set of projects is accepted by either method (SM 7.59). For mutually exclusive projects, see the NPV-IRR conflict.

Formats and formulas

Reproduce these from memory.

∑ **Formula**

NPV per rupee of investment

$$\text{NPV per ₹1} = \frac{\text{NPV}}{\text{Investment}} \qquad \text{PI} = 1 + \frac{\text{NPV}}{\text{Investment}}$$

The PI form holds when the whole outlay is made at year 0; both give the same ranking (SM 7.53–7.54).

∑ Formula

NPV of a part project

$$\text{NPV of the part taken} = \text{NPV} \times \frac{\text{Funds remaining}}{\text{Investment}}$$

Step 1. Ranking (SM 7.54)

Project	Investment (₹)	NPV (₹)	NPV per ₹1	Rank
A	xxx	xxx	x.xx	n
...				

Step 2 (divisible). Building up the programme

Project (in rank order)	Investment (₹)	NPV (₹)
Rank 1	xxx	xxx
...		
Last: part (fraction)	xxx	xxx
Total	**Budget**	**xxx**

Step 2 (indivisible). Combinations within the limit

Combination	Investment (₹)	Total NPV (₹)
A + B + ...	xxx	xxx
Best		**xxx**

Worked examples

Worked example **Indore Masala Mills: one shed, two ideas**

Indore Masala Mills Pvt Ltd, one of Tapri's spice suppliers, can spend ₹60 lakh this year. Its five proposals all have positive NPVs at its cost of capital. A and B are two alternative uses of the same empty shed, so at most one of them can be undertaken; the others are independent.

Project	Investment (₹ lakh)	NPV (₹ lakh)
A: cold-grind line (in the shed)	25	9.0
B: export packing hall (in the shed)	15	6.0
C: warehouse	20	6.4
D: solar roof	10	2.5
E: delivery vans	30	8.7

Required:

(i) Optimise the package of projects within the capital spending limit if the projects are divisible (A or B may be taken in part, but not both).

(ii) Optimise the package if the projects are not divisible.

(iii) A bank offers ₹15 lakh more for a one-time arrangement fee of ₹5 lakh. Advise under (i) and under (ii).

THE WORKING

(i) Divisible projects. Computation of NPV per ₹1 of investment and ranking:

Project	Investment (₹ lakh)	NPV (₹ lakh)	NPV per ₹1	Rank
A	25	9.0	0.36	2
B	15	6.0	0.40	1
C	20	6.4	0.32	3
D	10	2.5	0.25	5
E	30	8.7	0.29	4

A and B cannot both enter, so build the programme twice, once with each:

Programme with B (no A)	Investment (₹ lakh)	NPV (₹ lakh)
B	15	6.00
C	20	6.40
E (25/30)	25	7.25
Total	**60**	**19.65**

Programme with A (no B)	Investment (₹ lakh)	NPV (₹ lakh)
A	25	9.00
C	20	6.40
E (15/30, one-half)	15	4.35
Total	**60**	**19.75**

Undertake **A, C and one-half of E**; reject B and D. Total NPV ₹19.75 lakh. B has the best NPV per rupee, but A uses ₹10 lakh more than B and adds ₹3.0 lakh more NPV: 0.30 per rupee on those ₹10 lakh, which beats the 0.29 they would earn in E.

(ii) Indivisible projects. Combinations within ₹60 lakh, never A with B:

Combination	Investment (₹ lakh)	Total NPV (₹ lakh)
A + C + D	**55**	**17.9**
A + E	55	17.7
C + D + E	60	17.6
B + D + E	55	17.2
A + C	45	15.4
C + E	50	15.1

Undertake **A, C and D** (NPV ₹17.9 lakh) and reject B and E. C + D + E uses the whole ₹60 lakh but adds less: the test is total NPV, not an empty purse.

(iii) The extra ₹15 lakh (budget ₹75 lakh)

Particulars	Divisible (₹ lakh)	Indivisible (₹ lakh)
Best package at ₹75 lakh	A + C + E: 24.10	A + C + E: 24.10
Best package at ₹60 lakh	19.75	17.90
Extra NPV from ₹15 lakh	4.35	6.20
Less: Arrangement fee	(5.00)	(5.00)
Net gain / (loss)	**(0.65)**	**1.20**

With divisible projects the extra ₹15 lakh only completes E at 0.29 per rupee (₹4.35 lakh), less than the fee: **decline**. With indivisible projects it lets the mill swap D for the whole of E, adding ₹6.2 lakh: **accept**. (With B instead of A, ₹75 lakh buys B + C + D + E, NPV ₹23.6 lakh, which is lower.) The same ₹15 lakh is worth more when projects come only whole.

Worked example **Tapri's capex budget: NPVs first, then the cut**

Tapri's capital spending for 2026-27 is limited to ₹120 lakh. The cost of capital is 12%. Five independent proposals (cash inflows after tax, received evenly at the end of each year, no salvage):

Proposal	Outlay (₹ lakh)	Annual inflow (₹ lakh)	Life (years)
P: samosa line	50	14	6
Q: app features	25	13	3
R: airport kiosks	70	21	4
S: cold-chain vans	45	17	4
T: premix packing	60	29	3

PVAF at 12%: 3 years 2.402; 4 years 3.037; 6 years 4.111.

Required:

(i) Calculate the NPV and profitability index of each proposal.

(ii) Which proposals should Tapri undertake if they are divisible?

(iii) Which, if they are indivisible?

(iv) Kesar suggests using any money left over on the airport kiosks. Comment.

THE WORKING

(i) NPV and PI (₹ lakh)

Proposal	PV of inflows	NPV	PI	Rank by PI
P	14 × 4.111 = 57.554	7.554	1.151	3
Q	13 × 2.402 = 31.226	6.226	1.249	1
R	21 × 3.037 = 63.777	(6.223)	0.911	rejected
S	17 × 3.037 = 51.629	6.629	1.147	4
T	29 × 2.402 = 69.658	9.658	1.161	2

R has a negative NPV (PI below 1) and is rejected whatever the budget.

(ii) Divisible proposals

Proposal	Outlay (₹ lakh)	NPV (₹ lakh)
Q	25	6.226
T	60	9.658
P (35/50 = 70%)	35	5.288
Total	**120**	**21.172**

Undertake Q, T and 70% of P; S is not taken up. Total NPV **₹21.17 lakh**.

(iii) Indivisible proposals. Combinations within ₹120 lakh:

Combination	Outlay (₹ lakh)	Total NPV (₹ lakh)
P + Q + S	**120**	**20.409**
P + T	110	17.212
S + T	105	16.287
Q + T	85	15.884
P + S	95	14.183

Undertake **P, Q and S** (NPV ₹20.41 lakh). The package leaves out T, which has the highest NPV of all and the second-highest PI: taking T (₹60 lakh) leaves room for only one of the others. Taking projects in PI order (Q, then T) would have stopped at ₹15.88 lakh, and taking them in NPV order (T, then P) at ₹17.21 lakh.

(iv) Comment. Funds left idle are not a reason to accept a negative-NPV project. The airport kiosks earn less than the 12% cost of capital (NPV −₹6.22 lakh), so they would reduce shareholders' wealth; in (iii) the chosen package uses the whole ₹120 lakh in any case.

Exam angle

In the exam

Where capital rationing appeared

In the ICAI documents read for this lesson, capital rationing is rare:

- **New scheme (May 2024 to September 2026)**: no past paper, RTP or MTP among the 56 Paper 6 documents searched asks a capital rationing numerical. The topic appears once, in RTP January 2025 Q11(c): PI and NPV can conflict for mutually exclusive projects, and PI "is a better evaluation technique than NPV in a situation of capital rationing".
- **Old scheme**: November 2019 Q1(d) (5 marks) gave ₹1,00,000 and four projects with their investments and NPVs, and asked for the package "within the capital spending limit" if the projects are (i) divisible and (ii) not divisible. The suggested answer ranked by NPV per Re. 1 (C 0.50, E 0.48, D 0.35, F 0.30), took C, E and one-tenth of D (₹47,500), and for indivisible projects compared C + E (₹44,000), C + F (₹38,000) and D alone (₹35,000).
- **The Study Material's own questions**: MCQ 14 on the meaning of capital rationing (SM 7.73); Illustration 13 (SM 7.53–7.54); the PI limitations (SM 7.32), asked as "Describe the advantages and disadvantages of profitability index" (theory question 5).

In the exam

What earns the marks

The two-part layout of the suggested answers: a ranking table with NPV per rupee and the rank; a "building up" table ending with the fraction of the last project; then, for indivisible projects, a table of combinations with a one-line conclusion naming the projects to undertake and reject. In November 2019, C + E left ₹10,000 unused and still won: the test is total NPV, not using up the budget.

Trick

NPV per rupee from PI

NPV per ₹1 = PI − 1, so a PI of 1.249 is 24.9 paise of NPV per rupee. Condition: the whole outlay at year 0 (otherwise PI's denominator is the PV of all outflows).

Answer frame

"What is capital rationing? How are projects selected under it?"

1. Meaning: funds are restricted, and management has to choose from amongst available alternative investments, all with positive NPVs.
2. Objective: maximise the total NPV obtainable within the funds available (wealth maximisation).
3. Independent and divisible projects: modify the NPV rule; rank by NPV per rupee of capital (equivalently PI); accept in order, the last project in part.
4. Indivisible projects: rank by absolute NPV, examine combinations within the limit, and select the combination with the highest total NPV.
5. Note PI's limitation for indivisible projects, and that negative-NPV projects are never included.

Common mistakes

⚠ **Common mistake**

Hall of Shame: ranking divisible projects by the size of their NPV

Taking Tapri's projects in NPV order, C (17.5), K (16.0), then a third of V (4.5), gives ₹38.0 lakh against ₹43.0 lakh. **Fix:** a rationed budget is spent rupee by rupee; rank divisible projects by NPV per rupee.

⚠ **Common mistake**

Taking indivisible projects in PI order and stopping

Q, then T, then nothing fits: ₹15.88 lakh, with ₹35 lakh idle. **Fix:** list the combinations that fit and compare their total NPVs; here P + Q + S gives ₹20.41 lakh.

⚠ **Common mistake**

Spending the leftover on a negative-NPV project

"The money is there anyway" does not make the airport kiosks worth doing. **Fix:** a project with NPV < 0 (PI < 1) is rejected under rationing as it is without it.

⚠ **Common mistake**

Taking the full NPV of a part project

Half of E adds half of its NPV (₹4.35 lakh), not ₹8.7 lakh. **Fix:** NPV of the part = NPV × funds remaining ÷ investment.

Practice

Check yourself

1 Tapri has ₹50 lakh for three divisible, independent projects: A (investment ₹20 lakh, NPV ₹8 lakh), B (₹30 lakh, ₹9 lakh) and C (₹25 lakh, ₹12.5 lakh). What is the maximum total NPV, in ₹ lakh?

Answer: _____

2 Case scenario. Dev presents six positive-NPV proposals to Tapri's board. The capital budget is ₹1 crore, short of the ₹1.6 crore they need, and each proposal must be taken up whole or not at all. Ritu must recommend a method. Which of the following is correct?

- A Rank by profitability index and accept in that order until the next one does not fit
- B Examine the combinations that fit within ₹1 crore and select the one with the highest total NPV
- C Select the single proposal with the highest IRR, then the next highest, and so on
- D Accept every proposal with a positive NPV and borrow the shortfall

3 Under capital rationing with divisible projects, which statements are correct? Select all that apply.

- **A** Projects are ranked on NPV per rupee of capital
- **B** When the whole outlay is at year 0, ranking by PI gives the same order
- **C** A project with a negative NPV may be accepted to use up the budget
- **D** The last project selected may be taken up only in part, with that fraction of its NPV

More than one answer may be right.

4 Five indivisible projects: J (₹40 lakh, NPV ₹14 lakh), K (₹25 lakh, ₹10 lakh), L (₹35 lakh, ₹12.6 lakh), M (₹20 lakh, ₹6 lakh) and N (₹30 lakh, ₹9.3 lakh). The budget is ₹60 lakh. Which package should be chosen?

- **A** J and M
- **B** K and L
- **C** K and N
- **D** L and M

5 Which of these is a capital rationing problem?

- **A** Tapri's only plot in Nagpur can take a café or a premix warehouse, not both
- **B** The board caps next year's capital spending at ₹80 lakh, while the proposals with positive NPVs need ₹1.3 crore
- **C** A new fryer is useful only if the factory extension it needs is also built
- **D** An old delivery van is to be replaced by a new one that costs less to run

Answers: the answer key at the end of this lesson.

Recap

- Capital rationing: funds are restricted, so management chooses among positive-NPV projects; the aim is the highest total NPV within the budget.
- Independent, divisible projects: rank by NPV per rupee of capital (NPV ÷ investment), accept in order, take the last in part with the same fraction of its NPV.
- NPV per ₹1 = PI − 1 when the outlay is at year 0: the same ranking; PI is the better technique under capital rationing.
- Indivisible projects: rank by absolute NPV, examine combinations within the limit, select the highest total NPV; a lower-ranked project may replace a higher-ranked one (Illustration 13: E + B + D beats E + B + C).
- PI fails as a guide where projects are indivisible (SM 7.32).
- Negative-NPV projects are never accepted, even with idle funds; unused budget is not a reason to accept.
- (Loose sweets go in by value per kilo; pressure cookers go in by trying the sets.)

Answer key · Check yourself

1 **22**

NPV per ₹1: C 0.50, A 0.40, B 0.30. C and A use ₹45 lakh (₹20.5 lakh NPV); the last ₹5 lakh buys one-sixth of B, ₹1.5 lakh. Total ₹22.0 lakh.

2 **B Examine the combinations that fit within ₹1 crore and select the one with the highest total NPV**

With indivisible projects, PI fails as a guide (SM 7.32): the ranking order can leave funds idle or exclude a better package. The last option ignores the constraint the board has set.

3 **A Projects are ranked on NPV per rupee of capital**; **B When the whole outlay is at year 0, ranking by PI gives the same order**; **D The last project selected may be taken up only in part, with that fraction of its NPV**

NPV per ₹1 = PI − 1 when the outlay is at the start. A negative-NPV project reduces wealth whatever the budget.

4 **B K and L**

K + L uses ₹60 lakh for ₹22.6 lakh; J + M gives ₹20.0 lakh, K + N ₹19.3 lakh (₹55 lakh), L + M ₹18.6 lakh (₹55 lakh).

5 **B The board caps next year's capital spending at ₹80 lakh, while the proposals with positive NPVs need ₹1.3 crore**

Rationing means funds are restricted and management must choose among positive-NPV alternatives (SM 7.53). The plot is a mutually exclusive decision: the projects compete for the same use, not the same money. The fryer is a contingent decision (SM 7.7), and the van a replacement decision (SM 7.6).

CHAPTER 8

Dividend Decision

8.1 Forms of dividend, buy-back and determinants of dividend policy

How profit after tax reaches shareholders (cash dividend, buy-back, bonus shares), what a stock split does, the factors that shape a dividend policy, payout policies and Lintner's model.

In plain words

Kesar wants to say thank you

Tapri has had its best year. Profit after tax is ₹60 crore, and Kesar wants to thank the shareholders before Diwali.

> **Kesar:** Let's pay out everything. Shareholders love a fat dividend.
>
> **Dev:** And the 120 new cafés? Every rupee we pay out is a rupee we must borrow or raise again, with issue costs.
>
> **Ritu:** We could give them bonus shares instead. No cash leaves the company.
>
> **Dev:** Or buy back some shares. Same cash as a dividend, but only those who sell receive it.

Profit after tax can do only two things: go to shareholders as a **dividend**, or stay in the business as **retained earnings** to finance growth. Deciding the split is the **dividend decision**, the third of the finance manager's long-term decisions after the investment and financing decisions.

The money can reach shareholders in different **forms**. A **cash dividend** pays every share the same amount. A **share buy-back** (repurchase) uses the same cash to buy some shares back, so only the shareholders who sell receive cash, and the rest own a larger slice of the company. **Bonus shares** (a stock dividend) hand out new shares instead of cash: reserves become share capital and nothing leaves the company. A **stock split** is not a distribution at all: one share of ₹500 face value becomes five shares of ₹100.

How much to pay depends on the company's position: its investment opportunities, its cost of capital, the shareholders' expectations, the law, taxes and the share price. A café chain opening 120 outlets and earning more on retained money than its shareholders expect is a different animal from a mature company with spare cash and little to invest in.

Think of it like

The corporate pie

A share is a slice of the company's pie. A cash dividend cuts a piece off the pie and hands it out: the pie shrinks, the shareholders hold cash. A buy-back hands cash only to those who give back their slices: the remaining slices are each a little bigger share of a smaller pie. Bonus shares and stock splits cut the same pie into more slices; nobody gets more pie (James Porterfield's "division of corporate pie into a larger number of pieces", SM 8.6).

Key idea

Dividend is the part of profit after tax distributed to shareholders; the rest is retained for growth ($g = b \times r$). Cash dividends and buy-backs pay out cash, bonus shares only capitalise reserves, and stock splits only change the number of shares. The right payout depends on investment opportunities, cost of capital, shareholders' expectations, legal and tax constraints, and the share price.

How it works

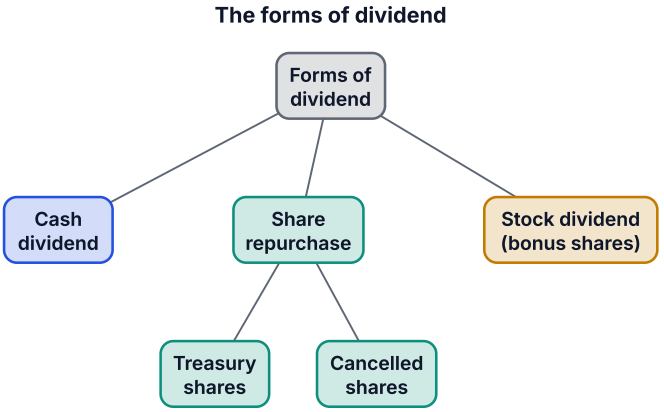
Definition

Dividend

Dividend is that part of Profit After Tax (PAT) which is distributed to the shareholders of the company. Profit after tax can be used for (i) distribution of dividend, or (ii) retaining as surplus for future growth. (SM 8.3)

Ex-dividend refers to shares that no longer carry the right to dividend; the price at which the shares trade immediately on the next day after declaration of dividend is the **ex-dividend price** (SM 8.3). With nothing else changing, it is the cum-dividend price less the dividend.

The forms of dividend (SM 8.3–8.6)



Blue: cash paid to every shareholder; teal: cash paid to those who sell, the bought-back shares kept for re-issue (treasury) or retired from issued capital (cancelled); amber: new shares instead of cash. A stock split is not on the tree: it distributes nothing.

1. **Cash dividend**: the most common form. Cash means cash, cheque, warrant, demand draft, pay order or directly through Electronic Clearing Service (ECS), but not in kind.
2. **Share repurchases**: a transaction in which the company buys back its own shares using corporate cash. The shares bought back are either **treasury shares**, kept for re-issuance in future, or **cancelled shares**, retired from issued share capital. Share repurchases are also viewed as one form of dividend distribution: keeping other things the same (such as tax considerations), the effect of a cash dividend and of a share repurchase on shareholders' wealth is the same.
3. **Stock dividend (bonus shares)**: a distribution of shares in lieu of cash dividend. New shares issued to existing shareholders without any consideration are bonus shares. They are distributed proportionately, so proportionate ownership is retained; equity share capital increases and

reserves and surplus decrease. The total net worth is not affected, as retained earnings are only capitalised.

Conditions for a bonus issue (SEBI, SM 8.5):

- Bonus shares are issued not in lieu of cash dividends.
- The issue should be authorised by the Articles of Association and not be declared unless all partly paid-up shares have been converted into fully paid-up shares.
- The company should not have defaulted in repayment of loan, interest and any statutory dues.
- Bonus shares are issued only from share premium and free reserves, not from capital reserve on account of revaluation of fixed assets.

Advantages of stock dividend (SM 8.5–8.6)

To shareholders	To the company
No tax on a stock dividend from a domestic company: it is treated as a capital asset, not as dividend	Conserves cash for profitable investment opportunities
If the fixed dividend per share continues after the bonus, total cash dividends rise in future	Suitable when cash is short, or lenders restrict cash dividends
Liquidity: lower-priced shares let shareholders sell a few and raise cash	

Limitations of stock dividend (SM 8.6): to shareholders, it does not affect their wealth and so has no value for them: it capitalises past earnings they already own, and their proportionate ownership does not change (though it creates a favourable psychological impact, as a sign of growth). To the company, stock dividends are costlier to administer than cash dividends, and periodic small stock dividends are a disadvantage.

The Study Material adds that bonus shares discourage immediate selling, because the holding period of bonus shares starts from their date of issue and short-term capital gains were taxed more heavily than long-term ones (it quotes 15% and 10%), which supports the share price (SM 8.5).

> ⚠ **Careful**
>
> Tax rates and rules quoted from the Study Material (capital gains rates, the treatment of stock dividends, dividend taxation from 1 April 2020) are reproduced as printed. Verify against the Study Material and Statutory Update for your attempt.

Stock splits and share buy-backs (SM 8.33–8.34)

A **stock split** means splitting one share into many, say one share of ₹500 into 5 shares of ₹100. It is a tool to regulate share prices: a price that rises from ₹50 to ₹1,000 over the years may go out of the range of many investors. **Advantages**: it makes the share affordable to small investors; more shares may mean more shareholders, so the potential of investment may increase. **Limitations**: additional expenditure on the process; a low share price may attract speculators or short-term investors, generally not preferred by any company.

A **share buy-back** means buying or repurchasing own shares by the company, resulting in a decrease in share capital: the shares bought back are cancelled, reducing the number of shares outstanding. It is also a form of shareholders' dividend: as the number of shares in the market falls, the dividend per share in future increases. The two main routes:

- **Open market**: the company buys through the secondary market.
- **Tender offer**: the company offers a fixed price at which all shareholders can participate or sell their shares.

Point	Bonus issue	Stock split
What changes	More shares, same face value	More shares, lower face value
Share capital and reserves	Reserves capitalised into share capital	Unchanged in total
A distribution of profits?	Yes, a form of dividend	No
Main purpose	Reward shareholders, conserve cash	Keep the price in a tradable range

Why dividend policy matters (SM 8.6–8.7)

Dividend policy is governed by two decisions:

1. **Long-term financing decision.** Equity can be raised externally (new shares) or internally (retained earnings), and retained earnings are preferable because they involve no flotation costs. Paying cash dividends reduces the funds for profitable investments. The decision turns on whether the organisation has opportunities to invest the profit if retained, and whether the return on such investment (ROI) will be higher than shareholders' expectation, Ke.
2. **Wealth maximisation decision**: what dividend payout ratio maximises the market price of the share? Because of market imperfections and uncertainty, shareholders give more importance to near dividends than to future dividends and capital gains; higher dividends tend to increase the value of shares. But retained earnings invested profitably raise future earnings per share. Management must divide net earnings between dividends and retained earnings in an optimum way.

Retained earnings and growth (SM 8.8)

$$g = b \times r$$

g = growth rate of the firm, b = retention ratio, r = rate of return on investment. In the Study Material's Example 1, two companies each earn 20% on ₹50,00,000: A Ltd distributes all its ₹10,00,000 and earns ₹10,00,000 again next year; B Ltd retains ₹5,00,000, employs ₹55,00,000 and earns ₹11,00,000, 10% more (b = 0.5, r = 20%). If Tapri retains 60% of its earnings and earns 18% on them, its earnings grow at 0.6 × 18% = 10.8% a year. How the price reacts is the subject of Walter's and Gordon's models.

Determinants of dividend decisions (SM 8.9–8.10)

ICAI's list, in ICAI's order:

1. **Availability of funds**: if the business needs funds, retained earnings are a good source, saving flotation cost and preventing the dilution of control that a public issue of equity shares causes.
2. **Cost of capital**: if financing is through debt (relatively cheaper), it is preferable to distribute more dividend; if through a fresh issue of equity shares, better to use retained earnings as much as possible.
3. **Capital structure**: an optimum debt-equity ratio should be considered.
4. **Stock price**: generally, higher dividends increase the market value of shares and low dividends decrease it.
5. **Investment opportunities in hand**: with opportunities in hand, the company may prefer to retain more earnings.
6. **Trend of industry**: investors depend on some industries for regular dividend income, so firms there must pay dividends to survive in the market.
7. **Expectation of shareholders**: some invest for regular income, others for growth; generally investors prefer current dividend over future growth.
8. **Legal constraints**: under Section 123 of the Companies Act, 2013, dividend shall be declared or paid only (a) out of the profits of the year arrived at after providing for depreciation, (b) out of undistributed profits of previous years after providing for depreciation, (c) out of both, or (d) out of money provided by the Central or a State Government in pursuance of its guarantee. Unrealised or notional gains, revaluation of assets and fair value changes are excluded in computing profits for dividends.
9. **Taxation**: before 1 April 2020, the company paid dividend distribution tax (Section 115-O) and the dividend was exempt for the shareholder (Section 10(34)); after the Finance Act 2020, dividend income from shares held as investment is taxable in the hands of the investor at the applicable slab rate.

Practical considerations (SM 8.10–8.13)

The policy depends on two questions: should there be a stable pattern of dividends over the years, or should each dividend decision be treated independently?

(a) Financial needs of the company. When the company's rate of return exceeds the return shareholders require, retaining earnings benefits shareholders; raising new shares brings flotation costs.

Mature companies	Growth companies
Few investment opportunities: high payout ratios	Need funds for fast-growing fixed assets: low payout ratios
Share prices sensitive to any change in payout	Retain earnings and declare bonus shares to offset shareholders' dividend expectations
Keep a small portion of earnings for emergent and occasional needs	Increase dividends gradually as profitable opportunities start falling

(b) Constraints on paying dividends: legal (the determinants above); liquidity (dividends are cash outflows; a growth company with good profits may still lack cash); access to the capital market (large dividends may force new issues that dilute control if existing shareholders cannot subscribe); and investment opportunities (if inadequate, pay dividends and raise external funds when needed).

(c) Payout policies can be maintained by fixing the amount or rate of dividend irrespective of the earnings of the company (SM 8.12–8.13). The Study Material gives two:

1. **Constant dividend policy**: shareholders are given a fixed amount of dividend irrespective of actual earnings, generally maintained for a considerable period. To keep paying it in poor years, the company creates a reserve such as a **Dividend Equalisation Reserve Fund**, earmarked by marketable securities. The policy treats equity shareholders at par with preference shareholders and is preferred by persons and institutions that depend on dividend income.
2. **Stable dividend policy**: the Study Material describes two forms under this heading.
   - A **constant payout ratio**: a fixed percentage of net earnings every year, so the dividend moves in direct proportion to earnings. At a 40% payout, EPS of ₹2 gives 80 paise and EPS of ₹1.50 gives 60 paise; for losses, no dividend. It has a conservative approach and provides a guarantee against over- or under-payment. Infosys's 2011 policy (up to 30% of consolidated PAT) is the Study Material's example.
   - Dividends based on a **long-term forecast of sustainable earnings**, increased when earnings have risen to a substantially higher level: the target payout ratio of Lintner's model.

Lintner's model: dividends that follow earnings slowly (SM 8.13–8.14)

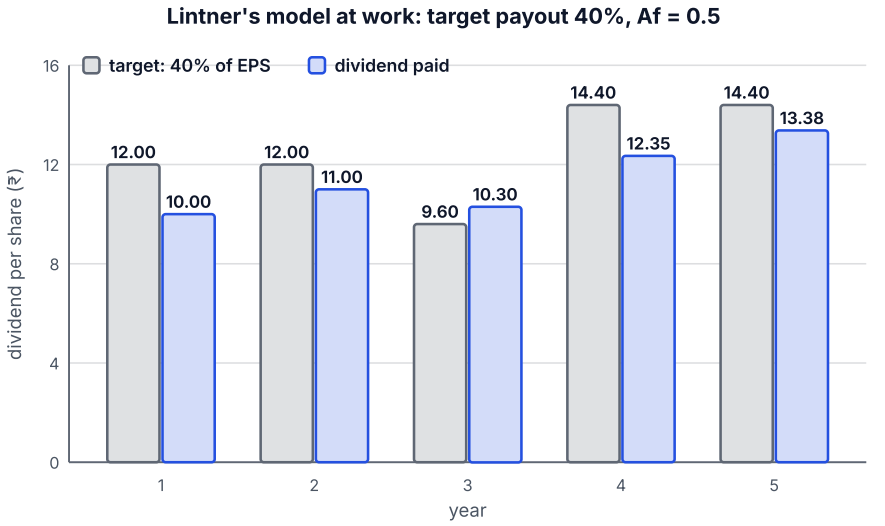
John Lintner built his model from interviews with corporate managers in the mid-1950s. It has **two parameters**: (i) the **target payout ratio**, and (ii) the **spread at which current dividends adjust to the target**. Its assumptions:

1. Firms have a long-term dividend payout ratio; mature companies with stable earnings may have high payouts, growth companies low ones.
2. Managers are more concerned with changes in dividends than with the absolute amounts.
3. Dividend changes follow changes in long-run sustainable earnings.
4. Managers are reluctant to effect dividend changes that may have to be reversed.

So this year's dividend depends on this year's earnings and last year's dividend:

$$D_1 = D_0 + [(\text{EPS} \times \text{Target payout}) - D_0] \times A_f$$

D_1 = dividend in year 1, D_0 = last year's dividend, EPS = earnings per share, A_f = adjustment factor or speed of adjustment. Its limitations (SM 8.41, MCQ 7): it does not offer a market price for the shares, and the adjustment factor is an arbitrary number, not based on any scientific criterion or method.



EPS moves 30, 30, 24, 36, 36 after a dividend of ₹8 on EPS of ₹20. Grey: the target dividend; blue: the dividend paid, which closes half the gap each year, so it rises slowly, dips only a little when EPS falls, and never jumps to the target.

Formats and formulas

Reproduce these from memory.

∑ **Formula**

Payout and retention

$$\text{Payout ratio} = \frac{\text{DPS}}{\text{EPS}} \qquad b = 1 - \text{Payout ratio}$$

DPS = dividend per share; b = retention ratio.

∑ **Formula**

Growth from retained earnings

$$g = b \times r$$

b = retention ratio, r = rate of return on investment (SM 8.8).

∑ **Formula**

Lintner's model

$$D_1 = D_0 + [(\text{EPS} \times \text{Target payout}) - D_0] \times A_f$$

Af = adjustment factor (speed of adjustment), between 0 and 1 (SM 8.14).

Σ Formula

Share buy-back

$$\text{Shares bought back} = \frac{\text{Amount}}{\text{Buy-back price}}$$

$$\text{BVPS after} = \frac{\text{Book value before} - \text{Amount paid}}{\text{Shares remaining}}$$

Cash paid out reduces net worth; with earnings unchanged, EPS after = earnings ÷ shares remaining (SM 8.54, problem 9).

Ex-dividend price = cum-dividend price − dividend per share (SM 8.4; 8.54, problem 8).

Definition

Stock split

Stock split means splitting one share into many, say, one share of ₹500 into 5 shares of ₹100. (SM 8.33)

Definition

Share buyback

Share buyback, in simple terms, means buying/repurchasing own shares by the company resulting into decrease in share capital of the company. Thus, the shares bought back are cancelled leading reduction in outstanding number of shares. (SM 8.33)

Worked examples

Worked example **Thirty crore rupees, three ways to hand them out**

Tapri Tea Co. Ltd has 1 crore equity shares quoted at ₹300 (cum-dividend) and ₹30 crore of cash to return to shareholders. The board considers three alternatives:

- Option 1: a cash dividend of ₹30 per share.
- Option 2: buying back ₹30 crore of shares at the market price, every shareholder tendering in proportion.
- Option 3: no cash, but a bonus issue of 1 share for every 4 held.

Required: Show the wealth of Ritu, who holds 100 shares, immediately after each option, ignoring taxes and transaction costs. Comment.

THE WORKING

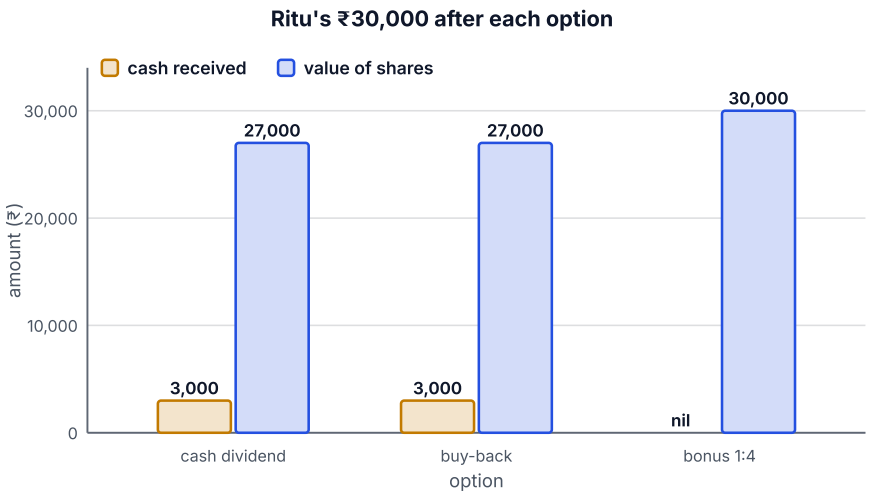
Option 1: cash dividend. Ex-dividend price = ₹300 − ₹30 = ₹270.

Option 2: buy-back. Shares bought back = ₹30 crore ÷ ₹300 = 10 lakh shares, 10% of the shares.

Particulars	Amount
Market capitalisation before buy-back	₹300 crore
Less: Shares repurchased	₹30 crore
Market capitalisation after buy-back	₹270 crore
Shares left (1 crore − 10 lakh)	90 lakh
Market price after buy-back	**₹300**

Option 3: bonus 1:4. Shares rise to 1.25 crore; net worth is not affected, so the theoretical price is ₹300 crore ÷ 1.25 crore = ₹240.

Ritu's position	Cash (₹)	Shares held	Value of shares (₹)	Wealth (₹)
Before	nil	100	30,000	30,000
Cash dividend	3,000	100	27,000	30,000
Buy-back	3,000	90	27,000	30,000
Bonus 1:4	nil	125	30,000	30,000



Amber: cash in hand; blue: the value of the shares still held. Each pair adds up to ₹30,000: the form changes what Ritu holds, not how much.

Comment. Keeping other things the same (taxes, costs), a cash dividend and a share repurchase have the same effect on shareholders' wealth (SM 8.4): Ritu has ₹30,000 either way, partly in cash. The bonus issue only divides the same pie into more pieces: no cash, no change in wealth, but the company conserves its ₹30 crore for investment, and the lower price per share improves liquidity.

Worked example **Tapri's buy-back: tender offer or open market?**

Tapri's board has earmarked ₹96 crore for a share buy-back. Its earnings are ₹60 crore, its EPS ₹30, its P/E

ratio 16 and its book value per share ₹240. Dev offers two routes:

- Route A: a tender offer at a premium of 25% to the current market price.
- Route B: open-market purchases at the current market price.

Assume the ₹96 crore would otherwise have earned nothing, so total earnings stay at ₹60 crore.

Required:

(i) Calculate the number of shares outstanding, the market price and the number of shares bought back under each route.

(ii) Calculate the EPS and the book value per share after the buy-back under each route.

(iii) At what buy-back price would the book value per share not change?

(iv) Comment on the results.

THE WORKING

(i) Shares and prices

1. Shares outstanding = ₹60 crore ÷ ₹30 = **2 crore**.
2. Market price = EPS × P/E = ₹30 × 16 = **₹480**; tender price = ₹480 × 1.25 = **₹600**.
3. Shares bought back: Route A = ₹96 crore ÷ ₹600 = **16 lakh**; Route B = ₹96 crore ÷ ₹480 = **20 lakh**.

(ii) After the buy-back

Particulars	Route A (₹600)	Route B (₹480)
Book value before (2 crore × ₹240)	₹480 crore	₹480 crore
Less: Amount paid	₹96 crore	₹96 crore
Book value after	₹384 crore	₹384 crore
Shares after	1.84 crore	1.80 crore
Book value per share	**₹208.70**	**₹213.33**
EPS (₹60 crore ÷ shares after)	**₹32.61**	**₹33.33**

(iii) Price that leaves book value per share unchanged. At the book value itself, ₹240: ₹96 crore buys 40 lakh shares, and ₹384 crore ÷ 1.60 crore shares = ₹240. Any price above book value lowers the book value per share; any price below it raises it.

(iv) Comment. Both routes pay more than the book value of ₹240, so book value per share falls, while EPS rises because the same earnings are shared by fewer shares. The 25% premium of the tender offer retires 4 lakh fewer shares for the same ₹96 crore, so its EPS gain is smaller (₹32.61 against ₹33.33). In return, a tender offer lets every shareholder sell at a fixed price, while open-market purchases go through the secondary market (SM 8.34). With fewer shares in circulation, the dividend per share in future can rise.

Worked example **Ritu smooths the dividend: Lintner's model**

Tapri paid a dividend of ₹8 per share last year. This year's EPS is ₹30. The board's target payout ratio is 40% and its speed of adjustment 0.5.

Required:

(i) Calculate this year's dividend per share using Lintner's model.

(ii) If EPS stays at ₹30 next year, what will next year's dividend be?

(iii) What speed of adjustment would have given a dividend of ₹11 this year?

THE WORKING

(i) Target dividend = ₹30 × 40% = ₹12.

$$D_1 = 8 + (12 - 8) \times 0.5 = 8 + 2 = ₹10$$

(ii) D_2 = 10 + (12 − 10) × 0.5 = **₹11**. The dividend closes half the remaining gap each year: ₹8, ₹10, ₹11, ₹11.50 ...

(iii) 11 = 8 + (12 − 8) × Af, so Af = 3 ÷ 4 = **0.75**. The higher the speed of adjustment, the faster the dividend moves to the target; at Af = 1 it would jump to ₹12 at once, and at Af = 0 it would never move.

Worked example **Case scenario: Kesar's Diwali dividend**

Tapri's board is setting the dividend for 2025-26. Profit after tax is ₹60 crore. The company plans 120 new cafés next year, needing ₹180 crore, and expects to earn 20% on them against the 14% its shareholders expect. A bank term-loan covenant limits cash dividends to 25% of profit after tax. Many retail shareholders, including Tapri's own staff, depend on a regular dividend, and the share price has risen to ₹2,400, beyond the reach of many small investors. Kesar proposes a cash dividend of 60% of profit "to keep everyone happy".

Required: Advise the board, with reasons, on Kesar's proposal and on whether bonus shares or a stock split could help.

MODEL ANSWER

Relevant considerations. The dividend decision is governed by the long-term financing decision (retained earnings are preferable as they involve no flotation cost, but only if the company has opportunities to invest them at a return above the shareholders' expectation, Ke) and the wealth maximisation decision. The determinants of dividend decisions include availability of funds, cost of capital, capital structure, stock price, investment opportunities in hand, trend of industry, expectation of shareholders, legal constraints and taxation (SM 8.6–8.10).

Facts of the case. Tapri needs ₹180 crore for expansion and expects a return of 20% against a Ke of 14%; a lender's covenant restricts cash dividends to 25% of profit after tax (₹15 crore); its shareholders expect regular income; its share price is high.

Analysis.

1. *Investment opportunities and availability of funds*: with profitable opportunities in hand (r > Ke), the company should retain more of its earnings; retention avoids the flotation cost and dilution of control of a fresh equity issue.
2. *Legal and contractual constraint*: a 60% payout (₹36 crore) would breach the covenant limit of ₹15 crore.
3. *Expectation of shareholders*: retail shareholders depending on dividend income argue for a regular, stable cash dividend, within what the company can afford.
4. *Stock dividend*: bonus shares conserve cash for the expansion and are suitable when lenders restrict cash dividends; they improve liquidity for shareholders, and if the dividend per share is maintained on the larger number of shares, future cash dividends increase.
5. *Stock price*: a stock split would bring the price within the reach of small investors and may increase the number of shareholders.

Conclusion. Kesar's proposal should not be accepted. The board should declare a regular cash dividend within the covenant limit (not exceeding ₹15 crore), retain the balance to finance the expansion, and may consider a bonus issue (subject to SEBI's conditions) or a stock split to reward shareholders and improve the marketability of the shares without parting with cash.

Exam angle

In the exam

Where these topics appeared

In the ICAI documents read for this lesson:

- **Theory, 2 to 5 marks**: any 2 advantages and limitations of stock splits (September 2025 Q4(a), 4 marks); one advantage of stock dividend to the company and to the investor (MTP Series II for January 2025 Q4(c), 2 marks, an OR choice); explain any common form of dividend (MTP Series II for January 2026 Q4(c), 2 marks, an OR choice); parameters of Lintner's model (MTP Series I for September 2024 Q3(b), 2 marks); inter-relationship between investment, financing and dividend decisions (MTP Series II for May 2024 Q4(a), 4 marks); old scheme: SEBI's conditions for a bonus issue (May 2023 Q6(a), 4 marks) and the determinants of dividend decisions (MTP Series II for May 2023 Q6(iii)).
- **Numerical**: book value per share before a repurchase of 20,000 shares at a 10% premium, from BVPS after of ₹193.20, earnings of ₹18,00,000, MPS ₹270 and P/E 18 (MTP Series II for September 2024 Q1(c), 5 marks; answer ₹210.50); last year's dividend from Lintner's model, ZED Ltd (January 2026 MCQ 8: D_1 ₹50, EPS ₹200, target 40%, Af 25%, D_0 = ₹40); old scheme: current dividend by Lintner's model, ₹6.727 (November 2023 Q1(c)(ii), 3 marks).
- **The Study Material's own questions**: theory questions 1, 2, 3 and 5 (factors governing the dividend decision, stock dividend, practical considerations, stock split); MCQs 5–8; problems 8 (reinvesting a dividend at the ex-dividend price) and 9 (book value after a buy-back).

In the exam

The traps ICAI used

- "Options other than cash used for distributing profits": the Study Material's answer is **bonus shares**, not "both bonus shares and stock split" (SM 8.41, MCQ 8; answer 8.47). A split distributes nothing.
- Lintner in reverse (January 2026 MCQ 8): 50 = D_0 + (80 − D_0) × 0.25, so 0.75 D_0 = 30 and D_0 = ₹40.
- Buy-backs at a premium: the price paid, not the market price, reduces book value (MTP Series II for September 2024).
- Reinvesting a dividend: buy at the ex-dividend price (SM problem 8: ₹50 − ₹10 = ₹40, so ₹10,00,000 buys 25,000 shares).

Trick

Lintner closes a fixed share of the gap

Each year the dividend moves Af of the way from last year's dividend to the target (EPS × target payout). With Af = 0.5 it halves the gap; reversed, D_0 = (D_1 − Af × target) ÷ (1 − Af). Condition: Af between 0 and 1.

Answer frame

"State dividend decision. Explain the factors which govern it" (SM theory question 1; ICAI's answer points to paragraphs 1, 2 and 6 of the chapter): the dividend decision as one of the three long-term finance decisions, with the financing and investment decisions; meaning (part of PAT distributed; the rest retained for growth); then the nine determinants in ICAI's order, one line each: availability of funds, cost of capital, capital structure, stock price, investment opportunities, trend of industry, expectation of shareholders, legal constraints (Section 123), taxation.

"Explain the advantages and disadvantages of stock dividend" (SM theory question 2): advantages to shareholders (three) and to the company (two), then limitations to shareholders and to the company, as in the tables above.

"Discuss the practical considerations in dividend policy" (SM theory question 3): (a) financial needs, with the mature and growth company contrast; (b) constraints: legal, liquidity, access to capital market, investment opportunities; (c) payout policies: (i) constant dividend policy (fixed amount, dividend equalisation reserve); (ii) stable dividend policy (constant payout ratio; dividends based on long-term sustainable earnings, Lintner's target payout).

"State the meaning of stock split; explain its advantages and disadvantages" (SM theory question 5; September 2025): ICAI's definition with the ₹500 into 5 × ₹100 example; two advantages; two limitations.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the stock split that "distributed profits"

Bunty lists bonus shares *and* stock splits as ways of distributing profits. **Fix:** bonus shares capitalise reserves and are a form of dividend; a stock split only divides each share into more shares of lower face value.

⚠ **Common mistake**

Using the market price instead of the buy-back price

Book value after = book value before − amount *paid*, at the buy-back price including any premium; and the number of shares retired is the amount ÷ the buy-back price. **Fix:** in Tapri's tender offer, ₹96 crore buys 16 lakh shares at ₹600, not 20 lakh at the market price of ₹480.

⚠ **Common mistake**

Lintner without the brake

Taking the target dividend (EPS × target payout) as this year's dividend. **Fix:** only the fraction Af of the gap is closed: ₹8 + (₹12 − ₹8) × 0.5 = ₹10, not ₹12.

⚠ **Common mistake**

Payout in place of retention in g = br

A company paying out 40% and earning 15% grows at 0.6 × 15% = 9%, not 0.4 × 15% = 6%. **Fix:** b is the retention ratio, 1 − payout.

Practice

Check yourself

1 Chai Express Ltd has 50 lakh shares with a book value of ₹120 each. It buys back 10 lakh shares at ₹180 each. What is its book value per share after the buy-back (₹)?

Answer: _____

2 A company paid ₹6 per share last year. EPS this year is ₹20, the target payout ratio 50% and the adjustment factor 0.4. What is this year's dividend under Lintner's model (₹)?

Answer: _____

3 Case scenario. Tapri's share price has climbed to ₹2,400. The board wants more small investors to be able to buy its shares, without paying out cash and without capitalising reserves. Kesar asks Dev which tool fits. Which of the following is correct?

- **A** A bonus issue of 1 for 1
- **B** A stock split, say one ₹10 share into five ₹2 shares
- **C** A buy-back through a tender offer
- **D** A cash dividend of ₹100 per share

4 Which of these are determinants of dividend decisions in ICAI's list? Select all that apply.

- **A** Availability of funds and investment opportunities in hand
- **B** Trend of industry and expectation of shareholders
- **C** The face value of the shares
- **D** Legal constraints under Section 123 of the Companies Act, 2013

More than one answer may be right.

5 Dona Packaging Ltd, a packaging supplier, earns ₹40 per share and pays a dividend of ₹14 per share. It earns 16% on the funds it retains. What growth rate does g = br give?

- **A** 5.6%
- **B** 10.4%
- **C** 16%
- **D** 35%

Answers: the answer key at the end of this lesson.

Recap

- Dividend: the part of profit after tax distributed to shareholders; the rest is retained for growth, $g = b \times r$.

- Forms: cash dividend; share repurchase (treasury or cancelled shares), same effect on wealth as a cash dividend; stock dividend (bonus shares), which capitalises reserves and leaves net worth unchanged.
- Bonus conditions (SEBI): not in lieu of cash dividend; authorised by AOA, all shares fully paid; no default on loans, interest or statutory dues; only from share premium and free reserves.
- Stock split: one share into many of lower face value; makes shares affordable; costs money and may attract speculators. Not a distribution of profits.
- Buy-back: open market or tender offer; reduces shares outstanding; BVPS after = (book value − amount paid) ÷ shares remaining.
- Determinants: funds, cost of capital, capital structure, stock price, investment opportunities, industry trend, shareholders' expectations, legal constraints, taxation.
- Payout policies: constant dividend (fixed amount, dividend equalisation reserve); stable dividend (constant payout ratio, or dividends based on long-term sustainable earnings). Lintner: $D_1 = D_0 + [(EPS \times \text{target payout}) - D_0] \times Af$, two parameters.
- (A bonus issue cuts the pie into more slices: nobody gets more pie.)

Answer key · Check yourself

1 **105**

Book value = ₹60 crore − (10 lakh × ₹180 = ₹18 crore) = ₹42 crore, over 40 lakh shares: ₹105. Buying above book value lowers it.

2 **7.6**

$D_1 = 6 + (20 \times 50\% - 6) \times 0.4 = 6 + 1.6 = ₹7.60$.

3 **B** **A stock split, say one ₹10 share into five ₹2 shares**

A split makes the share affordable to small investors and changes only the face value and number of shares. A bonus issue capitalises reserves; a buy-back and a cash dividend pay out cash.

4 **A** **Availability of funds and investment opportunities in hand**; **B** **Trend of industry and expectation of shareholders**; **D** **Legal constraints under Section 123 of the Companies Act, 2013**

The nine determinants: availability of funds, cost of capital, capital structure, stock price, investment opportunities, trend of industry, expectation of shareholders, legal constraints and taxation (SM 8.9–8.10).

5 **B** **10.4%**

Payout = 14 ÷ 40 = 35%, so b = 0.65 and g = 0.65 × 16% = 10.4%. 5.6% uses the payout ratio in place of the retention ratio; 35% is the payout ratio itself.

8.2 Dividend theories: traditional, Walter, Gordon and Modigliani-Miller

Does the payout change the share price? Walter and Gordon say yes, through r against Ke; Graham and Dodd weight dividends more; Modigliani and Miller say no, in a perfect market.

In plain words

Pay it out, or plant it?

Tapri's board has ₹10 of earnings per share to decide on.

> **Kesar:** Pay it all out. Shareholders like cash in hand.
>
> **Dev:** Every rupee we keep, we invest in new cafés at 18%. Our shareholders expect only 15%. A rupee kept is worth more to them than a rupee paid.
>
> **Ritu:** And if the cafés earned only 12%?
>
> **Dev:** Then pay it all out, and let them invest it themselves at 15%.

That exchange is **Walter's model** in four lines. Two numbers decide: **r**, the return the company earns on retained earnings (its internal rate of return), and **Ke**, the return shareholders expect (the cost of equity, or capitalisation rate). If r > Ke, retaining adds value and the best payout is zero; if r < Ke, paying out adds value and the best payout is 100%; if r = Ke, the payout does not matter. **Gordon's model** reaches the same conclusions with growth built in: retained earnings make dividends grow at g = b × r. Both say the dividend decision is **relevant** to the share price.

Modigliani and Miller (MM) disagree. In a perfect capital market, with no taxes and no issue costs, a shareholder who wants cash can sell a few shares (a "home-made dividend"), and a company that pays a dividend must raise the money back by issuing shares. The value of the firm depends only on its earning power and its investment policy, not on how the earnings are split: the dividend decision is **irrelevant**. The **traditional approach** of Graham and Dodd stands at the other end: the market puts considerable weight on dividends, so paying more raises the price.

Think of it like

The mango tree

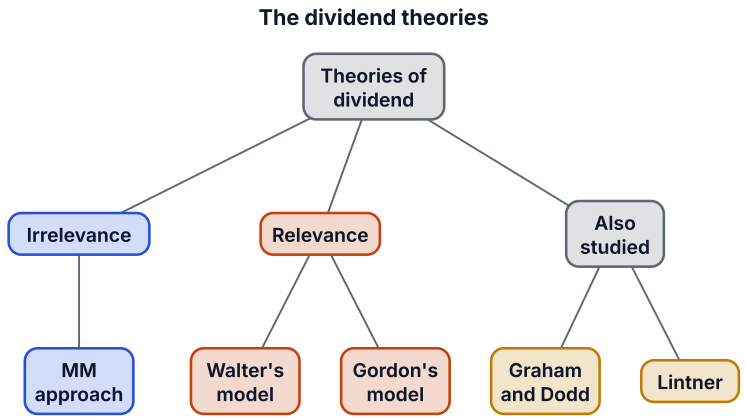
You own a mango tree. Eat this year's mangoes, or plant the seeds? If the seeds grow into more mangoes than the same money buys at the market (r > Ke), plant them all; if the market gives you more (r < Ke), eat them all and buy elsewhere. MM add: if there is a perfect market for mango trees, you can always sell a branch when you want mangoes, so how the tree splits its fruit cannot make you richer.

Key idea

Walter and Gordon: dividend policy is relevant; compare r with Ke. r > Ke: zero payout is optimum; r < Ke: 100%; r = Ke: no optimum (Walter: every payout is optimum). MM: dividend policy is irrelevant; value depends on earnings and investment policy. Graham and Dodd: P = m(D + E/3), so dividends weigh more than retained earnings.

How it works

Dividend theories fall under two categories: **irrelevance theory** (the MM approach) and **relevance theory** (Walter's, Gordon's and Lintner's models) (SM 8.2, 8.14). The Study Material also studies **Graham and Dodd's model**, by which the stock market places considerable weight on dividends than on retained earnings (SM 8.39), and Lintner's model of how dividends adjust to a target payout.



Blue: dividend policy has no effect on value (MM). Orange: it does (Walter, Gordon). Amber: the other two models in the chapter: Graham and Dodd's traditional weighting of dividends, and Lintner's target payout (SM 8.2, 8.14, 8.39).

1. The traditional approach: Graham and Dodd

According to Graham and Dodd, the stock market places considerable weight on dividends than on retained earnings (SM 8.39):

$$P = m\left(D + \frac{E}{3}\right)$$

P = market price per share, D = dividend per share, E = earnings per share, m = a multiplier. Writing E as D plus retained earnings R, P = m(4D + R) ÷ 3: a rupee paid out adds four times as much to the price as a rupee retained. With EPS of ₹12 and m = 10, a 50% payout gives P = 10(6 + 4) = ₹100, and a 75% payout ₹130.

2. Walter's model (SM 8.21–8.24)

Assumptions:

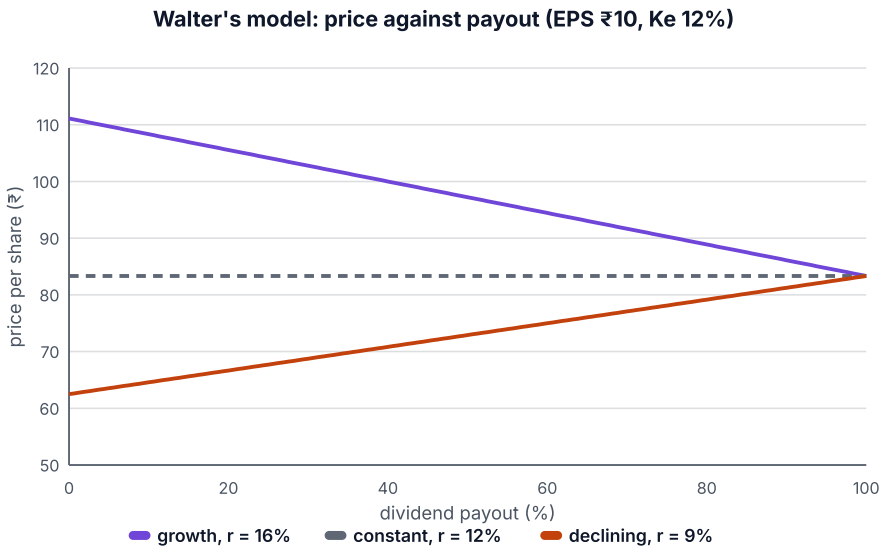
- All investment proposals of the firm are financed through retained earnings only.
- r (rate of return) and Ke (cost of capital) are constant.
- Perfect capital markets: all investors are rational and information is freely available to all.
- No taxes, or no tax discrimination between dividend income and capital appreciation (capital gain).
- No floatation or transaction cost.
- The firm has perpetual life.

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

P = market price of the share, E = earnings per share, D = dividend per share, Ke = cost of equity (rate of capitalisation, discount rate), r = internal rate of return (return on investment). Prof. James E. Walter argues that in the long run share prices reflect only the present value of expected dividends, and retentions influence prices only through their effect on future dividends. Two factors drive the price: the **dividend per share**, and the **relationship between r and Ke**.

Company	r vs Ke	Size of dividend and market price	Optimum payout
Growth	r > Ke	Negative correlation	Zero
Constant	r = Ke	No correlation	Every payout ratio is optimum
Decline	r < Ke	Positive correlation	100%

A growth company uses the funds better than its shareholders could, so they accept a low dividend; a declining company cannot cover its cost of capital, so shareholders prefer a higher dividend to invest elsewhere (SM 8.23). The model explains why shares of growing companies are priced high though their payout is low, and why some high-payout companies are also priced high. Retention involves an **opportunity cost**: the benefits shareholders forgo by leaving the funds in the business.



Violet (r > Ke): the price falls from ₹111.11 at zero payout to ₹83.33 at 100%. Grey dashed (r = Ke): ₹83.33 whatever the payout. Orange (r < Ke): the price rises from ₹62.50 to ₹83.33. At 100% payout every firm is worth E ÷ Ke.

Advantages (SM 8.23–8.24): the formula is simple to understand and easy to compute; it can envisage different market prices in different situations, considering the internal rate of return, the market capitalisation rate and the payout ratio. **Limitations**: it does not consider all the factors affecting dividend policy and share prices, and the market capitalisation rate is difficult to determine; it

ignores taxation, legal and contractual obligations, management policy and attitude towards dividend policy.

Interactive version: PassBook app, lesson Dividend theories: traditional, Walter, Gordon and Modigliani-Miller

3. Gordon's model (SM 8.26–8.33)

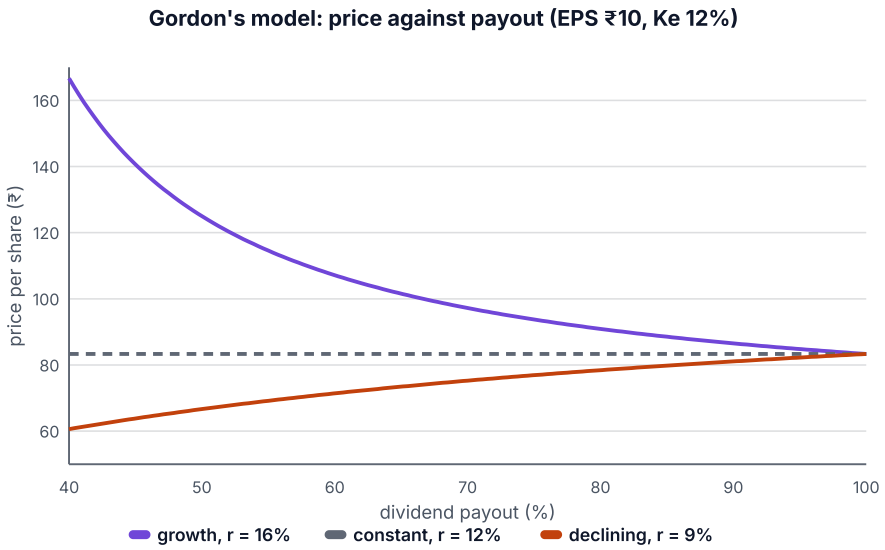
According to Gordon's model, dividend is relevant and dividend policy affects the value of the company. **Assumptions:**

- The firm is an all-equity firm (no debt).
- IRR remains constant, since a change in IRR would change the growth rate and the value.
- Ke remains constant, since a change in the discount rate would change the present value.
- The retention ratio (b), once decided upon, is constant: a constant dividend payout ratio is followed.
- The growth rate (g = br) is constant, since b and r are unchanged.
- **Ke > g**, necessary on the principle of the sum of a geometric progression for n years.
- All investment proposals are financed through retained earnings only.

$$P_0 = \frac{E_1(1-b)}{K_e - br}$$

P_0 = price per share, E_1 = earnings per share, b = retention ratio (1 − b = payout ratio), Ke = cost of capital, r = IRR, br = growth rate (g). When IRR is greater than the cost of capital, the price per share increases and dividend payout decreases (SM 8.27; MCQ 9); when IRR is lower, the price is highest at a 100% payout, as the table shows.

Company	r vs Ke	Optimum dividend payout ratio
Growth	r > Ke	Zero
Constant	r = Ke	There is no optimum ratio
Declining	r < Ke	100%



The same three firms as Walter's chart. Violet (r > Ke): ₹166.67 at a 40% payout, falling to ₹83.33 at 100%; the curve shoots up as retention approaches Ke ÷ r (75%), where g would equal Ke and the formula fails. Grey dashed: ₹83.33 throughout. Orange: ₹60.61 at 40%, rising to ₹83.33.

The bird-in-hand theory: Gordon's revised model. Gordon later considered risk and uncertainty, with two arguments: (i) investors are risk averse, and (ii) they put a premium on a certain return and a discount on an uncertain return. What is available at present is preferable to what may be available in the future, so investors pay a higher price for shares that pay current dividends and discount shares of a firm that postpones dividends (SM 8.28):

$$P_0 = \frac{D_0(1+g)}{K_e - g}$$

Po = market price per share (ex-dividend), Do = current year dividend, g = constant annual growth rate of dividends, Ke = cost of equity (expected rate of return).

The dividend discount model (DDM) values a share at the discounted value of its future dividends, at an appropriate risk-adjusted rate: intrinsic value = PV of dividends + PV of the sale price (SM 8.29):

$$\text{Intrinsic value} = \frac{D_1}{(1+K_e)^1} + \frac{D_2}{(1+K_e)^2} \quad +\cdots+ \frac{D_n}{(1+K_e)^n} + \frac{RV_n}{(1+K_e)^n}$$

Three situations (SM 8.30–8.31):

- **Zero growth**: every dividend the same, Po = D ÷ Ke (a ₹5 dividend at 10% is worth ₹50).
- **Constant growth** (Gordon's growth model): P = Do(1 + g) ÷ (Ke − g), with g = b × r.
- **Variable growth** (multi-stage): most commonly three stages, an initial high growth, a transition to slower growth, then a sustainable steady rate. Each phase is valued with the constant-growth method and the present values are added.

Advantages of Gordon's model (SM 8.32): the dividend discount model is a useful heuristic model that relates the present stock price to the present value of its future cash flows; and it is easy to understand. **Limitations of Gordon's model**: it depends on projections about the company's growth rate and future capitalisation rates, which may be difficult to calculate accurately; and the true intrinsic value of a stock is difficult to determine realistically.

Important considerations (SM 8.32–8.33): with dividends growing at g, the share price also grows at g; g is also called capital appreciation or capital yield; and the dividend yield $D_1 \div P_0$ stays constant.

Interactive version: PassBook app, lesson Dividend theories: traditional, Walter, Gordon and Modigliani-Miller

4. The Modigliani-Miller hypothesis (SM 8.14–8.21)

Proposed by Franco Modigliani and Merton Miller in 1961, the MM approach supports the **irrelevance** of dividends: a firm's dividend policy has no effect on either the price of its stock or its cost of capital. According to MM, the market value of equity shares depends solely on its earning power and is not influenced by the manner in which earnings are split between dividends and retained earnings; it is not affected by dividend size; and there is no meaningful distinction between dividends and share repurchases, both ways of returning cash to shareholders.

Assumptions:

- **Perfect capital markets**: all investors are rational and information is freely available to all.
- **No taxes**, or no tax discrimination between dividend income and capital gains.
- **Fixed investment policy**: all investment is financed through equity only.
- **No floatation or transaction cost.**
- **Risk of uncertainty does not exist**: investors forecast future prices and dividends with certainty, and one discount rate is appropriate for all securities and periods.

Three situations, in each of which the value of the firm is unchanged:

1. **Cash dividend from reserves and surplus**: cash moves from one pocket of the shareholders to another, with no net gain or loss.
2. **Cash dividend from a new issue of shares**: shareholders receive cash but suffer an equal capital loss through dilution of control and of earnings per share.
3. **No dividend**: a shareholder who wants cash sells part of the holding, the **home-made dividend**, a capital receipt matched by a capital loss through dilution.

MM's case rests on the **arbitrage argument**. The share price at the start equals the year-end price plus dividend, discounted at Ke:

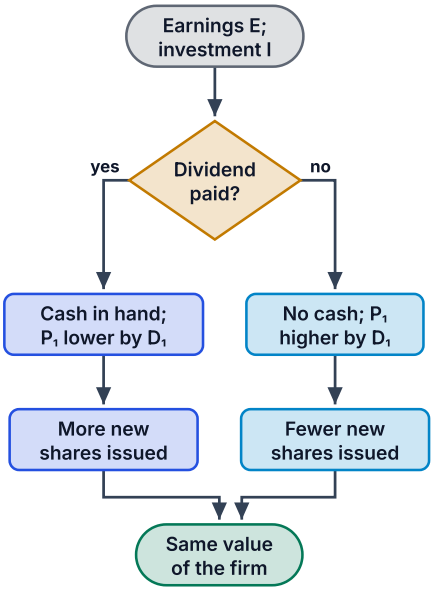
$$P_0 = \frac{P_1 + D_1}{1 + K_e}$$

Multiplying by n shares and allowing for the Δn new shares issued to fund the investment I from earnings E, the value of the firm is the same whether or not dividends are paid:

$$nP_0 = \frac{(n + \Delta n)P_1 - I + E}{1 + K_e} \qquad \Delta n = \frac{I - (E - nD_1)}{P_1}$$

D_1 disappears from the value of the firm: the dividend raises the funds to be found by new shares by exactly the dividend paid (SM 8.17–8.18).

MM: the shareholder's pocket stays the same



Blue: the dividend route; sky blue: no dividend. Either way the year-end value $(n + \Delta n)P_1$ plus dividends paid, less the investment, is the same, so nP_0 does not change.

Advantages (SM 8.19): the model is logically consistent, and it provides a satisfactory framework on dividend policy with the concept of the arbitrage process. **Limitations**: the validity of its assumptions is questionable, and it may not be valid under uncertainty. In the real world, market imperfections create problems for MM's irrelevance proposition (SM 8.21).

Interactive version: PassBook app, lesson Dividend theories: traditional, Walter, Gordon and Modigliani-Miller

The theories side by side

Point	Walter	Gordon	MM
Dividend policy	Relevant	Relevant	Irrelevant
Growth	Not compounded	g = br	–
Optimum, r > Ke	Zero	Zero	None
Optimum, r = Ke	Any payout	No optimum	None
Optimum, r < Ke	100%	100%	None

The formulas are in Formats and formulas below.

Formats and formulas

Reproduce these from memory; write the formula, then the figures.

∑ **Formula**

Graham and Dodd (traditional approach)

$$P = m\left(D + \frac{E}{3}\right)$$

m = a multiplier (SM 8.39).

∑ **Formula**

Walter's model

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

r = return on investment; Ke = capitalisation rate (SM 8.22).

∑ **Formula**

Gordon's model

$$P_0 = \frac{E_1(1-b)}{K_e - br} \qquad P_0 = \frac{D_0(1+g)}{K_e - g} = \frac{D_1}{K_e - g}$$

b = retention ratio; g = br; valid only if Ke > g (SM 8.27, 8.29).

∑ Formula

Zero growth and multi-stage DDM

$$P_0 = \frac{D}{K_e} \qquad P_0 = \sum_{t=1}^{n} \frac{D_t}{(1+K_e)^t} \qquad + \frac{D_{n+1}}{K_e - g} \times \frac{1}{(1+K_e)^n}$$

The second term is the price at the end of the high-growth years, discounted (SM 8.30; 8.51 and 8.54).

∑ Formula

Modigliani-Miller

$$P_0 = \frac{P_1 + D_1}{1 + K_e} \qquad nP_0 = \frac{(n + \Delta n)P_1 - I + E}{1 + K_e}$$

Δn = funds required ÷ P_1, where funds required = I − (E − nD_1) (SM 8.17–8.18).

The MM four steps (SM 8.19–8.21, Illustration 1)

Step	Dividend paid	Dividend not paid
1. Year-end price	$P_1 = P_0(1 + Ke) - D_1$	$P_1 = P_0(1 + Ke)$
2. Funds required	I − (E − nD_1)	I − E
3. New shares	Δn = funds required ÷ P_1	Δn = funds required ÷ P_1
4. Value of firm	[(n + Δn)P_1 − I + E] ÷ (1 + Ke)	Same formula: same value

Worked examples

Worked example **Walter and Gordon, before and after the boom**

Tapri's figures for 2026-27: profit after tax ₹72 crore; 9% preference share capital ₹200 crore; 6 crore equity shares; cost of equity 14%; dividend payout ratio 50%. While the new cafés are being built, Tapri earns 21% on retained earnings; once they mature, Dev expects only 10%.

Required:

(i) Calculate EPS, and the price per share by Walter's model and by Gordon's model, at a return of 21%.

(ii) Recalculate both prices at a return of 10%, and state the optimum payout ratio under each model at each return.

(iii) At what return on retained earnings would the payout ratio not affect the price, and what would the price then be?

(iv) While the return is 21%, Kesar wants to cut the payout to 30%. Calculate the price by Walter's model, and explain why Gordon's model cannot be applied.

THE WORKING

Working note: EPS

Particulars	₹ crore
Profit after tax	72
Less: Preference dividend (9% of ₹200 crore)	18
Earnings for equity shareholders	54
EPS (₹54 crore ÷ 6 crore shares)	**₹9**

At a 50% payout, D = ₹4.50 and b = 0.5.

(i) At r = 21%

$$\text{Walter: } P = \frac{4.50 + \frac{0.21}{0.14}(9 - 4.50)}{0.14} = \frac{4.50 + 6.75}{0.14} = ₹80.36$$

$$\text{Gordon: } P_0 = \frac{9(1 - 0.5)}{0.14 - 0.5 \times 0.21} = \frac{4.50}{0.035} = ₹128.57$$

(ii) At r = 10%

$$\text{Walter: } P = \frac{4.50 + \frac{0.10}{0.14}(9 - 4.50)}{0.14} = \frac{4.50 + 3.214}{0.14} = ₹55.10$$

$$\text{Gordon: } P_0 = \frac{4.50}{0.14 - 0.5 \times 0.10} = \frac{4.50}{0.09} = ₹50.00$$

Return (r)	Walter: optimum payout	Gordon: optimum payout
21%, r > Ke	Zero	Zero
10%, r < Ke	100%	100%

At 21%, Walter's price at zero payout is (1.5 × 9) ÷ 0.14 = ₹96.43. At 10%, both models give ₹9 ÷ 0.14 = ₹64.29 at a 100% payout.

(iii) When r = Ke = 14%, the payout does not affect the price: Walter's model gives every payout ratio as optimum and Gordon's has no optimum ratio. Both then give P = E ÷ Ke = ₹9 ÷ 0.14 = **₹64.29** at any payout.

(iv) Payout 30% (D = ₹2.70, b = 0.7):

$$\text{Walter: } P = \frac{2.70 + 1.5 \times 6.30}{0.14} \quad = \frac{12.15}{0.14} = ₹86.79$$

Under Gordon's model, g = br = 0.7 × 21% = 14.7%, which exceeds Ke of 14%. The model assumes Ke > g; with g above Ke the denominator is negative and the formula gives no meaningful price. Gordon's model can be applied here only while b stays below Ke ÷ r = 0.14 ÷ 0.21, a retention ratio of two-thirds.

Worked example

Two sister companies, two limits

Two listed companies in the Tapri group report the following. The risk-free rate is 6% and the market return 11%; each company's cost of equity is found by CAPM.

Particulars	Chai Express Ltd	Chulha Foods Ltd
EPS (₹)	15	12
Dividend per share (₹)	9	3
Return on retained earnings (r)	18%	9%
Beta	1.2	1.8

Required (Walter's model):

(i) Calculate each company's cost of equity and its price per share at the present payout.

(ii) State each company's optimum payout ratio and the price at it.

(iii) Chai Express's board will not let the payout fall below 30%, and Chulha Foods' lenders will not let it pay out more than 60% of its earnings. Calculate the highest price each can reach within its limit, and comment.

THE WORKING

(i) Cost of equity and present price

Chai Express: Ke = 6% + 1.2 × (11% − 6%) = **12%**; r ÷ Ke = 18 ÷ 12 = 1.5.

$$P = \frac{9 + 1.5 \times (15 - 9)}{0.12} \quad = \frac{9 + 9}{0.12} = ₹150.00$$

Chulha Foods: Ke = 6% + 1.8 × 5% = **15%**; r ÷ Ke = 9 ÷ 15 = 0.6.

$$P = \frac{3 + 0.6 \times (12 - 3)}{0.15} \quad = \frac{3 + 5.40}{0.15} = ₹56.00$$

(ii) Optimum payout

- Chai Express (r 18% > Ke 12%, a growth firm): the optimum payout is **zero**; P = (1.5 × 15) ÷ 0.12 = **₹187.50**.
- Chulha Foods (r 9% < Ke 15%, a declining firm): the optimum payout is **100%**; P = ₹12 ÷ 0.15 = **₹80.00**.

(iii) Within the limits

Chai Express at the 30% floor (D = ₹4.50):

$$P = \frac{4.50 + 1.5 \times 10.50}{0.12} \qquad = \frac{20.25}{0.12} = ₹168.75$$

Chulha Foods at the 60% cap (D = ₹7.20):

$$P = \frac{7.20 + 0.6 \times 4.80}{0.15} \qquad = \frac{10.08}{0.15} = ₹67.20$$

Comment. Under Walter's model the price moves in one direction with the payout: down for a growth firm, up for a declining firm. So each company should go as far towards its optimum as its limit allows: Chai Express to the 30% floor (₹168.75, against ₹150 now), Chulha Foods to the 60% cap (₹67.20, against ₹56 now). The limits cost ₹18.75 and ₹12.80 a share against the unconstrained optimum.

Worked example **Fading growth: the DDM, forwards and backwards**

Tapri paid a dividend of ₹6 per share for the year just ended. Dev expects dividends to grow by 15% next year, 12% the year after and 9% in the third year, and by 6% a year thereafter, indefinitely. Equity shareholders require 12%. The share is quoted at ₹120. PV factors at 12%: 0.893, 0.797, 0.712.

Required:

(i) Calculate the intrinsic value per share by the dividend discount model and state whether the share is over- or underpriced.

(ii) Keeping Dev's forecast for the first three years, calculate the long-run growth rate that the market price of ₹120 implies.

THE WORKING

(i) PV of the dividends of the first three years

Year	Dividend (₹)	PVF @ 12%	PV (₹)
1	6 × 1.15 = 6.900	0.893	6.162
2	6.900 × 1.12 = 7.728	0.797	6.159
3	7.728 × 1.09 = 8.424	0.712	5.998
	PV of dividends		**18.319**

Price at the end of year 3 (constant growth from year 4):

$$P_3 = \frac{D_4}{K_e - g} = \frac{8.424 \times 1.06}{0.12 - 0.06} \qquad = \frac{8.929}{0.06} = ₹148.82$$

PV of P_3 = ₹148.82 × 0.712 = ₹105.96.

Intrinsic value = ₹18.32 + ₹105.96 = **₹124.28** (three-decimal PV factors; with unrounded factors, ₹124.24).

The share is quoted at ₹120, below its intrinsic value: it is **underpriced** by ₹4.28.

(ii) Implied long-run growth. At the market price, the PV of P_3 must be ₹120 − ₹18.319 = ₹101.681, so P_3 = ₹101.681 ÷ 0.712 = ₹142.81. Then

$$142.81 = \frac{8.424(1+g)}{0.12 - g} \qquad \Rightarrow 17.137 - 142.81g = 8.424 + 8.424g$$

so 151.234g = 8.713 and **g = 5.76%**. The market is pricing long-run growth of about 5.8% against Dev's 6%: a quarter of a percentage point of growth explains the whole ₹4.28 gap, which shows how sensitive the value is to g when g is close to Ke.

Worked example **MM: three ways to spend one year's profit**

Sheetal Cold Chain Ltd, a listed refrigerated-transport company, has 4,00,000 equity shares quoted at ₹125. The capitalisation rate for its risk class is 12%. It expects net income of ₹70,00,000 for the year and plans new investments of ₹56,00,000. The board is choosing between three policies for the end of the year:

- Policy A: pay out only the surplus, ₹14,00,000, as a dividend of ₹3.50 per share.
- Policy B: pay a dividend of ₹15 per share and issue new shares for the shortfall.
- Policy C: pay no dividend and use the surplus to buy back shares at the year-end price.

Required (MM approach):

(i) Calculate the market price per share at the end of the year under each policy.

(ii) Calculate the number of shares issued or bought back under each policy.

(iii) Show that the value of the firm is the same under all three.

(iv) Kesar holds 1,000 shares and wants ₹7,000 in cash this year. Show his position at the year end under Policies B and C.

THE WORKING

(i) Year-end price: $P_1 = P_0 (1 + Ke) - D_1$ = ₹125 × 1.12 − D_1 = ₹140 − D_1.

Policy	Dividend per share (₹)	P_1 (₹)
A	3.50	136.50
B	15.00	125.00
C	nil	140.00

(ii) Shares issued or bought back: $\Delta n P_1 = I - (E - nD_1)$.

Particulars	A (₹)	B (₹)	C (₹)
Investment (I)	56,00,000	56,00,000	56,00,000
Net income (E)	70,00,000	70,00,000	70,00,000
Less: Total dividend (nD_1)	14,00,000	60,00,000	nil
Retained earnings	56,00,000	10,00,000	70,00,000
$\Delta n P_1 = I$ – retained earnings	nil	46,00,000	(14,00,000)
Δn	**nil**	**36,800 issued**	**10,000 bought back**

Δn: B ₹46,00,000 ÷ ₹125 = 36,800 new shares; C ₹14,00,000 ÷ ₹140 = 10,000 shares bought back.

(iii) Value of the firm: $nP_0 = [(n + \Delta n) P_1 - I + E] \div (1 + Ke)$.

Particulars	A	B	C
Shares at the year end	4,00,000	4,36,800	3,90,000
$(n + \Delta n) P_1$ (₹)	5,46,00,000	5,46,00,000	5,46,00,000
Add: E − I (₹)	14,00,000	14,00,000	14,00,000
Total (₹)	5,60,00,000	5,60,00,000	5,60,00,000
Value today (÷ 1.12) (₹)	**5,00,00,000**	**5,00,00,000**	**5,00,00,000**

The value of the firm is ₹5 crore (4,00,000 × ₹125) under every policy: the dividend decision does not change it.

(iv) Kesar's home-made dividend

- Policy B: he receives ₹15,000, keeps ₹7,000 and reinvests ₹8,000 in 64 new shares at ₹125. He ends with 1,064 shares × ₹125 = ₹1,33,000 plus ₹7,000 cash = **₹1,40,000**.
- Policy C: he receives nothing and sells 50 shares at ₹140 for ₹7,000. He ends with 950 shares × ₹140 = ₹1,33,000 plus ₹7,000 cash = **₹1,40,000**.

His holding is 1,064 ÷ 4,36,800 = 950 ÷ 3,90,000 = 0.2436% of the company either way. (Under Policy A his wealth is also 1,000 × ₹136.50 + ₹3,500 = ₹1,40,000.) He makes his own dividend, which is why MM call the policy irrelevant.

Exam angle

In the exam

Where the theories appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, a dividend model appeared in every past paper from May 2024 to May 2026 (the OCR text of the September 2026 paper shows none):

- **Walter**: January 2025 Q1(c) (5 marks: price ₹51.33, then the payout of 85.714% for a price of ₹48); May 2025 Q2(b) (2 marks: maximum price ₹26.67 at zero payout, minimum ₹20 at 100%); May 2026 Q3(a) (4 marks: price ₹45.33 at retention 20%, retention of 75% for a price of ₹60, Gordon's optimum payout zero). With a P/E ratio for Ke, in the same three parts each time (is the policy optimal; the P/E ratio at which dividend policy has no effect; does the decision change at another P/E ratio): RTP January 2025 Q8, MTP Series II for May 2025, MTP Series II for September 2026. Limitations: MTP Series I for January 2026 Q4(c) (2 marks).
- **Gordon and DDM**: May 2024 Q3(b) (Vista Ltd, two-stage DDM, ₹86.33); May 2025 Q1(b) (5 marks, retention for a target P/E); September 2025 Q2(b) (3 marks, DPS of ₹7 from a price of ₹159.09; no optimum when r = Ke); four assumptions of Gordon's model (September 2024 Q4(b), 4 marks); effect of a higher payout on price (RTP January 2025 MCQ 2: a fall of ₹444.45; MTP Series II for September 2025 MCQ 8).
- **MM**: January 2026 Q1(c) (5 marks: Gordon price ₹240, then MM year-end price ₹288 and the earnings required, ₹3,32,000); assumptions of MM (May 2026 Q4(a), 4 marks); value with and without dividend (RTP May 2024 Q6; MTP Series I for May 2025; MTP Series I for September 2026); retaining instead of paying leaves wealth unchanged (MTP Series I for January 2025 MCQ 8).
- **Graham and Dodd**: old scheme November 2023 Q1(c)(i) (2 marks, P = 5(36 + 20) = ₹280); the Study Material's MCQ 10.

In the exam

The traps ICAI used

- Preference dividend first: EPS = (net profit − preference dividend) ÷ equity shares (May 2026 Q3(a); Illustrations 3 and 4).
- "Retention ratio of 20%" means payout 80%: D = 0.8 × EPS (May 2026 Q3(a)).
- Gordon in its D form needs $D_1 = D_0(1 + g)$: January 2026 Q1(c) took 10 × 1.2 ÷ (0.25 − 0.20) = ₹240.
- Walter's optimum when r = Ke is "every payout ratio"; Gordon's is "no optimum ratio" (September 2025 Q2(b)).
- MM funds required subtract retained earnings ($E - nD_1$), not total earnings, when a dividend is paid.
- "The P/E ratio at which the dividend policy will have no effect": the one at which Ke = r, so P/E = 1 ÷ r (for r = 15%, a P/E of 6.67; SM 8.50).

Trick

The r-versus-Ke switch

One comparison answers every "optimum payout" question in Walter and Gordon: r > Ke, zero; r < Ke, 100%; r = Ke, payout does not matter. At 100% payout both models give P = E ÷ Ke. Condition: the models' assumptions (r and Ke constant, retained earnings the only finance).

Answer frame

"MM hypothesis is based on certain assumptions. What are they?" (May 2026 Q4(a), 4 marks): the five, in ICAI's order and words: perfect capital markets; no taxes (with the reason: universal applicability, since tax rates differ between countries); fixed investment policy; no floatation or transaction cost; no risk of uncertainty, one discount rate for all securities and periods.

"List any four assumptions of Gordon's model" (September 2024 Q4(b), 4 marks): all-equity firm; IRR constant; Ke constant; retention ratio constant; growth rate g = br constant; Ke > g; investments financed by retained earnings only. Give the reason ICAI attaches to each.

"Briefly explain the assumptions of Walter's model" (old scheme May 2022 Q6(b), 4 marks): the six assumptions above.

"State the limitations of Walter's model" (MTP Series I for January 2026, 2 marks): does not consider all factors, and Ke is difficult to determine; ignores taxation, legal and contractual obligations, management policy and attitude.

Numerical answers: working notes first (EPS after preference dividend, DPS, r and Ke, including Ke from a P/E ratio), then the formula, the figures and the result; end with the r against Ke conclusion in one sentence.

Common mistakes

⚠ **Common mistake**

Hall of Shame: forgetting the preference dividend

Dividing Tapri's ₹72 crore by 6 crore shares gives EPS of ₹12 and every price after it wrong. **Fix:** deduct the preference dividend first: (72 − 18) ÷ 6 = ₹9.

⚠ **Common mistake**

Last year's dividend in Gordon's numerator

P = 10 ÷ (0.25 − 0.20) = ₹200 for January 2026's PQR Ltd. **Fix:** the formula needs next year's dividend, $D_0(1 + g)$ = ₹12, so P = ₹240.

⚠ **Common mistake**

MM with the wrong funds required

With a dividend paid, the funds to be raised are $I - (E - nD_1)$, not I − E. **Fix:** in Sheetal Cold Chain's Policy B, ₹56,00,000 − (₹70,00,000 − ₹60,00,000) = ₹46,00,000, at the dividend-case price of ₹125; I − E would even say ₹14,00,000 is left over.

⚠ **Common mistake**

The optimum payout as a "balanced" 50%

Under Walter and Gordon the optimum is a corner: zero or 100%. **Fix:** compare r with Ke and write the corner; only r = Ke makes the payout irrelevant.

Practice

Check yourself

1 A company has EPS of ₹8 and pays a dividend of ₹4. Its return on investment is 15% and its cost of equity 12%. What is the price per share by Walter's model (₹)?

Answer: _____

2 EPS is ₹10, the retention ratio 40%, the return on retained earnings 15% and the cost of equity 12%. What is the price per share by Gordon's model (₹)?

Answer: _____

3 Case scenario. Masala Mills Ltd earns 11% on the funds it retains; its shareholders expect 14%. Its EPS is ₹20 and it pays out 50%. The managing director asks the CFO what Walter's model recommends. Which of the following is correct?

- **A** Retain all earnings, since retention always adds growth
- **B** Keep the 50% payout, since it balances growth and income
- **C** Pay out 100% of earnings, since r is less than Ke
- **D** The payout does not matter, since dividend policy is irrelevant

4 Under the MM approach, a share is quoted at ₹200 at the start of the year, the capitalisation rate is 10% and a dividend of ₹12 is expected at the year end. What is the year-end price if the dividend is paid (₹)?

Answer: _____

5 Which of the following are assumptions of the MM dividend irrelevance hypothesis? Select all that apply.

- **A** Perfect capital markets, with information freely available to all
- **B** No floatation or transaction costs
- **C** All investment financed by retained earnings only, with r and Ke constant
- **D** No risk of uncertainty: one discount rate for all securities and periods

More than one answer may be right.

6 By Graham and Dodd's model, with a multiplier of 8, EPS of ₹24 and a payout of 25%, what is the price per share (₹)?

Answer: _____

Answers: the answer key at the end of this lesson.

Recap

- Relevance: Walter and Gordon; irrelevance: MM; Graham and Dodd weight dividends more: $P = m(D + E/3)$.
- Walter: $P = [D + (r \div Ke)(E - D)] \div Ke$; $r > Ke$ zero payout, $r < Ke$ 100%, $r = Ke$ every payout optimum.
- Gordon: $P_0 = E_1(1 - b) \div (Ke - br) = D_1 \div (Ke - g)$; needs $Ke > g$; $r = Ke$ has no optimum ratio.
- Bird-in-hand: risk-averse investors prefer certain current dividends. DDM: zero growth $D \div Ke$; constant growth; multi-stage = PV of dividends + PV of the price at the end of the high-growth phase.
- MM: value depends on earning power and investment policy; $P_0 = (P_1 + D_1) \div (1 + Ke)$; $nP_0 = [(n + \Delta n)P_1 - I + E] \div (1 + Ke)$; $\Delta n = [I - (E - nD_1)] \div P_1$.
- MM assumptions: perfect markets, no taxes, fixed investment policy, no floatation or transaction cost, no uncertainty; dividends and repurchases are equivalent.
- Walter limitations: not all factors, Ke hard to determine, ignores taxes, legal and contractual obligations, management attitude.
- Preference dividend before EPS; D_1, not D_0, in Gordon; $Ke = 1 \div P/E$ when only P/E is given. (Plant the seeds only if they beat the market.)

Answer key · Check yourself

1 **75**

$P = [4 + (0.15 \div 0.12) \times (8 - 4)] \div 0.12 = (4 + 5) \div 0.12 = ₹75$.

2 **100**

$g = 0.4 \times 15\% = 6\%$; $P_0 = 10 \times 0.6 \div (0.12 - 0.06) = 6 \div 0.06 = ₹100$.

3 **C** **Pay out 100% of earnings, since r is less than Ke**

r (11%) < Ke (14%): a declining firm in Walter's terms; positive correlation between dividend and price, so the optimum payout is 100%. Irrelevance is MM's view, not Walter's.

4 **208**

$P_0 = (P_1 + D_1) \div (1 + Ke)$: $200 \times 1.10 = P_1 + 12$, so $P_1 = ₹208$ (₹220 with no dividend).

5 **A** **Perfect capital markets, with information freely available to all**; **B** **No floatation or transaction costs**; **D** **No risk of uncertainty: one discount rate for all securities and periods**

The third is Walter's (and Gordon's) assumption. MM's five: perfect markets, no taxes, fixed investment policy, no floatation or transaction cost, no uncertainty.

6 **112**

$D = 25\% \times ₹24 = ₹6$; $P = m(D + E \div 3) = 8 \times (6 + 8) = ₹112$.

CHAPTER 9

Management of Working Capital

9.1 Working capital, the operating cycle and financing policy

What working capital is (gross, net, permanent, temporary), what decides how much a business needs, the operating cycle R + W + F + D − C, and how investment and financing policies trade liquidity for profit.

In plain words

Record Diwali, empty bank account

Tapri's samosa factory in Indore has just had its best Diwali quarter: distributors ordered three times the usual frozen packs, and the quarter's accounts show a fat profit. On Friday, Dev cannot pay the milk and potato suppliers.

> **Kesar:** The accounts say we made a profit of ₹4 crore and the bank says we have nothing? Sab adjust ho jayega, just pay them.
>
> **Dev:** The profit is sitting in the cold store as samosas and in the distributors' unpaid bills. Cash pays the milkman on Friday, not profit.
>
> **Kesar:** Then how much money does the factory need just to keep running?
>
> **Dev:** As much as stays stuck on the wheel at any moment. Let me show you the wheel.

The money a business keeps tied up in its day-to-day running is its **working capital**. It sits in **current assets**, the things that turn into cash within the normal operating cycle: stocks of potatoes and flour, half-made samosas, packed samosas in the cold store, distributors who owe money, and cash itself. Part of it is paid for by other people: suppliers who give 25 days to pay, and staff who are paid at the end of the month. Those amounts owed are **current liabilities**. Current assets minus current liabilities is the **net working capital** the owners and lenders must provide.

The wheel is the **operating cycle**. A rupee leaves as cash to buy potatoes, waits 30 days as raw material, spends 12 days as samosas in the making, 18 days as packed stock, and 45 days as a distributor's unpaid bill before it comes home as cash. Suppliers carry the first 25 days. So Tapri's own money is out for 30 + 12 + 18 + 45 − 25 = 80 days, and the rupee can make the round only 4½ times in a 360-day year. The longer the round, the more money the factory needs to keep the wheel full.

Some of that money is needed all year: even in a quiet June the factory holds stock and waits for distributors. That base is **permanent working capital**. The Diwali bulge on top is **temporary working capital**: it comes and goes with the season. Sensible financing follows the same split: the permanent part from long-term money (shares, long-term loans), the bulge from short-term money (a bank limit used for three months).

Think of it like

The giant wheel at the mela

Each rupee boards a cabin as cash, rides past raw material, work-in-progress, finished stock and debtors, and gets off as cash again. Working capital is the number of cabins you must fill to keep the wheel turning. A faster wheel (a shorter cycle) needs fewer cabins for the same number of riders a year; suppliers' credit is a stretch of the ride in someone else's cabin. At Diwali the crowd swells, and you hire extra cabins for the season instead of buying them for the whole year.

Key idea

Working capital is the money stuck in the operating cycle: current assets minus current liabilities. The longer the cycle (days in raw material, work-in-progress, finished goods and debtors, less suppliers' credit), the more of it a business needs. Holding more is safer but earns less; holding less earns more but risks running dry. The permanent part is financed long-term, the temporary part short-term.

How it works

What counts as current

In accounting terms, working capital is the difference between current assets and current liabilities (SM 9.3).

Definition

Current assets and current liabilities (SM 9.3–9.4)

An **asset** is classified as current when (i) it is expected to be realised, or is intended to be sold or consumed, in the normal operating cycle of the entity or within twelve months after the reporting period, whichever is longer; and (ii) it is held primarily for the purpose of trading in the ordinary course of business.

A **liability** is classified as current when (i) it is expected to be settled in the normal operating cycle of the entity or within twelve months after the reporting period, whichever is longer; and (ii) it is settled either by the use of current assets or by creation of a new current liability.

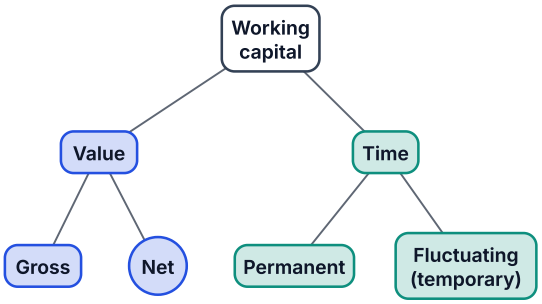
For working capital management ICAI groups them as follows (SM 9.3–9.4):

Current assets	Current liabilities
(a) Inventory: raw material, work-in-process, finished goods	(a) Payables: trade payables and bills payable
(b) Receivables: trade receivables and bills receivable	(b) Outstanding payments: wages and salary, overheads and other expenses
(c) Cash or cash equivalents, including short-term marketable securities	Others: short-term borrowings, current portion of long-term debt, short-term provisions such as provision for taxes
(d) Prepaid expenses; others: short-term loans or advances, accrued revenue	

Working capital management is a process designed to ensure that an organisation operates efficiently by monitoring and utilising its current assets and current liabilities to the best effect. Its primary objective is to enable a company to maintain sufficient cash flows to meet its day-to-day operating expenses and its short-term obligations (SM 9.4).

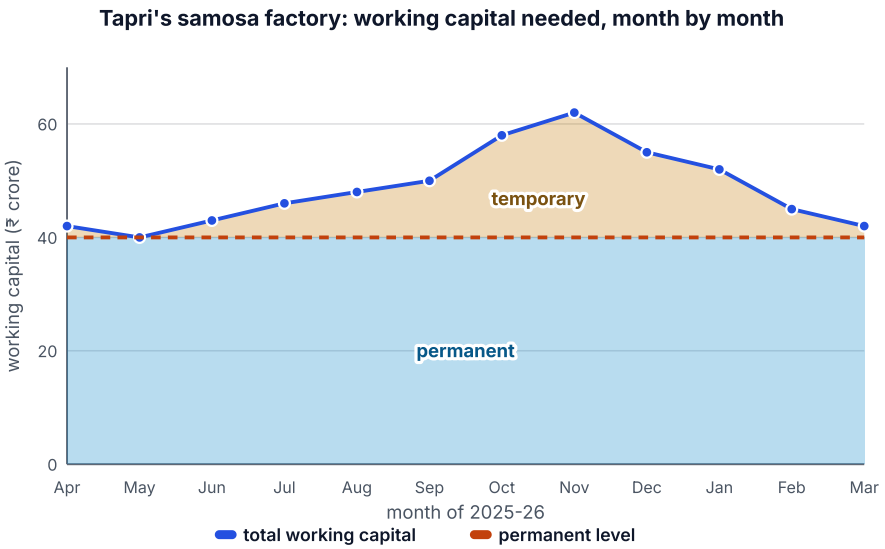
Two ways to look at the same money: value and time

The concepts of working capital



Blue: on the basis of value; teal: on the basis of time (SM 9.4). The Study Material calls the second kind fluctuating or temporary working capital.

- **Gross working capital** is the firm's investment in current assets. **Net working capital** is the difference between current assets and current liabilities. A positive net working capital indicates the company's ability to pay its short-term liabilities; a negative one shows inability to meet its short-term obligations (SM 9.4–9.5).
- **Permanent working capital** is the base: the minimum level of investment in current assets carried at all times to carry on day-to-day activities. It stays invested unless operations are scaled up or down permanently, and it is generally financed by long-term sources. **Temporary** (also called **variable** or **fluctuating**) working capital is the part required in addition to the permanent working capital, because of fluctuations in sales volume: more inventory for seasonal demand, for example. Both are necessary to facilitate production and sales through the operating cycle (SM 9.5).



Blue: the total requirement each month; orange dashed: the ₹40 crore that is never released. Sky blue shading: permanent working capital; amber: the temporary part, which peaks at ₹22 crore in November. If the factory grows, the permanent level itself slopes upward (the second picture at SM 9.5).

Neither too much nor too little

The finance manager must ensure that the working capital available is neither too large nor too small for the requirements (SM 9.6):

- **Too much** means idle funds. Funds have a cost: interest on the money locked up, or the opportunity cost of not investing it long term. Studies by the Bureau of Public Enterprises found that large funds locked up in working capital were one reason for the poor performance of public sector undertakings: **over-capitalisation**, too large funds for the requirements, a low rate of return.
- **Too little** brings the risk of insolvency: liabilities cannot be met, stock runs out, sales and reputation are lost.
- **Growth needs working capital too.** An investment decision must count the additional current assets that come with expansion: more production means more raw material and work-in-progress; more sales mean more debtors and finished goods (SM 9.6–9.7).

What is optimum? Traditionally, a **current ratio** (current assets ÷ current liabilities) of 2 for a manufacturing firm, supplemented by a **quick (acid test) ratio** (quick assets ÷ current liabilities, quick assets being current assets less inventory and prepaid expenses) of at least 1, are taken as indicators of a good working capital situation. As a thumb rule this may be adequate, but the optimum can be determined only with reference to the particular circumstances: a firm whose stock sells easily and whose debtors are as good as cash may be sound below 2, a restaurant with perishable stock cannot hold much, and a business with a long production time needs more (SM 9.7–9.8). Both excessive and inadequate working capital positions are dangerous. Ratios themselves are in liquidity ratios.

Nine things that decide how much you need

Working capital management is concerned with (a) maintaining adequate working capital (the level of each current asset and current liability) and (b) financing it (SM 9.8). The factors to consider while planning the requirement, in ICAI's order (SM 9.9–9.10):

1. **Need for cash**: the cash balance that meets day-to-day expenses while keeping holding costs (interest lost on surplus cash) low.
2. **Desired level of inventory**: enough for uninterrupted production, without excess investment in raw materials; techniques such as JIT and EOQ.
3. **Receivables**: credit terms that attract customers, such that the effect on cash flows and the cash conversion cycle is offset by more revenue and return on capital; tools such as early payment discounts and allowances.
4. **Short-term financing options**: inventory is ideally financed by supplier credit; depending on the cash conversion cycle, a bank loan or overdraft, or "converting debtors to cash" through factoring.
5. **Nature of business**: a restaurant sells mostly for cash and needs little; a pharmacy or bookstore holds more inventory.
6. **Market and demand conditions**: when demand far exceeds production, little finished stock piles up.
7. **Technology and manufacturing policies**: for seasonal demand, steady production all year or production only in the season.
8. **Operating efficiency**: eliminating waste, better coordination, process improvements reduce the need.
9. **Price level changes and exchange rate fluctuations**: rising prices need more funds for the same activity; an adverse exchange rate raises the cost of imported material.

Liquidity against profitability: the 3 Es

Working capital is compared to the life blood of a living body or the lubricant of an engine. Its management needs the **3 Es**: **economy** in financing, **efficiency** in utilisation and **effectiveness** in achieving the intended objectives. Its scope has two broad areas: (i) **liquidity and profitability**, and (ii) **investment and financing decisions** (SM 9.10–9.11).

Every rupee of capital has a cost. More working capital may raise revenue, but funds tied up in idle assets reduce both liquidity and the chance of a better return elsewhere; so a **trade-off between**

liquidity and profitability is needed, one that raises profitability without disturbing day-to-day functioning (SM 9.11). ICAI's table, component by component (SM 9.11–9.12):

Component	Higher side (profitability)	Trade-off tool	Lower side (liquidity)
Inventory	Fewer stock-outs increase profitability	EOQ, JIT to carry the optimum level	Less capital, but stock-outs and loss of goodwill
Receivables	Longer credit attracts customers and revenue (but more bad debts)	Evaluate the credit policy; use factoring agencies	Cash sales give liquidity but fail to boost sales
Prepaid expenses	Reduce uncertainty; profitable in inflation	Cost-benefit analysis	Improves or maintains liquidity
Cash and cash equivalents	Payables paid in time: goodwill, future discounts	Cash budgets and cash management techniques	Cash can be invested elsewhere
Payables and expenses	Capital can be used in other investment avenues	Evaluate the credit policy and its cost	Paid in time: goodwill, future discounts

How much to invest in current assets

The **investment decision** answers "how much" fund is tied up (effectiveness); the **financing decision** answers "where from", at the lowest possible cost (economy) (SM 9.12). There are no fixed rules for the level of investment; it depends on (SM 9.13): (a) **nature of industry** (construction, breweries: long gestation); (b) **types of products** (consumer durables hold more than perishables); (c) **manufacturing vs trading vs service** (a manufacturer holds raw material, work-in-process and finished goods; a trader only trading stock, a service entity only consumables); (d) **volume of sales** (high sales, high receivables); (e) **credit policy** (a liberal policy means high receivables and, depending on suppliers' credit, more capital for raw material).

Based on policy and the risk-return trade-off, there are three **approaches of working capital investment** (SM 9.13–9.14):

Approach	What the firm does	Advantages	Disadvantages
Aggressive	Minimal current assets: low inventory, strict credit, little cash	Less fund tied up, lower financial cost	Stock-outs; growth held back, lower use of fixed assets and long-term debt; may fall behind competitors
Conservative	High current assets: more inventory, liberal credit, cash to pay any current liability at once	Higher sales, more demand from liberal credit, suppliers' goodwill	Higher cost of capital, obsolescence, bad debts, liquidity shortage in the long run from longer cycles
Moderate	In between	Balance of risk and return	

A conservative policy implies greater liquidity and lower risk; an aggressive policy higher risk and poor liquidity; most firms follow a moderate policy, and a firm may change policy over time. Measured by the **current assets to fixed assets ratio** (with fixed assets constant), a higher ratio indicates a conservative policy and a lower one an aggressive policy (SM 9.14–9.15). ICAI's Illustration

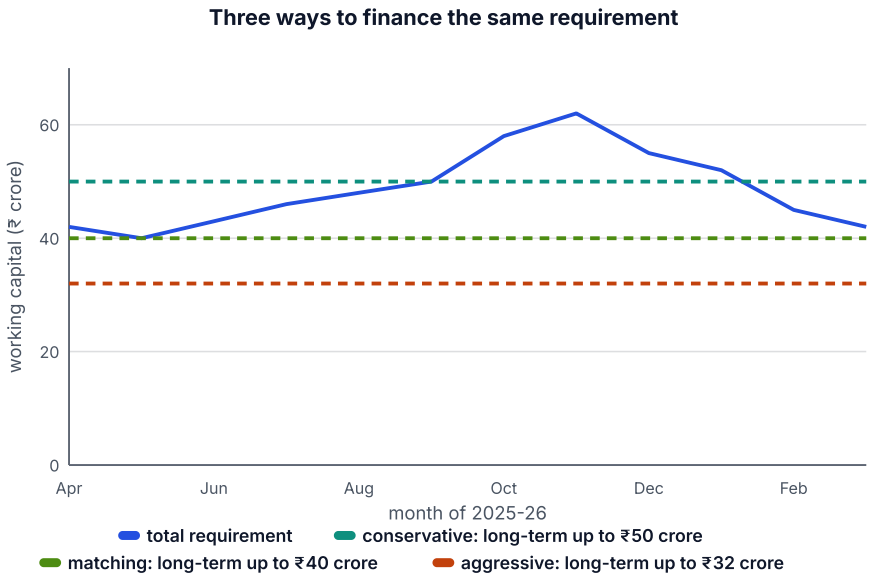
1 shows the inverse relationship between this ratio and the return on total assets: the firm balances the "profitability-solvency tangle" by minimising the total of the cost of liquidity and the cost of illiquidity (SM 9.16). The aggressive approach would better suit a highly integrated organisation with efficient processes (SM 9.14).

Where the money comes from: the matching principle

After determining the amount of working capital, the finance manager arranges the funds. ICAI advises bifurcating the requirement: the permanent working capital should be financed by long-term sources such as debt and equity; the temporary working capital may be financed by short-term sources (SM 9.102).

Sources are **spontaneous** (arising naturally in the course of business: trade credit, credit from employees and from suppliers of services) or **negotiated** (specifically negotiated with commercial banks, financial institutions, the general public). The parameters that guide the choice (SM 9.102–9.103): (i) cost factor; (ii) impact on credit rating; (iii) feasibility; (iv) reliability; (v) restrictions; (vi) the **hedging approach or matching approach**, i.e. financing of assets with the same maturity as of assets. ICAI's own MCQ puts it this way: under the hedging approach, each asset would be offset with a financing instrument of the same approximate maturity (SM 9.113). The sources themselves are in financing of working capital and bank credit.

ICAI's Mock Test Paper (May 2026, Series II, Q2(a)) grades **financing policies** the same way it grades investment: with the same assets, a **conservative** financing policy uses more long-term debt and less short-term debt, an **aggressive** one the reverse, and a **moderate** one sits between. On Tapri's curve, the level up to which long-term money is used decides the policy:



Blue: the requirement. Each dashed line is the level financed by long-term funds; everything above it is short-term borrowing. Matching (green) finances exactly the permanent ₹40 crore long-term. Conservative (teal) also finances part of the bulge long-term, leaving surplus cash in quiet months. Aggressive (orange) finances ₹8 crore of the permanent need with short-term money that must be renewed all year.

The effect, as ICAI's MTP answer computes it: more short-term debt means lower net working capital and a lower current ratio (less liquidity, and the risk that the loan is not renewed, or is renewed at a

higher rate), but cheaper funds and so a higher return on equity. Worked example 2 below does it with numbers.

The operating cycle

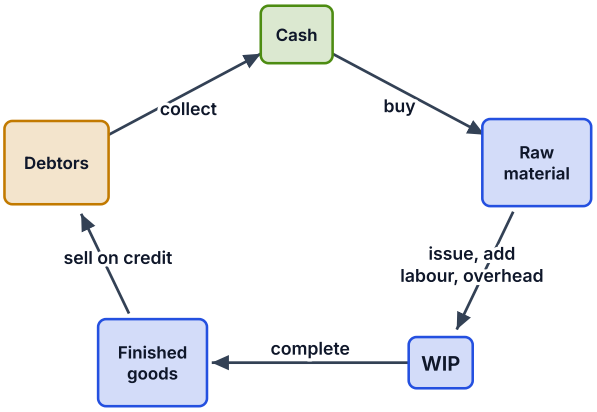
The **operating cycle** analyses the receivable, inventory and payable cycles in days: receivables by the average days to collect, inventory by the average days to turn over, payables by the average days to pay a supplier (SM 9.17).

Definition

Operating (working capital) cycle (SM 9.17)

Working capital cycle indicates the length of time between a company's paying for materials, entering into stock and receiving the cash from sales of finished goods. It can be determined by adding the number of days required for each stage in the cycle.

The operating cycle



Green: cash; blue: the three inventory stages (R, W, F); amber: receivables (D). Suppliers' credit (C) is the part of the round paid for by someone else, so it is subtracted (SM 9.18).

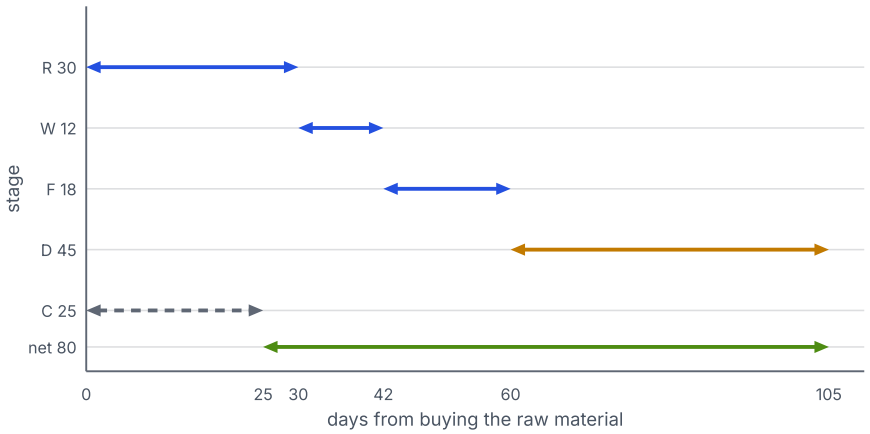
$$\text{Operating cycle} = R + W + F + D - C$$

R = raw material storage period; W = work-in-progress holding period; F = finished goods storage period; D = receivables (debtors) collection period; C = credit period allowed by suppliers (creditors) (SM 9.19). Then

$$\text{Number of operating cycles in a year} = \frac{360 \text{ or } 365}{\text{Operating cycle}}$$

and the more operating cycles a year, the better: it means a shorter cycle (SM 9.20). Tapri's factory from the opening story, on a timeline:

The factory's 80-day cycle on a timeline



Blue: raw material, WIP and finished goods (30 + 12 + 18 days); amber: debtors (45 days); grey dashed: the 25 days of suppliers' credit. Green: the net operating cycle of 80 days, from paying the supplier on day 25 to collecting from the distributor on day 105. 360 ÷ 80 = 4.5 cycles a year.

The length of the operating cycle is an indicator of the performance of management; the **net operating cycle** represents the time interval for which the firm has to negotiate for working capital from its lenders (SM 9.19). Most businesses cannot finance the inventory and receivable days with accounts payable alone; the shortfall is covered by internally generated profits, borrowed funds or both (SM 9.18). Each component has two dimensions, **time and money**: move money faster round the cycle or tie up less of it, and the business generates cash or borrows less (SM 9.18–9.19):

If you...	Then...
Collect receivables (debtors) faster	You release cash from the cycle
Collect receivables (debtors) slower	Your receivables soak up cash
Get better credit (duration or amount) from suppliers	You increase your cash resources
Shift inventory (stocks) faster	You free up cash
Move inventory (stocks) slower	You consume more cash

Interactive version: PassBook app, lesson Working capital, the operating cycle and financing policy

Turning these days into rupees, the working capital requirement, is the next lesson: estimating working capital requirement. Each part of the cycle then has a lesson of its own: inventory, receivables, payables and cash.

Formats and formulas

Reproduce the operating cycle formula and the five component formulas from memory: the denominators carry the marks.

∑ Formula

Working capital

$$\text{Net working capital} = \text{Current assets} - \text{Current liabilities}$$

$$\text{Gross working capital} = \text{Current assets}$$

∑ Formula

Operating cycle and number of cycles

$$\text{Operating cycle} = R + W + F + D - C \qquad \text{Number of cycles} = \frac{360 \text{ or } 365}{\text{Operating cycle}}$$

∑ Formula

The five components (SM 9.20)

$$R = \frac{\text{Average stock of raw material}}{\text{Average cost of raw material consumption per day}}$$

$$W = \frac{\text{Average work-in-progress inventory}}{\text{Average cost of production per day}}$$

$$F = \frac{\text{Average stock of finished goods}}{\text{Average cost of goods sold per day}}$$

$$D = \frac{\text{Average receivables}}{\text{Average credit sales per day}} \qquad C = \frac{\text{Average payables}}{\text{Average credit purchases per day}}$$

Average = (opening + closing) ÷ 2. Raw material consumed = opening stock + purchases − closing stock. Work-in-progress inventory may also be termed works cost (SM 9.19).

Item	Numerator	Denominator (per day)
R	Average raw material stock	Raw material consumed
W	Average WIP	Cost of production
F	Average finished goods	Cost of goods sold
D	Average receivables	Credit sales
C	Average payables	Credit purchases

Σ Formula

The ratio tests of the optimum (SM 9.7)

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$

$$\text{Quick ratio} = \frac{\text{Current assets} - \text{Inventory} - \text{Prepaid expenses}}{\text{Current liabilities}}$$

Thumb rule: current ratio 2 for a manufacturing firm, quick ratio at least 1; the real optimum depends on the business.

Σ Formula

Investment policy measures (SM 9.15–9.16)

$$\frac{\text{Current assets}}{\text{Fixed assets}} \qquad \text{Return on total assets} = \frac{\text{EBIT}}{\text{Total assets}}$$

Higher CA/FA ratio: conservative; lower: aggressive (fixed assets constant).

Definition

Permanent and temporary working capital (SM 9.5)

Permanent working capital refers to the base working capital, which is the minimum level of investment in the current assets that is carried by the entity at all times to carry its day-to-day activities. **Temporary working capital** refers to that part of total working capital which is required by an entity in addition to the permanent working capital; it is also called variable or fluctuating working capital.

Worked examples

Worked example **The samosa factory's cycle, from the balance sheet**

Tapri's samosa factory gives you the following information for 2025-26:

Balances (₹ lakh)	31 March 2025	31 March 2026
Raw material	38	46
Work-in-progress	20	24
Finished goods	50	58
Receivables	120	140
Payables for raw material	60	70

Annual figures for 2025-26	₹ lakh
Purchases of raw material	480
of which bought for cash at the Indore mandi	60
Cost of production	900
Cost of goods sold	892
Sales	1,260
of which cash sales at the factory outlet	165

Take a year as 365 days and round each period to the nearest day.

Required:

(i) Calculate the raw material storage period, WIP holding period, finished goods storage period, receivables collection period and credit period from suppliers.

(ii) Calculate the gross and net operating cycles and the number of operating cycles in the year.

(iii) Dev expects the new distributor portal to bring the collection period down to 30 days. Calculate the new net operating cycle and number of operating cycles.

THE WORKING

Working notes

1. Raw material consumed = opening stock + purchases − closing stock = 38 + 480 − 46 = **₹472 lakh**. All purchases count here, cash or credit: the material is consumed either way.
2. Credit purchases = 480 − 60 = **₹420 lakh**; credit sales = 1,260 − 165 = **₹1,095 lakh**.

(i) The five periods (averages: raw material 42, WIP 22, finished goods 54, receivables 130, payables 65)

Period	Working	Days
Raw material (R)	42 ÷ 472 × 365 = 32.48	32
WIP (W)	22 ÷ 900 × 365 = 8.92	9
Finished goods (F)	54 ÷ 892 × 365 = 22.10	22
Receivables (D)	130 ÷ 1,095 × 365 = 43.33	43
Payables (C)	65 ÷ 420 × 365 = 56.49	56

Each denominator is the flow that empties that stage: consumption for raw material, cost of production for WIP, cost of goods sold for finished goods, credit sales for receivables, credit purchases for payables. Cash sales never become receivables and cash purchases never become payables, so D and C use only the credit flows.

(ii) Gross operating cycle = R + W + F + D = 32 + 9 + 22 + 43 = **106 days**.

Net operating cycle = R + W + F + D − C = 106 − 56 = **50 days**.

Number of operating cycles = 365 ÷ 50 = **7.30 times**.

(iii) Collection in 30 days: net operating cycle = 32 + 9 + 22 + 30 − 56 = **37 days**; number of cycles = 365 ÷ 37 = **9.86 times**. Thirteen days less on the debtors' leg means the same costs need about a quarter less working capital (37 against 50 days of costs: 26% less).

Worked example **Premix division: how much to invest, and how to finance it**

For 2026-27, Tapri's chai premix division expects fixed assets of ₹300 lakh and spontaneous current liabilities (trade payables and accruals) of ₹150 lakh whatever the policy. Dev states three investment policies as current assets to fixed assets ratios; more stock and book debts support more sales and so more EBIT:

Investment policy	Current assets ÷ fixed assets	EBIT (₹ lakh)
Conservative	1.6	150
Moderate	1.3	143
Aggressive	1.0	126

The board adopts the moderate policy. Kesar wants the cheapest money; Dev wants to sleep soundly on the night the bank's renewal letter arrives. Its current assets are needed at ₹300 lakh all year round, plus ₹90 lakh on average for the season. Equity will provide ₹300 lakh; the rest of the assets not met by spontaneous liabilities will be borrowed, short-term at 10% or long-term at 14%. Tax is 25%; use average balances for the year. The three financing policies:

- **Conservative**: all borrowing long-term.
- **Matching (hedging)**: short-term debt finances exactly the seasonal ₹90 lakh; the rest is long-term.
- **Aggressive**: short-term debt finances the seasonal ₹90 lakh and ₹60 lakh of the permanent need.

Required:

(i) For each investment policy, calculate the current assets, net working capital, current ratio and return on total assets.

(ii) For each financing policy, calculate the short-term and long-term debt, the net working capital and current ratio (treating short-term debt as a current liability) and the return on equity.

(iii) On renewal, the bank raises its short-term rate to 16%. Recompute the return on equity under each financing policy and comment on the risk-return trade-off.

THE WORKING

(i) Investment policies (₹ lakh)

Particulars	Conservative	Moderate	Aggressive
Current assets (A) = ratio × 300	480	390	300
Fixed assets (B)	300	300	300
Total assets (A + B)	780	690	600
Current liabilities (D)	150	150	150
Net working capital (A − D)	330	240	150
Current ratio (A ÷ D)	3.20	2.60	2.00
EBIT	150	143	126
Return on total assets	**19.23%**	**20.72%**	**21.00%**

(ii) Financing policies (₹ lakh). Working note: borrowing = total assets 690 − spontaneous liabilities 150 − equity 300 = **240**.

Particulars	Conservative	Matching	Aggressive
Short-term debt	0	90	150
Long-term debt	240	150	90
EBIT	143.00	143.00	143.00
Less: interest on short-term debt @ 10%	0.00	9.00	15.00
Less: interest on long-term debt @ 14%	33.60	21.00	12.60
EBT	109.40	113.00	115.40
Less: tax @ 25%	27.35	28.25	28.85
EAT	82.05	84.75	86.55
Return on equity (EAT ÷ 300)	**27.35%**	**28.25%**	**28.85%**
Current liabilities incl. short-term debt	150	240	300
Net working capital (390 – above)	240	150	90
Current ratio (390 ÷ above)	2.60	1.63	1.30

(iii) Short-term rate 16% (₹ lakh)

Particulars	Conservative	Matching	Aggressive
Interest on short-term debt @ 16%	0.00	14.40	24.00
Interest on long-term debt @ 14%	33.60	21.00	12.60
EBT	109.40	107.60	106.40
EAT (after tax @ 25%)	82.05	80.70	79.80
Return on equity	**27.35%**	**26.90%**	**26.60%**

Comment. In investment, the aggressive policy earns the highest return on total assets (21.00%) on the thinnest cushion (net working capital ₹150 lakh, current ratio 2.00), with the risk of stock-outs and lost sales; the conservative policy is the safest and earns the least. The return on total assets moves inversely with the current assets to fixed assets ratio. In financing, cheaper short-term debt gives the aggressive policy the highest return on equity at 10% (28.85%), with the lowest liquidity (current ratio 1.30) and ₹150 lakh of loans to renew every year. When the short-term rate rises to 16% on renewal, the ranking reverses and the conservative policy, whose cost was fixed by long-term debt, earns the most (27.35%). That is the risk in short-term finance: it must be renewed at whatever rate the lender then asks. The matching policy strikes the balance between profitability and liquidity.

Exam angle

In the exam

Where this topic appeared, May 2024 to September 2026

Twelve of the ICAI documents read for this lesson examine it:

- **Operating cycle numericals.** Periods from balances, then the number of cycles and the working capital: September 2024 Q1(b) (5 marks, average stocks given); January 2025 Q2(b) (3 marks, periods given); RTP May 2026 Q9 (TYK-style, opening and closing balances); MTP January 2026 Series II Q2 (a working capital statement in which the cash-to-cash cycle is "30 days shorter than the gross operating cycle"). As a five-MCQ case: September 2025 MCQs 1–5 (back-solving cost of production, periods and purchases from the cycle). A single MCQ: MTP January 2026 Series I MCQ 7 (number of cycles).
- **Policies.** Investment and financing policies with returns and current ratios: MTP May 2026 Series II Q2(a) (8 marks).
- **Theory, 4 marks.** Factors in planning working capital, as a case (RTP September 2026 Q10(c)); trade-offs between components (RTP September 2025 Q10(c)); "Discuss the liquidity vs profitability issue" (MTP September 2026 Series II); current ratio raised by stockpiling (MTP September 2025 Series I Q4(a)); permanent and fluctuating working capital (old scheme, May 2023, 4 marks).
- **One-liners**: a contingency margin on net working capital (January 2025 MCQ 8). The chapter's own MCQs test gross and net working capital, the hedging approach and "conservative strategy = high current assets in relation to sales" (SM 9.110–9.114).

In the exam

The traps ICAI set

- Raw material storage uses **consumption**, not purchases: compute opening + purchases − closing first (TYK Problem 1, SM 9.129).
- WIP divides by **cost of production**, finished goods by **cost of goods sold** (September 2024 Q1(b)); receivables by **credit** sales and payables by **credit** purchases (SM 9.20).
- September 2025 MCQs ran the formulas backwards: from the number of cycles to the periods. Know each formula well enough to solve for its numerator.
- The paper's instruction: "Working notes should form part of the answer."

Interactive version: PassBook app, lesson Working capital, the operating cycle and financing policy

Trick

Which flow empties the tank?

Each period = average balance ÷ the daily flow that **leaves** that stage: raw material leaves by consumption, WIP by completion (cost of production), finished goods by sale (cost of goods sold), debtors by collection (credit sales), creditors by payment (credit purchases). Holds whenever the periods are measured in days of cost (or sales, for debtors).

Answer frame

"Discuss the factors to be considered while planning working capital requirement"

Name and explain the nine factors in ICAI's order, one line of example each: need for cash; desired level of inventory; receivables; short-term financing options; nature of business; market and demand conditions; technology and manufacturing policies; operating efficiency; price level changes and exchange rate fluctuations. In a case ("a pharmacy chain facing rising costs and uncertain demand"), tie each factor to a fact of the case.

"Discuss the liquidity vs profitability issue"

1. Each rupee of capital bears a cost; liquidity must be maintained with that cost in mind.
2. Higher working capital may raise revenue and profit, but funds tied up in idle assets reduce liquidity and the opportunity to earn better returns.
3. Hence a trade-off is required that increases profitability without disturbing day-to-day functioning, through the 3 Es: economy in financing, efficiency in utilisation, effectiveness in achieving objectives.
4. The component table: inventory, receivables, prepaid expenses, cash, payables, with the higher side, the trade-off tool and the lower side.

"Permanent and fluctuating working capital are both necessary. Describe."

Define each in ICAI's words (base level carried at all times, financed long-term; the part in addition to it, due to fluctuations in sales, e.g. seasonal inventory); a sketch of the two diagrams (flat and rising permanent line with the temporary bulge above); conclude that both facilitate production and sales through the operating cycle, and that the permanent part is financed by long-term and the temporary part by short-term sources.

"Explain the approaches of working capital investment"

Point	Aggressive	Conservative
Current assets	Minimal	High
Inventory, credit, cash	Low, strict, little	High, liberal, ample
Liquidity and risk	Poor liquidity, higher risk	Greater liquidity, lower risk
Return	Higher	Lower
Danger	Stock-outs, lost growth	Cost of capital, obsolescence, bad debts
CA ÷ FA	Lower	Higher

Add: moderate lies between and is what most firms follow.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the samosas measured against the wrong denominator

Dividing average WIP by cost of goods sold, or average finished goods by cost of production, or raw material by purchases, or creditors by all purchases. In Worked example 1, raw material on purchases gives 42 ÷ 480 × 365 = 31.94 days instead of 32.48; creditors on all purchases, cash ones included, give 49.43 days instead of 56.49, and the net cycle is overstated by a week. **Fix:** each stage over the flow that leaves it (the trick above); raw material consumed = opening + purchases − closing; only credit sales and credit purchases for D and C.

⚠ **Common mistake**

The creditors that were added

Writing the operating cycle as R + W + F + D + C. Suppliers' credit shortens the period for which the firm finances the cycle. **Fix:** R + W + F + D − C, and check: the net cycle is always shorter than the gross cycle.

⚠ **Common mistake**

"The higher the current ratio, the better"

A current ratio of 3.20 under a conservative policy is safer, not better: its return on total assets is the lowest. **Fix:** in theory answers say "neither too high nor too low"; the thumb rule of 2 (and quick ratio of 1) depends on the business.

⚠ **Common mistake**

Swapping aggressive and conservative

"Aggressive" sounds like "lots of stock". It is the opposite: aggressive means minimal current assets and, in financing, more short-term debt. **Fix:** aggressive takes risk to save funds; conservative pays for safety.

Practice

Check yourself

1 Tapri's kulhad unit holds raw material, work-in-progress, finished cups and debtors for 75 days in all (its gross operating cycle), and it completes 6 operating cycles in a 360-day year. How many days of credit do its suppliers allow?

Answer: _____

2 Case scenario. Tapri's cold-chain subsidiary keeps fixed assets constant at ₹10 crore. Its CFO proposes cutting current assets from ₹12 crore to ₹7 crore by tightening credit terms and holding less stock. The sales head warns that lost sales will cut EBIT from ₹4.40 crore to ₹3.74 crore. Which statement about the proposal is correct?

- A It is a conservative policy: liquidity rises and the return on total assets falls
- B Return on total assets falls, because EBIT falls by 15%
- C The current assets to fixed assets ratio rises, so risk falls
- D It is a more aggressive policy: the current assets to fixed assets ratio falls from 1.2 to 0.7, and the return on total assets still rises, from 20% to 22%

3 The finished goods storage period is calculated as average stock of finished goods divided by the average daily:

- A cost of production
- B cost of goods sold
- C credit sales
- D raw material consumption

4 Under the hedging (matching) approach, how should Tapri finance its permanent working capital of ₹40 crore and its Diwali peak of ₹22 crore?

- A Both from a short-term bank limit, since both are current assets
- B Both from long-term funds, to be safe
- C ₹40 crore from long-term sources and the ₹22 crore peak from short-term sources
- D ₹40 crore from short-term sources and the peak from equity

5 Which of these are among ICAI's factors to be considered while planning working capital requirement? Select all that apply.

- A Market and demand conditions
- B Price level changes and exchange rate fluctuations
- C The dividend payout ratio
- D Operating efficiency

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Net working capital = current assets − current liabilities; gross working capital = current assets (SM 9.3–9.4).

- Permanent: the base carried at all times, financed long-term; temporary (variable, fluctuating): the extra due to sales fluctuations, financed short-term.
- Too much: idle funds, low return, over-capitalisation. Too little: insolvency risk, lost sales. Thumb rule: current ratio 2, quick ratio 1.
- Nine determinants: cash, inventory, receivables, short-term financing, nature of business, market and demand, technology and manufacturing policy, operating efficiency, price and exchange rates.
- Investment approaches: aggressive (minimal current assets, high return, high risk), conservative (the reverse), moderate; higher CA ÷ FA = more conservative.
- Operating cycle = R + W + F + D − C; number of cycles = 360 (or 365) ÷ operating cycle.
- Denominators: consumption, cost of production, cost of goods sold, credit sales, credit purchases.
- Hedging (matching) approach: finance each asset with funds of the same maturity. (Diwali cabins are hired, not bought.)

Answer key · Check yourself

1 **15**

The number of cycles is 360 ÷ the net operating cycle, so the net cycle is 360 ÷ 6 = 60 days. Net cycle = gross cycle − suppliers' credit, so C = 75 − 60 = 15 days. Dividing 360 by the gross cycle gives 4.8 cycles, not 6: suppliers' credit is what makes the difference.

2 **D It is a more aggressive policy: the current assets to fixed assets ratio falls from 1.2 to 0.7, and the return on total assets still rises, from 20% to 22%**

Fewer current assets on the same fixed assets is the aggressive direction (lower CA ÷ FA). Total assets fall from ₹22 crore to ₹17 crore, faster than EBIT: ₹4.40 ÷ ₹22 = 20%, ₹3.74 ÷ ₹17 = 22%. The price is lower liquidity and the risk of stock-outs, and here lost sales too.

3 **B cost of goods sold**

Finished goods leave the store when sold, so the flow is cost of goods sold. Cost of production is the denominator for work-in-progress.

4 **C ₹40 crore from long-term sources and the ₹22 crore peak from short-term sources**

Matching finances each asset with funds of about the same maturity: the permanent base, needed all year, from long-term debt and equity; the seasonal part from short-term sources. Financing everything long-term is the conservative choice; financing part of the base short-term is aggressive.

5 **A Market and demand conditions; B Price level changes and exchange rate fluctuations; D Operating efficiency**

ICAI lists nine: need for cash, desired level of inventory, receivables, short-term financing options, nature of business, market and demand conditions, technology and manufacturing policies, operating efficiency, and price level changes and exchange rate fluctuations. Dividend policy is not one of them.

9.2 Estimating working capital requirement

Turning the operating cycle into rupees: the operating cycle shortcut, the item-by-item statement of working capital (ICAI's format), the cash cost basis, the half-period rule for WIP and the effect of double shift working.

In plain words

The bank wants a number

Tapri is opening a chai premix plant at Dewas, and the bank will sanction a working capital limit only against a proper estimate.

> **Kesar:** Write ₹1 crore. Banks like round numbers. Sab adjust ho jayega.
>
> **Dev:** Banks like workings. Ritu, how many weeks does each rupee sit in each place?
>
> **Ritu:** Per unit, please, then per week. Four weeks in raw material, two in process, three in the warehouse, six with the distributors. Suppliers wait four weeks.
>
> **Dev:** Then the estimate writes itself. Mostly.

The idea is one line long: **the money stuck in any stage = the cost flowing through that stage per week (or month, or day) × the time it stays there**. If a plant uses ₹2,50,000 of raw material a week and holds four weeks' stock, the store holds ₹10,00,000. Do that for raw material, work-in-progress, finished goods and debtors, add the cash balance, and you have the **current assets**. Do it again for what suppliers, workers and service providers let you owe, the **current liabilities**, and subtract. Add a **safety margin** for the unexpected, and that is the working capital to ask the bank for.

Two refinements carry most of the marks. Half-made goods have received all their material but, on average, only half their labour and overheads, because those build up evenly while the goods are in process. And a debtor's bill includes profit, which you never spent, and depreciation, which you never paid out in cash; so many experts measure debtors and finished goods at their **cash cost**, the money actually blocked.

Think of it like

The railway platform

How many people stand on the platform? Arrivals per minute × minutes each one waits. Six arrivals a minute, each waiting ten minutes: sixty people on the platform at any moment. Working capital is the same count in rupees: what flows in per day × the days it waits. Faster trains (a shorter cycle) mean an emptier platform with the same traffic.

Key idea

Each current asset = the cost flowing through it per period × the periods it is held; each current liability = the expense that flows into it per period × the periods it stays unpaid. Value each item at the cost actually paid in (for cash cost basis: no profit, no depreciation), take half the conversion cost for work-in-progress when it accrues evenly, subtract the liabilities and add the safety margin.

How it works

Three ways to estimate (SM 9.16–9.17)

The operating cycle is one of the most reliable methods; ICAI lists three:

1. **Current assets holding period**: working capital from the average holding period of current assets, related to costs on the basis of the previous year's experience. This is essentially the operating cycle concept.
2. **Ratio of sales**: working capital as a ratio of sales, on the assumption that current assets change with sales.
3. **Ratio of fixed investments**: working capital as a percentage of fixed investments.

The choice is affected by seasonal fluctuations, the accuracy of the sales forecast, investment cost and variability in sales price; the production cycle and the credit and collection policies must be given due weight. The operating cycle itself is built in working capital and the operating cycle.

The operating cycle shortcut

When a question gives the periods and the annual operating cost, the requirement is one division (TYK Problem 1, SM 9.130):

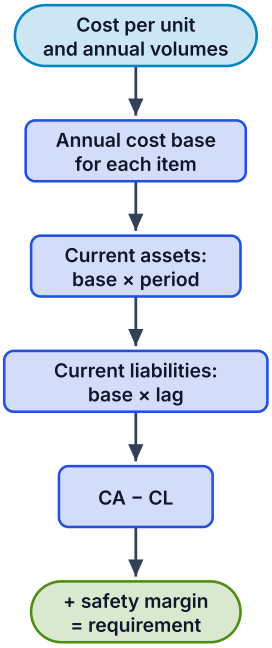
$$\text{Working capital required} = \frac{\text{Annual operating cost}}{\text{Number of operating cycles}}$$
$$= \text{Operating cost per day} \times \text{Net operating cycle (days)}$$

It assumes costs flow evenly and every rupee of operating cost waits the whole net cycle. On the cash cost basis, depreciation is taken out of the operating cost first.

The statement method, item by item (SM 9.22–9.24)

The holding period of each current asset and current liability can contract or expand the net operating cycle: the shorter the cycle, the lower the requirement. Each item is estimated separately, then set out in ICAI's statement:

Building the statement of working capital



The base differs by item: raw material at material cost, WIP at material plus part of conversion, finished goods at cost of production, debtors at cost of sales (or sales value), creditors at purchases. The period is the months, weeks or days held, divided by 12, 52 or 360 (365).

Current assets (production in units ÷ 12 months or 365 days × cost per unit × period):

- **Raw materials**: estimated production × estimated cost per unit × average raw material storage period.
- **Work-in-progress**: estimated production × estimated WIP cost per unit × average WIP holding period.
- **Finished goods**: estimated production × estimated cost of production per unit × average finished goods storage period.
- **Receivables**: estimated credit sales × estimated cost of sales (excluding depreciation) per unit × average collection period. Only cash cost is considered for debtors and finished goods: sales to debtors include cost and profit, and the funds required need not include profit; non-cash expenses like depreciation are also excluded (SM 9.23).
- **Cash and cash equivalents**: the minimum desired cash and bank balance is added.

Current liabilities are deducted, so the requirement is lowered to the extent of current liabilities (**other than bank credit**) arising in the normal course of business (SM 9.23–9.24):

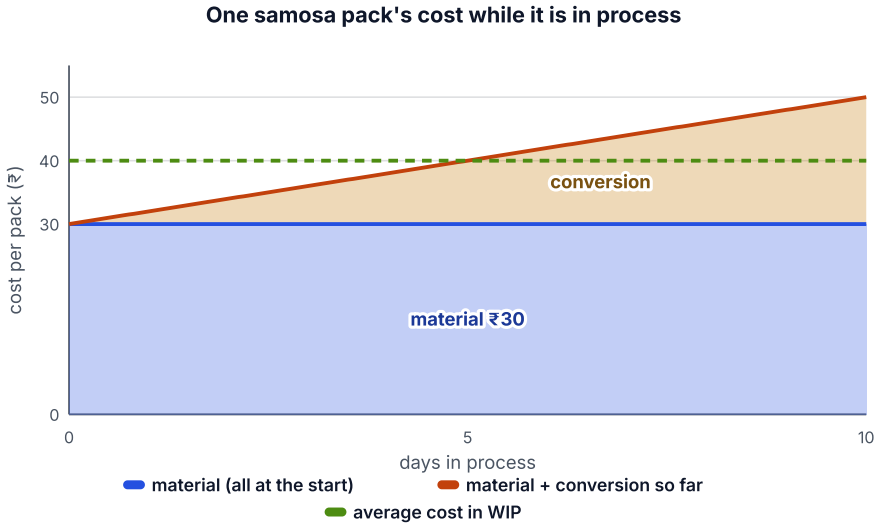
- **Trade payables**: estimated credit purchases × credit period allowed by suppliers.

- **Direct wages**: estimated labour hours × wage rate per hour × average time lag in payment of wages.
- **Overheads** (other than depreciation and amortisation): estimated overheads × average time lag in payment.

Number of days in a year may be taken as 365 or 360 (SM 9.24). Bank borrowing is not deducted: it is what the estimate is for.

Work-in-progress: the half-period rule

In ICAI's Illustration 3, each unit is in process for one month, the raw materials being fed into the pipeline immediately and the labour and overhead costs accruing evenly during the month. So WIP holds one month's raw material but, on average, ½ month's labour and overheads: on the first day of the month the labour in WIP is nil, on the last day a full month's (SM 9.26).



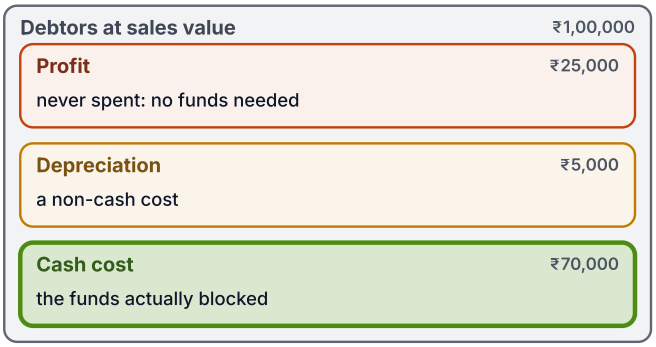
Material of ₹30 goes in on day 1; labour and overheads of ₹20 build up evenly over 10 days. A pack is worth ₹30 on entry and ₹50 on completion, ₹40 on average: full material plus half the conversion cost.

When a question states a degree of completion ("raw material 100%, other costs 50% complete", TYK Problem 2, SM 9.130), use it: that is the same rule written as percentages.

The cash cost basis (SM 9.28)

In current assets like sundry debtors and finished goods, the funds actually blocked are less than their book amount. ICAI's example:

₹1 lakh of debtors: what is really blocked



Cost of sales is ₹75,000 of the ₹1,00,000, the rest is profit; ₹5,000 of the cost is depreciation. Only ₹70,000 must be financed (SM 9.28). Finished goods, valued at cost, likewise leave out non-cash costs.

Under this approach debtors are calculated not as a percentage of sales value but as a percentage of **cash costs**, and finished goods are valued at cash costs too (SM 9.28). ICAI's Illustration 4 builds two working notes first: **cash cost of production** (materials + wages + cash manufacturing expenses) for finished goods, and **cash cost of sales** (cash cost of production + administrative + selling expenses) for debtors (SM 9.30).

Double shift working (SM 9.31)

Double shift's greatest economy is the greater use of fixed assets; little additional fund is needed for assets, but working capital changes:

1. Stocks rise with production, but perhaps not in proportion: the minimum levels may not be much higher, so stocks may not have to double when production doubles.
2. **Materials in process do not change**: work started in the first shift is completed in the second, so capital tied up in WIP stays the same, unless second-shift workers are paid more.
3. Larger purchases may bring economies (better terms from vendors), and the fixed costs may not all increase: **fixed cost per unit falls**.

ICAI's Illustration 5 values every item at the new, lower cost per unit, doubles the units of raw material, finished goods, debtors and liabilities, and keeps the units of WIP unchanged (SM 9.32–9.33).

Formats and formulas

Reproduce ICAI's statement from memory: the headings, the order and "Safety Margin" carry marks.

Estimation of Working Capital Requirements (SM 9.24–9.25)	₹	₹
I. Current Assets:		
Inventories: Raw materials; Work-in-process; Finished goods	---	---
Receivables: Trade debtors; Bills	---	---
Prepaid expenses		---
Minimum cash balance		---
Gross Working Capital		**---**
II. Current Liabilities:		
Trade payables; Bills payable	---	
Wages payable; Payables for overheads	---	---
III. Excess of current assets over current liabilities [I − II]		**---**
IV. Safety margin		---
V. Net Working Capital [III + IV]		**---**

Interactive version: PassBook app, lesson Estimating working capital requirement

∑ **Formula**

Operating cycle method

$$\text{Working capital} = \frac{\text{Annual operating cost}}{\text{Number of operating cycles}}$$

Cash cost basis: annual operating cost less depreciation.

∑ **Formula**

Current assets

$$\text{Item} = \frac{\text{Annual cost base}}{12 \text{ months (or } 365 \text{ days)}} \times \text{Holding period}$$

Item	Annual cost base (total cost basis)	Cash cost basis
Raw materials	Material consumed	Same
Work-in-progress	Material + share of conversion cost	Without depreciation
Finished goods	Cost of production	Cash cost of production
Debtors	Cost of sales or sales value	Cash cost of sales
Prepaid expenses	The expense paid in advance	Same
Trade payables	Credit purchases	Same
Wages, overheads payable	The expense paid in arrear	Same (no depreciation)

∑ **Formula**

Cash costs (Illustration 4, SM 9.30)

$$\text{Cash cost of production} = \text{Materials} + \text{Wages} \quad + \text{Cash manufacturing expenses}$$

$$\text{Cash cost of sales} = \text{Cash cost of production} \quad + \text{Administrative expenses} \quad + \text{Selling expenses}$$

Adjust for opening and closing finished stock when they differ (TYK Problem 5, SM 9.139).

∑ **Formula**

Work-in-progress (material at the start, conversion accruing evenly)

$$\text{WIP} = \text{Material for the period in process} \quad + \frac{1}{2} \times \text{Conversion cost for that period}$$

Worked examples

Worked example **Kulhad Works: the operating cycle shortcut, from turnover ratios**

Tapri Kulhad Works, Tapri's own pottery unit, which makes clay cups for its cafés, gives you its turnover ratios for 2025-26:

Turnover ratio	Times a year
Raw material (on materials consumed)	9
Work-in-progress (on cost of production)	24
Finished goods (on cost of goods sold)	18
Debtors (on credit sales)	8
Creditors (on credit purchases)	6

Its annual operating cost (cost of goods sold plus administration and selling expenses) is ₹57,60,000, including depreciation of ₹4,32,000. Assume a 360-day year.

Required:

(i) Calculate the net operating cycle and the number of operating cycles in the year.

(ii) Calculate the working capital requirement by the operating cycle method, on total cost and on cash cost basis.

(iii) Ritu has persuaded the clay suppliers to allow 75 days' credit. How much working capital does this release on each basis?

THE WORKING

Working note: periods from turnover ratios. A stock that turns over n times a year is held 360 ÷ n days (in months, 12 ÷ n: ICAI's collection period from the receivables turnover ratio, SM 9.80).

Period	Working	Days
Raw material (R)	360 ÷ 9	40
WIP (W)	360 ÷ 24	15
Finished goods (F)	360 ÷ 18	20
Debtors (D)	360 ÷ 8	45
Creditors (C)	360 ÷ 6	60

(i) Net operating cycle = R + W + F + D − C = 40 + 15 + 20 + 45 − 60 = **60 days**. **Number of operating cycles** = 360 ÷ 60 = **6 times**.

(ii) Working capital required

$$\text{Total cost basis} = \frac{57,60,000}{6} = ₹9,60,000$$

$$\text{Cash cost basis} = \frac{57,60,000 - 4,32,000}{6} = ₹8,88,000$$

(iii) Creditors at 75 days: net operating cycle = 40 + 15 + 20 + 45 − 75 = 45 days; 360 ÷ 45 = 8 cycles.

Basis	Before (₹)	After (₹)	Released (₹)
Total cost (57,60,000 ÷ 6, ÷ 8)	9,60,000	7,20,000	2,40,000
Cash cost (53,28,000 ÷ 6, ÷ 8)	8,88,000	6,66,000	2,22,000

Each extra day of suppliers' credit releases one day's operating cost: 15 days × (₹57,60,000 ÷ 360 = ₹16,000) = ₹2,40,000.

Worked example **Dewas premix plant: the statement, week by week**

The Dewas plant will produce 31,200 cartons of chai premix in 2026-27, evenly over 52 weeks. Estimated cost and price per carton:

Particulars	₹ per carton
Raw materials	480
Direct wages	120
Overheads (no depreciation)	240
Total cost	**840**
Selling price	1,000

Further information:

- Raw materials remain in stock for 4 weeks; suppliers allow 4 weeks' credit.
- Each carton is in process for 2 weeks: materials are issued at the start, wages and overheads accrue evenly.
- Finished cartons stay in the warehouse for 3 weeks; distributors are allowed 6 weeks' credit; all sales are on credit.
- Wages are paid 1 week in arrear and overheads 2 weeks in arrear.
- A cash balance of ₹1,00,000 is to be kept, and a safety margin of 10% added to net working capital.

Required:

(i) Prepare a statement of working capital requirement, taking debtors at cost.

(ii) Ritu proposes producing to order: finished cartons would stay 1 week instead of 3, but raw materials would need 7 weeks' stock instead of 4. Keeping creditors at 4 weeks of consumption, calculate the new requirement after the safety margin, and the change.

THE WORKING

Working notes (weekly production = 31,200 ÷ 52 = 600 cartons)

1. Raw material stock = 600 × ₹480 × 4 weeks = **₹11,52,000**.
2. WIP: 2 weeks in process = 1,200 cartons. Materials 1,200 × ₹480 = ₹5,76,000; wages 1,200 × ₹120 × ½ = ₹72,000; overheads 1,200 × ₹240 × ½ = ₹1,44,000; total **₹7,92,000**.
3. Finished goods = 600 × ₹840 × 3 weeks = **₹15,12,000**.
4. Debtors at cost = 600 × ₹840 × 6 weeks = **₹30,24,000**.
5. Creditors = 600 × ₹480 × 4 weeks = ₹11,52,000 (production is even, so purchases equal consumption). Wages payable = 600 × ₹120 × 1 week = ₹72,000. Overheads payable = 600 × ₹240 × 2 weeks = ₹2,88,000.

(i) Statement of working capital requirement

Particulars	₹	₹
A. Current assets		
Raw materials (WN 1)	11,52,000	
Work-in-progress (WN 2)	7,92,000	
Finished goods (WN 3)	15,12,000	
Debtors at cost (WN 4)	30,24,000	
Cash	1,00,000	65,80,000
B. Current liabilities		
Creditors for materials	11,52,000	
Wages payable	72,000	
Overheads payable	2,88,000	15,12,000
Net working capital (A − B)		**50,68,000**
Add: Safety margin @ 10%		5,06,800
Working capital requirement		**55,74,800**

(ii) Producing to order

Item	Now (₹)	Producing to order (₹)
Raw materials (600 × ₹480 × 4; × 7)	11,52,000	20,16,000
Finished goods (600 × ₹840 × 3; × 1)	15,12,000	5,04,000
Other current assets (WIP, debtors, cash)	39,16,000	39,16,000
Less: current liabilities	15,12,000	15,12,000
Net working capital	50,68,000	49,24,000
Add: safety margin @ 10%	5,06,800	4,92,400
Working capital requirement	**55,74,800**	**54,16,400**

The requirement falls by **₹1,58,400**: two weeks less of finished goods at ₹840 a carton (₹10,08,000) outweigh three weeks more of raw material at ₹480 (₹8,64,000), and the margin is 10% of the ₹1,44,000 difference.

Had the debtors been taken at selling price, they would be 600 × ₹1,000 × 6 = ₹36,00,000, which is ₹5,76,000 more: six weeks' profit (600 × ₹160 × 6), money the plant never spent. That is why the Study Material's formula takes debtors at cost (SM 9.23); when a question says which basis it wants, follow the question.

Worked example **Samosa factory: cash cost basis, with cash sales and work-in-progress**

Tapri's samosa factory projects the following for 2026-27:

Particulars	₹
Sales: 25% for cash at the factory outlet, the rest to distributors at 1½ months' credit	1,20,00,000
Materials consumed (suppliers allow one month's credit)	45,00,000
Wages (paid ½ month in arrear)	18,00,000
Manufacturing overheads, including depreciation of ₹6,00,000 (the cash part is paid one month in arrear)	27,00,000
Administrative expenses, including warehouse rent of ₹2,40,000 paid half-yearly in advance (the rest is paid one month in arrear)	7,20,000
Selling expenses (paid one month in arrear)	3,60,000

Further information:

- Raw materials are held for one month; finished goods for half a month, valued at cash cost of production.
- Each pack is in process for half a month: materials are issued at the start, wages and overheads accrue evenly.
- A cash balance of ₹2,50,000 is to be kept, and a safety margin of 10% added.

Required: Prepare a statement of working capital requirement on cash cost basis, and state how much higher the debtors would be at sales value.

THE WORKING

Working notes

1. Cash manufacturing overheads = 27,00,000 − 6,00,000 = ₹21,00,000. **Cash cost of production** = 45,00,000 + 18,00,000 + 21,00,000 = **₹84,00,000**.
2. Work-in-progress (half a month in process): materials 45,00,000 × ½ ÷ 12 = ₹1,87,500; wages and cash overheads (18,00,000 + 21,00,000) × ½ ÷ 12 × ½ = ₹81,250; total **₹2,68,750**.
3. **Cash cost of sales** = 84,00,000 + 7,20,000 + 3,60,000 = **₹94,80,000**. Cash and credit sales carry the same price, so 75% of it belongs to the credit sales: ₹71,10,000. Debtors = 71,10,000 × 1.5/12 = **₹8,88,750**.
4. Rent paid half-yearly in advance: six months are prepaid, 2,40,000 × 6/12 = **₹1,20,000** (ICAI takes the whole advance period, as for the sales promotion paid quarterly in advance in Illustration 4, SM 9.29).
5. Administrative expenses outstanding exclude the rent: (7,20,000 − 2,40,000) × 1/12 = **₹40,000**.

Statement of working capital requirement (cash cost basis)

Particulars	₹	₹
A. Current assets		
Raw materials (45,00,000 × 1/12)	3,75,000	
Work-in-progress (WN 2)	2,68,750	
Finished goods (84,00,000 × ½ ÷ 12)	3,50,000	
Debtors (WN 3)	8,88,750	
Prepaid rent (WN 4)	1,20,000	
Cash	2,50,000	22,52,500
B. Current liabilities		
Creditors (45,00,000 × 1/12)	3,75,000	
Wages outstanding (18,00,000 × ½ ÷ 12)	75,000	
Manufacturing overheads outstanding (21,00,000 × 1/12)	1,75,000	
Administrative expenses outstanding (WN 5)	40,000	
Selling expenses outstanding (3,60,000 × 1/12)	30,000	6,95,000
Net working capital (A − B)		**15,57,500**
Add: Safety margin @ 10%		1,55,750
Working capital requirement		**17,13,250**

At sales value, debtors would be 90,00,000 × 1.5/12 = ₹11,25,000 against ₹8,88,750: ₹2,36,250 higher, the profit and depreciation inside the bills. The cash sales stay out of debtors on either basis.

Assumptions: all purchases are on credit and the raw material stock is constant, so purchases equal consumption; depreciation is left out of work-in-progress and finished goods as well as debtors.

Worked example

Kulhad Works goes double shift

Tapri Kulhad Works makes 30,000 crates of kulhads a year in one shift and sells them at ₹24 a crate. Cost per crate:

Per crate	₹
Materials	8.00
Wages (of which ₹3.00 is fixed: supervisors' salaries)	6.00
Overheads (of which ₹4.50 is fixed)	6.00
Total cost	**20.00**

Its working capital norms: raw materials, 2 months' consumption; work-in-progress, 1,500 crates in the kilns at any time, valued at materials plus wages; finished goods, 1 month's output; debtors, 2 months' sales, valued at total cost; creditors, 2 months' purchases; wages and overheads paid half a month in arrear.

To supply Tapri's new cafés, production and sales will double with a night shift: the kilns will never cool, and Ritu's spreadsheet will never close. Night-shift workers get 20% more than the variable day wage; fixed wages and fixed overheads stay the same in total, and so do material prices. Because the minimum stock does not double, Ritu expects the raw material stock to rise by only half, to 7,500 crates' worth; the other norms stay as they are.

Required: Prepare a comparative statement of working capital requirement and compute the additional working capital needed for double shift working.

THE WORKING

Working notes

1. Fixed costs in total: wages ₹3.00 × 30,000 = ₹90,000; overheads ₹4.50 × 30,000 = ₹1,35,000.
2. Variable wage per crate: day ₹3.00, night ₹3.00 × 1.20 = ₹3.60; on double shift the average is ₹3.30.
3. Cost per crate:

Element	Single shift (₹)	Double shift (₹)
Materials	8.00	8.00
Wages: variable	3.00	3.30
Wages: fixed (₹90,000)	3.00	1.50
Overheads: variable	1.50	1.50
Overheads: fixed (₹1,35,000)	4.50	2.25
Total cost	**20.00**	**16.55**
Materials + wages (for WIP)	14.00	12.80

4. Quantities. Single shift (2,500 crates a month): raw materials 5,000; WIP 1,500; finished goods 2,500; debtors 5,000; creditors 5,000; wages and overheads payable 1,250. Double shift (5,000 a month): raw materials 7,500; WIP 1,500; finished goods 5,000; debtors 10,000; creditors 10,000; wages and overheads payable 2,500.

Comparative statement of working capital requirement

Particulars	Single: crates × ₹	₹	Double: crates × ₹	₹
Raw materials	5,000 × 8.00	40,000	7,500 × 8.00	60,000
Work-in-progress	1,500 × 14.00	21,000	1,500 × 12.80	19,200
Finished goods	2,500 × 20.00	50,000	5,000 × 16.55	82,750
Debtors (at cost)	5,000 × 20.00	1,00,000	10,000 × 16.55	1,65,500
Current assets (A)		**2,11,000**		**3,27,450**
Creditors for materials	5,000 × 8.00	40,000	10,000 × 8.00	80,000
Wages payable	1,250 × 6.00	7,500	2,500 × 4.80	12,000
Overheads payable	1,250 × 6.00	7,500	2,500 × 3.75	9,375
Current liabilities (B)		**55,000**		**1,01,375**
Working capital (A − B)		**1,56,000**		**2,26,075**

Additional working capital = ₹2,26,075 − ₹1,56,000 = **₹70,075**.

Note

Why the numbers move as they do

WIP stays at 1,500 crates because the night shift finishes what the day shift started (SM 9.31). Its value per crate falls from ₹14.00 to ₹12.80 although night workers are paid more, because the fixed wage is now spread over twice the crates. Creditors are taken at 2 months of consumption, the steady state of ICAI's first, policy approach to Illustration 5; in the first year of the night shift, purchases would also include the 2,500 crates added to stock (ICAI's second approach, SM 9.34–9.35). Production doubles; working capital rises by about 45%.

Exam angle

In the exam

Where estimation appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, estimating working capital came up 12 times in 11 documents.

- **Operating cycle method, 3 to 5 marks**: September 2024 Q1(b) (periods from averages, then the amount); January 2025 Q2(b) (periods given; the suggested answer also shows a cash cost alternative excluding depreciation); MTP September 2025 Series I (with a 10% contingency); RTP May 2026 Q9.
- **Cash cost statements, 5 marks**: September 2026 Q1(c) (sales at 2 months' credit, sales promotion quarterly in advance, cash at 50% of current liabilities, 15% safety margin); MTP May 2024 Series II (15% margin, ignore WIP); RTP May 2025 (domestic and export sales at different prices).
- **As five-MCQ case scenarios, 2 marks each**: May 2025 MCQs 1–5 (a working capital loan proposal on cash cost basis: cash costs, finished goods and debtors, current assets, current liabilities, the requirement); RTP January 2026 MCQs (gross working capital, requirement, current and quick ratios); September 2025 MCQs 1–5 (back from the number of cycles).
- **Newly formed company applying to a bank, 6 marks**: MTP January 2025 Series II Q2(a) (WIP at prime cost, 90% material, 60% conversion).
- **One-liner**: January 2025 MCQ 8 (a 20% contingency on net working capital).

In the exam

The traps ICAI set

- "Safety margin 15% of the net working capital **inclusive of the margin amount**" (MTP January 2025 Series II): the requirement is (CA − CL) ÷ 0.85, not × 1.15.
- Cash "to the extent of 50% of current liabilities" (September 2026 Q1(c)): compute the liabilities first.
- Depreciation inside "cost of production" or "manufacturing overheads": take it out for the cash cost basis (May 2025 MCQs; RTP May 2025).
- The paper's instruction: "Working notes should form part of the answer." Every line of the statement needs its working.

Interactive version: PassBook app, lesson Estimating working capital requirement

Trick

Months, weeks or days: divide by the same unit

Annual base ÷ 12 × months, ÷ 52 × weeks, or ÷ 360 (365) × days, whichever unit the question uses for the periods. Never mix: a "1½ months" credit with a 360-day year is 45 days ÷ 360, the same as 1.5 ÷ 12.

Answer frame

"Discuss the estimation of working capital need based on the operating cycle process" (SM 9.115, Q3)

1. The operating cycle is one of the most reliable methods of computing working capital (the others: ratio of sales, ratio of fixed investments); it estimates working capital from the average holding period of current assets, related to costs.

2. Operating cycle = R + W + F + D − C, each component defined, with its formula (average balance ÷ daily cost or sales).
3. Number of operating cycles = 360 (365) ÷ operating cycle; working capital = annual operating cost ÷ number of cycles.
4. Item by item: raw materials, WIP, finished goods, debtors (at cash cost, excluding profit and depreciation), cash; less trade payables, wages and overheads payable.
5. Add a safety margin; the shorter the cycle, the lower the requirement.

The numerical, three ways

Loses marks

A statement with bare figures: "Debtors 8,88,750; Creditors 3,75,000..."

No base, no period: an examiner cannot give method marks to a wrong number.

Passes

Each line shows base × period ("94,80,000 × 1.5/12"), but the cash sales are left inside debtors and depreciation inside the finished goods.

Two of the traps cost the marks.

Full marks

Working notes first (cash cost of production, WIP, cash cost of sales and the credit sales' share, the prepaid rent), then ICAI's statement with every base and period visible, safety margin last, assumptions stated.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the overdraft that was counted as cash

TYK Problem 7's "₹2,50,000 available including the overdraft limit of ₹75,000 not yet utilised" written as cash ₹2,50,000. The overdraft is bank credit: it finances working capital, it is not part of it. **Fix:** cash = ₹1,75,000 (SM 9.143); and do not deduct bank borrowing as a current liability.

⚠ **Common mistake**

Profit smuggled into the debtors

In Worked example 3, writing debtors as 1½ months of all sales (₹15,00,000) instead of 1½ months of the cash cost of the credit sales (₹8,88,750): the profit, the depreciation and the cash sales all smuggled in. **Fix:** build the cash cost of sales working note first, take the credit sales' share, and the debtors come from it.

⚠ **Common mistake**

WIP charged with a full period of conversion

Taking WIP as 2 weeks of the full ₹840 cost (₹10,08,000) when wages and overheads accrue evenly. **Fix:** full material, half conversion: ₹7,92,000. Use the stated percentage when the question gives a degree of completion.

⚠ **Common mistake**

WIP doubled in double shift

Doubling the WIP units with production. **Fix:** the quantity in process is unchanged, because work started in the first shift is completed in the second (SM 9.31); only its value per unit changes (a lower fixed wage per unit, a higher night-shift wage, cheaper material if there is a discount).

Practice

Check yourself

1 Tapri's packaging unit has an annual operating cost of ₹45,00,000, and its working capital requirement by the operating cycle method is ₹9,00,000 (360-day year). What is its net operating cycle, in days?

Answer: _____

2 Case scenario. Tapri's new tea-bag line makes 2,400 boxes a month evenly. Material of ₹50 a box is issued at the start of processing; labour and overheads of ₹30 a box accrue evenly over the one month each box spends in process. What is the work-in-progress to be included in the working capital statement?

A ₹1,92,000

B ₹1,56,000

C ₹1,20,000

D ₹96,000

3 Annual cash cost of sales is ₹72,00,000 and annual sales ₹90,00,000. Customers get two months' credit. On cash cost basis, debtors in the working capital statement are:

A ₹15,00,000

B ₹12,00,000

C ₹18,00,000

D ₹6,00,000

4 A company's current assets are ₹60,00,000 and current liabilities ₹24,00,000. It wants a safety margin of 10% of the net working capital required, inclusive of the margin itself. What is the working capital requirement, in rupees?

Answer: _____

5 Which of these is NOT deducted as a current liability when estimating the working capital requirement?

A Wages payable for the last half month

B Creditors for raw materials

C The cash credit the company is applying for

D Manufacturing expenses paid one month in arrear

Answers: the answer key at the end of this lesson.

Recap

- Three methods: current assets holding period (operating cycle), ratio of sales, ratio of fixed investments (SM 9.16–9.17).
- Operating cycle method: working capital = annual operating cost ÷ number of operating cycles; cash basis: depreciation out first.
- Each item = annual base ÷ 12 (52, 360, 365) × period held; current liabilities other than bank credit are deducted.
- WIP with material at the start and conversion accruing evenly: full material + half conversion for the period in process.
- Cash cost basis: finished goods at cash cost of production, debtors at cash cost of sales (no profit, no depreciation).
- ICAI's statement: current assets (gross working capital) − current liabilities = excess; + safety margin = net working capital.
- Double shift: WIP units unchanged; other stocks rise, though not always in proportion; every item valued at the new cost per unit (lower fixed cost per unit).
- Overdraft limits are not cash; a margin "inclusive of itself" divides. (Count the platform, not the trains.)

Answer key · Check yourself

1 **72**

Number of operating cycles = annual operating cost ÷ working capital = ₹45,00,000 ÷ ₹9,00,000 = 5; net operating cycle = 360 ÷ 5 = 72 days. Check: one day's operating cost, ₹12,500, × 72 days = ₹9,00,000.

2 **B** **₹1,56,000**

One month in process = 2,400 boxes. Material in full: 2,400 × ₹50 = ₹1,20,000; conversion at half: 2,400 × ₹30 × ½ = ₹36,000; total ₹1,56,000. ₹1,92,000 charges full conversion; ₹1,20,000 forgets it; ₹96,000 halves everything.

3 **B** **₹12,00,000**

Cash cost basis: ₹72,00,000 × 2/12 = ₹12,00,000. ₹15,00,000 is at sales value; ₹18,00,000 uses three months; ₹6,00,000 one month.

4 **4000000**

Excess of CA over CL = ₹36,00,000, which is 90% of the requirement: ₹36,00,000 ÷ 0.90 = ₹40,00,000. Adding 10% (₹39,60,000) answers a different question.

5 **C** **The cash credit the company is applying for**

Current liabilities other than bank credit are deducted (SM 9.23); the bank limit is what the estimate is meant to justify.

9.3 Management of receivables, credit policy, factoring and forfaiting

How a firm decides whom to give credit, on what terms and how to collect; evaluating credit policies by the total and incremental approaches; and turning receivables into cash through pledging, factoring and forfaiting.

In plain words

The distributors want 90 days

BrewBucks has launched an instant coffee and is giving distributors 60 days to pay. Tapri's premix distributors now want 90.

> **Kesar:** Give them 90 days. More credit, more sales. Simple.
>
> **Dev:** More sales, yes. Also more money stuck with distributors, more bad debts, more phone calls to chase them.
>
> **Kesar:** So how many days is right?
>
> **Dev:** The number of days where the extra profit stops paying for the extra cost. I have a table for that. Two, actually. Even my optimism is discounted at 12%.

Selling on credit creates **receivables** (debtors): money customers owe the firm. **Management of receivables** is planning and controlling that debt, also called trade credit management. Credit brings sales, but every rupee waiting with a customer is a rupee the firm has to finance, some customers never pay (**bad debts**), and chasing them costs money. Too little credit, on the other hand, and competitors take the customers.

So the finance manager decides three things: the **credit policy** (to whom, for how long, with what discount for early payment), the **credit analysis** (how risky this particular customer is) and the **control** of receivables (following up, collecting). To compare policies, Dev writes down for each one the profit it earns, less the cost of the money locked up in debtors; the policy with the biggest net figure wins.

When cash is short, the debts themselves can be turned into cash. Tapri can borrow against them (**pledging**), sell them to a specialist who collects them for a fee (**factoring**), or, for export bills, sell the importer's bills to a bank that takes the whole risk of non-payment (**forfaiting**).

Think of it like

The kirana shop's udhaar diary

The corner shop keeps a diary of who owes what. The shopkeeper decides whose name goes in the diary (credit analysis), how long they may run up a bill and whether a quick payer gets a few rupees off (credit policy), and when to send the boy round with a reminder (control). If he needs cash for Diwali stock, he can borrow against the diary, or hand it to a recovery agent who pays him most of it now and collects the rest himself for a cut. That agent is a factor.

Key idea

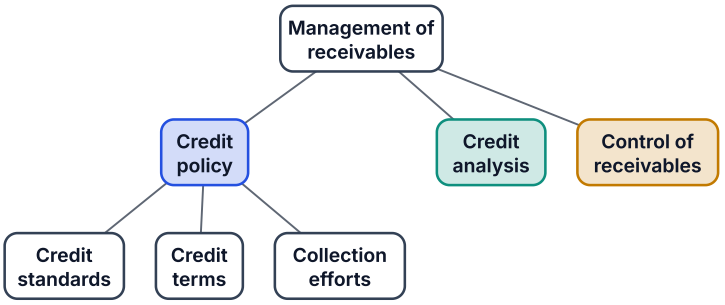
Credit is worth giving only up to the point where the extra contribution from extra sales exceeds the extra bad debts, cash discount, collection cost and the opportunity cost of the money locked in receivables. Factoring and forfaiting turn receivables into cash: factoring is worth it when its effective cost is below the firm's cost of borrowing.

How it works

Meaning and the three aspects

The basic objective of management of receivables is to optimise the return on investment on these assets (SM 9.70). Large receivables bring bad debts and collection costs; small ones may restrict sales when competitors offer more liberal terms. There are basically three aspects (SM 9.70–9.71):

The three aspects of management of receivables



Blue: the credit policy, which covers credit standards, credit terms and collection efforts; teal: credit analysis of each customer; amber: control (follow-up and collection policy).

1. **Credit policy**: a balanced policy covering credit standards, credit terms and collection efforts. It involves a trade-off between the profits on additional sales from extending credit and the cost of carrying those debtors and bad debt losses; it decides the credit period, cash discount and other matters. The credit period is stated in net days: "net 50" means customers repay not later than 50 days. The cash discount policy specifies (a) the rate of cash discount, (b) the cash discount period and (c) the net credit period: "3/15 net 60" means a 3% discount if the customer pays within 15 days; otherwise payment within 60 days.
2. **Credit analysis**: how risky it is to advance credit to a particular party: due diligence or reputation check of the customer's creditworthiness.
3. **Control of receivables**: following up debtors and deciding a suitable collection policy; both laying down credit policies and executing them.

The cost of maintaining receivables (SM 9.71): (i) the cost of additional funds blocked in receivables, as interest (loan funds) or opportunity cost (own funds); (ii) administrative costs: record keeping, investigation of creditworthiness; (iii) collection costs; (iv) defaulting costs.

What decides the size of receivables

The credit policy determines both the quantity and quality of receivables. The factors that determine the size of the investment (SM 9.71–9.72): (i) the effect of credit on the volume of sales; (ii) credit

terms; (iii) cash discount; (iv) policies and practices for selecting credit customers; (v) paying practices and habits of the customers; (vi) the firm's policy and practice of collection; (vii) the degree of operating efficiency in billing, record keeping and adjustment. Rising interest, collection and bad debt costs depress the size of the investment.

A **lenient** policy sells on credit on very liberal terms and standards; a **stringent** policy sells on credit only to customers of proper creditworthiness who are financially sound. More credit means more sales but also more financing, investigation, collection effort and bad debts; less credit may mean lost sales to competitors (SM 9.72).

The **finance manager's** involvement (SM 9.72): (a) supervising the administration of credit; (b) contributing to top management decisions on the best credit policies; (c) deciding the criteria for selecting credit applications; (d) speeding up the conversion of receivables into cash by an aggressive collection policy. In summary, a balance between the cost of increased investment in receivables and the profits from higher sales.

Evaluating credit policies: total and incremental

ICAI gives two methods, the **total approach** and the **incremental approach**, with a format for each (SM 9.73–9.75; reproduced under Formats and formulas):

- **Total approach**: for every policy, expected profit (credit sales − variable costs − fixed costs − bad debts − cash discount, less tax) minus the **opportunity cost of the investment in receivables**. The policy with the highest net benefit is adopted.
- **Incremental approach**: for every proposed policy, the incremental profit over the present policy minus the required return on the incremental investment in receivables.
- A third presentation in Illustration 12: the **expected rate of return** = incremental expected profit ÷ incremental investment in receivables, compared with the required rate (SM 9.78–9.79).

Both approaches give the same ranking. The two working formulas (SM 9.74):

$$\text{Total fixed cost} = (\text{Average cost per unit} - \text{Variable cost per unit}) \times \text{Units sold on credit under present policy}$$

$$\text{Opportunity cost} = \text{Total cost of credit sales} \times \frac{\text{Collection period (days)}}{365 \text{ or } 360} \times \frac{\text{Required rate of return}}{100}$$

Note

Which cost to lock up

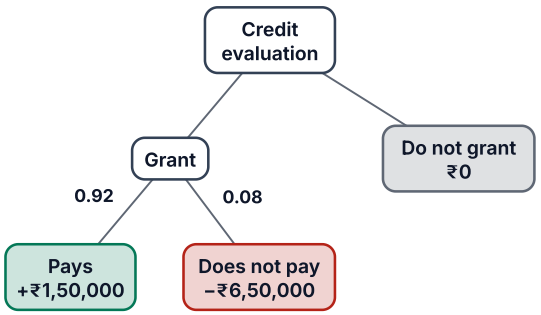
ICAI's format uses the **total cost** of credit sales (variable + fixed) for the investment in receivables (Illustration 12, SM 9.77; in Illustration 13 no fixed cost is given, so the variable cost is the whole cost, SM 9.80). When only the contribution ratio is given, ICAI uses the **variable cost**, "only relevant or variable costs are considered" (Illustration 14, SM 9.81). Illustration 16 values the new debtors at new variable cost plus fixed cost at the existing level (SM 9.97). State the basis you use.

Interactive version: PassBook app, lesson Management of receivables, credit policy, factoring and forfaiting

One customer: grant credit or not?

Granting credit to a customer is a cost-benefit decision: if the customer pays, the firm earns its profit; if not, the whole cost of the goods is lost. With probabilities of payment and default, the expected net benefit decides (SM 9.93). A café chain orders ₹8 lakh of premix that costs Tapri ₹6.5 lakh; credit evaluation puts the chance of payment at 0.92:

Credit-granting decision tree

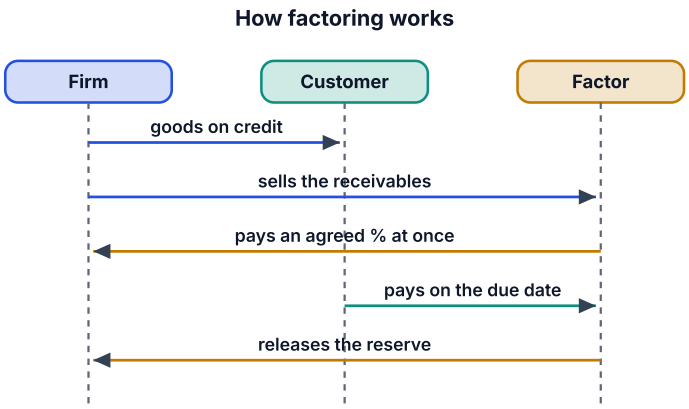


Green: the profit if the customer pays; rose: the cost lost if it defaults. Expected net benefit of granting = 0.92 × ₹1,50,000 − 0.08 × ₹6,50,000 = ₹1,38,000 − ₹52,000 = ₹86,000, above nil: grant the credit.

Financing receivables: pledging and factoring

Pledging and factoring have emerged as important sources of financing receivables (SM 9.82):

- **Pledging** is the use of a firm's receivables to secure a short-term loan. After cash, receivables are the most liquid assets and serve as prime collateral. The lender scrutinises them, selects acceptable accounts, creates a lien and fixes the percentage financed, around 50 to 90%. Advantages: ease and flexibility, regular financing. Drawbacks: high cost; being a loan, it raises debt and affects the debt-equity ratio.
- **Factoring** is the outright sale of receivables to a factor, a firm that acquires the receivables of other firms. The factoring agreement lays down the conditions; the factoring agency bears the risk of collection and services the accounts for a fee.



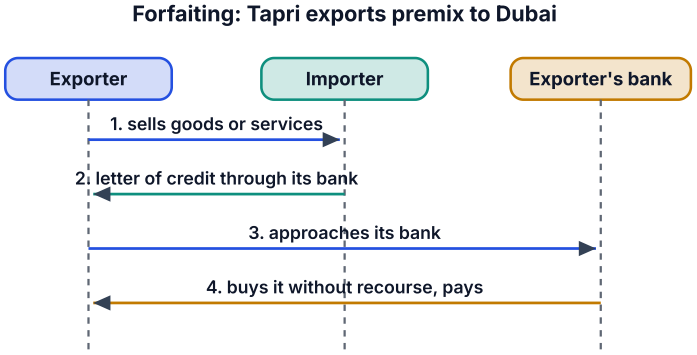
Based on the diagram at SM 9.82 (goods to the customer, an agreed percentage paid by the factor, customers paying the factor). The advance is net of the commission, the reserve and the interest; the reserve is released when the customers pay.

Factoring may be **with recourse** (if the factor cannot collect, it turns the receivable back to the firm, generally to be replaced by new receivables) or **non-recourse** (the factor bears the ultimate risk of loss, and so charges a higher commission) (SM 9.83). Its biggest advantages are the immediate conversion of receivables into cash and a predicted pattern of cash flows. It gives liquidity without creating a net liability, so there is no impact on the debt-equity ratio; it is not considered a loan, involves no debt repayment, and provides timely funds, efficient record keeping and effective management of collection (SM 9.83).

Advise (SM 9.84): avail factoring if the rate of effective cost of factoring is less than the existing cost of borrowing, or if it gives positive net annual benefits; do not avail it if the effective cost is more than the cost of borrowing.

Forfaiting

"Forfait" is a French term meaning "relinquish a right". **Forfaiting** is an arrangement of bill discounting in which a financial institution or bank buys the trade bills or receivables from exporters, the exporter relinquishing the right to receive payment from the importer. The bank provides immediate finance on a **without recourse** basis, so the risks and rewards pass to it. The overseas buyer can open a letter of credit (or other negotiable instrument) in favour of the exporter and import on deferred payment terms (SM 9.86).



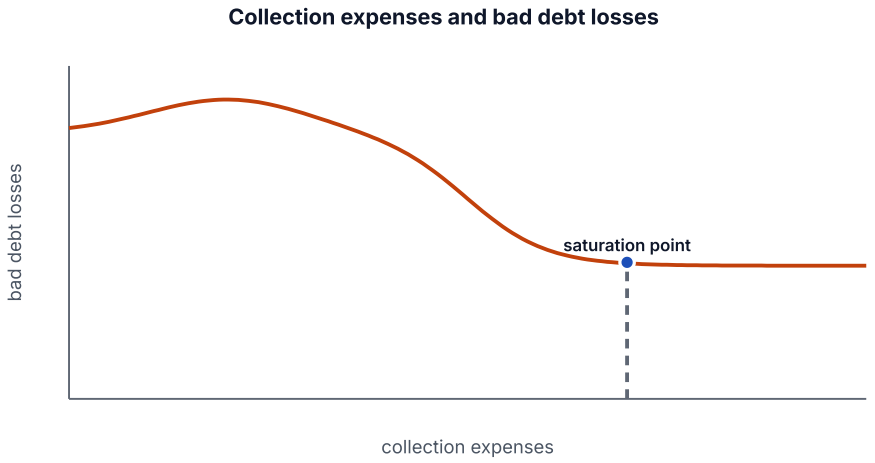
The four functions of forfaiting in ICAI's order (SM 9.86–9.87). The importer's bank issues the letter of credit; from step 4 the risk of non-payment is the bank's.

Features (SM 9.87): it motivates exporters to explore new geographies as payment is assured; the importer can import on deferred payment terms; the exporter enjoys reduced transaction costs and complexities; the exporter can compete internationally and keep its working capital at work to scale up; importers avail of it from international institutions to finance imports at competitive rates. **Example**: Exim Bank of India's **Buyer's Credit** gives credit to overseas buyers to import from India, providing non-recourse finance to Indian exporters by converting a deferred credit contract into a cash contract (SM 9.87–9.88).

Innovations in receivable management (SM 9.89–9.94)

1. **Re-engineering the receivable process**: (a) centralisation of high-volume receivable and payable transactions; (b) alternative payment strategies: direct debit, Integrated Voice Response (payment over the phone), collection by a third party, lock box processing (an outsourced partner captures cheques and invoice data), internet payments (RTGS, NEFT, UPI, app-based); (c) customer orientation for strategically important customers.
2. **Evaluation of risk**: assess risks, then introduce controls to contain or eliminate them.
3. **Use of latest technology**: e-commerce (EDI, e-mail, EFT, electronic catalogues) and automated accounts receivable management systems.
4. **Receivable collection practices**: issue of invoice; open account or open-end credit; credit terms or time limits; periodic statements and follow-ups; payment incentives and penalties; record keeping and continuous audit; export factoring; business process outsourcing.
5. **Financial tools and techniques**: credit analysis and credit rating (sources: trade references, bank references, credit bureau reports, past experience, published financial statements, salesman's interview and reports; then a credit limit, raised as experience is gained); the credit-granting decision tree; control of receivables; collection policy.

The **collection policy** trades collection expense against lower bad debts and investment in debtors; it must answer how long a balance may run before collection starts, how and how often reminders go, whether personal calls are made, and how doubtful accounts are handled (legal action or an escalation matrix) (SM 9.94). Spending more on collection cuts bad debts only up to a point:



Redrawn from SM 9.96. The first rupees of collection effort barely move bad debt losses; then they fall; beyond the saturation point, more expense achieves almost nothing.

Monitoring receivables (SM 9.94–9.96)

1. **Average age of receivables**: the average collection period.
2. **Ageing schedule**: receivables classified by age, for closer control over the quality of individual accounts, compared with earlier schedules of the same firm and with other firms; a tendency for old accounts to accumulate is a warning.
3. **Debt collection programme**: (a) monitoring the state of receivables; (b) intimation to customers when the due date approaches; (c) e-mail and telephonic advice on the due date; (d) reminding the legal recourse on overdue accounts and following an escalation matrix; (e) legal action on overdue accounts.

Tapri's premix receivables at two quarter-ends:

Age (days)	30 June 2026 (₹)	%	30 September 2026 (₹)	%
1–30	18,00,000	40.9	26,40,000	60.0
31–60	12,60,000	28.6	11,00,000	25.0
61–90	7,20,000	16.4	4,00,000	9.1
91–120	3,60,000	8.2	1,76,000	4.0
Over 120	2,60,000	5.9	84,000	1.9
Total	**44,00,000**	**100.0**	**44,00,000**	**100.0**

The total is unchanged, but the share over 60 days fell from 30.5% to 15.0%: a substantial improvement in the liquidity of receivables, as in ICAI's own schedule (SM 9.95–9.96).

Formats and formulas

Reproduce ICAI's evaluation formats from memory: the line order is what the examiner checks.

Statement showing the evaluation of credit policies (total approach) (SM 9.73)

Particulars	Present policy	Proposed policy I	Proposed policy II
A. Expected profit:			
(a) Credit sales	...	...	...
(b) Total cost other than bad debts: (i) variable costs; (ii) fixed costs	...	...	...
(c) Bad debts	...	...	...
(d) Cash discount	...	...	...
(e) Expected net profit before tax (a − b − c − d)	...	...	...
(f) Less: tax; (g) expected profit after tax	...	...	...
B. Opportunity cost of investments in receivables	...	...	...
Net benefits (A − B)	...	...	...

Incremental approach (SM 9.74–9.75): A. incremental expected profit [(a) incremental credit sales, less (b) incremental variable and fixed costs, (c) incremental bad debts, (d) incremental cash discount, (e) incremental expected profit, (f) tax, (g) after tax]; B. required return on incremental investments [(a) cost of credit sales, (b) collection period, (c) investment in receivables = a × b ÷ 365 (360), (d) incremental investment, (e) required rate of return, (f) required return = d × e]; **incremental net benefits (A − B)**.

Statement showing the evaluation of a factoring proposal (SM 9.83–9.84)

Particulars	₹
A. Annual savings (benefit) on taking factoring service	
Cost of credit administration saved	...
Bad debts avoided	...
Interest saved due to reduction in average collection period: cost of annual credit sales × rate of interest × (present − new collection period) ÷ 360	...
B. Annual cost of factoring to the firm	
Factoring commission: annual credit sales × % of commission	...
Interest charged by factor on advance: amount available for advance × collection period ÷ 360 × rate of interest (annualised)	...
C. Net annual benefits / cost of factoring (A − B)	...

Σ Formula

Effective cost of factoring (SM 9.84)

$$\text{Rate of effective cost} = \frac{\text{Net annual cost of factoring}}{\text{Advances to be paid}} \times 100$$

$$\text{or} \quad \frac{\text{Net annual cost}}{\text{Amount available for advance}} \times 100$$

Amount available for advance = receivables − factoring commission − factoring reserve; advances to be paid = amount available for advance − interest deducted by the factor. A year is taken as 360 days.

Σ Formula

Cash discount and average collection period

$$\text{Cash discount} = \text{Credit sales} \quad \times \text{Share of customers taking the discount} \quad \times \text{Rate}$$

Average collection period = the sum, over the groups of customers, of each group's share of credit sales × the days that group takes to pay. From a receivables turnover ratio: collection period in months = 12 ÷ turnover (Illustration 13, SM 9.80).

Σ Formula

Expected rate of return (Illustration 12)

$$\text{Expected rate of return} = \frac{\text{Incremental expected profit}}{\text{Incremental investment}} \times 100$$

Definition

Factoring (SM 9.82, 9.106)

The outright sale of accounts receivables to a factor or financial agency; a continuous arrangement in which the factor purchases the client's trade debts, with or without recourse, exercises control over the credit extended to customers and administers the client's sales ledger.

Definition

Forfaiting (SM 9.86)

An arrangement of bill discounting in which a financial institution or bank buys the trade bills or receivables from exporters, the exporter relinquishing the right to receive payment from the importer, providing immediate finance on a without recourse basis.

Worked examples

Worked example **Premix distributors: longer credit, or a discount?**

Tapri sells chai premix to distributors at ₹200 a carton, all on credit. Present terms are net 30: credit sales are ₹1,20,00,000 a year, every distributor pays on the 30th day, and bad debts are 1% of sales. Variable cost is

₹140 a carton; at the present volume the average cost is ₹160 a carton, and fixed costs will not change. Tax is 25%, and Tapri wants a post-tax return of 12% on its investment in receivables. Dev's consultant estimates:

Policy	Terms	Credit sales (₹)	Distributors pay	Bad debts
A	net 60	1,32,00,000	on day 60	2%
B	net 90	1,44,00,000	on day 90	3.5%
C	2/10 net 90	1,44,00,000	half on day 10	2.25%

Under C, half the distributors (by value) take the discount and pay on the 10th day; the other half pay on the 90th day.

Required: Evaluate the policies by the total approach and recommend one; check the result with the incremental approach. Assume a 360-day year.

THE WORKING

Working notes

1. Units at present = ₹1,20,00,000 ÷ ₹200 = 60,000 cartons. Fixed cost = (₹160 − ₹140) × 60,000 = **₹12,00,000**.
2. Variable cost = 70% of sales (₹140 ÷ ₹200).
3. Policy C: average collection period = 50% × 10 + 50% × 90 = **50 days**; cash discount = ₹1,44,00,000 × 50% × 2% = **₹1,44,000**.
4. Opportunity cost = total cost × collection period ÷ 360 × 12%. Present: ₹96,00,000 × 30/360 × 12% = ₹96,000.

Statement showing the evaluation of credit policies (total approach) (₹)

Particulars	Present	A	B	C
Collection period	30 days	60 days	90 days	50 days
(a) Credit sales	1,20,00,000	1,32,00,000	1,44,00,000	1,44,00,000
(b)(i) Variable costs	84,00,000	92,40,000	1,00,80,000	1,00,80,000
(b)(ii) Fixed costs	12,00,000	12,00,000	12,00,000	12,00,000
(c) Bad debts	1,20,000	2,64,000	5,04,000	3,24,000
(d) Cash discount	—	—	—	1,44,000
(e) Profit before tax	22,80,000	24,96,000	26,16,000	26,52,000
(f) Less: tax @ 25%	5,70,000	6,24,000	6,54,000	6,63,000
A. Profit after tax	17,10,000	18,72,000	19,62,000	19,89,000
Investment in receivables (total cost × days ÷ 360)	8,00,000	17,40,000	28,20,000	15,66,667
B. Opportunity cost @ 12%	96,000	2,08,800	3,38,400	1,88,000
Net benefits (A − B)	**16,14,000**	**16,63,200**	**16,23,600**	**18,01,000**

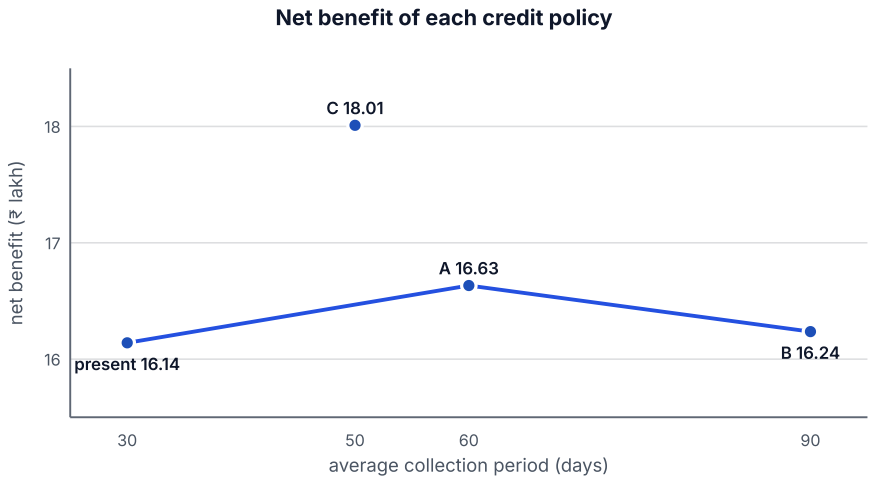
Recommendation: Policy C (2/10 net 90) should be adopted, since its net benefits are the highest.

Check: incremental approach (over the present policy, ₹)

Particulars	A	B	C
Incremental credit sales	12,00,000	24,00,000	24,00,000
Less: incremental variable costs	8,40,000	16,80,000	16,80,000
Less: incremental bad debts	1,44,000	3,84,000	2,04,000
Less: cash discount	—	—	1,44,000
Incremental profit before tax	2,16,000	3,36,000	3,72,000
Less: tax @ 25%	54,000	84,000	93,000
Incremental profit after tax	1,62,000	2,52,000	2,79,000
Incremental investment in receivables	9,40,000	20,20,000	7,66,667
Required return @ 12%	1,12,800	2,42,400	92,000
Incremental net benefits	**49,200**	**9,600**	**1,87,000**
Expected rate of return	17.23%	12.48%	36.39%

The incremental net benefits are the differences between the total-approach figures. Of the plain credit periods, 60 days is the best: B's extra sales barely pay for ₹10,80,000 more locked in receivables and the jump in bad debts. C gets B's sales with less money locked up than even A, because half the

distributors pay on day 10; the ₹1,44,000 discount is more than paid for by lower bad debts and the smaller lock-up.



The blue line joins the plain net-day policies: net benefit rises to 60 days, then falls. Policy C, plotted at its 50-day average collection period, sits above the line: the discount brings in the 90-day sales with much less money locked up.

Worked example **Institutional sales: what is a quick payment worth?**

Tapri's institutional division (offices and colleges) has credit sales of ₹60,00,000 a year on terms of net 45, but its customers actually pay, on average, on the 60th day; bad debts are 2.5% of sales. Dev proposes terms of 2/10 net 45. He expects no change in sales; 60% of customers (by value) to take the discount and pay on the 10th day; the rest to keep paying on the 60th day; and bad debts to arise only on the sales to customers who do not take the discount, at 2.5% of those sales. The contribution to sales ratio is 30%, the tax rate 25% and the opportunity cost of funds 16% before tax. Assume a 360-day year.

Required:

(i) Advise whether the division should change its credit terms.

(ii) Calculate the highest rate of cash discount at which the change would still pay, if the same customers take it.

THE WORKING

Working notes

1. New average collection period = 60% × 10 + 40% × 60 = 6 + 24 = **30 days**.
2. Cash discount = ₹60,00,000 × 60% × 2% = **₹72,000**.
3. Bad debts: present 2.5% × ₹60,00,000 = ₹1,50,000; proposed 2.5% × 40% × ₹60,00,000 = **₹60,000**.
4. The statement is after tax, so the pre-tax opportunity cost is converted: 16% × (1 − 0.25) = **12%** (as in Illustration 14, SM 9.81). Investment in receivables at variable cost (only relevant costs): present ₹42,00,000 × 60/360 = ₹7,00,000; proposed ₹42,00,000 × 30/360 = ₹3,50,000.

Statement showing the evaluation of credit policies (₹)

Particulars	Present (paid in 60 days)	Proposed (2/10 net 45)
Credit sales	60,00,000	60,00,000
Less: variable cost @ 70%	42,00,000	42,00,000
Less: bad debts (WN 3)	1,50,000	60,000
Less: cash discount (WN 2)	—	72,000
Profit before tax	16,50,000	16,68,000
Less: tax @ 25%	4,12,500	4,17,000
Profit after tax	12,37,500	12,51,000
Less: opportunity cost @ 12% (WN 4)	84,000	42,000
Net benefit	**11,53,500**	**12,09,000**

(i) Advice: The proposed terms should be adopted: the net benefit increases by ₹12,09,000 – ₹11,53,500 = **₹55,500**. The discount costs ₹72,000, but it saves ₹90,000 of bad debts and halves the money locked in receivables.

(ii) The highest discount. With the same customers taking it, the discount can cost at most what the change saves before tax: bad debts ₹90,000, plus the opportunity cost saved, ₹42,000 after tax = ₹42,000 ÷ 0.75 = ₹56,000 before tax; ₹1,46,000 in all. The discount is paid on 60% × ₹60,00,000 = ₹36,00,000 of sales, so

$$\text{Highest rate} = \frac{1{,}46{,}000}{36{,}00{,}000} \times 100 = 4.06\%$$

Any discount below about 4.06% leaves the division better off, on the assumption that the same 60% take it up.

Worked example **Modern trade receivables: which factoring scheme?**

Tapri's credit sales to supermarket chains are ₹4,80,00,000 a year, collected on average in 72 days. Bad debts are 1.5% of credit sales. Credit administration costs ₹9,00,000 a year, of which ₹6,00,000 would be avoided if the receivables were factored.

A factor offers two schemes. Under both it withholds a reserve of 10% and advances the balance at 15% p.a., the interest deducted in advance for the collection period, which stays at 72 days:

- **Scheme R, with recourse**: commission 1%; bad debts remain Tapri's.
- **Scheme N, without recourse**: commission 2%; the factor bears the bad debts.

Tapri's bank cash credit costs 13% p.a.

Required:

(i) Calculate the rate of effective cost of each scheme, on the advance to be paid.

(ii) Advise Tapri.

Assume a 360-day year.

THE WORKING

Working notes

1. Average receivables = ₹4,80,00,000 × 72/360 = **₹96,00,000**; they turn over 360 ÷ 72 = 5 times a year.
2. Advance from the factor:

Particulars	Scheme R (₹)	Scheme N (₹)
Average receivables	96,00,000	96,00,000
Less: commission @ 1% and 2%	96,000	1,92,000
Less: reserve @ 10%	9,60,000	9,60,000
Amount available for advance	85,44,000	84,48,000
Less: interest @ 15% for 72 days	2,56,320	2,53,440
Advance to be paid	**82,87,680**	**81,94,560**

(i) Statement showing the evaluation of the factoring proposals

Particulars	Scheme R (₹)	Scheme N (₹)
A. Annual savings		
Credit administration saved	6,00,000	6,00,000
Bad debts avoided (1.5% of ₹4,80,00,000)	—	7,20,000
Total	6,00,000	13,20,000
B. Annual cost of factoring		
Factoring commission (× 360/72)	4,80,000	9,60,000
Interest charges (× 360/72)	12,81,600	12,67,200
Total	17,61,600	22,27,200
C. Net annual cost to the firm (B − A)	**11,61,600**	**9,07,200**
Rate of effective cost (C ÷ advance to be paid)	**14.02%**	**11.07%**

On the amount available for advance, the rates are 13.60% and 10.74%.

(ii) Advice: Scheme N's effective cost (11.07%) is below the bank's 13%; Scheme R's (14.02%) is above it. Tapri should avail factoring under Scheme N. The cheaper commission is the dearer scheme: with recourse, the ₹7,20,000 of bad debts stays with Tapri, far more than the extra 1% commission

(₹4,80,000 a year) that buys them off. Either way the commission works out as a percentage of annual credit sales: 2% of the ₹96,00,000 outstanding, five times a year, is 2% of ₹4,80,00,000.

Exam angle

In the exam

Where receivables appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, receivables management came up 17 times in 16 documents.

- **Factoring, 5 marks**: May 2024 Q1(c) (non-recourse, guaranteed 45 days, a one-time redundancy payment, effective cost on the advance); January 2026 Q1(b) (average debtors ₹90 lakh, 2% commission, 15% reserve: "should not be accepted"); RTP September 2025 Q9 and RTP September 2026 Q8 (terms 2/40 net 120, half the customers on each date, factoring against a bank rate); MTP May 2024 Series I as a five-MCQ case ending with "the effective cost of factoring on advance received".
- **Credit policy evaluation, 5 to 8 marks**: September 2024 Q3(a) (total approach, two policies, extra creditors); MTP September 2026 Series II Q1(a) (1/15 net 45 to 2/15 net 60, after-tax return of 12%); MTP September 2025 Series II Q3(a) ("2/10, net 30", 8 marks); RTP September 2024 Q11; MTP September 2024 Series II Q2(a); MTP September 2026 Series I (credit to three customers).
- **Cash discount to cut the collection period**: May 2026 Q3(b) (6 marks, the saving in minimum cash against the after-tax cost of the discount); May 2026 MCQ 6 (incremental investment in debtors and post-tax effect).
- **Theory**: the difference between factoring and forfaiting (MTP January 2025 Series II Q4(a), 4 marks); the ageing schedule (MTP May 2026 Series II Q2(b), 2 marks); the three aspects of management of receivables (MTP May 2026 Series I); features of forfaiting (old scheme, July 2021, 4 marks).

In the exam

The traps ICAI set

- A **pre-tax** opportunity cost against an **after-tax** profit: Illustration 14 uses 70% of 15%. Read whether the rate is pre-tax or post-tax (May 2026 MCQ 6 says "post-tax").
- **Average collection period from discount takers**: "50% pay in 40 days, the rest in 120" is 80 days (RTP September 2026 Q8).
- **Annualising the factor's charges**: commission and interest for one collection period × (360 ÷ collection period).
- **Two rates of effective cost**: ICAI shows both, on advances to be paid and on the amount available for advance; give the first and mention the second.
- The suggested answers end with a sentence of advice. Without it, the last mark is lost.

Interactive version: PassBook app, lesson Management of receivables, credit policy, factoring and forfaiting

Trick

Credit pays when the margin beats the lock-up

A longer credit period pays only if extra contribution − extra bad debts − extra cash discount exceeds the extra investment in debtors × required return. This is just the difference between two policies' net benefits, so it always agrees with the total approach. Conditions: fixed costs unchanged, and profit and the required return on the same (pre- or post-tax) basis. Still work out every policy: a step that fails can be followed by one that passes.

Answer frame

"Explain the three aspects of management of receivables"

1. **Credit policy**: credit standards, credit terms and collection efforts; the trade-off between profit on additional sales and the cost of carrying debtors and bad debts; credit period ("net 50") and cash discount policy (rate, discount period, net credit period: "3/15 net 60").
2. **Credit analysis**: how risky it is to advance credit to a party; due diligence or reputation check.
3. **Control of receivables**: follow-up and a suitable collection policy; laying down policies and executing them.

"Describe factoring" / "Describe the salient features of forfaiting"

Factoring: definition (outright sale of receivables to a factor, who bears the risk of collection and services the accounts for a fee); with or without recourse; advantages (immediate cash, predictable cash flows, no impact on debt-equity, not a loan, efficient record keeping and collection); the evaluation (savings against commission and interest; accept if the effective cost is below the cost of borrowing).

Forfaiting: meaning ("relinquish a right"; bank buys exporters' receivables without recourse); the four functions in order; the five features; Exim Bank's Buyer's Credit as the example.

"Explain the ageing schedule"

Receivables classified by age from the date of invoice (0–30, 31–60, 61–90, 91–120 days and more); compared with earlier schedules and with other firms; gives closer control over the quality of individual accounts; shows old accounts accumulating; supplements the average collection period. A two-date table makes the point.

Common mistakes

Common mistake

Hall of Shame: the profit locked up in debtors

Computing the investment in receivables as sales × days ÷ 360 when ICAI's format wants the cost of credit sales. In Worked example 1, policy C at sales value gives ₹20,00,000 instead of ₹15,66,667, and an opportunity cost ₹52,000 too high. **Fix:** total cost (or variable cost, when only the contribution ratio is given), and say which; ICAI uses sales only when no cost is given (TYK Problem 11, SM 9.151).

⚠ **Common mistake**

The discount rate applied to everybody

Cash discount taken as 2% of all credit sales when only 60% of customers take it: ₹1,20,000 instead of ₹72,000 in Worked example 2. **Fix:** credit sales × share taking the discount × rate.

⚠ **Common mistake**

Commission for one period compared with an annual saving

Setting Scheme N's ₹1,92,000 commission for one 72-day collection period against an annual bad-debt saving of ₹7,20,000. **Fix:** put everything on an annual basis: × 360 ÷ 72.

⚠ **Common mistake**

Factoring and forfaiting treated as the same thing

Writing "forfaiting may be with or without recourse". **Fix:** forfaiting is always without recourse and for export receivables; factoring may be either way and domestic or international.

Practice

Check yourself

1 A firm's credit sales are ₹36,00,000; 40% of customers (by value) take a discount of 1% by paying on the 10th day, and the rest pay on the 50th day. What is the average collection period in days?

Answer: _____

2 Case scenario. Kesar's cousin runs Chai Point Exports and sells masala chai premix to a buyer in Muscat on six months' deferred terms against a letter of credit. He wants the full invoice value now and does not want to carry any risk of the buyer not paying. Which arrangement suits him?

- A Factoring with recourse
- B Pledging the receivable with his bank
- C Forfaiting
- D Bills discounting with recourse

3 Under the total approach, a policy shows an expected profit of ₹18,00,000; the total cost of its credit sales is ₹72,00,000, the collection period 45 days and the required return 20%. What is its net benefit, in rupees? (Assume 360 days.)

Answer: _____

4 A factor offers non-recourse factoring at an effective cost of 11.2% on the advances it pays. The company's bank overdraft costs 12.5%. According to ICAI's advice, the company should:

- **A** avail the factoring service
- **B** not avail it, because non-recourse factoring carries a higher commission
- **C** avail it only if the factoring is with recourse
- **D** not avail it, because factoring is a loan that raises the debt-equity ratio

5 Which of these are among the costs of maintaining receivables in ICAI's list? Select all that apply.

- **A** Interest or opportunity cost on funds blocked in receivables
- **B** Collection costs
- **C** Cost of carrying finished goods inventory
- **D** Defaulting costs

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Objective: optimise the return on investment in receivables; three aspects: credit policy, credit analysis, control of receivables.
- "3/15 net 60": 3% discount within 15 days, otherwise pay within 60; cash discount = credit sales × share taking it × rate.
- Costs of maintaining receivables: funds blocked (interest or opportunity cost), administration, collection, default.
- Total approach: expected profit − opportunity cost of investment in receivables; incremental approach: incremental profit − required return on incremental investment; both pick the same policy.
- Opportunity cost = total cost of credit sales × collection period ÷ 360 (365) × required rate; fixed cost = (average − variable cost per unit) × present units.
- Factoring: sale of receivables, with or without recourse, no impact on debt-equity; accept if effective cost < cost of borrowing.
- Effective cost of factoring = net annual cost ÷ advances to be paid (or ÷ amount available for advance) × 100.
- Forfaiting: exporters' receivables sold to a bank without recourse; Exim Bank's Buyer's Credit. Ageing schedule: receivables by age, compared over time. (Hand the diary to the agent only if his cut is cheaper than the bank.)

Answer key · Check yourself

1 **34**

40% × 10 + 60% × 50 = 4 + 30 = 34 days. (The discount itself costs ₹36,00,000 × 40% × 1% = ₹14,400.)

2 **C** **Forfaiting**

Forfaiting is for export receivables, always without recourse, and pays 100% of the value: the bank buys the letter of credit and takes the risk. Factoring with recourse and pledging leave the risk with him; pledging is also a loan that raises debt.

3 **1620000**

Investment in receivables = ₹72,00,000 × 45/360 = ₹9,00,000; opportunity cost @ 20% = ₹1,80,000; net benefit = ₹18,00,000 − ₹1,80,000 = ₹16,20,000.

4 **A** **avail the factoring service**

ICAI: avail factoring if its rate of effective cost is less than the existing cost of borrowing (11.2% against 12.5%), or if it gives positive net annual benefits. A higher commission for non-recourse factoring is already inside the 11.2%. Factoring is not a loan and does not affect the debt-equity ratio; pledging does.

5 **A** **Interest or opportunity cost on funds blocked in receivables**; **B** **Collection costs**; **D** **Defaulting costs**

ICAI's four: the cost of additional funds blocked, administrative costs, collection costs and defaulting costs. Inventory carrying cost belongs to inventory management.

9.4 Treasury and cash management

What the treasury does, why a firm holds cash, the cash budget with a minimum balance, the four floats and how to cut them, Baumol and Miller-Orr for the optimum balance, and where surplus cash goes.

In plain words

₹38 lakh, fast asleep

It is the second week of August. Tapri's premix division has ₹38 lakh in its current account, earning nothing. In September it must pay for a new packing machine, an instalment of advance tax and the festive stock, all in the same month.

> **Kesar:** ₹38 lakh in the bank! Finally we are rich.
>
> **Dev:** Rich today. Ask me again after the machine, the tax and the festive stock are paid for.
>
> **Kesar:** So leave it all in the current account till then. Safe hai.
>
> **Dev:** Safe, and asleep. The present value of an idle rupee is very disappointing.

Cash management is deciding how much cash to hold, when, and what to do with the rest. A firm holds cash to pay its day-to-day bills, to grab a bargain that needs ready money, and as a cushion against surprises. Every rupee beyond that could have earned interest, or repaid a loan.

So the finance manager does four things. He **plans**: a **cash budget** shows, month by month, when cash will pile up and when it will run short, so the surplus can be parked in safe short-term investments (**marketable securities**) and the shortage arranged in advance. He **speeds up** collections: the days between a sale and the customer's money becoming usable (the invoice going out, the cheque in the post, on the firm's desk, in bank clearing: the **float**) are days the firm's own money is out of reach. He **slows down** payments to the due date. And he decides **how much** to keep: the **Baumol model** when payments are predictable, the **Miller-Orr model** when they are not.

The **treasury** is the department that does all this, and more: it also manages foreign currency, raises funds, deals with the banks and helps with acquisitions.

Think of it like

The wallet and the ATM

Every trip to the ATM costs a fee and twenty minutes. A fat wallet saves trips, but the money in it earns nothing and could be lost. Fill the wallet too little and you live at the ATM; fill it too much and you lose the interest. The Baumol model finds the wallet size at which fees plus lost interest are smallest. The Miller-Orr model is a rule instead: if the wallet ever gets too fat, put the excess back in the bank; if it ever runs empty, withdraw enough to bring it back to a set level; in between, do nothing.

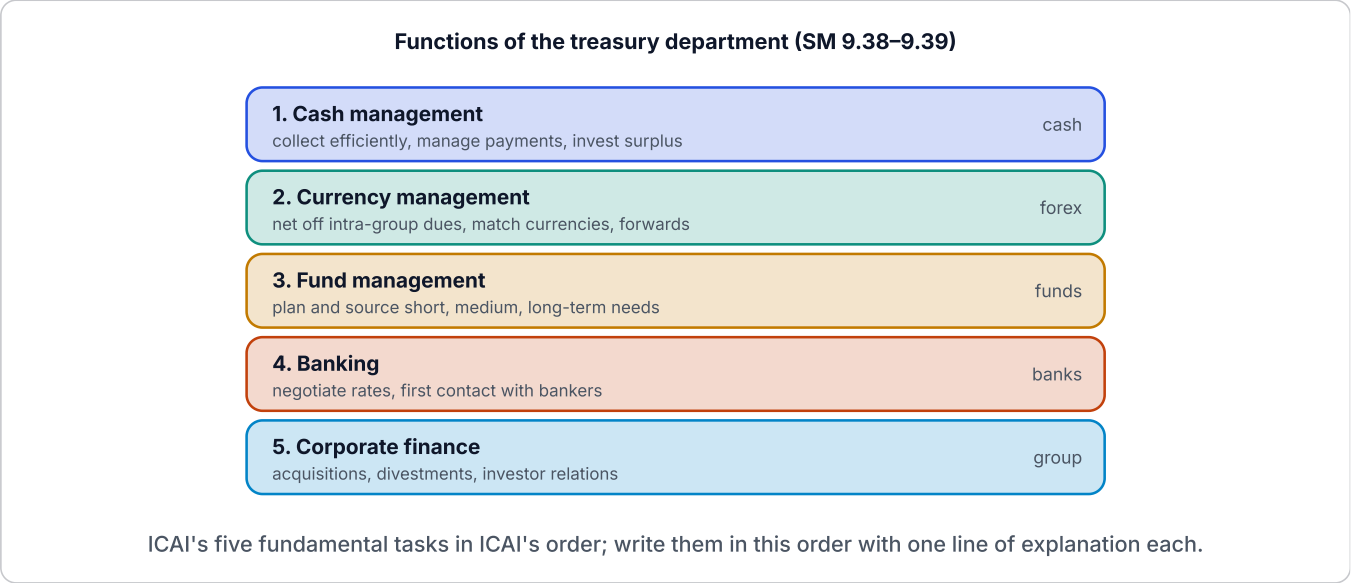
Key idea

Cash management is a balance between liquidity and cost: enough cash to meet every payment on time, no rupee left idle, and every surplus rupee parked where it is safe, short-term and quickly saleable.

How it works

What the treasury department does

Treasury management encompasses planning, organising and controlling the funds and working capital of an enterprise in order to ensure best use of funds, maintain liquidity, reduce the overall cost of funds and mitigate operational and financial risk (SM 9.37). It mainly deals with **working capital management** and **financial risk management** (forex and interest rate management). Its key goals: maximise the return on the available cash; minimise interest cost on borrowings; mobilise as much cash as possible for corporate ventures for maximum returns; and deal effectively in forex, money and commodity markets to reduce the risks from fluctuating exchange rates, interest rates and prices (SM 9.37).



ICAI's five fundamental tasks in ICAI's order; write them in this order with one line of explanation each.

In ICAI's words, shortened (SM 9.38–9.39):

1. **Cash management**: efficient cash collection and management of payments inside the organisation and to third parties; complete centralisation in a group treasury, or advice to subsidiaries on collection and payment periods, discounts; management of surplus funds in an investment portfolio, considering future needs for liquid funds and acceptable levels of risk.
2. **Currency management**: the foreign currency risk exposure. In a large MNC the first step is to set off intra-group indebtedness; matching receipts and payments in the same currency saves transaction costs and unfavourable exchange movements; net exposures are managed by company policy, using forward contracts to buy or sell currency forward.
3. **Fund management**: planning and sourcing the short, medium and long-term cash needs; temporary investment of surplus funds by mapping the gap between inflows and outflows; participating in the capital structure decision and forecasting interest and currency rates.
4. **Banking**: keeping a good relationship with bankers, negotiating interest and exchange rates, acting as the initial point of contact; short-term finance from bank loans or the sale of commercial paper.
5. **Corporate finance**: acquisition and divestment activities within the group, and often investor relations, since share-price performance affects the ability to make acquisitions or to resist a

hostile bid.

Three flows, three objectives, three motives

Management of cash is concerned with managing (i) cash flows into and out of the firm, (ii) cash flows within the firm, and (iii) cash balances held at a point of time, by financing a deficit or investing surplus cash (SM 9.39). Its main objectives:

- provide adequate cash to each of its units as per requirements;
- no funds are blocked in idle cash; and
- the surplus cash (if any) is invested to maximise returns for the business.

A cash management scheme is therefore a delicate balance between the twin objectives of liquidity and costs (SM 9.39).

The need for cash. Lord Keynes gave three basic considerations (SM 9.39–9.40):

Need	Why the cash is held
Transaction need	Day-to-day expenses and debt payments; when operating inflows are temporarily blocked, only a reserve balance pays on time
Speculative need	To take advantage of profitable opportunities that would be lost for want of ready cash
Precautionary need	Safety against unexpected events

Cash planning and the cash budget

Cash planning is a technique to plan and control the use of cash: a projected cash statement from a forecast of expected inflows and outflows for a period, made daily, weekly or monthly depending on the size of the firm and the philosophy of management (SM 9.40). Its first tool is the **cash budget**, "the most significant device to plan for and control cash receipts and payments" (SM 9.40). Its purposes:

1. To coordinate the timings of cash needs, identifying the periods of shortage or of an abnormally large requirement.
2. To pinpoint the periods of excess cash.
3. To let a firm with enough cash take advantage of cash discounts on its payables.
4. To plan and arrange needed funds on favourable terms, avoiding both excess and shortage.

On its basis the firm invests surplus cash in marketable securities and arranges any shortage by overdraft or credit arrangements with banks (SM 9.41). Preparing it involves (a) selecting the period, which defines the planning horizon, and (b) selecting the factors that bear on cash flows: **operating** (from the firm's operations) and **financial** (from its financial activities) (SM 9.41).

Three methods (SM 9.41–9.42): the **receipts and payments method** (all expected receipts and payments, including capital expenditure; the one commonly used in business, and the one ICAI's month-by-month cash budgets use), the **adjusted income method** (sales and costs adjusted for changes in debtors and creditors, non-cash items such as depreciation eliminated) and the **adjusted**

balance sheet method (a budgeted balance sheet built from percentages of sales shows whether extra finance is needed). The capital budget is considered too.

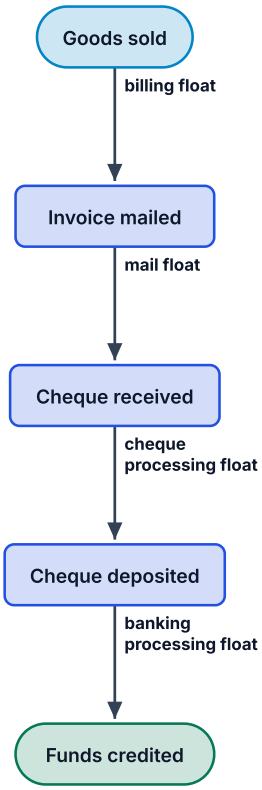
For a **short-period** (month-by-month) budget you need, on the receipts side, collections from debtors, cash sales and any other receipts (say, a dividend from a subsidiary); on the payments side, purchases, expenses, periodic payments (debenture interest, advance income tax, GST) and special payments (dividends, redemptions, loan repayments, assets bought) (SM 9.42–9.43). The format is under Formats and formulas; the Paper 4 lesson on the cash budget drills the working notes.

Collect faster, pay on the due date: the four floats

Once the budget is set, the actual cash flows must not stray from it. The twin objectives (SM 9.53): **accelerate cash collections** as much as possible, and **decelerate cash disbursements** within the permissible time frame.

The **float** is the period that affects cash as it moves through the stages of the collection process. ICAI names four (SM 9.53–9.54):

From sale to spendable money: the four floats



Each arrow is one float, in ICAI's order. Concentration banking shortens the time from the customer mailing a remittance to spendable funds; a lock box eliminates the cheque processing float.

- **Billing float**: the time between the sale and the mailing of the invoice.
- **Mail float**: the time the cheque is being carried by post, messenger or other means.

- **Cheque processing float**: the time the seller takes to sort, record and deposit the cheque after receiving it.
- **Banking processing float**: the time from the deposit of the cheque to the crediting of funds in the seller's account.

The firm attacks them by invoicing quickly, by cutting the lag between the customer paying and the cheque being collected, and by making the funds available sooner (SM 9.54), using two decentralised collection systems:

Concentration banking. The company sets up strategic collection centres in different regions instead of one at head office. Payments received are deposited with local banks, which transfer surplus funds to the **concentration bank** of head office, where the company has its major account. It reduces the period between the customer mailing the remittance and its becoming spendable funds; ICAI calls it "one important and popular way of reducing the size of the float" (SM 9.54).

Lock box system. The company rents local post-office boxes and authorises its bank to pick up the remittances several times a day and deposit the cheques; customers are told to mail their payments to the lock boxes. Its purpose is to eliminate the time between the company receiving a remittance and depositing it: the **cheque processing float** disappears. Its drawback is cost: the bank's charge is almost directly proportional to the number of cheques, so lock boxes are usually not profitable if the average remittance is small (SM 9.54–9.55).

The rule for both (SM 9.55): compare the added cost of the most efficient system with the marginal income that can be generated from the released funds. If the cost is less than the income, the system is profitable; otherwise it is not worth undertaking.

Controlling payments (SM 9.55–9.56): pay on the due date, and use drafts (bills of exchange) rather than cheques. **Playing the float**: estimate accurately when the cheques issued will be presented, issue more cheques, and keep in the bank only enough to honour those actually expected on each date. Paying outstation suppliers by cheque sent by mail also lengthens the float.

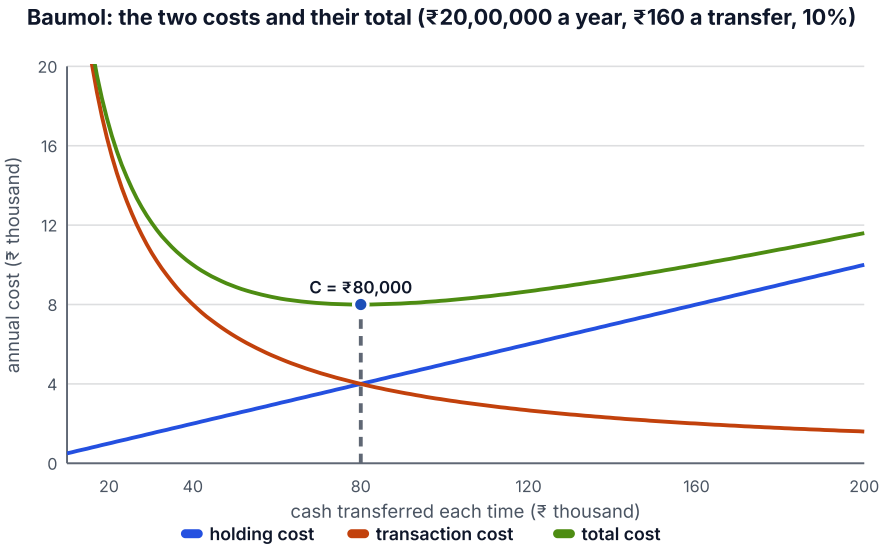
How much cash to keep: Baumol and Miller-Orr

The firm should hold the **optimum cash balance**: neither too much (to avoid the opportunity cost) nor too little (to settle day-to-day payments); the level depends on the risk-return trade-off (SM 9.59). The mathematical models fall into two categories: **inventory type models**, where cash flows are predictable (Baumol), and **stochastic models**, where the demand for cash is not known in advance (Miller-Orr) (SM 9.59).

William J. Baumol's model (1952) treats cash like inventory. The optimum cash level is the level where the carrying costs and transaction costs are the minimum (SM 9.59–9.60):

- **Carrying cost**: the cost of holding cash, namely the opportunity cost or interest foregone on marketable securities.
- **Transaction cost**: the cost of converting marketable securities into cash when the firm falls short: clerical, brokerage, registration and other costs.

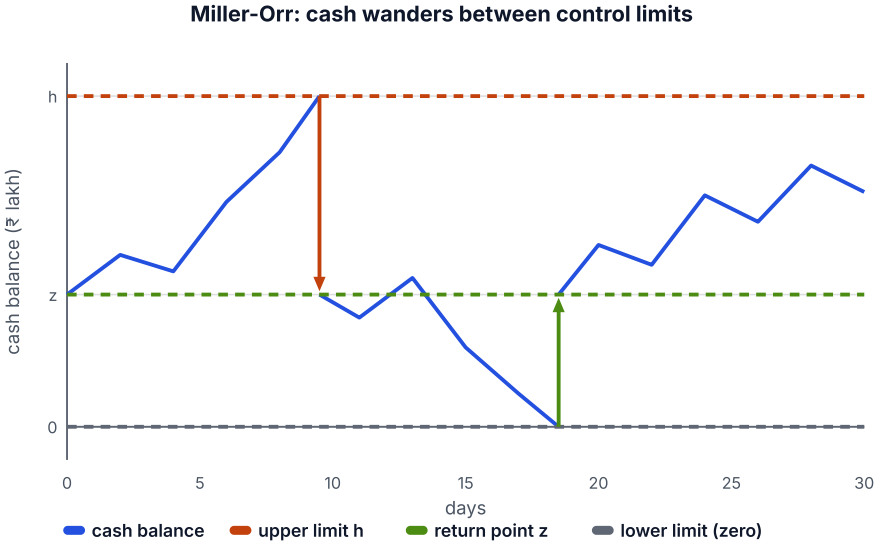
A big transfer means few transactions but a large average balance; a small one, the reverse. Take a café with annual disbursements of ₹20,00,000, a cost of ₹160 per transfer and securities yielding 10%:



Blue: holding cost, the average balance (half the transfer) × 10%. Orange: transaction cost, which falls as bigger transfers mean fewer of them. Green: their total, lowest at ₹80,000 (25 transfers), where the two costs are equal at ₹4,000 each.

The model assumes (SM 9.60): (i) the cash needs of the firm are known with certainty; (ii) cash is used uniformly over the period, also known with certainty; (iii) the holding cost is known and constant; (iv) the transaction cost also remains constant. Its **limitation**: it does not allow cash flows to fluctuate, while firms in practice neither use cash uniformly nor predict daily inflows and outflows (SM 9.61).

The Miller-Orr model (1966) takes the net cash flow as completely stochastic and applies control theory. It decides the time and size of transfers between an investment account and the cash account by setting **control limits**: **h** the upper limit, **z** the return point and zero the lower limit (SM 9.61–9.62).



Blue: the cash balance moving at random. At the orange dashed upper limit h, h − z is invested in securities (orange arrow); at the grey dashed lower limit, zero in the Study Material, securities are sold to bring cash back to the green dashed return point z (green arrow). In between, nothing is done.

The rules (SM 9.62):

- When the cash balance reaches the upper limit, cash equal to **h − z** is invested in the marketable securities account.
- When it touches the lower limit, a transfer is made from the marketable securities account to the cash account.
- While the balance stays between (h, z) and (z, 0), no transactions are made between the two accounts.

The limits are set on the basis of the **fixed cost of securities transactions**, the **opportunity cost of holding cash** and the **degree of likely fluctuations in cash balances**; they satisfy the demands for cash at the lowest possible total cost. The model is more realistic, since it allows the balance to vary between the limits, and the finance manager can set them according to the firm's liquidity requirements, its minimum and maximum balances (SM 9.62).

Interactive version: PassBook app, lesson Treasury and cash management

Recent developments in cash management

Technology and the desire to cut operating costs brought seven techniques (SM 9.62–9.67):

Technique	What it does
Electronic fund transfer	Networked bank branches: instant updating of accounts, quick transfer of funds, instant information on exchange rates
Zero balance account	Each day the firm totals the cheques presented; any excess balance buys marketable securities, a shortage is met by selling them
Money market operations	The treasury invests surplus funds in the money market, from overnight to one year, at competitive rates that can move hourly
Petty cash imprest system	A week's estimated petty expenses kept separate; each week starts with the same predetermined balance
Temporary cash surplus	Invested in short-term deposits, short-term debt instruments, long-term debt with flexible maturity, blue chip shares
Electronic cash management system	Funds and data moved electronically between collection, transfer and payment points, sometimes with limited third-party access
Virtual banking	Banking services through information technology without the customer going to the bank: ATMs, net banking, mobile banking

Electronic cash management results in (SM 9.65), in ICAI's order: significant saving in time; increase in interest earned and decrease in interest expense; less paper-work and manpower; greater accounting accuracy (easy detection of book-keeping errors); more control over time and funds; support for electronic payments; faster transfer of funds between locations; speedy conversion of instruments into cash; funds available wherever and whenever required; reduction of idle float to the maximum possible extent; no idle funds anywhere in the organisation; easier inter-bank balancing of

funds; a true form of centralised cash management; faster electronic reconciliation; fewer cheques issued.

Advantages of virtual banking (SM 9.66–9.67): (i) lower cost of handling a transaction; (ii) increased speed of response to customer requirements; (iii) lower cost of operating the branch network and reduced staff costs; (iv) improved and a range of services available to the customer rapidly, accurately and at his convenience. Its popularity with customers is due to the **speed, convenience and round the clock access** it offers.

Where the surplus goes: marketable securities

Excess funds are parked in short-term securities that can be liquidated when cash is needed. Three principles guide the choice (SM 9.67):

1. **Safety**: since the aim is liquidity, minimum risk is the criterion.
2. **Maturity**: match maturity with forecasted cash needs; long-term securities fluctuate more with interest rates, so short-term ones are preferred.
3. **Marketability**: the convenience, speed and cost of converting the security into cash without loss of time and price.

The choice is mainly limited to government treasury bills, deposits with banks and inter-corporate deposits; units of UTI, commercial papers, deposits with sister or associate concerns and money market mutual funds (MMMFs) are other avenues (SM 9.67).

Formats and formulas

Reproduce the cash budget format and the Baumol formula from memory.

ICAI's cash budget format (SM 9.43–9.44)

Heading: (name of company) Co. Ltd., Cash Budget, Period (...).

Particulars	Month 1	Month 2	... to Month 12
Receipts: 1. Opening balance			
2. Collection from debtors			
3. Cash sales			
4. Loans from banks			
5. Proceeds from issue of shares, debentures etc.			
6. Miscellaneous receipts			
7. Other items			
Total			
Payments: 1. Payments to creditors			
2. Wages			
3. Overheads (factory rent, administration, selling and distribution)			
4. Interest			
5. Dividend			
6. Corporate tax			
7. Capital expenditure			
8. Other items			
Total			
Closing balance [Surplus (+) / Shortfall (–)]			

The layout with a minimum balance (ICAI Illustration 6, SM 9.46)

When a minimum balance is kept and surpluses are invested or deficits financed at each month end, ICAI's solution adds these lines below the receipts and payments:

Line	How it is found
Total cash available (A)	Opening balance + receipts
Total payments (B)	All cash payments
Minimum cash balance desired	Given
Total cash needed (C)	B + minimum balance
Surplus – deficit (A – C)	Positive: invest; negative: liquidate or borrow
Investment / financing (D)	Investments in brackets; liquidations and borrowings positive
Closing cash balance	A + D – B, which equals the minimum

∑ Formula

Baumol: optimum cash balance (SM 9.60)

$$C = \sqrt{\frac{2U \times P}{S}}$$

C = optimum cash balance (the amount transferred each time); U = annual (or monthly) cash disbursement; P = fixed cost per transaction; S = opportunity cost of one rupee p.a. (or p.m.). U and S must be for the same period.

∑ Formula

What follows from C

$$\text{Number of transfers} = \frac{U}{C} \qquad \text{Average cash balance} = \frac{C}{2} \qquad \text{Total cost} = \frac{U}{C} \times P + \frac{C}{2} \times S$$

The first two are how ICAI's suggested answer to November 2022 Q1(a) (old scheme) worked; the interval between transfers is days in the year ÷ number of transfers. At the optimum the two costs are equal.

∑ Formula

Cash cycle and minimum operating cash (Illustration 11, SM 9.68)

$$\text{Cash cycle} = \text{Debtors period} + \text{Inventory period} - \text{Creditors period}$$

$$\text{Cash turnover} = \frac{360 \text{ days}}{\text{Cash cycle}}$$

$$\text{Minimum operating cash} = \frac{\text{Total annual operating outlay}}{\text{Cash turnover}}$$

∑ Formula

Is a faster collection system worth it? (SM 9.55)

$$\text{Funds released} = \text{Collections per day} \times \text{Days of float saved}$$

$$\text{Accept if: Funds released} \times \text{Rate of return} > \text{Added cost of the system}$$

Definition

Treasury management

Planning, organizing and controlling the funds and working capital of an enterprise in order to ensure best use of funds, maintain liquidity, reduce overall cost of funds and mitigate operational and financial risk. (SM 9.37)

Definition

Float

The periods that affect cash as it moves through the different stages of the collection process: billing float, mail float, cheque processing float and banking processing float. (SM 9.53–9.54)

Definition

Optimum cash balance (Baumol)

That level of cash where the carrying costs and transactions costs are the minimum. (SM 9.59)

Worked examples

Worked example **Premix division, August to November: a minimum balance, treasury bills and a cash credit**

Tapri's premix division sells to modern trade, distributors and its own app. Dev gives you its sales (₹ lakh):

Month	Sales	Month	Sales
June (actual)	90	October	160
July (actual)	100	November	90
August	80	December	60
September	120		

1. 20% of sales are collected in the month of sale, 50% in the following month and 30% in the second month after the sale. There are no bad debts.
2. Ingredients and packing material are bought one month before they are sold, at 60% of their selling value; half is paid in the month of purchase and half in the following month.
3. Wages and salaries are ₹8 lakh a month; a festival bonus of ₹6 lakh is paid in October.
4. Other expenses are ₹10 lakh a month, including depreciation of ₹2 lakh, paid in the same month.
5. An instalment of advance tax of ₹12 lakh and a new packing machine costing ₹25 lakh are both paid in September.
6. The cash balance on 1 August is ₹20 lakh, and the division keeps a minimum balance of ₹10 lakh. At each month end, a surplus above the minimum first repays any cash credit drawn and the rest buys treasury bills; a shortfall is met first by selling treasury bills held and then by drawing on the cash credit. Ignore interest and yield.

Required: Prepare the monthly cash budget for August to November, showing the investment and financing each month, and state the treasury bills and cash credit outstanding at the end of November.

THE WORKING

Working note 1: collections from debtors (₹ lakh)

Particulars	Aug	Sep	Oct	Nov
20% of the month's sales	16	24	32	18
50% of last month's sales	50	40	60	80
30% of sales two months back	27	30	24	36
Total collections	**93**	**94**	**116**	**134**

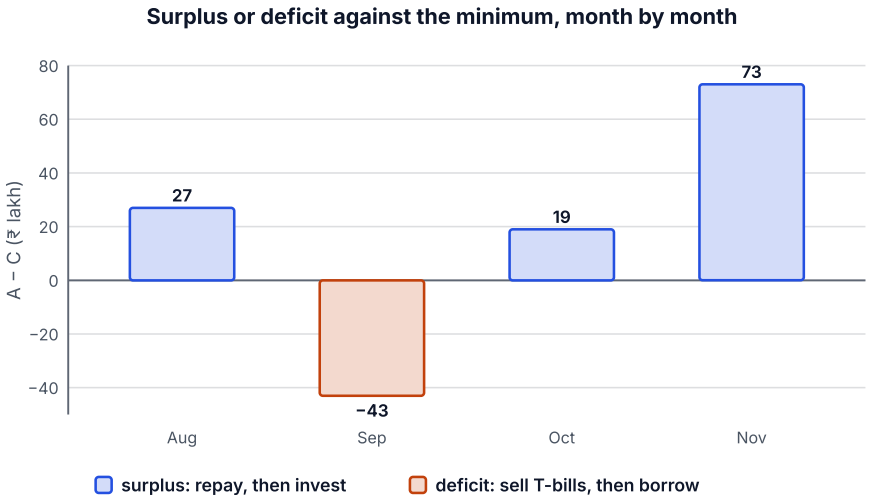
Working note 2: payments for purchases (₹ lakh)

Purchases of a month are 60% of the next month's sales: July 48 (for August's 80), August 72, September 96, October 54, November 36.

Particulars	Aug	Sep	Oct	Nov
Half of this month's purchases	36	48	27	18
Half of last month's purchases	24	36	48	27
Total paid	**60**	**84**	**75**	**45**

Cash budget, August to November (₹ lakh)

Particulars	Aug	Sep	Oct	Nov
Opening balance	20	10	10	10
Collections (WN 1)	93	94	116	134
Total cash available (A)	**113**	**104**	**126**	**144**
Purchases (WN 2)	60	84	75	45
Wages, salaries, bonus	8	8	14	8
Other expenses (10 − 2 depreciation)	8	8	8	8
Advance tax	—	12	—	—
Packing machine	—	25	—	—
Total payments (B)	**76**	**137**	**97**	**61**
Minimum cash balance	10	10	10	10
Total cash needed (C)	**86**	**147**	**107**	**71**
Surplus − deficit (A − C)	**27**	**(43)**	**19**	**73**
Treasury bills bought	(27)	—	(3)	(73)
Treasury bills sold	—	27	—	—
Cash credit drawn	—	16	—	—
Cash credit repaid	—	—	(16)	—
Investment / financing (D)	**(27)**	**43**	**(19)**	**(73)**
Closing balance (A + D − B)	**10**	**10**	**10**	**10**



Blue bars: surpluses above the ₹10 lakh minimum; orange bar: September's deficit. The 43 is met by selling all 27 of treasury bills and drawing 16 on the cash credit; October's 19 repays the 16 first and invests only 3.

Outstanding at the end of November: treasury bills ₹3 lakh + ₹73 lakh = **₹76 lakh**; cash credit **nil** (₹16 lakh drawn in September, repaid in October).

Points that carry marks: depreciation is left out (only ₹8 lakh is paid); September's deficit is measured against the minimum, so the financing is ₹43 lakh, not ₹33 lakh; and October repays the cash credit before buying bills.

Worked example

The cafés' disbursement account: Baumol against the monthly habit

Tapri's kulhad cafés pay their staff and vendors from a separate disbursement account. Disbursements are ₹30 lakh a month, spread evenly. The cafés' surplus sits in a liquid fund yielding 10% a year, and each transfer from the fund to the disbursement account costs ₹500 (bank charges and treasury staff time). At present the treasury transfers ₹30 lakh on the first of every month. Assume 360 days in a year.

Required:

(i) Calculate the optimum transfer size by the Baumol model, the number of transfers a year and the interval between them.

(ii) Calculate the annual holding and transaction costs at the optimum, and under the present policy. How much does the change save?

(iii) The bank offers an electronic sweep that cuts the cost per transfer to ₹125. Calculate the new optimum and its total cost, and comment on how the optimum responds to the transfer cost.

(iv) Comment on whether the Baumol model suits the cafés' cash flows.

THE WORKING

(i) Optimum transfer

U = ₹30,00,000 × 12 = ₹3,60,00,000 a year; P = ₹500; S = 0.10.

$$C = \sqrt{\frac{2 \times 3{,}60{,}00{,}000 \times 500}{0.10}} \qquad = \sqrt{3{,}60{,}00{,}00{,}00{,}000} \qquad = ₹6{,}00{,}000$$

Number of transfers = ₹3,60,00,000 ÷ ₹6,00,000 = **60 a year**; interval = 360 ÷ 60 = **6 days**.

(ii) Costs

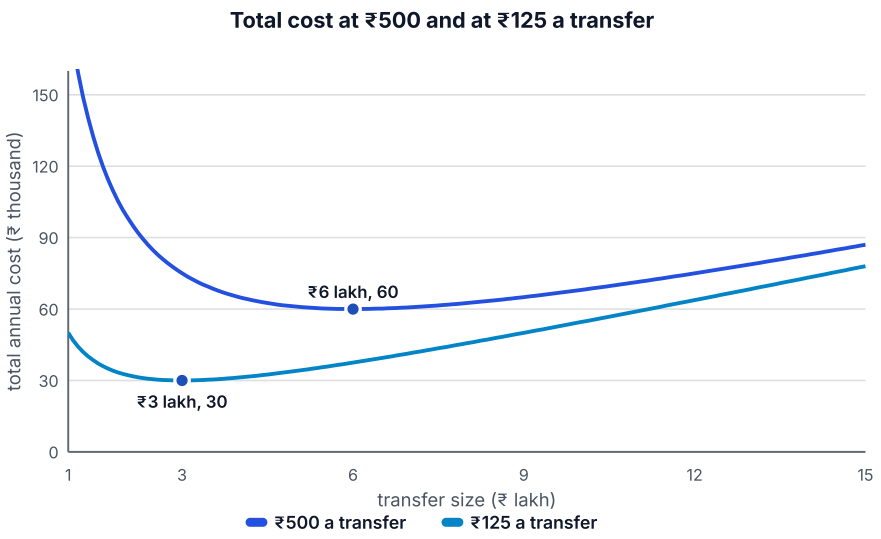
Particulars	Baumol optimum	Present policy
Transfer size (₹)	6,00,000	30,00,000
Transfers a year	60	12
Average balance (₹)	3,00,000	15,00,000
Holding cost at 10% (₹)	30,000	1,50,000
Transaction cost at ₹500 (₹)	30,000	6,000
Total cost (₹)	**60,000**	**1,56,000**

The change saves **₹96,000** a year: the present policy saves on transfers but leaves an average of ₹15 lakh earning nothing.

(iii) Transfer cost of ₹125

$$C = \sqrt{\frac{2 \times 3{,}60{,}00{,}000 \times 125}{0.10}} \qquad = ₹3{,}00{,}000$$

120 transfers, every 3 days; average balance ₹1,50,000; holding cost ₹15,000 + transaction cost ₹15,000 = **₹30,000**.



Blue: total cost at ₹500 a transfer, lowest at ₹6 lakh (₹60,000). Sky blue: at ₹125, lowest at ₹3 lakh (₹30,000). A quarter of the transfer cost halves both the optimum and the total cost.

Comment. C varies with the square root of P: a transfer cost one quarter as large gives an optimum one half as large ($\sqrt{1/4} = 1/2$), and the total cost also halves. Cheaper transfers mean more frequent, smaller ones.

(iv) Suitability. Baumol assumes cash needs known with certainty and used uniformly. Staff and vendor payments of a fixed monthly amount fit it reasonably. The cafés' **receipts**, though, rise and fall with footfall, festivals and the weather; for the cafés' main account, where net flows are random, the Miller-Orr model, with control limits set by the finance manager, is the more realistic choice (SM 9.61–9.62).

Worked example **Distributors' cheques: concentration banking or a lock box?**

Tapri's premix distributors mail their cheques to the head office in Indore. Collections are ₹180 crore a year, received evenly over 360 days. On average a cheque spends 4 days in the post (mail float), 2 days at head office before it is deposited (cheque processing float) and 2 days in clearing (banking processing float). Released funds would reduce the cash credit, which costs 11% a year. Two proposals:

1. **Concentration banking**: six regional collection centres depositing with local banks, which transfer surplus funds to head office's bank. The mail float falls to 1.5 days and the cheque processing float to 1 day. Added cost ₹12,00,000 a year.
2. **Lock box**: post-office boxes in six cities, emptied several times a day by the bank, which deposits the cheques directly. The mail float falls to 1.5 days and the cheque processing float is eliminated. The bank charges ₹3,00,000 a year plus ₹20 per cheque. The 1,200 distributors each send 4 cheques a month.

Required:

(i) Evaluate each proposal and advise.

(ii) Distributors plan to pay weekly in smaller amounts, so that the average cheque falls to ₹10,000. Re-evaluate the lock box.

(iii) Name the float each proposal attacks, and the rule ICAI gives for deciding.

THE WORKING

Working notes

1. Collections per day = ₹180,00,00,000 ÷ 360 = **₹50,00,000**.
2. Cheques a year = 1,200 × 4 × 12 = **57,600**; average cheque = ₹180 crore ÷ 57,600 = ₹31,250.

(i) Evaluation

Particulars	Concentration banking	Lock box
Mail float saved (days)	2.5	2.5
Cheque processing float saved (days)	1	2
Days saved	**3.5**	**4.5**
Funds released (₹)	1,75,00,000	2,25,00,000
Income at 11% (₹)	19,25,000	24,75,000
Added cost (₹)	12,00,000	14,52,000
Net benefit (₹)	**7,25,000**	**10,23,000**

Lock box cost = ₹3,00,000 + 57,600 × ₹20 = ₹14,52,000. **Advice:** both are profitable; the lock box gives the larger net benefit, ₹10,23,000 a year, because it also removes the two days head office takes to deposit the cheques.

(ii) Average cheque of ₹10,000

Cheques = ₹180 crore ÷ ₹10,000 = 1,80,000 a year; lock box cost = ₹3,00,000 + 1,80,000 × ₹20 = ₹39,00,000, against income of ₹24,75,000: a **net loss of ₹14,25,000**. Concentration banking, whose cost does not depend on the number of cheques, still gains ₹7,25,000 and becomes the better choice. This is ICAI's warning that lock box arrangements are usually not profitable if the average remittance is small (SM 9.55); here the lock box stays ahead of concentration banking only while its cost is below ₹24,75,000 – ₹7,25,000 = ₹17,50,000, that is, up to 72,500 cheques (an average cheque of about ₹24,828).

(iii) Both shorten the **mail float** by collecting regionally; the lock box also eliminates the **cheque processing float**. The rule (SM 9.55): compare the added cost of the most efficient system with the marginal income generated from the released funds; if the cost is less than the income, the system is profitable, otherwise it is not worth undertaking.

Exam angle

In the exam

Where cash management appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, it was examined 11 times in 9 new-scheme documents:

- **Baumol, numerical.** January 2026 MCQ 7 (2 marks) runs the model backwards: given C, P and S, find the annual disbursement ($U = C^2S/2P$). RTP January 2025 Q1 (a case, MCQ (i)) needs ₹1.50 lakh a month turned into ₹18 lakh a year before the 12% rate is applied.
- **Cash budget.** January 2025 Q1(a) (5 marks): three months, a minimum balance of ₹1 lakh, investment above ₹2 lakh, borrowing below the minimum. RTP September 2026 Q9: a quarter's budget against a target balance.
- **Cash cycle and minimum operating cash.** May 2026 Q3(b) (6 marks): cash cycle, minimum cash and a cash-discount scheme judged by the saving on the reduced balance. MTP January 2026 Series II Q2 (10 marks) gives the cash turnover (9 times) and asks for the rest of the working capital statement.
- **Theory, 2 to 4 marks.** Functions of the treasury department: MTP May 2026 Series II Q4(b) (4 marks) and RTP May 2026 Q10(b). Virtual banking: January 2025 Q4(c) (two advantages, 2 marks) and January 2026 Q4(b) (why popular, and the advantages, 4 marks). Features of the Miller-Orr model: September 2026 Q4(b) (4 marks; suggested answers not yet published).
- **Old scheme, for practice:** November 2022 Q1(a) (Baumol: average balance, conversions, interval; 5 marks); MTP May 2022 Series II (Baumol and EOQ inside a working capital estimate); MTP December 2021 Series I (short note on electronic fund transfer).

In the exam

The traps ICAI used

- **Period of U and S.** U monthly with S annual gives a wrong C; convert both to a year (RTP January 2025).
- **Cash outlay, not sales.** In May 2026 Q3(b) the annual operating outlay was sales less net profit (₹200 lakh – ₹20 lakh), not sales.
- **"60% of debtors avail the discount".** The May 2026 suggested answer showed two readings: everyone's period falls by 30 days, or a weighted period (60% × 30 + 40% × 60 = 42 days). State your reading.
- **The minimum balance band.** January 2025: borrow up to the minimum of ₹1 lakh, invest only the excess over ₹2 lakh. Read the instruction for each side.

Trick

Square root, so a quarter gives a half

In Baumol, C moves with the square root of U and P and inversely with the square root of S: four times the disbursement or the transfer cost doubles C; four times the interest rate halves it. Use it to check an answer in seconds. Condition: the model's assumptions (certain, uniform use; constant costs).

Answer frame

"Explain briefly the functions of the treasury department" (4 marks)

The five tasks, in ICAI's order, one or two lines each: **cash management** (collections, payments, surplus funds in an investment portfolio), **currency management** (set off intra-group indebtedness, match receipts and payments, forward contracts), **fund management** (plan and source short, medium and long-term needs; temporary investment of surplus; capital structure; forecast rates), **banking** (relationship and negotiations with bankers; bank loans, commercial paper), **corporate finance** (acquisitions and divestments; investor relations).

"Explain Baumol's model of cash management"

1. Optimum cash level: where carrying costs and transaction costs are the minimum.
2. Carrying cost = opportunity cost (interest foregone on marketable securities); transaction cost = cost of converting securities into cash (clerical, brokerage, registration).
3. Formula $C = \sqrt{2UP/S}$ with every symbol defined.
4. The diagram: holding cost rising, transaction cost falling, total cost U-shaped, optimum at its lowest point.
5. The four assumptions; the limitation (no fluctuation allowed), which Miller-Orr overcomes.
6. "With example": a three-line calculation.

"Discuss the Miller-Orr cash management model"

1. Net cash flow completely stochastic; a control limit model; decides the time and size of transfers between an investment account and the cash account.
2. Limits: h (upper), z (return point), zero (lower).
3. At h, invest h − z in marketable securities; at the lower limit, transfer from securities to cash; in between, no transactions.
4. Limits set by the fixed cost of securities transactions, the opportunity cost of holding cash and the degree of likely fluctuations.
5. More realistic than Baumol; the finance manager can set the limits to the firm's liquidity requirements.
6. A labelled sketch.

Short notes

- **Virtual banking**: meaning, the four advantages, the reason for its popularity (speed, convenience, round the clock access).

- **Kinds of float**: the four, one line each, then concentration banking and the lock box as the cures.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the monthly U with the annual S

Disbursements of ₹2,00,000 a month, a 12% annual yield, ₹150 a transfer: $\sqrt{2 \times 2{,}00{,}000 \times 150/0.12}$ gives ₹22,361. **Fix:** put U and S on the same period: $\sqrt{2 \times 24{,}00{,}000 \times 150/0.12} = ₹77{,}460$ (or the monthly U with a monthly S of 1%, which gives the same).

⚠ **Common mistake**

Borrowing to zero instead of to the minimum

A balance of −₹33 lakh before financing, with a ₹10 lakh minimum, is financed with ₹33 lakh. **Fix:** finance to the minimum: ₹43 lakh. ICAI's layout (total cash needed = payments + minimum) makes the slip impossible.

⚠ **Common mistake**

Depreciation paid in cash

"Other expenses ₹10 lakh, including depreciation ₹2 lakh" entered at ₹10 lakh. **Fix:** a cash budget has only cash: ₹8 lakh. The same goes for provisions and write-offs.

⚠ **Common mistake**

Calling the optimum balance the average balance

Baumol's C is the amount transferred each time; the average balance is C ÷ 2, and the holding cost is charged on the average. **Fix:** holding cost = (C ÷ 2) × S; transaction cost = (U ÷ C) × P.

Practice

Check yourself

1 Tapri's Ujjain café pays out ₹40,00,000 a year evenly from its disbursement account. Each transfer from its liquid fund costs ₹180, and the fund yields 10% a year. If it follows the Baumol model, what is its total annual cost of holding cash and making transfers, in rupees?

Answer: _____

2 Case scenario. Tapri's southern distributors mail their cheques to Indore and complain that their payments take a week to show. Dev proposes collection centres in Chennai, Bengaluru and Kochi, each banking with a local branch that passes the surplus funds daily to the bank where head office keeps its main account. Which of the following is correct?

- **A** It is a lock box system, and its main effect is to remove the cheque processing float
- **B** It is playing the float, and it delays Tapri's payments to its suppliers
- **C** It is concentration banking, and it mainly shortens the time between the distributor mailing the remittance and its becoming spendable funds
- **D** It is a zero balance account, and it invests the day's surplus in marketable securities

3 Under the Miller-Orr model, the firm's cash balance touches the upper control limit h. What does the firm do?

- **A** Invests an amount equal to h − z in marketable securities
- **B** Invests the whole balance h in marketable securities
- **C** Sells marketable securities worth h − z
- **D** Nothing, until the balance returns to z by itself

4 Tapri's cold-chain unit keeps a minimum cash balance of ₹5 lakh. For March its cash budget shows a balance of −₹2 lakh before any financing, and it holds no investments. How much must it borrow at the end of March?

- **A** ₹2 lakh
- **B** ₹5 lakh
- **C** ₹3 lakh
- **D** ₹7 lakh

5 A samosa supplier spends ₹54 lakh a year at a constant rate and keeps a minimum operating cash of ₹9 lakh, worked out by ICAI's cash cycle method. It collects from its debtors in 40 days and pays its creditors in 30 days. Taking 360 days, how many days does it hold inventory?

Answer: _____

6 Which of these are among ICAI's three principles for selecting marketable securities for surplus cash? Select all that apply.

- **A** Safety
- **B** Highest possible return
- **C** Maturity
- **D** Marketability

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Treasury management: planning, organising and controlling funds and working capital for best use of funds, liquidity, lower cost of funds and less risk; five tasks: cash, currency, fund management, banking, corporate finance.
- Cash management balances liquidity against cost; Keynes's three needs: transaction, speculative, precautionary.
- Cash budget: the most significant device for planning cash; with a minimum balance, total cash needed = payments + minimum, and the surplus or deficit is invested or financed (ICAI Illustration 6 layout). Only cash items; no depreciation.
- Four floats: billing, mail, cheque processing, banking processing. Concentration banking cuts the time from remittance to spendable funds; a lock box eliminates the cheque processing float but costs per cheque. Accept if the income on released funds exceeds the added cost.
- Baumol: $C = \sqrt{2UP/S}$; transfers U ÷ C; average balance C ÷ 2; at the optimum, holding cost = transaction cost. Assumes certain, uniform cash use.
- Miller-Orr: random flows; limits h, z and zero; invest h − z at the top, sell securities at the bottom, nothing in between.
- Minimum operating cash = annual operating outlay ÷ cash turnover, where cash turnover = 360 ÷ cash cycle.
- Surplus goes to securities chosen for safety, maturity and marketability. (Fill the wallet once a week, not once a year.)

Answer key · Check yourself

1 **12000**

C = √(2 × 40,00,000 × 180 ÷ 0.10) = ₹1,20,000. Holding cost = (₹1,20,000 ÷ 2) × 10% = ₹6,000; transaction cost = (₹40,00,000 ÷ ₹1,20,000) × ₹180 = ₹6,000; total ₹12,000. Charging the holding cost on the full ₹1,20,000 gives ₹18,000, the commonest slip.

2 **C** **It is concentration banking, and it mainly shortens the time between the distributor mailing the remittance and its becoming spendable funds**

Regional collection centres depositing in local banks that transfer surplus funds to the head office's concentration bank is concentration banking (SM 9.54). A lock box needs post-office boxes emptied by the bank; playing the float concerns payments, not collections.

3 **A** **Invests an amount equal to h − z in marketable securities**

At h, cash equal to h − z is invested, bringing the balance back to the return point z (SM 9.62). Securities are sold only at the lower limit; between the limits nothing is done.

4 **D** **₹7 lakh**

The borrowing must restore the minimum: ₹5 lakh − (−₹2 lakh) = ₹7 lakh. ₹2 lakh only reaches zero; ₹3 lakh subtracts the deficit from the minimum.

5 **50**

Cash turnover = ₹54 lakh ÷ ₹9 lakh = 6, so the cash cycle = 360 ÷ 6 = 60 days. Debtors + inventory − creditors = 60, so inventory = 60 − 40 + 30 = 50 days (Illustration 11 method, SM 9.68). Forgetting to add back the creditors gives 20 days.

6 **A Safety; C Maturity; D Marketability**

Safety, maturity and marketability (SM 9.67). Return and risk go hand in hand, and since the aim is liquidity, minimum risk is the criterion, not the highest return.

9.5 Management of inventory

Inventory as money on a shelf: what inventory management covers, the stock-out against over-stock trade-off, the Paper 4 control tools seen through a finance lens, and what cutting stock days does to working capital.

In plain words

The godown that ate the cash credit

Tapri's Indore godown is stacked to the roof with cardamom, because Kesar is afraid of a shortage before the festive season. Across the road, at the bank, the cash credit is fully drawn.

> **Kesar:** Full godown, happy heart. If the cardamom runs out, there is no premix.
>
> **Dev:** Full godown, empty bank. Every sack in there is sitting on our cash credit at 12%.
>
> **Ritu:** And I have a formula that says how much to order at a time. Per kilo, please.

Inventory is the stock a business holds: raw materials, work-in-progress (half-made goods) and finished goods. It is one of the biggest parts of working capital, and it has two opposite dangers. Too little, and production stops, orders are lost and customers go elsewhere: a **stock-out**. Too much, and money is locked on the shelf, paying interest, rent, insurance and the risk that the stock goes stale: **over-stocking**.

Paper 4 teaches how a store controls its stock: how much to order, when to reorder, which items to watch closely. Paper 6A asks the finance manager's question: how much money should be invested in stock, and what does each extra day of stock cost? The answer is priced at the **cost of the funds** tied up in it: the interest on the cash credit that finances the stock, or whatever rate the question says the money could otherwise earn. Every day of stock removed from the operating cycle is cash released.

Think of it like

The fridge before a family wedding

Buy all the vegetables for the week on Monday and the fridge overflows, half of it wilts by Saturday, and the money left your wallet on Monday. Go to the market every morning and nothing spoils, but you pay the auto fare and lose an hour every day. The sensible family buys in a few lots, keeps a little extra in case guests arrive early, and knows the sabziwala who can deliver at short notice. Ordering in economic lots is EOQ; the little extra is safety stock; the sabziwala at the door is just-in-time.

Key idea

Inventory is working capital on a shelf. Hold enough to avoid stock-outs and no more, and price every extra rupee and every extra day of stock at the cost of the funds that finance it.

How it works

What inventory management covers

Inventories constitute a major element of working capital, so the investment in them must be properly controlled; the objectives of inventory management are, to a great extent, similar to those of cash management (SM 9.69). In ICAI's words, inventory management covers a large number of problems (SM 9.69):

1. Fixation of minimum and maximum levels.
2. Determining the size of inventory to be carried.
3. Deciding about the issues, receipts and inspection procedures.
4. Determining the economic order quantity.
5. Proper storage facilities.
6. Keeping check over obsolescence.
7. Ensuring control over movement of inventories.

The Paper 6A Study Material then refers the student to Chapter 2 (Material Cost) of Paper 4, where these tools are taught in detail (SM 9.69). This lesson keeps to what the finance manager needs from them; the techniques themselves are in the Paper 4 lessons linked below.

The trade-off: stock-outs against locked-up capital

ICAI's table of working capital trade-offs gives inventory this row (SM 9.11):

Advantage of the higher side (profitability)	The trade-off	Advantage of the lower side (liquidity)
Fewer stock-outs increase the profitability	Use techniques like EOQ, JIT etc. to carry the optimum level of inventory	Lower inventory requires less capital, but endangers stock-out and loss of goodwill

Among the factors to be considered while planning working capital, the **desired level of inventory** is the level which allows for uninterrupted production but reduces the investment in raw materials and hence increases cash flow; techniques like Just in Time (JIT) and Economic Order Quantity (EOQ) are used for it (SM 9.9). And inventory is ideally financed by credit granted by the supplier (SM 9.10); beyond that, banks sanction cash credit limits against the security of tradable goods by way of pledge or hypothecation (FM Study Material 2.27; see financing of working capital and bank credit).

The CIMA definition the Paper 4 Study Material uses says the same in one line: **inventory control** is "the function of ensuring that sufficient goods are retained in stock to meet all requirements without

carrying unnecessarily large stocks" (CMA Study Material 2.21).

Inventory in the operating cycle

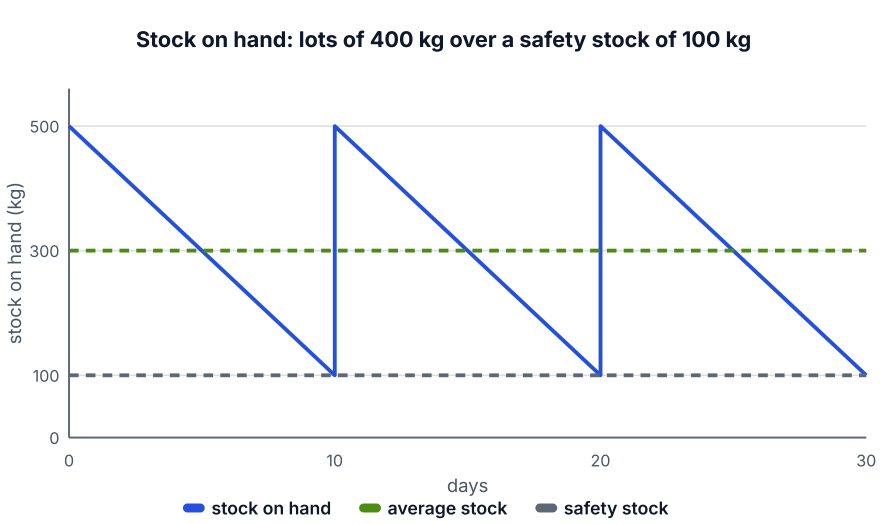
Three of the five stages of the operating cycle are inventory (SM 9.19–9.20): **R** (raw material storage period), **W** (work-in-progress holding period) and **F** (finished goods storage period), in

$$\text{Operating cycle} = R + W + F + D - C$$

Each is measured against its own cost base: average raw material stock against the average cost of raw material consumed per day, average WIP against the average cost of production per day, and average finished goods against the average cost of goods sold per day (SM 9.20). The Study Material's rule of thumb (SM 9.19): **shift inventory faster and you free up cash; move it slower and you consume more cash.** The cycle itself is built in working capital and the operating cycle.

What a shelf of stock holds, in rupees

When a firm orders a fixed lot and keeps a safety stock, its stock falls steadily from the top of each lot to the safety stock and jumps back when the next lot arrives. Take lots of 400 kg, a safety stock of 100 kg and use of 40 kg a day:



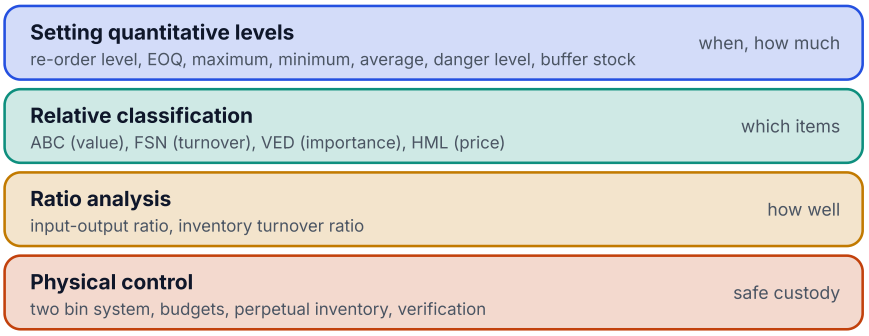
Blue: stock on hand, falling 40 kg a day from 500 to 100 and refilled by each 400 kg lot. Green dashed: the average, half a lot plus the safety stock (200 + 100 = 300 kg). Grey dashed: the safety stock, which is never used in a normal cycle.

The money invested in the item is the **average stock** × its cost: half the order quantity plus the safety stock. That is why smaller, more frequent lots (as under JIT) or a smaller safety stock free cash, and why the carrying cost is charged on the average stock and not on the lot.

The control tools, from Paper 4

The Paper 4 Study Material groups inventory control into four bases (CMA Study Material 2.22):

The four bases of inventory control (CMA Study Material 2.22)



Each base answers a different question; the right-hand column names it. The detail of each is in the Paper 4 lessons linked below.

Tool	What it decides	Rule (CMA Study Material)
Re-order level	When to order	Maximum consumption × maximum re-order period (2.22)
Economic order quantity	How much to order	Order size at which ordering + carrying cost is minimum (2.23–2.24)
Maximum, minimum, danger levels	How much to hold	Formulas at 2.27; see stock levels
ABC analysis	Which items need tight control	About 10% of items, 70% of value are A (2.35–2.36); see ABC and JIT
Just in Time	Whether to hold stock at all	Buy only when actually required for production (2.35)
Inventory turnover ratio	Is stock moving?	Low ratio: over-investment, working capital locked up (2.42); see turnover ratios

Three Paper 4 definitions the finance manager uses most (CMA Study Material 2.23–2.24, 2.35):

- **Ordering costs**: costs associated with the purchase or order of materials: inviting quotations, purchase orders, procurement staff, transportation and inspection.
- **Carrying costs**: costs of holding inventories in store: the cost of fund invested in inventories, storage, insurance, obsolescence.
- **Just in Time (JIT)**: a system of inventory management with an approach to have zero inventories in stores; material is purchased only when actually required for production. Its two principles: produce goods only when required, and deliver products to customers only when they want them. It is also called a "demand pull" or "pull through" system.

The EOQ and its quantity-discount tests are worked in full in economic order quantity.

The finance manager's lens

Three habits distinguish a finance answer from a stores answer:

1. **The rate.** The cost of funds inside the carrying cost is the firm's borrowing rate (the cash credit) or its cost of capital; a higher rate shrinks the EOQ and makes every day of stock dearer.
2. **The base.** Stock is valued at cost, not at selling price: raw materials at material cost, work-in-progress at its cost to date, finished goods at the cost of production (SM 9.22–9.23); the estimation is in estimating working capital requirement.
3. **The cash.** A change in stock policy is judged by its total cost (purchase price, ordering, carrying) and by the working capital it releases or absorbs; but the interest on released funds is already inside the carrying cost, so it is counted once.

The Paper 4 list of physical controls includes one that is pure finance: **a system of budgets**, an inventories requirement budget from the production plan, which "will discourage the unnecessary investment in inventories" (CMA Study Material 2.45).

Formats and formulas

Reproduce the EOQ formula and the cost comparison layout from memory.

Σ **Formula**

Economic order quantity (CMA Study Material 2.24)

$$\text{EOQ} = \sqrt{\frac{2 \times A \times O}{C}}$$

A = annual requirement (units); O = cost per order; C = carrying cost per unit per annum, which includes the cost of funds invested in the stock.

Σ **Formula**

Average stock and the money in it

$$\text{Average stock} = \frac{Q}{2} + \text{Safety stock}$$

$$\text{Investment} = \text{Average stock} \times \text{Unit cost}$$

Q = order quantity.

Σ **Formula**

Total annual cost of a stock policy

$$\text{Total cost} = \text{Purchase cost} + \frac{A}{Q} \times O \qquad + \left(\frac{Q}{2} + \text{Safety stock}\right) \times C$$

Q = order quantity. Compare policies on this total; the purchase cost matters whenever the policies change the price.

Σ Formula

Inventory holding periods (SM 9.20)

$$R = \frac{\text{Average raw material stock}}{\text{Raw material consumed per day}}$$

$$F = \frac{\text{Average finished goods stock}}{\text{Cost of goods sold per day}}$$

In ICAI's words: average stock of raw material ÷ average cost of raw material consumption per day; average stock of finished goods ÷ average cost of goods sold per day; W = average work-in-progress inventory ÷ average cost of production per day.

Σ Formula

Working capital released by fewer days of stock

$$\text{Funds released} = \text{Cost per day} \times \text{Days cut}$$

$$\text{Annual saving} = \text{Funds released} \times \text{Carrying rate}$$

Cost per day on the right base: material consumed per day for raw materials, cost of production per day for finished goods.

Comparison statement for two stock policies

Particulars	Policy I (₹)	Policy II (₹)
Purchase cost (units × price)	xxx	xxx
Ordering cost (number of orders × cost per order)	xxx	xxx
Carrying cost (average stock × carrying cost per unit)	xxx	xxx
Total annual cost	**xxx**	**xxx**

Definition

Inventory control (CIMA, as quoted by ICAI)

The function of ensuring that sufficient goods are retained in stock to meet all requirements without carrying unnecessarily large stocks. (CMA Study Material 2.21)

Definition

Economic order quantity

The size of an order for which total of ordering and carrying cost are minimum. (CMA Study Material 2.23)

Worked examples

Worked example **Cardamom: economic lots or weekly deliveries?**

Tapri's premix plant uses 7,200 kg of cardamom a year, evenly. It buys from a Kerala grower at ₹2,000 per kg. Each order costs ₹7,500 (documentation, freight booked per consignment, inspection). The carrying cost is 15% a year of the stock's cost: 12% is the interest on the cash credit that finances the stock and 3% is storage and insurance. The plant keeps a safety stock of 200 kg.

The grower proposes a just-in-time arrangement: weekly deliveries of 150 kg (48 a year) at ₹2,010 per kg, under a standing order that cuts the cost per delivery to ₹1,200. Because supply becomes regular, Ritu thinks the safety stock can come down to 50 kg.

Required:

(i) Calculate the EOQ, the number of orders a year, the average stock and the average investment in cardamom under the present system.

(ii) Calculate the total annual cost under the present system ordering the EOQ, and under the JIT proposal.

(iii) Advise Tapri, and state the working capital the proposal releases.

(iv) Kesar says the released funds will also save 12% interest on the cash credit, so that saving should be added. Comment.

(v) State one risk of the proposal that the figures do not show.

THE WORKING

(i) Present system

Carrying cost per kg = 15% × ₹2,000 = ₹300 a year.

$$\text{EOQ} = \sqrt{\frac{2 \times 7{,}200 \times 7{,}500}{300}} \qquad = \sqrt{3{,}60{,}000} = 600 \text{ kg}$$

Orders a year = 7,200 ÷ 600 = **12** (one a month). Average stock = 600 ÷ 2 + 200 = **500 kg**; average investment = 500 × ₹2,000 = **₹10,00,000**.

(ii) Total annual cost

Under JIT, carrying cost per kg = 15% × ₹2,010 = ₹301.50; average stock = 150 ÷ 2 + 50 = 125 kg.

Working notes: purchase cost = 7,200 kg × ₹2,000 (present) or × ₹2,010 (JIT); ordering cost = 12 × ₹7,500 or 48 × ₹1,200; cycle stock carried = 300 kg × ₹300 or 75 kg × ₹301.50; safety stock carried = 200 kg × ₹300 or 50 kg × ₹301.50.

Particulars	Present (₹)	JIT (₹)
Purchase cost	1,44,00,000	1,44,72,000
Ordering cost	90,000	57,600
Carrying: cycle stock	90,000	22,613
Carrying: safety stock	60,000	15,075
Total annual cost	**1,46,40,000**	**1,45,67,288**

(iii) Advice. The JIT proposal costs **₹72,712 a year less**: the ₹72,000 price premium is more than paid for by ₹32,400 less ordering cost and ₹1,12,312 less carrying cost. Accept it, subject to (v). Average investment falls from ₹10,00,000 to 125 × ₹2,010 = ₹2,51,250, releasing **₹7,48,750** of working capital, which can be used to reduce the cash credit.

(iv) Kesar's extra saving. It would be counted twice. The 15% carrying cost already includes 12% interest on the funds invested in stock: the interest on the present ₹10,00,000 (₹1,20,000) and on the JIT ₹2,51,250 (₹30,150) is inside the two carrying cost lines. Adding ₹89,850 again would overstate the benefit.

(v) Risk. With only 50 kg of safety stock (about 2.5 days' use), a single late delivery stops the premix line; the cost of a stock-out (lost production, sales and goodwill) is not in the table. The grower's reliability, an alternative supplier and a penalty clause should be settled before the standing order is signed.

Worked example

The samosa factory: fewer days of stock, worth the software?

Tapri's frozen samosa factory consumes raw materials costing ₹7.20 crore a year, and its conversion costs (wages and cash overheads) are ₹3.60 crore a year. Raw materials are held for 30 days; work-in-progress takes 6 days, with materials issued at the start and conversion costs 50% complete on average; finished goods are held for 15 days. Debtors pay in 30 days and suppliers are paid in 45 days. Assume 360 days in a year.

Ritu proposes ABC-based stock control with automated reordering, which would cut the raw material period to 18 days and the finished goods period to 9 days, at a cost of ₹6,00,000 a year (software licence and an analyst). Stock is financed by the cash credit at 12%, and storage and insurance cost 3% a year of the value of stock held.

Required:

(i) Calculate the investment in each type of inventory before and after the proposal.

(ii) Calculate the change in the net operating cycle.

(iii) Evaluate the proposal on financial grounds and comment.

THE WORKING

Working notes

1. Cost of production = ₹7.20 crore + ₹3.60 crore = **₹10.80 crore**; per day ₹3,00,000.

2. Raw material consumed per day = ₹7.20 crore ÷ 360 = ₹2,00,000; conversion cost per day = ₹1,00,000.

(i) Investment in inventory (₹)

Raw materials = ₹2,00,000 a day × 30 days (18 after); WIP materials = ₹2,00,000 × 6 days; WIP conversion = ₹1,00,000 × 6 days × 50%; finished goods = ₹3,00,000 × 15 days (9 after).

Particulars	Before	After
Raw materials	60,00,000	36,00,000
WIP: materials	12,00,000	12,00,000
WIP: conversion	3,00,000	3,00,000
Finished goods	45,00,000	27,00,000
Total inventory	**1,20,00,000**	**78,00,000**

Finished goods are valued at the cost of production, not at selling price (SM 9.23).

(ii) Net operating cycle

Before: R + W + F + D − C = 30 + 6 + 15 + 30 − 45 = **36 days**. After: 18 + 6 + 9 + 30 − 45 = **18 days**. The cycle halves.

(iii) Evaluation

Particulars	Amount (₹)
Inventory released (1,20,00,000 − 78,00,000)	42,00,000
Interest saved on the cash credit at 12%	5,04,000
Storage and insurance saved at 3%	1,26,000
Total annual saving	**6,30,000**
Less: annual cost of the proposal	6,00,000
Net annual benefit	**30,000**

Comment. The proposal is only just worth it: a net ₹30,000 a year on ₹6,00,000 of cost. On interest alone (₹5,04,000) it would not pay; the storage saving tips it. The decision is sensitive to the rate: at a higher cash credit rate it improves, and if the lower stocks cause even one costly stock-out it fails. With so thin a margin, the case rests as much on control as on cost: ABC analysis exists to put tight control on the few high-value A items, a benefit the figures do not show.

Exam angle

In the exam

How inventory management was examined, May 2024 to September 2026

In the ICAI Paper 6A documents read for this lesson, inventory came up 4 times in 4 new-scheme documents, always inside working capital, never as a stand-alone EOQ question:

- **Theory.** RTP September 2025 Q9(c): analyse the trade-offs between the components of working capital; the suggested answer reproduces ICAI's table, with the inventory row (fewer stock-outs, EOQ and JIT, less capital but risk of stock-out). RTP September 2026 Q10(c): the factors to be considered while planning working capital, including the desired level of inventory (JIT and EOQ).
- **Case scenario.** MTP May 2025 Series II, MCQ 3: rank three proposals to cut the inventory of a washing machine maker (discounts, an exchange scheme, delivery after advance booking) by their benefit.
- **Operating cycle.** MTP September 2025 Series I Q1(c) (5 marks): a just-in-time manufacturer that decides to hold slightly higher inventories after supply disruptions; the net operating cycle from opening and closing stocks of raw material, WIP and finished goods, then the working capital with a 10% contingency reserve.
- **Old scheme, for practice:** May 2022 Q1(c) (5 marks): re-order level, EOQ and a 1% quantity discount in Paper 8; MTP May 2022 Series II Q3 (10 marks): raw material stock equal to the EOQ inside a working capital estimate.

In the exam

What earns the marks

- Inventory periods use **cost** bases (consumption, cost of production, cost of goods sold), never sales.
- In a stock-policy comparison, show purchase, ordering and carrying cost for each policy, then the difference.
- In theory answers, name the trade-off in ICAI's words: profitability (fewer stock-outs) against liquidity (less capital locked up), resolved by EOQ and JIT.

Trick

Half the lot plus the cushion

Money in an item = (order quantity ÷ 2 + safety stock) × cost per unit. Condition: steady usage and lots arriving as the previous lot runs down to the safety stock.

Answer frame

"Explain the scope of inventory management" or "What does inventory management cover?"

1. Inventories are a major element of working capital; their investment must be properly controlled; the objectives are similar to those of cash management.
2. The seven problems in ICAI's order: minimum and maximum levels; size of inventory; issues, receipts and inspection procedures; EOQ; storage; obsolescence; control over movement.
3. The aim: sufficient stock for all requirements without unnecessarily large stocks.

"Discuss the trade-off between profitability and liquidity in inventory"

Point	Higher inventory	Lower inventory
Effect	Fewer stock-outs; uninterrupted production and sales	Less capital required; cash released
Risk	Funds locked up; carrying cost; obsolescence	Stock-out and loss of goodwill
Tools	EOQ, JIT, stock levels: carry the optimum level	EOQ, JIT, stock levels: carry the optimum level

Common mistakes

⚠ **Common mistake**

Hall of Shame: the interest that was saved twice

The carrying cost is 15%, of which 12% is the cost of funds, and the answer adds "interest saved on released stock" on top of the lower carrying cost. **Fix:** the cost of funds is inside the carrying cost; count it once.

⚠ **Common mistake**

The whole lot on the shelf all year

Taking the average stock as the EOQ (600 kg) instead of half of it plus the safety stock (500 kg), or forgetting the safety stock altogether (300 kg). **Fix:** average stock = Q ÷ 2 + safety stock; the investment and the carrying cost follow from it.

⚠ **Common mistake**

Finished goods at selling price

Valuing finished goods stock at sales value inflates the working capital by the profit margin. **Fix:** finished goods at the cost of production (cash cost where the question uses the cash cost basis), WIP at its cost to date (SM 9.22–9.23).

⚠ **Common mistake**

The cheaper order that ignored the price

Comparing JIT and EOQ on ordering and carrying cost alone, when JIT comes with a price premium. **Fix:** when a policy changes the price, the purchase cost is a relevant cost and enters the comparison.

Practice

Check yourself

1 Tapri's café chain buys 4,000 cartons of tea a year at ₹400 each. Each order costs ₹1,440, and the carrying cost is 18% a year of the stock's cost. It keeps a safety stock of 100 cartons and orders the EOQ. What is the average investment in this tea, in rupees?

Answer: _____

2 Case scenario. After two late deliveries of potato flakes, the samosa factory's management doubles the safety stock of raw materials. Usage, the order quantity, prices and all credit periods are unchanged. Which of the following is correct?

- **A** The EOQ rises, so orders are placed less often
- **B** The raw material storage period, the operating cycle and the working capital requirement all rise
- **C** The working capital requirement is unchanged, because safety stock is never used
- **D** The operating cycle falls, because stock-outs are avoided

3 In ICAI's table of working capital trade-offs, what is the advantage of the lower side for inventory?

- **A** Fewer stock-outs increase profitability
- **B** It allows quantity discounts from suppliers
- **C** It removes the need for EOQ and JIT
- **D** Lower inventory requires less capital, though it endangers stock-out and loss of goodwill

4 The carrying cost of a component is 16% a year: 13% interest on the cash credit and 3% storage. A new ordering policy releases ₹5,00,000 of stock, and the carrying cost saving has been computed at 16% of it. Kesar wants to add a further ₹65,000 for "interest saved on the cash credit". The correct treatment is:

- **A** Do not add it: the 13% interest is already part of the 16% carrying cost saving
- **B** Add it: interest and carrying cost are separate items
- **C** Add it, but only after tax
- **D** Replace the carrying cost saving with the interest saving

5 Which of these are carrying costs in ICAI's definition? Select all that apply.

- **A** Cost of fund invested in inventories
- **B** Insurance of the stock
- **C** Cost of inviting quotations
- **D** Obsolescence
- **E** Inspection of materials received

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Inventories are a major element of working capital; their objectives are similar to those of cash management (SM 9.69).

- Scope, in ICAI's order, is seven problems: minimum and maximum levels, size of inventory, issues, receipts and inspection, EOQ, storage, obsolescence and control over movement.
- Trade-off: more stock means fewer stock-outs (profitability), less stock means less capital but a risk of stock-out and loss of goodwill (liquidity), and EOQ and JIT find the optimum (SM 9.11).
- EOQ = √(2AO ÷ C); C includes the cost of funds invested in the stock.
- Average stock = Q ÷ 2 + safety stock; money invested = average stock × cost per unit.
- R, W and F are measured on cost bases; shifting stock faster frees cash.
- Compare stock policies on total cost (purchase + ordering + carrying), and count the interest on released funds only once.
- Inventory is ideally financed by supplier credit; banks lend against it as a cash credit. (Buy for the week, not for the wedding season.)

Answer key · Check yourself

1 **120000**

C = 18% × ₹400 = ₹72; EOQ = √(2 × 4,000 × 1,440 ÷ 72) = √1,60,000 = 400 cartons. Average stock = 400 ÷ 2 + 100 = 300 cartons; investment = 300 × ₹400 = ₹1,20,000. Taking the whole EOQ gives ₹1,60,000; forgetting the safety stock, ₹80,000.

2 **B** **The raw material storage period, the operating cycle and the working capital requirement all rise**

The EOQ depends on annual usage, ordering cost and carrying cost, not on the safety stock. The extra safety stock raises the average stock, so R, the operating cycle and the funds invested all rise, even though the safety stock is not used in a normal cycle.

3 **D** **Lower inventory requires less capital, though it endangers stock-out and loss of goodwill**

SM 9.11: the higher side gives fewer stock-outs (profitability); the lower side requires less capital (liquidity) but risks stock-out and loss of goodwill; EOQ and JIT are the trade-off tools.

4 **A** **Do not add it: the 13% interest is already part of the 16% carrying cost saving**

₹5,00,000 × 16% = ₹80,000 already includes ₹65,000 of interest and ₹15,000 of storage. Adding ₹65,000 again counts it twice.

5 **A** **Cost of fund invested in inventories**; **B** **Insurance of the stock**; **D** **Obsolescence**

Carrying costs: cost of funds invested, storage, insurance, obsolescence (CMA Study Material 2.23). Inviting quotations and inspection on receipt are ordering costs.

9.6 Management of payables

Trade credit as a spontaneous source of finance, its hidden costs and the costs of not using it, and how to price a supplier's cash discount: ICAI's simple and compound formulas and the refuse-or-accept comparison.

In plain words

"Free money", said Kesar

Tapri buys elaichi from a spice trader in Kochi on terms of 3% off if it pays within 15 days, otherwise the full amount within 60 days. Kesar always pays on day 60.

> **Kesar:** Trade credit is free money. The trader waits two months and charges nothing.
>
> **Dev:** He charges 3%. Whenever we skip the discount, we are borrowing ₹97 from him for 45 days and paying ₹3 for it.
>
> **Kesar:** 3% only. The bank charges 12%.
>
> **Dev:** The bank's 12% is for a year. His 3% is for 45 days. Same shop, very different price tag.

When a supplier delivers goods and lets you pay later, he is lending to you: that is **trade credit**, and the amount you owe him is a **payable** (a creditor). It arises on its own, out of ordinary buying, which is why it is called a **spontaneous** source of finance, and it often looks free.

It is free only when there is no discount for paying early. Once there is, every day you hold on to the supplier's money after the discount period costs you the discount. Turn that cost into a yearly rate and compare it with what your own money costs (the cash credit rate) or earns: if skipping the discount is dearer, pay early and take it; if it is cheaper, keep the cash and pay on the last day of the credit period, never later, because paying late costs goodwill and supplies.

Managing payables also means looking after the supplier. Slow payment creates ill-feeling, can disrupt supplies and spoils the firm's image.

Think of it like

Two prices at the sweet shop

The halwai says: "₹97 today, or ₹100 when you settle your account in six weeks." Settling later is not free: you are paying ₹3 to keep ₹97 for six weeks. Six weeks fit into a year more than eight times, so that ₹3 works out to well over twice a 12% bank loan: a very expensive way to borrow.

Key idea

Trade credit costs nothing only when no discount is lost. When there is a cash discount, convert it into an annual rate for the extra days of credit and compare it with the cost (or return) of your own funds: take the discount when forgoing it is dearer; otherwise pay on the last day of credit.

How it works

Why payables need managing

"If you can buy well then you can sell well": management of creditors and suppliers is just as important as the management of debtors (SM 9.98). **Trade creditor** is a spontaneous, short-term source of finance, in the sense that it arises from ordinary business transactions. But slow payment by you may create ill-feeling, your supplies could be disrupted, and it creates a bad image for your company; creditors are a vital part of effective cash management and should be managed carefully to enhance the cash position (SM 9.98). Used judiciously, trade credit reduces the burden that growth places on investment in working capital (SM 9.99).

Trade credit matters more than its informality suggests: ICAI says it contributes about one-third of the total short-term requirements, and credit obtained without much formality is called **open account trade credit** (SM 9.103). The other spontaneous sources, bills payable and accrued expenses, are in financing of working capital and bank credit.

Both sides of the ledger: costs of availing and of not availing

ICAI lists the costs on each side (SM 9.98–9.99), and the lists are asked as they stand:

Cost of availing trade credit	Cost of not taking trade credit
Price: the discount the firm forgoes, since it is available only for prompt payment; it can translate into a high implicit cost	**Impact of inflation**: borrowers gain over lenders, as a fixed amount paid later is worth less; later purchases are at higher prices anyway
Loss of goodwill: if credit is overstepped, suppliers may discriminate against delinquent customers when supplies are short; the effect depends on the relative market strengths	**Interest**: trade credit is a type of interest free loan, so failing to use it has an interest cost, higher when interest rates are higher
Cost of managing: administrative and accounting costs of managing creditors	**Inconvenience**: it may inconvenience a supplier geared to deferred payment
Conditions: suppliers may insist on a minimum order size, or on orders on a regular basis	

Pricing the discount: two formulas

The question is the cost of **not** taking the discount. ICAI gives the nominal annual cost first (SM 9.99):

$$\text{Nominal annual cost} = \frac{d}{100 - d} \times \frac{365 \text{ days}}{t}$$

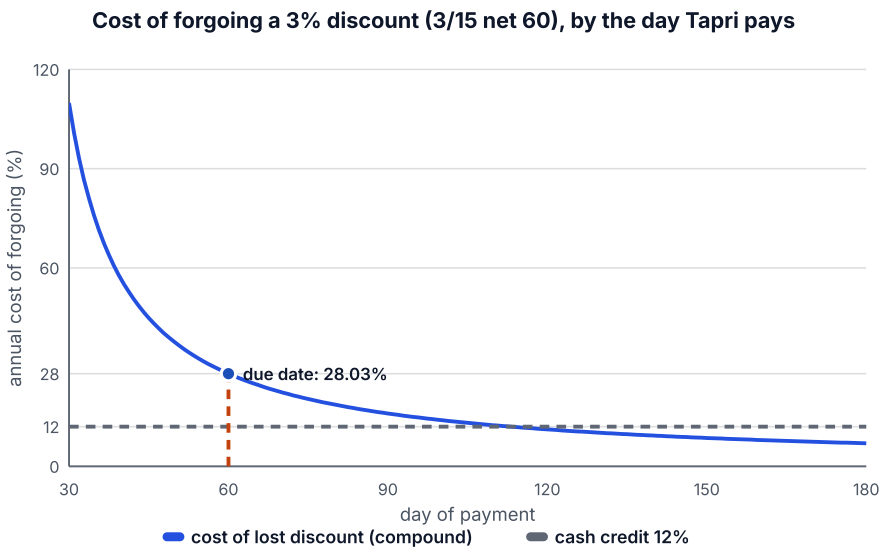
That formula ignores compounding, so the true cost is higher still; the **cost of lost cash discount** is estimated by (SM 9.99):

$$\text{Cost of lost cash discount} = \left(\frac{100}{100 - d}\right)^{365/t} - 1$$

Here **d** is the size of the discount (for 6%, d = 6) and **t** is the reduction in the payment period, in days, necessary to obtain the early discount: days credit outstanding − discount period (SM 9.99).

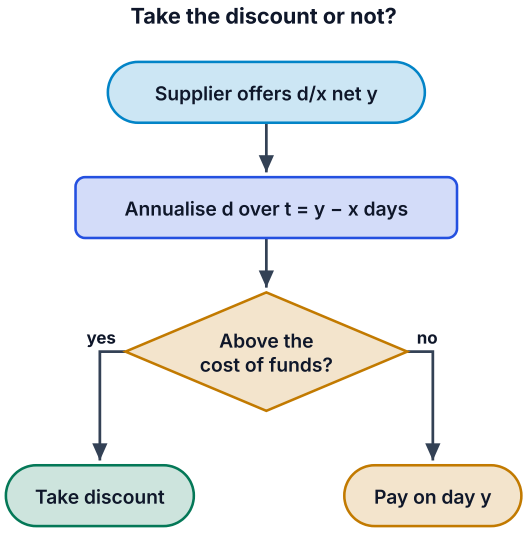
Why d ÷ (100 − d) and not d ÷ 100? Because the money you keep by not paying early is the discounted amount (₹97 on ₹100), and the price of keeping it is the discount (₹3). And why t and not the whole credit period? Because the first 15 days are free either way; only the extra days are bought with the discount.

The price falls as the payment day moves out, which is the temptation to pay late. For the elaichi trader's 3/15 net 60:



Blue: the compound cost of forgoing the discount, ((100 ÷ 97)^(365 ÷ (day − 15)) − 1). At the due date, day 60, it is 28.03%, far above the grey dashed 12% cash credit: take the discount. It sinks below 12% only after about day 113, long past the due date (orange dashed line): a figure reached only by stretching the supplier, at the price of his goodwill.

The decision



The cost of funds is the borrowing rate (if early payment needs a loan) or the return the cash would earn if kept. Green: pay by day x and take the discount; amber: use the full credit period, paying on day y and not later.

ICAI's two illustrations give three rules:

- **Compare like with like.** A 2% discount must not be compared with a 12% annual bank rate: the 2% is earned for a few days' earlier payment, while the 12% is for a year. Annualise the discount first (Illustration 18(a), SM 9.100–9.101).
- **If the discount is not taken, use the maximum credit period.** When bank finance is not available and the firm must rely on trade credit, it should pay on the last day of the credit period to reduce the annual cost (Illustration 18(b), SM 9.101).
- **The refuse-or-accept comparison.** For an invoice, compare the payment if the discount is accepted with the payment if it is refused, less the return from investing the discounted amount between the discount day and the due date (Illustration 17, SM 9.100). The layout is under Formats and formulas.

Formats and formulas

Reproduce both formulas and the comparison statement from memory.

∑ **Formula**

Nominal annual cost of not taking the discount (SM 9.99)

$$\frac{d}{100-d} \times \frac{365 \text{ days}}{t}$$

d = size of the discount (for 6%, d = 6); t = days credit outstanding − discount period. Multiply by 100 for a percentage.

∑ **Formula**

Cost of lost cash discount, with compounding (SM 9.99)

$$\left(\frac{100}{100-d}\right)^{365/t} - 1$$

Always above the nominal cost. Use 360 for 365 when the question says so.

∑ **Formula**

Funds needed to take a discount

$$\text{Funds} = \text{Purchases} \times \left(1 - \frac{d}{100}\right) \times \frac{t}{365}$$

Funds = the average extra funds tied up; Purchases = annual purchases. Paying t days earlier on every invoice ties up t days of discounted purchases. Net benefit = discount earned − cost of these funds.

The refuse-or-accept statement (ICAI Illustration 17 layout, SM 9.100)

Particulars	Refuse discount (₹)	Accept discount (₹)
Payment to supplier	invoice value	invoice value × (1 – d%)
Less: return from investing the discounted amount between the discount day and the due date	(xxx)	—
Net cost	**xxx**	**xxx**

The lower net cost wins. The return is on the discounted amount (₹98 out of ₹100), for t days, at the rate the firm can earn (or saves on borrowing).

Definition

Trade credit

A spontaneous source of finance normally extended to the purchaser organisation by the sellers or service providers; credit obtained without completing much formality is open account trade credit. (SM 9.103)

Definition

t in the discount formulas

The reduction in the payment period in days, necessary to obtain the early discount, or days credit outstanding – discount period. (SM 9.99)

Worked examples

Worked example **Four suppliers, one cash credit limit**

Tapri's premix division buys from four suppliers. The unused limit on its cash credit, which costs 12% a year, is ₹90 lakh; the division has no other spare funds. Assume 365 days in a year.

Supplier	Terms	Annual purchases (₹)
S (packaging)	1.5/10 net 30	3,65,00,000
P (cardamom)	2/12 net 50	5,47,50,000
Q (milk powder)	1/15 net 45	4,38,00,000
R (sugar)	1/10 net 90	2,92,00,000

At present the division takes no discounts and pays every supplier on the last day of credit.

Required:

(i) Calculate the nominal annual cost and the cost of lost cash discount (compound) of not taking each supplier's discount.

(ii) If funds were unlimited at 12%, which discounts should the division take?

(iii) With only ₹90 lakh available, which discounts should it take, and what is the annual net benefit? Comment.

THE WORKING

(i) Cost of not taking each discount

Supplier	t (days)	Nominal (%)	Compound (%)
S	20	27.79	31.76
P	38	19.60	21.42
Q	30	12.29	13.01
R	80	4.61	4.69

For S: nominal = 1.5 ÷ 98.5 × 365 ÷ 20 = 27.79%; compound = (100 ÷ 98.5)^(365 ÷ 20) – 1 = 31.76%. The others follow in the same way.

(ii) Unlimited funds at 12%. Take the discounts whose cost of forgoing exceeds 12%: **S, P and Q**. Do not take R's: forgoing it costs only about 4.6% a year, so its 80 extra days of credit are cheaper than the cash credit.

(iii) Funds needed and the benefit of each

Supplier	Extra funds (₹)	Discount earned (₹)
S	19,70,000	5,47,500
P	55,86,000	10,95,000
Q	35,64,000	4,38,000
R	63,36,000	2,92,000

Supplier	Interest at 12% (₹)	Net benefit (₹)
S	2,36,400	3,11,100
P	6,70,320	4,24,680
Q	4,27,680	10,320
R	7,60,320	(4,68,320)

Extra funds = annual purchases × (1 – d%) × t ÷ 365; for S, ₹3,65,00,000 × 0.985 × 20 ÷ 365 = ₹19,70,000.

Each rupee of headroom should go first where forgoing the discount is dearest: S (27.79%), then P (19.60%), then Q (12.29%). S and P together need ₹75,56,000, leaving ₹14,44,000 of the ₹90 lakh; Q would need ₹35,64,000.

Advice. Take the discounts of **S and P** in full: annual net benefit ₹3,11,100 + ₹4,24,680 = **₹7,35,780**. Q's discount earns only 0.29% a year above the cash credit rate: on the ₹14,44,000 left it would add about

₹4,181. That is not worth using up the last of the cash credit, which is better kept as the division's liquidity cushion. Never take R's discount on these rates.

Worked example **The potters' co-operative and the day-60 habit**

The Gwalior Kumhar Sahkari Samiti, a potters' co-operative, supplies Tapri's cafés with kulhads on terms of 1.5/7 net 30. Each monthly invoice is ₹5,00,000. The cafés' surplus cash is kept in a liquid fund yielding 9% a year. Kesar suggests paying on day 60 instead of day 30, because "the samiti never complains". Assume 365 days in a year.

Required:

(i) Calculate the nominal annual cost and the cost of lost cash discount if Tapri pays on day 30.

(ii) Recalculate both if it pays on day 60.

(iii) Prepare ICAI's comparison of net cost for one invoice: accept the discount, refuse and pay on day 30, refuse and pay on day 60.

(iv) Advise Tapri, including the factors the figures leave out.

(v) At what return on the liquid fund would refusing the discount and paying on day 30 cost the same as accepting it? Use simple interest.

THE WORKING

(i) Paying on day 30: t = 30 − 7 = 23 days.

$$\text{Nominal} = \frac{1.5}{98.5} \times \frac{365}{23} = 24.17\% \qquad \text{Compound} = \left(\frac{100}{98.5}\right)^{365/23} - 1 = 27.11\%$$

(ii) Paying on day 60: t = 60 − 7 = 53 days; nominal = 1.5 ÷ 98.5 × 365 ÷ 53 = **10.49%**; compound = (100 ÷ 98.5)^(365 ÷ 53) − 1 = **10.97%**.

(iii) Net cost of one invoice

The discounted amount is ₹5,00,000 × 98.5% = ₹4,92,500.

Columns: accept and pay on day 7; refuse and pay on day 30; refuse and pay on day 60.

Particulars	Accept (₹)	Day 30 (₹)	Day 60 (₹)
Payment	4,92,500	5,00,000	5,00,000
Less: return	—	(2,793)	(6,436)
Net cost	**4,92,500**	**4,97,207**	**4,93,564**

Return for 23 days = ₹4,92,500 × 9% × 23 ÷ 365 = ₹2,793; for 53 days, ₹6,436.

(iv) Advice. Accept the discount and pay within 7 days. Refusing costs ₹4,707 more per invoice if Tapri pays on day 30, and still ₹1,064 more even if it stretches to day 60, because 9% is below both 24.17% and 10.49%. Stretching to day 60 also oversteps the agreed credit: ICAI's costs of availing trade credit include the **loss of goodwill** (a supplier may discriminate against delinquent customers when supplies are short) and possible **conditions** on future supply. A small co-operative of potters depends on prompt payment; "never complains" is not the same as "does not mind".

(v) Break-even return (simple): ₹7,500 ÷ (₹4,92,500 × 23 ÷ 365) = **24.17%**, which is the nominal cost from (i). Refusing would pay only if the cafés could earn more than 24.17% a year, risk-free, on their cash.

Exam angle

In the exam

How payables appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, payables came up 5 times in 5 new-scheme documents:

- **Numerical.** MTP January 2026 Series II Q2(ii)–(iii), inside a 10-mark working capital question: the implied annual cost of forgoing a 2/15 net 60 discount (compound formula, 360 days) against the firm's return of 20%, then the refuse-or-accept statement for a ₹10,000 invoice. The firm refused the discount: its return was higher.
- **Theory, 2 to 4 marks.** "DISCUSS the costs of availing trade credit": MTP September 2024 Series I Q4(a) (4 marks), answered with ICAI's four costs word for word. "EXPLAIN the importance of trade credit and accruals as source of working capital. What is the cost of these sources?": RTP May 2024 Q8(ii). "SHOW the difference between trade credit and bank overdraft": RTP September 2025 Q9(b). Spontaneous sources of finance: RTP January 2026 Q11(b).
- **Old scheme, for practice:** RTP December 2021 Q8: the real annual cost of trade credit (3/30 net 90) compared with a bank loan with a compensating balance and factoring.

In the exam

Wording that decides the answer

- "Implied cost", "cost of lost cash discount": the compound formula. "Nominal cost": the simple one.
- "Net 60" means payment not later than day 60 (SM 9.110, MCQ 2); "3/15 net 60" means 3% off within 15 days, otherwise full payment within 60 days (SM 9.110, MCQ 1).
- Read whether the firm would borrow to pay early (compare with the borrowing rate) or would invest the cash if it pays late (compare with the return).

Trick

The rule of thumb for a quick check

The nominal cost ≈ d% × 365 ÷ t: 2% for 20 days is about 2 × 18 = 36% a year. Use it only to check the size of an answer; write ICAI's formula in the exam. Condition: small discounts (the d ÷ (100 − d) correction is then minor).

Answer frame

"Discuss the costs of availing trade credit" (4 marks)

Open with ICAI's sentence: "Normally it is considered that the trade credit does not carry any cost. However, it carries the following costs:" then the four, each with one line: **(i) Price** (forgone discount, a high implicit cost), **(ii) Loss of goodwill** (discrimination against delinquent customers when supplies are short), **(iii) Cost of managing** (administrative and accounting costs), **(iv) Conditions** (minimum order size or regular orders). If the question asks for both sides, add the three costs of not taking trade credit: inflation, interest, inconvenience.

"Show the difference between trade credit and bank overdraft" (RTP September 2025 Q9(b))

Point	Trade credit	Bank overdraft
Meaning	Credit extended by suppliers, allowing the firm to delay payment for goods or services	Customers may withdraw in excess of the credit balance in their current account
Duration	Usually 15 to 90 days	Repayable on demand, but generally continued for long periods by annual renewal of limits
Cost	A type of interest free loan; failure to use it has an interest cost	Interest charged on daily balances
How it arises	Generated automatically in the course of business	Borrower applies to the bank; operated like cash credit and current accounts
Size	Grows automatically with the volume of business	A fixed limit is granted within which the borrower may overdraw

"Explain the importance of trade credit and accruals, and their cost"

1. Credit from suppliers in the normal course of trade; a short-term source; small firms depend heavily on it.
2. Advantages: easy availability, flexibility, informality.
3. Not cost-free: the supplier's opportunity cost of funds in receivables is passed on as a higher price or as a cash discount not extended; the forgone discount is an implicit cost.
4. Accrued expenses (wages, salaries, taxes paid at the end of the period) are built-in and automatic, with no explicit or implicit cost (SM 9.103–9.104).

Common mistakes

⚠ **Common mistake**

Hall of Shame: the 2% that was compared with 12%

"The discount is only 2%; the bank charges 12%, so we skip it." The 2% buys a few days of credit, the 12% a whole year. **Fix:** annualise the discount over t days first, then compare.

⚠ **Common mistake**

t taken as the whole credit period

For 1.5/7 net 30, using t = 30 gives a nominal cost of 18.53% instead of 24.17%. **Fix:** t = days credit outstanding − discount period = 23; the first 7 days are free either way.

⚠ **Common mistake**

d ÷ 100 instead of d ÷ (100 – d)

The firm keeps the discounted amount, not the invoice value. **Fix:** 1.5 ÷ 98.5, not 1.5 ÷ 100; and in the comparison statement, the return is on ₹4,92,500, not on ₹5,00,000.

⚠ **Common mistake**

Stretching sold as a saving

Paying on day 60 on net-30 terms lowers the annualised cost on paper (10.49% instead of 24.17%) and the answer recommends it. **Fix:** a payment beyond the agreed period oversteps the credit; ICAI lists the loss of goodwill and supply discrimination as costs. If the discount is not taken, pay on the last day of credit, not later.

Practice

Check yourself

1 A supplier of paper cups offers 1.5/15 net 30. Using ICAI's nominal formula and 365 days, what is the annual cost of not taking the discount, in % (two decimals)?

Answer: _____

2 Case scenario. Tapri's cash credit costs 14% a year. Its packaging supplier, keen to win a bigger share of Tapri's orders, changes its terms from 2/10 net 30 to 2/10 net 60. Using ICAI's compound formula and 365 days, which of the following is correct?

- **A** Under the new terms Tapri should forgo the discount, since 2% for 50 days is cheaper than 14%
- **B** The discount is 2% under both terms, so the decision cannot change
- **C** Under the old terms Tapri should forgo the discount and under the new terms take it
- **D** Tapri should take the discount under both terms, although the new terms make forgoing it far cheaper

3 Which of the following is NOT one of ICAI's costs of availing trade credit?

- **A** Loss of goodwill, if the credit is overstepped
- **B** Impact of inflation
- **C** Conditions such as a minimum order size
- **D** Cost of managing creditors

4 For credit terms of 2/15 net 45, what is t in ICAI's formulas?

- **A** 45 days
- **B** 15 days
- **C** 30 days
- **D** 60 days

5 Which of these are spontaneous sources of finance in ICAI's classification? Select all that apply.

- A Trade credit
- B Bills payable
- C Commercial paper
- D Accrued expenses
- E Cash credit

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Trade creditor: a spontaneous short-term source arising from ordinary business; slow payment causes ill-feeling, disrupted supplies and a bad image (SM 9.98).
- Costs of availing trade credit: price (forgone discount), loss of goodwill, cost of managing, conditions.
- Costs of not taking trade credit: impact of inflation, interest, inconvenience.
- Nominal cost of not taking the discount = d ÷ (100 − d) × 365 ÷ t; cost of lost cash discount = (100 ÷ (100 − d))^(365/t) − 1; t = credit days − discount days.
- Take the discount when forgoing it costs more than the cost (or return) of your funds; otherwise pay on the last day of credit, not later.
- Refuse-or-accept: payment refused less the return on the discounted amount for t days, against the discounted payment.
- Never compare a period discount with an annual rate. (₹3 to keep ₹97 for six weeks is not cheap money.)

Answer key · Check yourself

1 **37.0 to 37.1**

t = 30 − 15 = 15 days; 1.5 ÷ 98.5 × 365 ÷ 15 × 100 = 37.06%. Using t = 30 gives 18.53%, the commonest slip.

2 **D Tapri should take the discount under both terms, although the new terms make forgoing it far cheaper**

Old terms, t = 20: (100 ÷ 98)^(365 ÷ 20) − 1 = 44.59%. New terms, t = 50: (100 ÷ 98)^(7.3) − 1 = 15.89%. Both exceed 14%, so take the discount; the longer credit period has cut the cost of forgoing it by almost two-thirds.

3 **B Impact of inflation**

Inflation appears on the other list: it is a cost of NOT taking trade credit, since under inflation borrowers gain from paying a fixed amount later (SM 9.99). The costs of availing are price, loss of goodwill, cost of managing and conditions (SM 9.98).

4 **C 30 days**

t = days credit outstanding − discount period = 45 − 15 = 30 days: the days of credit bought by forgoing the discount.

5 **A Trade credit**; **B Bills payable**; **D Accrued expenses**

Spontaneous sources arise naturally in the course of business: trade credit, bills payable, accrued expenses (SM 9.102–9.104). Commercial paper and cash credit have to be specifically negotiated.

9.7 Financing of working capital and bank credit

Where working capital money comes from: spontaneous and negotiated sources, commercial paper, ICDs, bills discounting and factoring; bank finance after MPBF; the seven forms of bank credit; and the effective cost of each source.

In plain words

Four doors to the same ₹2 crore

The festive season is ten weeks away, and Tapri's premix division needs ₹2 crore for 90 days to build stock.

> **Kesar:** Just take ₹2 crore from the bank. Money is money.
>
> **Dev:** Money has a price tag, and some tags are hidden in the small print. A bank loan that makes us keep part of it idle in a current account. Discounting our distributors' bills. Commercial paper. Skipping a supplier's discount.
>
> **Kesar:** Four doors, all leading to ₹2 crore?
>
> **Dev:** Four doors, four rents. I will calculate each one per rupee we can actually spend.

Once the finance manager knows how much working capital the business needs, the next step is to **arrange the funds**. First he splits the need in two. The part that is always there whatever the season, the **permanent working capital**, should come from long-term sources such as equity and long-term debt. The part that comes and goes with the season, the **temporary working capital**, can come from short-term sources.

Short-term sources are of two kinds. **Spontaneous** sources arise by themselves in the course of business: suppliers' credit, wages and taxes not yet paid. **Negotiated** sources have to be arranged with a lender: loans from other companies, commercial paper, public deposits, bill discounting, factoring and, above all, **bank credit**.

Banks are the biggest lenders of working capital. For years they assessed a borrower's need under guidelines evolved by two committees (Tandon and Chore), using a concept called maximum permissible bank finance; RBI withdrew it in 1997, and each bank now uses its own Board-approved method. Bank credit comes in seven forms, from the cash credit (draw what you need up to a limit) to the bank guarantee (the bank promises a third party on your behalf).

The price of each source is not just its quoted rate: it is what you pay divided by what you can actually use.

Think of it like

Paying for a family wedding

Some of the money comes on its own: the caterer will be paid after the function, the relatives' help asks no interest (spontaneous). The rest has to be asked for: a loan from a cousin's business, a gold loan, a credit card you draw on as bills arrive and repay when the gifts come in. A rich uncle might take your signed IOU, but only because your family's name is excellent: that is commercial paper. And a loan that makes you keep a fixed deposit with the same bank is dearer than its rate says.

Key idea

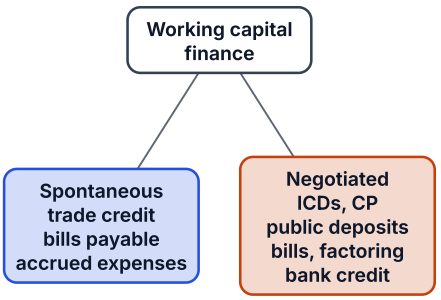
Finance permanent working capital from long-term sources and temporary working capital from short-term ones. Among short-term sources, compare the effective cost per rupee you can actually use, and weigh credit rating, feasibility, reliability and restrictions before choosing.

How it works

Two needs, two kinds of source

After determining the amount of working capital required, the next step is to arrange the funds (SM 9.102). The finance manager should bifurcate the requirement between **permanent** working capital, always needed irrespective of sales fluctuations and hence financed by long-term sources such as debt and equity, and **temporary** working capital, which may be financed by short-term sources (SM 9.102). This is the matching (hedging) approach of working capital and its financing policy.

Sources of working capital finance (SM 9.102–9.106)



Blue: spontaneous sources, which arise naturally in the course of business. Orange: negotiated sources, which have to be specifically negotiated with lenders (banks, financial institutions, the public).

Spontaneous sources naturally arise in the course of business operations: trade credit, credit from employees, credit from suppliers of services. **Negotiated sources** have to be specifically negotiated with lenders, say commercial banks, financial institutions or the general public (SM 9.102). The choice of a source, or a combination, is guided by (SM 9.102–9.103): (i) cost factor; (ii) impact on credit rating; (iii) feasibility; (iv) reliability; (v) restrictions; (vi) the hedging or matching approach, i.e. financing assets with the same maturity as the assets.

The sources, one by one

Source	What ICAI says (SM 9.103–9.106)
Trade credit	Extended by sellers or service providers; about one-third of total short-term requirements; credit without much formality is open account trade credit; see management of payables
Bills payable	The purchaser gives a written promise to pay on demand or at a fixed future date; simple, easily available, lesser explicit cost; helpful to small and medium enterprises
Accrued expenses	Services availed but not yet paid for (wages, salaries, taxes, duties); built-in, automatic, interest free, with no explicit or implicit cost
Inter-corporate loans and deposits	Firms with surplus funds lend for a short period; rate higher than the bank rate, depending on the borrower's financial soundness; reduces dependence on banks
Commercial paper	An unsecured promissory note for a short period, for highly rated borrowers; details below
Funds from operations	Profit and depreciation; working capital increases by the funds generated
Public deposits	For well-established big companies with a huge capital base; short and medium term
Bills discounting	The supplier draws a bill of exchange on the buyer, gets it accepted, and discounts it
Bill rediscounting scheme	Introduced by RBI from 1 November 1970 to develop a bill market; all licensed scheduled banks may offer eligible bills to RBI for rediscount
Factoring	The firm sells its trade debts at a discount to a factor, with or without recourse; see management of receivables

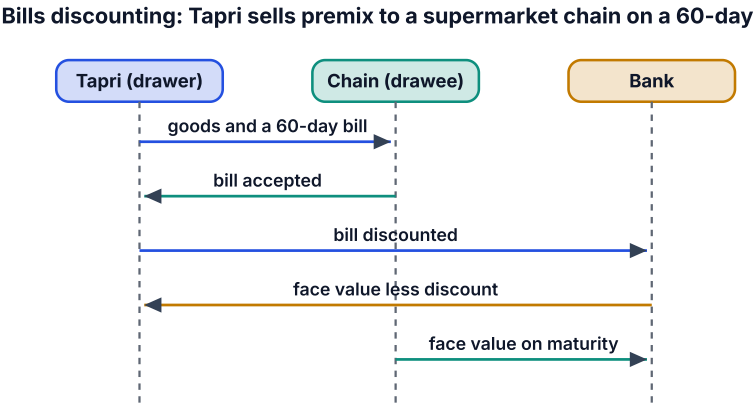
Commercial paper (CP) is an unsecured promissory note issued by a firm to raise funds for a short period. It enables highly rated corporate borrowers to borrow short term and gives investors an instrument with a freely negotiable interest rate. Maturity ranges from a minimum of **7 days to less than 1 year** from the date of issue; it is issued in denominations of **₹5 lakh or multiples** (SM 9.104).

Advantages to the issuer (SM 9.104)	Limitations (SM 9.104–9.105)
(a) Sold on an unsecured basis, with no restrictive conditions	(i) Only highly rated firms can use it; new and moderately rated firms generally cannot issue CP
(b) Maturing CP can be repaid by selling new CP: a continuous source of funds	(ii) CP can neither be redeemed before maturity nor extended beyond maturity
(c) Maturity can be tailored to the issuer's requirement	
(d) It can be issued even when the money market is tight	
(e) Its cost is generally lower than that of commercial bank loans	

⚠ **Careful**

CP denominations, maturities and the bill rediscounting date are stated as the Study Material gives them. Verify against the Study Material and Statutory Update for your attempt.

Bills discounting in pictures: the seller is paid now, the bank waits.



The bank earmarks a bill discounting limit for Tapri (SM 9.107); the discount is the bank's charge for paying 60 days early.

Bank finance: from MPBF to the bank's own method

Banks in India constitute the major suppliers of working capital credit; some term-lending institutions also offer it. Two committees, the **Tandon Committee** and the **Chore Committee**, evolved the guidelines and parameters that laid the foundations of working capital financing (SM 9.106). What applies now (SM 9.106–9.107):

1. The Reserve Bank of India **withdrew** the prescription for assessing working capital needs on the basis of **Maximum Permissible Bank Finance (MPBF)** in **April 1997**.
2. Banks are now free to evolve, **with the approval of their Boards**, methods for assessing the working capital requirements of borrowers, **within the prudential guidelines and exposure norms** prescribed.
3. They must still take into account RBI's instructions on **directed credit** (priority sector, export, etc.) and on **prohibition of credit** (bridge finance, rediscounting of bills earlier discounted by NBFCs) while framing their lending policies.
4. All instructions relating to MPBF stand withdrawn; the guidelines on lending discipline in appraisal, sanction, monitoring and utilisation cease to be mandatory, though banks may incorporate those they consider necessary in their own policies.

> ⚠ **Careful**
>
> RBI norms for bank credit are version-sensitive. Write them as the Study Material states them, and verify against the Study Material and Statutory Update for your attempt.

A bank's own method can still be stated as a percentage: in the Study Material's case scenario the bank grants 75% of net current assets as advance against working capital, and asks for 10% of the loan to be kept as a minimum average balance in a current account (SM 9.127). ICAI's answer still labels that 75% "Maximum Permissible Bank Finance" (SM 9.156), so an MCQ that asks for MPBF wants the question's own stipulation applied. The rest of the net current assets, the borrower's **margin**, is not bank-financed: it is met from long-term funds.

The seven forms of bank credit

In ICAI's order (SM 9.107–9.108):

Form	ICAI's description
Cash credit	Credit facility given on a continuous basis; the borrower is not allowed to exceed the limits sanctioned
Bank overdraft	Short-term borrowing for urgent needs, within limits; quickly and easily repaid when no longer required; the bank can call it in at short notice
Bills discounting	The seller draws a bill on the buyer, who accepts it; the seller discounts it with his banker, who earmarks a discounting limit
Bills acceptance	The company draws a bill of exchange on the bank; the bank accepts it, promising to pay the amount at a specified future date
Line of credit	A commitment by a bank to lend a certain amount of funds on demand, specifying the maximum amount
Letter of credit	The issuing bank, on a customer's instructions or on its own behalf, undertakes to pay, accept or negotiate (or authorises another bank to do so) against stipulated documents, subject to the specified terms and conditions
Bank guarantee	A facility the bank extends on behalf of its client in favour of a third party, the beneficiary of the guarantee

Chapter 2 of the Study Material adds the bank's other short-term facilities (short-term loans, clean overdrafts, advances against goods, bills purchased, export finance): short-term sources of finance. A cash credit, for instance, charges interest only on the amount actually availed, not on the full limit, and is usually secured by pledge or hypothecation of goods (FM Study Material 2.27).

The price per usable rupee

Every short-term source is compared on one yardstick, the **effective annual cost**: what the source costs in a year, divided by the funds the firm can actually use.

- **A loan with a minimum balance.** If part of the loan must sit idle in a current account, the interest is paid on the whole loan but only the rest is usable. The Study Material's case: a 16.5% loan with 10% kept as a minimum balance costs 16.5 ÷ 90% = **18.33%** (SM 9.156). Only a balance the firm **would not otherwise maintain** counts; ICAI's old-scheme question says so in those words (RTP December 2021 Q8).
- **A discounted instrument** (a discounted bill, commercial paper): the charge is deducted at the start, so the cost is the charge divided by the **net proceeds**, annualised for the days it runs.
- **Trade credit**: the cost of the discount forgone (management of payables).
- **Factoring**: the net cost (commission and interest less savings) on the advance received (management of receivables).

Formats and formulas

Reproduce the effective cost formulas and the comparison layout from memory.

Σ Formula

Loan with a compensating (minimum) balance

$$\text{Effective cost} = \frac{\text{Interest rate}}{1 - b}$$

b = the minimum balance as a fraction of the loan, counted only if the firm would not otherwise keep it. SM 9.156: 16.5% ÷ (1 − 0.10) = 18.33%. Amount to borrow for a usable sum S: S ÷ (1 − b).

Σ Formula

Discounted bill or commercial paper

$$\text{Cost} = \frac{\text{Discount} + \text{Charges}}{\text{Net proceeds}} \times \frac{365}{n}$$

Cost = effective annual cost; n = days to maturity; net proceeds = face value − discount − charges. Never divide by the face value.

Σ Formula

Bank finance as a share of net current assets (SM case scenario)

$$\text{Bank finance} = 75\% \times \text{NCA}$$

$$\text{Margin} = \text{NCA} - \text{Bank finance}$$

NCA (net current assets) = current assets − current liabilities other than the bank borrowing. The percentage is the bank's stipulation in the question.

Comparison statement of short-term sources

Source	Effective annual cost
Bank loan with minimum balance	Rate ÷ (1 − b)
Bills discounting	Discount and charges ÷ net proceeds × 365 ÷ n
Commercial paper	(Face − net proceeds) ÷ net proceeds × 365 ÷ n
Trade credit (discount forgone)	d ÷ (100 − d) × 365 ÷ t

Rank the sources, choose the cheapest that is feasible, and comment on rating, reliability and restrictions.

Definition

Commercial paper

An unsecured promissory note issued by a firm to raise funds for a short period; an instrument that enables highly rated corporate borrowers to borrow short term. (SM 9.104)

Definition

Line of credit

A commitment by a bank to lend a certain amount of funds on demand, specifying the maximum amount. (SM 9.108)

Definition

Letter of credit

An arrangement by which the issuing bank, on the instructions of a customer or on its own behalf, undertakes to pay or accept or negotiate, or authorizes another bank to do so, against stipulated documents subject to compliance with specified terms and conditions. (SM 9.108)

Definition

Bank guarantee

A facility that commercial banks extend on behalf of their clients in favour of third parties, who are the beneficiaries of the guarantees. (SM 9.108)

Worked examples

Worked example **₹2 crore for 90 days: four sources compared**

Tapri's premix division needs about ₹2 crore of usable funds for 90 days to build festive stock. Tapri is a listed company and its rating makes it eligible to issue commercial paper. Four options are available (assume 365 days):

1. **Bank loan** at 11% a year. The bank requires a minimum balance of 15% of the loan in a current account, which Tapri would not otherwise keep.
2. **Forgo a supplier's discount.** The main ingredient supplier offers 1.5/10 net 50, and Tapri usually takes the discount; paying on day 50 instead would provide the funds.
3. **Bills discounting.** Distributors accept 90-day bills; the bank discounts them at 10% a year on the face value and charges a processing fee of 0.25% of the face value.
4. **Commercial paper** for 91 days, issued at ₹98 per ₹100 of face value; rating, issuing and paying agent and stamp costs come to ₹0.20 per ₹100 of face value.

Required:

(i) Calculate the effective annual cost of each option.

(ii) Calculate the amount Tapri must borrow under option 1, and the face value of CP it must issue under option 4, to have ₹2 crore to use.

(iii) Rank the options and advise, commenting on the factors other than cost.

THE WORKING

(i) Effective annual cost

Option 1, bank loan: 11% ÷ (1 − 0.15) = **12.94%**.

Option 2, forgone discount, t = 50 − 10 = 40 days: 1.5 ÷ 98.5 × 365 ÷ 40 = **13.90%** nominal (compound: (100 ÷ 98.5)^(365 ÷ 40) − 1 = 14.79%).

Option 3, bills discounting, per ₹100 of face value: discount = 100 × 10% × 90 ÷ 365 = ₹2.466; fee ₹0.25; net proceeds = ₹97.284.

$$\text{Cost} = \frac{2.466 + 0.25}{97.284} \times \frac{365}{90} \qquad = 11.32\%$$

Option 4, commercial paper, per ₹100 of face value: net proceeds = ₹98 − ₹0.20 = ₹97.80.

$$\text{Cost} = \frac{100 - 97.80}{97.80} \times \frac{365}{91} \qquad = 9.02\%$$

(ii) Amounts

- Bank loan: ₹2,00,00,000 ÷ 0.85 = **₹2,35,29,412**, of which ₹35,29,412 sits idle; interest for 90 days = ₹2,35,29,412 × 11% × 90 ÷ 365 = ₹6,38,195.
- CP: ₹2,00,00,000 ÷ 0.978 = ₹2,04,49,898 of face value; CP is issued in multiples of ₹5 lakh, so Tapri issues **₹2,05,00,000** (41 lots), receiving ₹2,05,00,000 × 0.978 = ₹2,00,49,000 and repaying ₹2,05,00,000 on maturity.

(iii) Ranking and advice

Rank	Source	Effective annual cost
1	Commercial paper	9.02%
2	Bills discounting	11.32%
3	Bank loan with 15% minimum balance	12.94%
4	Forgone supplier discount	13.90%

Advice. Issue commercial paper: it is the cheapest by about 2.3 percentage points, unsecured and free of restrictive conditions (SM 9.104). Points to weigh:

- CP can neither be redeemed before maturity nor extended beyond it (SM 9.105): if festive sales come in early, Tapri cannot repay early; if they come in late, it must roll over by a fresh issue or fall back on the bank. A bank cash credit, where interest runs only on the amount used, is the natural standby.
- Bills discounting, the second choice, depends on distributors agreeing to accept bills, and it is limited by the bank's discounting limit.
- The bank loan's quoted 11% is really 12.94%, because ₹35 lakh must lie idle.
- Forgoing the discount is the dearest, and it raises the price of every purchase while it lasts.

Worked example

Narmada Bank assesses Tapri Cold Chain

Tapri Cold Chain Pvt Ltd, Tapri's refrigerated logistics subsidiary, applies to Narmada Bank for working capital finance. Its projected current assets and current liabilities (other than bank borrowing) are:

Particulars	Amount (₹)
Current assets	
Inventory of spares and fuel	1,80,00,000
Receivables	2,40,00,000
Cash and bank	20,00,000
Prepaid expenses	10,00,000
Current liabilities	
Trade creditors	90,00,000
Outstanding expenses	30,00,000
Advances from customers	10,00,000

The bank will grant 75% of net current assets as working capital finance, at 12% a year, on condition that 12.5% of the loan is kept as a minimum average balance in a current account with it.

Required:

(i) Calculate the working capital finance the bank will grant and the margin the company must find.

(ii) State how the margin should be financed.

(iii) Calculate the effective cost of the bank finance, assuming the minimum balance is entirely extra.

(iv) The company's own operating cash of ₹20,00,000 can be kept in this current account. Recalculate the effective cost.

(v) At its peak the company needs ₹2,20,00,000 of bank funds it can actually spend. Is the bank's finance enough under (iii) and under (iv)?

THE WORKING

(i) Bank finance and margin

Particulars	Amount (₹)
Current assets	4,50,00,000
Less: current liabilities (other than bank)	1,30,00,000
Net current assets	**3,20,00,000**
Bank finance at 75%	**2,40,00,000**
Margin (25%)	**80,00,000**

(ii) The margin is the part of net current assets the bank will not fund. The current assets are financed by the current liabilities, the bank finance and long-term funds; with the first two already counted, the margin of ₹80,00,000 must come from long-term sources such as equity or long-term debt, the sources the Study Material assigns to the permanent part of working capital (SM 9.102).

(iii) Minimum balance entirely extra

Minimum balance = 12.5% × ₹2,40,00,000 = ₹30,00,000; usable funds = ₹2,10,00,000; interest = ₹2,40,00,000 × 12% = ₹28,80,000.

$$\text{Effective cost} = \frac{28{,}80{,}000}{2{,}10{,}00{,}000} = 13.71\% \qquad \left(= \frac{12\%}{1 - 0.125}\right)$$

(iv) Own cash counted towards the balance

The company would keep ₹20,00,000 anyway, so only ₹10,00,000 is extra idle money; usable funds = ₹2,40,00,000 − ₹10,00,000 = ₹2,30,00,000.

$$\text{Effective cost} = \frac{28{,}80{,}000}{2{,}30{,}00{,}000} = 12.52\%$$

(v) Is the finance enough? Under (iii) the usable funds are ₹2,10,00,000, short of the ₹2,20,00,000 peak by **₹10,00,000**, which the company must find elsewhere (or the bank must enhance the limit). Under (iv) they are ₹2,30,00,000, which covers the peak with ₹10,00,000 to spare.

Comment. A compensating balance raises the cost, and cuts the usable funds, only to the extent the borrower would not otherwise maintain it. The effect of the extra balance on the assessment itself has been ignored.

Worked example **Four needs, four forms of bank credit**

Kesar has four requests for Narmada Bank this year:

(a) Indian Railways has invited tenders for kulhad cafés at 30 stations; the tender requires a performance security in the railway's favour, payable if Tapri fails to perform.

(b) Tapri is importing a tea-packing line from Germany; the German supplier will ship only if a bank undertakes to pay against the shipping documents.

(c) The cafés' working capital need swings between ₹2 crore and ₹6 crore over the year; Tapri wants to draw only what it needs, within a limit, and repay as cash comes in.

(d) A supermarket chain buys premix on 60-day bills of exchange, but Tapri wants the money now.

Required: Advise the form of bank credit for each request, with reasons.

MODEL ANSWER

(a) Bank guarantee. A bank guarantee is a facility that commercial banks extend on behalf of their clients in favour of third parties, who are the beneficiaries of the guarantees (SM 9.108). Here the bank would issue the guarantee on Tapri's behalf in favour of Indian Railways, the beneficiary, which can claim on it if Tapri fails to perform; no funds are lent unless the guarantee is invoked.

(b) Letter of credit. Under a letter of credit the issuing bank, on the instructions of its customer, undertakes to pay or accept or negotiate, or authorises another bank to do so, against stipulated documents, subject to compliance with specified terms and conditions (SM 9.108). Narmada Bank, on Tapri's instructions, would undertake to pay the German supplier against the shipping documents, giving the supplier the assurance it needs.

(c) Cash credit. A cash credit is a credit facility given on a continuous basis, within a sanctioned limit that the borrower is not allowed to exceed (SM 9.107); the borrower draws only to the extent of his requirements and deposits surplus funds, and interest is charged on the amount actually availed (FM Study Material 2.27). A fluctuating need of ₹2 crore to ₹6 crore suits a cash credit limit of about ₹6 crore. (If Tapri wanted only a standby commitment, a **line of credit**, a commitment by the bank to lend up to a specified maximum on demand, would serve.)

(d) Bills discounting. Tapri draws a bill on the chain, which accepts it; Tapri then discounts the bill with its banker, which will generally earmark a discounting limit (SM 9.107). Tapri receives the face value less the discount at once, and the bank collects from the chain on maturity.

Exam angle

In the exam

How financing of working capital appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, it was examined 6 times in 6 new-scheme documents:

- **Forms of bank credit.** May 2024 Q4(b) (4 marks): "describe any four forms of bank credit"; the suggested answer listed the seven of SM 9.107–9.108 and added short-term loans from Chapter 2.
- **MPBF and its cost.** RTP May 2024 Q1 (a case, MCQ (v)): the Study Material's case scenario, 75% of net current assets and an annualised cost of 16.5 ÷ 90% = 18.33%.
- **Spontaneous sources.** RTP January 2026 Q11(b): meaning and the three sources. RTP September 2025 Q9(b): trade credit against bank overdraft, five points.
- **Bank finance in other chapters.** May 2026 Q4(b) (4 marks): identify post-shipment finance and explain three forms. RTP May 2025 Q9: a letter of credit cuts the export debtors' period in a working capital estimate.
- **Old scheme, for practice:** May 2022 Q1(b) and RTP May 2023 (Tandon's three methods); RTP December 2021 Q8 (bank loan with a 10% compensating balance, 15 ÷ 90 = 16.67%, against trade credit and factoring); four MTPs on the features of commercial paper.

In the exam

Traps

- **Usable funds, not loan.** Divide interest by the funds the firm can use; a compensating balance counts only if it would not otherwise be kept.
- **Net proceeds, not face value**, for any discounted instrument.
- **MPBF is withdrawn.** A theory answer that presents Tandon's norms as current RBI rules contradicts the Study Material.
- **ICAI's order** of the seven forms: cash credit, bank overdraft, bills discounting, bills acceptance, line of credit, letter of credit, bank guarantee.

Trick

Divide by what you can spend

Any source whose price is deducted up front or whose funds are partly locked away: effective cost = annual cost in rupees ÷ rupees you can actually spend. Condition: compare all sources over the same year basis (365 or 360, as the question says).

Answer frame

"Describe the forms of bank credit" (4 marks, any four)

Write each form with ICAI's one-line description, in ICAI's order: cash credit (continuous facility within a sanctioned limit); bank overdraft (short-term, urgent needs, within limits, callable at short notice); bills discounting (seller discounts the accepted bill; discounting limit); bills acceptance (company draws on the bank, which accepts it); line of credit (commitment to lend on demand up to a maximum); letter of credit (bank undertakes to pay against stipulated documents); bank guarantee (on behalf of the client in favour of a third-party beneficiary). Four well-explained forms earn the marks; ICAI's answer also accepted short-term loans.

"Explain commercial paper: its advantages and limitations"

1. Definition: an unsecured promissory note for a short period; for highly rated corporate borrowers; freely negotiable interest rate.
2. Maturity 7 days to less than 1 year; denominations of ₹5 lakh or multiples.
3. Advantages (a) to (e) in ICAI's order: unsecured, no restrictive conditions; rollover; tailored maturity; available in a tight money market; cheaper than bank loans.
4. Limitations: only highly rated firms; neither redeemable before nor extendable beyond maturity.

"Assessment of working capital by banks after 1997"

The four points of SM 9.106–9.107 in order: MPBF withdrawn (April 1997); banks' own Board-approved methods within prudential guidelines and exposure norms; RBI's directed credit and prohibition of credit still apply; MPBF instructions withdrawn and lending-discipline guidelines optional.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the 11% loan that cost 11%

Quoting the bank loan at its headline rate when 15% of it must lie idle. **Fix:** 11% ÷ 0.85 = 12.94%, and borrow ₹2,35,29,412 to have ₹2 crore to use.

⚠ **Common mistake**

Discount divided by the face value

For a bill or CP, dividing the discount by the face value understates the cost (2.20 ÷ 100 × 365 ÷ 91 = 8.82% instead of 9.02%). **Fix:** divide by the net proceeds.

⚠ **Common mistake**

MPBF written as today's rule

"Banks lend 75% of the working capital gap as per RBI's Tandon norms." **Fix:** RBI withdrew MPBF in April 1997; banks use their own Board-approved methods. Use a percentage only when the question stipulates it.

⚠ **Common mistake**

The CP that was repaid early

Planning to retire commercial paper as soon as cash comes in. **Fix:** CP can neither be redeemed before maturity nor extended beyond it; plan the maturity to the need, with a bank limit as standby.

Practice

Check yourself

1 A bank lends at 12% a year and insists that 20% of the loan be kept as a minimum balance in a current account the firm would not otherwise maintain. What is the effective annual cost of the loan, in %?

Answer: _____

2 Case scenario. Narmada Bank's new credit officer tells Tapri Cold Chain that RBI's maximum permissible bank finance norms of the Tandon Committee bind the bank, so it cannot lend more than those norms allow. Tapri's CFO disagrees. As per the Study Material, which of the following is correct?

A The officer is right: MPBF is the RBI norm for all working capital assessment

B The officer is right, but only for borrowers in the priority sector

C The CFO is right: RBI withdrew MPBF in April 1997, and banks assess working capital by Board-approved methods within prudential guidelines and exposure norms

D Both are wrong: banks may now lend without any RBI instructions at all

3 Which of the following is NOT an advantage of commercial paper to the issuing company, per ICAI?

- A It is sold on an unsecured basis and contains no restrictive conditions
- B Maturing CP can be repaid by selling new CP
- C It can be redeemed before maturity when the company has surplus funds
- D It can be issued even when the money market is tight

4 Tapri discounts a ₹10,00,000 bill due in 73 days. The bank's discount and charges total ₹20,000. Using 365 days, what is the effective annual cost?

- A 10.00%
- B 10.20%
- C 2.04%
- D 2.00%

5 Which of these are forms of bank credit in ICAI's list (SM 9.107–9.108)? Select all that apply.

- A Cash credit
- B Commercial paper
- C Letter of credit
- D Bank guarantee
- E Public deposits

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Permanent working capital comes from long-term sources (debt, equity), temporary working capital from short-term sources (SM 9.102).
- Spontaneous sources arise naturally (trade credit, bills payable, accrued expenses); negotiated sources are arranged with lenders.
- Choice of source: cost, impact on credit rating, feasibility, reliability, restrictions, matching of maturities.
- Commercial paper: unsecured promissory note, highly rated firms, 7 days to less than 1 year, ₹5 lakh or multiples; cheaper than bank loans; cannot be redeemed early or extended.
- Tandon and Chore Committees set the old norms; RBI withdrew MPBF in April 1997; banks now use Board-approved methods within prudential guidelines and exposure norms.
- Seven forms of bank credit: cash credit, overdraft, bills discounting, bills acceptance, line of credit, letter of credit, bank guarantee.

- Effective cost = annual cost ÷ usable funds: a loan with a minimum balance costs rate ÷ (1 − balance %); a discounted bill or CP costs its charge ÷ net proceeds, annualised. (Four doors, four rents: read the small print.)

Answer key · Check yourself

1 **15**

12% ÷ (1 − 0.20) = 15%. The interest is paid on the whole loan but only 80% of it can be used.

2 **C The CFO is right: RBI withdrew MPBF in April 1997, and banks assess working capital by Board-approved methods within prudential guidelines and exposure norms**

SM 9.106–9.107. Banks are free to evolve their own methods, but must still follow RBI's instructions on directed credit and prohibition of credit, so "without any RBI instructions" is wrong too.

3 **C It can be redeemed before maturity when the company has surplus funds**

CP can neither be redeemed before maturity nor extended beyond maturity: one of its two limitations (SM 9.105). The other three are among its advantages (SM 9.104).

4 **B 10.20%**

Net proceeds = ₹9,80,000; cost = 20,000 ÷ 9,80,000 × 365 ÷ 73 = 10.20%. Dividing by the face value gives 10.00%; the 2% figures are not annualised.

5 **A Cash credit; C Letter of credit; D Bank guarantee**

The seven forms: cash credit, bank overdraft, bills discounting, bills acceptance, line of credit, letter of credit, bank guarantee. Commercial paper and public deposits are negotiated sources raised from the market and the public, not bank credit.